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Singularities in global pluripotential theory

– Lectures at Zhejiang University –

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Preface

This book is an expanded version of my lecture notes at the Institute for Advanced Study in Mathematics (IASM) at Zhejiang University. My initial goal was modest: to write a self-contained reference for the participants of the lectures. But I soon realized that many results circulating in the area had never been rigorously proved anywhere in the literature. Each attempt to resolve these loose ends produced new ones, and the notes grew steadily until they became the present book.

Three pluripotential theories.

The aim of this book is to present, from a unified point of view, three different but interrelated theories that all deserve to be called a *global pluripotential theory*:

(1) the pluripotential theory on compact Kähler manifolds;
(2) the pluripotential theory on the Berkovich analytification of a projective variety;
(3) the toric pluripotential theory on a projective toric variety.

The three theories are bound together by a common circle of ideas. Many results were originally proved in one setting and have analogues, sometimes evident and sometimes far from evident, in the others.

The central question: classifying singularities.

Let us begin with the analytic side. Fix a compact Kähler manifold X. The central objects are the *quasi-plurisubharmonic functions* on X — functions which are locally the sum of a smooth function and a plurisubharmonic function. We are mostly interested in their *singularities*: the locus where $\varphi = -\infty$ and the rate at which φ tends to $-\infty$ near this locus.

Singularities are not pathological exceptions; they are forced upon us by geometry. In typical applications, X is the compactification of the moduli space of some class of geometric objects, with a Zariski-open locus $U \subseteq X$ parameterizing smooth objects. The natural metric on the associated polarizing line bundle is then smooth only on U, and the Grauert–Remmert extension theorem (Theorem B.2.2) extends it to all of X at the cost of introducing singularities along $X \setminus U$. Understanding such singularities is therefore not an aesthetic refinement but a prerequisite for any moduli-theoretic application.

The local theory of these singularities is intricate, and a complete classification is essentially out of reach: locally, (quasi-)plurisubharmonic functions exhibit extremely complicated behavior, and the natural local invariants (Lelong numbers and multiplier ideal sheaves) capture only their rough features. For example, a function with log-log singularities has the same local invariants as a bounded one.

The picture changes drastically when one passes to the global setting. On a compact manifold, there are three natural equivalence relations on $\mathrm{QPSH}(X)$, of decreasing fineness:

(1) the *singularity-type* relation $\sim$, identifying φ, ψ whose difference $\varphi - \psi$ is bounded;
(2) the *P-equivalence* $\sim_P$, two potentials with $\varphi \leq \psi$ are identified if the mixed non-pluripolar Monge–Ampère masses

$$\int_X \omega_\varphi^j \wedge \omega_\phi^{n-j} = \int_X \omega_\psi^j \wedge \omega_\phi^{n-j}, \qquad j = 0, \dots, n,$$

agree against every reference potential ϕ. Here ω is a sufficiently large Kähler form. In other words, φ and ψ behave similarly with respect to the global non-pluripolar mass (Definition 6.1.1);
(3) the *$\mathcal{I}$-equivalence* $\sim_{\mathcal{I}}$, identifying φ, ψ for which the multiplier ideal sheaves agree, $\mathcal{I}(k\varphi) = \mathcal{I}(k\psi)$ for every $k \in \mathbb{Z}_{>0}$ — equivalently, φ and ψ have the same generic Lelong numbers along every prime divisor over every modification of X (Definition 6.1.2).

The first relation records the finest information one can hope to capture by global invariants; $\sim_P$ retains the global *mass* side, while $\sim_{\mathcal{I}}$ retains the algebraic-geometric side. They are studied systematically in Chapter 6.

A natural plan to classify singularities is the three-step decomposition:

(1) classify the $\mathcal{I}$-equivalence classes;
(2) classify the P-equivalence classes inside a given $\mathcal{I}$-class;
(3) classify the singularity types inside a given P-class.

Step 3 has been the subject of much activity in the past decade under the name of *pluripotential theory with prescribed singularities*, and is by now reasonably well understood. The first goal of this book is to give a complete answer to Step 1, and a satisfactory answer to Step 2 in the cases that arise in geometry.

$\mathcal{I}$-good singularities.

Step 1 is the subject of Chapter 7. We characterize $\mathcal{I}$-equivalence classes very explicitly through approximations by an explicit class of singularities, known as *analytic singularities*; the analogous problem of approximating a plurisubharmonic function by ones with analytic singularities was studied in detail by Demailly.

The class of $\mathcal{I}$-good potentials is the central protagonist of this book. In a single equivalence theorem (Theorem 7.1.1), $\mathcal{I}$-goodness admits three intuitive descriptions, each capturing a different facet of what it means for a singularity to be "well-behaved." In the non-degenerate case, a quasi-plurisubharmonic function φ with $\theta_\varphi = \theta + \mathrm{dd^c}\varphi \geq 0$ for some closed smooth real $(1,1)$-form θ is $\mathcal{I}$-good if and only if any of the following hold:

(1) φ is the limit, with respect to the d_S-pseudometric introduced in Chapter 6, of a sequence of potentials with analytic singularities. (*Topological description*: $\mathcal{I}$-good potentials are precisely those reachable as limits of the explicit, well-understood class of analytic singularities.)

(2) The non-pluripolar Monge–Ampère mass of φ equals the algebraic volume introduced in Definition 3.2.3:

$$\int_X \theta_\varphi^n = \operatorname{vol}\theta_\varphi .$$

(*Mass–volume identity*: no mass is lost to pluripolar sets, so the analytic mass and the algebraic volume coincide.)

(3) The P-envelope and the $\mathcal{I}$-envelope of φ introduced in Chapter 3 agree:

$$P_\theta[\varphi] = P_\theta[\varphi]_{\mathcal{I}} .$$

(*Analytic–algebraic envelope agreement*: the two natural least-singular representatives, one constructed from the P-equivalence relation and one from multiplier ideals, coincide.)

The reader is invited to keep these three pictures in mind throughout the book; we will use whichever is most convenient at any given moment.

Step 2 is the more delicate one, and its difficulty turns out to be a feature, not a bug.

A given $\mathcal{I}$-equivalence class can consist of an enormous number of P-equivalence classes; constructing such examples is easy. By contrast, in the toric and non-Archimedean settings, Step 2 is essentially trivial: every $\mathcal{I}$-equivalence class consists of a single P-equivalence class. In the toric case, an $\mathcal{I}$- (or P-) equivalence class is a sub-convex body of the Newton body. In the non-Archimedean case, it is a homogeneous plurisubharmonic metric.

The discrepancy between the analytic and the toric/non-Archimedean theories suggests that, in the pluripotential theory on compact Kähler manifolds, a great many $\mathcal{I}$-classes are pathological, and that what one really wants is to single out within each $\mathcal{I}$-class a canonical P-class. The functions in this canonical class will be called

$\mathcal{I}$*-good*. The development of the theory of $\mathcal{I}$-good singularities, in Chapter 7, is the technical core of this book. As the reader will see, almost every singularity that arises naturally in geometric applications is $\mathcal{I}$-good.

A useful guiding analogy is with measure theory. There exist many non-measurable functions, but unless one constructs one by hand using the axiom of choice, one only encounters measurable functions in practice. Similarly, although the space of non-$\mathcal{I}$-good singularities is non-empty, it is essentially invisible to geometric applications. Once one accepts the framework of $\mathcal{I}$-good singularities, Step 2 becomes essentially solved.

Tools, applications, conventions.

The classification of singularities, in the form just described, is somewhat abstract. To put it to use, we develop two general techniques that allow induction on $\dim X$. For a prime divisor Y in general position, the *analytic Bertini theorem* relates quasi-plurisubharmonic functions on X and on Y. For non-generic Y, the *trace operator* associates with φ a quasi-plurisubharmonic function on the normalization of Y, well-defined modulo $\mathcal{I}$-equivalence. These tools occupy Chapter 8.

The toric pluripotential theory on ample line bundles is mostly classical and is recalled, with full proofs, in Chapter 5. The corresponding theory for big toric line bundles, which has never been written down in the literature, requires the partial Okounkov body machinery developed in Chapter 10 and is the content of Chapter 12.

The applications to non-Archimedean pluripotential theory occupy Chapter 13, building on the test-curve theory of Chapter 9. The convergence of partial Bergman kernels with arbitrary singular weight, generalising the Berman–Boucksom–Witt Nyström theorem, is proved in Chapter 14.

Reading guide.

This book is not an elementary introduction to pluripotential theory. It should not be the reader's first textbook in this field. The reader is expected to be comfortable with the basic pluripotential theory at the level of [GZ17], which is essentially the only prerequisite for the main body of the book.

This book is not "a compilation of all the author's works", as an anonymous referee who had almost no knowledge in this field and barely read this book said, rather, we present a new point of view to this theory, with new results and new insights. A reader who has gone through all my related papers is still encouraged to read at least part of this book.

Results in this book are presented in the greatest possible generality. An expert reader may use this book as a convenient reference for the most general statements, and skip the proofs of the more general results if they are already familiar with them in a special case.

The reader is expected to read Part II linearly, and consult the necessary preliminaries in Part I when necessary.

A reader interested only in one application chapter may skip directly to it once the relevant Part II prerequisites are in place. Chapter 5 is needed only for Chapter 12; it can otherwise be skipped without loss.

Some chapters in this book draw on more specialised background, recorded in the appendices: Appendix A for convex analysis, Appendix B for pluripotential theory on unibranch Kähler spaces, and Appendix C for the almost-semigroup machinery underlying the partial Okounkov body construction.

In order to develop the theory of b-divisors as in Chapter 11, we need to take limits with respect to the net of all modifications of X. In general, this net does not admit a cofinal sequence. As a consequence, nets are widely used in this book, and the reader is expected to be familiar with their basic properties. [Kel75] is a good reference for the basic theory. It is essential to remember that the dominated convergence theorem and Fatou's lemma both fail for nets. While a version of the monotone convergence theorem holds, as in [Fol99, Proposition 7.12].

I made a great effort to make sure that every single expression involving $(-\infty)-(-\infty)$ is explained. This point is often overlooked in the literature. In most cases, the fix is simple, but in some cases, it requires a non-trivial amount of work. As a consequence, a number of our proofs seem more complicated than in the original papers. I hope that the reader will appreciate the effort and find the book more readable as a result.

Comments and questions are always welcome, and I will maintain an errata page on my homepage as long as I remain in academia.

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in Hangzhou, March 2024

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夏銘辰，職業乱法選手。

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The clerks are French, and, like most French people, are in a bad temper till they have eaten their lunch.
— George Orwell (1903–1950)

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The book was written in the pre-AI era, but during the long publication process, the AI tools have developed rapidly, and I have been using them to check for typos and mistakes, and to improve the writing. I am grateful to Claude, Codex and ChatGPT for their help in improving the quality of the book. I am grateful to Xu Wang and Ragnar Sigurðsson for their invitations to Europe, making global internet stably accessible to me. At the same time, I am extremely disappointed by the decision of Claude of blocking users from mainland China, including myself.

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Contents

Conventions

In the whole book, we adopt the following conventions:

- A complex space is always assumed to be *reduced*, *paracompact* and *Hausdorff*.
- A *modification* of a complex space X is a proper bimeromorphic morphism $\pi\colon Y \to X$ that is locally obtained from a finite composition of blow-ups with smooth centers.
- A *subnet* of a net refers to a Kelley subnet.
- A *domain* in $\mathbb{C}^n$ refers to a connected open subset.
- A *complex manifold* is assumed to be paracompact.
- A *submanifold* of a complex manifold means a closed complex submanifold.
- A *neighborhood* is not necessarily open.
- The set $\mathbb{N}$ of natural numbers includes 0.
- *Increasing functions* and *decreasing functions* are not necessarily strictly monotone.
- The function $\mathbb{R} \ni x \mapsto x^2$ is *convex* instead of concave. This is different from the convention in [CLS11].

We will use the following notations throughout the book:

- If I is a non-empty set, then $\mathrm{Fin}(I)$ denotes the net of finite non-empty subsets of I, ordered by inclusion.
- $\mathrm{dd^c}$ means $(2\pi)^{-1}\mathrm{i}\partial\overline{\partial}$.

Part I
Preliminaries

This part collects the foundational material on which the rest of the book builds. A reader already familiar with the pluripotential theory of compact Kähler manifolds may skim this part on a first reading and refer back to it as needed; we have included it both to fix our notational conventions (which are not uniform across the literature) and to keep the book reasonably self-contained. The classical material is presented concisely, with references to fuller accounts at each step. The non-classical material, in particular the treatment of P- and $\mathcal{I}$-envelopes in Chapter 3 and the geodesic theory between singular potentials in Chapter 4, is presented in full.

Chapters 1 and 2 recall the basic theory of quasi-plurisubharmonic functions on a compact Kähler manifold X and the non-pluripolar product of a family of closed positive $(1,1)$-currents. The presentation is brief; we follow the conventions of [GZ17], to which we refer for the missing proofs. By the end of these two chapters the reader should be comfortable with the spaces $\mathrm{PSH}(X,\theta)$ and $\mathrm{QPSH}(X)$, the preorder $\preceq$, the non-pluripolar Monge–Ampère measure θ_φ^n, the Lelong number $\nu(\varphi,x)$, and the multiplier ideal sheaf $\mathcal{I}(\varphi)$.

Chapter 3 develops the theory of *envelope operators*. Two envelopes play a central role throughout the book: the P-envelope $P_\theta[\varphi]$, which extracts the singularity-type "hull" of φ relative to non-pluripolar mass, and the $\mathcal{I}$-envelope $P_\theta[\varphi]_{\mathcal{I}}$, which extracts the corresponding hull relative to multiplier ideals. Their basic properties are established here. Many of these results are known and scattered across the literature; this chapter assembles them into a single coherent framework.

Chapter 4 develops the theory of geodesics in the space $\mathrm{PSH}(X,\theta)$ with prescribed (and possibly singular) boundary data. Most results are known to varying degrees, but not in the fully general form presented here, where the boundary data are allowed to have arbitrary singularities. The proofs follow the literature in spirit but require careful treatment of the singular boundary; in particular, one cannot reduce to the smooth case by regularization. The chapter introduces the Darvas d_1-metric, the space of geodesic rays $\mathcal{R}^1$, and the basic results that underpin the metric theory of singularities developed in Chapter 6.

Chapter 5 recalls the toric pluripotential theory on an ample line bundle associated with a Delzant lattice polytope P. The toric–convex dictionary developed here will be generalized to the big setting in Chapter 12. The proof of the toric Riemann–Roch formula Theorem 5.2.2 is the main novelty. Chapter 5 is otherwise independent of the rest of the book; a reader interested only in the non-toric theory may proceed directly from Chapter 4 to Part II.

Chapter 1
Plurisubharmonic functions

Once Frigyes Riesz[a] *gave a brilliant explanation of why scientific work is easy. "Everyone has ideas, both right ideas and wrong ideas," he said. "Scientific work consists merely of separating them."*
— Istvan Vincze

[a] Frigyes Riesz (1880–1956), known as Frédéric Riesz in French and Frederic Riesz in English, was the first mathematician to define the general notion of subharmonic functions, and he also gave these functions a Frenglish name from the very beginning — *fonctions subharmoniques*.

The aim of this chapter is to recall the basic theory of plurisubharmonic functions and to fix the conventions used throughout the rest of the book. Since the material is largely standard, most proofs are only sketched or omitted; we refer the reader to [GZ17], [Dem12a] and [Dem12b] for thorough treatments. Two topics, however, merit closer attention. The first is the plurifine topology, for which the proofs in the literature are scattered and in places incomplete — in particular the Bedford–Taylor base theorem of [BT87] was stated without proof. The second is the choice of normalization for Lelong numbers and multiplier ideal sheaves, for which several mutually inconsistent conventions are in use.

The chapter is organized as follows.

In Section 1.1 we recall the definition of plurisubharmonic functions on domains in $\mathbb{C}^n$ and on complex manifolds. We also include the 0-dimensional case, which is occasionally convenient in inductive arguments and is absent from the standard references. The basic properties of plurisubharmonic functions (closedness under upper envelopes and decreasing limits, Choquet's lemma, local L^p-integrability, the Grauert–Remmert extension theorems, and Kiselman's principle) are collected in the section that follows.

Section 1.3 is devoted to the plurifine topology. Here we provide complete proofs, including a proof of the Bedford–Taylor base theorem (Theorem 1.3.2). Our presentation largely follows [EMW06].

In Section 1.4 we recall Lelong numbers and multiplier ideal sheaves. The conventions we adopt — chosen so that the formulas in later chapters carry no spurious factors of 2 — differ from those of [GZ17, Dem12a], and we explain the conversion in Remark 1.4.1 and Remark 1.4.2. We also state Siu's semicontinuity theorem, the strong openness theorem of Guan–Zhou, the Ohsawa–Takegoshi extension theorem, and the valuative characterization of multiplier ideals.

The remaining sections develop the global theory needed for the rest of the book. In Section 1.5 we introduce θ-plurisubharmonic and quasi-plurisubharmonic functions, the preorder "more singular than" (denoted $\preceq$, see Definition 1.5.2), and the notion of singularity type. Section 1.6 discusses analytic singularities, the existence of log resolutions, and the quasi-equisingular approximation theorem of [DPS01]. Section 1.7 recalls closed positive $(1, 1)$-currents, the pseudo-effective and big cones, Kähler currents, Siu's decomposition, and the non-Kähler locus, non-nef locus and modified nef classes; these objects will play a central role from Chapter 3 onwards.

Finally, Section 1.8 translates the picture into the language of line bundles. We recall (singular) Hermitian and plurisubharmonic metrics, Fubini–Study metrics, and we state the version of the Ohsawa–Takegoshi extension theorem (Theorem 1.8.1) that will be used repeatedly in later chapters.

A reader already familiar with this material may safely skim the chapter on a first pass and return to it as needed; we do however recommend checking the conventions of Section 1.4 against those of any reference being consulted, since the choices we make differ from several common alternatives.

1.1 The definition of plurisubharmonic functions

In this section, we recall the notion of plurisubharmonic functions. We will also take care of the 0-dimensional case, which makes a number of induction arguments easier to carry out. None of our references treats the 0-dimensional case, but the reader can easily verify that the results in this section hold in this exceptional case.

1.1.1 The 1-dimensional case

Let Ω be a domain (a connected open subset) in $\mathbb{C}$.

Definition 1.1.1 A *subharmonic function* on Ω is a function $\varphi\colon \Omega \to [-\infty, \infty)$ satisfying the following three conditions:

(1) $\varphi \not\equiv -\infty$;
(2) φ is upper semicontinuous;
(3) φ satisfies the *sub-mean value inequality*: For any $a \in \Omega$ and $r > 0$ such that $\overline{B_1(a, r)} \subseteq \Omega$, we have

$$\varphi(a) \leq \frac{1}{2\pi} \int_0^{2\pi} \varphi(a + r\mathrm{e}^{\mathrm{i}\theta})\, \mathrm{d}\theta.$$[1]

We will denote the set of subharmonic functions on Ω by $\mathrm{SH}(\Omega)$.

Here, $B_1(a, r) \coloneqq \{z \in \mathbb{C} : |z - a| < r\}$ denotes the open ball with center a and radius r.

[1] Condition (2) guarantees that φ is measurable and locally bounded from above, and hence the integral in Condition (3) makes sense.

In fact, for each $a \in \Omega$, in (3), it suffices to require the sub-mean value inequality for all small enough $r > 0$.

Intuitively, at a specific point $a \in \Omega$, Condition (2) says $\varphi(a)$ is no smaller than the upper limit of nearby values, while Condition (3) says $\varphi(a)$ is no larger than the mean. This intuition leads to the following rigidity theorem:

Theorem 1.1.1 *Let $\varphi\colon \Omega \to [-\infty, \infty)$ be a measurable function. Then the following are equivalent:*

(1) *φ is locally integrable and $\Delta\varphi \geq 0$.*
(2) *φ coincides almost everywhere with a subharmonic function ψ on Ω.*

Moreover, the subharmonic function ψ in (2) is unique.

Here in Condition (1), $\Delta\varphi = 2\mathrm{i}\partial\bar{\partial}\varphi$ is the Laplacian in the sense of currents. This is a special case of Theorem 1.1.2 below.

This theorem gives a very useful way of constructing subharmonic functions.

1.1.2 The higher dimensional case

We will fix $n \in \mathbb{N}$ and a domain Ω (a connected open subset) in $\mathbb{C}^n$.

Definition 1.1.2 When $n \geq 1$, a *plurisubharmonic function* on Ω is a function $\varphi\colon \Omega \to [-\infty, \infty)$ satisfying the following three conditions:

(1) $\varphi \not\equiv -\infty$;
(2) φ is upper semicontinuous;
(3) for any complex line $L \subseteq \mathbb{C}^n$ and any connected component U of $L \cap \Omega$, the restriction $\varphi|_U$ is either subharmonic or constantly $-\infty$.[2]

When $n = 0$, the only domain Ω is the singleton. In this case, a *plurisubharmonic function* on Ω is a real-valued function on Ω.

The set of plurisubharmonic functions on Ω is denoted by $\mathrm{PSH}(\Omega)$.

A plurisubharmonic function is also called a psh function for short. The relevant notations are indicated in Fig. 1.1.[3]

Example 1.1.1 When $n = 0$, we have a canonical bijection $\mathrm{PSH}(\Omega) \cong \mathbb{R}$.

Example 1.1.2 When $n = 1$, we have $\mathrm{PSH}(\Omega) = \mathrm{SH}(\Omega)$.

Similar to Theorem 1.1.1, we have a rigidity theorem for plurisubharmonic functions as well.

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[2] An extremely common mistake in the literature is to replace (3) by the condition that φ is locally integrable and $\mathrm{dd}^{\mathrm{c}}\varphi \geq 0$ in the sense of currents. For a concrete counterexample, consider a function φ that takes a constant value 0 at all but one single point, at which the value of φ is 1.

[3] Recall that most figures in this book are somewhat misleading: We usually draw a complex dimension as a real dimension. The figures should not be read literally!

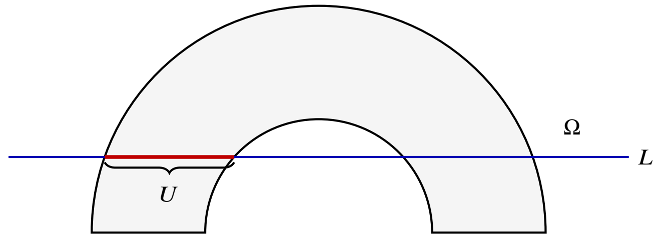

Fig. 1.1 A domain Ω cut by a complex line L.

Theorem 1.1.2 *Let $\varphi\colon \Omega \to [-\infty, \infty)$ be a measurable function. Then the following are equivalent:*

(1) *φ is locally integrable and $\mathrm{dd^c}\varphi \geq 0$;*
(2) *φ coincides almost everywhere with a plurisubharmonic function ψ on Ω.*

Moreover, the plurisubharmonic function ψ is unique.

Here, the operator $\mathrm{dd^c}$ is normalized so that

$$\mathrm{dd^c} = \frac{\mathrm{i}}{2\pi}\partial\bar{\partial}.$$

For the proof, we refer to [GZ17, Proposition 1.43].

Plurisubharmonic functions have nice functorialities:

Proposition 1.1.1 *Let $n' \in \mathbb{N}$ and $\Omega' \subseteq \mathbb{C}^{n'}$ be a domain. Given any holomorphic map $f\colon \Omega \to \Omega'$ and any $\varphi \in \mathrm{PSH}(\Omega')$, exactly one of the following cases occurs:*

(1) *$f^*\varphi \equiv -\infty$;*
(2) *$f^*\varphi \in \mathrm{PSH}(\Omega)$.*

We refer to [GZ17, Proposition 1.44] for the proof[4].

For each $n \in \mathbb{N}$, $a \in \mathbb{C}^n$ and $r > 0$, we write

$$B_n(a,r) = \{z \in \mathbb{C}^n : |z-a| < r\}. \tag{1.1}$$

Proposition 1.1.2 *Let $\varphi \in \mathrm{PSH}(B_n(a, r_0))$ for some $r_0 > 0$. Then the function*

$$(-\infty, \log r_0) \to \mathbb{R}, \quad \log r \mapsto \sup_{B_n(a,r)} \varphi$$

is convex and increasing.

See [Hör07, Theorem 4.1.13] for the case $n > 1$ and [Bou17, Corollary 2.4] for the general case.

[4] Recall that the statement of [GZ17, Proposition 1.44] is flawed. One has to reduce to the case where Case (1) does not occur before following their proof.

Proposition 1.1.3 *Let $a < b$ be two real numbers. Let $f\colon (a,b) \to [-\infty,\infty)$ be a function. Define*

$$g\colon \{z \in \mathbb{C} : a < \operatorname{Re} z < b\} \to [-\infty,\infty), \quad z \mapsto f(\operatorname{Re} z).$$

Suppose that g is subharmonic. Then f is convex. In particular, f takes real values only.

See [HK76, Theorem 2.12] for a more general result.

1.1.3 The manifold case

Let X be a complex manifold. In the whole book, complex manifolds are assumed to be paracompact, namely, all connected components have countable bases.

Definition 1.1.3 A *plurisubharmonic function* on X is a function $\varphi\colon X \to [-\infty,\infty)$ such that for any $x \in X$, there exists an open neighborhood U of x in X, an integer $n \in \mathbb{N}$, a domain $\Omega \subseteq \mathbb{C}^n$ and a biholomorphic map $F\colon \Omega \to U$ such that $F^*(\varphi|_U) \in \operatorname{PSH}(\Omega)$.

The set of plurisubharmonic functions on X is denoted by $\operatorname{PSH}(X)$.

Example 1.1.3 When X is a domain in $\mathbb{C}^n$, the notions of plurisubharmonic functions in Definition 1.1.3 and in Definition 1.1.2 coincide.

Example 1.1.4 Write $\{X_i\}_{i\in I}$ for the set of connected components of X. Then we have a natural bijection

$$\operatorname{PSH}(X) \cong \prod_{i\in I} \operatorname{PSH}(X_i).$$

Here the product is in the category of sets. In particular, if $X = \varnothing$, then $\operatorname{PSH}(X)$ is a singleton.

This example allows us to reduce to the case of connected manifolds when studying general plurisubharmonic functions.

Proposition 1.1.4 *Let Y be a complex manifold and $f : Y \to X$ be a holomorphic map. Then for any $\varphi \in \operatorname{PSH}(X)$, exactly one of the following cases occurs:*

(1) *$f^*\varphi$ is identically $-\infty$ on some connected component of Y;*
(2) *$f^*\varphi \in \operatorname{PSH}(Y)$.*

This proposition follows from Proposition 1.1.1 applied separately on each connected component of Y.

Theorem 1.1.2 implies immediately the general form of the rigidity theorem:

Theorem 1.1.3 *Let $\varphi\colon X \to [-\infty,\infty)$ be a measurable function. Then the following are equivalent:*

(1) *φ is locally integrable and* $\mathrm{dd^c}\varphi \geq 0$*;*
(2) *φ coincides almost everywhere with a plurisubharmonic function ψ on X.*

Moreover, the plurisubharmonic function ψ in (2) is unique.

Definition 1.1.4 A subset $E \subseteq X$ is *pluripolar* if for any $x \in X$, there is an open neighborhood U of x in X and a function $\psi \in \mathrm{PSH}(U)$ such that

$$\psi|_{E\cap U} \equiv -\infty.$$

A subset $E \subseteq X$ is *non-pluripolar* if E is not pluripolar.
A subset $F \subseteq X$ is *co-pluripolar* if $X \setminus F$ is pluripolar.

When X has dimension 1, a pluripolar set is called a *polar set*. We say some property about objects on X holds *quasi-everywhere* if it holds outside a pluripolar set.

Theorem 1.1.4 (Josefson's theorem) *Let $E \subseteq \mathbb{C}^n$ be a pluripolar set. Then there is $\varphi \in \mathrm{PSH}(\mathbb{C}^n)$ such that $\varphi|_E \equiv -\infty$.*

See [GZ17, Corollary 4.41] for the proof of a more general result.
There is also a global version of Josefson's theorem:

Theorem 1.1.5 *Assume that X is a compact complex manifold and $E \subseteq X$ is a pluripolar set. Then there is a quasi-plurisubharmonic function φ on X with $\varphi|_E \equiv -\infty$.*

For a proof, see [Vu19]. The notion of quasi-plurisubharmonic functions will be recalled in Definition 1.5.1 below.

1.2 Properties of plurisubharmonic functions

In this section, we explore the basic properties of plurisubharmonic functions.
Let X be a complex manifold.

Proposition 1.2.1 *Let $\varphi \in \mathrm{PSH}(X)$. Then for any $p \geq 1$, $\varphi \in L^p_{\mathrm{loc}}(X)$.*

See [GZ17, Theorem 1.46, Theorem 1.48].

Proposition 1.2.2

(1) *Assume that $(\varphi_i)_{i\in I}$ is a non-empty family in $\mathrm{PSH}(X)$ that is locally uniformly bounded from above. Then $\sup_{i\in I}{}^*\varphi_i \in \mathrm{PSH}(X)$.*
(2) *Assume that $(\varphi_i)_{i\in I}$ is a decreasing net in $\mathrm{PSH}(X)$ such that $\varphi := \inf_{i\in I}\varphi_i$ is not identically $-\infty$ on any connected component of X. Then $\varphi \in \mathrm{PSH}(X)$. Furthermore, $\varphi_i \xrightarrow{L^1_{\mathrm{loc}}} \varphi$.*

Here $\sup^*$ denotes the upper semicontinuous regularization of the supremum. When I is a finite family, observe that

$$\sup_{i\in I}{}^*\varphi_i = \sup_{i\in I}\varphi_i.$$

When $I = \{1, \dots, m\}$, we write

$$\varphi_1 \vee \cdots \vee \varphi_m := \sup_{i\in I}\varphi_i.$$

We refer to [GZ17, Proposition 1.28, Proposition 1.40][5].

Proposition 1.2.3 (Choquet's lemma) *Assume that X has countably many connected components. Assume that $(\varphi_i)_{i\in I}$ is a non-empty family in $\mathrm{PSH}(X)$ that is locally uniformly bounded from above. There exists a countable subset $J \subseteq I$ such that*

$$\sup_{i\in I}{}^*\varphi_i = \sup_{j\in J}{}^*\varphi_j.$$

In the sequel, we say a subset $B \subseteq X$ is a *holomorphic ball* if B has smooth boundary in X, and B is biholomorphic to an open ball in $\mathbb{C}^n$.

Proof We may assume that X is connected. Since by our convention, the complex manifold X is paracompact, it can be covered by countably many holomorphic balls, so we can easily reduce to the case where X is an open ball in $\mathbb{C}^n$. In this case, the result is proved in [GZ17, Lemma 4.31]. □

Proposition 1.2.4 *A pluripolar set $E \subseteq X$ is a Lebesgue null set.*

Proof This is a trivial consequence of Proposition 1.2.1. □

Proposition 1.2.5 *Let $(\varphi_i)_{i\in I}$ be a non-empty family in $\mathrm{PSH}(X)$ that is locally uniformly bounded from above. Then the set*

$$\left\{x \in X : \sup_{i\in I}\varphi_i < \sup_{i\in I}{}^*\varphi_i\right\}$$

is pluripolar and hence Lebesgue null.

See [GZ17, Corollary 4.28].

Proposition 1.2.6 *Suppose that $\varphi, \psi \in \mathrm{PSH}(X)$. If $\varphi \leq \psi$ almost everywhere, then $\varphi \leq \psi$ everywhere.*

In particular, if $\varphi = \psi$ almost everywhere, then $\varphi = \psi$ everywhere.

Proof Replace φ by $\varphi \vee \psi$. Then φ, ψ are two plurisubharmonic functions that coincide almost everywhere. The uniqueness statement in Theorem 1.1.3 gives $\varphi = \psi$. □

[5] In [GZ17, Proposition 1.28], the second part is only stated for sequences, the net version is obvious using the sub-mean value inequality and the monotone convergence theorem for nets as in [Fol99, Proposition 7.12].

Proposition 1.2.7 *Let $(E_i)_{i\in\mathbb{Z}_{>0}}$ be a sequence of pluripolar sets in X. Then*

$$E := \bigcup_{i=1}^{\infty} E_i$$

is also pluripolar.

Proof The problem is local, so we may assume that $X \subseteq \mathbb{C}^n$ is a bounded domain. In this case, by Theorem 1.1.4 for each $i \in \mathbb{Z}_{>0}$ we can choose $\psi_i \in \mathrm{PSH}(\mathbb{C}^n)$ such that

$$\psi_i|_{E_i} \equiv -\infty, \quad \psi_i|_X \leq 0$$

for all $i > 0$. By Proposition 1.2.1, we have $\psi_i|_X \in L^1(X)$ for all $i > 0$. After rescaling, we may also assume that $\|\psi_i\|_{L^1(X)} \leq 1$ for all $i > 0$.

We then define

$$\psi = \sum_{i=1}^{\infty} 2^{-i}\psi_i|_X.$$

Then $\psi \in \mathrm{PSH}(X)$ according to Proposition 1.2.2 and $\psi|_E = -\infty$. □

Corollary 1.2.1 *Let $(\varphi_j)_{j\in\mathbb{Z}_{>0}}$ be a sequence in* $\mathrm{PSH}(X)$ *such that $\varphi_j \xrightarrow{L^1_{\mathrm{loc}}} \varphi \in$* $\mathrm{PSH}(X)$. *Then*

$$\varphi = \inf_{j>0} \sup_{k\geq j}{}^{*} \varphi_k, \tag{1.2}$$

and the set

$$\left\{x \in X : \varphi(x) \neq \varlimsup_{j\to\infty} \varphi_j(x)\right\}$$

is pluripolar.

Proof We may assume that X is connected.

We first observe that $(\varphi_j)_j$ is locally uniformly bounded from above. This follows from [GZ17, Exercise 1.20].

For each $j \geq 1$, let

$$\psi_j = \sup_{k\geq j}{}^{*} \varphi_k.$$

Then $\psi_j \in \mathrm{PSH}(X)$ by Proposition 1.2.2. Moreover, $(\psi_j)_j$ is a decreasing sequence and $\psi_j \geq \varphi_j$ for all j. After considering a subsequence of $(\varphi_j)_j$ which converges to φ almost everywhere, we get $\varphi \leq \psi := \inf_j \psi_j$ almost everywhere. In particular, $\psi \in \mathrm{PSH}(X)$ by Proposition 1.2.2.

On the other hand, by Proposition 1.2.5, there exist pluripolar sets $Z_j \subseteq X$ such that

$$\psi_j = \sup_{k\geq j} \varphi_k$$

on $X \setminus Z_j$. Let

$$Z = \bigcup_{j=1}^{\infty} Z_j.$$

Then Z is a pluripolar set by Proposition 1.2.7, and for any $x \in X \setminus Z$, we have $\psi(x) = \overline{\lim}_j \varphi_j(x)$. Moreover, $\psi = \varphi$ almost everywhere by [GZ17, Theorem 1.46], and hence everywhere by Proposition 1.2.6. Hence we conclude (1.2). Moreover, for all $x \in X \setminus Z$,

$$\varphi(x) = \varlimsup_{j\to\infty} \varphi_j(x).$$

This completes the proof. □

Theorem 1.2.1 (Brelot, Grauert–Remmert) *Let Z be an analytic subset in X and $\varphi \in \mathrm{PSH}(X \setminus Z)$. Then the function φ admits an extension to $\mathrm{PSH}(X)$ in the following two cases:*

(1) *The set Z has codimension at least 2 everywhere.*
(2) *The set Z has codimension at least 1 everywhere and φ is locally bounded from above near each point of Z.*

In both cases, the extension is unique and is given by

$$\varphi(x) = \varlimsup_{X\setminus Z \ni y \to x} \varphi(y), \quad x \in Z. \tag{1.3}$$

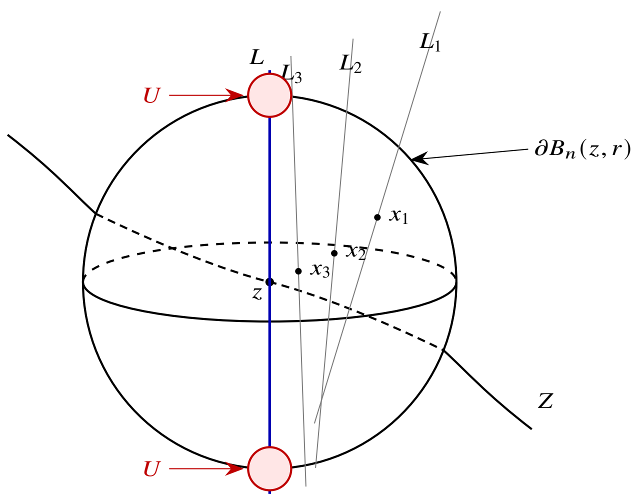

Fig. 1.2 Notation in the proof of Theorem 1.2.1 (1).

Proof The extension is unique thanks to Proposition 1.2.6. We shall prove the existence.

(2) Thanks to the uniqueness of the extension, the problem is local, so we may assume that X is a domain in $\mathbb{C}^n$ with $n > 0$ and there is a non-zero holomorphic function f on X vanishing on Z. For each $\epsilon > 0$, we claim that the function φ_ϵ defined by

$$\varphi_\epsilon(x) := \begin{cases} \varphi(x) + \epsilon \log |f(x)|^2, & x \in X \setminus Z; \\ -\infty, & x \in Z \end{cases}$$

is plurisubharmonic on X. By Definition 1.1.2, it suffices to verify the case $n = 1$. In this case, we may assume that $Z = \{0\}$. It is clear that $\varphi_\epsilon \in \mathrm{SH}(X \setminus Z)$. It suffices to verify the sub-mean value inequality at 0, which is immediate.

Next observe that the net φ_ϵ is increasing as $\epsilon \searrow 0$ and φ_ϵ is locally uniformly bounded from above. It follows from Proposition 1.2.2 that $\tilde{\varphi} := \sup_{\epsilon>0}{}^*\varphi_\epsilon \in \mathrm{PSH}(X)$. Moreover, $\tilde{\varphi}$ clearly extends φ. Note that (1.3) follows from the construction.

(1) We invite the reader to consult Fig. 1.2 for the notations used in the proof.

Thanks to (2), it suffices to verify that φ is locally bounded from above near each point of Z. The problem is local, so we may assume that X is a domain in $\mathbb{C}^n$ with $n \geq 2$.

Suppose the assertion fails. Then there exist $z \in Z$ and a sequence $(x_j)_{j\geq 1}$ in $X \setminus Z$ converging to z such that

$$\lim_{j\to\infty} \varphi(x_j) = \infty. \tag{1.4}$$

Since Z has codimension at least 2[6], we may choose a complex line L through z which meets Z in a discrete set; after shrinking X we may further assume that

$$L \cap Z = \{z\}.$$

Pick an open ball $B_n(z,r) \Subset X$. After adding a constant to φ we may arrange $\varphi < 0$ on $L \cap \partial B_n(z,r)$. Since φ is upper semicontinuous, there is an open neighborhood $U \subseteq X \setminus Z$ of $L \cap \partial B_n(z,r)$ such that

$$\varphi|_U < 0.$$

For each $j \geq 1$ we now choose a complex line L_j through x_j, disjoint from $Z \cap B_n(z,r)$, with $L_j \to L$ as $j \to \infty$ (the convergence being in the Grassmannian of affine complex lines in $\mathbb{C}^n$). The existence of such L_j is precisely where the assumption $\operatorname{codim} Z \geq 2$ enters: the set of complex directions through x_j that meet $Z \cap B_n(z,r) \setminus \{x_j\}$ is the image of $Z \cap B_n(z,r) \setminus \{x_j\}$ under the linear projection with w mapped to the line containing w and x_j in $\mathbb{P}^{n-1}$, hence has complex dimension at most $\dim Z \leq n-2$. It is therefore contained in a proper analytic subset of $\mathbb{P}^{n-1}$, and its complement contains directions arbitrarily close to that of L.

For j large enough we now have

$$L_j \cap \partial B_n(z,r) \subseteq U,$$

so applying the sub-mean value inequality to $\varphi|_{L_j \cap X}$ at x_j yields $\varphi(x_j) < 0$, contradicting (1.4). □

[6] In fact, codimension at least 1 suffices for this step.

Lemma 1.2.1 *Let $\varphi \in \mathrm{PSH}((\Delta^*)^n)$ be an $(S^1)^n$-invariant plurisubharmonic function. Then φ is finite everywhere.*

Here
$$\Delta^* := \{z \in \mathbb{C} : 0 < |z| < 1\}$$
denotes the punctured unit disk.

Proof It clearly suffices to handle the case $n = 1$. In this case, by [HK76, Theorem 2.12], the map
$$\log r \mapsto \int_0^1 \varphi\left(r e^{2\pi \mathrm{i}\theta}\right) \mathrm{d}\theta = \varphi(r)$$
is a convex function of $\log r$. So, the set $\{r \in (0,1) : \varphi(r) = -\infty\}$ is convex. But φ is almost everywhere finite by Proposition 1.2.1. Since φ is S^1-invariant, $\varphi|_{(0,1)}$ is almost everywhere finite. It follows from the convexity that it is everywhere finite.□

Proposition 1.2.8 (Kiselman's principle) *Let $\Omega \subseteq \mathbb{C}^m \times \mathbb{C}^n$ be a pseudoconvex domain. Assume that for each $z \in \mathbb{C}^m$, the set*
$$\Omega_z := \left\{w \in \mathbb{C}^n : (z,w) \in \Omega\right\}$$
has the form $E + \mathrm{i}\mathbb{R}^n$, where $E \subseteq \mathbb{R}^n$ is a subset. Let $\varphi \in \mathrm{PSH}(\Omega)$, assume that φ is independent of the imaginary part of the variable in $\mathbb{C}^n$. Let Ω' be the projection of Ω to $\mathbb{C}^m$. Define $\psi : \Omega' \to [-\infty, \infty)$ as follows:
$$\psi(z) = \inf_{w \in \Omega_z} \varphi(z,w).$$
Then either $\psi \equiv -\infty$ or $\psi \in \mathrm{PSH}(\Omega')$.

See [Dem12b, Theorem 7.5] for the proof as well as the notion of pseudoconvex domains.

Lemma 1.2.2 *Let $\Omega \subseteq \mathbb{C}^n$ be a domain and $\Omega' \subseteq \Omega$ be a subdomain. Consider $\varphi \in \mathrm{PSH}(\Omega)$ and $\psi \in \mathrm{PSH}(\Omega')$. Assume that*
$$\varliminf_{\substack{\Omega' \ni y \to x, \\ \psi(y) \neq -\infty}} \big(\varphi(y) - \psi(y)\big) \geq 0 \tag{1.5}$$
for any $x \in \Omega \cap \partial\Omega'$. Define
$$\eta(z) = \begin{cases} \varphi(z) \vee \psi(z), & \text{if } z \in \Omega', \\ \varphi(z), & \text{if } z \in \Omega \setminus \Omega'. \end{cases}$$
Then $\eta \in \mathrm{PSH}(\Omega)$.

This is essentially [GZ17, Proposition 1.30]. Since the statement in the reference is slightly misleading for our purposes, we include the proof for clarification.

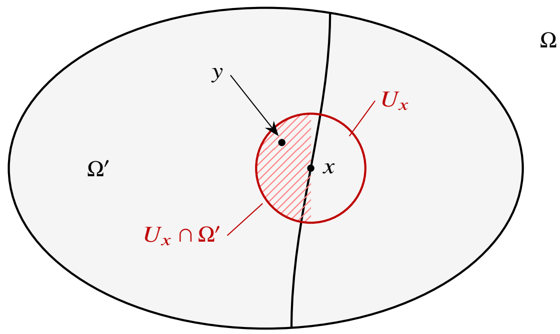

Fig. 1.3 Gluing procedure

Proof See Fig. 1.3 for the notations used in the proof.

Take $\epsilon > 0$. We first define

$$\eta_\epsilon(z) = \begin{cases} \varphi(z) \vee (\psi(z) - 2\epsilon), & \text{if } z \in \Omega'; \\ \varphi(z), & \text{if } z \in \Omega \setminus \Omega'. \end{cases}$$

We claim that

$$\eta_\epsilon \in \mathrm{PSH}(\Omega).$$

By our assumption, for each $x \in \Omega \cap \partial\Omega'$, we can find an open neighborhood $U_x \subseteq \Omega$ such that for any $y \in U_x \cap \Omega'$, we have $\varphi(y) \geq \psi(y) - \epsilon$. Therefore, by taking a union of the U_x's for various x, there is an open neighborhood U of $\Omega \cap \partial\Omega'$ such that

$$\varphi(y) \geq \psi(y) - \epsilon, \quad \forall y \in U \cap \Omega'.$$

Since U contains $\Omega \cap \partial\Omega'$, the set $(\Omega \setminus \Omega') \cup U$ is open in Ω. On this open set we have $\eta_\epsilon = \varphi$, hence η_ϵ is plurisubharmonic there. It is plurisubharmonic on Ω' by Proposition 1.2.2. These two descriptions agree on the overlap $U \cap \Omega'$, so our claim follows from the local nature of plurisubharmonicity.

Next we observe that as ϵ decreases to 0, the functions η_ϵ increase to η. The family $(\eta_\epsilon)_{0<\epsilon\leq 1}$ is locally uniformly bounded from above. This is clear away from $\Omega \cap \partial\Omega'$. Near $\Omega \cap \partial\Omega'$, applying the previous estimate with $\epsilon = 1$ gives an open neighborhood U_1 of $\Omega \cap \partial\Omega'$ such that $\psi \leq \varphi + 1$ on $U_1 \cap \Omega'$, hence $\eta_\epsilon \leq \varphi + 1$ on U_1 for all $0 < \epsilon \leq 1$. Therefore, $\eta^* \in \mathrm{PSH}(\Omega)$ by Proposition 1.2.2.

It remains to observe that η is upper semicontinuous. This is only non-trivial on the boundary of Ω'. Take $x \in \Omega \cap \partial\Omega'$ and let $(y_i)_{i>0}$ be a sequence in Ω' with limit x. We first show that

$$\varlimsup_{i\to\infty} \psi(y_i) \leq \varphi(x). \tag{1.6}$$

We may assume that $\psi(y_i) \neq -\infty$ for all $i > 0$ and the left-hand side of (1.6) is not $-\infty$. Then we can compute

$$\varlimsup_{i\to\infty} \psi(y_i) \le \varlimsup_{i\to\infty} \psi(y_i) + \varliminf_{i\to\infty} \big(\varphi(y_i) - \psi(y_i)\big) \le \varlimsup_{i\to\infty} \varphi(y_i) \le \varphi(x).$$

Combining this with the upper semicontinuity of φ, we get

$$\varlimsup_{i\to\infty} \big(\varphi(y_i) \vee \psi(y_i)\big) \le \varphi(x) = \eta(x).$$

This proves upper semicontinuity across $\Omega \cap \partial\Omega'$. Therefore, $\eta = \eta^* \in \mathrm{PSH}(\Omega)$. □

Note that the boundary condition (1.5) enters the proof only through (1.6), therefore, we can formulate a variant of Lemma 1.2.2:

Corollary 1.2.2 *Let $\Omega \subseteq \mathbb{C}^n$ be a domain and $\Omega' \subseteq \Omega$ be a subdomain*[7]*. Consider $\varphi \in \mathrm{PSH}(\Omega)$ and $\psi \in \mathrm{PSH}(\Omega')$. Assume that*

$$\varlimsup_{\Omega' \ni y \to x} \psi(y) \le \varphi(x)$$

for any $x \in \Omega \cap \partial\Omega'$. Define

$$\eta(z) = \begin{cases} \varphi(z) \vee \psi(z), & \text{if } z \in \Omega'; \\ \varphi(z), & \text{if } z \in \Omega \setminus \Omega'. \end{cases}$$

Then $\eta \in \mathrm{PSH}(\Omega)$.

Using the ideas of Lemma 1.2.2 and the solution of the homogeneous Monge–Ampère equations on a strong pseudoconvex domain, one can prove the following:

Proposition 1.2.9 *Let $\Omega \subseteq \mathbb{C}^n$ be a domain, let $D \Subset \Omega$ be a strongly pseudoconvex domain, and let $\varphi \in \mathrm{PSH}(\Omega)$ be locally bounded. Then there exists a unique locally bounded $\psi \in \mathrm{PSH}(\Omega)$ such that*

$$\begin{cases} (\mathrm{dd^c}\psi)^n = 0 & \text{on } D, \\ \psi = \varphi & \text{on } \Omega \setminus D. \end{cases}$$

Moreover, $\psi \ge \varphi$.

See [BT82, Proposition 9.1] for the proof. We sometimes say ψ is the *balayage* of φ on D.

Let us also recall the following domination principle:

Proposition 1.2.10 *Let $\Omega \subseteq \mathbb{C}^n$ be a bounded domain. Let $\varphi, \psi \in \mathrm{PSH}(\Omega)$ be locally bounded. Assume that*

(1) *for each $z \in \partial\Omega$, we have*

---

[7] In [GZ17, Proposition 3.10], Guedj–Zeriahi assumed in addition that Ω' is relatively compact in Ω. This assumption is not necessary for the proof, and the applications of this proposition in [GZ17] usually do not satisfy this condition.

$$\varliminf_{\Omega \ni w \to z} \Big(\varphi(w) - \psi(w) \Big) \geq 0;$$

(2) $\varphi \leq \psi$ *a.e. with respect to* $(\mathrm{dd^c}\psi)^n$,

Then $\varphi \leq \psi$.

This is [GZ17, Corollary 3.31].[8]

1.3 Plurifine topology

In this section, we introduce the notion of plurifine topology. Unlike other sections in this chapter, this section contains full details. This is mainly due to the unfortunate omissions and numerous minor problems in the foundational paper [BT87]. In particular, the key theorem Theorem 1.3.2 was claimed by Bedford–Taylor without proof. Our presentation largely follows [EMW06]. We also include enough details so that this section is readable for those who are not familiar with the classical potential theory.

Very recently (after the submission of this book to Springer), El Kadiri–Fuglede published a book [EKF25][9] about classical fine potential theory, containing essentially all results in this section. The interested reader may consult their book for further details.

1.3.1 Plurifine topology on domains

Let $\Omega \subseteq \mathbb{C}^n$ $(n \in \mathbb{N})$ be a domain.

Definition 1.3.1 The *plurifine topology* on Ω is the weakest topology making all $\mathbb{R}$-valued plurisubharmonic functions on Ω continuous.

We want to distinguish the Euclidean topology from the plurifine topology. In the whole book, topological notions without adjectives refer to those with respect to the Euclidean topology. We include the symbol $\mathcal{F}$ in order to denote those with respect to the plurifine topology. For example, we will say $\mathcal{F}$-open subset, $\mathcal{F}$-neighborhood, $\mathcal{F}$-closure, etc. The $\mathcal{F}$-closure of a set $E \subseteq \Omega$ will be denoted by $\overline{E}^{\mathcal{F}}$.

Recall that in the whole book, we follow Bourbaki's convention: a neighborhood is not necessarily open. Similarly, an $\mathcal{F}$-neighborhood is not necessarily $\mathcal{F}$-open.

A priori, we should include Ω into the notations as well, but as we will see shortly in Corollary 1.3.1, this is usually unnecessary.

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[8] The statement in the reference is flawed, but the proof is correct.

[9] An anonymous referee kindly pointed me to this reference in a rather enigmatic way: instead of revealing the title or the authors of the book, they merely informed me that, because of the existence of a book on this subject, my entire section was unnecessary.

Proposition 1.3.1 *The plurifine topology on Ω is finer than the Euclidean topology.*

Proof It suffices to show that the unit ball $\{z \in \mathbb{C}^n : |z| < 1\}$ is $\mathcal{F}$-open. This follows from the observation that this set can be written as

$$\{\psi < 0\} \quad \text{with} \quad \psi(z) := (\log|z|) \vee (-1).$$

This completes the proof. □

Example 1.3.1 Let $\varphi \in \mathrm{PSH}(\Omega)$ and $C \in \mathbb{R}$. Then the sets $\{\varphi > C\}$ and $\{\varphi < C\}$ are both $\mathcal{F}$-open.

In fact, the latter case follows from Proposition 1.3.1, while the former follows from the observation

$$\{\varphi > C\} = \{\varphi \vee (C-1) > C\}.$$

Definition 1.3.2 A subset $E \subseteq \Omega$ is *thin*[10] at $x \in \Omega$ if one of the following conditions holds:

(1) $x \notin \overline{E}$;
(2) $x \in \overline{E}$ and there is an open neighborhood $U \subseteq \Omega$ of x and $\varphi \in \mathrm{PSH}(U)$ such that

$$\varlimsup_{(U\cap E)\setminus\{x\}\ni y\to x} \varphi(y) < \varphi(x).$$

We say E is *thin* if it is thin at all $x \in \Omega$.

Remark 1.3.1 In the second case, we can always arrange that

$$\varphi|_{(E\setminus\{x\})\cap U} = 0.$$

In fact, we may assume that $\varphi \leq 0$ and $C < 0$ is such that

$$\varlimsup_{(U\cap E)\setminus\{x\}\ni y\to x} \varphi(y) < C < \varphi(x).$$

We let

$$\psi = (-C)^{-1}(\varphi \vee C) + 1.$$

Then ψ satisfies our requirements for a smaller U.

In the second case, the function φ can be very much improved.

Proposition 1.3.2 (Bedford–Taylor) *Consider a set $E \subseteq \Omega$ and $x \in \overline{E}$. Assume that E is thin at x. Then there is $\varphi \in \mathrm{PSH}(\mathbb{C}^n)$ with the following properties:*

(1) *φ is locally bounded outside x;*
(2) *$\varphi(x) > -\infty$;*
(3)

$$\lim_{E\setminus\{x\}\ni y\to x} \varphi(y) = -\infty.$$

[10] A more proper name would be *plurithin*. But since we will never need the classical notion of thin sets à la Cartan in this book, we prefer omitting the prefix *pluri-*.

Proof [11] By Remark 1.3.1, there is an open neighborhood $U \subseteq \Omega$ of x and $\psi \in \mathrm{PSH}(U)$ such that

$$\psi|_{U\cap(E\setminus\{x\})} = -1 \quad \text{and} \quad \psi(x) = 0.$$

Without loss of generality we may assume $x = 0$ and that $U = B_n(0, 1)$ is the open unit ball.

Since ψ is upper semicontinuous at 0 with $\psi(0) = 0$, we may choose a strictly decreasing sequence $(\delta_j)_{j\geq 1} \subseteq (0, \mathrm{e}^{-1})$ such that

$$\psi(y) < 2^{-j-2} \quad \text{whenever } |y| < \delta_j. \tag{1.7}$$

In particular $|\log\delta_j| > 1$ for every $j \geq 1$, so the series of constants

$$\sum_{j=1}^{\infty} \frac{2^{-j-1}}{|\log\delta_j|} \;\leq\; \sum_{j=1}^{\infty} 2^{-j-1} = \tfrac{1}{2} \tag{1.8}$$

converges. We further set

$$\gamma_j \coloneqq \exp\bigl(2^{j+1}\log\delta_j\bigr) = \delta_j^{2^{j+1}} \in (0, 1),$$

which is also decreasing in j.

For each $j \geq 1$ define

$$\varphi_j(z) \coloneqq \begin{cases} \dfrac{2^{-j-1}}{|\log\delta_j|}\log|z| \;\vee\; \bigl(\psi(z) - 2^{-j}\bigr), & |z| < \delta_j, \\ \dfrac{2^{-j-1}}{|\log\delta_j|}\log|z|, & |z| \geq \delta_j. \end{cases}$$

We check that $\varphi_j \in \mathrm{PSH}(\mathbb{C}^n)$ via the gluing Lemma 1.2.2.

By construction, $\varphi_j \leq 0$ on U, $\varphi_j(0) = -2^{-j}$, and for $z \in E \setminus \{0\}$ with $|z| < \gamma_j$ we have $\varphi_j(z) \leq -1$, since

$$\frac{2^{-j-1}}{|\log\delta_j|}\log|z| < \frac{2^{-j-1}}{|\log\delta_j|}\,(2^{j+1}\log\delta_j) = -1$$

and, on $E \setminus \{0\}$, $\psi(z) - 2^{-j} = -1 - 2^{-j} < -1$.

We then define

$$\varphi \coloneqq \sum_{j=1}^{\infty} \varphi_j.$$

On the unit ball U each $\varphi_j \leq 0$, so the partial sums $S_N \coloneqq \sum_{j=1}^{N} \varphi_j$ form a decreasing sequence of plurisubharmonic functions with

[11] The original argument in [BT82, Proposition 10.2] was quite intriguing: Neither the auxiliary functions φ_j's nor the simple computations were correct. However, I believe that Bedford–Taylor had a correct proof in mind. Something more than a typo, but not yet a mistake, could be properly called a *thinkpo*, a terminology invented by R. Berman.

$$S_N(0) = -\sum_{j=1}^{N} 2^{-j} \to -1 > -\infty, \quad \text{as } N \to \infty$$

so $\varphi|_U \in \mathrm{PSH}(U)$ by Proposition 1.2.2. On $\mathbb{C}^n \setminus B_n(0,\delta_1)$ each φ_j equals $\frac{2^{-j-1}}{|\log \delta_j|} \log|z|$, and (1.8) shows S_N converges uniformly on every compact subset to $\left(\sum_{j\geq 1} \frac{2^{-j-1}}{|\log \delta_j|}\right) \log|z|$, a plurisubharmonic limit. Both formulas agree on $U \setminus B_n(0,\delta_1)$, so altogether $\varphi \in \mathrm{PSH}(\mathbb{C}^n)$.

We now verify the three claims.

Property (2). This is the equality $\varphi(0) = -\sum_{j\geq 1} 2^{-j} = -1 > -\infty$ obtained above.

Property (3). Fix $j_0 \geq 1$. For $z \in E \setminus \{0\}$ with $|z| < \gamma_{j_0}$, the bound $\varphi_j(z) \leq -1$ holds for every $j \leq j_0$ (since the γ_j are decreasing), while $\varphi_j(z) \leq 0$ for $j > j_0$. Hence $\varphi(z) \leq -j_0$, and letting $j_0 \to \infty$ yields

$$\overline{\lim_{E\setminus\{0\}\ni y\to 0}} \varphi(y) = -\infty.$$

Property (1). Fix any $z_0 \in \mathbb{C}^n$ with $z_0 \neq 0$ and choose $J \geq 1$ such that $\delta_J < |z_0|/2$. On the open set $W = \{w \in \mathbb{C}^n : |w| > |z_0|/2\}$, all φ_j with $j \geq J$ lie in the outer regime, and (1.8) shows $\sum_{j\geq J} \varphi_j(w) = \left(\sum_{j\geq J} \frac{2^{-j-1}}{|\log \delta_j|}\right) \log|w|$ is locally bounded on W. The remaining finitely many terms $\varphi_1, \dots, \varphi_{J-1}$ are plurisubharmonic on $\mathbb{C}^n$, hence locally bounded above on W, while $\varphi_j(w) \geq \frac{2^{-j-1}}{|\log \delta_j|} \log|w|$ keeps them locally bounded below as well. Therefore φ is locally bounded on W. Since $z_0 \neq 0$ was arbitrary, φ is locally bounded on $\mathbb{C}^n \setminus \{0\}$, which is property (1). □

Lemma 1.3.1 *Let $E_1, E_2 \subseteq \Omega$. Assume that E_1, E_2 are both thin at $x \in \Omega$. Then so is $E_1 \cup E_2$.*

Proof We may assume that $x \in \overline{E_1} \cap \overline{E_2}$. Take an open neighborhood $U \subseteq \Omega$ of x and $\varphi_1, \varphi_2 \in \mathrm{PSH}(U)$ such that

$$\overline{\lim_{y\to x, y\in E_i\setminus\{x\}}} \varphi_i(y) < \varphi_i(x), \quad i = 1, 2.$$

Then $\varphi_1 + \varphi_2 \in \mathrm{PSH}(U)$ and

$$\overline{\lim_{y\to x, y\in(E_1\cup E_2)\setminus\{x\}}} (\varphi_1 + \varphi_2)(y) < \varphi_1(x) + \varphi_2(x).$$

In particular, $E_1 \cup E_2$ is thin at x. □

Theorem 1.3.1 (H. Cartan) *Consider a set $E \subseteq \Omega$ and $x \in E$. Then the following are equivalent:*

(1) *E is an $\mathcal{F}$-neighborhood of x;*
(2) *$\Omega \setminus E$ is thin at x.*

Proof (2) $\implies$ (1). Assume (2). We may assume that $x \in \overline{\Omega \setminus E}$. Otherwise, our assertion follows from Proposition 1.3.1.

By Proposition 1.3.2, there is an open neighborhood U of x in Ω and $\varphi \in \mathrm{PSH}(\mathbb{C}^n)$ such that

$$\varphi(x) > \sup_{y \in U \cap (\Omega \setminus E)} \varphi(y) =: \lambda.$$

Let $F = \{y \in \Omega : \varphi(y) > \lambda\}$. Then $x \in F$ and F is $\mathcal{F}$-open by Example 1.3.1. Moreover, $U \cap F \subseteq E$. By Proposition 1.3.1, we conclude (1).

(1) $\implies$ (2). Assume (1). We may always replace E by smaller $\mathcal{F}$-neighborhoods of x. In particular, we may assume that E has the following form

$$\{y \in U : \varphi_1(y) > \lambda_1, \dots, \varphi_m(y) > \lambda_m\},$$

where $U \subseteq \Omega$ is an open neighborhood of x, and $\varphi_1, \dots, \varphi_m$ are $\mathbb{R}$-valued psh functions on Ω, and $\lambda_1, \dots, \lambda_m \in \mathbb{R}$. Since a finite union of thin sets is still thin by Lemma 1.3.1, we may assume that $m = 1$. In this case, $\Omega \setminus E$ is clearly thin at x. □

Theorem 1.3.2 *A base of the plurifine topology on Ω is given by sets of the following form:*

$$\{x \in U : \varphi(x) > 0\}, \tag{1.9}$$

where $U \subseteq \Omega$ is an open subset and $\varphi \in \mathrm{PSH}(U)$.

Proof Observe that sets of the form (1.9) are $\mathcal{F}$-open.[12] By Theorem 1.3.1, it suffices to show its complement in Ω is thin at each point of (1.9), which is obvious.

Now consider $x \in \Omega$ and an $\mathcal{F}$-open neighborhood $V \subseteq \Omega$ of x. We want to find a set of the form (1.9) contained in V and containing x.

Write $E = \Omega \setminus V$. In case $x \in \operatorname{Int} V$, there is nothing to prove. So we may assume that $x \in \overline{E}$. By Theorem 1.3.1, E is thin at x. By definition, there is an open neighborhood $U \subseteq \Omega$ of x and $\varphi \in \mathrm{PSH}(U)$ such that

$$\varlimsup_{y \to x, y \in U \cap (E \setminus \{x\})} \varphi(y) < \varphi(x).$$

We may assume that $\varphi|_{E \cap U} \leq 0 < \varphi(x)$. Then the set $\{y \in U : \varphi(y) > 0\}$ suffices for our purpose. □

Remark 1.3.2 Recall that in general, an $\mathcal{F}$-open set is *not* a *countable* union of sets of the form (1.9). In fact, an $\mathcal{F}$-open set is not a Borel set in general. See [EK23] for a concrete example.

Corollary 1.3.1 *Let $\Omega_1 \subseteq \Omega_2 \subseteq \mathbb{C}^n$ be two non-empty open subsets. Then the plurifine topology on Ω_1 is the same as the subspace topology induced from the plurifine topology on Ω_2.*

In particular, when we talk about an $\mathcal{F}$-open set U in $\mathbb{C}^n$, we no longer have to specify the domain $\Omega \supseteq U$.

Corollary 1.3.2 *Let $\Omega \subseteq \mathbb{C}^n$ be an $\mathcal{F}$-open subset and $x \in \Omega$. Then x has a compact $\mathcal{F}$-neighborhood contained in Ω.*

[12] This is not entirely obvious by definition, as φ is not defined on the whole Ω.

Proof By Theorem 1.3.2, we may assume that there is an open set $U \subseteq \mathbb{C}^n$ and a plurisubharmonic function φ on U such that $\Omega = \{y \in U : \varphi(y) > 0\}$.

Take a compact neighborhood K of x in U. Now $\{y \in K : \varphi(y) \geq \varphi(x)/2\}$ is a compact $\mathcal{F}$-neighborhood of x contained in Ω. □

Corollary 1.3.3 *Let $\Omega \subseteq \mathbb{C}^n$, $\Omega' \subseteq \mathbb{C}^{n'}$ be two domains and $F : \Omega' \to \Omega$ be a holomorphic map. Then F is $\mathcal{F}$-continuous.*

Proof It suffices to show that the inverse image $F^{-1}(U)$ of each $\mathcal{F}$-open subset $U \subseteq \Omega$ is $\mathcal{F}$-open. By Theorem 1.3.2, after possibly shrinking Ω and Ω', we may assume that U has the form $\{x \in \Omega : \psi(x) > 0\}$, where $\psi \in \mathrm{PSH}(\Omega)$. Clearly, we may assume that $F^{-1}(U)$ is non-empty. Then $F^*\psi \in \mathrm{PSH}(\Omega')$ by Proposition 1.1.4, we find that

$$F^{-1}(U) = \{y \in \Omega' : F^*\psi(y) > 0\}$$

is $\mathcal{F}$-open. □

1.3.2 Plurifine topology on manifolds

Let X be a complex manifold.

Definition 1.3.3 The *plurifine topology* on X is the topology with a base consisting of sets of the form $F^{-1}(V)$, where $U \subseteq X$ is an open subset, $F : U \to \Omega$ is a biholomorphic map, $\Omega \subseteq \mathbb{C}^n$ is a domain for some $n \in \mathbb{N}$, and $V \subseteq \Omega$ is $\mathcal{F}$-open.

Note that these sets form a topological base thanks to Corollary 1.3.3. Moreover, it also follows from Corollary 1.3.3 that the plurifine topologies on domains defined in Definition 1.3.3 and in Definition 1.3.1 coincide.

We refer to Definition 1.5.1 for the notion of quasi-plurisubharmonic functions.

Proposition 1.3.3 *Let $\varphi \in \mathrm{QPSH}(X)$. Then φ is $\mathcal{F}$-continuous.*

Proof It suffices to show that $\varphi|_{\{\varphi \neq -\infty\}}$ is $\mathcal{F}$-continuous, since $\{\varphi = -\infty\}$ is $\mathcal{F}$-closed.

The problem is local, so we may assume that $X \subseteq \mathbb{C}^n$ is a domain and $\varphi = \psi + g$, where $\psi \in \mathrm{PSH}(X)$ and $g \in C^\infty(X)$ and $|g| \leq C$ for some $C > 0$. Take an open interval $(a, b) \subseteq \mathbb{R}$, it suffices to show that

$$U := \{x \in X : a < \varphi(x) < b\} = \{x \in X : a - g(x) < \psi(x) < b - g(x)\}$$

is $\mathcal{F}$-open. Take $x \in U$, we can find an open neighborhood V of x in U such that

$$\sup_{y \in V} (a - g(y)) < \psi(x) < \inf_{y \in V} (b - g(y)) .$$

Therefore,

$$\left\{ z \in V : \sup_{y \in V} (a - g(y)) < \psi(z) < \inf_{y \in V} (b - g(y)) \right\}$$

is an $\mathcal{F}$-open neighborhood of x in U. We conclude that U is $\mathcal{F}$-open. □

Corollary 1.3.4 *Let $\varphi, \psi \in \mathrm{QPSH}(X)$. Then the set*

$$\{x \in X : \varphi(x) > \psi(x)\}$$

is $\mathcal{F}$-open.

Proof It suffices to show that for any $x \in X$ such that $\varphi(x) > \psi(x)$, the same holds on an $\mathcal{F}$-neighborhood U of x. Observe that $\varphi(x) \neq -\infty$. If $\psi(x) \neq -\infty$, then it suffices to apply Proposition 1.3.3. Otherwise, take

$$U \coloneqq \{y \in X : \varphi(y) > \varphi(x) - 1\} \cap \{y \in X : \psi(y) < \varphi(x) - 1\}.$$

This is an $\mathcal{F}$-open neighborhood of x contained in $\{\varphi > \psi\}$. □

Lemma 1.3.2 *Let $Z \subseteq X$ be a pluripolar subset. Then*

$$\overline{X \setminus Z}^{\mathcal{F}} = X.$$

Proof The problem is local, so we may assume that X is a domain in $\mathbb{C}^n$ and that there is $\varphi \in \mathrm{PSH}(X)$ with $Z \subseteq \{\varphi = -\infty\}$. It suffices to show that $\{\varphi > -\infty\}$ is $\mathcal{F}$-dense.

Let $x \in X$ be a point with $\varphi(x) = -\infty$ and $U \subseteq X$ be an $\mathcal{F}$-open neighborhood of x in X. We need to show that $U \cap \{\varphi > -\infty\} \neq \emptyset$.

Thanks to Theorem 1.3.2, after shrinking U, we may assume that there is $\psi \in \mathrm{PSH}(X)$ such that $U = \{\psi > 0\}$. Observe that U is not a pluripolar set: Otherwise, $\psi \leq 0$ almost everywhere by Proposition 1.2.4, and hence everywhere by Proposition 1.2.6. So $\varphi|_U \not\equiv -\infty$. We conclude that $U \cap \{\varphi > -\infty\} \neq \emptyset$, as desired. □

Corollary 1.3.5 *Let $\varphi, \psi \in \mathrm{QPSH}(X)$. Set*

$$W = \{x \in X : \varphi(x) = -\infty\} \text{ or } W = \{x \in X : \psi(x) = -\infty\}.$$

Then for any pluripolar set $Z \subseteq X$, we have

$$\sup_{X \setminus W} (\varphi - \psi) = \sup_{X \setminus (W \cup Z)} (\varphi - \psi), \quad \inf_{X \setminus W} (\varphi - \psi) = \inf_{X \setminus (W \cup Z)} (\varphi - \psi).$$

In particular, taking $\psi = 0$, we find that

$$\sup_{X \setminus Z} \varphi = \sup_X \varphi.$$

Proof This is an immediate consequence of Lemma 1.3.2 and Proposition 1.3.3. □

In the literature about pluripotential theory, one often finds the careless expressions like $\sup_X (\varphi - \psi)$. The issue is that $\varphi - \psi$ is not defined everywhere, and hence this expression does not make sense if you read it literally. Corollary 1.3.5 tells you that you do not have to worry too much about the details on a pluripolar set. In other words, sup and inf could always be understood as a kind of essential supremum and essential infimum modulo pluripolar sets.

There is a convenient way to fix this issue in the literature — just replace the suprema by quasi-suprema:

Definition 1.3.4 Let $Z \subseteq X$ be a pluripolar set, and $f\colon X \setminus Z \to [-\infty, \infty]$ be a function. We define the *quasi-supremum* and the *quasi-infimum* of f as follows:

$$\begin{aligned} \operatorname*{q-sup}_X f &:= \inf\left\{ \sup_{X\setminus W} f : W \supseteq Z, W \text{ is pluripolar} \right\}, \\ \operatorname*{q-inf}_X f &:= \sup\left\{ \inf_{X\setminus W} f : W \supseteq Z, W \text{ is pluripolar} \right\}. \end{aligned}$$

For two functions f and g equal quasi-everywhere, the quasi-suprema and quasi-infima of them are equal, as is clear from the definition.

For a quasi-plurisubharmonic function φ, we have

$$\operatorname*{q-sup}_X \varphi = \sup_X \varphi,$$

as a consequence of Corollary 1.3.5.

1.4 Lelong numbers and multiplier ideal sheaves

In this section, we briefly recall the notions of Lelong numbers and multiplier ideal sheaves. Our presentation is by no means intended to be complete. Readers are encouraged to read the textbooks [GZ17, Section 2.3] and [Dem12a].

Let X be a complex manifold.

Definition 1.4.1 Let $\varphi \in \mathrm{PSH}(X)$ and $x \in X$. The *Lelong number* $\nu(\varphi, x)$ of φ at x is defined as follows: Take an open neighborhood U of x in X and a biholomorphic map $F\colon U \to \Omega$, where Ω is a domain in $\mathbb{C}^n$. Then we define

$$\nu(\varphi, x) := \sup\left\{ \gamma \in \mathbb{R}_{\geq 0} : \varphi(F^{-1}(y)) \leq \gamma \log |y - F(x)|^2 + O(1) \text{ as } y \to F(x) \right\}. \tag{1.10}$$

Observe that $\nu(\varphi, x)$ does not depend on the choices of U and F, see [Dem12b, Page 167, Corollary 7.2]. Furthermore, it follows from Proposition 1.4.1 below that the supremum in (1.10) is a maximum.

Remark 1.4.1 Our definition of the Lelong number is not standard. It differs from the standard definition by a factor of 2. As a mnemonic, just remember

$$\nu\left(\log|z|^2, 0\right) = 1 \quad \text{(instead of 2)}.$$

Our convention of the Lelong numbers together with the convention of the multiplier ideal sheaves below make sure that (1.13) has no extra factors.

Proposition 1.4.1 *Let* $\varphi \in \mathrm{PSH}(B_n(0,1))$. *Then*

$$\nu(\varphi, 0) = \lim_{r\to 0+} \frac{\sup_{B_n(0,r)} \varphi}{\log r^2} \in [0, \infty). \tag{1.11}$$

Proof It follows from Proposition 1.1.2 that the limit in (1.11) exists and is finite. We shall denote the limit by $\nu'(\varphi, 0)$ for the time being.

We first observe that by Proposition 1.1.2, for any fixed $r_1 \in (0, 1)$ there is a constant C_{r_1} such that

$$\varphi(x) \leq \nu'(\varphi, 0) \log|x|^2 + C_{r_1} \tag{1.12}$$

when $x \in B_n(0, r_1) \setminus \{0\}$. In particular, $\nu(\varphi, 0) \geq \nu'(\varphi, 0)$.

In order to argue the reverse inequality, we may assume that $\nu(\varphi, 0) > 0$.

Next observe that by (1.10), for each small enough $\epsilon > 0$, we can find $r_0 \in (0, 1)$ and $C > 0$ so that for all $x \in B_n(0, r_0) \setminus \{0\}$, we have

$$\varphi(x) \leq (\nu(\varphi, 0) - \epsilon) \log|x|^2 + C.$$

It follows that $\nu'(\varphi, 0) \geq \nu(\varphi, 0) - \epsilon$. Letting $\epsilon \to 0+$, we conclude that $\nu'(\varphi, 0) \geq \nu(\varphi, 0)$, which combined with the previous inequality gives (1.11). □

We recall Siu's semicontinuity theorem.

Theorem 1.4.1 (Siu) *Let* $\varphi \in \mathrm{PSH}(X)$. *Then the map* $X \ni x \mapsto \nu(\varphi, x)$ *is upper semicontinuous with respect to the Zariski topology.*

For an elegant proof we refer to [Dem12a, Theorem 2.10].

Proposition 1.4.2 *Let* $\varphi, \psi \in \mathrm{PSH}(X)$, $\lambda \in \mathbb{R}_{\geq 0}$ *and* $x \in X$. *Then*

$$\begin{aligned}
\nu(\varphi \vee \psi, x) =& \min\{\nu(\varphi, x), \nu(\psi, x)\},\\
\nu(\varphi + \psi, x) =& \nu(\varphi, x) + \nu(\psi, x),\\
\nu(\lambda\varphi, x) =& \lambda\nu(\varphi, x).
\end{aligned}$$

Proof All properties are local, so we may assume that $X = B_n(0, 1)$ for some $n \in \mathbb{N}$. All properties follow directly from Proposition 1.4.1. □

Corollary 1.4.1 *Let* $(\varphi_i)_{i\in I}$ *be a non-empty family in* $\mathrm{PSH}(X)$ *locally uniformly bounded from above and* $x \in X$. *Then*

$$\nu\left(\sup_{i\in I}{}^{*}\varphi_i, x\right) = \inf_{i\in I} \nu(\varphi_i, x).$$

Proof We may assume that X is connected. Write $\varphi = \sup_{i\in I}{}^*\varphi_i$. Then $\varphi \in \mathrm{PSH}(X)$ by Proposition 1.2.2.

We observe that the $\leq$ inequality is trivial. It remains to argue the reverse inequality.

It follows from Proposition 1.2.3 that we may assume that I is countable. When I is finite, this is already proved in Proposition 1.4.2. Otherwise, we may further assume that $I = \mathbb{Z}_{>0}$. Thanks to Proposition 1.4.2, we may further assume that $(\varphi_i)_{i\in\mathbb{Z}_{>0}}$ is an increasing sequence. Furthermore, since the problem is local, we may assume that $x = 0$, that $X = B_n(0,1)$ for some $n \in \mathbb{N}$, and that $(\varphi_i)_i$ is uniformly bounded from above on some ball $B_n(0,r_1)$ with $0 < r_1 < 1$. Since the sequence $(\varphi_i)_i$ is increasing, the sequence $\nu(\varphi_i, 0)$ is decreasing. Thus the constants in (1.12) can be chosen independently of i, and we have

$$\varphi_i(x) \leq \nu(\varphi_i, 0)\log|x|^2 + C$$

for all $x \in B_n(0,r_1) \setminus \{0\}$ and all $i \geq 1$, where C is a constant independent of i. In particular, thanks to Proposition 1.2.5, for almost all $x \in B_n(0,r_1)$, we have

$$\varphi(x) \leq \lim_{i\to\infty} \nu(\varphi_i, 0)\log|x|^2 + C.$$

Thanks to Proposition 1.2.6, the same holds for all $x \neq 0$ near 0, and hence

$$\nu(\varphi, 0) \geq \lim_{i\to\infty} \nu(\varphi_i, 0).$$

This completes the proof. □

Definition 1.4.2 Let $F \subseteq X$ be a non-empty analytic subset. Then we define the *generic Lelong number* of φ along F as

$$\nu(\varphi, F) \coloneqq \min_{x\in F} \nu(\varphi, x).$$

Note that the minimum is attained by Theorem 1.4.1.

Definition 1.4.3 Let $\varphi \in \mathrm{PSH}(X)$. Let E be a prime divisor over X (see Definition B.1.1). Take a proper bimeromorphic morphism $\pi\colon Y \to X$ from a complex manifold Y such that E is a prime divisor on Y, then we define the *generic Lelong number* of φ along E as

$$\nu(\varphi, E) \coloneqq \nu(\pi^*\varphi, E).$$

It follows from Theorem 1.4.1 that $\nu(\varphi, E)$ does not depend on the choice of π.

Definition 1.4.4 Let $\varphi \in \mathrm{PSH}(X)$, the *multiplier ideal sheaf* $\mathcal{I}(\varphi)$ of φ is by definition the ideal sheaf given by

$$\Gamma(U, \mathcal{I}(\varphi)) = \left\{ f \in \mathcal{O}_X(U) : |f|^2 \exp(-\varphi) \in L^1_{\mathrm{loc}}(U) \right\}$$

for any open subset $U \subseteq X$.

Remark 1.4.2 This definition is different from a few references, where instead of $\exp(-\varphi)$, they use $\exp(-2\varphi)$. The conventions adopted in the current book are the most convenient ones as far as I know. They simplify a number of formulae. As a mnemonic, for any real $\lambda > 0$, we have

$$\mathcal{I}(\lambda \log |z|^2) = \mathcal{O}_{\mathbb{C}}(-\lfloor \lambda \rfloor \{0\}) \quad \Big(\text{instead of } \mathcal{O}_{\mathbb{C}}(-\lfloor 2\lambda \rfloor \{0\})\Big),$$

where z is a variable in $\mathbb{C}$ and $\{0\}$ is the divisor defined by $0 \in \mathbb{C}$.

Proposition 1.4.3 (Nadel) *Let $\varphi \in \mathrm{PSH}(X)$. Then $\mathcal{I}(\varphi)$ is coherent.*

See [Dem12a, Proposition 5.7].

Theorem 1.4.2 *Let $\varphi, \psi \in \mathrm{PSH}(X)$. Then*

$$\mathcal{I}(\varphi + \psi) \subseteq \mathcal{I}(\varphi) \cdot \mathcal{I}(\psi).$$

See [DEL00], [Dem12a, Theorem 14.2].

The two invariants are related by the following simple result:

Proposition 1.4.4 *Let $\varphi \in \mathrm{PSH}(X)$ and E be a prime divisor over X. Then*

$$\nu(\varphi, E) = \lim_{k \to \infty} \frac{1}{k} \operatorname{ord}_E \mathcal{I}(k\varphi). \tag{1.13}$$

See [DX24b, Proposition 2.14].

Recall that this particular form of the formula is compatible with our conventions of ν and $\mathcal{I}$. As a consistency check, consider $\varphi = \log |z|^2$ with $z \in \mathbb{C}$ and let E be the divisor $\{0\}$. Then both sides of (1.13) are equal to 1. See Remark 1.4.1 and Remark 1.4.2.

Also observe the following simple lemma:

Lemma 1.4.1 *Let $x \in X$ and $\varphi \in \mathrm{PSH}(X)$. Let $\pi : Y \to X$ be the blow-up of X at x with exceptional divisor E. Then*

$$\nu(\varphi, x) = \nu(\varphi, E),$$

See [Bou02b, Corollaire 1.1.8].

Conversely, the information of the generic Lelong numbers determines the multiplier ideal sheaves:

Theorem 1.4.3 *Let $\varphi \in \mathrm{PSH}(X)$. Let $x \in X$ and $f \in \mathcal{O}_{X,x}$. Then the following are equivalent:*

(1) *$f \in \mathcal{I}(\varphi)_x$;*
(2) *there exists $\epsilon > 0$ such that for any prime divisor E over X such that x is contained in the center of E on X, we have*

$$\operatorname{ord}_E(f) \geq (1 + \epsilon)\nu(\varphi, E) - A_X(E). \tag{1.14}$$

In case φ has analytic singularities and $\pi\colon Y \to X$ is a log resolution of φ (see Definition 1.6.3 for the definition) with finitely many exceptional divisors $\{E_i\}$ whose centers on X contain x, one may replace (1.14) by

$$\operatorname{ord}_{E_i}(f) > \nu(\varphi, E_i) - A_X(E_i) \quad \forall i. \tag{1.15}$$

Here $A_X(\bullet)$ denotes the log discrepancy of prime divisors. We refer to [Bou17, Corollary 10.18, Proposition 10.12] for the proof and the precise definition of A_X.

The notion of analytic singularities is recalled in Section 1.6.

Recall the strong openness theorem:

Theorem 1.4.4 (Guan–Zhou) *Let $\varphi, \psi_j \in \mathrm{PSH}(X)$ ($j \in \mathbb{Z}_{>0}$) such that ψ_j is an increasing sequence converging to φ almost everywhere. Then for any $x \in X$, the germs satisfy*

$$\mathcal{I}\left(\psi_j\right)_x = \mathcal{I}(\varphi)_x$$

when j is large enough.

See [GZ15, Hie14] for the proof.

Proposition 1.4.5 *Let $\pi\colon Y \to X$ be a smooth morphism between complex manifolds. Assume that $\varphi \in \mathrm{PSH}(X)$. Then*

$$\mathcal{I}(\pi^*\varphi) = \pi^*\mathcal{I}(\varphi).$$

Proof It follows from [Gro60, Théorème 3.10] that locally π can be written as the composition of an étale morphism and a projection. It suffices to handle the two cases separately.

Recall that in the complex analytic setting, an étale morphism is locally biholomorphic, so there is nothing to prove in this case.

Next, assume that $Y = X \times U$, where $U \subseteq \mathbb{C}^n$ is a domain and π is the natural projection. It follows from Fubini's theorem that

$$\mathcal{I}(\pi^*\varphi) \supseteq \pi^*\mathcal{I}(\varphi).$$

The reverse inequality is proved in [Dem12a, Proposition 14.3][13]. □

Definition 1.4.5 Given a coherent ideal sheaf $\mathcal{I}$ on X and a closed complex subspace $Y \subseteq X$, the *restriction* $\mathrm{Res}_Y \mathcal{I}$ is the inverse image ideal sheaf given by

$$\mathrm{Res}_Y \mathcal{I} := \mathcal{I}/(\mathcal{I} \cap \mathcal{I}_Y), \tag{1.16}$$

where $\mathcal{I}_Y$ is the ideal sheaf defining Y.

In the literature, it is common to denote this sheaf by the misleading notation $\mathcal{I}|_Y$.

There is a natural morphism

[13] In [Dem12a, Proposition 14.3], Demailly used the highly non-standard notation $f^*\mathcal{I}(\varphi)$ to denote the image of $f^*\mathcal{I}(\varphi) \to \mathcal{O}_X$, even when f is not flat.

$$i_Y^* \mathcal{I} = \mathcal{I}/(\mathcal{I} \cdot \mathcal{I}_Y) \to \operatorname{Res}_Y \mathcal{I}, \tag{1.17}$$

where $i_Y : Y \to X$ is the inclusion.

Theorem 1.4.5 (Ohsawa–Takegoshi) *Let Y be a connected submanifold of X and $\varphi \in \mathrm{PSH}(X)$. Assume that $\varphi|_Y \not\equiv -\infty$. Then*

$$\mathcal{I}(\varphi|_Y) \subseteq \operatorname{Res}_Y \mathcal{I}(\varphi).$$

See [Dem12a, Theorem 14.1].

1.5 Quasi-plurisubharmonic functions

In practice, it is important to consider a variant of plurisubharmonic functions. We will fix a complex manifold X.

Definition 1.5.1 Let θ be a closed real smooth $(1,1)$-form on X.

A *θ-plurisubharmonic function* or *plurisubharmonic function!θ-psh function* for short on X is a function $\varphi\colon X \to [-\infty, \infty)$ such that for each $x \in X$ and each open neighborhood U of x in X satisfying the condition that $\theta = \mathrm{dd^c} g$ for some smooth function g on U, we have $g + \varphi|_U \in \mathrm{PSH}(U)$. The set of θ-psh functions on X is denoted by $\mathrm{PSH}(X, \theta)$.

A *quasi-plurisubharmonic function* on X is a function $\varphi\colon X \to [-\infty, \infty)$ such that there exists a smooth closed real $(1,1)$-form θ' on X such that $\varphi \in \mathrm{PSH}(X, \theta')$. The set of quasi-plurisubharmonic functions on X is denoted by $\mathrm{QPSH}(X)$.

There is a natural preorder on $\mathrm{QPSH}(X)$ defined as follows:

Definition 1.5.2 Assume that X is compact. Given $\varphi, \psi \in \mathrm{QPSH}(X)$, we say that φ is *more singular* than ψ and write $\varphi \preceq \psi$ [14] if there is $C \in \mathbb{R}$ such that $\varphi \leq \psi + C$. We also say ψ is less singular than φ and write $\psi \succeq \varphi$.

In case $\varphi \preceq \psi$ and $\psi \preceq \varphi$, we say φ and ψ have the same *singularity type*. We write $\varphi \sim \psi$ in this case.

When X is not compact, one can still define similar notions, but the generalization is not unique, and we shall not consider them in this book.

Remark 1.5.1 The preceding results concerning plurisubharmonic functions can be extended *mutatis mutandis* to quasi-plurisubharmonic functions. We will apply these extensions without further explanations.

Proposition 1.5.1 *Assume that X is compact. Let θ be a closed real smooth $(1,1)$-form on X. Then for any $a, b \in \mathbb{R}$, $a \leq b$, the set*

$$\left\{\varphi \in \mathrm{PSH}(X, \theta) : \sup_X \varphi \in [a, b]\right\}$$

[14] Some people write $\psi \preceq \varphi$.

is compact with respect to the L^1-topology. Moreover, $\varphi \mapsto \sup_X \varphi$ is L^1-continuous for $\varphi \in \mathrm{PSH}(X,\theta)$.

This is an immediate consequence of [GZ17, Proposition 8.5, Exercise 1.20].

Remark 1.5.2 More generally, if $K \subseteq X$ is a closed non-pluripolar subset, then

$$\left\{\varphi \in \mathrm{PSH}(X,\theta) : \sup_K \varphi \in [a,b]\right\}$$

is relatively compact with respect to the L^1-topology. See [GZ05, Corollary 4.3].

Proposition 1.5.2 *Assume that X is compact. Let θ be a closed real smooth $(1,1)$-form on X and E be a prime divisor over X. Then*

$$\sup\left\{\nu(\varphi,E) : \varphi \in \mathrm{PSH}(X,\theta)\right\} < \infty.$$

Proof It follows from the proof of [GZ17, Exercise 2.7] that $\nu(\bullet,E)$ is upper semicontinuous with respect to the L^1-topology on $\mathrm{PSH}(X,\theta)$. Since adding a constant does not change Lelong numbers, we may restrict to the compact set $\{\varphi \in \mathrm{PSH}(X,\theta) : \sup_X \varphi = 0\}$ from Proposition 1.5.1. The desired upper bound follows. ◻

Proposition 1.5.3 *Let $\pi\colon Y \to X$ be a proper bimeromorphic morphism from a compact Kähler manifold Y. Let θ be a closed real smooth $(1,1)$-form on X. Then the pull-back gives a bijection*

$$\pi^*\colon \mathrm{PSH}(X,\theta) \xrightarrow{\sim} \mathrm{PSH}(Y,\pi^*\theta).$$

This follows from a more general result Theorem B.1.1.

Since we shall constantly need the balayage argument of Proposition 1.2.9, we explain explicitly the global version of it.

Proposition 1.5.4 *Fix a closed smooth real $(1,1)$-form θ on X. Let $\varphi \in \mathrm{PSH}(X,\theta)$, and $B \Subset B' \subseteq X$ be two holomorphic balls. Assume that φ is locally bounded on B'. Then there is a unique $\psi \in \mathrm{PSH}(X,\theta)$ so that ψ is bounded on an open neighborhood of $\overline{B}$, and*

$$\begin{cases} (\theta|_B + \mathrm{dd^c}\psi|_B)^n = 0, \\ \psi = \varphi \quad \text{on } X \setminus B. \end{cases}$$

Moreover, $\psi \geq \varphi$.

We shall call ψ the *balayage* of φ on B.

Proof By the Dolbeault–Grothendieck lemma, after possibly shrinking B', we may write $\theta = \mathrm{dd^c} g$ for some bounded function $g \in C^\infty(B')$. We can then use g to translate θ-psh functions to psh functions, and our assertion follows immediately from Proposition 1.2.9. ◻

1.6 Analytic singularities

The simplest type of plurisubharmonic singularities is given by the so-called *analytic singularities*. The notion is fairly subtle and there are several mutually *incompatible* definitions in the literature, as we shall explain below.

Let X be a complex manifold.

Definition 1.6.1 We say $\varphi \in \mathrm{QPSH}(X)$ has *analytic singularities* if for each $x \in X$, we can find a connected open neighborhood U of x such that $\varphi|_U$ has the form:

$$c\log(|f_1|^2 + \cdots + |f_N|^2) + R, \tag{1.18}$$

where $f_1, \ldots, f_N$ are holomorphic functions on U, $c \in \mathbb{Q}_{>0}$ and R is a bounded function on U.[15]

When R can be taken to be smooth[16], we say φ has *neat analytic singularities*.

Suppose that there is a coherent ideal $\mathcal{I} \subseteq \mathcal{O}_X$ on X such that we can choose U so that the $f_1, \ldots, f_N$ can be chosen as the generators of $\Gamma(U, \mathcal{I})$ and c is independent of the choice of U, we say φ has analytic singularities of *type* $(c, \mathcal{I})$.

Each potential with analytic singularities has a type. The type is not uniquely determined. We refer to [Bou02b] and [Bou02a] for the details.

Some authors allow $c \in \mathbb{R}_{>0}$ in (1.18). We keep $c \in \mathbb{Q}_{>0}$ because the following closure property, which is essential in constructing Demailly approximations, would otherwise fail.

Proposition 1.6.1 *Let $\varphi, \psi \in \mathrm{QPSH}(X)$ be potentials with analytic singularities. Then so are $\lambda\varphi$ ($\lambda \in \mathbb{Q}_{>0}$), $\varphi + \psi$ and $\varphi \vee \psi$.*

Proof The assertion for $\lambda\varphi$ follows by multiplying the rational coefficient in (1.18). The assertion for $\vee$ is proved in [Dem15, Proposition 4.1.8]. For the addition assertion, take local presentations as in (1.18) and multiply the corresponding finite systems of holomorphic generators. □

Definition 1.6.2 Let D be an effective $\mathbb{Q}$-divisor[17] on X. We say $\varphi \in \mathrm{QPSH}(X)$ has *log singularities* (along D) on X if for each $x \in X$, there is an open neighborhood U of x such that

(1) $D|_U$ has finitely many irreducible components and can be written as

$$D|_U = \sum_{i=1}^{N} a_i D_i$$

[15] Note that since $\varphi|_U \not\equiv -\infty$ by assumption, it is implicit in the definition that $\{f_1 = \cdots = f_N = 0\} \subseteq U$ is a proper subset.

[16] The decomposition (1.18) is highly non-unique. Here we mean for any x, there is an open neighborhood U and a decomposition of the form (1.18) with R smooth. In the non-trivial cases, R cannot be smooth for all decompositions (1.18).

[17] Divisors and $\mathbb{Q}$-divisors are implicitly assumed to have locally finite coefficients as usual.

with D_i being prime divisors on U, $a_i \in \mathbb{Q}_{>0}$ and there is a holomorphic function s_i on U defining D_i, and

(2) we have
$$\varphi|_U = \sum_{i=1}^{N} a_i \log |s_i|^2 + R, \tag{1.19}$$
where R is a bounded function on U.

By Proposition 1.6.1, φ has analytic singularities.

Lemma 1.6.1 *Suppose that X is a compact Kähler manifold, θ is a closed smooth real $(1,1)$-form on X, and $\varphi \in \mathrm{PSH}(X, \theta)$. Suppose that φ has log singularities along an effective $\mathbb{Q}$-divisor D on X. Then the cohomology class $\{\theta\} - \{D\}$ is nef.*

Moreover, if in addition θ_φ is a Kähler current[18]*, then the cohomology class $\{\theta\} - \{D\}$ is ample.*

Here and in the sequel, we write θ_φ for $\theta + \mathrm{dd}^c\varphi$.

Proof The current $\theta_\varphi - [D]$ has bounded local potentials. Therefore, the class $\{\theta\} - \{D\}$ is nef.

If θ_φ is a Kähler current, say $\theta_\varphi \geq \omega$ for a Kähler form ω, then $\theta_\varphi - [D] - \omega$ is positive and has bounded local potentials, and therefore represents a nef class. We conclude that $\{\theta\} - \{D\}$ is ample. □

The following proposition follows immediately from the definitions:

Proposition 1.6.2 *Let $\pi : Y \to X$ be a proper bimeromorphic morphism from a complex manifold Y. Suppose that $\varphi \in \mathrm{QPSH}(X)$ has analytic singularities (resp. has log singularities along an effective $\mathbb{Q}$-divisor D). Then $\pi^*\varphi$ has analytic singularities (resp. has log singularities along π^*D).*

Definition 1.6.3 Let $\varphi \in \mathrm{QPSH}(X)$ be a potential with analytic singularities. A *log resolution* of φ is a modification $\pi : Y \to X$ such that $\pi^*\varphi$ has log singularities.

See Definition B.1.3 for the notion of modifications.

Theorem 1.6.1 *Assume that X is compact. Suppose that $\varphi \in \mathrm{QPSH}(X)$ has analytic singularities. Then there is a log resolution of φ.*

For a proof, we refer to the arguments on [MM07, Page 104].

A general quasi-plurisubharmonic function can be nicely approximated by those with analytic singularities. We need a few preliminary definitions.

Definition 1.6.4 Let X be a compact Kähler manifold and θ be a closed real smooth $(1,1)$-form on X. Consider $\varphi \in \mathrm{PSH}(X, \theta)$. A sequence $(\varphi_j)_{j \in \mathbb{Z}_{>0}}$ in $\mathrm{QPSH}(X)$ is *quasi-equisingular approximation* of φ if

(1) φ_j has analytic singularities for each j;

[18] That is, there is a Kähler form ω on X such that $\theta_\varphi \geq \omega$ in the sense of currents.

(2) φ_j is decreasing with limit φ;
(3) there is a decreasing sequence $\epsilon_j \geq 0$ with limit 0 and a Kähler form ω on X such that $\varphi_j \in \mathrm{PSH}(X, \theta + \epsilon_j \omega)$;
(4) for each $\lambda' > \lambda > 0$, there is $j > 0$ such that

$$\mathcal{I}(\lambda' \varphi_j) \subseteq \mathcal{I}(\lambda \varphi). \tag{1.20}$$

We also say $(\theta_{\varphi_j})_j$ is a quasi-equisingular approximation of θ_φ.

Definition 1.6.5 Let $\mathcal{I} \subseteq \mathcal{O}_X$ be a coherent ideal sheaf and $c \in \mathbb{Q}_{>0}$. A function $\varphi \in \mathrm{QPSH}(X)$ is said to have *gentle analytic singularities* (of type $(c, \mathcal{I})$) if

(1) φ has analytic singularities of type $(c, \mathcal{I})$;
(2) $\mathrm{e}^{\varphi/c} \colon X \to \mathbb{R}_{\geq 0}$ is a smooth function;
(3) there is a proper bimeromorphic morphism $\pi \colon \widetilde{X} \to X$ from a Kähler manifold $\widetilde{X}$ and an effective $\mathbb{Z}$-divisor D on $\widetilde{X}$ such that one can write $\pi^*\varphi$ locally as

$$\pi^*\varphi = c \log |g|^2 + h,$$

where g is a local equation of the divisor D and h is smooth.

Theorem 1.6.2 *Let X be a compact Kähler manifold and θ be a closed real smooth $(1,1)$-form on X. Then any $\varphi \in \mathrm{PSH}(X, \theta)$ admits a quasi-equisingular approximation $(\varphi_j)_{j \in \mathbb{Z}_{>0}}$.*

Moreover, we can guarantee that for all $j > 0$, φ_j has gentle analytic singularities of type $\left(2^{-j}, \mathcal{I}(2^j \varphi)\right)$.

We refer to [Dem92, DPS01] for the proof.

Quasi-equisingular approximations are essentially unique in the following sense:

Proposition 1.6.3 *Let X be a compact Kähler manifold and θ be a closed real smooth $(1,1)$-form on X. Consider $\varphi \in \mathrm{PSH}(X, \theta)$. Let $(\varphi_j)_j$ and $(\psi_j)_j$ be two quasi-equisingular approximations of φ. Then for any $\epsilon > 0$ and any $j > 0$, we can find $k_0 > 0$ such that for any $k \geq k_0$, we have*

$$\psi_k \preceq (1 - \epsilon) \varphi_j.$$

See [Dem15, Corollary 4.1.7].

Definition 1.6.6 Let $\varphi \in \mathrm{QPSH}(X)$ be a potential with analytic singularities. Then we define $\mathcal{I}_\infty(\varphi)$ as the ideal sheaf consisting of germs f of holomorphic functions such that $|f|^2 \exp(-\varphi)$ is locally bounded.

By definition, $\mathcal{I}_\infty(\varphi) \subseteq \mathcal{I}(\varphi)$.

Lemma 1.6.2 *Assume that X is compact. Let $\varphi \in \mathrm{QPSH}(X)$ be a potential with analytic singularities. The sheaf $\mathcal{I}_\infty(\varphi)$ is coherent.*

Proof By Theorem 1.6.1, we may find a modification $\pi\colon Y \to X$ such that $\pi^*\varphi$ has log singularities. Observe that

$$\mathcal{I}_\infty(\varphi) = \pi_*\mathcal{I}(\pi^*\varphi),$$

so we may replace X and φ by Y and $\pi^*\varphi$ and assume that φ has log singularities along an effective $\mathbb{Q}$-divisor D. We decompose D into its irreducible components:

$$D = \sum_{i=1}^N a_i D_i.$$

In this case, observe that

$$\mathcal{I}_\infty(\varphi) = \mathcal{O}_X\left(-\sum_{i=1}^N (\lceil a_i \rceil D_i)\right)$$

is clearly coherent. □

The multiplier ideal sheaf $\mathcal{I}$ and the sheaf $\mathcal{I}_\infty$ are not very different in the asymptotic sense, as shown by the following lemma:

Lemma 1.6.3 *Assume that X is compact. Let $\varphi \in \mathrm{QPSH}(X)$ be a potential with analytic singularities. Then for any $\epsilon > 0$, we can find $k_0 > 0$ such that for each $k \geq k_0$, we have*

$$\mathcal{I}\left(k(1+\epsilon)\varphi\right) \subseteq \mathcal{I}_\infty(k\varphi). \tag{1.21}$$

Proof We shall prove a more precise local result. Take $x \in X$, take an open neighborhood $U \subseteq X$ of x, on which

$$\varphi = c\log\left(|g_1|^2 + \cdots + |g_N|^2\right) + O(1),$$

where $c \in \mathbb{Q}_{>0}$ and $g_1, \ldots, g_N$ are holomorphic functions on U. Then we claim that for any $\lambda > 0$, we have

$$\mathcal{I}(\lambda\varphi)_x \subseteq \mathcal{I}_\infty\left(\left(\lambda - c^{-1}n\right)_+ \varphi\right)_x, \tag{1.22}$$

where for any $a \in \mathbb{R}$, a_+ means $a \vee 0$. Note that (1.21) is a straightforward consequence of (1.22).

Fix a smooth volume form $\mathrm{d}V$ on X. Take $f \in \mathcal{I}(\lambda\varphi)$, by strong openness Theorem 1.4.4, we can find $\epsilon > 0$ so that $f \in \mathcal{I}((\lambda + \epsilon/c)\varphi)$. Therefore, we can find an open neighborhood $W \subseteq U$ of x so that

$$\int_W |f|^2 \left(|g_1|^2 + \cdots + |g_N|^2\right)^{-c\lambda-\epsilon} \,\mathrm{d}V < \infty.$$

But it follows from the Briançon–Skoda division theorem ([Sko72], [SBc74], [Dem12a, Theorem 11.17]) that

$$f_x \in \left(g_{1,x}, \ldots, g_{N,x}\right)^{\alpha},$$

where $\alpha = (\lfloor c\lambda \rfloor - n + 1)_+$. Here $\left(g_{1,x}, \ldots, g_{N,x}\right)$ denotes the ideal in $\mathcal{O}_{X,x}$ generated by the germs of $g_1, \ldots, g_N$ at x. Note that $\alpha \geq (c\lambda - n)_+$.

It follows that on a neighborhood of x, we have

$$\log |f|^2 \leq \alpha \log \left(|g_1|^2 + \cdots + |g_N|^2\right) + O(1) \leq c^{-1}\alpha\varphi + O(1).$$

Hence (1.22) follows. □

Theorem 1.6.3 *Let X be a connected compact Kähler manifold and $Y \subseteq X$ be a connected closed submanifold. Take a Kähler form ω on X and $\varphi \in \mathrm{PSH}(Y, \omega|_Y)$ such that $\omega|_Y + \mathrm{dd^c}\varphi$ is a Kähler current and e^{φ} is a Hölder continuous function on Y. Then there exists $\tilde{\varphi} \in \mathrm{PSH}(X, \omega)$ satisfying*

(1) $\tilde{\varphi}|_Y = \varphi$;
(2) $\omega_{\tilde{\varphi}}$ *is a Kähler current.*

In addition, if φ has analytic singularities, then so does $\tilde{\varphi}$.

See [DRWN+26, Theorem 6.1].

1.7 The space of currents

Let X be a connected compact Kähler manifold of dimension n and $\alpha \in \mathrm{H}^{1,1}(X, \mathbb{R})$.

Definition 1.7.1 Let Y be a complex manifold of dimension N and $m \in \mathbb{N}$. We say an (m, m)-current T on Y is *positive*[19] if either $m > N$ or for any smooth $(1, 0)$-forms $\beta_1, \ldots, \beta_{N-m}$ on Y, the measure

$$T \wedge \mathrm{i}\beta_1 \wedge \overline{\beta_1} \wedge \cdots \wedge \mathrm{i}\beta_{N-m} \wedge \overline{\beta_{N-m}}$$

is positive.

The basic properties of positive currents can be found in [Dem12b, Section III.1]. Recall that a positive current is necessarily real.

Definition 1.7.2 We say α is *pseudo-effective* if there is a closed positive $(1, 1)$-current in α.

We say α is *big* if there is a closed positive $(1, 1)$-current T in α dominating a Kähler form. Such currents are called *Kähler currents*.

Given classes $\alpha, \beta \in \mathrm{H}^{1,1}(X, \mathbb{R})$, we say $\alpha \leq \beta$ if $\beta - \alpha$ is pseudo-effective.

Definition 1.7.3 We introduce the following notations:

[19] This notion is sometimes known as *weak positivity*.

(1) $\mathcal{Z}_+(X)$ denotes the space of closed positive $(1,1)$-currents on X;
(2) given a pseudo-effective $(1,1)$-class α on X, we write $\mathcal{Z}_+(X,\alpha)$ for the set of $T \in \mathcal{Z}_+(X)$ such that $[T] = \alpha$.

Here $[T]$ denotes the cohomology class represented by T.

Definition 1.5.2 has a natural analogue for currents.

Definition 1.7.4 Given $T, T' \in \mathcal{Z}_+(X)$, we write $T \preceq T'$ and say T is *more singular* than T' if when we write $T = \theta + \mathrm{dd^c}\varphi$, $T' = \theta' + \mathrm{dd^c}\varphi'$, we have $\varphi \preceq \varphi'$. We write $T \sim T'$ if $T \preceq T'$ and $T' \preceq T$. In this case, we say T and T' have the same *singularity type*.

Remark 1.7.1 Observe that

$$\mathcal{Z}_+(X)/\!\sim \,\cong \mathrm{QPSH}(X)/\!\sim$$

canonically. The correspondence sends the class of a closed positive current $\theta_\varphi = \theta + \mathrm{dd^c}\varphi$ to the class of φ.

We will adopt the following convention: Whenever we have a notion for quasi-plurisubharmonic functions which depends only on the singularity type, we use the same notation and the same definition for closed positive $(1,1)$-currents.

Example 1.7.1 As an important example of Remark 1.7.1, given $T = \theta + \mathrm{dd^c}\varphi \in \mathcal{Z}_+(X)$ and $x \in X$, we define

$$\nu(T,x) = \nu(\varphi,x). \tag{1.23}$$

Again, as in Remark 1.4.1, this differs from the definitions in some literature by a factor of 2. But given our normalization

$$\mathrm{dd^c} = \frac{\mathrm{i}}{2\pi}\partial\overline{\partial},$$

(1.23) seems to be the most natural choice.

The key example to keep in mind is the following:

$$\nu([0],0) = 1,$$

where $[0]$ is the current of integration at $0 \in \mathbb{P}^1$. In fact, as a simple application of the Green's second identity, one can verify that

$$\frac{\mathrm{i}}{2\pi}\partial\overline{\partial}\log|z|^2 = \delta_0,$$

where the right-hand side is the Dirac delta distribution at $0 \in \mathbb{C}$.

Definition 1.7.5 Given $T \in \mathcal{Z}_+(X)$, represent T as $\theta + \mathrm{dd^c}\varphi$ for some closed smooth real $(1,1)$-form θ on X and $\varphi \in \mathrm{PSH}(X,\theta)$. The *polar locus* of T is defined as the set $\{\varphi = -\infty\}$.

It is clear that the polar locus of T is independent of the choices of θ and φ.

Definition 1.7.6 Assume that α is big. The *non-Kähler locus* $\mathrm{nK}(\alpha)$ of α is the intersection of the polar loci of all Kähler currents with analytic singularities in α.

Theorem 1.7.1 (Boucksom) *Assume that α is big. There is a Kähler current $T \in \alpha$ with analytic singularities, such that the polar locus of T is exactly* $\mathrm{nK}(\alpha)$. *In particular,* $\mathrm{nK}(\alpha)$ *is a proper Zariski closed subset of X.*

See [Bou02b, Théorème 2.1.20].

Definition 1.7.7 Assume that α is big. The *non-nef locus* $\mathrm{nn}(\alpha)$ of α is the following set:

$$\mathrm{nn}(\alpha) := \{x \in X : \nu(T_{\min}, x) > 0\},$$

where $T_{\min}$ is a current with minimal singularities in α.

Note that $\mathrm{nn}(\alpha) \subseteq \mathrm{nK}(\alpha)$. Thanks to Theorem 1.4.1, $\mathrm{nn}(\alpha)$ is a countable union of proper Zariski closed subsets of X. The non-Kähler locus and non-nef locus are studied in detail in [Bou02b].

Lemma 1.7.1 (Siu's decomposition) *Let E be a prime divisor on X. Then for any closed positive $(1,1)$-current T on X, the difference $T - \nu(T, E)[E]$ is a closed positive $(1,1)$-current.*

Here $[E]$ is the current of integration associated with E.[20] See [GH94, Page 386, Example 1] for the precise definition. See [Dem12a, Lemma 2.17] for the proof.

As a consequence, for each closed positive $(1,1)$-current T on X, we can write

$$T = \operatorname{Reg} T + \sum_i c_i [E_i], \tag{1.24}$$

where $\{E_i\}$ is a countable collection of prime divisors on X, $c_i > 0$.

Definition 1.7.8 A closed positive $(1,1)$-current T on X is *non-divisorial* (resp. *divisorial*) if $T = \operatorname{Reg} T$ (resp. $\operatorname{Reg} T = 0$).

It is helpful to check that our conventions are always consistent: There is no extra factor of 2 or $1/2$ anywhere. One can verify this using our favorite example as in Example 1.7.1.

Next we recall the notion of modified nef classes.

Definition 1.7.9 A class $\alpha \in \mathrm{H}^{1,1}(X, \mathbb{R})$ is *modified nef* if the following condition holds: Fix a reference Kähler metric ω on X, then for any $\epsilon > 0$, we can find a closed $(1,1)$-current $T \in \alpha$ such that

(1) $T + \epsilon\omega \geq 0$;

[20] We have also used $[E]$ to denote the cohomology class of E. Whenever there is a risk of confusion, we shall denote the cohomology class by $\{E\}$ instead.

(2) $\nu(T+\epsilon\omega, D)=0$ for any prime divisor D on X.

This definition is independent of the choice of ω.

These classes are called *nef en codimension* 1 in Boucksom's thesis [Bou02b], where they were introduced for the first time. They are also known as *movable classes* in other references. Modified nef classes form a closed convex cone in $\mathrm{H}^{1,1}(X,\mathbb{R})$. Note that a modified nef class is necessarily pseudo-effective. A nef class is obviously modified nef.

Recall the multiplicity of a cohomology class as defined in [Bou02b, Section 2.1.3]:

Definition 1.7.10 Let $\alpha \in \mathrm{H}^{1,1}(X,\mathbb{R})$ be a pseudo-effective class and D be a prime divisor on X. We define the *Lelong number* $\nu(\alpha, D)$ as follows:

(1) When α is big, define $\nu(\alpha, D)=\nu(T,D)$ for any closed positive $(1,1)$-current $T\in\alpha$ with minimal singularities.
(2) In general, define
$$\nu(\alpha, D) \coloneqq \lim_{\epsilon\to 0+} \nu(\alpha+\epsilon\{\omega\}, D).$$

When α is big, (2) is compatible with (1) and the definition is independent of the choice of ω.

By definition, a pseudo-effective class α is modified nef if and only if $\nu(\alpha, D)=0$ for all prime divisors D on X.

Let us recall the behavior of several cones under modifications.

Proposition 1.7.1 *Let $\pi\colon Y\to X$ be a proper bimeromorphic morphism from a Kähler manifold Y.*

(1) *For any nef class $\alpha\in \mathrm{H}^{1,1}(X,\mathbb{R})$, $\pi^*\alpha$ is nef.*
(2) *For any modified nef class $\beta\in\mathrm{H}^{1,1}(Y,\mathbb{R})$, $\pi_*\beta$ is modified nef.*

Proof Only (2) requires a proof. Fix a Kähler class γ on Y. Since the modified nef cone is closed, replacing β by $\beta+\epsilon\gamma$ and then letting $\epsilon\to 0+$ reduces the proof to the case where β is big as well. Let T (resp. S) be a current with minimal singularities in $\pi_*\beta$ (resp. in β), and let D be a prime divisor on X. It suffices to show that

$$\nu(T,D)=0,$$

Since β is modified nef, S is non-divisorial. Hence by Lemma 1.7.2 below, $\nu(\pi_*S, D)=0$. Since T has minimal singularities in $\pi_*\beta$, we have $T\succeq\pi_*S$, and the desired equality follows. □

Recall that non-divisorial currents are introduced in Definition 1.7.8.

Lemma 1.7.2 *Let $\pi\colon Y\to X$ be a proper bimeromorphic morphism from a Kähler manifold Y. Let T be a non-divisorial current on Y. Then π_*T is non-divisorial.*

Conversely, if S is a non-divisorial current on X, π^*S could have divisorial part. As a simple example, consider S on $\mathbb{P}^2$, whose local potential near $0\in\mathbb{C}^2_{z,w}$ looks like $\log(|z|^2+|w|^2)$.

Proof Let D be a prime divisor on X. It follows from Zariski's main theorem Theorem B.1.1 that D is not contained in the exceptional locus of π. Let D' be the strict transform of D. Thanks to Siu's semicontinuity theorem Theorem 1.4.1, we have

$$\nu(\pi_* T, D) = \nu(T, D') = 0.$$

Hence $\pi_* T$ is non-divisorial. □

1.8 Plurisubharmonic metrics on line bundles

A natural source of quasi-plurisubharmonic functions is the metrics on line bundles.

Let X be a connected Kähler manifold and L be a holomorphic line bundle on X. Usually, we do not distinguish L from the associated invertible sheaf $\mathcal{O}_X(L)$.

Definition 1.8.1 Let V be a 1-dimensional complex linear space. A *Hermitian form* h on V is a map $h\colon V \times V \to \mathbb{C}$ such that

(1) h is $\mathbb{C}$-linear in the second variable and conjugate linear in the first, and
(2)
$$|v|_h^2 := h(v, v) \in \mathbb{R}_{>0}$$
for each $v \in V \setminus \{0\}$.

We usually identify h with the quadratic form $V \to \mathbb{R}$ sending v to $|v|_h^2$. We write $|v|_h = \sqrt{|v|_h^2}$ for any $v \in V$.

The *singular Hermitian form* on V is the map $V \to \{0, \infty\}$ sending 0 to 0 and other elements to ∞.

Definition 1.8.2 Let V_1 and V_2 be 1-dimensional complex linear spaces. Given two maps $h_i\colon V_i \to [0, \infty]$ $(i = 1, 2)$, each of which is either a Hermitian form or a singular Hermitian form, we define the *tensor product* $h_1 \otimes h_2\colon V_1 \otimes V_2 \to [0, \infty]$ as follows:

(1) If either h_1 or h_2 is singular, we define $h_1 \otimes h_2$ as the singular Hermitian form;
(2) otherwise, define $h_1 \otimes h_2$ as the usual tensor product: For any $v_1 \in V_1$, $v_2 \in V_2$, set
$$h_1 \otimes h_2(v_1 \otimes v_2) = h_1(v_1) h_2(v_2).$$

Definition 1.8.3 A *Hermitian metric* h on L is a family of Hermitian forms $(h_x)_{x \in X}$, such that

(1) for each $x \in X$, h_x is a Hermitian form on L_x, and
(2) for each local holomorphic section s of $\mathcal{O}_X(L)$, the map $x \mapsto |s(x)|_{h_x}^2$ is smooth.

The pair (L, h) is called a *Hermitian line bundle*. We shall write $\mathrm{dd^c} h = c_1(L, h)$[21] for the first Chern form of h[22], normalized so that

$$[c_1(L, h)] = c_1(L).$$

The map $x \mapsto |s(x)|_{h_x}$ will be denoted by $|s|_h$.

To be more precise, if $U \subseteq X$ is an open subset on which L admits a nowhere vanishing holomorphic section s, then we define

$$(\mathrm{dd^c} h)|_U = \mathrm{dd^c}\left(-\log |s|_h^2\right).$$

Proposition 1.8.1 (Lelong–Poincaré) *Let $s \in \mathrm{H}^0(X, L)$ be non-zero and h be a Hermitian metric on L. Then*

$$c_1(L, h) + \mathrm{dd^c} \log |s|_h^2 = [Z(s)], \tag{1.25}$$

where $Z(s)$ is the zero divisor defined by s and $[\bullet]$ denote the associated current of integration.

See [Dem12a, (3.11)]. Again, we want to check that our conventions are compatible by investigating the following simple example.

Example 1.8.1 Let $X = \mathbb{P}^1$ and $L = \mathcal{O}_{\mathbb{P}^1}(1)$. The homogeneous coordinates on $\mathbb{P}^1$ will be denoted by $[X_0 : X_1]$. At a point $x = [X_0 : X_1] \in \mathbb{P}^1$, the fiber L_x is identified with the dual of $[x]$, where $[x] \subseteq \mathbb{C}^2$ is the line represented by x.

In order to introduce the Hermitian metric h on L, we fix the standard Hermitian norm $\| \bullet \|$ on $\mathbb{C}^2$. Then given $\lambda \in L_x = [x]^\vee$, we introduce

$$|\lambda|_{h_x} = \frac{|\lambda(\tilde{x})|}{\|\tilde{x}\|},$$

where $\tilde{x}$ is an arbitrary non-zero element in $[x]$. The reader can easily verify that h is indeed a Hermitian metric on L. The Hermitian metric h is known as the *Fubini–Study metric*.

A holomorphic section $s \in \mathrm{H}^0(X, L)$ can be formally identified with a linear form $a_0 X_0 + a_1 X_1$: At $x \in X$, the corresponding linear form on $[x]$ is given by sending (X_0, X_1) to $a_0 X_0 + a_1 X_1$.

Next we compute $\mathrm{dd^c} h = c_1(L, h)$. For this purpose, we cover $\mathbb{P}^1$ by $\mathbb{C} = \mathbb{P}^1 \setminus \{\infty\}$ and $\mathbb{P}^1 \setminus \{0\}$. Both are holomorphic coordinate charts with coordinate function $z = X_0/X_1$ and $z^{-1} = X_1/X_0$ respectively.

We claim that on $\mathbb{C}$,

$$\mathrm{dd^c} h = \mathrm{dd^c} \log(1 + |z|^2). \tag{1.26}$$

[21] The unusual notation $\mathrm{dd^c} h$ is sometimes referred to as the *Göteborg notation* because it is widely used by complex geometers in Göteborg (usually spelled as Gothenburg in English, the second largest (yet very poorly known) city in Sweden). As I identify myself as *Göteborgare*, I do not feel guilty about this notation.

[22] In the literature, people sometimes define the *curvature form* of (L, h) as $\Theta_h = -2\pi i \mathrm{dd^c} h$.

In fact, let t be the nowhere vanishing section of L on $\mathbb{C}$ corresponding to X_1. Then for $z \in \mathbb{C}$, we have an obvious lift $(z, 1) \in [z]$, so

$$|t|_h^2(z) = \frac{1}{|z|^2 + 1}.$$

So (1.26) follows.

In order to obtain a non-trivial case of the Lelong–Poincaré formula, we need to consider a section which vanishes at some points in $\mathbb{C}$. Let s be the holomorphic section of L corresponding to X_0. Then

$$\log |s|_h^2(z) = \log \frac{|z|^2}{|z|^2 + 1}$$

for any $z \in \mathbb{C}$ using the same argument as above. Therefore, we find that restricted to $\mathbb{C}$, we have

$$c_1(L, h) + \mathrm{dd^c} \log |s|_h^2 = \mathrm{dd^c} f = [0],$$

where $f(z) = \log |z|^2$. So the Lelong–Poincaré formula (1.25) is verified in this case.

The Kähler form $\mathrm{dd^c} h$ on $\mathbb{P}^1$ is also known as the *Fubini–Study metric*.

Definition 1.8.4 A (singular) *plurisubharmonic metric* (or *psh metric* for short)[23] h on L is a family $(h_x)_{x \in X}$ such that

(1) for each $x \in X$, h_x is either a Hermitian form on L_x or the singular Hermitian form on L_x, and
(2) there is a Hermitian metric h_0 on L and $\varphi \in \mathrm{PSH}(X, c_1(L, h_0))$ such that for each $x \in X$ and each $v \in L_x$, we have

$$|v|_{h_x}^2 = \begin{cases} 0, & \text{if } v = 0; \\ |v|_{h_{0,x}}^2 \mathrm{e}^{-\varphi(x)}, & \text{if } v \neq 0. \end{cases} \tag{1.27}$$

The *(first) Chern current* of h is by definition

$$\mathrm{dd^c} h = c_1(L, h) \coloneqq c_1(L, h_0) + \mathrm{dd^c} \varphi.$$

The set of psh metrics on L will be denoted by $\mathrm{PSH}(L)$.

We shall write the plurisubharmonic metric defined by (1.27) as $h_0 \exp(-\varphi)$[24]. With these conventions, $c_1(L, h)$ does not depend on the choice of h_0.

Given a local holomorphic section s of L, we shall use the following notations interchangeably to denote the map $x \mapsto h_x(s)$:

$$|s|_h^2 = h(s) = h(s, s).$$

[23] In the literature, people usually refer to such metrics as *positively curved singular Hermitian metrics*. I dislike this terminology, as having positive curvature only determines a plurisubharmonic metric almost everywhere, not everywhere.

[24] Be careful, this is not $h_0^2 \exp(-\varphi)$, as I prefer to think of h_0 as a quadratic form.

Remark 1.8.1 In the literature, some people prefer the convention that in (1.27), neither side has the square. Our choice seems to be the most natural one given our normalization of $\mathrm{dd^c}$.

Observe that once a Hermitian metric h_0 on L is given, the construction in (2) gives a bijection between $\mathrm{PSH}(X, c_1(L, h_0))$ and the set of plurisubharmonic metrics on L.

Based on this correspondence, plurisubharmonic metrics have the same functoriality property as quasi-plurisubharmonic functions in Proposition 1.1.4. In particular, given a connected closed submanifold Y of X, $h|_Y$ is a plurisubharmonic metric as long as it is not singular everywhere.

Definition 1.8.5 Given two holomorphic line bundles L_1, L_2 on X and plurisubharmonic metrics h_1 on L_1 and h_2 on L_2, we define the *tensor product* plurisubharmonic metric $h_1 \otimes h_2$ on $L_1 \otimes L_2$ as follows: for each $x \in X$, define

$$(h_1 \otimes h_2)_x = h_{1,x} \otimes h_{2,x}$$

in the sense of Definition 1.8.2.

We can easily verify that $h_1 \otimes h_2$ is indeed a plurisubharmonic metric on $L_1 \otimes L_2$.

Example 1.8.2 We continue with our example Example 1.8.1. Let $X = \mathbb{P}^1$ and $L = \mathcal{O}_{\mathbb{P}^1}(1)$. Let h^0 denote the Fubini–Study metric on L as defined in Example 1.8.1. Note that we have changed the notation from h to h^0. Let $\omega = \mathrm{dd^c} h^0$.

We construct $\varphi \in \mathrm{PSH}(X, \omega)$ as follows: On $\mathbb{C}$, define

$$\varphi(z) = \log \frac{|z|^2}{1 + |z|^2}. \tag{1.28}$$

Then $\varphi \in \mathrm{PSH}(\mathbb{C}, \omega|_{\mathbb{C}})$ by (1.26). Setting $\varphi(\infty) = 0$, we can easily verify that $\varphi \in \mathrm{PSH}(\mathbb{P}^1, \omega)$.[25]

We then get a plurisubharmonic metric $h = h^0 \exp(-\varphi)$. To be more explicit, the fiber metric h^0 at 0 is singular, $h_\infty = h^0_\infty$, while for $z \in \mathbb{C} \setminus \{0\}$ and $\lambda \in [z]^\vee$, we have

$$|\lambda|_{h_z} = \frac{|\lambda(z, 1)|}{|z|}.$$

In the remainder of this section, we assume that X is compact.

Definition 1.8.6 Assume that L is a pseudo-effective line bundle on X. A *Fubini–Study metric* on L is a psh metric h on L of the following form: There exists $m \in \mathbb{Z}_{>0}$, finitely many sections $s_1, \ldots, s_N \in \mathrm{H}^0(X, L^m)$ and $\lambda_1, \ldots, \lambda_N \in \mathbb{Q}$ such that for any local nowhere vanishing holomorphic section s of L, we have

$$|s|_h^2 = \min_{i=1,\ldots,N} \left| \frac{s^{\otimes m}}{\mathrm{e}^{\lambda_i/2} s_i} \right|^{2m^{-1}}.$$

[25] This can also be verified using the Grauert–Remmert extension theorem Theorem 1.2.1.

We write $\mathrm{FS}(L)$ for the set of Fubini–Study metrics on L.

If we fix a reference smooth Hermitian metric h_0 on L with $\theta = \mathrm{dd^c} h_0$, we can write $h = h_0 \exp(-\varphi)$ with

$$\varphi = \frac{1}{m} \max_{i=1,\dots,N} \left(\log |s_i|^2_{h_0^m} + \lambda_i \right).$$

Similarly, we write $\mathrm{FS}(X, \theta)$ for the set of such functions.

Definition 1.8.7 Assume that L is a pseudo-effective line bundle on X. The set $\widetilde{\mathrm{FS}}(L)$ of *generalized Fubini–Study metrics* is the smallest subset of $\mathrm{PSH}(L)$ containing $\mathrm{FS}(L)$ which is closed under the following two operations:

(1) $\mathbb{Q}$-convex combinations: if $h_1, h_2 \in \widetilde{\mathrm{FS}}(L)$ and $t \in (0, 1)$, then

$$h_1^t \otimes h_2^{1-t} \in \widetilde{\mathrm{FS}}(L);$$

(2) minima: if $h_1, h_2 \in \widetilde{\mathrm{FS}}(L)$, then

$$\min\{h_1, h_2\} \in \widetilde{\mathrm{FS}}(L).$$

We shall need the following Ohsawa–Takegoshi type extension theorem.

Theorem 1.8.1 *Assume that L is big and T is a holomorphic line bundle on X. Fix a Hermitian metric h_T on T. Fix a Hermitian metric h on L and let $\theta = c_1(L, h)$. Take a Kähler form ω on X. Let $Y \subseteq X$ be a connected closed submanifold of dimension m. Suppose that $\varphi \in \mathrm{PSH}(X, \theta - \delta\omega)$ for some $\delta > 0$ and $\varphi|_Y \not\equiv -\infty$. Then there exists $k_0(\delta, h_T) > 0$ such that for all $k \geq k_0$ and $s \in \mathrm{H}^0(Y, T \otimes L|_Y^k \otimes \mathcal{I}(k\varphi|_Y))$[26], there exists an extension $\tilde{s} \in \mathrm{H}^0(X, T \otimes L^k \otimes \mathcal{I}(k\varphi))$ such that*

$$\int_X |\tilde{s}|^2_{h^k \otimes h_T} \mathrm{e}^{-k\varphi}\, \omega^n \leq C \int_Y |s|^2_{(h^k \otimes h_T)|_Y} \mathrm{e}^{-k\varphi|_Y}\, \omega|_Y^m,$$

where $C > 0$ is an absolute constant, independent of the data (φ, s, k).

This is a special case of [His12, Theorem 1.4].

Let us also recall a simple result about the Bergman kernel approximations:

Proposition 1.8.2 *Let (L, h) be a Hermitian line bundle on X and set $\theta = c_1(L, h)$. Let (T, h_T) be a Hermitian line bundle on X. Assume that $\varphi \in \mathrm{PSH}(X, \theta)$ is a potential with analytic singularities such that θ_φ is a Kähler current. Fix a Kähler form ω on X. For each $k \geq 1$, we let*

$$\varphi_k \coloneqq \frac{1}{k} \log \sup_{\substack{s \in \mathrm{H}^0(X, L^k \otimes T) \\ \int_X |s|^2_{h^k \otimes h_T} \mathrm{e}^{-k\varphi} \omega^n \leq 1}} |s|^2_{h^k \otimes h_T}. \tag{1.29}$$

[26] Here and in the sequel, we usually abbreviate $\otimes k$ in the superscript as k to save space.

Then for all sufficiently large k,

$$\varphi \preceq \varphi_k \preceq \alpha_k \varphi,$$

where $\alpha_k \in (0,1)$ is an increasing sequence with limit 1.

Proof First we prove $\varphi \preceq \varphi_k$. Since θ_φ is a Kähler current, there is $\delta > 0$ such that $\varphi \in \mathrm{PSH}(X, \theta - \delta\omega)$. Let $x \in X$ with $\varphi(x) > -\infty$. Applying Theorem 1.8.1 to the point $Y = \{x\}$, for all sufficiently large k and every unit vector $e_x \in (L^k \otimes T)_x$, there is a section $\tilde{s} \in \mathrm{H}^0(X, L^k \otimes T \otimes \mathcal{I}(k\varphi))$ with $\tilde{s}(x) = e_x$ and

$$\int_X |\tilde{s}|^2_{h^k \otimes h_T} \mathrm{e}^{-k\varphi} \omega^n \leq C\mathrm{e}^{-k\varphi(x)}.$$

After rescaling $\tilde{s}$, the definition of φ_k gives

$$\varphi_k(x) \geq \varphi(x) - \frac{\log C}{k}.$$

This proves $\varphi \preceq \varphi_k$.

The other part of the inequality follows from Lemma 1.6.3 and (1.22).

More precisely, if $s_1, \ldots, s_N$ is an orthonormal basis of $\mathrm{H}^0(X, L^k \otimes T \otimes \mathcal{I}(k\varphi))$, with respect to the L^2-norm:

$$s \mapsto \left(\int_X |s|^2_{h^k \otimes h_T} \mathrm{e}^{-k\varphi} \, \omega^n \right)^{1/2},$$

then by (1.22),

$$s_i \in \mathrm{H}^0 \left(X, L^k \otimes T \otimes \mathcal{I}_\infty \Big((k - c^{-1}n)_+ \varphi \Big) \right),$$

where $c > 0$ depends only on φ. So

$$\sum_{i=1}^N |s_i|^2_{h^k \otimes h_T} \mathrm{e}^{-(k-c^{-1}n)_+\varphi} \leq C.$$

In other words,

$$\varphi_k \preceq \frac{1}{k} \left(k - c^{-1}n \right)_+ \varphi.$$

It suffices to take $\alpha_k = (1 - k^{-1}c^{-1}n)_+$. $\square$

Chapter 2
Non-pluripolar products

Pour exprimer d'une manière frappante que le monument que j'élève sera placé sous l'invocation de la Science, j'ai décidé d'inscrire en lettres d'or sur la grande frise du premier étage et à la place d'honneur, les noms des plus grands savants[a] *qui ont honoré la France depuis 1789 jusqu'à nos jours.*
— Gustave Eiffel, 1889

[a] Gaspard Monge, Comte de Péluse (1746–1818), known oddly by his family name instead of *de Péluse*, is one of the 72 names scribed on the Eiffel tower. He was both a mathematician and a politician, active mainly after the French Revolution.

Let X be a complex manifold and $\varphi_1, \dots, \varphi_p \in \mathrm{PSH}(X)$ ($p \in \mathbb{N}$). When the functions $\varphi_1, \dots, \varphi_p$ are all smooth, there is an obvious definition of a differential form

$$\mathrm{dd^c}\varphi_1 \wedge \cdots \wedge \mathrm{dd^c}\varphi_p \tag{2.1}$$

by the usual differential calculus. This product is known as the *Monge–Ampère product*, and a great deal of pluripotential theory turns on extending it meaningfully to the case where the φ_i are no longer smooth.

Several distinct approaches have been developed in the literature. In this book we adopt the *non-pluripolar theory* of Bedford, Taylor, Guedj, Zeriahi, Boucksom and Eyssidieux. Its appeal is that, among the available approaches, it is the only one combining both features that we shall need: it is defined for arbitrary psh singularities (at least in the global setting), and it satisfies a monotonicity theorem.

The chapter is organized as follows. Section 2.1 recalls the Bedford–Taylor theory of locally bounded psh functions; Section 2.2 introduces the non-pluripolar product; Section 2.3 deals with quasi-continuous functions; and Section 2.4 collects the key properties of the resulting product that are used throughout the rest of the book. A reader unfamiliar with this material is encouraged to consult the original article [BEGZ10] and the survey [DDNL25].

2.1 Bedford–Taylor theory

Let X be a complex manifold and $\varphi_1, \dots, \varphi_p \in \mathrm{PSH}(X)$ ($p \in \mathbb{N}$) be locally bounded plurisubharmonic functions on X[1]. In this case, there is a canonical definition of the Monge–Ampère type product (2.1).

[1] In the literature, some people use $\mathrm{PSH}(X) \cap L^\infty_{\mathrm{loc}}(X)$ to denote the set of such functions, which is an abuse of notation. However, this is legitimate thanks to the rigidity Theorem 1.1.3.

Definition 2.1.1 We define the closed positive (p,p)-current (2.1) on X as follows: We proceed by induction on $p \geq 0$. When $p=0$, we define (2.1) as the $(0,0)$-current $[X]$. When $p>0$, we let

$$\mathrm{dd^c}\varphi_1 \wedge \cdots \wedge \mathrm{dd^c}\varphi_p := \mathrm{dd^c}\left(\varphi_1\, \mathrm{dd^c}\varphi_2 \wedge \cdots \wedge \mathrm{dd^c}\varphi_p\right).$$

We call this product the *Bedford–Taylor product* .

Remark 2.1.1 There is also a slightly more general version of this construction. Given a closed positive current T, one can also define the product

$$\mathrm{dd^c}\varphi_1 \wedge \cdots \wedge \mathrm{dd^c}\varphi_p \wedge T$$

in a very similar way.

Proposition 2.1.1 *The product* $\mathrm{dd^c}\varphi_1 \wedge \cdots \wedge \mathrm{dd^c}\varphi_p$ *is a closed positive* (p,p)*-current on* X*. Moreover, the product is symmetric in the* φ_i*'s.*

See [GZ17, Proposition 3.3, Corollary 3.12]. The proof relies crucially on an important estimate, known as the *Chern–Levine–Nirenberg inequality*. See [GZ17, Theorem 3.9].

The Bedford–Taylor theory has many satisfactory properties.

Theorem 2.1.1 *Let* $(\varphi_i^j)_{j\in\mathbb{Z}_{>0}}$ *be decreasing sequences (resp. increasing sequences) of locally bounded psh functions on* X *converging (resp. converging a.e.) to locally bounded psh functions* φ_i*'s, where* $i=0,\ldots,p$*. Then*

$$\varphi_0^j\, \mathrm{dd^c}\varphi_1^j \wedge \cdots \wedge \mathrm{dd^c}\varphi_p^j \rightharpoonup \varphi_0\, \mathrm{dd^c}\varphi_1 \wedge \cdots \wedge \mathrm{dd^c}\varphi_p$$

as $j\to\infty$*. In particular, if* $(\varphi_0^j)_j$ *is the constant sequence* 1*, we have*

$$\mathrm{dd^c}\varphi_1^j \wedge \cdots \wedge \mathrm{dd^c}\varphi_p^j \rightharpoonup \mathrm{dd^c}\varphi_1 \wedge \cdots \wedge \mathrm{dd^c}\varphi_p.$$

Here the notation $\rightharpoonup$ denotes the weak-* convergence of currents.

We refer to [GZ17, Theorem 3.18, Theorem 3.23] for the proofs.

By contrast, we emphasize that the Bedford–Taylor product is not continuous with respect to the L^1_{loc}-convergence in general. A simple example can be found in [GZ17, Example 3.25].

2.2 The non-pluripolar products

The proof of all results in this section can be found in [BEGZ10].

Let X be a complex manifold.

Definition 2.2.1 Let $\varphi_1,\ldots,\varphi_p \in \mathrm{PSH}(X)$. We set

$$O_k := \bigcap_{j=1}^{p} \{\varphi_j > -k\}, \quad k \in \mathbb{N}.$$

We say that $\mathrm{dd^c}\varphi_1 \wedge \cdots \wedge \mathrm{dd^c}\varphi_p$ is *well-defined* if for each connected open subset $U \subseteq X$, any smooth positive $(1,1)$-form ω on U, and each compact subset $K \subseteq U$, we have

$$\sup_{k\geq 0} \int_{K\cap O_k} \left(\bigwedge_{j=1}^{p} \mathrm{dd^c}(\varphi_j \vee (-k)) \right)\Bigg|_U \wedge \omega^{\dim U - p} < \infty. \tag{2.2}$$

In this case, we define the *non-pluripolar product* $\mathrm{dd^c}\varphi_1 \wedge \cdots \wedge \mathrm{dd^c}\varphi_p$ by

$$\mathbb{1}_{O_k} \mathrm{dd^c}\varphi_1 \wedge \cdots \wedge \mathrm{dd^c}\varphi_p = \mathbb{1}_{O_k} \bigwedge_{j=1}^{p} \mathrm{dd^c}\left(\varphi_j \vee (-k)\right) \tag{2.3}$$

on $\bigcup_{k\geq 0} O_k$ and extend it by zero to X.

As recalled in Section 1.3, an $\mathcal{F}$-open subset means an open subset with respect to the plurifine topology.

Proposition 2.2.1 *Let $\varphi_1, \dots, \varphi_p \in \mathrm{PSH}(X)$.*

(1) *The product $\mathrm{dd^c}\varphi_1 \wedge \cdots \wedge \mathrm{dd^c}\varphi_p$ is local with respect to the plurifine topology in the following sense: Let $O \subseteq X$ be an $\mathcal{F}$-open subset and $\psi_1, \dots, \psi_p \in \mathrm{PSH}(X)$. Assume that*

$$\varphi_j|_O = \psi_j|_O, \quad j = 1, \dots, p,$$

and that

$$\bigwedge_{j=1}^{p} \mathrm{dd^c}\varphi_j \text{ and } \bigwedge_{j=1}^{p} \mathrm{dd^c}\psi_j$$

are both well-defined, then

$$\bigwedge_{j=1}^{p} \mathrm{dd^c}\varphi_j\Bigg|_O = \bigwedge_{j=1}^{p} \mathrm{dd^c}\psi_j\Bigg|_O. \tag{2.4}$$

If furthermore O is open in the usual topology, then the product

$$\bigwedge_{j=1}^{p} \mathrm{dd^c}\varphi_j|_O$$

on O is well-defined and

$$\bigwedge_{j=1}^{p} \mathrm{dd^c}\varphi_j\Bigg|_O = \bigwedge_{j=1}^{p} \mathrm{dd^c}\varphi_j|_O. \tag{2.5}$$

Let $\mathcal{U}$ be an open covering of X. Then $\mathrm{dd^c}\varphi_1 \wedge \cdots \wedge \mathrm{dd^c}\varphi_p$ is well-defined if and only if each product

$$\bigwedge_{j=1}^{p} \mathrm{dd^c}\varphi_j|_U, \quad U \in \mathcal{U}.$$

(2) *The current $\mathrm{dd^c}\varphi_1 \wedge \cdots \wedge \mathrm{dd^c}\varphi_p$ and the fact that it is well-defined depend only on the currents $\mathrm{dd^c}\varphi_j$, not on the choice of the φ_j's nor on the ordering of the φ_j's.*

(3) *When $\varphi_1, \dots, \varphi_p \in L^\infty_{\mathrm{loc}}(X)$, the product $\mathrm{dd^c}\varphi_1 \wedge \cdots \wedge \mathrm{dd^c}\varphi_p$ is well-defined and is equal to the Bedford–Taylor product.*

(4) *Assume that $\mathrm{dd^c}\varphi_1 \wedge \cdots \wedge \mathrm{dd^c}\varphi_p$ is well-defined. Then $\mathrm{dd^c}\varphi_1 \wedge \cdots \wedge \mathrm{dd^c}\varphi_p$ puts no mass on pluripolar sets.*

(5) *Assume that $\mathrm{dd^c}\varphi_1 \wedge \cdots \wedge \mathrm{dd^c}\varphi_p$ is well-defined. Then $\bigwedge_{j=1}^{p} \mathrm{dd^c}\varphi_j$ is a closed positive (p, p)-current on X.*

(6) *The product is multilinear: Let $\psi_1 \in \mathrm{PSH}(X)$, $a, b > 0$ then*

$$\mathrm{dd^c}(a\varphi_1 + b\psi_1) \wedge \bigwedge_{j=2}^{p} \mathrm{dd^c}\varphi_j = a\,\mathrm{dd^c}\varphi_1 \wedge \bigwedge_{j=2}^{p} \mathrm{dd^c}\varphi_j + b\,\mathrm{dd^c}\psi_1 \wedge \bigwedge_{j=2}^{p} \mathrm{dd^c}\varphi_j \tag{2.6}$$

in the sense that the left-hand side is well-defined if and only if both terms on the right-hand side are well-defined, and the equality holds in that case.

In view of (3), we do not need to specify whether our product $\mathrm{dd^c}\varphi_1 \wedge \cdots \wedge \mathrm{dd^c}\varphi_p$ is the Bedford–Taylor product or the non-pluripolar product when the φ_i's are all locally bounded.

Definition 2.2.2 Let $T_1, \dots, T_p$ be closed positive $(1, 1)$-currents on X. We say that $T_1 \wedge \cdots \wedge T_p$ is *well-defined* if there exists an open covering $\mathcal{U}$ of X such that on each $U \in \mathcal{U}$, we can find $\varphi_j^U \in \mathrm{PSH}(U)$ $(j = 1, \dots, p)$ such that

$$\mathrm{dd^c}\varphi_j^U = T_j, \quad j = 1, \dots, p$$

and $\mathrm{dd^c}\varphi_1^U \wedge \cdots \wedge \mathrm{dd^c}\varphi_p^U$ is well-defined. In this case, we define the *non-pluripolar product* $T_1 \wedge \cdots \wedge T_p$ as the closed positive (p, p)-current on X defined by

$$\left(T_1 \wedge \cdots \wedge T_p\right)|_U = \mathrm{dd^c}\varphi_1^U \wedge \cdots \wedge \mathrm{dd^c}\varphi_p^U, \quad U \in \mathcal{U}. \tag{2.7}$$

The product $T_1 \wedge \cdots \wedge T_p$ is independent of the choices we made thanks to Proposition 2.2.1 (1) and (2).

Proposition 2.2.1 can be formulated in terms of currents without any difficulty.

Remark 2.2.1 Similar to Remark 2.1.1, there is also an extension of the non-pluripolar theory allowing us to define

$$T_1 \wedge \cdots \wedge T_p \cap T$$

for any closed positive current T. This is the *relative non-pluripolar product* introduced by Vu [Vu21]. Unlike the relative Bedford–Taylor products, the relative non-pluripolar products present some pathological behaviors. For example, they are not linear in general.

Remark 2.2.2 Another possible generalization of the non-pluripolar products is motivated by Proposition 2.2.1. One could begin by defining a generalized notion of plurisubharmonic functions on $\mathcal{F}$-open sets, called *$\mathcal{F}$-plurisubharmonic functions* and define their non-pluripolar products. See [EKFW11, EKW14].

Proposition 2.2.2 *Let X be a compact Kähler manifold and let $T_1, \dots, T_p$ be closed positive $(1,1)$-currents on X. Then $T_1 \wedge \cdots \wedge T_p$ is well-defined.*

This proposition explains why we usually work in the setting of compact Kähler manifolds.

2.3 Quasi-continuous functions

Let X be a compact Kähler manifold of dimension n and θ be a closed real smooth $(1,1)$-form on X representing a big cohomology class.

Definition 2.3.1 Let $A \subseteq X$ be a Borel subset. The *θ-capacity* $\mathrm{Cap}_\theta(A)$ of A is defined as

$$\mathrm{Cap}_\theta(A) \coloneqq \sup\left\{\int_A \theta_\varphi^n : \varphi \in \mathrm{PSH}(X,\theta), V_\theta - 1 \le \varphi \le V_\theta\right\}.$$

Here V_θ is defined in (2.9) below. As we will recall below, the non-pluripolar product θ_φ^n is well-defined.

The capacity is not very sensitive to the choice of θ:

Theorem 2.3.1 *Let θ' be another closed real smooth $(1,1)$-form on X representing a big cohomology class. Then there are continuous functions $f, g\colon [0,\infty) \to [0,\infty)$, with $f(0) = g(0) = 0$, such that for any Borel subset $E \subseteq X$, we have*

$$\mathrm{Cap}_\theta(E) \le f\left(\mathrm{Cap}_{\theta'}(E)\right), \quad \mathrm{Cap}_{\theta'}(E) \le g\left(\mathrm{Cap}_\theta(E)\right).$$

A more general result is proved in [Lu21]. Similar comparison results hold between θ-capacity and the classical Bedford–Taylor capacity, see [GZ17, Section 9.2] for the proof. As a consequence, we can freely apply the results in [GZ17, Section 4.2], even though capacity has a different meaning there.

Definition 2.3.2 Let U be an open subset of X. A function $f\colon U \to [-\infty,\infty]$ is *quasi-continuous* if it is Borel measurable and for any $\epsilon > 0$, there is an open subset $G \subseteq U$ such that

(1) $\mathrm{Cap}_\theta(G) \le \epsilon$;

(2) $f|_{U\setminus G}$ is real-valued and continuous.

Thanks to Theorem 2.3.1, the notion of quasi-continuous functions is independent of the choice of θ. Note that if $f, g\colon U \to [-\infty, \infty]$ are two Borel measurable functions equal quasi-everywhere (see Definition 1.1.4), then f is quasi-continuous if and only if g is.

Theorem 2.3.2 *Let $A \subseteq X$ be a Borel set. Then the following are equivalent:*

(1) *A is pluripolar;*
(2) $\mathrm{Cap}_\theta(A) = 0$.

See [GZ17, Theorem 4.40] for the proof.

Example 2.3.1 A quasi-plurisubharmonic function on an open subset $U \subseteq X$ is always quasi-continuous. See [GZ17, Theorem 4.20] for the proof.

More generally, if φ, ψ are two quasi-plurisubharmonic functions on U, then the following function

$$f(x) := \begin{cases} \varphi(x) - \psi(x), & \text{if } (\varphi \vee \psi)(x) \neq -\infty; \\ \infty, & \text{otherwise} \end{cases}$$

is quasi-continuous.[2]

Definition 2.3.3 Let U be an open subset of X. Let $(f_i)_{i\in I}$ be a net of Borel measurable functions $f_i\colon U \to [-\infty, \infty]$, and $f\colon U \to [-\infty, \infty]$ be a Borel measurable function. We say $(f_i)_{i\in I}$ converges to f *in capacity* if for any $\delta > 0$, we have

$$\lim_{i\in I} \mathrm{Cap}_\theta\Big(\{f_i > f + \delta\}\Big) = 0, \quad \lim_{i\in I} \mathrm{Cap}_\theta\Big(\{f_i < f - \delta\}\Big) = 0^{3}.$$

We sometimes write $f_i \xrightarrow{\mathrm{C}} f$.

Note that f is not uniquely determined by the net $(f_i)_{i\in I}$. Thanks to Theorem 2.3.1, the notion of convergence in capacity is independent of the choice of θ.

Proposition 2.3.1 *Let U be an open subset of X. Let $(\varphi_i)_{i\in I}$ be a net in* $\mathrm{PSH}(U, \theta)$ *and $\varphi \in$* $\mathrm{PSH}(U, \theta)$. *Assume one of the following conditions holds:*

(1) *$(\varphi_i)_{i\in I}$ is decreasing and φ is the limit of the net;*
(2) *$(\varphi_i)_{i\in I}$ is increasing and converges almost everywhere to φ.*

Then $(\varphi_i)_{i\in I}$ converges to φ in capacity.

See [GZ17, Proposition 4.25] for the proof. The reference concerns only the sequence case, but the proof works for nets as well.

---

[2] In the literature, people usually say carelessly that $\varphi - \psi$ is quasi-continuous.

[3] It is very tempting to write $\lim_{i\in I} \mathrm{Cap}_\theta\left(\{|f_i - f| > \delta\}\right) = 0$, as in [GZ17, Definition 4.23] for example. But the set where $f_i - f$ is not defined is not a pluripolar set in general. Hence this abuse of notation is not acceptable.

Theorem 2.3.3 *Let U be an open subset of X. Let $(\chi_j : U \to [0,\infty))_j$ be a sequence of quasi-continuous functions. Assume that the sequence is uniformly bounded and converges in capacity to $\chi : U \to [0,\infty)$. For each $j = 1,\dots,p$, let $(\varphi_j^i)_i$ be uniformly bounded sequences in $\mathrm{PSH}(U,\theta_j)$ converging in capacity to bounded $\varphi_j \in \mathrm{PSH}(U,\theta_j)$. Then*

$$\chi_j \bigwedge_{j=1}^{p} \theta_{j,\varphi_j^i} \rightharpoonup \chi \bigwedge_{j=1}^{p} \theta_{j,\varphi_j}$$

weakly as currents on U.

See [GZ17, Theorem 4.26].

2.4 Properties of non-pluripolar products

Let X be a connected compact Kähler manifold of dimension n and $\theta,\theta_1,\dots,\theta_n$ be closed real smooth $(1,1)$-forms on X.

We write

$$\mathrm{PSH}(X,\theta)_{>0} = \left\{\varphi \in \mathrm{PSH}(X,\theta) : \int_X \theta_\varphi^n > 0\right\}. \tag{2.8}$$

The non-pluripolar product θ_φ^n is well-defined thanks to Proposition 2.2.2.

Remark 2.4.1 Suppose that X is a connected complex manifold of dimension 0, namely, X is a single point. In this case, by definition, the non-pluripolar product θ_φ^n is given by the current of integration at the unique point. So $\mathrm{PSH}(X,\theta)_{>0} = \mathrm{PSH}(X,\theta) \cong \mathbb{R}$ in this case and $\int_X \theta_\varphi^n = 1$ for all $\varphi \in \mathrm{PSH}(X,\theta)$.

Recall the following basic result:

Proposition 2.4.1 *Assume that $\mathrm{PSH}(X,\theta)_{>0}$ is non-empty. Then the cohomology class $[\theta]$ is big.*

See [BEGZ10, Proposition 1.22].

We recall a few basic facts about the non-pluripolar masses.

Proposition 2.4.2 *Let $\pi : Y \to X$ be a proper bimeromorphic morphism from a Kähler manifold Y and $\varphi_i \in \mathrm{PSH}(X,\theta_i)$ for $i = 1,\dots,n$. Then*

$$\int_Y (\pi^*\theta_1)_{\pi^*\varphi_1} \wedge \cdots \wedge (\pi^*\theta_n)_{\pi^*\varphi_n} = \int_X \theta_{1,\varphi_1} \wedge \cdots \wedge \theta_{n,\varphi_n}.$$

Proof The assertion follows from Proposition 2.2.1 (1) on the Zariski open set where π is an isomorphism. The exceptional locus is pluripolar, and both non-pluripolar products put no mass on pluripolar sets by Proposition 2.2.1 (4). □

Theorem 2.4.1 *Let $\varphi_0, \varphi_1 \in \mathrm{PSH}(X, \theta)$. Then the map*

$$[0,1] \ni t \mapsto \log \int_X \theta^n_{t\varphi_1+(1-t)\varphi_0}$$

is concave.

More generally, given closed positive $(1,1)$-currents $T_1, \dots, T_n$ on X, we have

$$\int_X T_1 \wedge \cdots \wedge T_n \geq \prod_{i=1}^n \left(\int_X T_i^n \right)^{1/n}.$$

See [DDNL21a] for the proof.

Remark 2.4.2 Here and in the sequel, when we write expressions like $t\varphi + (1-t)\psi$ for $\varphi, \psi \in \mathrm{QPSH}(X)$, we will follow the convention that when $t = 0$, the value is ψ and when $t = 1$, the value is φ.

We shall write

$$V_\theta = \sup \{\varphi \in \mathrm{PSH}(X, \theta) : \varphi \leq 0\}. \tag{2.9}$$

Note that the upper semicontinuous regularization V_θ^* of V_θ is itself a candidate of (2.9) if $\mathrm{PSH}(X, \theta) \neq \varnothing$ as follows from Proposition 1.2.2, therefore, $V_\theta = V_\theta^* \in \mathrm{PSH}(X, \theta)$ if $\mathrm{PSH}(X, \theta) \neq \varnothing$.

The function V_θ should be regarded as a canonical representative of the least singular potentials in $\mathrm{PSH}(X, \theta)$. We recall the following result:

Lemma 2.4.1 *Suppose that $\{\theta\}$ is big. The potential V_θ is locally bounded on $X \setminus \mathrm{nK}(\{\theta\})$.*

Recall that $\mathrm{nK}(\{\theta\})$ is the non-Kähler locus introduced in Definition 1.7.6.

Proof By definition of $\mathrm{nK}(\{\theta\})$, it suffices to show that if $\varphi \in \mathrm{PSH}(X, \theta)$ has analytic singularities and θ_φ is a Kähler current, then V_θ is locally bounded outside $\{\varphi = -\infty\}$. After replacing φ by $\varphi - \sup_X \varphi$, the function φ is a candidate in (2.9); hence $V_\theta \geq \varphi$ and $V_\theta \leq 0$. Since φ is locally bounded outside its pole set, so is V_θ. □

The non-pluripolar product has a lower semicontinuity property.

Theorem 2.4.2 (Semicontinuity theorem) *Let $\varphi_j, \varphi_j^k \in \mathrm{PSH}(X, \theta_j)$ ($k \in \mathbb{N}$, $j = 1, \dots, n$). Let χ_k, χ ($k \in \mathbb{N}$) be non-negative uniformly bounded quasi-continuous functions on X such that $(\chi_k)_k$ converges to χ in capacity. Assume that for any $j = 1, \dots, n$, as $k \to \infty$, either φ_j^k decreases to $\varphi_j \in \mathrm{PSH}(X, \theta_j)$ or increases to $\varphi_j \in \mathrm{PSH}(X, \theta_j)$ almost everywhere. Then*

$$\varliminf_{k\to\infty} \int_X \chi_k\, \theta_{1,\varphi_1^k} \wedge \cdots \wedge \theta_{n,\varphi_n^k} \geq \int_X \chi\, \theta_{1,\varphi_1} \wedge \cdots \wedge \theta_{n,\varphi_n}. \tag{2.10}$$

If in addition,

$$\lim_{k\to\infty} \int_X \theta_{1,\varphi_1^k} \wedge \cdots \wedge \theta_{n,\varphi_n^k} = \int_X \theta_{1,\varphi_1} \wedge \cdots \wedge \theta_{n,\varphi_n},$$

then $\chi_k\, \theta_{1,\varphi_1^k} \wedge \cdots \wedge \theta_{n,\varphi_n^k}$ *converges to* $\chi\, \theta_{1,\varphi_1} \wedge \cdots \wedge \theta_{n,\varphi_n}$ *weakly as measures*[4]. *In particular, the limit in* (2.10) *exists and equality holds in* (2.10).

See [DDNL25, Theorem 2.6] for the proof.

The non-pluripolar mass is a monotone quantity with respect to the singularity type.

Theorem 2.4.3 (Monotonicity theorem) *Let* $\varphi_j, \psi_j \in \mathrm{PSH}(X, \theta_j)$ *for* $j = 1, \dots, n$. *Assume that* $\varphi_j \succeq \psi_j$[5] *for every* j, *then*

$$\int_X \theta_{1,\varphi_1} \wedge \cdots \wedge \theta_{n,\varphi_n} \geq \int_X \theta_{1,\psi_1} \wedge \cdots \wedge \theta_{n,\psi_n}.$$

In particular, if $\varphi, \psi \in \mathrm{PSH}(X, \theta)$ *with* $\varphi \succeq \psi$, *then*

$$\int_X \theta_\varphi^n \geq \int_X \theta_\psi^n.$$

In other words, the non-pluripolar mass increases as the potentials become less singular.

See [DDNL18b, Theorem 1.1]. We will prove a vast extension of this theorem in Proposition 6.1.4.

Thanks to this theorem, the non-pluripolar mass $\int_X \theta_\varphi^n$ could be used as a rough measure of the singularities of $\varphi \in \mathrm{PSH}(X, \theta)$. In Section 3.1, we shall refine this measure by defining the notion of P-envelope.

As a corollary, we obtain that

Corollary 2.4.1 *Fix a directed set* I. *For each* $j = 1, \dots, n$, *take an increasing net* $(\varphi_j^i)_{i \in I}$ *in* $\mathrm{PSH}(X, \theta_j)$, *uniformly bounded from above. Set*

$$\varphi_j := \sup_{i \in I}{}^* \varphi_j^i, \quad j = 1, \dots, n.$$

Then

$$\lim_{i \in I} \int_X \theta_{1,\varphi_1^i} \wedge \cdots \wedge \theta_{n,\varphi_n^i} = \int_X \theta_{1,\varphi_1} \wedge \cdots \wedge \theta_{n,\varphi_n}. \tag{2.11}$$

In particular, if the nets are indeed sequences, we have

$$\theta_{1,\varphi_1^i} \wedge \cdots \wedge \theta_{n,\varphi_n^i} \rightharpoonup \theta_{1,\varphi_1} \wedge \cdots \wedge \theta_{n,\varphi_n}.$$

Proof The final assertion follows from (2.11) together with Theorem 2.4.2. It remains to establish (2.11).

We may assume that I is infinite as there is nothing to prove otherwise. Thanks to Theorem 2.4.3, we already know the $\leq$ inequality in (2.11). We prove the reverse

[4] Recall that the weak convergence of Radon measures is stronger than the weak convergence as currents in general. When the Radon measures have uniformly bounded total variation, they are equivalent.

[5] See Definition 1.5.2 for the notation.

inequality. When $I \cong \mathbb{Z}_{>0}$ as directed sets, the reverse inequality follows from Theorem 2.4.2. In general, by Choquet's lemma Proposition 1.2.3, we can find a countable infinite subset $R \subseteq I$ such that

$$\sup_{r\in R}{}^{*}\varphi_j^r = \sup_{i\in I}{}^{*}\varphi_j^i$$

for all $j = 1, \dots, n$. We fix a bijection $R \cong \mathbb{Z}_{>0}$. For any $j = 1, \dots, n$, we will then denote elements φ_j^r $(r \in R)$ by $\varphi_j^1, \varphi_j^2, \dots$. We shall write

$$\psi_j^a = \varphi_j^1 \vee \dots \vee \varphi_j^a$$

for each $a \in \mathbb{Z}_{>0}$.

For every a we may choose $i_a \in I$ dominating the finitely many indices corresponding to $\varphi_j^1, \dots, \varphi_j^a$ for all j. It then follows from the monotonicity theorem that

$$\lim_{i\in I} \int_X \theta_{1,\varphi_1^i} \wedge \dots \wedge \theta_{n,\varphi_n^i} \geq \lim_{a\to\infty} \int_X \theta_{1,\psi_1^a} \wedge \dots \wedge \theta_{n,\psi_n^a}.$$

From the special case mentioned above, we know that the right-hand side is exactly the right-hand side of (2.11), so we conclude. □

We prove an interesting inequality about the Monge–Ampère measure of the maximum of two potentials.

Lemma 2.4.2 *Let $\varphi, \psi \in \mathrm{PSH}(X, \theta)$. Then*

$$\theta_{\varphi\vee\psi}^n \geq \mathbb{1}_{\{\varphi\geq\psi\}}\, \theta_\varphi^n + \mathbb{1}_{\{\varphi<\psi\}}\, \theta_\psi^n. \tag{2.12}$$

In particular, if $\varphi \leq \psi$, then

$$\mathbb{1}_{\{\varphi=\psi\}}\, \theta_\varphi^n \leq \mathbb{1}_{\{\varphi=\psi\}}\, \theta_\psi^n.$$

At a first sight, (2.12) might seem trivial, and it is if we replace $\{\varphi \geq \psi\}$ by $\{\varphi > \psi\}$. The difficulty really lies on understanding the contact set $\{\varphi = \psi\}$.

Proof For each $k \in \mathbb{N}$, we set

$$\psi_k = \psi \vee (V_\theta - k), \quad \varphi_k = \varphi \vee (V_\theta - k).$$

For each $t > 0$ and each $k \in \mathbb{N}$, we have

$$\begin{aligned}\theta_{\psi_k\vee(\varphi_k+t)}^n \geq & \mathbb{1}_{\{\psi_k>\varphi_k+t\}}\theta_{\psi_k\vee(\varphi_k+t)}^n + \mathbb{1}_{\{\psi_k<\varphi_k+t\}}\theta_{\psi_k\vee(\varphi_k+t)}^n \\ = & \mathbb{1}_{\{\psi_k>\varphi_k+t\}}\theta_{\psi_k}^n + \mathbb{1}_{\{\psi_k<\varphi_k+t\}}\theta_{\varphi_k}^n,\end{aligned}$$

where the equality follows from the plurifine locality of non-pluripolar products Proposition 2.2.1(1). We observe that as $t \to 0+$, the measures $\theta_{\psi_k\vee(\varphi_k+t)}^n$ converge weakly to $\theta_{\psi_k\vee\varphi_k}^n$ as a consequence of Theorem 2.4.2 and Theorem 2.4.3.

Now letting $t \to 0+$, we conclude that

$$\theta^n_{\psi_k\vee\varphi_k} \geq \mathbb{1}_{\{\psi_k>\varphi_k\}}\theta^n_{\psi_k} + \mathbb{1}_{\{\psi_k\leq\varphi_k\}}\theta^n_{\varphi_k}.$$

In particular, multiplying both sides by $\mathbb{1}_{\{\min\{\varphi,\psi\}>V_\theta-k\}}$ and applying Proposition 2.2.1(1) again, we find that

$$\begin{aligned}\mathbb{1}_{\{\min\{\varphi,\psi\}>V_\theta-k\}}\theta^n_{\psi\vee\varphi} \geq & \mathbb{1}_{\{\min\{\varphi,\psi\}>V_\theta-k\}\cap\{\psi\leq\varphi\}}\theta^n_{\varphi} \\ & + \mathbb{1}_{\{\min\{\varphi,\psi\}>V_\theta-k\}\cap\{\psi>\varphi\}}\theta^n_{\psi}.\end{aligned}$$

Letting $k \to \infty$, we conclude (2.12). □

Corollary 2.4.2 *Let $(\varphi_i)_{i\in I}$ be a net in $\mathrm{PSH}(X,\theta)$, and $\varphi \in \mathrm{PSH}(X,\theta)$ so that $\varphi_i \to \varphi$ in L^1. Assume that $\varphi_i \leq 0$ for all $i \in I$. Then*

$$\varlimsup_{i\in I}\int_{\{\varphi_i=0\}}\theta^n_{\varphi_i} \leq \int_{\{\varphi=0\}}\theta^n_{\varphi}. \tag{2.13}$$

Proof Observe that $\varphi \leq 0$ by Proposition 1.5.1.

Step 1. We first assume that $I \cong \mathbb{Z}_{>0}$ as directed sets. Namely, $(\varphi_i)_{i>0}$ is a sequence.

For each $k > 0$, let

$$\psi_k = \sup_{j\geq k}{}^*\varphi_j.$$

Then $(\psi_k)_k$ is a decreasing sequence in $\mathrm{PSH}(X,\theta)$ with limit φ, see Corollary 1.2.1 for example.

Thanks to Lemma 2.4.2, for each $k > 0$, we have

$$\mathbb{1}_{\{\varphi_k=\psi_k\}}\theta^n_{\varphi_k} \leq \mathbb{1}_{\{\varphi_k=\psi_k\}}\theta^n_{\psi_k}.$$

Multiplying both sides with $\mathbb{1}_{\{\varphi_k=0\}}$, we find that

$$\mathbb{1}_{\{\varphi_k=0\}}\theta^n_{\varphi_k} \leq \mathbb{1}_{\{\varphi_k=0\}}\theta^n_{\psi_k}.$$

Take $b, C > 0$, for each $k > 0$, we have

$$\begin{aligned}\int_{\{\varphi_k=0\}}\theta^n_{\varphi_k} &\leq \int_{\{\varphi_k=0\}}\theta^n_{\psi_k} \leq \int_{\{\psi_k=0\}}\theta^n_{\psi_k} \\ &= \int_{\{\psi_k=0\}}\theta^n_{\psi_k\vee(V_\theta-C)} \leq \int_X \mathrm{e}^{b\psi_k}\theta^n_{\psi_k\vee(V_\theta-C)},\end{aligned}$$

where the equality part follows from Proposition 2.2.1(1).

Letting $k \to \infty$ with the help of Theorem 2.4.2, we conclude that

$$\varlimsup_{k\to\infty}\int_{\{\varphi_k=0\}}\theta^n_{\varphi_k} \leq \int_X \mathrm{e}^{b\varphi}\theta^n_{\varphi\vee(V_\theta-C)}.$$

Letting $b \to \infty$ and then $C \to \infty$, we conclude (2.13).

Step 2. We now handle the general net case.

Fix a volume form on X, so that we can talk about the L^1-norm on X in the sequel.

Suppose that (2.13) fails. Then there is $\delta > 0$ such that, for every $i_0 \in I$, one can find $i \geq i_0$ satisfying

$$\int_{\{\varphi_i=0\}} \theta^n_{\varphi_i} > \int_{\{\varphi=0\}} \theta^n_{\varphi} + \delta.$$

Since $\varphi_i \longrightarrow \varphi$ in L^1, for every $k > 0$ there is $a_k \in I$ such that

$$\|\varphi_i - \varphi\|_{L^1} < k^{-1}, \quad i \geq a_k.$$

We can then recursively choose an increasing sequence $(i_k)_{k>0}$ in I with $i_k \geq a_k$ and

$$\int_{\{\varphi_{i_k}=0\}} \theta^n_{\varphi_{i_k}} > \int_{\{\varphi=0\}} \theta^n_{\varphi} + \delta.$$

This sequence converges to φ in L^1, contradicting Step 1. □

Next we introduce an envelope construction, which will be repeatedly used in the sequel.

Definition 2.4.1 Given a function $f : X \longrightarrow [-\infty, \infty]$, we define $P_\theta(f)$ as follows:

$$P_\theta(f) := \sup{}^* \{\varphi \in \mathrm{PSH}(X, \theta) : \varphi \leq f \text{ quasi-everywhere}\}\ ^{6}. \tag{2.14}$$

The function $P_\theta(f)$ is either constantly $\pm\infty$ or lies in $\mathrm{PSH}(X, \theta)$. Moreover, given another function $g : X \longrightarrow [-\infty, \infty]$, equal to f quasi-everywhere, we have $P_\theta(f) = P_\theta(g)$. In particular, it makes sense to talk about $P_\theta(f)$ even if f is only defined outside a pluripolar set.

We also observe that

$$P_\theta(f) \leq f \quad \text{quasi-everywhere}. \tag{2.15}$$

This is a consequence of Proposition 1.2.3 and Proposition 1.2.5.

Lemma 2.4.3 *Fix a closed smooth real $(1, 1)$-form θ on X representing a big cohomology class. Let $f : X \longrightarrow \mathbb{R}$ be a lower semicontinuous function. Assume that $P_\theta(f) \not\equiv \infty$. Then $P_\theta(f)$ is locally bounded outside* $\mathrm{nK}(\{\theta\})$. *Set $U := \{P_\theta(f) < f\} \setminus \mathrm{nK}(\{\theta\})$, then*

$$\left(\theta|_U + \mathrm{dd^c} P_\theta(f)|_U\right)^n = 0. \tag{2.16}$$

Proof First observe that $V_\theta + \inf_X f$ is a candidate in the definition (2.14). So

[6] In the original article [DDNL25], the authors required that $\varphi \leq f$ everywhere. But in the proof of their Theorem 2.7 (Theorem 2.4.4 below), they actually relied on the current definition. In the most relevant case, if f is $\mathcal{F}$-upper semicontinuous (for example, if f is upper semicontinuous), then these two definitions coincide. This is due to Lemma 1.3.2 and Proposition 1.3.3.

$$P_\theta(f) \geq V_\theta + \inf_X f,$$

and in particular $P_\theta(f) \not\equiv -\infty$.

In particular, $P_\theta(f) \in \mathrm{PSH}(X,\theta)$. It is therefore bounded from above. By Lemma 2.4.1, it is locally bounded away from nK($\{\theta\}$).

Now fix $x \in U$, and $\epsilon > 0$ so that

$$P_\theta(f)(x) < f(x) - \epsilon.$$

Take holomorphic balls $B \Subset B' \Subset U$ containing x. After shrinking B, we can guarantee that

$$P_\theta(f)|_B < f(x) - \epsilon, \quad f|_B > f(x) - \epsilon. \tag{2.17}$$

Let $\varphi \in \mathrm{PSH}(X,\theta)$ be the balayage of $P_\theta(f)$ on B, as constructed in Proposition 1.5.4. By the domination principle Proposition 1.2.10, we find that

$$\varphi|_B \leq f(x) - \epsilon < f|_B.$$

Since $\varphi = P_\theta(f)$ outside B, we conclude that

$$\varphi \leq f \quad \text{quasi-everywhere.}$$

From this it follows that $\varphi \leq P_\theta(f)$, and hence equality holds. But then (2.16) holds automatically on B since φ is constructed by balayage. Since x and B are arbitrary, we conclude (2.16). □

Theorem 2.4.4 *Let $f\colon X \to [-\infty,\infty]$ be a quasi-continuous function.*
Assume that $P_\theta(f) \not\equiv \pm\infty$. Then $P_\theta(f) \in \mathrm{PSH}(X,\theta)$ and

$$\int_{\{P_\theta(f)<f\}} \theta^n_{P_\theta(f)} = 0. \tag{2.18}$$

Thanks to (2.15), we can rewrite (2.18) as

$$\int_{\{P_\theta(f)\neq f\}} \theta^n_{P_\theta(f)} = 0.$$

We shall say that $\theta^n_{P_\theta(f)}$ is carried by the contact set $\{P_\theta(f) = f\}$. [7]

Proof **Step 1**. We first reduce to the case where f is bounded.

Step 1.1. Reduce to the case where $f \leq 0$.

Take $C \in \mathbb{R}$ so that $P_\theta(f) \leq C$. Then

---

[7] It is tempting to say that $\theta^n_{P_\theta(f)}$ is supported on the contact set $\{P_\theta(f) = f\}$, as people are already doing in the literature. It should be mentioned that this is an abuse of the language, since the support of $\theta^n_{P_\theta(f)}$ (as a closed subset of X) could be much larger in general. One could probably introduce a notion of plurifine support, similar to what Fuglede did in the classical potential theory [Fug72].

$$P_\theta(f) = P_\theta\Big(\min\{f, C\}\Big).$$

Thus we can replace f by $\min\{f, C\}$. Replacing f by $f - \sup_X f$, we reduce easily to the case where $f \leq 0$.

Step 1.2. Reduce to the case where f is bounded. Assume that in this case, the theorem is known. We prove (2.18) for $f \leq 0$.

For each $j > 0$, we have

$$\int_{\{P_\theta(f_j)<f_j\}} \theta^n_{P_\theta(f_j)} = 0, \tag{2.19}$$

where $f_j = f \vee (-j)$.

Now fix $C > 0$ and two open sets $G' \Subset G \Subset X \setminus \mathrm{nK}(\{\theta\})$. Fix a smooth function $\chi\colon X \to [0,1]$ so that $\chi|_{G'} \equiv 1$ and χ is supported in G.

Define

$$U_C := G \cap \{P_\theta(f) > V_\theta - C\}, \quad U'_C := G' \cap \{P_\theta(f) > V_\theta - C\}.$$

Then U_C is $\mathcal{F}$-open. It follows from Proposition 2.2.1(1) that for any $j > 0$,

$$\mathbb{1}_{U_C}\, \theta^n_{P_\theta(f_j)\vee(V_\theta - C)} = \mathbb{1}_{U_C}\, \theta^n_{P_\theta(f_j)}.$$

In particular, thanks to (2.19),

$$\int_{U_C} \left(1 - \mathrm{e}^{P_\theta(f_j)-f_j}\right) \theta^n_{P_\theta(f_j)\vee(V_\theta-C)} = 0. \tag{2.20}$$

Note that the $P_\theta(f_j) \vee (V_\theta - C)$'s for various j are uniformly bounded from below on U_C, thanks to Lemma 2.4.1.

Next we claim that $(P_\theta(f_j))_j$ is decreasing and converges to $P_\theta(f)$.

The sequence is decreasing; let us compute its limit. In fact, $P_\theta(f_j) \leq f_j$ quasi-everywhere. It follows that $\inf_j P_\theta(f_j) \leq f$ quasi-everywhere and hence

$$\inf_{j>0} P_\theta(f_j) \leq P_\theta(f).$$

The reverse inequality is trivial, and our assertion follows.

Since $P_\theta(f) \not\equiv -\infty$, we know that the set $\{f = -\infty\}$ is pluripolar. It follows that $P_\theta(f_j) - f_j$ converges to the following function $g\colon X \to [-\infty, \infty)$ in capacity:

$$g(x) = \begin{cases} P_\theta(f)(x) - f(x), & \text{if } f(x) \neq -\infty; \\ -\infty, & \text{otherwise.} \end{cases}$$

Therefore,

$$1 - \mathrm{e}^{P_\theta(f_j)-f_j} \xrightarrow{\mathrm{C}} 1 - \mathrm{e}^{g}.$$

Next we claim that

$$\int_{U'_C} (1 - \mathrm{e}^g)\ \theta^n_{P_\theta(f)} = 0. \tag{2.21}$$

Since $g \le 0$ quasi-everywhere, the left-hand side of (2.21) is non-negative. It suffices to prove the $\le$ direction.

We wish to let $j \to \infty$ directly in (2.20), but since U_C is not open in general, this cannot be done directly. In the sequel, we shall slightly enlarge U_C to get an open set and then take the limit.

Fix $\epsilon > 0$, we can find an open subset $W \Subset X \setminus \mathrm{nK}(\{\theta\})$, so that

$$\mathrm{Cap}_{C^{-1}\theta}(W \setminus U_C) < \epsilon. \tag{2.22}$$

For example, we can construct W as follows. By Example 2.3.1, we can find an open set $A \subseteq G$ so that $\mathrm{Cap}_{C^{-1}\theta}(A) < \epsilon$ and $P_\theta(f) - V_\theta$ is continuous on $G \setminus A$. Then $U_C \cap (G \setminus A)$ is relatively open in $G \setminus A$, so we may choose an open set $W_0 \subseteq G$ such that

$$W_0 \cap (G \setminus A) = U_C \cap (G \setminus A).$$

Taking $W = W_0 \cup A$, we get $U_C \subseteq W$ and $W \setminus U_C \subseteq A$, hence (2.22).

Thanks to (2.22) and (2.20), we have

$$\int_W \chi \left(1 - \mathrm{e}^{P_\theta(f_j) - f_j}\right)\ \theta^n_{P_\theta(f_j) \vee (V_\theta - C)} \le C^n \epsilon.$$

Letting $j \to \infty$ and applying Theorem 2.3.3, we find that

$$\int_{U'_C} (1 - \mathrm{e}^g)\ \theta^n_{P_\theta(f) \vee (V_\theta - C)} \le \int_W \chi\, (1 - \mathrm{e}^g)\ \theta^n_{P_\theta(f) \vee (V_\theta - C)} \le C^n \epsilon.$$

Since $\epsilon > 0$ is arbitrary, (2.21) follows.

Now letting $C \to \infty$ in (2.21), we find

$$\int_{G'} (1 - \mathrm{e}^g)\ \theta^n_{P_\theta(f)} = 0.$$

Since $G' \Subset X \setminus \mathrm{nK}(\{\theta\})$ is arbitrary, we finally conclude

$$\int_X (1 - \mathrm{e}^g)\ \theta^n_{P_\theta(f)} = 0.$$

In other words, $P_\theta(f) = f$ almost everywhere with respect to $\theta^n_{P_\theta(f)}$. This proves (2.18).

Step 2. We now assume that $-C' \le f \le 0$ for some $C' > 0$.

For each $j > 0$, take an open subset $V_j \subseteq X$ so that

(1) $\mathrm{Cap}_\theta(V_j) \le 2^{-j-1}$, and
(2) $f|_{X \setminus V_j}$ is continuous.

Without loss of generality, we may assume that the V_j's are decreasing. Take a continuous function $f_j \colon X \to [-C', 0]$ extending $f|_{X \setminus V_j}$. This is always possible thanks to Tietze's extension theorem. For each $j > 0$, we let

$$h_j \coloneqq \sup_{k \geq j} f_k.$$

Then h_j is lower semicontinuous and h_j agrees with f outside V_j. Set

$$h = \inf_{j>0} h_j.$$

Then the set $\{h \neq f\}$ is contained in the intersection of the V_j's and hence is a pluripolar set, thanks to Theorem 2.3.2. In particular, $P_\theta(f) = P_\theta(h)$.

Now we can apply Lemma 2.4.3 to conclude that

$$\int_{X\setminus \mathrm{nK}(\{\theta\})} \left(1 - \mathrm{e}^{P_\theta(h_j)-h_j}\right) \theta^n_{P_\theta(h_j)} = 0$$

for each $j > 0$.

Fix two open subsets $G' \Subset G \Subset X \setminus \mathrm{nK}(\{\theta\})$. Note that $-C' \leq h_j \leq 0$. In particular,

$$V_\theta - C' \leq P_\theta(h_j) \leq V_\theta.$$

Hence,

$$\begin{aligned}
&\int_G \left|1 - \mathrm{e}^{P_\theta(h_j)-f}\right| \theta^n_{P_\theta(h_j)} \\
=&\int_{G\cap V_j} \left|1 - \mathrm{e}^{P_\theta(h_j)-f}\right| \theta^n_{P_\theta(h_j)} + \int_{G\setminus V_j} \left|1 - \mathrm{e}^{P_\theta(h_j)-h_j}\right| \theta^n_{P_\theta(h_j)} \\
=&\int_{G\cap V_j} \left|1 - \mathrm{e}^{P_\theta(h_j)-f}\right| \theta^n_{P_\theta(h_j)} \\
\leq&\int_{G\cap V_j} \left(1 + \mathrm{e}^{C'}\right) \theta^n_{P_\theta(h_j)} \\
\leq&\left(1 + \mathrm{e}^{C'}\right) \cdot 2^{-j-1} C'^n.
\end{aligned}$$

Now we can apply the same arguments as in Step 1.2 to conclude that

$$\int_{G'} \left(1 - \mathrm{e}^{P_\theta(f)-f}\right) \theta^n_{P_\theta(f)} = 0.$$

Since $G' \Subset X \setminus \mathrm{nK}(\{\theta\})$, (2.18) follows. □

The following lemma is striking in that we begin only with an upper bound of φ, but at the end of the day, we get a lower bound almost for free. This powerful method will be employed again and again in the whole book.

Lemma 2.4.4 *Let $\varphi, \psi \in \mathrm{PSH}(X, \theta)$, $\varphi \leq \psi$ and $\int_X \theta^n_\varphi > 0$. Then for any*

$$a \in \left(1, \left(\frac{\int_X \theta_\psi^n}{\int_X \theta_\psi^n - \int_X \theta_\varphi^n}\right)^{1/n}\right) \text{[8]}, \tag{2.23}$$

there is $\eta \in \mathrm{PSH}(X,\theta)_{>0}$ *such that*

$$a^{-1}\eta + \left(1 - a^{-1}\right)\psi \leq \varphi. \tag{2.24}$$

We write

$$\begin{aligned} P_\theta\Big(a\varphi + (1-a)\psi\Big) :=& \sup{}^* \left\{\eta \in \mathrm{PSH}(X,\theta) : a^{-1}\eta + \left(1 - a^{-1}\right)\psi \leq \varphi\right\} \\ &\in \mathrm{PSH}(X,\theta) \cup \{-\infty\}. \end{aligned} \tag{2.25}$$

Note that if we regard $a\varphi + (1-a)\psi$ as a function defined outside the pluripolar set $\{\varphi = \psi = -\infty\}$ on X, then (2.25) coincides with the envelope in the sense of (2.14).

Observe that

$$a^{-1}P_\theta\Big(a\varphi + (1-a)\psi\Big) + (1 - a^{-1})\psi \leq \varphi. \tag{2.26}$$

In fact, this equation holds outside a pluripolar set by Proposition 1.2.5, hence it holds everywhere by Proposition 1.2.6.

Finally, observe that

$$P_\theta\Big(a\varphi + (1-a)\psi\Big) \leq P_\theta\Big(a\varphi + (1-a)\varphi\Big) = \varphi.$$

Proof [9] Without loss of generality, we may assume that $\varphi \leq \psi \leq 0$.

Step 1. We show the existence of $\eta \in \mathrm{PSH}(X,\theta)$ satisfying (2.24).

Step 1.1. We first consider the case where $\psi \sim \varphi$. In this case, we can take $\eta = \varphi - C$ enough $C > 0$. But since by (2.26), we also have $P_\theta(a\varphi + (1-a)\psi) \leq \psi$, we find that

$$P_\theta\Big(a\varphi + (1-a)\psi\Big) \sim \varphi \sim \psi$$

in this case.

Step 1.2. We make a first reduction of the problem.

Now for each $\gamma \in \mathrm{PSH}(X,\theta)$, we let

$$P_\theta[\gamma] := \sup{}^*\Big\{\eta \in \mathrm{PSH}(X,\theta) : \eta \leq \min\{\gamma + C, 0\} \text{ for some } C > 0\Big\} \text{[10]}.$$

---

[8] The fraction in (2.23) is understood as ∞ if either $\int_X \theta_\psi^n = \int_X \theta_\varphi^n$ or $n = 0$. Thanks to the monotonicity theorem Theorem 2.4.3, the interval (2.23) is non-empty.

[9] The original proof in [DDNL25, Theorem 3.10] is flawed: Right below their (3.4), when they applied Lemma 3.5, the assumption $\varphi_j \leq \nu$ was not satisfied. Here one needs not only Lemma 3.5, but also Theorem 3.14 proved later. But the latter theorem relies on Theorem 3.10. This explains why we need the complicated transfinite induction below.

Observe that due to Corollary 2.4.1 and Theorem 2.4.3, we have

$$\int_X \theta^n_{P_\theta[\gamma]} = \int_X \theta^n_\gamma. \tag{2.27}$$

Further, $0 \geq P_\theta[\gamma] \geq \gamma$ if $\gamma \leq 0$.

We now define an increasing transfinite sequence $(\psi_\alpha)_\alpha$ consisting of non-positive functions in $\mathrm{PSH}(X,\theta)$ as follows:

(1) $\psi_0 = \psi$;
(2) $\psi_{\alpha+1} = P_\theta[\psi_\alpha]$;
(3) if α is a limit ordinal, we define

$$\psi_\alpha = \sup_{\beta<\alpha}{}^* \psi_\beta.$$

Observe that

$$\int_X \theta^n_{\psi_\alpha} = \int_X \theta^n_\psi$$

by Corollary 2.4.1 and (2.27). Since $\mathrm{PSH}(X,\theta)$ is indeed a set, the transfinite sequence must stabilize, say at some $\phi \in \mathrm{PSH}(X,\theta)$. Note that $0 \geq \phi \geq \psi$, and $P_\theta[\phi] = \phi$.

Note that the interval (2.23) remains the same if we replace ψ by ϕ. Take a in this interval, it suffices to prove the existence of $\eta \in \mathrm{PSH}(X,\theta)$ so that

$$a^{-1}\eta + \left(1 - a^{-1}\right)\phi \leq \varphi. \tag{2.28}$$

Let us record the following observation for later use. Suppose that $\tau \in \mathrm{PSH}(X,\theta)$ and $\tau \leq \phi$. Then

$$\sup_{\{\phi\neq-\infty\}} (\tau - \phi) = \sup_X \tau. \tag{2.29}$$

Observe that $\phi \leq 0$, so on the set $\{\phi \neq -\infty\}$, we have

$$\tau - \phi \geq \tau.$$

So the $\geq$ direction in (2.29) follows from Corollary 1.3.5. Conversely, by assumption, we can find a constant $C > 0$ so that

$$\tau - \sup_X \tau \leq \min\{\phi + C, 0\}.$$

It follows that $\tau - \sup_X \tau \leq P_\theta[\phi] = \phi$. Therefore, the reverse inequality follows, and (2.29) follows.

For each $k > 0$, we introduce

$$\varphi_k = \varphi \vee (\phi - k), \quad \eta_k \coloneqq P_\theta\Big(a\varphi_k + (1-a)\phi\Big).$$

[10] This is precisely the P-envelope to be introduced later in Definition 3.1.2.

Since $\varphi_k \sim \phi$, using Step 1.1, we have $\eta_k \in \mathrm{PSH}(X,\theta)$ and $\eta_k \sim \phi$ as well.

Note that $(\eta_k)_k$ is decreasing in k. Let

$$\eta \coloneqq \inf_{k\in\mathbb{N}} \eta_k .$$

Note that η automatically satisfies (2.28). It remains to show that $\eta \not\equiv -\infty$.

Step 1.3. We prove that $\eta \not\equiv -\infty$.

Assume by contrary that $\sup_X \eta_k \to -\infty$. For each $k > 0$, let

$$\gamma_k \coloneqq a^{-1}\eta_k + \left(1 - a^{-1}\right)\phi, \quad D_k \coloneqq \{\gamma_k = \varphi_k\}\,.$$

We claim that

$$\theta^n_{\eta_k} \leq a^n \mathbb{1}_{D_k}\,\theta^n_{\varphi_k}. \tag{2.30}$$

Since $\gamma_k \leq \varphi_k$ with equality on D_k, Lemma 2.4.2 gives

$$\mathbb{1}_{D_k}\,\theta^n_{\gamma_k} \leq \mathbb{1}_{D_k}\,\theta^n_{\varphi_k}.$$

On the other hand,

$$\theta_{\gamma_k} = a^{-1}\theta_{\eta_k} + \left(1 - a^{-1}\right)\theta_\phi,$$

so positivity and multilinearity of the non-pluripolar product imply

$$a^{-n}\theta^n_{\eta_k} \leq \theta^n_{\gamma_k}.$$

Finally, it follows from Theorem 2.4.4 that

$$\mathbb{1}_{D_k}\,\theta^n_{\eta_k} = \theta^n_{\eta_k}.$$

Putting these results together, we conclude (2.30).

Fix $j > k > 0$. Note that

$$\int_{\{\varphi\leq\phi-k\}} \theta^n_{\varphi_j} = \int_X \theta^n_{\varphi_j} - \int_{\{\varphi>\phi-k\}} \theta^n_{\varphi_j} = \int_X \theta^n_\phi - \int_{\{\varphi>\phi-k\}} \theta^n_\varphi, \tag{2.31}$$

where we applied Theorem 2.4.3 and Proposition 2.2.1(1) in the second equality.

Next, we compute

$$
\begin{aligned}
\int_{\{\eta_j\leq\phi-ak\}}\theta^n_{\eta_j} \leq & a^n\int_{\{\eta_j\leq\phi-ak\}}\mathbb{1}_{D_j}\,\theta^n_{\varphi_j} && \text{by (2.30)}\\
\leq & a^n\int_{\{a\varphi_j+(1-a)\phi\leq\phi-ak\}\cap\{\phi\neq-\infty\}}\theta^n_{\varphi_j}\\
= & a^n\int_{\{\varphi_j\leq\phi-k\}}\theta^n_{\varphi_j}\\
= & a^n\int_{\{\varphi\leq\phi-k\}}\theta^n_{\varphi_j}\\
\leq & a^n\left(\int_X\theta^n_\phi-\int_{\{\varphi>\phi-k\}}\theta^n_\varphi\right) && \text{by (2.31).}
\end{aligned}
$$

Next thanks to (2.29),

$$
\sup_{\{\phi\neq-\infty\}}(\eta_j-\phi)=\sup_X\eta_j\to-\infty.
$$

In particular, for a fixed k, if j is large enough, we have

$$
\left\{\eta_j\leq\phi-ak\right\}=X.
$$

Therefore, for a fixed k, for any large enough j,

$$
\int_X\theta^n_\phi=\int_X\theta^n_{\eta_j}\leq a^n\left(\int_X\theta^n_\phi-\int_{\{\varphi>\phi-k\}}\theta^n_\varphi\right).
$$

Letting $k\to\infty$, we find

$$
\int_X\theta^n_\phi\leq a^n\left(\int_X\theta^n_\phi-\int_X\theta^n_\varphi\right).
$$

This is a contradiction with our choice of a.

Step 2. Next we argue that $P_\theta(a\varphi+(1-a)\psi)\in\mathrm{PSH}(X,\theta)_{>0}$. Choose

$$
a'\in\left(a,\left(\frac{\int_X\theta^n_\psi}{\int_X\theta^n_\psi-\int_X\theta^n_\varphi}\right)^{1/n}\right).
$$

It follows from (2.25) that

$$
P_\theta\Big(a\varphi+(1-a)\psi\Big)\geq\frac{a}{a'}P_\theta\Big(a'\varphi+(1-a')\psi\Big)+\frac{a'-a}{a'}\varphi. \tag{2.32}
$$

Therefore, by Theorem 2.4.3, we have

$$
\int_X\theta^n_{P_\theta(a\varphi+(1-a)\psi)}\geq\frac{(a'-a)^n}{a'^n}\int_X\theta^n_\varphi>0. \tag{2.33}
$$

We conclude the proof. □

When φ and ψ have the same mass, we can say more:

Corollary 2.4.3 *Let $\varphi, \psi \in \mathrm{PSH}(X, \theta)_{>0}$, $\varphi \leq \psi$. Assume that $\int_X \theta_\varphi^n = \int_X \theta_\psi^n$. Then for any $\epsilon \in (0, 1)$, there is $\eta \in \mathrm{PSH}(X, \theta)$ such that*

(1) $\int_X \theta_\eta^n = \int_X \theta_\varphi^n$;
(2) $\epsilon\eta + (1 - \epsilon)\psi \leq \varphi$.

Proof Fix $\epsilon \in (0, 1)$, we define

$$\eta = P_\theta\left(\epsilon^{-1}\varphi + (1 - \epsilon^{-1})\psi\right).$$

By Lemma 2.4.4, $\eta \in \mathrm{PSH}(X, \theta)_{>0}$. Observe that (2) is satisfied by (2.26).

Thanks to (2.33), for each $a' > \epsilon^{-1}$, we have

$$\int_X \theta_\eta^n > \left(\frac{a' - \epsilon^{-1}}{a'}\right)^n \int_X \theta_\varphi^n.$$

Letting $a' \to \infty$, we conclude that

$$\int_X \theta_\eta^n \geq \int_X \theta_\varphi^n.$$

On the other hand, since $\eta \leq \psi$, using the monotonicity theorem Theorem 2.4.3 we find that

$$\int_X \theta_\eta^n \leq \int_X \theta_\psi^n = \int_X \theta_\varphi^n.$$

Hence,

$$\int_X \theta_\eta^n = \int_X \theta_\varphi^n.$$

Therefore, (1) holds as well. □

Next we prove a domination principle.

Theorem 2.4.5 (Domination principle) *Let $\varphi, \psi \in \mathrm{PSH}(X, \theta)_{>0}$. Assume that*

$$\exists \phi \in \mathrm{PSH}(X, \theta) \quad \text{s.t.} \quad \varphi \leq \phi, \quad \psi \leq \phi, \quad \int_X \theta_\varphi^n = \int_X \theta_\psi^n = \int_X \theta_\phi^n, \tag{2.34}$$

and

$$\int_{\{\varphi < \psi\}} \theta_\varphi^n = 0. \tag{2.35}$$

Then $\varphi \geq \psi$.

Remark 2.4.3 Using the terminologies to be introduced in Chapter 3, we can reformulate a special case of Theorem 2.4.5 as follows: Suppose that $\phi \in \mathrm{PSH}(X, \theta)_{>0}$ is a model potential and $\varphi, \psi \in \mathcal{E}(X, \theta; \phi)$. Assume that (2.35) holds then $\varphi \geq \psi$.

Equivalently, using the terminologies of Chapter 6, the assumption (2.34) can be reformulated as $\varphi \sim_P \psi$.

Proof Thanks to the monotonicity theorem Theorem 2.4.3,

$$\int_X \theta^n_{\varphi\vee\psi} = \int_X \theta^n_{\phi}.$$

We may replace ψ by $\varphi \vee \psi$ and assume that $\varphi \leq \psi$. Without loss of generality, we may also assume that $\psi \leq 0$.

Now fix $a > 1$, we define

$$\eta_a \coloneqq P_\theta\Big(a\varphi + (1-a)\psi\Big).$$

Note that $\eta_a \in \mathrm{PSH}(X,\theta)$ by Corollary 2.4.3 and

$$\int_X \theta^n_{\eta_a} = \int_X \theta^n_{\phi}.$$

Furthermore, $\eta_a \leq P_\theta(a\varphi + (1-a)\varphi) = \varphi$.

Define

$$\gamma_a \coloneqq a^{-1}\eta_a + (1-a^{-1})\psi, \quad D_a \coloneqq \{\gamma_a = \varphi\}.$$

Then $\gamma_a \leq \varphi$ with equality on D_a. Therefore,

$$\begin{aligned} a^{-n}\,\theta^n_{\eta_a} \leq & a^{-n}\,\theta^n_{\eta_a} + \mathbb{1}_{D_a}(1-a^{-1})^n\,\theta^n_\psi & \\ = & \mathbb{1}_{D_a} a^{-n}\,\theta^n_{\eta_a} + \mathbb{1}_{D_a}(1-a^{-1})^n\,\theta^n_\psi & \text{by Theorem 2.4.4} \\ \leq & \mathbb{1}_{D_a}\theta^n_{\gamma_a} & \\ \leq & \mathbb{1}_{D_a}\theta^n_{\varphi} & \text{by Lemma 2.4.2} \\ = & \mathbb{1}_{D_a\cap\{\varphi=\psi\}}\,\theta^n_\varphi & \text{by (2.35).} \end{aligned}$$

Note that on $D_a \cap \{\varphi = \psi\}$, we have $\eta_a = \varphi$. We deduce that

$$\theta^n_{\eta_a} = \mathbb{1}_{\{\eta_a=\varphi\}}\,\theta^n_{\eta_a} \leq \theta^n_\varphi,$$

where the inequality follows from Lemma 2.4.2.

But the two ends have the same mass, and hence

$$\theta^n_{\eta_a} = \theta^n_\varphi = \mathbb{1}_{\{\eta_a=\varphi\}}\,\theta^n_\varphi.$$

Therefore,

$$\int_X \mathrm{e}^{\eta_a}\,\theta^n_\varphi = \int_X \mathrm{e}^{\varphi}\,\theta^n_\varphi > 0.$$

Note that η_a is decreasing in a. The above equation shows that

$$\eta \coloneqq \inf_{a>1} \eta_a \not\equiv -\infty.$$

On the other hand, if $x \in X$ is such that $\varphi(x) < \psi(x)$, we then have

$$\eta_a(x) \leq a\varphi(x) - (a-1)\psi(x) = \psi(x) + a\Big(\varphi(x) - \psi(x)\Big).$$

Letting $a \to \infty$, we find that $\eta(x) = -\infty$. Therefore, $\{\varphi \neq \psi\}$ is pluripolar, and hence empty by Proposition 1.2.6. Our assertion follows. □

Lemma 2.4.5 *For any $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$, there is $\psi \in \mathrm{PSH}(X,\theta)$ such that*

(1) *θ_ψ is a Kähler current, and*
(2) *$\psi \leq \varphi$.*

In particular, there is an increasing sequence $(\varphi_i)_{i>0}$ in $\mathrm{PSH}(X,\theta)$ converging almost everywhere to φ such that θ_{φ_i} is a Kähler current for all $i \geq 1$.

Proof Using Lemma 2.4.4, we can find $\epsilon \in (0,1)$ and $\gamma \in \mathrm{PSH}(X,\theta)$ such that

$$\epsilon V_\theta + (1-\epsilon)\gamma \leq \varphi.$$

We observe that the cohomology class $\{\theta\}$ is big as a consequence of Proposition 2.4.1. Therefore, we can take $\eta \in \mathrm{PSH}(X,\theta)$ such that θ_η is a Kähler current and $\eta \leq 0$. Since $\eta \leq V_\theta$, we may take

$$\psi \coloneqq \epsilon\eta + (1-\epsilon)\gamma.$$

Then ψ clearly satisfies (1) and (2).

For the latter claim, it suffices to take

$$\varphi_i = \Big(1 - (i+1)^{-1}\Big)\varphi + (i+1)^{-1}\psi$$

for each $i > 0$. □

Lemma 2.4.6 *Let L be a holomorphic line bundle on X with $\theta \in c_1(L)$. Assume that $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$. Then there exists $k_0 > 0$ such that for each $k \geq k_0$, we have*

$$\mathrm{H}^0\Big(X, L^k \otimes \mathcal{I}(k\varphi)\Big) \neq 0.$$

Proof By Lemma 2.4.5, there is $\psi \in \mathrm{PSH}(X,\theta)$ with θ_ψ a Kähler current and $\psi \leq \varphi$. Since $\mathcal{I}(k\psi) \subseteq \mathcal{I}(k\varphi)$, it suffices to prove the assertion with ψ in place of φ. In this case, the result follows from Hörmander's L^2-estimate ([H65̈]), see [Dem12a, Theorem 13.21]. □

In the remainder of this section, we sharpen Theorem 2.4.4 for $C^{1,\bar{1}}$ barrier functions.

This part relies on various techniques not covered in this book.

Theorem 2.4.6 (Tosatti, Chu–Zhou) *Let X be a compact Kähler manifold with a Kähler form ω and let $f \in C^{1,1}(X)$. Then*

$$P_\omega(f) \in C^{1,1}(X),$$

and there is a constant $C = C(X, \omega, \|f\|_{C^{1,1}})$ *such that*

$$\|P_\omega(f)\|_{C^{1,1}} \leq C.$$

This is the main result of [Tos18, CZ19]. The proof of this theorem relies on some PDE techniques far beyond the scope of the current book. Weaker results were due to [Ber18, Ber19].

Proposition 2.4.3 (Darvas–Rubinstein) *Let* ω *be a Kähler form on* X*, and let* $f_1, f_2 \in C^{1,\bar{1}}(X)$*. Then* $P_\omega(\min\{f_1, f_2\}) \in C^{1,\bar{1}}(X)$*.*

This is [DR16, Theorem 2.5]. The original proof relies unfortunately on [BD12], which can be replaced by Theorem 2.4.6.

Lemma 2.4.7 *Assume that* ω *is a Kähler form on* X *and let* $f_1, f_2 \in C^{1,\bar{1}}(X)$*. Then the functions* f_1*,* f_2 *and* $P_\omega(\min\{f_1, f_2\})$ *coincide up to second order at almost every point of the contact set* $\{P_\omega(\min\{f_1, f_2\}) = f_1 = f_2\}$*. In particular,*

$$\mathbb{1}_{\{P_\omega(\min\{f_1,f_2\})=f_1=f_2\}}\, \omega_{f_1}^n = \mathbb{1}_{\{P_\omega(\min\{f_1,f_2\})=f_1=f_2\}}\, \omega_{f_2}^n.$$

We also have for $j \in \{1, 2\}$*,*

$$\begin{aligned} \omega^n_{P_\omega(\min\{f_1,f_2\})} =&\mathbb{1}_{\{P_\omega(\min\{f_1,f_2\})=f_1\}}\, \omega_{f_1}^n + \mathbb{1}_{\{P_\omega(\min\{f_1,f_2\})=f_2\}}\, \omega_{f_2}^n \\ &- \mathbb{1}_{\{P_\omega(\min\{f_1,f_2\})=f_1=f_2\}}\, \omega_{f_j}^n \\ \geq& \mathbb{1}_{\{P_\omega(\min\{f_1,f_2\})=f_j\}}\, \omega_{f_j}^n. \end{aligned} \tag{2.36}$$

Here $\omega_{f_i} := \omega + \mathrm{dd^c} f_i \in L^n(X)$, since $f_i \in C^{1,\bar{1}}(X)$. Therefore, $\omega_{f_i}^n$ makes sense as an absolutely continuous Radon measure.

Proof Set $\varphi := P_\omega(\min\{f_1, f_2\})$. Since $\min\{f_1, f_2\} \in C^0(X)$, we have $\varphi \leq \min\{f_1, f_2\}$. By Proposition 2.4.3, $\varphi \in C^{1,\bar{1}}(X)$. As $f_1, f_2, \varphi \in C^{1,\bar{1}}(X)$, the measures $\omega_{f_1}^n$, $\omega_{f_2}^n$ and ω_φ^n are all absolutely continuous.

Step 1: We show that at almost every contact point, φ and f_i agree to order two.

For $i \in \{1, 2, 3\}$ set

$$\Psi_1 := \varphi - f_1, \quad \Psi_2 := \varphi - f_2, \quad \Psi_3 := f_1 - f_2.$$

Each Ψ_i lies in $C^{1,\bar{1}}(X)$.

We claim that for $i = 1, 2, 3$, we have

$$\Psi_i \text{ vanishes to second order at almost every point of } \{\Psi_i = 0\}. \tag{2.37}$$

To prove this, write $\{\Psi_i = 0\}$ as the disjoint union $A_i \sqcup B_i \sqcup C_i \sqcup D_i$ of the following four sets:

- $A_i := \{\Psi_i = 0\} \cap \{\mathrm{d}\Psi_i \neq 0\}$;
- $B_i := \{\Psi_i = 0\} \cap \{$the classical real Hessian of Ψ_i does not exist$\}$;
- $C_i := \{\Psi_i = 0\} \cap \{\mathrm{d}\Psi_i = 0\} \cap \{$the classical real Hessian exists and is non-zero$\}$;
- $D_i := \{\Psi_i = 0\} \cap \{\mathrm{d}\Psi_i = 0\} \cap \{$the classical real Hessian exists and equals zero$\}$.

It suffices to prove A_i, B_i, C_i are Lebesgue null sets.

The set A_i is a C^1 real hypersurface in X, hence has Lebesgue measure zero.

Since $\Psi_i \in W^{2,1}_{\mathrm{loc}}$, the classical pointwise real Hessian of Ψ_i exists almost everywhere, so B_i is Lebesgue null.

Finally C_i is Lebesgue null by Stampacchia's lemma. This proves (2.37).

Specialising (2.37) to Ψ_1, Ψ_2, the functions φ and f_i agree to order two at almost every point of the contact set $\{\varphi = f_i\}$. Consequently $\omega + \mathrm{dd^c}\varphi$ and $\omega + \mathrm{dd^c} f_i$ have the same matrix of coefficients almost everywhere on $\{\varphi = f_i\}$, and therefore so do their n-fold wedge powers:

$$\mathbb{1}_{\{\varphi = f_i\}}\, \omega_\varphi^n = \mathbb{1}_{\{\varphi = f_i\}}\, \omega_{f_i}^n, \qquad i \in \{1, 2\}. \tag{2.38}$$

The analogous statement applied to Ψ_3 gives

$$\mathbb{1}_{\{f_1 = f_2\}}\, \omega_{f_1}^n = \mathbb{1}_{\{f_1 = f_2\}}\, \omega_{f_2}^n. \tag{2.39}$$

Step 2. The barrier $\min\{f_1, f_2\}$ is continuous, hence quasi-continuous, so Theorem 2.4.4 yields

$$\int_{X \setminus (\{\varphi = f_1\} \cup \{\varphi = f_2\})} \omega_\varphi^n = \int_{\{\varphi < \min\{f_1, f_2\}\}} \omega_\varphi^n = 0. \tag{2.40}$$

Set $V_1 := \{f_1 < f_2\}$ and $V_2 := \{f_1 > f_2\}$, both open. We may decompose

$$\{\varphi = f_1\} \cup \{\varphi = f_2\} = \big(V_1 \cap \{\varphi = f_1\}\big) \sqcup \big(V_2 \cap \{\varphi = f_2\}\big) \sqcup \{\varphi = f_1 = f_2\}.$$

This combined with (2.40) and (2.38) yields

$$\begin{aligned} \omega_\varphi^n &= \mathbb{1}_{V_1 \cap \{\varphi = f_1\}}\, \omega_\varphi^n + \mathbb{1}_{V_2 \cap \{\varphi = f_2\}}\, \omega_\varphi^n + \mathbb{1}_{\{\varphi = f_1 = f_2\}}\, \omega_\varphi^n \\ &= \mathbb{1}_{V_1 \cap \{\varphi = f_1\}}\, \omega_{f_1}^n + \mathbb{1}_{V_2 \cap \{\varphi = f_2\}}\, \omega_{f_2}^n + \mathbb{1}_{\{\varphi = f_1 = f_2\}}\, \omega_{f_j}^n. \end{aligned}$$

On the other hand,

$$\mathbb{1}_{V_j \cap \{\varphi = f_j\}}\, \omega_{f_j}^n = \mathbb{1}_{\{\varphi = f_j\}}\, \omega_{f_j}^n - \mathbb{1}_{\{\varphi = f_1 = f_2\}}\, \omega_{f_j}^n.$$

Therefore,

$$\omega_\varphi^n = \mathbb{1}_{\{\varphi = f_1\}}\, \omega_{f_1}^n + \mathbb{1}_{\{\varphi = f_2\}}\, \omega_{f_2}^n + \mathbb{1}_{\{\varphi = f_1 = f_2\}}\, \big(\omega_{f_j}^n - \omega_{f_1}^n - \omega_{f_2}^n\big).$$

Together with (2.39) we obtain (2.36). □

Theorem 2.4.7 (Di Nezza–Trapani) *Let $\theta_1, \dots, \theta_n$ be closed real smooth $(1,1)$-forms on X representing pseudo-effective cohomology classes. For each $i = 1, \dots, n$,* *let $\varphi_i \in \mathrm{PSH}(X,\theta_i)$ and $f_i \in C^{1,\bar{1}}(X)$ be functions with $\varphi_i \leq f_i$. Then*

$$\mathbb{1}_{\bigcap_i \{\varphi_i = f_i\}}\, \theta_{1,\varphi_1} \wedge \cdots \wedge \theta_{n,\varphi_n} = \mathbb{1}_{\bigcap_i \{\varphi_i = f_i\}}\, \theta_{1,f_1} \wedge \cdots \wedge \theta_{n,f_n}. \tag{2.41}$$

Proof **Step 1**. We first reduce to the case where $\theta_1 = \cdots = \theta_n$, $\varphi_1 = \cdots = \varphi_n$ and $f_1 = \cdots = f_n$.

Assume for the moment that this special case is known. Let us handle the general case.

For $\lambda = (\lambda_1, \dots, \lambda_n) \in (\mathbb{R}_{>0})^n$, we set

$$\theta_\lambda \coloneqq \sum_{j=1}^n \lambda_j \theta_j, \quad \varphi_\lambda \coloneqq \sum_{j=1}^n \lambda_j \varphi_j, \quad f_\lambda \coloneqq \sum_{j=1}^n \lambda_j f_j.$$

Then $\varphi_\lambda \in \mathrm{PSH}(X,\theta_\lambda)$, $f_\lambda \in C^{1,\bar{1}}(X)$, $\varphi_\lambda \leq f_\lambda$, and

$$\bigcap_{j=1}^n \{\varphi_j = f_j\} \subseteq \{\varphi_\lambda = f_\lambda\}.$$

So

$$\mathbb{1}_{\bigcap_j \{\varphi_j = f_j\}} (\theta_\lambda)^n_{\varphi_\lambda} = \mathbb{1}_{\bigcap_j \{\varphi_j = f_j\}} (\theta_\lambda)^n_{f_\lambda}.$$

Both sides are polynomials in $\lambda_1, \dots, \lambda_n$; comparing the coefficient of $\lambda_1 \cdots \lambda_n$ gives (2.41).

From now on, we assume

$$\theta = \theta_1 = \cdots = \theta_n, \quad \varphi = \varphi_1 = \cdots = \varphi_n \in \mathrm{PSH}(X,\theta),$$

and

$$f = f_1 = \cdots = f_n \in C^{1,\bar{1}}(X)$$

with

$$\varphi \leq f.$$

We wish to prove that

$$\mathbb{1}_{\{\varphi = f\}}\, \theta^n_\varphi = \mathbb{1}_{\{\varphi = f\}}\, \theta^n_f. \tag{2.42}$$

Step 2. We first handle the special case where $f \in \mathrm{PSH}(X,\theta)$.

Step 2.1. We assume that θ is a Kähler form, denoted by ω for clarity.

Pick a decreasing sequence $(\phi_j)_j$ in $C^\infty(X)$ with pointwise limit φ and set

$$\psi_j \coloneqq P_\omega\left(\min\{\phi_j, f\}\right).$$

Then $\psi_j \in \mathrm{PSH}(X,\omega)$, and

$$\varphi \leq \psi_j \leq \min\{\phi_j, f\} \leq f,$$

and hence $\psi_j \searrow \varphi$. Applying Lemma 2.4.7 with $f_1 = \phi_j$ and $f_2 = f$ gives

$$\omega_{\psi_j}^n \geq \mathbb{1}_{\{\psi_j=f\}}\,\omega_f^n \geq \mathbb{1}_{\{\varphi=f\}}\,\omega_f^n. \tag{2.43}$$

Fix $C > 0$ with $\min_X f > -C$ and $\epsilon > 0$. Let

$$h_j^{C,\epsilon} := \frac{(\psi_j + C) \vee 0}{\big((\psi_j + C) \vee 0\big) + \epsilon}, \qquad h^{C,\epsilon} := \frac{(\varphi + C) \vee 0}{(\varphi + C) \vee 0) + \epsilon}.$$

These functions are both quasi-continuous with values in $[0, 1]$. Observe that $h_j^{C,\epsilon}$ converges in capacity to $h^{C,\epsilon}$ as $j \to \infty$.

Now fix $g \in C^0(X)$, $g \geq 0$. Note that $h_j^{C,\epsilon} = 0$ on the set $\{\psi_j \leq -C\}$ and $h^{C,\epsilon} = 0$ on the set $\{\varphi \leq -C\}$. Using locality of the non-pluripolar product Proposition 2.2.1 (1), we have

$$h_j^{C,\epsilon}\, g\, \omega_{\psi_j}^n = h_j^{C,\epsilon}\, g\, \omega_{\psi_j \vee (-C)}^n, \quad h^{C,\epsilon}\, g\, \omega_{\varphi}^n = h^{C,\epsilon}\, g\, \omega_{\varphi \vee (-C)}^n.$$

Therefore,

$$\begin{aligned}
\int_X g\, \omega_\varphi^n \geq{}& \int_X h^{C,\epsilon}\, g\, \omega_\varphi^n \\
={}& \int_X h^{C,\epsilon}\, g\, \omega_{\varphi\vee(-C)}^n \\
={}& \lim_{j\to\infty} \int_X h_j^{C,\epsilon}\, g\, \omega_{\psi_j\vee(-C)}^n \qquad \text{by Theorem 2.3.3} \\
={}& \lim_{j\to\infty} \int_X h_j^{C,\epsilon}\, g\, \omega_{\psi_j}^n \\
\geq{}& \lim_{j\to\infty} \int_{\{\varphi=f\}} h_j^{C,\epsilon}\, g\, \omega_f^n \qquad \text{by (2.43)} \\
={}& \int_{\{\varphi=f\}} h^{C,\epsilon}\, g\, \omega_f^n,
\end{aligned}$$

where the last equality follows from the dominated convergence theorem. By our choice of C, we have $h^{C,\epsilon} > 0$ on $\{\varphi = f\}$, and hence when $\epsilon \to 0+$, $h^{C,\epsilon} \to 1$ everywhere on this set. Letting $\epsilon \to 0+$, we find

$$\int_X g\, \omega_\varphi^n \geq \int_{\{\varphi=f\}} g\, \omega_f^n$$

by the dominated convergence theorem.

By the Riesz representation theorem, we conclude that

$$\omega_\varphi^n \geq \mathbb{1}_{\{\varphi=f\}}\, \omega_f^n.$$

Then by Lemma 2.4.2, the reverse inequality also holds, and (2.42) follows in this case.

Step 2.2. We no longer assume that θ is a Kähler form.

Choose a Kähler form ω on X so that $\theta+\omega$ is a Kähler form and $f \in \mathrm{PSH}(X, \theta+\omega)$. Fix $g \in C^0(X)$ and consider, for $t \geq 0$,

$$Q(t) := \int_{\{\varphi=f\}} g\,(\theta + t\omega + \mathrm{dd^c}\varphi)^n - \int_{\{\varphi=f\}} g\,(\theta + t\omega + \mathrm{dd^c} f)^n.$$

By multilinearity of the non-pluripolar product, Q is a polynomial in t. Step 2.1 applied to $\theta + t\omega$ for $t > 1$ gives $Q(t) \equiv 0$ for $t > 1$, hence $Q \equiv 0$, and in particular $Q(0) = 0$. Since g is arbitrary, (2.42) holds.

Step 3. We handle the general case. We no longer assume that $f \in \mathrm{PSH}(X, \theta)$.

Choose a Kähler form ω on X so that $\theta + \omega$ is a Kähler form. Then

$$\varphi \leq P_{\theta+\omega}(f) \leq f.$$

By Lemma 2.4.7 applied to $f_1 = f_2 = f$, we find that f and $P_{\theta+\omega}(f)$ agree to the second order on the contact set $\{P_{\theta+\omega}(f) = f\}$, so

$$\mathbb{1}_{\{P_{\theta+\omega}(f)=f\}}\, \theta^n_{P_{\theta+\omega}(f)} = \mathbb{1}_{\{P_{\theta+\omega}(f)=f\}}\, \theta^n_f.$$

However, Step 2 implies that

$$\mathbb{1}_{\{P_{\theta+\omega}(f)=\varphi\}}\, \theta^n_\varphi = \mathbb{1}_{\{P_{\theta+\omega}(f)=\varphi\}}\, \theta^n_{P_{\theta+\omega}(f)}.$$

Combining these results, we conclude (2.42). □

Corollary 2.4.4 *Assume that* $\mathrm{PSH}(X, \theta) \neq \varnothing$. *For any* $f \in C^{1,\bar{1}}(X)$, *we have*

$$\theta^n_{P_\theta(f)} = \mathbb{1}_{\{P_\theta(f)=f\}}\, \theta^n_f.$$

In particular,

$$\theta^n_{V_\theta} = \mathbb{1}_{\{V_\theta=0\}}\, \theta^n.$$

Proof From Theorem 2.4.7, we find

$$\mathbb{1}_{\{P_\theta(f)=f\}}\, \theta^n_{P_\theta(f)} = \mathbb{1}_{\{P_\theta(f)=f\}}\, \theta^n_f.$$

By the concentration property Theorem 2.4.4, our assertion follows. □

Theorem 2.4.8 (Di Nezza–Trapani) *Let θ be a smooth closed real* $(1,1)$*-form on X representing a big cohomology class. Let $f \in C^{1,\bar{1}}(X)$. Then*

$$P_\theta(f) \in C^{1,\bar{1}}\big(X \setminus \mathrm{nK}(\{\theta\})\big).$$

In particular, $V_\theta \in C^{1,\bar{1}}\big(X \setminus \mathrm{nK}(\{\theta\})\big)$.

We refer to [DNT24] for the proof.

Chapter 3
The envelope operators

Politiques et scientifiques ont le sens des réalités, mais ce ne sont pas les mêmes. Il en résulte — et ce sera là un principe que le général de Gaulle fera sien que l'activité de recherche ne peut être évaluée, quant à sa qualité propre, que par des hommes qui la pratiquent eux-mêmes.
— Pierre Lelong[a], 1999

[a] Pierre Lelong (1912–2011) was the husband of another famous mathematician Jacqueline Ferrand. During their marriage (1947–1977), the latter published under the name of Jacqueline Lelong-Ferrand.

In this chapter, we study two envelope operators lying at the heart of the whole theory. The first envelope, called the P-envelope, is built from the rooftop construction and is characterized by non-pluripolar masses; the second, called the $\mathcal{I}$-envelope, is defined using multiplier ideal sheaves. The corresponding theories are developed in Section 3.1 and Section 3.2 respectively.

Later, in Chapter 6, we develop the corresponding P- and $\mathcal{I}$-preorders associated with these envelopes, allowing us to compare the singularities.

For the convenience of the reader, we reproduce a number of proofs that are also given, often in greater detail, in the survey of Darvas–Di Nezza–Lu [DDNL25]. The reader is welcome to skim these on a first reading and to consult [DDNL25] for a more leisurely treatment.

3.1 The P-envelope

In this section, X will denote a connected compact Kähler manifold of dimension n.

3.1.1 Rooftop operator and the definition of the P-envelope

Fix a smooth closed real $(1,1)$-form θ on X.

Definition 3.1.1 Given $\varphi, \psi \in \mathrm{PSH}(X, \theta)$, we define their *rooftop operator* as follows:

$$\varphi \wedge \psi = \sup\{\eta \in \mathrm{PSH}(X, \theta) : \eta \le \varphi, \eta \le \psi\}\,. \tag{3.1}$$

For simplicity of notation, we extend the definition to the case where φ or ψ is constantly $-\infty$, in which case we simply set

$$\varphi \wedge \psi = -\infty.$$

When we want to be more specific, we may also write $\varphi \wedge_\theta \psi$.

Proposition 3.1.1 *The operator $\wedge$ is a well-defined commutative, associative binary operator*

$$\big(\mathrm{PSH}(X,\theta) \cup \{-\infty\}\big) \times \big(\mathrm{PSH}(X,\theta) \cup \{-\infty\}\big) \to \mathrm{PSH}(X,\theta) \cup \{-\infty\}.$$

Proof We first show that the map is well-defined. For this purpose, take $\varphi, \psi \in \mathrm{PSH}(X,\theta)$. When the set in (3.1) is empty, there is nothing to prove. So let us assume that the set is not empty.

Define

$$\gamma = \sup{}^* \{\eta \in \mathrm{PSH}(X,\theta) : \eta \leq \varphi, \eta \leq \psi\}.$$

Then by Proposition 1.2.2, we find that $\gamma \in \mathrm{PSH}(X,\theta)$ and hence γ is a candidate for the supremum in (3.1). Therefore, $\gamma \leq \varphi \wedge \psi$. The reverse inequality follows from the definition of γ, so

$$\varphi \wedge \psi = \gamma \in \mathrm{PSH}(X,\theta).$$

The commutativity and the associativity of $\wedge$ follow directly from the defining maximality property. □

Note that from the proof, we have

$$\varphi \wedge \psi = P_\theta\Big(\min\{\varphi,\psi\}\Big).$$

See Definition 2.4.1 for the notation $P_\theta(\bullet)$.

Lemma 3.1.1 *Let $\varphi, \psi \in \mathrm{PSH}(X,\theta)$. Assume that $\varphi \wedge \psi \in \mathrm{PSH}(X,\theta)$. Then*

$$\theta^n_{\varphi\wedge\psi} \leq \mathbb{1}_{\{\varphi\wedge\psi=\varphi\}}\,\theta^n_\varphi + \mathbb{1}_{\{\varphi\wedge\psi=\psi\}}\,\theta^n_\psi. \tag{3.2}$$

In particular, $\theta^n_{\varphi\wedge\psi}$ is carried by the union $\{\varphi\wedge\psi=\varphi\} \cup \{\varphi\wedge\psi=\psi\}$.

Proof We first observe that as a consequence of Lemma 2.4.2, we have

$$\mathbb{1}_{\{\varphi\wedge\psi=\varphi\}}\,\theta^n_{\varphi\wedge\psi} \leq \mathbb{1}_{\{\varphi\wedge\psi=\varphi\}}\,\theta^n_\varphi, \quad \mathbb{1}_{\{\varphi\wedge\psi=\psi\}}\,\theta^n_{\varphi\wedge\psi} \leq \mathbb{1}_{\{\varphi\wedge\psi=\psi\}}\,\theta^n_\psi.$$

Applying Theorem 2.4.4 to $\min\{\varphi,\psi\}$, we conclude that

$$\theta^n_{\varphi\wedge\psi} \leq \mathbb{1}_{\{\varphi\wedge\psi=\varphi\}}\,\theta^n_{\varphi\wedge\psi} + \mathbb{1}_{\{\varphi\wedge\psi=\psi\}}\,\theta^n_{\varphi\wedge\psi} \leq \mathbb{1}_{\{\varphi\wedge\psi=\varphi\}}\,\theta^n_\varphi + \mathbb{1}_{\{\varphi\wedge\psi=\psi\}}\,\theta^n_\psi,$$

and (3.2) is established. □

We recall that the relations $\preceq$ and $\sim$ are introduced in Definition 1.5.2.

Definition 3.1.2 Given $\varphi \in \mathrm{PSH}(X,\theta)$, we define its *P-envelope* as follows:

$$P_\theta[\varphi] \coloneqq \sup{}^* \{\psi \in \mathrm{PSH}(X,\theta) : \psi \leq 0, \psi \preceq \varphi\}. \tag{3.3}$$

Note that $\varphi - \sup_X \varphi$ is always a candidate for the supremum. By Proposition 1.2.2, we have $P_\theta[\varphi] \in \mathrm{PSH}(X,\theta)$ and $P_\theta[\varphi] \leq 0$. Moreover, the definition can be equivalently described as

$$P_\theta[\varphi] = \sup_{C\in\mathbb{Z}_{>0}}{}^*(\varphi + C) \wedge V_\theta. \tag{3.4}$$

Recall that V_θ is introduced in (2.9). For any $C \in \mathbb{R}$, we have $(\varphi+C)\wedge V_\theta \in \mathrm{PSH}(X,\theta)$ and

$$(\varphi + C) \wedge V_\theta \sim \varphi.$$

In other words, in (3.3), we may replace the condition $\psi \leq \varphi$ by $\psi \sim \varphi$.

The idea lying behind the definition of $P_\theta[\varphi]$ is that we choose the least singular element out of all potentials with the same singularity type as φ. As we shall see in Example 3.1.1 below, $P_\theta[\varphi]$ does not necessarily have the same singularity type as φ. This forces us to define a rougher equivalence relation in Definition 6.1.1.

The envelope depends on the choice of θ, but the dependence is easy to understand:

Proposition 3.1.2 *Let $\theta' = \theta + \mathrm{dd^c} g$ for some $g \in C^\infty(X)$. Then for any $\varphi \in \mathrm{PSH}(X,\theta)$, we have $\varphi' := \varphi - g \in \mathrm{PSH}(X,\theta')$ and*

$$P_\theta[\varphi] \sim P_{\theta'}[\varphi'].$$

Proof By symmetry, it suffices to show that

$$P_\theta[\varphi] \leq P_{\theta'}[\varphi'].$$

We may assume that $g \geq 0$ (after replacing g by $g+C$ with $C \geq -\inf_X g$, which leaves θ' unchanged and shifts φ' by an additive constant). Then for any $\psi \in \mathrm{PSH}(X,\theta)$ with $\psi \leq \varphi$ and $\psi \leq 0$, we set $\psi' := \psi - g \in \mathrm{PSH}(X,\theta')$. Then $\psi' \leq \varphi'$ and $\psi' \leq 0$, so $\psi' \leq P_{\theta'}[\varphi']$. Since ψ is arbitrary, it follows that

$$P_\theta[\varphi] - \sup_X g \leq P_\theta[\varphi] - g \leq P_{\theta'}[\varphi'],$$

which gives $P_\theta[\varphi] \leq P_{\theta'}[\varphi']$. □

The P-envelope preserves the non-pluripolar masses:

Proposition 3.1.3 *Suppose that $\theta_1, \ldots, \theta_n$ are smooth closed real $(1,1)$-forms on X. Let $\varphi_i \in \mathrm{PSH}(X,\theta_i)$ for each $i = 1, \ldots, n$. Then*

$$\int_X \theta_{1,P_{\theta_1}[\varphi_1]} \wedge \cdots \wedge \theta_{n,P_{\theta_n}[\varphi_n]} = \int_X \theta_{1,\varphi_1} \wedge \cdots \wedge \theta_{n,\varphi_n}. \tag{3.5}$$

This proposition together with Theorem 2.4.3 will be combined to a common generalization in Proposition 6.1.4 after introducing the P-preorder.

Proof For each $C \in \mathbb{Z}_{>0}$ and each $i = 1, \ldots, n$, we have

$$(\varphi_i + C) \wedge V_{\theta_i} \sim \varphi_i .$$

It follows from the monotonicity theorem Theorem 2.4.3 that

$$\int_X \theta_{1,(\varphi_1+C)\wedge V_{\theta_1}} \wedge \cdots \wedge \theta_{n,(\varphi_n+C)\wedge V_{\theta_n}} = \int_X \theta_{1,\varphi_1} \wedge \cdots \wedge \theta_{n,\varphi_n} .$$

So (3.5) follows from (3.4) and Corollary 2.4.1. □

Conversely, Proposition 3.1.3 characterizes the P-envelope, as we will see in a moment in Theorem 3.1.2. We need some preparation for the proof. We first show that the P-envelope can be regarded as a concentration of the non-pluripolar mass:

Theorem 3.1.1 *Let $\varphi \in \mathrm{PSH}(X, \theta)$. Then*

$$\theta^n_{P_\theta[\varphi]} \leq \mathbb{1}_{\{P_\theta[\varphi]=0\}}\, \theta^n . \tag{3.6}$$

In fact, equality holds as we will see in Proposition 3.1.7 below.

Proof Thanks to Lemma 3.1.1, for each $C > 0$, we have

$$\begin{aligned}\theta^n_{(\varphi+C)\wedge V_\theta} &\leq \mathbb{1}_{\{(\varphi+C)\wedge V_\theta = \varphi+C\}}\, \theta^n_\varphi + \mathbb{1}_{\{(\varphi+C)\wedge V_\theta = V_\theta\}}\, \theta^n_{V_\theta} \\ &\leq \mathbb{1}_{\{\varphi+C \leq V_\theta\}}\, \theta^n_\varphi + \mathbb{1}_{\{P_\theta[\varphi]=V_\theta\}}\, \theta^n_{V_\theta} .\end{aligned}$$

We wish to let $C \to \infty$. The dominated convergence theorem assures that the first term $\mathbb{1}_{\{\varphi+C\leq V_\theta\}}\theta^n_\varphi$ converges weakly to 0, while the semicontinuity theorem Corollary 2.4.1 guarantees that $\theta^n_{(\varphi+C)\wedge V_\theta}$ converges weakly to $\theta^n_{P_\theta[\varphi]}$. So we conclude that

$$\theta^n_{P_\theta[\varphi]} \leq \mathbb{1}_{\{P_\theta[\varphi]=V_\theta\}}\, \theta^n_{V_\theta} .$$

Taking Corollary 2.4.4 into consideration, we conclude (3.6). □

Using essentially the same proof, we arrive at the following conclusion:

Corollary 3.1.1 *Let $\varphi, \psi \in \mathrm{PSH}(X, \theta)$. Let*

$$\eta \coloneqq \sup_{C>0}{}^{*} (\varphi + C) \wedge \psi .[1]$$

Assume that $\eta \not\equiv -\infty$ (for example when $\varphi \leq \psi$). Then

$$\theta^n_\eta \leq \mathbb{1}_{\{\eta=\psi\}}\, \theta^n_\psi .$$

Proof Thanks to Lemma 3.1.1, for each $C > 0$, we have

$$\begin{aligned}\theta^n_{(\varphi+C)\wedge\psi} &\leq \mathbb{1}_{\{(\varphi+C)\wedge\psi = \varphi+C\}}\, \theta^n_\varphi + \mathbb{1}_{\{(\varphi+C)\wedge\psi=\psi\}}\, \theta^n_\psi \\ &\leq \mathbb{1}_{\{\varphi+C\leq\psi\}}\, \theta^n_\varphi + \mathbb{1}_{\{\eta=\psi\}}\, \theta^n_\psi .\end{aligned}$$

[1] Using the terminology to be introduced in Definition 3.1.4, we can also write $\eta = P_\theta[\varphi](\psi)$.

We wish to let $C \to \infty$. The dominated convergence theorem assures that the first term $\mathbb{1}_{\{\varphi+C\leq\psi\}}\theta^n_\varphi$ converges weakly to 0, while the semicontinuity theorem Corollary 2.4.1 guarantees that $\theta^n_{(\varphi+C)\wedge\psi}$ converges weakly to θ^n_η. So we conclude that

$$\theta^n_\eta \leq \mathbb{1}_{\{\eta=\psi\}}\,\theta^n_\psi.$$

This completes the proof. □

Theorem 3.1.2 *Assume that $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$. Then*

$$P_\theta[\varphi] = \sup\left\{\psi \in \mathrm{PSH}(X,\theta) : \psi \leq 0, \varphi \leq \psi, \int_X \theta^n_\varphi = \int_X \theta^n_\psi\right\}. \tag{3.7}$$

In particular, in this case,

$$P_\theta\Big[P_\theta[\varphi]\Big] = P_\theta[\varphi]. \tag{3.8}$$

Note that in (3.7) and (3.3), the auxiliary function ψ lies on different sides of φ.

Proof Let ψ be a candidate of the right-hand side of (3.7). It follows from Theorem 3.1.1 that

$$\int_{\{P_\theta[\varphi]<\psi\}} \theta^n_{P_\theta[\varphi]} \leq \int_{\{P_\theta[\varphi]<\psi\}\cap\{P_\theta[\varphi]=0\}} \theta^n = 0.$$

On the other hand, we have

$$P_\theta[\varphi] \leq P_\theta[\psi], \quad \psi \leq P_\theta[\psi]$$

and all these potentials have the same mass due to Proposition 3.1.3. So the domination principle Theorem 2.4.5 is applicable and gives

$$P_\theta[\varphi] \geq \psi.$$

Hence we get the $\geq$ direction in (3.7).

On the other hand, $P_\theta[\varphi]$ itself is a candidate of the right-hand side of (3.7), as a consequence of Proposition 3.1.3. Therefore, (3.7) follows.

As for (3.8), the $\geq$ direction is trivial. For the other direction, let $\psi \in \mathrm{PSH}(X,\theta)$ be a potential satisfying

$$\psi \leq 0, \quad P_\theta[\varphi] \leq \psi, \quad \int_X \theta^n_{P_\theta[\varphi]} = \int_X \theta^n_\psi.$$

Then it follows from Proposition 3.1.3 that

$$\psi \leq 0, \quad \varphi \leq \psi, \quad \int_X \theta^n_\varphi = \int_X \theta^n_\psi.$$

In view of (3.7), we conclude the $\leq$ direction of (3.8). □

In general, we do not know if (3.8) holds when $\int_X \theta_\varphi^n = 0$. We expect it to be wrong. According to our general philosophy, the P-envelope operator is the correct object only when the non-pluripolar mass is positive. We will avoid using the degenerate case in the whole book.

Definition 3.1.3 If $\varphi = P_\theta[\varphi]$ and $\int_X \theta_\varphi^n > 0$, we say φ is a *model potential*.

Recall that the notion of model potentials depends heavily on the choice of θ. When there is a risk of confusion, we also say φ is a model potential in $\mathrm{PSH}(X,\theta)$.

Remark 3.1.1 Definition 3.1.3 is different from the common definition in the literature: We impose the extra condition $\int_X \theta_\varphi^n > 0$. The author believes that this is the only case where this notion is natural. We sometimes emphasize this point by saying $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$ is a model potential.

There are plenty of model potentials:

Corollary 3.1.2 *Let $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$. Then $P_\theta[\varphi]$ is a model potential in $\mathrm{PSH}(X,\theta)$. Moreover,*

$$\int_X \theta^n_{P_\theta[\varphi]} = \int_X \theta_\varphi^n.$$

Proof This follows immediately from Theorem 3.1.2 and Proposition 3.1.3. □

As we have already seen in the proof of Lemma 2.4.4, we have the following interesting property:

Proposition 3.1.4 *Let $\varphi \in \mathrm{PSH}(X,\theta)$. Consider a model potential $\phi \in \mathrm{PSH}(X,\theta)_{>0}$. Assume that $\varphi \leq \phi$. Then*

$$\sup_{\{\phi\neq-\infty\}} (\varphi - \phi) = \sup_X \varphi.$$

Proof Observe that $\phi \leq 0$, so on the set $\{\phi \neq -\infty\}$, we have

$$\varphi - \phi \geq \varphi.$$

So the $\geq$ direction follows from Corollary 1.3.5. Conversely, by assumption, we can find a constant $C > 0$ so that

$$\varphi - \sup_X \varphi \leq \min\{\phi + C, 0\}.$$

It follows that $\varphi - \sup_X \varphi \leq P_\theta[\phi] = \phi$. Therefore, the reverse inequality follows. □

Example 3.1.1 We continue our favorite example Example 1.8.1. Let $X = \mathbb{P}^1$ and ω be the Fubini–Study metric. We define $\varphi \in \mathrm{PSH}(X,\omega)$ as follows: For $z \in \mathbb{C}$, we let

$$\varphi(z) = \begin{cases} -\log\left(|z|^2+1\right) + \left(-\log\left(-\log|z|^2\right)\right) \vee \left(2+\log|z|^2\right), & \text{if } |z| < 1/\sqrt{2}, \\ 2 + \log\frac{|z|^2}{|z|^2+1}, & \text{otherwise,} \end{cases}$$

while $\varphi(\infty) = 2$. Note that φ is ω-psh on $X = \mathbb{P}^1$ by the gluing lemma Lemma 1.2.2. The singularity of φ only occurs at $z = 0$, close to which, φ is approximately $-\log\left(-\log |z|^2\right)$. This type of singularity is therefore called the *log-log type singularity*.

We claim that

$$P_\omega[\varphi] = 0. \tag{3.9}$$

In particular, we find that φ and $P_\omega[\varphi]$ have different singularity types.

By Theorem 3.1.2, in order to verify (3.9), it suffices to verify that

$$\int_X \omega_\varphi^1 = 1. \tag{3.10}$$

Here ω_φ^1 is taken in the non-pluripolar sense. Since $\{0, \infty\} \subseteq \mathbb{P}^1$ is pluripolar, this reduces to showing that

$$\int_{\mathbb{C}^*} \mathrm{dd^c}\psi = \frac{1}{4\pi}\int_{\mathbb{C}^*} (\Delta\psi)\,\mathrm{d}\mu = 1,$$

where $\psi(z) = \varphi(z) + \log(|z|^2 + 1)$ and μ is the standard Lebesgue measure on $\mathbb{C}$.

Note that the Laplacian vanishes outside $\overline{B(0, 0.7)}$ since $\psi(z) = 2 + \log |z|^2$ there, which is harmonic. Therefore,

$$\int_{\mathbb{C}^*} \mathrm{dd^c}\psi = \frac{1}{4\pi}\int_{|z|<1/\sqrt{2}} (\Delta\psi)(z)\,\mathrm{d}\mu.$$

It is an elementary exercise to see that the right-hand side is exactly equal to 1. If you are familiar with toric geometry, this is more or less trivial since

$$\nabla_r\Big((-\log(-r)) \vee (2 + r)\Big)(-\infty, -\log 2) = (0, 1].$$

Otherwise, just try to evaluate the integral using Green's identities. Therefore, (3.10) is proved and our assertion (3.9) follows.

Next we give a criterion on when the rooftop operator is not identically $-\infty$.

Proposition 3.1.5 *Assume that $\varphi, \psi \in \mathrm{PSH}(X, \theta)$ and*

$$\int_X \theta_\varphi^n + \int_X \theta_\psi^n > \int_X \theta_{\varphi\vee\psi}^n. \tag{3.11}$$

Then $\varphi \wedge \psi \in \mathrm{PSH}(X, \theta)$.

Thanks to the monotonicity theorem Theorem 2.4.3, we may also replace $\varphi \vee \psi$ on the right-hand side of (3.11) by any $\gamma \in \mathrm{PSH}(X, \theta)$ such that $\varphi \vee \psi \leq \gamma$.

Proof Without loss of generality, we may assume that $\varphi, \psi \leq 0$. For simplicity, we write

$$\eta = P_\theta[\varphi \vee \psi].$$

Note that due to our assumption (3.11), $\varphi,\psi\in \mathrm{PSH}(X,\theta)_{>0}$, and hence so is $\varphi\vee\psi$ thanks to the monotonicity theorem Theorem 2.4.3. By Corollary 3.1.2, η is a model potential, and $\int_X \theta_\eta^n=\int_X\theta^n_{\varphi\vee\psi}>0$.

Take $C>0$ large enough, so that

$$\int_{\{\varphi>\eta-C\}}\theta_\varphi^n+\int_{\{\psi>\eta-C\}}\theta_\psi^n>\int_X\theta_\eta^n. \tag{3.12}$$

This is possible thanks to Proposition 2.2.1(4). Fix $C'>C$. Write

$$\gamma_{C'}:=\Big(\varphi\vee(\eta-C')\Big)\wedge\Big(\psi\vee(\eta-C')\Big).$$

Then observe that $\gamma_{C'}\sim\eta$, and

$$\inf_{C'>C}\gamma_{C'}=\varphi\wedge\psi.$$

Assume by contradiction that $\varphi\wedge\psi\equiv-\infty$, then we have

$$\lim_{C'\to\infty}\sup_X\gamma_{C'}=-\infty.$$

Thanks to Proposition 3.1.4, for each $C'>C$,

$$\sup_X\gamma_{C'}=\sup_{\{\eta\neq-\infty\}}(\gamma_{C'}-\eta),$$

since η is a model potential. It follows that

$$\lim_{C'\to\infty}\sup_{\{\eta\neq-\infty\}}(\gamma_{C'}-\eta)=-\infty. \tag{3.13}$$

In particular, we can take C' large enough so that

$$\gamma_{C'}\leq\eta-C.$$

For such C', we have

$$\begin{aligned}\int_X\theta^n_{\gamma_{C'}}=&\int_{\{\gamma_{C'}\leq\eta-C\}}\theta^n_{\gamma_{C'}}\\ \leq&\int_{\{\varphi\vee(\eta-C')\leq\eta-C\}}\theta^n_{\varphi\vee(\eta-C')}+\int_{\{\psi\vee(\eta-C')\leq\eta-C\}}\theta^n_{\psi\vee(\eta-C')}\\ =&2\int_X\theta_\eta^n-\int_{\{\varphi>\eta-C\}}\theta^n_\varphi-\int_{\{\psi>\eta-C\}}\theta^n_\psi\\ <&\int_X\theta^n_\eta,\end{aligned}$$

where the second line follows from Lemma 3.1.1, the fourth line follows from (3.12). This contradicts the fact that $\gamma_{C'}\sim\eta$ in view of Theorem 2.4.3. □

Proposition 3.1.6 *Let $\varphi, \psi \in \mathrm{PSH}(X, \theta)$. Assume that*

$$\varphi = P_\theta[\varphi], \quad \psi = P_\theta[\psi], \quad \varphi \wedge \psi \not\equiv -\infty. \tag{3.14}$$

Then

$$P_\theta[\varphi \wedge \psi] = \varphi \wedge \psi. \tag{3.15}$$

Proof Observe that by Proposition 3.1.10 (3),

$$P_\theta[\varphi \wedge \psi] \leq P_\theta[\varphi] = \varphi, \quad P_\theta[\varphi \wedge \psi] \leq P_\theta[\psi] = \psi.$$

So the $\leq$ direction in (3.15) holds. The reverse direction is trivial. □

Lemma 3.1.2 *Let $\varphi, \psi \in \mathrm{PSH}(X, \theta)$. Assume that* (3.14) *holds and*

$$\int_X \theta^n_{\varphi \wedge \psi} > 0.$$

Suppose that $\gamma, \eta \in \mathrm{PSH}(X, \theta)$ and satisfy

$$\gamma \leq \varphi, \quad \eta \leq \psi, \quad \int_X \theta^n_\gamma = \int_X \theta^n_\varphi, \quad \int_X \theta^n_\eta = \int_X \theta^n_\psi.$$

Then $\gamma \wedge \eta \not\equiv -\infty$ and

$$\int_X \theta^n_{\gamma \wedge \eta} = \int_X \theta^n_{\varphi \wedge \psi}. \tag{3.16}$$

Later on in Chapter 6, we shall reformulate and generalize this lemma in terms of the P-preorder, which is a more natural language to express the result. See Proposition 6.1.7.

Proof Without loss of generality, we may assume that

$$\gamma \leq \varphi, \quad \eta \leq \psi.$$

Step 1. We first show that $\gamma \wedge \psi \not\equiv -\infty$ and

$$\int_X \theta^n_{\gamma \wedge \psi} = \int_X \theta^n_{\varphi \wedge \psi}. \tag{3.17}$$

For any $a > 1$, we define

$$\gamma_a := P_\theta\Big(a\gamma + (1-a)\varphi\Big).$$

When $a = 1$, we simply write $\gamma_1 = \gamma$.

Thanks to Corollary 2.4.3, for any $a \geq 1$, we have $\gamma_a \leq \varphi$ and

$$\int_X \theta^n_{\gamma_a} = \int_X \theta^n_\varphi.$$

By our assumption, for any $a \geq 1$, we have

$$\int_X \theta_{\gamma_a}^n + \int_X \theta_{\varphi\wedge\psi}^n > \int_X \theta_\varphi^n, \quad \gamma_a \leq \varphi, \quad \varphi \wedge \psi \leq \varphi.$$

So Proposition 3.1.5 gives

$$\gamma_a \wedge \varphi \wedge \psi = \gamma_a \wedge \psi \in \mathrm{PSH}(X, \theta).$$

In particular, for $a = 1$, we find $\gamma \wedge \psi \not\equiv -\infty$. It remains to verify (3.17).

For $a > 1$, we have

$$\gamma \geq a^{-1}\gamma_a + \left(1 - a^{-1}\right)\varphi,$$

thanks to the definition of γ_a (cf. (2.26)). Hence

$$\gamma \wedge \psi \geq a^{-1}\left(\gamma_a \wedge \psi\right) + \left(1 - a^{-1}\right)(\varphi \wedge \psi).$$

Therefore, by the monotonicity theorem Theorem 2.4.3,

$$\int_X \theta_{\gamma\wedge\psi}^n \geq \left(1 - a^{-1}\right)^n \int_X \theta_{\varphi\wedge\psi}^n.$$

Letting $a \to \infty$ and taking Theorem 2.4.3 into account, (3.17) follows.

Step 2. We complete the proof.

For any $a > 1$, we let

$$\eta_a \coloneqq P_\theta\Big(a\eta + (1-a)\psi\Big).$$

When $a = 1$, we set $\eta_1 = \eta$. Thanks to Corollary 2.4.3, for each $a \geq 1$, we have $\eta_a \leq \psi$ and

$$\int_X \theta_{\eta_a}^n = \int_X \theta_\psi^n.$$

From Step 1, we have

$$\int_X \theta_{\gamma\wedge\psi}^n + \int_X \theta_{\eta_a}^n = \int_X \theta_{\varphi\wedge\psi}^n + \int_X \theta_\psi^n > \int_X \theta_\psi^n.$$

Applying Proposition 3.1.5 again, we find

$$\gamma \wedge \psi \wedge \eta_a = \gamma \wedge \eta_a \in \mathrm{PSH}(X, \theta).$$

When $a = 1$, we find $\gamma \wedge \eta \in \mathrm{PSH}(X, \theta)$. It remains to prove (3.16).

By definition of η_a, we have

$$\eta \geq a^{-1}\eta_a + \left(1 - a^{-1}\right)\psi.$$

Therefore,

$$\gamma \wedge \eta \geq a^{-1}(\gamma \wedge \eta_a) + \left(1 - a^{-1}\right)(\gamma \wedge \psi)\,.$$

By the monotonicity theorem Theorem 2.4.3,

$$\int_X \theta^n_{\gamma\wedge\eta} \geq \left(1 - a^{-1}\right)^n \int_X \theta^n_{\gamma\wedge\psi}.$$

Letting $a \to \infty$ and applying Theorem 2.4.3 again, we arrive at (3.16). □

There is an interesting diamond-like inequality regarding the non-pluripolar masses:

Theorem 3.1.3 *Assume that* $\varphi, \psi \in \mathrm{PSH}(X, \theta)$ *and* $\varphi \wedge \psi \in \mathrm{PSH}(X, \theta)$. *Then*

$$\int_X \theta^n_\varphi + \int_X \theta^n_\psi \leq \int_X \theta^n_{\varphi\vee\psi} + \int_X \theta^n_{\varphi\wedge\psi}. \tag{3.18}$$

Proof We may assume that $\varphi, \psi \in \mathrm{PSH}(X, \theta)_{>0}$, as otherwise (3.18) follows immediately from the monotonicity theorem Theorem 2.4.3.

Step 1. We claim that it suffices to prove (3.18) with $P_\theta[\varphi]$ and $P_\theta[\psi]$ in place of φ and ψ.

In fact, as $\varphi \wedge \psi \not\equiv -\infty$, we also have $P_\theta[\varphi] \wedge P_\theta[\psi] \not\equiv -\infty$. Moreover,

$$\int_X \theta^n_{P_\theta[\varphi]} = \int_X \theta^n_\varphi, \quad \int_X \theta^n_{P_\theta[\psi]} = \int_X \theta^n_\psi$$

by Proposition 3.1.3. On the other hand, as $C \to \infty$, the potentials

$$\left((\varphi + C) \wedge V_\theta\right) \vee \left((\psi + C) \wedge V_\theta\right)$$

converge to $P_\theta[\varphi] \vee P_\theta[\psi]$ almost everywhere. Therefore, thanks to Corollary 2.4.1 and Theorem 2.4.3,

$$\int_X \theta^n_{P_\theta[\varphi]\vee P_\theta[\psi]} = \int_X \theta^n_{\varphi\vee\psi}.$$

Finally, we have

$$\int_X \theta^n_{P_\theta[\varphi]\wedge P_\theta[\psi]} = \int_X \theta^n_{\varphi\wedge\psi}$$

as well. When the left-hand side vanishes, this follows from Theorem 2.4.3. Otherwise, it follows from Lemma 3.1.2.

In particular, (3.18) is equivalent to the corresponding result with $P_\theta[\varphi]$ and $P_\theta[\psi]$ in place of φ and ψ.

Step 2. We shall assume that φ and ψ are model potentials in the sequel.

We write $\gamma = \varphi \vee \psi$ for simplicity. Note that $\gamma \leq 0$.

For each $C > 0$, we introduce

$$\varphi_C = \varphi \vee (\gamma - C), \quad \psi_C = \psi \vee (\gamma - C).$$

Note that $\varphi_C \sim \psi_C \sim \gamma$. Observe that

$$\inf_{C>0} \varphi_C \wedge \psi_C = \varphi \wedge \psi.$$

The $\geq$ direction is trivial, and the $\leq$ direction follows from the observation that the left-hand side is dominated by both φ_C and ψ_C for any $C > 0$, and therefore by both φ and ψ.

For $C > 0$, we compute

$$\begin{aligned}
\theta^n_{\varphi_C} =& \mathbb{1}_{\{\varphi>\gamma-C\}}\,\theta^n_{\varphi_C} + \mathbb{1}_{\{\varphi\leq\gamma-C\}}\,\theta^n_{\varphi_C} \\
=& \mathbb{1}_{\{\varphi>\gamma-C\}}\,\theta^n_{\varphi} + \mathbb{1}_{\{\varphi\leq\gamma-C\}}\,\theta^n_{\varphi_C} \qquad \text{by Proposition 2.2.1 (1)} \\
=& \theta^n_{\varphi} + \mathbb{1}_{\{\varphi\leq\gamma-C\}}\,\theta^n_{\varphi_C} \qquad \text{by Theorem 3.1.1.}
\end{aligned} \tag{3.19}$$

In the last line we use that φ is model, so θ^n_φ is carried by $\{\varphi = 0\} \subseteq \{\varphi > \gamma - C\}$. We also get a similar formula for $\theta^n_{\psi_C}$.

Therefore, taking the monotonicity theorem Theorem 2.4.3 into consideration, we find

$$\int_X \theta^n_\gamma - \int_X \theta^n_\varphi = \int_{\{\varphi\leq\gamma-C\}} \theta^n_{\varphi_C}, \quad \int_X \theta^n_\gamma - \int_X \theta^n_\psi = \int_{\{\psi\leq\gamma-C\}} \theta^n_{\psi_C}. \tag{3.20}$$

On the other hand, using Lemma 3.1.1 and (3.19), we find

$$\begin{aligned}
\theta^n_{\varphi_C\wedge\psi_C} \leq& \mathbb{1}_{\{\varphi_C\wedge\psi_C=\varphi_C\}}\,\theta^n_{\varphi_C} + \mathbb{1}_{\{\varphi_C\wedge\psi_C=\psi_C\}}\,\theta^n_{\psi_C} \\
\leq& \mathbb{1}_{\{\varphi_C\wedge\psi_C=\varphi_C=0\}}\,\theta^n_{\varphi} + \mathbb{1}_{\{\varphi_C\wedge\psi_C=\psi_C=0\}}\,\theta^n_{\psi} + \mathbb{1}_{\{\varphi\leq\gamma-C\}}\,\theta^n_{\varphi_C} \\
&+ \mathbb{1}_{\{\psi\leq\gamma-C\}}\,\theta^n_{\psi_C}.
\end{aligned}$$

In particular,

$$\mathbb{1}_{\{\varphi_C\wedge\psi_C<0\}}\,\theta^n_{\varphi_C\wedge\psi_C} \leq \mathbb{1}_{\{\varphi\leq\gamma-C\}}\,\theta^n_{\varphi_C} + \mathbb{1}_{\{\psi\leq\gamma-C\}}\,\theta^n_{\psi_C}.$$

Integrating, we find

$$\begin{aligned}
\int_{\{\varphi_C\wedge\psi_C=0\}} \theta^n_{\varphi_C\wedge\psi_C} \geq& \int_X \theta^n_\gamma - \int_{\{\varphi\leq\gamma-C\}} \theta^n_{\varphi_C} - \int_{\{\psi\leq\gamma-C\}} \theta^n_{\psi_C} \\
=& \int_X \theta^n_\varphi + \int_X \theta^n_\psi - \int_X \theta^n_\gamma,
\end{aligned}$$

where on the first line we used the fact that $\varphi_C \wedge \psi_C \sim \gamma$, while on the second line, we used (3.20).

Letting $C \to \infty$, and using Corollary 2.4.2, we conclude (3.18). □

For later use, we also introduce a generalized envelope operator:

Definition 3.1.4 Given $\varphi \in \mathrm{PSH}(X,\theta)$ and a function $f\colon X \to [-\infty,\infty]$, we define

$$P_\theta[\varphi](f) := \sup{}^*\left\{\eta \in \mathrm{PSH}(X,\theta) : \eta \leq f \text{ quasi-everywhere}, \ \eta \leq \varphi\right\}.$$

In general, we cannot replace $\eta \leq f$ quasi-everywhere by everywhere.

Example 3.1.2 When $f = 0$, we have $P_\theta[\varphi](f) = P_\theta[\varphi]$.

When $\varphi = V_\theta$, we have $P_\theta[V_\theta](f) = P_\theta(f)$.

So Definition 3.1.4 is a common generalization of Definition 3.1.2 and Definition 2.4.1.

Proposition 3.1.7 *Let $\varphi \in \mathrm{PSH}(X,\theta)$ and $f \in C^{1,\bar{1}}(X)$ with $\varphi \leq f$. Then*

$$\theta^n_{P_\theta[\varphi](f)} = \mathbb{1}_{\{P_\theta[\varphi](f)=f\}}\,\theta^n_f .$$

In particular,

$$\theta^n_{P_\theta[\varphi]} = \mathbb{1}_{\{P_\theta[\varphi]=0\}}\,\theta^n .$$

This is a vast generalization of Theorem 3.1.1.

Proof By definition, we have

$$P_\theta[\varphi](f) = P_\theta[\varphi]\Big(P_\theta(f)\Big)$$

and $\varphi \leq P_\theta(f)$. By Corollary 3.1.1 and Corollary 2.4.4,

$$\theta^n_{P_\theta[\varphi](f)} \leq \mathbb{1}_{\{P_\theta[\varphi](f)=P_\theta(f)\}}\,\theta^n_{P_\theta(f)} = \mathbb{1}_{\{P_\theta[\varphi](f)=f\}}\,\theta^n_f .$$

The reverse inequality follows immediately from Theorem 2.4.7. ◻

Proposition 3.1.8 *Let $\varphi \in \mathrm{PSH}(X,\theta)$ and $f \in C^{1,\bar{1}}(X)$ with $\varphi \leq f$. Then*

$$\begin{aligned}\mathbb{1}_{\{\varphi=P_\theta(f)\}}\,\theta^n_{P_\theta(f)} &= \mathbb{1}_{\{\varphi=P_\theta(f)\}}\,\theta^n_\varphi,\\ \mathbb{1}_{\{\varphi=P_\theta[\varphi](f)\}}\,\theta^n_{P_\theta[\varphi](f)} &= \mathbb{1}_{\{\varphi=P_\theta[\varphi](f)\}}\,\theta^n_\varphi.\end{aligned}$$

Proof By Corollary 2.4.4 and Theorem 2.4.7, we have

$$\mathbb{1}_{\{\varphi=P_\theta(f)\}}\,\theta^n_{P_\theta(f)} = \mathbb{1}_{\{\varphi=P_\theta(f)=f\}}\,\theta^n_f = \mathbb{1}_{\{\varphi=P_\theta(f)=f\}}\,\theta^n_\varphi \leq \mathbb{1}_{\{\varphi=P_\theta(f)\}}\,\theta^n_\varphi.$$

The reverse inequality follows from Lemma 2.4.2. The first equality follows.

As for the second, we compute using Proposition 3.1.7 and Theorem 2.4.7 that

$$\begin{aligned}\mathbb{1}_{\{\varphi=P_\theta[\varphi](f)\}}\,\theta^n_{P_\theta[\varphi](f)} &= \mathbb{1}_{\{\varphi=P_\theta[\varphi](f)=f\}}\,\theta^n_f\\ &= \mathbb{1}_{\{\varphi=P_\theta[\varphi](f)=f\}}\,\theta^n_\varphi \leq \mathbb{1}_{\{\varphi=P_\theta[\varphi](f)\}}\,\theta^n_\varphi.\end{aligned}$$

Since $\varphi \leq P_\theta[\varphi](f)$, Lemma 2.4.2 gives the opposite inequality. The second equality follows. ◻

3.1.2 Properties of the P-envelope

Let $\theta, \theta_1, \theta_2$ be smooth closed real $(1,1)$-forms on X.

Proposition 3.1.9 *Let $\pi\colon Y \to X$ be a proper bimeromorphic morphism from a Kähler manifold Y to X. Then for any $\varphi \in \mathrm{PSH}(X,\theta)$, we have*

$$P_{\pi^*\theta}[\pi^*\varphi] = \pi^* P_\theta[\varphi].$$

In particular, a potential $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$ is model if and only if $\pi^*\varphi \in \mathrm{PSH}(Y,\pi^*\theta)_{>0}$ is model.

Proof This follows immediately from Proposition 1.5.3. □

We have the following concavity property of the P-envelope.

Proposition 3.1.10

(1) *Suppose that $\varphi \in \mathrm{PSH}(X,\theta)$ and $\lambda \in \mathbb{R}_{>0}$. Then*

$$P_{\lambda\theta}[\lambda\varphi] = \lambda P_\theta[\varphi].$$

(2) *Suppose that $\varphi_1 \in \mathrm{PSH}(X,\theta_1)$ and $\varphi_2 \in \mathrm{PSH}(X,\theta_2)$. Then*

$$P_{\theta_1+\theta_2}[\varphi_1+\varphi_2] \geq P_{\theta_1}[\varphi_1] + P_{\theta_2}[\varphi_2].$$

(3) *Suppose that $\varphi,\psi \in \mathrm{PSH}(X,\theta)$ and $\varphi \leq \psi$. Then*

$$P_\theta[\varphi] \leq P_\theta[\psi].$$

Proof (1) This follows by rescaling the candidates in the definition of the P-envelope.

(2) Suppose that $\psi_1 \in \mathrm{PSH}(X,\theta_1)$ and $\psi_2 \in \mathrm{PSH}(X,\theta_2)$ satisfy

$$\psi_i \leq 0, \quad \psi_i \leq \varphi_i$$

for $i = 1, 2$. Then

$$\psi_1+\psi_2 \leq 0, \quad \psi_1+\psi_2 \leq \varphi_1+\varphi_2.$$

It follows from (3.3) that

$$\psi_1+\psi_2 \leq P_{\theta_1+\theta_2}[\varphi_1+\varphi_2].$$

Since ψ_1 and ψ_2 are arbitrary, we conclude.

(3) This follows by inclusion of the defining candidate sets. □

Proposition 3.1.11 *Let $(\varphi_j)_{j\in I}$ be a decreasing net of potentials in $\mathrm{PSH}(X,\theta)$ satisfying $P_\theta[\varphi_j] = \varphi_j$ for each $j \in I$. Set $\varphi := \inf_{j\in I}\varphi_j$. Then $P_\theta[\varphi] = \varphi$.*

Proof Since $P_\theta[\varphi_j] = \varphi_j$, we have $\sup_X \varphi_j = 0$. By the L^1-compactness of normalized θ-psh functions (Proposition 1.5.1), the decreasing limit φ cannot be identically $-\infty$. It follows from Proposition 1.2.2 that $\varphi \in \mathrm{PSH}(X,\theta)$. Therefore, for each $j \in I$,

$$\varphi \leq P_\theta[\varphi] \leq P_\theta[\varphi_j] = \varphi_j.$$

Therefore, $\varphi = P_\theta[\varphi]$. □

Proposition 3.1.12 *Let $(\epsilon_i)_{i\in I}$ be a decreasing net in $\mathbb{R}_{\geq 0}$ with limit 0. Take a Kähler form ω on X. Consider a decreasing net $(\varphi_i)_{i\in I}$ with $\varphi_i \in \mathrm{PSH}(X, \theta+\epsilon_i\omega)$ satisfying*

$$P_{\theta+\epsilon_i\omega}[\varphi_i] = \varphi_i \tag{3.21}$$

for every $i \in I$. Set $\varphi := \inf_{i\in I} \varphi_i$. Then $P_\theta[\varphi] = \varphi$, and

$$\lim_{i\in I} \int_X (\theta + \epsilon_i\omega)^n_{\varphi_i} = \int_X \theta^n_\varphi. \tag{3.22}$$

Moreover, if $\int_X \theta^n_\varphi > 0$, then for any prime divisor E over X, we have

$$\lim_{i\in I} \nu(\varphi_i, E) = \nu(\varphi, E). \tag{3.23}$$

Note that both (3.22) and (3.23) fail without the assumption (3.21).

Proof We first observe that $\varphi \in \mathrm{PSH}(X, \theta)$. In fact, for each $i_0 \in I$, the tail $(\varphi_i)_{i\geq i_0}$ lies in $\mathrm{PSH}(X, \theta + \epsilon_{i_0}\omega)$ and decreases to φ. Since $\sup_X \varphi_i = 0$ by (3.21), the limit is not identically $-\infty$. Thus $\varphi \in \mathrm{PSH}(X, \theta + \epsilon_{i_0}\omega)$ for every $i_0 \in I$. Since $\epsilon_{i_0} \to 0$, it follows that $\varphi \in \mathrm{PSH}(X, \theta)$. For every $i \in I$ we have

$$P_\theta[\varphi] \leq P_{\theta+\epsilon_i\omega}[\varphi] \leq P_{\theta+\epsilon_i\omega}[\varphi_i] = \varphi_i.$$

Therefore, $P_\theta[\varphi] \leq \varphi$. The reverse inequality is trivial. Hence $P_\theta[\varphi] = \varphi$.

By the monotonicity theorem Theorem 2.4.3, we have

$$\varliminf_{i\in I} \int_X (\theta + \epsilon_i\omega)^n_{\varphi_i} \geq \varliminf_{i\in I} \int_X (\theta + \epsilon_i\omega)^n_{\varphi} = \int_X \theta^n_\varphi.$$

We now argue the reverse inequality.

Fix $i_0 \in I$. Since passing to the tail $i \geq i_0$ does not change the limsup, we have

$$\begin{aligned}\varlimsup_{i\in I} \int_X (\theta + \epsilon_i\omega)^n_{\varphi_i} &= \varlimsup_{i\geq i_0} \int_{\{\varphi_i=0\}} (\theta + \epsilon_i\omega)^n_{\varphi_i} \\ &\leq \varlimsup_{i\geq i_0} \int_{\{\varphi_i=0\}} \left(\theta + \epsilon_{i_0}\omega\right)^n_{\varphi_i} \\ &\leq \int_{\{\varphi=0\}} \left(\theta + \epsilon_{i_0}\omega\right)^n_{\varphi},\end{aligned}$$

where in the first line we used (3.21) and Theorem 3.1.1, and in the last line we used the fact that $\varphi_i \searrow \varphi$ on the tail $i \geq i_0$ and Corollary 2.4.2. Taking the limit with respect to i_0, we arrive at the desired conclusion:

$$\varlimsup_{i\in I} \int_X (\theta + \epsilon_i\omega)^n_{\varphi_i} \leq \varliminf_{i_0\in I} \int_{\{\varphi=0\}} \left(\theta + \epsilon_{i_0}\omega\right)^n_{\varphi} = \int_{\{\varphi=0\}} \theta^n_\varphi \leq \int_X \theta^n_\varphi.$$

This finishes the proof of (3.22).

It remains to argue (3.23). Fix $i_0 \in I$. By Proposition 1.5.2, there is a constant $B > 0$ such that

$$\nu(\psi, E) \leq B, \quad \psi \in \mathrm{PSH}(X, \theta + \epsilon_{i_0}\omega).$$

By Lemma 2.4.4 and (3.22), for any $\delta \in (0, 1)$ there exists $i_\delta \geq i_0$ such that for all $i \geq i_\delta$ there exists $\psi_i \in \mathrm{PSH}(X, \theta + \epsilon_i\omega)$ satisfying $(1-\delta)\varphi_i + \delta\psi_i \leq \varphi$. This implies that for $i \geq i_\delta$ we have

$$(1-\delta)\nu(\varphi_i, E) + \delta\nu(\psi_i, E) \geq \nu(\varphi, E) \geq \nu(\varphi_i, E).$$

Since $\psi_i \in \mathrm{PSH}(X, \theta + \epsilon_{i_0}\omega)$ for $i \geq i_\delta$, we get

$$\varliminf_{i \in I} \nu(\varphi_i, E) \geq \frac{\nu(\varphi, E) - \delta B}{1-\delta}.$$

Letting $\delta \searrow 0$ gives $\varliminf_{i\in I} \nu(\varphi_i, E) \geq \nu(\varphi, E)$. The reverse inequality follows from $\varphi \leq \varphi_i$, and hence we conclude (3.23). □

Corollary 3.1.3 *Let $\varphi \in \mathrm{PSH}(X, \theta)_{>0}$ be a model potential. Let ω be a Kähler form on X. Then*

$$\varphi = \inf_{\epsilon>0} P_{\theta+\epsilon\omega}[\varphi].$$

Proof Clearly, we have the $\leq$ direction, and the right-hand side is non-positive. So by Theorem 3.1.2, it suffices to show that they have the same mass, which follows from Proposition 3.1.12 applied to the decreasing net $(P_{\theta+\epsilon\omega}[\varphi])_{\epsilon\searrow 0}$, together with Proposition 3.1.3. □

Proposition 3.1.13 *Let $(\varphi_i)_{i\in I}$ be an increasing net of potentials in $\mathrm{PSH}(X, \theta)_{>0}$ uniformly bounded from above. Let $\varphi := \sup_{i\in I}{}^*\varphi_i$. Then*

$$\sup_{i\in I}{}^* P_\theta[\varphi_i] = P_\theta[\varphi].$$

In particular, if φ_i is model for all $i \in I$, then so is φ.

Proof We may assume that I is infinite since otherwise, there is nothing to prove.

We write

$$\eta := \sup_{i\in I}{}^* P_\theta[\varphi_i].$$

Then it is clear that $\eta \leq P_\theta[\varphi]$.

By Corollary 2.4.1, we have

$$\lim_{i\in I} \int_X \theta^n_{\varphi_i} = \int_X \theta^n_\varphi > 0.$$

So by Lemma 2.4.4, we can find a decreasing net $\epsilon_i \searrow 0$ ($i \in I$) with $\epsilon_i \in (0, 1)$ and $\psi_i \in \mathrm{PSH}(X, \theta)$ ($i \in I$) such that for all $i \in I$,

$$(1-\epsilon_i)\varphi + \epsilon_i\psi_i \leq \varphi_i.$$

By the concavity property of the envelope Proposition 3.1.10, we have

$$P_\theta[\varphi] + \epsilon_i P_\theta[\psi_i] \leq (1-\epsilon_i) P_\theta[\varphi] + \epsilon_i P_\theta[\psi_i] \leq \eta.$$

Each $P_\theta[\psi_i]$ has supremum 0, hence the family is bounded in L^1 by Proposition 1.5.1. Thus $\epsilon_i P_\theta[\psi_i] \to 0$ in L^1. The previous inequality then gives $P_\theta[\varphi] \leq \eta$ almost everywhere, and hence everywhere by Proposition 1.2.6. □

3.1.3 Relative full mass classes

Let θ be a smooth closed real $(1,1)$-form on X representing a big cohomology class. Fix a model potential $\phi \in \mathrm{PSH}(X,\theta)_{>0}$.

Definition 3.1.5 We define

$$\begin{aligned}
\mathrm{PSH}(X,\theta;\phi) &:= \{\eta \in \mathrm{PSH}(X,\theta) : \eta \leq \phi\}, \\
\mathcal{E}^\infty(X,\theta;\phi) &:= \{\eta \in \mathrm{PSH}(X,\theta) : \eta \sim \phi\}, \\
\mathcal{E}(X,\theta;\phi) &:= \left\{\eta \in \mathrm{PSH}(X,\theta;\phi) : \int_X \theta_\eta^n = \int_X \theta_\phi^n\right\}, \\
\mathcal{E}^1(X,\theta;\phi) &:= \left\{\eta \in \mathcal{E}(X,\theta;\phi) : \int_X |\phi-\eta|\,\theta_\eta^n < \infty\right\}.
\end{aligned}$$

Potentials in the last three classes are said to have *relatively minimal singularities*, *full mass* and *finite energy* relative to ϕ respectively.

We have the following inclusions:

$$\mathcal{E}^\infty(X,\theta;\phi) \subseteq \mathcal{E}^1(X,\theta;\phi) \subseteq \mathcal{E}(X,\theta;\phi) \subseteq \mathrm{PSH}(X,\theta;\phi). \tag{3.24}$$

The only non-trivial part is the first inclusion, which follows from the monotonicity theorem Theorem 2.4.3.

Remark 3.1.2 Note that this integral

$$\int_X |\phi-\eta|\,\theta_\eta^n$$

is defined: The locus where $\phi-\eta$ is undefined is a pluripolar set, while the product θ_η^n puts no mass on pluripolar sets (Proposition 2.2.1).

Similar remarks apply when we talk about similar integrals in the sequel.

When $\phi = V_\theta$, we usually write

$$\begin{aligned}
\mathcal{E}^\infty(X,\theta;V_\theta) &= \mathcal{E}^\infty(X,\theta), \\
\mathcal{E}(X,\theta;V_\theta) &= \mathcal{E}(X,\theta), \\
\mathcal{E}^1(X,\theta;V_\theta) &= \mathcal{E}^1(X,\theta).
\end{aligned}$$

Potentials in the three classes are said to have *minimal singularities*, *full mass* and *finite energy* respectively. The relation (3.24) can be written as

$$\mathcal{E}^\infty(X,\theta) \subseteq \mathcal{E}^1(X,\theta) \subseteq \mathcal{E}(X,\theta)$$

in this case.

The P-envelope can be used to characterize the full mass classes:

Proposition 3.1.14 *Let $\varphi \in \mathrm{PSH}(X,\theta)$. Then the following are equivalent:*

(1) $\varphi \in \mathcal{E}(X,\theta;\phi)$;
(2) $P_\theta[\varphi] = \phi$.

Proof (2) $\implies$ (1). This follows from Proposition 3.1.3.

(1) $\implies$ (2). By (1), $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$. Note that ϕ is a candidate of $P_\theta[\varphi]$ as in (3.7). So $P_\theta[\varphi] \geq \phi$. The reverse inequality is trivial. □

We have the following comparison principle.

Proposition 3.1.15 (Comparison principle) *Fix $j \in \{0,\dots,n\}$. Let $\theta_1,\dots,\theta_j$ be closed smooth real $(1,1)$-forms on X and $\psi_i \in \mathrm{PSH}(X,\theta_i)$ for $i = 1,\dots,j$. Suppose that $\varphi,\psi \in \mathcal{E}(X,\theta;\phi)$. Then*

$$\int_{\{\varphi<\psi\}} \theta_\varphi^{n-j} \wedge \theta_{1,\psi_1} \wedge \cdots \wedge \theta_{j,\psi_j} \geq \int_{\{\varphi<\psi\}} \theta_\psi^{n-j} \wedge \theta_{1,\psi_1} \wedge \cdots \wedge \theta_{j,\psi_j}. \tag{3.25}$$

As a consequence, we also have

$$\int_{\{\varphi\geq\psi\}} \theta_\varphi^{n-j} \wedge \theta_{1,\psi_1} \wedge \cdots \wedge \theta_{j,\psi_j} \leq \int_{\{\varphi\geq\psi\}} \theta_\psi^{n-j} \wedge \theta_{1,\psi_1} \wedge \cdots \wedge \theta_{j,\psi_j}.$$

Proof Observe that $\varphi \vee \psi \in \mathcal{E}(X,\theta;\phi)$, as a consequence of Theorem 2.4.3. We compute

$$\begin{aligned}
&\int_X \theta_{\varphi\vee\psi}^{n-j} \wedge \theta_{1,\psi_1} \wedge \cdots \wedge \theta_{j,\psi_j}\\
\geq& \int_{\{\varphi>\psi\}} \theta_\varphi^{n-j} \wedge \theta_{1,\psi_1} \wedge \cdots \wedge \theta_{j,\psi_j} + \int_{\{\varphi<\psi\}} \theta_\psi^{n-j} \wedge \theta_{1,\psi_1} \wedge \cdots \wedge \theta_{j,\psi_j}\\
=& \int_X \theta_\varphi^{n-j} \wedge \theta_{1,\psi_1} \wedge \cdots \wedge \theta_{j,\psi_j} - \int_{\{\varphi\leq\psi\}} \theta_\varphi^{n-j} \wedge \theta_{1,\psi_1} \wedge \cdots \wedge \theta_{j,\psi_j}\\
&+ \int_{\{\varphi<\psi\}} \theta_\psi^{n-j} \wedge \theta_{1,\psi_1} \wedge \cdots \wedge \theta_{j,\psi_j}\\
=& \int_X \theta_{\varphi\vee\psi}^{n-j} \wedge \theta_{1,\psi_1} \wedge \cdots \wedge \theta_{j,\psi_j} - \int_{\{\varphi\leq\psi\}} \theta_\varphi^{n-j} \wedge \theta_{1,\psi_1} \wedge \cdots \wedge \theta_{j,\psi_j}\\
&+ \int_{\{\varphi<\psi\}} \theta_\psi^{n-j} \wedge \theta_{1,\psi_1} \wedge \cdots \wedge \theta_{j,\psi_j},
\end{aligned}$$

where in the last step, we applied Proposition 3.1.3. Therefore,

$$\int_{\{\varphi\leq\psi\}} \theta_\varphi^{n-j}\wedge\theta_{1,\psi_1}\wedge\cdots\wedge\theta_{j,\psi_j} \geq \int_{\{\varphi<\psi\}} \theta_\psi^{n-j}\wedge\theta_{1,\psi_1}\wedge\cdots\wedge\theta_{j,\psi_j}.$$

Next we replace φ by $\varphi+\epsilon$ and let $\epsilon\searrow 0$, then we conclude (3.25). □

The full mass potentials are essential in resolving the Monge–Ampère equations. We recall the following two Aubin–Yau type theorems.

Theorem 3.1.4 *Let μ be a non-pluripolar measure on X with $\mu(X)=\int_X\theta_\phi^n$. Then there is a unique $\varphi\in\mathrm{PSH}(X,\theta;\phi)$ such that*

$$\theta_\varphi^n=\mu,\quad \sup_X\varphi=0.$$

Recall that a measure μ on X is *non-pluripolar* if it is a Radon measure and $\mu(K)=0$ for each pluripolar set $K\subseteq X$.[2]

Theorem 3.1.5 *Fix $\lambda>0$. Let μ be a non-pluripolar measure on X with $\mu(X)>0$. Then there is a unique $\varphi\in\mathcal{E}(X,\theta;\phi)$ such that*

$$\theta_\varphi^n=\mathrm{e}^{\lambda\varphi}\mu.$$

Furthermore, for any $\psi\in\mathcal{E}(X,\theta;\phi)$ satisfying

$$\theta_\psi^n\geq\mathrm{e}^{\lambda\psi}\mu,$$

we have $\varphi\geq\psi$.

The proofs are beyond the scope of this book; we refer to [DDNL21a].

In order to handle the finite energy classes, it is convenient to introduce the following quantity:

Definition 3.1.6 We define the *Monge–Ampère energy* $E_\theta^\phi\colon\mathcal{E}^\infty(X,\theta;\phi)\to\mathbb{R}$ as follows

$$E_\theta^\phi(\varphi)\coloneqq\frac{1}{n+1}\sum_{j=0}^n\int_X(\varphi-\phi)\,\theta_\varphi^j\wedge\theta_\phi^{n-j}. \tag{3.26}$$

More generally, we extend E_θ^ϕ to a functional $E_\theta^\phi\colon\mathrm{PSH}(X,\theta;\phi)\to[-\infty,\infty)$ as follows

$$E_\theta^\phi(\varphi)\coloneqq\inf\left\{E_\theta^\phi(\psi):\psi\in\mathcal{E}^\infty(X,\theta;\phi),\varphi\leq\psi\right\}. \tag{3.27}$$

We write E_θ instead of E_θ^ϕ when $\phi=V_\theta$.

Note that

$$E_\theta^\phi(\varphi+C)=E_\theta^\phi(\varphi)+C\int_X\theta_\phi^n \tag{3.28}$$

for any $\varphi\in\mathrm{PSH}(X,\theta;\phi)$ and $C\in\mathbb{R}$.

[2] Here we identify as usual a Radon measure with its completion.

Lemma 3.1.3 *The functional $E_\theta^\phi\colon \mathcal{E}^\infty(X,\theta;\phi)\to\mathbb{R}$ is increasing. In particular, the extended definition (3.27) agrees with (3.26) on $\mathcal{E}^\infty(X,\theta;\phi)$, and is increasing as well.*

Proof Let $\varphi,\psi\in\mathcal{E}^\infty(X,\theta;\phi)$, we have

$$
\begin{aligned}
&(n+1)E_\theta^\phi(\varphi)-(n+1)E_\theta^\phi(\psi)-\sum_{j=0}^n\int_X(\varphi-\psi)\,\theta_\varphi^j\wedge\theta_\psi^{n-j}\\
=&\sum_{j=0}^n\int_X(\varphi-\phi)\,\theta_\varphi^j\wedge\theta_\phi^{n-j}-\sum_{j=0}^n\int_X(\psi-\phi)\,\theta_\psi^j\wedge\theta_\phi^{n-j}-\sum_{j=0}^n\int_X(\varphi-\psi)\,\theta_\varphi^j\wedge\theta_\psi^{n-j}\\
=&\sum_{j=0}^n\int_X(\varphi-\psi)\,\left(\theta_\varphi^j\wedge\theta_\phi^{n-j}-\theta_\varphi^j\wedge\theta_\psi^{n-j}\right)\\
&+\sum_{j=0}^n\int_X(\psi-\phi)\,\left(\theta_\varphi^j\wedge\theta_\phi^{n-j}-\theta_\psi^j\wedge\theta_\phi^{n-j}\right)\\
=&\sum_{\substack{j+a+b=n-1,\\ j,a,b\geq 0}}(n-j)\int_X(\psi-\phi)\,\left(\theta_\phi^a\wedge\theta_\psi^{b+1}\wedge\theta_\varphi^j-\theta_\phi^a\wedge\theta_\psi^b\wedge\theta_\varphi^{j+1}\right)\\
&+\sum_{\substack{j+a+b=n-1,\\ j,a,b\geq 0}}(n-j)\int_X(\psi-\phi)\,\left(\theta_\phi^j\wedge\theta_\varphi^{a+1}\wedge\theta_\psi^b-\theta_\phi^j\wedge\theta_\varphi^a\wedge\theta_\psi^{b+1}\right)\\
=&0,
\end{aligned}
$$

where the third equality follows from the integration by parts formula. See [Xia19, Lu21] for the proof in the context of non-pluripolar products.

In other words,

$$
E_\theta^\phi(\varphi)-E_\theta^\phi(\psi)=\frac{1}{n+1}\sum_{j=0}^n\int_X(\varphi-\psi)\,\theta_\varphi^j\wedge\theta_\psi^{n-j}. \tag{3.29}
$$

The monotonicity follows. □

Lemma 3.1.4 *Let $\varphi,\psi\in\mathcal{E}^\infty(X,\theta;\phi)$. Then*

$$
\int_X(\varphi-\psi)\,\theta_\psi^n\geq E_\theta^\phi(\varphi)-E_\theta^\phi(\psi)\geq\int_X(\varphi-\psi)\,\theta_\varphi^n. \tag{3.30}
$$

Proof Thanks to (3.28), we may assume that $\varphi\geq\psi$.

As we have seen in (3.29), the middle term can be written as

$$
\frac{1}{n+1}\sum_{j=0}^n\int_X(\varphi-\psi)\,\theta_\varphi^j\wedge\theta_\psi^{n-j}.
$$

We claim that for each individual j, the inequality (3.30) holds:

$$\int_X (\varphi-\psi)\,\theta_\psi^n \geq \int_X (\varphi-\psi)\,\theta_\varphi^j\wedge\theta_\psi^{n-j} \geq \int_X (\varphi-\psi)\,\theta_\varphi^n.$$

Thanks to Proposition 3.1.15, we have

$$\begin{aligned}\int_X (\varphi-\psi)\,\theta_\varphi^j\wedge\theta_\psi^{n-j} &= \int_0^\infty \int_{\{\varphi>\psi+t\}} \theta_\varphi^j\wedge\theta_\psi^{n-j}\,\mathrm{d}t\\ &\geq \int_0^\infty \int_{\{\varphi>\psi+t\}} \theta_\varphi^n\,\mathrm{d}t\\ &= \int_X (\varphi-\psi)\,\theta_\varphi^n.\end{aligned}$$

The other inequality is similar. ◻

Lemma 3.1.5 *The function $E_\theta^\phi\colon \mathcal{E}^\infty(X,\theta;\phi)\to\mathbb{R}$ is concave.*

Proof Let $\varphi_0,\varphi_1\in\mathcal{E}^\infty(X,\theta;\phi)$. We write $\varphi_t=t\varphi_1+(1-t)\varphi_0$ for each $t\in(0,1)$.

We claim that for each $t\in[0,1]$, we have

$$\frac{\mathrm{d}}{\mathrm{d}t}E_\theta^\phi(\varphi_t)=\int_X (\varphi_1-\varphi_0)\;\theta_{\varphi_t}^n. \tag{3.31}$$

At the boundary $t=0,1$, the derivative is understood as the one-side derivative.

By symmetry, it suffices to compute the right-derivative for $t\in[0,1)$. It clearly suffices to handle the case $t=0$. In this case we compute for any small $s>0$ that

$$\begin{aligned}(n+1)\left(E_\theta^\phi(\varphi_s)-E_\theta^\phi(\varphi_0)\right)&=\sum_{j=0}^n s\int_X(\varphi_1-\varphi_0)\,\theta_{\varphi_s}^j\wedge\theta_{\varphi_0}^{n-j}\\ &=s\sum_{j=0}^n\int_X(\varphi_1-\varphi_0)\,\theta_{\varphi_0}^n+O(s^2).\end{aligned}$$

Therefore, (3.31) follows.

Now it suffices to show that the derivative (3.31) is decreasing in t. Take $0\leq s<t\leq 1$. We wish to show that

$$\int_X(\varphi_1-\varphi_0)\,\theta_{\varphi_t}^n\leq\int_X(\varphi_1-\varphi_0)\,\theta_{\varphi_s}^n.$$

This can be equivalently written as

$$\int_X(\varphi_t-\varphi_s)\,\theta_{\varphi_t}^n\leq\int_X(\varphi_t-\varphi_s)\,\theta_{\varphi_s}^n.$$

In particular, replacing φ_0 by φ_s and φ_1 by φ_t, we are reduced to proving the following

$$\int_X(\varphi_1-\varphi_0)\,\theta_{\varphi_1}^n\leq\int_X(\varphi_1-\varphi_0)\,\theta_{\varphi_0}^n.$$

But this is just a consequence of Lemma 3.1.4. □

Proposition 3.1.16 *Let $(\varphi_i)_{i>0}$ be a decreasing sequence in* $\mathrm{PSH}(X,\theta;\phi)$ *with limit $\varphi \not\equiv -\infty$. Then*

$$\lim_{i\to\infty} E^{\phi}_{\theta}(\varphi_i) = E^{\phi}_{\theta}(\varphi). \tag{3.32}$$

Proof Thanks to Lemma 3.1.3, we know that $E^{\phi}_{\theta}(\varphi_i)$ is decreasing in i. So the limit in (3.32) exists and the $\geq$ inequality holds. Conversely, let $\psi \in \mathcal{E}^{\infty}(X,\theta;\phi)$ and $\varphi \leq \psi$. We need to show that

$$E^{\phi}_{\theta}(\psi) \geq \lim_{i\to\infty} E^{\phi}_{\theta}\left(\varphi_i \vee \psi\right).$$

Replacing the sequence $(\varphi_i)_i$ by $(\varphi_i \vee \psi)_i$, we are reduced to proving the $\leq$ direction of (3.32) when $\varphi \in \mathcal{E}^{\infty}(X,\theta;\phi)$.

In this case, thanks to Lemma 3.1.4, we have

$$E^{\phi}_{\theta}(\varphi_i) - E^{\phi}_{\theta}(\varphi) \leq \int_X (\varphi_i - \varphi)\,\theta^n_{\varphi}.$$

Our assertion follows from the monotone convergence theorem. □

Proposition 3.1.17 *Let $\varphi \in \mathcal{E}(X,\theta;\phi)$. The following are equivalent:*

(1) *$\varphi \in \mathcal{E}^1(X,\theta;\phi)$;*
(2) *$E^{\phi}_{\theta}(\varphi) > -\infty$.*

Proof Fix $\varphi \in \mathcal{E}(X,\theta;\phi)$. Without loss of generality, we may assume that $\varphi \leq 0$. In particular, $\varphi \leq \phi$.

(2) $\implies$ (1). Assume (2). Observe that $\mathbb{1}_{\{\varphi>\phi-C\}}\theta^n_{\varphi\vee(\phi-C)} = \mathbb{1}_{\{\varphi>\phi-C\}}\theta^n_{\varphi}$ converges *strongly*[3] to θ^n_{φ} as $C \to \infty$ by the dominated convergence theorem. Therefore, for a fixed $D > 0$,

$$\begin{aligned}
\int_X \Big(\varphi \vee (\phi - D) - \phi\Big)\,\theta^n_{\varphi} &= \lim_{C\to\infty} \int_{\{\varphi>\phi-C\}} \Big(\varphi \vee (\phi - D) - \phi\Big)\,\theta^n_{\varphi\vee(\phi-C)} \\
&\geq \varlimsup_{C\to\infty} \int_{\{\varphi>\phi-C\}} \Big(\varphi \vee (\phi - C) - \phi\Big)\,\theta^n_{\varphi\vee(\phi-C)} \\
&\geq \varlimsup_{C\to\infty} \int_X \Big(\varphi \vee (\phi - C) - \phi\Big)\,\theta^n_{\varphi\vee(\phi-C)} \\
&\geq \varlimsup_{C\to\infty} (n+1)E^{\phi}_{\theta}\Big(\varphi \vee (\phi - C)\Big) \\
&\geq (n+1)E^{\phi}_{\theta}(\varphi).
\end{aligned}$$

Here the last but one inequality follows from (3.26) and the fact that $\varphi \vee (\phi - C) \leq \phi$. Letting $D \to \infty$, we conclude (1).

(1) $\implies$ (2). Assume (1). Thanks to Lemma 3.1.4, we know that for each $C > 0$,

[3] Namely, the convergence holds after integrating against any L^{∞} function.

$$
\begin{aligned}
E_\theta^\phi\Big(\varphi\vee(\phi-C)\Big) &\geq \int_X \Big(\varphi\vee(\phi-C)-\phi\Big)\,\theta^n_{\varphi\vee(\phi-C)}\\
&= \int_{\{\varphi>\phi-C\}}\Big(\varphi-\phi\Big)\,\theta^n_{\varphi\vee(\phi-C)} + \int_{\{\varphi\leq\phi-C\}}(-C)\,\theta^n_{\varphi\vee(\phi-C)}\\
&= \int_{\{\varphi+C>\phi\}}(\varphi+C-\phi)\ \theta^n_\varphi - C\int_X\theta^n_\phi\\
&\geq \int_X(\varphi+C-\phi)\ \theta^n_\varphi - C\int_X\theta^n_\phi\\
&= \int_X(\varphi-\phi)\,\theta^n_\varphi.
\end{aligned}
$$

Due to Proposition 3.1.16, we have

$$
\lim_{C\to\infty} E_\theta^\phi\Big(\varphi\vee(\phi-C)\Big) = E_\theta^\phi(\varphi),
$$

so (2) holds. □

Corollary 3.1.4 *The function* $E_\theta^\phi\colon \mathrm{PSH}(X,\theta;\phi)\to[-\infty,\infty)$ *is concave.*

Proof Let $\varphi,\psi\in\mathrm{PSH}(X,\theta;\phi)$ and $t\in(0,1)$. We need to show that

$$
E_\theta^\phi\Big(t\varphi+(1-t)\psi\Big) \geq tE_\theta^\phi(\varphi)+(1-t)E_\theta^\phi(\psi). \tag{3.33}
$$

For this purpose, observe that when $C\to\infty$, we have

$$
t\Big(\varphi\vee(\phi-C)\Big)+(1-t)\Big(\psi\vee(\phi-C)\Big)\searrow t\varphi+(1-t)\psi.
$$

By Lemma 3.1.5, for each $C>0$, we have

$$
E_\theta^\phi\bigg(t\Big(\varphi\vee(\phi-C)\Big)+(1-t)\Big(\psi\vee(\phi-C)\Big)\bigg) \geq tE_\theta^\phi\Big(\varphi\vee(\phi-C)\Big)+(1-t)E_\theta^\phi\Big(\psi\vee(\phi-C)\Big).
$$

Letting $C\to\infty$, we conclude (3.33) using Proposition 3.1.16. □

Corollary 3.1.5 *The sets* $\mathrm{PSH}(X,\theta;\phi)$*,* $\mathcal{E}(X,\theta;\phi)$*,* $\mathcal{E}^1(X,\theta;\phi)$*,* $\mathcal{E}^\infty(X,\theta;\phi)$ *are all convex.*

Proof The sets $\mathrm{PSH}(X,\theta;\phi)$ and $\mathcal{E}^\infty(X,\theta;\phi)$ are trivially convex.

We next handle $\mathcal{E}(X,\theta;\phi)$. In this case, suppose that $\varphi,\psi\in\mathcal{E}(X,\theta;\phi)$ and $t\in(0,1)$. Then we compute

$$
\begin{aligned}
\int_X\theta^n_{t\varphi+(1-t)\psi} &= \sum_{j=0}^n\binom{n}{j}t^j(1-t)^{n-j}\int_X\theta^j_\varphi\wedge\theta^{n-j}_\psi\\
&= \sum_{j=0}^n\binom{n}{j}t^j(1-t)^{n-j}\int_X\theta^n_\phi = \int_X\theta^n_\phi,
\end{aligned}
$$

where we applied Proposition 3.1.3 in the second equality. Therefore, $t\varphi + (1-t)\psi \in \mathcal{E}(X,\theta;\phi)$.

Finally, let us consider $\mathcal{E}^1(X,\theta;\phi)$. In this case, suppose that $\varphi,\psi \in \mathcal{E}^1(X,\theta;\phi)$ and $t \in (0,1)$. We have already shown that $t\varphi + (1-t)\psi \in \mathcal{E}(X,\theta;\phi)$. Furthermore, the energy of this potential is finite due to Corollary 3.1.4. Therefore, $t\varphi + (1-t)\psi \in \mathcal{E}^1(X,\theta;\phi)$ in view of Proposition 3.1.17. □

Lemma 3.1.6 *Let $\varphi,\psi,\gamma \in \mathcal{E}(X,\theta;\phi)$. Assume that $\gamma \geq \varphi \vee \psi$. Then*

$$\int_X (\gamma-\varphi)\,\theta_\psi^n \leq 2\int_X (\gamma-\varphi)\,\theta_\varphi^n + 2\int_X (\gamma-\psi)\,\theta_\psi^n.$$

Proof Observe that

$$\{\gamma > \varphi + 2t\} \subseteq \{\gamma > \psi + t\} \cup \{\psi > \varphi + t\}\,. \tag{3.34}$$

So

$$\begin{aligned}
\int_X (\gamma-\varphi)\,\theta_\psi^n =& 2\int_0^\infty \int_{\{\gamma>\varphi+2t\}} \theta_\psi^n \,\mathrm{d}t \\
\leq& 2\int_0^\infty \int_{\{\gamma>\psi+t\}} \theta_\psi^n \,\mathrm{d}t + 2\int_0^\infty \int_{\{\psi>\varphi+t\}} \theta_\psi^n \,\mathrm{d}t \\
\leq& 2\int_X (\gamma-\psi)\,\theta_\psi^n + 2\int_0^\infty \int_{\{\psi>\varphi+t\}} \theta_\varphi^n \,\mathrm{d}t \\
\leq& 2\int_X (\gamma-\psi)\,\theta_\psi^n + 2\int_0^\infty \int_{\{\gamma>\varphi+t\}} \theta_\varphi^n \,\mathrm{d}t \\
=& 2\int_X (\gamma-\psi)\,\theta_\psi^n + 2\int_X (\gamma-\varphi)\,\theta_\varphi^n,
\end{aligned}$$

where the second line follows from (3.34), while the third line follows from Proposition 3.1.15. □

Proposition 3.1.18 *Given $\varphi,\psi \in \mathcal{E}^1(X,\theta;\phi)$, we have the following cocycle equality*

$$E_\theta^\phi(\psi) - E_\theta^\phi(\varphi) = \frac{1}{n+1}\sum_{j=0}^n \int_X (\psi-\varphi)\,\theta_\psi^j \wedge \theta_\varphi^{n-j}. \tag{3.35}$$

As a consequence of (3.35), for $\varphi \in \mathcal{E}^1(X,\theta;\phi)$, (3.26) continues to hold.

Proof Since the conclusion is known when $\varphi,\psi \in \mathcal{E}^\infty(X,\theta;\phi)$ (see (3.29)), it suffices to prove that for each $k=0,\dots,n$, we have

$$\lim_{C\to\infty}\int_X \Big(\varphi\vee(\phi-C) - \psi\vee(\phi-C)\Big)\,\theta_{\varphi\vee(\phi-C)}^k \wedge \theta_{\psi\vee(\phi-C)}^{n-k} = \int_X (\varphi-\psi)\,\theta_\varphi^k \wedge \theta_\psi^{n-k}. \tag{3.36}$$

For this purpose, we may assume that $\varphi,\psi \leq \phi$ and it suffices to establish the following:

$$\lim_{C\to\infty}\int_X \Big(\varphi\vee(\phi-C)-\phi\Big)\,\theta^k_{\varphi\vee(\phi-C)}\wedge\theta^{n-k}_{\psi\vee(\phi-C)}=\int_X(\varphi-\phi)\,\theta^k_\varphi\wedge\theta^{n-k}_\psi. \tag{3.37}$$

Fix $C>0$, we compute the difference as follows:

$$\begin{aligned}
&\int_X \Big(\varphi\vee(\phi-C)-\phi\Big)\,\theta^k_{\varphi\vee(\phi-C)}\wedge\theta^{n-k}_{\psi\vee(\phi-C)}-\int_X(\varphi-\phi)\,\theta^k_\varphi\wedge\theta^{n-k}_\psi\\
=&\int_{\{\min\{\varphi,\psi\}>\phi-C\}}(\varphi-\phi)\ \theta^k_\varphi\wedge\theta^{n-k}_\psi-\int_X(\varphi-\phi)\,\theta^k_\varphi\wedge\theta^{n-k}_\psi\\
&+\int_{\{\min\{\varphi,\psi\}\leq\phi-C\}}\Big(\varphi\vee(\phi-C)-\phi\Big)\,\theta^k_{\varphi\vee(\phi-C)}\wedge\theta^{n-k}_{\psi\vee(\phi-C)}\\
=&-\int_{\{\min\{\varphi,\psi\}\leq\phi-C\}}(\varphi-\phi)\ \theta^k_\varphi\wedge\theta^{n-k}_\psi+\\
&+\int_{\{\min\{\varphi,\psi\}\leq\phi-C\}}\Big(\varphi\vee(\phi-C)-\phi\Big)\,\theta^k_{\varphi\vee(\phi-C)}\wedge\theta^{n-k}_{\psi\vee(\phi-C)}
\end{aligned}$$

We first handle the first term. In order to show that it converges to 0 when $C\to\infty$, it suffices to show that

$$\int_X(\phi-\varphi)\ \theta^k_\varphi\wedge\theta^{n-k}_\psi<\infty.$$

For this purpose, it suffices to prove a stronger inequality:

$$\int_X(\phi-\varphi)\ \theta^n_{(\varphi+\psi)/2}<\infty. \tag{3.38}$$

From Corollary 3.1.5, we know that $(\varphi+\psi)/2\in\mathcal{E}^1(X,\theta;\phi)$. Therefore, (3.38) follows from Lemma 3.1.6.

It remains to handle the second term. It then suffices to prove the following:

$$\begin{aligned}
\lim_{C\to\infty}C\int_{\{\varphi\leq\phi-C\}}\theta^k_{\varphi\vee(\phi-C)}\wedge\theta^{n-k}_{\psi\vee(\phi-C)}&=0,\\
\lim_{C\to\infty}C\int_{\{\psi\leq\phi-C\}}\theta^k_{\varphi\vee(\phi-C)}\wedge\theta^{n-k}_{\psi\vee(\phi-C)}&=0.
\end{aligned} \tag{3.39}$$

By symmetry, it suffices to handle the first. Observe the following inclusions:

$$\{\varphi\leq\phi-C\}\subseteq\left\{\varphi\vee(\phi-C)\leq\frac{1}{2}\Big(\psi\vee(\phi-C)+\phi-C\Big)\right\}\subseteq\{\varphi\leq\phi-C/2\}.$$

By Proposition 3.1.15, we find

$$
\begin{aligned}
&C\int_{\{\varphi\le\phi-C\}}\theta^k_{\varphi\vee(\phi-C)}\wedge\theta^{n-k}_{\psi\vee(\phi-C)}\\
\le&C\int_{\left\{\varphi\vee(\phi-C)\le\frac{1}{2}\left(\psi\vee(\phi-C)+\phi-C\right)\right\}}\theta^k_{\varphi\vee(\phi-C)}\wedge\theta^{n-k}_{\psi\vee(\phi-C)}\\
\le&2^{n-k}C\int_{\left\{\varphi\vee(\phi-C)\le\frac{1}{2}\left(\psi\vee(\phi-C)+\phi-C\right)\right\}}\theta^k_{\varphi\vee(\phi-C)}\wedge\theta^{n-k}_{\frac{1}{2}\left(\psi\vee(\phi-C)+\phi-C\right)}\\
\le&2^{n-k}C\int_{\{\varphi\vee(\phi-C)\le\frac{1}{2}\left(\psi\vee(\phi-C)+\phi-C\right)\}}\theta^n_{\varphi\vee(\phi-C)}\\
\le&2^{n-k}C\int_{\{\varphi\le\phi-C/2\}}\theta^n_{\varphi\vee(\phi-C)}\\
=&2^{n-k}C\int_{\{\varphi\le\phi-C/2\}}\theta^n_{\varphi}.
\end{aligned}
$$

Here on the final step, we have applied the following argument:

$$
\begin{aligned}
&\int_{\{\varphi\le\phi-C/2\}}\theta^n_{\varphi\vee(\phi-C)}\\
=&\int_X\theta^n_{\varphi\vee(\phi-C)}-\int_{\{\varphi>\phi-C/2\}}\theta^n_{\varphi\vee(\phi-C)}\\
=&\int_X\theta^n_{\varphi}-\int_{\{\varphi>\phi-C/2\}}\theta^n_{\varphi} && \text{by Proposition 2.2.1 (1)}\\
=&\int_{\{\varphi\le\phi-C/2\}}\theta^n_{\varphi}.
\end{aligned}
$$

On the other hand,

$$
\varlimsup_{C\to\infty}C\int_{\{\varphi\le\phi-C/2\}}\theta^n_\varphi\le 2\varlimsup_{C\to\infty}\int_{\{\varphi\le\phi-C\}}(\phi-\varphi)\,\theta^n_\varphi=0,
$$

since $\varphi\in\mathcal{E}^1(X,\theta;\phi)$. We conclude (3.39). □

Proposition 3.1.19 *Let $\varphi,\psi\in\mathcal{E}^1(X,\theta;\phi)$. Then*

$$
\int_X(\psi-\varphi)\,\theta^n_\psi\le E^\phi_\theta(\psi)-E^\phi_\theta(\varphi)\le\int_X(\psi-\varphi)\,\theta^n_\varphi. \tag{3.40}
$$

Proof Thanks to (3.36), this can be reduced to the case where $\varphi,\psi\in\mathcal{E}^\infty(X,\theta;\phi)$, which is established in Lemma 3.1.4. □

Proposition 3.1.20 *Let $(\varphi_j)_{j\ge1}$ be a sequence in $\mathcal{E}^1(X,\theta;\phi)$. Assume that $\varphi\in\mathcal{E}^1(X,\theta;\phi)$ and either one of the following conditions holds:*

(1) *$(\varphi_j)_j$ is a decreasing sequence with limit φ;*
(2) *$(\varphi_j)_j$ is increasing with almost everywhere limit φ.*

Then

$$\lim_{j\to\infty} E_\theta^\phi(\varphi_j) = E_\theta^\phi(\varphi). \tag{3.41}$$

Proof The decreasing case is already proved in Proposition 3.1.16, so let us focus on the case where $(\varphi_j)_j$ is increasing.

Step 1. We first prove the result when $\varphi_j, \varphi \in \mathcal{E}^\infty(X, \theta; \phi)$ for each $j \geq 1$.

In fact, in this case, by (3.29), we have

$$\begin{aligned} 0 \leq & E_\theta^\phi(\varphi) - E_\theta^\phi(\varphi_j) \\ = & \frac{1}{n+1} \sum_{k=0}^n \int_X (\varphi - \varphi_j)\, \theta_\varphi^k \wedge \theta_{\varphi_j}^{n-k}. \end{aligned}$$

We then apply Theorem 2.4.2 to conclude that the right-hand side converges to 0.

Step 2. We prove the general case. The $\leq$ direction in (3.41) follows from the monotonicity of E_θ^ϕ, as proved in Lemma 3.1.3. It remains to prove the $\geq$ direction.

After subtracting a constant from all φ_j and from φ, which shifts all the energies by the same constant in view of (3.28), we may assume that $\varphi \leq \phi$. For each $C > 0$ and $j \geq 1$, we let

$$\varphi_C \coloneqq \varphi \vee (\phi - C), \quad \varphi_{j,C} \coloneqq \varphi_j \vee (\phi - C).$$

By Step 1, for each $C > 0$, we have

$$\lim_{j\to\infty} E_\theta^\phi(\varphi_{j,C}) = E_\theta^\phi(\varphi_C) \geq E_\theta^\phi(\varphi).$$

It suffices therefore to show that

$$\lim_{C\to\infty} E_\theta^\phi(\varphi_{j,C}) - E_\theta^\phi(\varphi_j) = 0 \quad \text{uniformly for } j \text{ in a tail}. \tag{3.42}$$

Fix $i \geq 1$. For $j \geq i$, we compute

$$
\begin{aligned}
&E^{\phi}_{\theta}(\varphi_{j,C}) - E^{\phi}_{\theta}(\varphi_j) \\
\leq &\int_X \left(\varphi_{j,C} - \varphi_j\right)\, \theta^n_{\varphi_j} && \text{by Proposition 3.1.19} \\
\leq &\int_{\{\varphi_j \leq \phi - C\}} (\phi - C - \varphi_j)\, \theta^n_{\varphi_j} \\
= &\int_C^{\infty} \int_{\{\varphi_j \leq \phi - t\}} \theta^n_{\varphi_j}\, \mathrm{d}t \\
\leq &\int_C^{\infty} \int_{\{\varphi_i \leq (\varphi_j + \phi - t)/2\}} \theta^n_{\varphi_j}\, \mathrm{d}t \\
\leq 2^n &\int_C^{\infty} \int_{\{\varphi_i \leq (\varphi_j + \phi - t)/2\}} \theta^n_{(\varphi_j + \phi - t)/2}\, \mathrm{d}t \\
\leq 2^n &\int_C^{\infty} \int_{\{\varphi_i \leq (\varphi_j + \phi - t)/2\}} \theta^n_{\varphi_i}\, \mathrm{d}t && \text{by Proposition 3.1.15} \\
\leq 2^n &\int_C^{\infty} \int_{\{\varphi_i \leq \phi - t/2\}} \theta^n_{\varphi_i}\, \mathrm{d}t \\
= 2^{n+1} &\int_{\{\varphi_i < \phi - C/2\}} (\phi - \varphi_i - C/2)\, \theta^n_{\varphi_i} \\
= 2^{n+1} &\int_{\{\varphi_i < \phi - C/2\}} (\phi - \varphi_i)\, \theta^n_{\varphi_i} - C 2^n \int_{\{\varphi_i < \phi - C/2\}} \theta^n_{\varphi_i}.
\end{aligned}
$$

As $C \to \infty$, the first term converges to 0 since $\varphi_i \in \mathcal{E}^1(X, \theta; \phi)$. As for the second term,

$$\lim_{C \to \infty} C \int_{\{\varphi_i < \phi - C/2\}} \theta^n_{\varphi_i} \leq 2 \lim_{C \to \infty} \int_{\{\varphi_i < \phi - C/2\}} (\phi - \varphi_i)\, \theta^n_{\varphi_i} = 0.$$

We conclude (3.42). □

Next we want to prove that $\mathcal{E}^1(X, \theta; \phi)$ is closed under the rooftop operator. For this purpose, we shall need a few preliminary results.

We shall approximate the rooftop operator by solutions of certain Monge–Ampère equations.

Lemma 3.1.7 *Let $\varphi, \psi \in \mathcal{E}^{\infty}(X, \theta; \phi)$. Then there is $\gamma \in \mathcal{E}^{\infty}(X, \theta; \phi)$ such that*

$$\theta^n_{\gamma} = \mathrm{e}^{\gamma - \varphi}\, \theta^n_{\varphi} + \mathrm{e}^{\gamma - \psi}\, \theta^n_{\psi}. \tag{3.43}$$

It is not clear if $\int_X \mathrm{e}^{-\varphi} \theta^n_{\varphi} < \infty$, hence we cannot say that $\mathrm{e}^{-\varphi} \theta^n_{\varphi} + \mathrm{e}^{-\psi} \theta^n_{\psi}$ is a Radon measure, so Theorem 3.1.5 is not directly applicable.

Proof For each $j \geq 1$, let $\varphi_j := \varphi \vee (-j)$, $\psi_j := \psi \vee (-j)$. Let

$$\mu_j = \mathrm{e}^{-\varphi_j} \theta^n_{\varphi} + \mathrm{e}^{-\psi_j} \theta^n_{\psi}.$$

By Theorem 3.1.5, we can find $\gamma_j \in \mathcal{E}(X, \theta; \phi)$ such that

$$\theta^n_{\gamma_j} = \mathrm{e}^{\gamma_j}\mu_j. \tag{3.44}$$

Take a constant $C > 0$ so that $\psi - 2C \leq \varphi \leq \psi + 2C$. Let

$$\eta \coloneqq \frac{\varphi + \psi}{2} - C - n\log 2.$$

Then $\eta \in \mathcal{E}^\infty(X, \theta; \phi)$ and we have the following inequalities:

$$\begin{aligned}\theta^n_\eta =&\theta^n_{(\varphi+\psi)/2}\\ \geq&2^{-n}\theta^n_\varphi + 2^{-n}\theta^n_\psi\\ \geq&2^{-n}\mathrm{e}^{(\varphi+\psi)/2}\mathrm{e}^{-C}\left(\mathrm{e}^{-\varphi}\theta^n_\varphi + \mathrm{e}^{-\psi}\theta^n_\psi\right)\\ \geq&2^{-n}\mathrm{e}^{(\varphi+\psi)/2}\mathrm{e}^{-C}\left(\mathrm{e}^{-\varphi_j}\theta^n_\varphi + \mathrm{e}^{-\psi_j}\theta^n_\psi\right)\\ =&\mathrm{e}^\eta\mu_j.\end{aligned}$$

Hence, $\gamma_j \geq \eta$ by Theorem 3.1.5. By Theorem 3.1.5 again, γ_j is decreasing in j; let $\gamma = \inf_{j>0}\gamma_j$. Then $\gamma \geq \eta$, hence $\gamma \in \mathcal{E}^\infty(X, \theta; \phi)$.

Now observe that as $j \to \infty$, $\theta^n_{\gamma_j}$ converges weakly to θ^n_γ, as a consequence of Theorem 2.4.2. Finally observe that there is a constant $C' > 0$ so that

$$\gamma_1 \leq \varphi + C', \quad \gamma_1 \leq \psi + C'.$$

since $\varphi, \psi \in \mathcal{E}^\infty(X, \theta; \phi)$. Therefore, for each $j > 0$,

$$\gamma_j \leq \varphi_j + C', \quad \gamma_j \leq \psi_j + C'.$$

Hence, (3.43) follows from (3.44) by letting $j \to \infty$. □

We also need a few integral estimates.

Lemma 3.1.8 *Let $\varphi, \psi, \gamma \in \mathcal{E}(X, \theta; \phi)$. Assume that $\varphi \leq \psi \leq \gamma$. Then*

$$\int_X (\gamma - \psi)\,\theta^n_\psi \leq 2^{n+1}\int_X (\gamma - \varphi)\,\theta^n_\varphi.$$

Proof Observe that for any $t \geq 0$,

$$\begin{aligned}\{\gamma > \psi + 2t\} \setminus \{\psi = -\infty\} &\subseteq \{(\gamma + \psi)/2 > \psi + t\}\\ &\subseteq \{(\gamma + \psi)/2 > \varphi + t\} \subseteq \{\gamma > \varphi + t\}.\end{aligned} \tag{3.45}$$

So

$$
\begin{aligned}
\int_X (\gamma-\psi)\,\theta_\psi^n =& 2\int_0^\infty \int_{\{\gamma-\psi>2t\}} \theta_\psi^n \,\mathrm{d}t \\
\leq& 2\int_0^\infty \int_{\{(\gamma+\psi)/2>\psi+t\}} \theta_\psi^n \,\mathrm{d}t && \text{by (3.45)} \\
\leq& 2^{n+1}\int_0^\infty \int_{\{(\gamma+\psi)/2>\varphi+t\}} \theta_{(\gamma+\psi)/2}^n \,\mathrm{d}t \\
\leq& 2^{n+1}\int_0^\infty \int_{\{(\gamma+\psi)/2>\varphi+t\}} \theta_\varphi^n \,\mathrm{d}t && \text{by Proposition 3.1.15} \\
\leq& 2^{n+1}\int_0^\infty \int_{\{\gamma>\varphi+t\}} \theta_\varphi^n \,\mathrm{d}t && \text{by (3.45)} \\
=& 2^{n+1}\int_X (\gamma-\varphi)\,\theta_\varphi^n && .
\end{aligned}
$$

This completes the proof. □

Lemma 3.1.9 *Let $\varphi_j, \gamma \in \mathcal{E}^1(X,\theta;\phi)$ ($j \in \mathbb{Z}_{>0}$). Assume that $\varphi_j \leq \gamma$ for each j and that φ_j converges to $\varphi \in \mathrm{PSH}(X,\theta)$ with respect to the L^1-topology. Assume that there is a constant $A > 0$ such that for any $j > 0$,*

$$\int_X (\varphi_j-\gamma)\,\theta_{\varphi_j}^n \geq -A.$$

Then $\varphi \in \mathcal{E}^1(X,\theta;\phi)$ and

$$\int_X (\varphi-\gamma)\,\theta_\varphi^n \geq -2^{n+3}A. \tag{3.46}$$

Proof Without loss of generality, we may assume that $\gamma \leq 0$. In particular, $\gamma \leq \phi$.

Step 1. Assume that $(\varphi_j)_j$ is decreasing. In this case, we prove

$$\int_X (\varphi-\gamma)\,\theta_\varphi^n \geq -4A. \tag{3.47}$$

By Lemma 3.1.6, for any $j, k > 0$,

$$\int_X (\varphi_j-\gamma)\,\theta_{\varphi_k}^n \geq -4A.$$

For any $C > 0$,

$$\int_X \Big(\varphi_j \vee (\gamma - C) - \gamma\Big)\,\theta_{\varphi_k}^n \geq \int_X (\varphi_j-\gamma)\,\theta_{\varphi_k}^n \geq -4A.$$

Letting $k \to \infty$, by Theorem 2.4.2, we find

$$\int_X \Big(\varphi_j \vee (\gamma - C) - \gamma\Big)\,\theta_{\varphi}^n \geq -4A.$$

Letting $j \to \infty$, by the monotone convergence theorem, we get

$$\int_X \Big(\varphi \vee (\gamma - C) - \gamma\Big)\, \theta_\varphi^n \geq -4A.$$

Then we let $C \to \infty$, again by the monotone convergence theorem, we conclude (3.47).

Step 2. In general, let

$$\psi_j = \sup_{k\geq j}{}^* \varphi_k.$$

Then $\varphi = \inf_{j>0} \psi_j$ due to Corollary 1.2.1.

For each $C > 0$, let

$$\psi_{j,C} = \psi_j \vee (\gamma - C), \quad \varphi_C = \varphi \vee (\gamma - C).$$

Observe that $\psi_{j,C}$ decreases to φ_C as $j \to \infty$. Moreover,

$$\gamma \geq \psi_{j,C} \geq \psi_j \geq \varphi_j.$$

By Lemma 3.1.8,

$$\int_X (\psi_{j,C} - \gamma)\, \theta_{\psi_{j,C}}^n \geq -2^{n+1} A.$$

By Step 1,

$$\int_X (\varphi_C - \gamma)\, \theta_{\varphi_C}^n \geq -2^{n+3} A. \tag{3.48}$$

In particular,

$$\int_{\{\varphi > \gamma - C\}} (\varphi - \gamma)\, \theta_\varphi^n \geq -2^{n+3} A.$$

Letting $C \to \infty$, we conclude (3.46). As a consequence,

$$\int_X (\phi - \varphi)\, \theta_\varphi^n < \infty.$$

In order to conclude that $\varphi \in \mathcal{E}^1(X, \theta; \phi)$, we still have to prove that $\varphi \in \mathcal{E}(X, \theta; \phi)$. In fact, by (3.48), for any $C > 0$,

$$\int_{\{\varphi \leq \gamma - C\}} \theta_{\varphi_C}^n \leq \frac{1}{C} \int_X (\gamma - \varphi_C)\, \theta_{\varphi_C}^n \leq 2^{n+3} C^{-1} A.$$

Using the monotonicity theorem Theorem 2.4.3, we find

$$\int_X \theta_\phi^n = \int_X \theta_{\varphi_C}^n = \int_{\{\varphi \leq \gamma - C\}} \theta_{\varphi_C}^n + \int_{\{\varphi > \gamma - C\}} \theta_\varphi^n.$$

Letting $C \to \infty$, we conclude that

$$\int_X \theta_\varphi^n = \int_X \theta_\phi^n.$$

This completes the proof. □

Proposition 3.1.21 *Assume that $\varphi, \psi \in \mathcal{E}(X, \theta; \phi)$ (resp. $\mathcal{E}^1(X, \theta; \phi)$, $\mathcal{E}^\infty(X, \theta; \phi)$), then so is $\varphi \wedge \psi$.*

Proof The case of $\mathcal{E}^\infty(X, \theta; \phi)$ is trivial.

We consider the case $\mathcal{E}(X, \theta; \phi)$. It follows from Proposition 3.1.5 that $\varphi \wedge \psi \in \mathrm{PSH}(X, \theta)$. By Theorem 3.1.3, we have

$$\int_X \theta_{\varphi \wedge \psi}^n \geq \int_X \theta_\phi^n.$$

By the monotonicity theoremTheorem 2.4.3, equality holds. By Theorem 3.1.2, we conclude that

$$P_\theta[\varphi \wedge \psi] = \phi.$$

Finally, assume that $\varphi, \psi \in \mathcal{E}^1(X, \theta; \phi)$. We may assume that $\varphi, \psi \leq \phi$. For each $j \geq 1$, consider the approximations:

$$\varphi_j := \varphi \vee (\phi - j), \quad \psi_j := \psi \vee (\phi - j).$$

By Lemma 3.1.7 below, we can take $\gamma_j \in \mathcal{E}^\infty(X, \theta; \phi)$ solving the following equation:

$$\theta_{\gamma_j}^n = \mathrm{e}^{\gamma_j - \varphi_j}\, \theta_{\varphi_j}^n + \mathrm{e}^{\gamma_j - \psi_j}\, \theta_{\psi_j}^n.$$

In particular,

$$\theta_{\gamma_j}^n \geq \mathrm{e}^{\gamma_j - \varphi_j}\, \theta_{\varphi_j}^n.$$

We claim that this implies $\gamma_j \leq \varphi_j$. Indeed, by Proposition 3.1.15,

$$\int_{\{\varphi_j < \gamma_j\}} \theta_{\varphi_j}^n \geq \int_{\{\varphi_j < \gamma_j\}} \theta_{\gamma_j}^n \geq \int_{\{\varphi_j < \gamma_j\}} \mathrm{e}^{\gamma_j - \varphi_j}\, \theta_{\varphi_j}^n.$$

Hence

$$\int_{\{\varphi_j < \gamma_j\}} (\mathrm{e}^{\gamma_j - \varphi_j} - 1)\, \theta_{\varphi_j}^n = 0.$$

It follows that $\int_{\{\varphi_j < \gamma_j\}} \theta_{\varphi_j}^n = 0$, and the domination principle Theorem 2.4.5 gives $\gamma_j \leq \varphi_j$. Similarly, $\gamma_j \leq \psi_j$. In particular, $\gamma_j \leq \varphi_j \wedge \psi_j$. We claim that

$$\int_X (\gamma_j - \phi)\, \theta_{\gamma_j}^n > -C \tag{3.49}$$

for some C independent of j.

Assume the claim is true for now. We get immediately that

$$\sup_X \gamma_j = \sup_{X\setminus\{\phi=-\infty\}} (\gamma_j - \phi) \geq -C/\int_X \theta_\phi^n,$$

where the first equality follows from Proposition 3.1.4. By Proposition 1.5.1, after possibly extracting a subsequence, we may assume that $\gamma_j \to \gamma \in \mathrm{PSH}(X,\theta)$ in L^1-topology. Then $\gamma \in \mathcal{E}^1(X,\theta;\phi)$ by Lemma 3.1.9. Moreover, since $\gamma_j \leq \varphi_j \wedge \psi_j$, we know that $\gamma \leq \varphi \wedge \psi$. In particular, $\varphi \wedge \psi \in \mathrm{PSH}(X,\theta)$. Now by Lemma 3.1.8, $\varphi \wedge \psi \in \mathcal{E}^1(X,\theta;\phi)$.

Now we prove the claim (3.49). By symmetry, it suffices to prove

$$\int_X (\phi - \gamma_j)\mathrm{e}^{\gamma_j - \varphi_j}\, \theta_{\varphi_j}^n \leq C.$$

But note that

$$\int_X (\phi - \gamma_j)\mathrm{e}^{\gamma_j - \varphi_j}\, \theta_{\varphi_j}^n = \int_X (\phi - \varphi_j)\mathrm{e}^{\gamma_j - \varphi_j}\, \theta_{\varphi_j}^n + \int_X (\varphi_j - \gamma_j)\mathrm{e}^{\gamma_j - \varphi_j}\, \theta_{\varphi_j}^n.$$

But $x\mathrm{e}^{-x} \leq \mathrm{e}^{-1}$ for $x \geq 0$ (with maximum at $x = 1$), so the second term is bounded; it remains to prove

$$\int_X (\phi - \varphi_j)\mathrm{e}^{\gamma_j - \varphi_j}\, \theta_{\varphi_j}^n \leq C.$$

As $\gamma_j \leq \varphi_j$, it suffices to prove

$$\int_X (\phi - \varphi_j)\, \theta_{\varphi_j}^n \leq C. \tag{3.50}$$

We compute, using (3.26) and the fact that the $k < n$ terms $\int_X (\varphi_j - \phi)\theta_{\varphi_j}^k \wedge \theta_\phi^{n-k}$ are all non-positive (since $\varphi_j \leq \phi$),

$$\int_X (\varphi_j - \phi)\, \theta_{\varphi_j}^n \geq (n+1)E_\theta^\phi(\varphi_j).$$

Since $\varphi_j \searrow \varphi$ and E_θ^ϕ is continuous along decreasing sequences (Proposition 3.1.16), $E_\theta^\phi(\varphi_j)$ is bounded below uniformly in j. Thus, (3.50) follows. □

Proposition 3.1.22 *Let $\varphi, \psi \in \mathrm{PSH}(X,\theta)$ be potentials such that $\varphi \leq \psi \leq \phi$. Assume that $\varphi \in \mathcal{E}(X,\theta;\phi)$ (resp. $\mathcal{E}^1(X,\theta;\phi)$, $\mathcal{E}^\infty(X,\theta;\phi)$), then so is ψ.*

Proof We may assume that $\varphi \leq \psi$.

The case $\mathcal{E}^\infty(X,\theta;\phi)$ is trivial. The case $\mathcal{E}(X,\theta;\phi)$ follows from Theorem 2.4.3. The case $\mathcal{E}^1(X,\theta;\phi)$ follows from the characterization of $\mathcal{E}^1(X,\theta;\phi)$ in Proposition 3.1.17 and the monotonicity of E_θ^ϕ proved in Lemma 3.1.3. □

Proposition 3.1.23 *Let $(\varphi_i)_{i\in I}$ be a uniformly bounded from above non-empty family in $\mathcal{E}(X,\theta;\phi)$ (resp. $\mathcal{E}^1(X,\theta;\phi)$, $\mathcal{E}^\infty(X,\theta;\phi)$), then so is $\sup^*_{i\in I}\varphi_i$.*

Proof Thanks to Proposition 3.1.22, it suffices to show that

$$\sup_{i\in I}{}^*\varphi_i \leq \phi.$$

Since ϕ is model and $\varphi_i \leq \phi$, we know that

$$\varphi_i - \sup_X \varphi_i \leq \phi$$

for any $i \in I$. Since $(\varphi_i)_{i\in I}$ is uniformly bounded from above by assumption, our assertion follows. □

Proposition 3.1.24 *Let $\varphi, \psi \in \mathcal{E}(X, \theta; \phi)$. Then*

$$P_\theta[\varphi](\psi) = \sup_{C>0}{}^*(\varphi + C) \wedge \psi = \psi.$$

Proof The first equality follows from the definition of $P_\theta[\varphi](\psi)$. We focus on the second inequality.

Since for each $C \geq 0$,

$$(\varphi \wedge \psi + C) \wedge \psi \leq (\varphi + C) \wedge \psi \leq \psi,$$

we may replace φ by $\varphi \wedge \psi$ (cf. Proposition 3.1.21) and assume that $\varphi \leq \psi$.

Let

$$\gamma := \sup_{C>0}{}^*(\varphi + C) \wedge \psi.$$

Observe that $\gamma \leq \psi$, and $\gamma \in \mathcal{E}(X, \theta; \phi)$ due to Proposition 3.1.21 and Proposition 3.1.23. Then Corollary 3.1.1 guarantees that

$$\int_{\{\gamma<\psi\}} \theta_\gamma^n = 0.$$

Therefore, we can apply the domination principle Theorem 2.4.5 to conclude that $\gamma = \psi$. □

Lemma 3.1.10 *Let $\varphi, \psi \in \mathcal{E}^\infty(X, \theta; \phi)$. Define*

$$\eta_t := \Big((1-t)\varphi + t\psi\Big) \wedge \psi, \quad t \in [0, 1].$$

Then $E_\theta^\phi(\eta_t)$ is differentiable for $t \in [0, 1]$ and

$$\frac{\mathrm{d}}{\mathrm{d}t} E_\theta^\phi(\eta_t) = \int_X \Big(\psi - \min\{\varphi, \psi\}\Big)\, \theta_{\eta_t}^n. \tag{3.51}$$

Proof Let us prove (3.51) with right-derivative instead of derivative, and $t \in [0, 1)$. The left-derivative case is completely parallel.

For each $t \in [0, 1]$, we let

$$f_t = \min\{(1-t)\varphi + t\psi, \psi\}.$$

Fix $t \in [0,1)$ and $s > 0$ small enough so that $t+s < 1$. Thanks to Proposition 3.1.19, we have

$$\begin{aligned} E_\theta^\phi(\eta_{t+s}) - E_\theta^\phi(\eta_t) &\leq \int_X (\eta_{t+s} - \eta_t)\ \theta_{\eta_t}^n \\ &= \int_X (\eta_{t+s} - f_t)\ \theta_{\eta_t}^n \qquad\qquad \text{by Lemma 3.1.1} \\ &\leq \int_X (f_{t+s} - f_t)\ \theta_{\eta_t}^n \\ &= s \int_X \Big(\psi - \min\{\varphi,\psi\}\Big)\, \theta_{\eta_t}^n. \end{aligned}$$

Similarly, we get

$$E_\theta^\phi(\eta_{t+s}) - E_\theta^\phi(\eta_t) \geq s \int_X \Big(\psi - \min\{\varphi,\psi\}\Big)\, \theta_{\eta_{t+s}}^n.$$

Observe that η_{t+s} converges in capacity (in fact, even uniformly) to η_t as $s \to 0+$. The desired result (3.51) is then just a consequence of Theorem 2.3.3. □

In particular, we have a diamond-like inequality.

Corollary 3.1.6 *Let $\varphi, \psi \in \mathcal{E}^1(X,\theta;\phi)$. Then*

$$E_\theta^\phi(\varphi \vee \psi) + E_\theta^\phi(\varphi \wedge \psi) \geq E_\theta^\phi(\psi) + E_\theta^\phi(\varphi). \tag{3.52}$$

Proof **Step 1**. We first reduce to the case where $\varphi, \psi \in \mathcal{E}^\infty(X,\theta;\phi)$.

Assume that (3.52) is known in that case. For each $C > 0$, we let

$$\varphi_C = \varphi \vee (\phi - C), \quad \psi_C = \psi \vee (\phi - C).$$

Then

$$E_\theta^\phi(\varphi_C \vee \psi_C) + E_\theta^\phi(\varphi_C \wedge \psi_C) \geq E_\theta^\phi(\psi_C) + E_\theta^\phi(\varphi_C).$$

Letting $C \to \infty$ and applying Proposition 3.1.16, we conclude (3.52).

Step 2. We assume that $\varphi, \psi \in \mathcal{E}^\infty(X,\theta;\phi)$.

Thanks to Lemma 3.1.10, we have

$$\begin{aligned} E_\theta^\phi(\varphi \vee \psi) - E_\theta^\phi(\varphi) &= \int_0^1 \int_X (\varphi \vee \psi - \varphi)\, \theta_{(1-t)\varphi + t(\varphi\vee\psi)}^n\, \mathrm{d}t \\ &= \int_0^1 \int_{\{\psi > \varphi\}} (\psi - \varphi)\, \theta_{(1-t)\varphi + t\psi}^n\, \mathrm{d}t. \end{aligned}$$

On the other hand,

$$
\begin{aligned}
& E_\theta^\phi(\psi) - E_\theta^\phi(\varphi\wedge\psi) \\
= & \int_0^1 \int_X (\psi - \min\{\varphi,\psi\})\, \theta^n_{((1-t)\varphi+t\psi)\wedge\psi}\,\mathrm{d}t \\
\leq & \int_0^1 \int_{\{\varphi<\psi\}} (\psi-\varphi)\,\theta^n_{(1-t)\varphi+t\psi}\,\mathrm{d}t \qquad\qquad \text{by Lemma 3.1.1.}
\end{aligned}
$$

Adding these equations together, we conclude (3.52). □

3.2 The $\mathcal{I}$-envelope

From the algebraic point of view, a more natural envelope operator is given by the $\mathcal{I}$-envelope.

In this section, X will denote a connected compact Kähler manifold of dimension n.

3.2.1 $\mathcal{I}$-equivalence

Proposition 3.2.1 *Given $\varphi,\psi \in \mathrm{QPSH}(X)$, the following are equivalent:*

(1) *For any $k\in\mathbb{Z}_{>0}$, we have*
$$\mathcal{I}(k\varphi)=\mathcal{I}(k\psi);$$

(2) *for any $\lambda\in\mathbb{R}_{>0}$, we have*
$$\mathcal{I}(\lambda\varphi)=\mathcal{I}(\lambda\psi);$$

(3) *for any modification $\pi\colon Y\to X$ and any $y\in Y$, we have*
$$\nu(\pi^*\varphi,y)=\nu(\pi^*\psi,y);$$

(4) *for any proper bimeromorphic morphism $\pi\colon Y\to X$ from a Kähler manifold and any $y\in Y$, we have*
$$\nu(\pi^*\varphi,y)=\nu(\pi^*\psi,y);$$

(5) *for any prime divisor E over X, we have*
$$\nu(\varphi,E)=\nu(\psi,E).$$

See Definition B.1.1 for the definition of prime divisors over X. Recall that in the whole book, a *modification* of a compact complex space means a finite composition of blow-ups with smooth centers. This terminology is highly non-standard.

For simplicity, we usually write

$$\nu(\varphi, y) = \nu(\pi^*\varphi, y)$$

when π is obvious from the context.

Proof (4) $\iff$ (5). This follows from Lemma 1.4.1.

(3) $\iff$ (5). This follows from Corollary B.1.1.

(1) $\implies$ (5). This follows from Proposition 1.4.4.

(5) $\implies$ (2). This follows from Theorem 1.4.3.

(2) $\implies$ (1). This is the restriction to positive integral values of λ. ◻

Definition 3.2.1 Given $\varphi, \psi \in \mathrm{QPSH}(X)$, we say they are *$\mathcal{I}$-equivalent* and write $\varphi \sim_{\mathcal{I}} \psi$ if the equivalent conditions in Proposition 3.2.1 are satisfied.

Clearly, $\sim_{\mathcal{I}}$ is an equivalence relation on $\mathrm{QPSH}(X)$.

Proposition 3.2.2 *Let $\pi\colon Y \to X$ be a proper bimeromorphic morphism from a Kähler manifold Y to X. For $\varphi, \psi \in \mathrm{QPSH}(X)$, the following are equivalent:*

(1) $\varphi \sim_{\mathcal{I}} \psi$;
(2) $\pi^*\varphi \sim_{\mathcal{I}} \pi^*\psi$.

Proof (1) $\implies$ (2). This follows from Proposition 3.2.1(4).

(2) $\implies$ (1). This follows from the simple fact that

$$\mathcal{I}(k\varphi) = \pi_*\left(\omega_{Y/X} \otimes \mathcal{I}(k\pi^*\varphi)\right), \quad \mathcal{I}(k\psi) = \pi_*\left(\omega_{Y/X} \otimes \mathcal{I}(k\pi^*\psi)\right)$$

for any $k \in \mathbb{Z}_{>0}$, where $\omega_{Y/X}$ is the relative dualizing sheaf. See [Dem12a, Proposition 5.8]. ◻

Proposition 3.2.3 *Let $\varphi, \varphi', \psi, \psi' \in \mathrm{QPSH}(X)$ and $\lambda > 0$. Assume that $\varphi \sim_{\mathcal{I}} \psi$ and $\varphi' \sim_{\mathcal{I}} \psi'$. Then*

$$\varphi \vee \varphi' \sim_{\mathcal{I}} \psi \vee \psi', \quad \varphi + \varphi' \sim_{\mathcal{I}} \psi + \psi', \quad \lambda\varphi \sim_{\mathcal{I}} \lambda\psi.$$

Similarly, if $(\varphi_i)_{i\in I}$, $(\psi_i)_{i\in I}$ are two non-empty uniformly bounded from above families in $\mathrm{PSH}(X, \theta)$ for some closed smooth real $(1,1)$-form θ on X such that $\varphi_i \sim_{\mathcal{I}} \psi_i$ for all $i \in I$, then

$$\sup_{i\in I}{}^*\varphi_i \sim_{\mathcal{I}} \sup_{i\in I}{}^*\psi_i.$$

Proof This follows from Proposition 1.4.2 and Corollary 1.4.1. ◻

3.2.2 The definition of the $\mathcal{I}$-envelope

Fix a smooth closed real $(1,1)$-form θ on X.

Definition 3.2.2 Given $\varphi \in \mathrm{PSH}(X, \theta)$, we define its $\mathcal{I}$*-envelope* as follows:

$$P_\theta[\varphi]_{\mathcal{I}} \coloneqq \sup{}^*\{\psi \in \mathrm{PSH}(X, \theta) : \psi \leq 0, \psi \sim_{\mathcal{I}} \varphi\}. \tag{3.53}$$

If $\varphi = P_\theta[\varphi]_{\mathcal{I}}$, we say φ is an $\mathcal{I}$*-model potential* (in $\mathrm{PSH}(X, \theta)$).

Note that by Proposition 1.2.2, $P_\theta[\varphi]_{\mathcal{I}} \in \mathrm{PSH}(X, \theta)$.

Proposition 3.2.4 *Let $\theta' = \theta + \mathrm{dd^c} g$ for some $g \in C^\infty(X)$. Then for any $\varphi \in \mathrm{PSH}(X, \theta)$, we have $\varphi' \coloneqq \varphi - g \in \mathrm{PSH}(X, \theta')$ and*

$$P_\theta[\varphi]_{\mathcal{I}} \sim P_{\theta'}[\varphi']_{\mathcal{I}}.$$

Proof By symmetry, it suffices to show that

$$P_\theta[\varphi]_{\mathcal{I}} \preceq P_{\theta'}[\varphi']_{\mathcal{I}}.$$

After replacing g by $g + C$ for a suitable constant C, which only changes φ' by an additive constant, we may assume that $g \geq 0$. Let $\psi \in \mathrm{PSH}(X, \theta)$ be such that $\psi \leq 0$ and $\psi \sim_{\mathcal{I}} \varphi$. Then $\psi' \coloneqq \psi - g$ lies in $\mathrm{PSH}(X, \theta')$, satisfies $\psi' \leq 0$, and is $\mathcal{I}$-equivalent to φ' since adding a smooth function does not change multiplier ideals. Hence

$$\psi - g = \psi' \leq P_{\theta'}[\varphi']_{\mathcal{I}}.$$

It follows that

$$\psi - \sup_X g \leq P_{\theta'}[\varphi']_{\mathcal{I}}.$$

Taking the upper envelope over all such ψ gives

$$P_\theta[\varphi]_{\mathcal{I}} - \sup_X g \leq P_{\theta'}[\varphi']_{\mathcal{I}},$$

which proves the desired comparison. □

Proposition 3.2.5 *Let $\pi \colon Y \to X$ be a proper bimeromorphic morphism from a Kähler manifold Y to X. Then for $\varphi \in \mathrm{PSH}(X, \theta)$, we have*

$$P_{\pi^*\theta}[\pi^*\varphi]_{\mathcal{I}} = \pi^* P_\theta[\varphi]_{\mathcal{I}}.$$

Proof Let

$$U = P_{\pi^*\theta}[\pi^*\varphi]_{\mathcal{I}}.$$

First take a candidate $\psi \in \mathrm{PSH}(X, \theta)$ in (3.53), so that $\psi \leq 0$ and $\psi \sim_{\mathcal{I}} \varphi$. Then $\pi^*\psi \leq 0$ and $\pi^*\psi \sim_{\mathcal{I}} \pi^*\varphi$ by Proposition 3.2.2; hence $\pi^*\psi \leq U$. Taking the upper envelope over all such ψ gives

$$\pi^* P_\theta[\varphi]_{\mathcal{I}} \leq U.$$

Conversely, let $\chi \in \mathrm{PSH}(Y, \pi^*\theta)$ be a candidate in the definition of U, so that $\chi \leq 0$ and $\chi \sim_{\mathcal{I}} \pi^*\varphi$. By Proposition 1.5.3, there is a unique $\psi \in \mathrm{PSH}(X, \theta)$

such that $\chi = \pi^*\psi$. Since π is surjective, $\psi \leq 0$. By Proposition 3.2.2, the relation $\chi \sim_{\mathcal{I}} \pi^*\varphi$ implies $\psi \sim_{\mathcal{I}} \varphi$. Thus $\psi \leq P_\theta[\varphi]_{\mathcal{I}}$, and hence

$$\chi \leq \pi^* P_\theta[\varphi]_{\mathcal{I}}.$$

Taking the upper envelope over all such χ gives the reverse inequality. □

Proposition 3.2.6 *Let* $\varphi \in \mathrm{PSH}(X, \theta)$. *Then*

$$\varphi \sim_{\mathcal{I}} P_\theta[\varphi]_{\mathcal{I}}.$$

In particular,

$$P_\theta\Big[P_\theta[\varphi]_{\mathcal{I}}\Big]_{\mathcal{I}} = P_\theta[\varphi]_{\mathcal{I}}$$

and the upper semicontinuous regularization in (3.53) *is not necessary.*

Proof In view of Proposition 3.2.1, it suffices to show that for $k \in \mathbb{Z}_{>0}$, we have

$$\mathcal{I}(k\varphi) = \mathcal{I}(kP_\theta[\varphi]_{\mathcal{I}}). \tag{3.54}$$

By Proposition 1.2.3, we can find $\psi_i \in \mathrm{PSH}(X, \theta)$ ($i \in \mathbb{Z}_{>0}$) such that $\psi_i \leq 0$, $\psi_i \sim_{\mathcal{I}} \varphi$ for all $i \geq 1$ and

$$\sup_{i>0}{}^* \psi_i = P_\theta[\varphi]_{\mathcal{I}}.$$

By Proposition 3.2.3, we may replace ψ_i by $\psi_1 \vee \cdots \vee \psi_i$ and assume that the sequence ψ_i is increasing. In this case, it follows from the strong openness theorem Theorem 1.4.4 that for each $k \in \mathbb{Z}_{>0}$, we have

$$\mathcal{I}(k\varphi) = \mathcal{I}(k\psi_j) = \mathcal{I}(kP_\theta[\varphi]_{\mathcal{I}})$$

for j large enough. □

Definition 3.2.3 Let $\varphi \in \mathrm{PSH}(X, \theta)$, we define the *volume*[4] $\mathrm{vol}(\theta, \varphi)$ as

$$\mathrm{vol}(\theta, \varphi) = \int_X \left(\theta + \mathrm{dd^c} P_\theta[\varphi]_{\mathcal{I}}\right)^n.$$

Proposition 3.2.7 *Let* $\theta' = \theta + \mathrm{dd^c} g$ *for some* $g \in C^\infty(X)$. *Then for any* $\varphi \in \mathrm{PSH}(X, \theta)$*, we have* $\varphi' = \varphi - g \in \mathrm{PSH}(X, \theta')$ *and*

$$\mathrm{vol}(\theta, \varphi) = \mathrm{vol}(\theta', \varphi').$$

Proof This follows immediately from Proposition 3.2.4 and Theorem 2.4.3. □

[4] We choose to call this quantity the *volume* instead of the $\mathcal{I}$*-volume* so that the terminology is consistent with the line bundle case.

In view of Proposition 3.2.7, the volume $\mathrm{vol}(\theta, \varphi)$ depends only on the current θ_φ, and we may write

$$\mathrm{vol}\,\theta_\varphi = \mathrm{vol}(\theta, \varphi). \tag{3.55}$$

Definition 3.2.4 Let $\alpha \in \mathrm{H}^{1,1}(X, \mathbb{R})$ be a pseudo-effective class. The *volume* $\mathrm{vol}\,\alpha$ of α is defined as

$$\mathrm{vol}\,\alpha = \mathrm{vol}\,T_{\min},$$

where $T_{\min}$ is a current with minimal singularities in α. Note that $\mathrm{vol}\,\alpha$ is independent of the choice of $T_{\min}$, thanks to Theorem 2.4.3.

Let us recall the following elementary result for later use.

Proposition 3.2.8 *Let $\pi\colon Y \to X$ be a proper bimeromorphic morphism from a Kähler manifold Y.*

(1) *For any big class $\alpha \in \mathrm{H}^{1,1}(X, \mathbb{R})$, $\pi^*\alpha$ is big. Moreover,* $\mathrm{vol}\,\pi^*\alpha = \mathrm{vol}\,\alpha$.
(2) *For any big class $\beta \in \mathrm{H}^{1,1}(Y, \mathbb{R})$, $\pi_*\beta$ is big. Moreover,* $\mathrm{vol}\,\pi_*\beta \geq \mathrm{vol}\,\beta$.

Proof (1) Take a current $T_{\min}$ with minimal singularities in α, then $\pi^*T_{\min}$ is a current with minimal singularities in $\pi^*\alpha$. Our assertion follows.

(2) Take a current $S_{\min}$ with minimal singularities in β, then $\pi_*S_{\min}$ is a current in $\pi_*\beta$, so our assertion follows from the monotonicity theorem Theorem 2.4.3. □

The $\mathcal{I}$-envelope and the P-envelope are related in a simple manner.

Proposition 3.2.9 *Let $\varphi \in \mathrm{PSH}(X, \theta)$. Then*

$$P_\theta[\varphi] \leq P_\theta[\varphi]_{\mathcal{I}}, \quad \varphi \sim_{\mathcal{I}} P_\theta[\varphi].$$

Proof It suffices to show that $\varphi \sim_{\mathcal{I}} P_\theta[\varphi]$. Namely, for each $k \in \mathbb{Z}_{>0}$, we have

$$\mathcal{I}(k\varphi) = \mathcal{I}\Big(kP_\theta[\varphi]\Big). \tag{3.56}$$

Fix k for now. It follows from (3.4) and the strong openness theorem Theorem 1.4.4 that

$$\mathcal{I}\Big(kP_\theta[\varphi]\Big) = \mathcal{I}\Big((k\varphi + C) \wedge kV_\theta\Big),$$

when C is large enough. Since $(k\varphi + C) \wedge kV_\theta \sim k\varphi$, we have

$$\mathcal{I}\Big((k\varphi + C) \wedge kV_\theta\Big) = \mathcal{I}(k\varphi)$$

and (3.56) follows. □

In particular, we obtain an interesting relation between the non-pluripolar mass and the volume.

Corollary 3.2.1 *Let $\varphi \in \mathrm{PSH}(X,\theta)$. Then*

$$\int_X \theta_\varphi^n \leq \operatorname{vol}\theta_\varphi.$$

The reverse inequality fails in general, see Example 6.1.3.

Proof This follows from Proposition 3.2.9, Theorem 2.4.3 and Proposition 3.1.3. □

We note the following special case:

Proposition 3.2.10 *Let $\varphi \in \mathrm{PSH}(X,\theta)$. Assume that φ has analytic singularities. Then*

$$\varphi \sim P_\theta[\varphi] \sim P_\theta[\varphi]_{\mathcal{I}}. \tag{3.57}$$

Hence, $P_\theta[\varphi]$ and $P_\theta[\varphi]_{\mathcal{I}}$ both have analytic singularities.

In particular,

$$\int_X \theta_\varphi^n = \operatorname{vol}\theta_\varphi. \tag{3.58}$$

Proof First observe that (3.58) follows from (3.57) and Theorem 2.4.3. It remains to establish (3.57).

In view of Proposition 3.2.9, it suffices to show that

$$P_\theta[\varphi]_{\mathcal{I}} \leq \varphi. \tag{3.59}$$

By Proposition 3.2.5, Proposition 3.1.9 and Theorem 1.6.1, we may assume that φ has log singularities along an effective $\mathbb{Q}$-divisor D. By rescaling using Proposition 3.2.11, we may assume that D is a $\mathbb{Z}$-divisor. Take quasi-equisingular approximations $(\eta_j)_j$ and $(\varphi_j)_j$ of $P_\theta[\varphi]_{\mathcal{I}}$ and of φ respectively. Recall that by Theorem 1.6.2, we can guarantee that η_j and φ_j both have the singularity type $(2^{-j}, \mathcal{I}(2^j\varphi))$ and hence $\eta_j \sim \varphi_j$ for all large enough j. On the other hand, it is clear that $\varphi_j \sim \varphi$ for all $j \geq 1$. So (3.59) follows. □

3.2.3 Properties of the $\mathcal{I}$-envelope

Let $\theta, \theta_1, \theta_2$ be smooth closed real $(1,1)$-forms on X.

We have the following properties of the $\mathcal{I}$-envelope.

Proposition 3.2.11

(1) *Suppose that $\varphi \in \mathrm{PSH}(X,\theta)$ and $\lambda \in \mathbb{R}_{>0}$. Then*

$$P_{\lambda\theta}[\lambda\varphi]_{\mathcal{I}} = \lambda P_\theta[\varphi]_{\mathcal{I}}.$$

(2) *Suppose that $\varphi_1 \in \mathrm{PSH}(X,\theta_1)$ and $\varphi_2 \in \mathrm{PSH}(X,\theta_2)$. Then*

$$P_{\theta_1+\theta_2}[\varphi_1+\varphi_2]_{\mathcal{I}} \geq P_{\theta_1}[\varphi_1]_{\mathcal{I}} + P_{\theta_2}[\varphi_2]_{\mathcal{I}}.$$

(3) *Suppose that* $\varphi_1 \in \mathrm{PSH}(X, \theta_1)$ *and* $\varphi_2 \in \mathrm{PSH}(X, \theta_2)$. *Then*

$$P_{\theta_1+\theta_2}[\varphi_1 + \varphi_2]_{\mathcal{I}} \sim_{\mathcal{I}} P_{\theta_1}[\varphi_1]_{\mathcal{I}} + P_{\theta_2}[\varphi_2]_{\mathcal{I}}.$$

(4) *Suppose that* $\varphi_1, \varphi_2 \in \mathrm{PSH}(X, \theta)$. *Then*

$$P_\theta[\varphi_1 \vee \varphi_2]_{\mathcal{I}} \sim_{\mathcal{I}} P_\theta[\varphi_1]_{\mathcal{I}} \vee P_\theta[\varphi_2]_{\mathcal{I}}.$$

Proof (1) This follows by rescaling the candidates in the definition of the $\mathcal{I}$-envelope.

(2) Suppose that $\psi_1 \in \mathrm{PSH}(X, \theta_1)$ and $\psi_2 \in \mathrm{PSH}(X, \theta_2)$ satisfy

$$\psi_i \leq 0, \quad \psi_i \sim_{\mathcal{I}} \varphi_i$$

for $i = 1, 2$. Then thanks to Proposition 3.2.3,

$$\psi_1 + \psi_2 \leq 0, \quad \psi_1 + \psi_2 \sim_{\mathcal{I}} \varphi_1 + \varphi_2.$$

It follows that

$$\psi_1 + \psi_2 \leq P_{\theta_1+\theta_2}[\varphi_1 + \varphi_2]_{\mathcal{I}}.$$

Since ψ_1 and ψ_2 are arbitrary, we conclude.

(3) and (4): These follow easily from Proposition 3.2.6 and Proposition 3.2.3. □

Lemma 3.2.1 *Let* $\varphi, \psi \in \mathrm{PSH}(X, \theta)$. *Assume that* $\varphi \leq \psi$. *Then*

$$P_\theta[\varphi]_{\mathcal{I}} \leq P_\theta[\psi]_{\mathcal{I}}.$$

Proof We may assume that $\psi \leq 0$. Observe that $P_\theta[\varphi]_{\mathcal{I}} \vee \psi \sim_{\mathcal{I}} \psi$ as a consequence of Proposition 1.4.2 and Proposition 3.2.6. So

$$P_\theta[\varphi]_{\mathcal{I}} \vee \psi \leq P_\theta[\psi]_{\mathcal{I}}.$$

The desired inequality follows. □

Proposition 3.2.12 *Consider a decreasing sequence* $(\varphi_i)_{i>0}$ *of model potentials in* $\mathrm{PSH}(X, \theta)_{>0}$. *Suppose that* $\varphi := \inf_{i>0} \varphi_i \not\equiv -\infty$ *and* $\int_X \theta_\varphi^n > 0$. *Then*

$$\inf_{i>0} P_\theta[\varphi_i]_{\mathcal{I}} = P_\theta[\varphi]_{\mathcal{I}}.$$

Proof Let $\eta = \inf_{i>0} P_\theta[\varphi_i]_{\mathcal{I}}$. We have $\eta \geq P_\theta[\varphi]_{\mathcal{I}}$ as a consequence of Lemma 3.2.1. In particular, $\eta \in \mathrm{PSH}(X, \theta)$.

By Proposition 3.1.12, we have

$$\lim_{i>0} \int_X \theta_{\varphi_i}^n = \int_X \theta_\varphi^n > 0.$$

So by Lemma 2.4.4, we can find a decreasing sequence $\epsilon_i \searrow 0$ $(i > 0)$ with $\epsilon_i \in (0, 1)$ and $\psi_i \in \mathrm{PSH}(X, \theta)$ such that for all $i > 0$,

$$(1-\epsilon_i)\varphi_i + \epsilon_i\psi_i \le \varphi.$$

By Proposition 3.2.11 and Lemma 3.2.1, we have

$$\eta+\epsilon_i P_\theta[\psi_i]_{\mathcal{I}} \le (1-\epsilon_i)\eta+\epsilon_i P_\theta[\psi_i]_{\mathcal{I}} \le (1-\epsilon_i)P_\theta[\varphi_i]_{\mathcal{I}}+\epsilon_i P_\theta[\psi_i]_{\mathcal{I}} \le P_\theta[\varphi]_{\mathcal{I}}.$$

Each $P_\theta[\psi_i]_{\mathcal{I}}$ has supremum 0, so Proposition 1.5.1 shows that the family is bounded in L^1. Hence $\epsilon_i P_\theta[\psi_i]_{\mathcal{I}} \to 0$ in L^1, and the preceding inequality gives $\eta \le P_\theta[\varphi]_{\mathcal{I}}$ almost everywhere. By Proposition 1.2.6, the inequality holds everywhere. □

Proposition 3.2.13 *Let $(\varphi_i)_{i\in I}$ be a decreasing net of $\mathcal{I}$-model potentials in* $\mathrm{PSH}(X,\theta)$*. Set $\varphi := \inf_{i\in I}\varphi_i$, then φ is also $\mathcal{I}$-model in* $\mathrm{PSH}(X,\theta)$.

Proof Observe that $\varphi_i \le 0$ and $\sup_X \varphi_i = 0$ for each $i \in I$ by the definition of the $\mathcal{I}$-envelope. By Proposition 1.5.1, the decreasing limit φ is not identically $-\infty$, hence $\varphi \in \mathrm{PSH}(X,\theta)$ by Proposition 1.2.2. Let $\eta \in \mathrm{PSH}(X,\theta)$ with $\eta \sim_{\mathcal{I}} \varphi$ and $\eta \le 0$. Then for each $i \in I$, using Proposition 3.2.3, we have $\eta \vee \varphi_i \sim_{\mathcal{I}} \varphi_i$. Therefore,

$$\eta \le \eta \vee \varphi_i \le \varphi_i.$$

It follows that $\eta \le \varphi$. Hence $\varphi = P_\theta[\varphi]_{\mathcal{I}}$. □

Proposition 3.2.14 *Let $(\varphi_i)_{i\in I}$ be an increasing net in $\mathrm{PSH}(X,\theta)_{>0}$ uniformly bounded from above. Let $\varphi := \sup^*_{i\in I}\varphi_i$. Then*

$$\sup_{i\in I}{}^* P_\theta[\varphi_i]_{\mathcal{I}} = P_\theta[\varphi]_{\mathcal{I}}.$$

In particular, if the φ_i's are all $\mathcal{I}$-model, then so is φ.

Proof Let $\eta = \sup^*_{i\in I} P_\theta[\varphi_i]_{\mathcal{I}}$. Then $\eta \le P_\theta[\varphi]_{\mathcal{I}}$ as a consequence of Lemma 3.2.1.

By Corollary 2.4.1, we have

$$\lim_{i\in I}\int_X \theta^n_{\varphi_i} = \int_X \theta^n_\varphi > 0.$$

So by Lemma 2.4.4, we can find a decreasing net $\epsilon_i \searrow 0$ ($i \in I$) with $\epsilon_i \in (0,1)$ and $\psi_i \in \mathrm{PSH}(X,\theta)$ such that for all $i \in I$,

$$(1-\epsilon_i)\varphi + \epsilon_i\psi_i \le \varphi_i.$$

By Proposition 3.2.11 and Lemma 3.2.1, we have

$$P_\theta[\varphi]_{\mathcal{I}} + \epsilon_i P_\theta[\psi_i]_{\mathcal{I}} \le (1-\epsilon_i)P_\theta[\varphi]_{\mathcal{I}} + \epsilon_i P_\theta[\psi_i]_{\mathcal{I}} \le P_\theta[\varphi_i]_{\mathcal{I}} \le \eta.$$

As above, $\epsilon_i P_\theta[\psi_i]_{\mathcal{I}} \to 0$ in L^1. It follows that $P_\theta[\varphi]_{\mathcal{I}} \le \eta$ almost everywhere, hence everywhere by Proposition 1.2.6. □

Remark 3.2.1 One could also define the following interpolation between the $\mathcal{I}$-envelope and the P-envelope: Suppose $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$, $j \in \{0,\dots,n\}$. Then we let

$$
\begin{aligned}
P_{\theta,j}[\varphi] \coloneqq \sup{}^* \Big\{ \psi \in \mathrm{PSH}(X,\theta) : \psi \le 0, \varphi \preceq \psi, \int_X \theta_\varphi^j \wedge \theta^{n-j}_{P_\theta[\varphi]_{\mathcal{I}}} \\
= \int_X \theta_\psi^j \wedge \theta^{n-j}_{P_\theta[\psi]_{\mathcal{I}}} \Big\}.
\end{aligned}
$$

Based on the techniques developed in Chapter 6, one could show that $P_{\theta,j}[\bullet]$ is a projection operator. The cases $j = 0$ and $j = n$ recover the $\mathcal{I}$- and P-envelopes respectively; the intermediate operators $P_{\theta,j}$ have not, to our knowledge, been systematically studied in the literature.

Chapter 4
Geodesic rays in the space of potentials

In den Dreißiger Jahren besuchte ich regelmäßig die Schweiz, teils um mich auch auf den Viertausendern zu tummeln, zum großen Teil aber auch, um Emigrantenblätter zu lesen und mich mit Kollegen über Naziverbrechen zu unterhalten. Aber auch die Schweizer schauten sich, wenn sie offen reden wollten, ebenso ängstlich um wie das bei uns üblich war.[a]
— Oskar Perron[b]

[a] The recent policy of ETH against Chinese students makes me feel that nothing has changed in Switzerland after the collapsing of Nazi for almost 80 years.

[b] Oskar Perron (1880–1975), after earning an *Eisernes Kreuz* during WWI, obtained a position in München in 1922, initiating the glorious period there. Among his colleagues were Carathéodory, Tietze and Sommerfield.

When ω is a Kähler form, the space of smooth strictly ω-plurisubharmonic potentials carries the Mabuchi L^2-metric, whose geodesic equation is the homogeneous complex Monge–Ampère equation on $X \times \{0 < \operatorname{Re} z < 1\}$. Subsolutions to this equation are precisely the subgeodesics, and the geodesic between two given boundary potentials is recovered as the supremum of all subgeodesics with the prescribed boundary data — a construction which, unlike the smooth Mabuchi geodesic, retains a meaning beyond the smooth Kähler setting.

The goal of this chapter is to develop the theory of *subgeodesics* and *geodesics* in $\mathrm{PSH}(X, \theta)$ for θ a smooth closed real $(1, 1)$-form in a big cohomology class, with prescribed (and possibly singular) boundary data $\varphi_0, \varphi_1 \in \mathrm{PSH}(X, \theta)$. The boundary data may be unsmooth or even unbounded, so one cannot reduce to the smooth case by a simple regularization argument. We define subgeodesics by plurisubharmonicity of the complexification $\Phi(x, z) = \varphi_{\operatorname{Re} z}(x)$, and the geodesic from φ_0 to φ_1 as the Perron envelope of subgeodesics with these endpoints; in the smooth Kähler case this coincides with the classical Mabuchi geodesic.

The chapter is organized as follows. Section 4.1 sets up the basic theory of subgeodesics. Section 4.2 develops the existence theory for geodesics with prescribed (possibly singular) endpoints, constructing the geodesic as the Perron envelope of subgeodesics with the given boundary data. We also define and study the notion of *geodesic rays*. Section 4.3 introduces the Darvas d_1-metric on the finite-energy class $\mathcal{E}^1(X, \theta; \phi)$ and on the space of geodesic rays $\mathcal{R}^1(X, \theta; \phi)$, and proves completeness of $(\mathcal{E}^1, d_1)$.

The material developed in this chapter is foundational for the rest of the book: the d_1-metric on rays underlies the pseudometric d_S on the space $\mathrm{PSH}(X, \theta)$ studied in Chapter 6, while the geodesic rays themselves provide the bridge between psh singularities and non-Archimedean potentials in Chapter 13.

4.1 Subgeodesics

Let X be a connected compact Kähler manifold of dimension n and θ be a smooth closed real $(1,1)$-form on X representing a big cohomology class.

Definition 4.1.1 Let us fix $\varphi_0, \varphi_1 \in \mathrm{PSH}(X, \theta)$. A *subgeodesic* from φ_0 to φ_1 is a family $(\varphi_t)_{t\in(0,1)}$ in $\mathrm{PSH}(X, \theta)$ such that

(1) if we define

$$\Phi\colon X \times \left\{z \in \mathbb{C} : \operatorname{Re} z \in (0,1)\right\} \to [-\infty, \infty), \quad (x,z) \mapsto \varphi_{\operatorname{Re} z}(x),$$

then Φ is $p_1^*\theta$-psh, where $p_1\colon X \times \{z \in \mathbb{C} : \operatorname{Re} z \in (0,1)\} \to X$ is the natural projection;

(2) when $t \to 0+$ (resp. $t \to 1-$), φ_t converges to φ_0 (resp. φ_1) with respect to the L^1-topology.

We also say $(\varphi_t)_{t\in[0,1]}$ is a subgeodesic.[1]

We call Φ the *complexification* of the subgeodesic $(\varphi_t)_t$.

When we do not want to specify φ_0 and φ_1, we shall say $(\varphi_t)_{t\in(0,1)}$ is a subgeodesic.

More generally, a family $(\psi_t)_{t\in[a,b]}$ (resp. $(\psi_t)_{t\in(a,b)}$) in $\mathrm{PSH}(X, \theta)$ for some $a \leq b$ is called a *subgeodesic* if $(\psi_{tb+(1-t)a})_{t\in[0,1]}$ (resp. $(\psi_{tb+(1-t)a})_{t\in(0,1)}$) is a subgeodesic.

Remark 4.1.1 Note that given a subgeodesic $(\psi_t)_{t\in(a,b)}$, the endpoints ψ_a and ψ_b are uniquely determined, in view of Proposition 1.2.6.

Remark 4.1.2 In the literature, people sometimes regard Φ as a function defined on $X \times \{z \in \mathbb{C} : \mathrm{e}^{-1} < |z| < 1\}$, with $\Phi(x,z) = \varphi_{-\log|z|}(x)$. We sometimes also use this convention; the ambient strip or annulus will make the convention clear.

In general, there are no subgeodesics from φ_0 to φ_1. In fact, by Proposition 4.1.2 below, the existence of a subgeodesic from φ_0 to φ_1 forces $\varphi_0 \wedge \varphi_1 \not\equiv -\infty$, a condition that may fail (see Example 5.2.3).

We first note that the subgeodesics are well-behaved under the change of θ:

Proposition 4.1.1 *Let g be a smooth real function on X. Let $\theta' = \theta + \mathrm{dd^c} g$. Suppose that $(\varphi_t)_{t\in[0,1]}$ is a subgeodesic in $\mathrm{PSH}(X, \theta)$. Then $(\varphi_t - g)_{t\in[0,1]}$ is a subgeodesic in $\mathrm{PSH}(X, \theta')$.*

Proof This follows directly from the definition of the complexification. □

Example 4.1.1 Let $\varphi_0 \in \mathrm{PSH}(X, \theta)$, $C \in \mathbb{R}$. Let

$$\varphi_t = \varphi_0 + tC, \quad t \in [0,1].$$

[1] Note that it is implicit in the definition that φ_t as $t \to 0+$ or $t \to 1-$ admits an L^1-limit.

Then $(\varphi_t)_{t\in[0,1]}$ is a subgeodesic. Indeed, since $\operatorname{Re} z$ is harmonic in z, the complexification $\Phi(x,z)=\varphi_0(x)+C\ \operatorname{Re} z$ is $p_1^*\theta$-psh.

As a consequence, the constant $(\varphi_0)_{t\in[0,1]}$ is a subgeodesic, called the *constant subgeodesic* at φ_0.

A more general version is as follows: Suppose that $(\varphi_t)_{t\in[0,1]}$ is a subgeodesic in $\operatorname{PSH}(X,\theta)$, $C_1,C_2\in\mathbb{R}$. Then $(\varphi_t+C_1t+C_2)_{t\in[0,1]}$ is also a subgeodesic.

Proposition 4.1.2 *Let $\varphi_0,\varphi_1\in\operatorname{PSH}(X,\theta)$ and $(\varphi_t)_{t\in(0,1)}$ be a subgeodesic from φ_0 to φ_1. Then for each $x\in X$, $[0,1]\ni t\mapsto\varphi_t(x)$ is a convex function. Furthermore,*

$$\inf_{t\in(0,1)}\varphi_t\in\operatorname{PSH}(X,\theta),\qquad \inf_{t\in(0,1)}\varphi_t\leq\varphi_0\wedge\varphi_1.$$

As a consequence of the convexity, we find

$$\varlimsup_{t\to 0+}\varphi_t\leq\varphi_0,\qquad \varlimsup_{t\to 1-}\varphi_t\leq\varphi_1. \tag{4.1}$$

These are not equalities in general. For example, consider an arbitrary subgeodesic from an unbounded $\varphi_0\in\mathcal{E}(X,\omega)$ to a bounded $\varphi_1\in\operatorname{PSH}(X,\omega)$ for some Kähler form ω on X.

Proof Let Φ be the complexification of $(\varphi_t)_{t\in(0,1)}$.

For each $x\in X$, the map

$$\left\{z\in\mathbb{C}:\operatorname{Re} z\in(0,1)\right\}\to[-\infty,\infty),\qquad z\mapsto\Phi(x,z)$$

is either subharmonic or constantly $-\infty$, as follows from Definition 4.1.1 (1) and Proposition 1.1.4. In the latter case, the convexity of $[0,1]\ni t\mapsto\varphi_t(x)$ is trivial. In the former case, the convexity on the interval $(0,1)$ follows from Proposition 1.1.3.

In order to verify the convexity at the boundary, let us fix $s\in(0,1)$. We need to show that

$$\varphi_s(x)\leq s\varphi_1(x)+(1-s)\varphi_0(x) \tag{4.2}$$

for all $x\in X$. Thanks to Proposition 1.2.6, it suffices to prove this for almost all x.

Choose sequences $a_j\searrow 0$ and $b_j\nearrow 1$ such that $\varphi_{a_j}\to\varphi_0$ and $\varphi_{b_j}\to\varphi_1$ almost everywhere. Outside a set $Z\subseteq X$ of zero Lebesgue measure, all these convergences hold and all the values involved are finite. For $x\in X\setminus Z$ and j large enough that $a_j<s<b_j$, the convexity on $(0,1)$ gives

$$\varphi_s(x)\leq\frac{b_j-s}{b_j-a_j}\varphi_{a_j}(x)+\frac{s-a_j}{b_j-a_j}\varphi_{b_j}(x).$$

Letting $j\to\infty$ gives (4.2) almost everywhere.

Let us prove the last assertion. Let

$$\varphi\coloneqq\inf_{t\in(0,1)}\varphi_t.$$

By Kiselman's principle Proposition 1.2.8, we know that $\varphi \in \mathrm{PSH}(X,\theta) \cup \{-\infty\}$. Since the complexification Φ is not identically $-\infty$, Fubini's theorem shows that for almost every $x \in X$, the function $t \mapsto \varphi_t(x)$ is finite for almost every $t \in (0,1)$. For such an x, the convexity just proved implies that $t \mapsto \varphi_t(x)$ is a finite convex function on $(0,1)$, hence it is bounded from below on $(0,1)$. Thus $\varphi \not\equiv -\infty$, and so $\varphi \in \mathrm{PSH}(X,\theta)$.

For any $t \in (0,1)$, using the convexity established above, we have

$$\varphi \leq (1-t)\varphi_0 + t\varphi_1.$$

It follows that $\varphi \leq \varphi_0$ and $\varphi \leq \varphi_1$ almost everywhere and hence everywhere by Proposition 1.2.6. Our assertion follows. □

Corollary 4.1.1 *Let $(\varphi_t)_{t\in[0,1]}$ be a subgeodesic in $\mathrm{PSH}(X,\theta)$. Then*

$$\sup_X \inf_{s\in(0,1)} \varphi_s \leq \sup_X \varphi_t \leq \sup_X \varphi_0 \vee \sup_X \varphi_1 \tag{4.3}$$

for all $t \in (0,1)$.

Furthermore, $[0,1] \ni t \mapsto \varphi_t$ is continuous with respect to the L^1-topology.

In particular, in view of Proposition 1.5.1, the family $\{\varphi_t\}_{t\in[0,1]}$ is relatively compact with respect to the L^1-topology on $\mathrm{PSH}(X,\theta)$.

Proof (4.3) follows immediately from Proposition 4.1.2. Let us verify the continuity of $t \mapsto \varphi_t$. The continuity at $t=0$ or $t=1$ follows from the definition of subgeodesic. While the continuity on $(0,1)$ follows from the convexity of $t \mapsto \varphi_t$, and the fact that $\varphi_t \in L^1(X)$ for each $t \in (0,1)$. □

Proposition 4.1.3 *Let $(\varphi_0^i)_{i\in I}$, $(\varphi_1^i)_{i\in I}$ be two non-empty uniformly bounded from above families in $\mathrm{PSH}(X,\theta)$. Let $(\varphi_t^i)_{t\in(0,1)}$ be subgeodesics from φ_0^i to φ_1^i for each $i \in I$. Then*

$$\left(\sup_{i\in I}{}^*\varphi_t^i\right)_{t\in(0,1)}$$

is a subgeodesic from $\sup_i{}^*\varphi_0^i$ to $\sup_i{}^*\varphi_1^i$.

Proof We may assume that $\varphi_0^i, \varphi_1^i \leq 0$ for all $i \in I$. Then it follows that $\varphi_t^i \leq 0$ for all $t \in (0,1)$ and all $i \in I$ by Proposition 4.1.2.

We define

$$\varphi_t \coloneqq \sup_{i\in I}{}^*\varphi_t^i \in \mathrm{PSH}(X,\theta)$$

for all $t \in [0,1]$. Observe that $[0,1] \ni t \mapsto \varphi_t$ is convex by the same argument leading to (4.2).

Let $(\psi_t)_{t\in(0,1)}$ be the subgeodesic whose complexification Φ_ψ corresponds to $\sup_i{}^*\Phi_{\varphi^i}$, where Φ_{φ^i} is the complexification of $(\varphi_t^i)_{t\in(0,1)}$. Then clearly, $\varphi_t \leq \psi_t$ for each $t \in (0,1)$. On the other hand, by Proposition 1.2.5,

$$\psi_t = \sup_{i\in I} \varphi_t^i = \varphi_t \quad \text{almost everywhere}$$

for almost all $t \in (0,1)$. Therefore, using Proposition 1.2.6, we find $\psi_t = \varphi_t$ for almost all $t \in (0,1)$. Since both functions are convex in t, we conclude that $\psi_t = \varphi_t$ for all $t \in (0,1)$.

It remains to argue that $\varphi_t \xrightarrow{L^1} \varphi_0$ as $t \to 0+$ and $\varphi_t \xrightarrow{L^1} \varphi_1$ as $t \to 1-$. By symmetry, it suffices to argue the former.

Thanks to Proposition 1.2.3, we may further assume that I is a countable set. We know that for any $t \in (0,1)$ and any $j \in I$,

$$\varphi_t^j \leq \varphi_t \leq t\varphi_1 + (1-t)\varphi_0.$$

Letting $t \to 0+$, we find that

$$\varphi_0^j \leq \varlimsup_{t \to 0+} \varphi_t \leq \varphi_0$$

almost everywhere. Since I is countable, we conclude that

$$\varphi_0 = \varlimsup_{t \to 0+} \varphi_t \tag{4.4}$$

almost everywhere.

Fix $i_0 \in I$. Recall that by Proposition 4.1.2, for each $t \in (0,1)$, we have

$$\inf_{t \in (0,1)} \sup_X \varphi_t \geq \inf_{t \in (0,1)} \sup_X \varphi_t^{i_0} \geq \sup_X \left(\varphi_0^{i_0} \land \varphi_1^{i_0}\right) > -\infty,$$

so the set $\{\varphi_t\}_{t \in (0,1)}$ is relatively compact with respect to the L^1-topology by Proposition 1.5.1. Let ψ be a cluster point as $t \to 0+$. It suffices to show that $\psi = \varphi_0$. By Corollary 1.2.1 and (4.4), this holds almost everywhere. Therefore, it holds everywhere by Proposition 1.2.6. □

Proposition 4.1.4 *Let* $(\varphi_t)_{t \in [0,1]}$ *be a subgeodesic in* $\mathrm{PSH}(X,\theta)$*. Then for any* $0 \leq a \leq b \leq 1$*, the segment* $(\varphi_t)_{t \in [a,b]}$ *is a subgeodesic.*

Proof It suffices to show that

$$\varphi_{tb+(1-t)a} \xrightarrow{L^1} \varphi_a, \quad \varphi_{tb+(1-t)a} \xrightarrow{L^1} \varphi_b$$

as $t \to 0+$ and $t \to 1-$ respectively. In other words, we need to show that for any $c \in (0,1)$, we have

$$\varphi_t \xrightarrow{L^1} \varphi_c$$

as $t \to c$. For this purpose, observe that by Proposition 4.1.2,

$$\sup_X \inf_{s \in (0,1)} \varphi_s \leq \sup_X \varphi_t \leq \left(\sup_X \varphi_0\right) \lor \left(\sup_X \varphi_1\right)$$

for any $t \in (0,1)$. Therefore, $\{\varphi_t\}_{t \in (0,1)}$ is a relatively compact family with respect to the L^1-topology on $\mathrm{PSH}(X,\theta)$ by Proposition 1.5.1. It suffices to show that any cluster point ψ of φ_t as $t \to c$ is equal to φ_c. By Corollary 1.2.1 and the convexity Proposition 4.1.2, we have $\varphi_c = \psi$ almost everywhere and hence everywhere by Proposition 1.2.6. ◻

For the purpose of this chapter, it is useful to reformulate the gluing lemma Corollary 1.2.2 in the following way, which allows us to glue subgeodesics directly:

Lemma 4.1.1 *Let $(\varphi_t)_{t\in[0,1]}$ be a subgeodesic in* $\mathrm{PSH}(X,\theta)$*. Let $0 \le a < b \le 1$ and let $(\psi_t)_{t\in[a,b]}$ be a subgeodesic such that $\psi_a = \varphi_a$, $\psi_b = \varphi_b$, and*

$$\psi_t \ge \varphi_t, \quad t \in (a,b).$$

Define

$$\gamma_t = \begin{cases} \psi_t, & \text{if } t \in (a,b), \\ \varphi_t, & \text{if } t \in [0,1] \setminus (a,b). \end{cases}$$

Then $(\gamma_t)_{t\in[0,1]}$ is a subgeodesic.

Proof The L^1-convergence of γ_t to φ_0 at 0 and φ_1 at 1 follows immediately from the corresponding property for $(\varphi_t)_t$ and $(\psi_t)_t$. It remains to prove that the complexification of $(\gamma_t)_t$ is $p_1^*\theta$-psh.

Let

$$S = \{z \in \mathbb{C} : 0 < \operatorname{Re} z < 1\}, \quad S' = \{z \in \mathbb{C} : a < \operatorname{Re} z < b\}.$$

Let Φ and Ψ be the complexifications of $(\varphi_t)_t$ and $(\psi_t)_t$ on the corresponding strips. Since $\Psi \ge \Phi$ on $X \times S'$, the complexification of $(\gamma_t)_t$ is equal to $\Phi \vee \Psi$ on $X \times S'$ and to Φ on $X \times (S \setminus S')$. By the gluing lemma Corollary 1.2.2, it is enough to verify that

$$\varlimsup_{\substack{S' \ni z' \to z \\ x' \to x}} \Psi(x', z') \le \Phi(x, z)$$

at every point $(x,z) \in X \times S$ with $\operatorname{Re} z = a$ or $\operatorname{Re} z = b$.

Suppose for instance that $a > 0$, and let (x_j, z_j) tend to (x,z) with $\operatorname{Re} z = a$ and $a < \operatorname{Re} z_j < b$. Set $t_j = \operatorname{Re} z_j$. Since $(\psi_t)_t$ is a subgeodesic from φ_a to φ_b, the sequence ψ_{t_j} converges to φ_a in L^1. Applying Corollary 1.2.1 to this sequence gives

$$\varphi_a = \inf_{m>0} \left(\sup_{j\ge m}{}^* \psi_{t_j} \right).$$

For fixed m, the upper semicontinuity of $\sup_{j\ge m}{}^* \psi_{t_j}$ gives

$$\varlimsup_{j\to\infty} \psi_{t_j}(x_j) \le \left(\sup_{j\ge m}{}^* \psi_{t_j} \right)(x).$$

Letting $m \to \infty$, we get

$$\varlimsup_{j\to\infty} \psi_{t_j}(x_j) \le \varphi_a(x).$$

Since $\Phi(x,z) = \varphi_a(x)$ when $\operatorname{Re} z = a$, this is the desired boundary inequality along $\{\operatorname{Re} z = a\}$. The same argument applies along $\{\operatorname{Re} z = b\}$ when $b < 1$. Hence Corollary 1.2.2 shows that the pasted complexification is $p_1^*\theta$-psh. □

Definition 4.1.2 A ray $\ell = (\ell_t)_{t\geq 0}$ is a *subgeodesic ray* in $\mathrm{PSH}(X,\theta)$ if for any $0 \leq a \leq b$, the segment $(\ell_t)_{t\in[a,b]}$ is a subgeodesic in $\mathrm{PSH}(X,\theta)$. We say ℓ *emanates* from ℓ_0.

The *complexification* of a subgeodesic ray ℓ is defined as the potential

$$\Phi\colon X \times \{z \in \mathbb{C} : \operatorname{Re} z > 0\} \to [-\infty,\infty), \quad (x,z) \mapsto \ell_{\operatorname{Re} z}(x).$$

Note that Φ is $p_1^*\theta$-psh, where $p_1\colon X \times \{z \in \mathbb{C} : \operatorname{Re} z > 0\} \to X$ is the natural projection.

Remark 4.1.3 As in Remark 4.1.2, one may also define the complexification as a function $X \times \{z \in \mathbb{C} : 0 < |z| < 1\} \to [-\infty,\infty)$.

4.2 Geodesics in the space of potentials

Let X be a connected compact Kähler manifold of dimension n and θ be a smooth closed real $(1,1)$-form on X representing a big cohomology class. Fix a model potential $\phi \in \mathrm{PSH}(X,\theta)_{>0}$. See Definition 3.1.3 for the definition.

Definition 4.2.1 Let $\varphi_0,\varphi_1 \in \mathrm{PSH}(X,\theta)$. The *geodesic* $(\varphi_t)_{t\in(0,1)}$ from φ_0 to φ_1 is the family of potentials $\varphi_t \in \mathrm{PSH}(X,\theta) \cup \{-\infty\}$ such that

$$\begin{aligned}\varphi_t = \sup{}^* \{\psi_t : (\psi_s)_s &\text{ is a subgeodesic from } \psi_0 \text{ to } \psi_1, \\ &\psi_0,\psi_1 \in \mathrm{PSH}(X,\theta),\ \psi_0 \leq \varphi_0, \psi_1 \leq \varphi_1\}.\end{aligned} \tag{4.5}$$

More generally, let $(\varphi_t)_{t\in[a,b]}$ ($a,b \in \mathbb{R}$, $a \leq b$) be a curve in $\mathrm{PSH}(X,\theta)$. We say $(\varphi_t)_{t\in[a,b]}$ is a *geodesic* if the curve $(\varphi_{tb+(1-t)a})_{t\in(0,1)}$ is a geodesic from φ_a to φ_b.

We also say $(\varphi_t)_{t\in(a,b)}$ is a *geodesic* in $\mathrm{PSH}(X,\theta)$ from φ_a to φ_b.

The envelopes of the form (4.5) are usually referred to as the *Perron envelopes*. In general, the geodesic defined by (4.5) fails to have the correct limit when $t \to 0+$ or $t \to 1-$. Therefore, *a priori* it is not a subgeodesic from φ_0 to φ_1. In the sequel, we use the word *geodesic* only in situations where the endpoint behavior has been verified.

Lemma 4.2.1 *Let $\varphi_0,\varphi_1 \in \mathrm{PSH}(X,\theta)$. Assume that there is a subgeodesic from φ_0 to φ_1. Then the geodesic $(\varphi_t)_{t\in(0,1)}$ from φ_0 to φ_1 is a subgeodesic from φ_0 to φ_1.*

Proof This is an immediate consequence of Proposition 4.1.3. □

Example 4.2.1 Let $\varphi_0 \in \mathrm{PSH}(X, \theta)$ and $C \in \mathbb{R}$. Then the subgeodesic $(\varphi_0 + tC)_{t\in[0,1]}$ studied in Example 4.1.1 is a geodesic. This follows easily from Proposition 4.1.2.

In particular, when $C = 0$, we find that the constant subgeodesic at φ_0 is indeed a geodesic, which we call the *constant geodesic* at φ_0.

More generally, suppose that $(\varphi_t)_{t\in[0,1]}$ is a geodesic and $C_1, C_2 \in \mathbb{R}$, then $(\varphi_t + C_1 t + C_2)_{t\in[0,1]}$ is also a geodesic. This follows immediately from Example 4.1.1.

Next we want to show that under mild assumptions, there exist subgeodesics between two potentials. The assumption below turns out to be necessary as well, as we shall prove in Theorem 6.1.1 later.

Proposition 4.2.1 *Given $\varphi_0, \varphi_1 \in \mathcal{E}(X, \theta; \phi)$, the geodesic $(\varphi_t)_{t\in(0,1)}$ from φ_0 to φ_1 defined by* (4.5) *is a subgeodesic from φ_0 to φ_1 and $\varphi_t \in \mathcal{E}(X, \theta; \phi)$ for each $t \in (0, 1)$.*

Moreover, for any $0 \le a \le b \le 1$, the restriction $(\varphi_t)_{t\in[a,b]}$ is a geodesic.

If furthermore $\varphi_0, \varphi_1 \in \mathcal{E}^1(X, \theta; \phi)$ (resp. $\mathcal{E}^\infty(X, \theta; \phi)$), then $\varphi_t \in \mathcal{E}^1(X, \theta; \phi)$ (resp. $\mathcal{E}^\infty(X, \theta; \phi)$) for all $t \in (0, 1)$.

We refer to Section 3.1.3 for the definition of $\mathcal{E}(X, \theta; \phi)$. Our assumption means that $P_\theta[\varphi_0] = P_\theta[\varphi_1] = \phi$.

The restriction $(\varphi_t)_{t\in[a,b]}$ is also a subgeodesic since $\varphi_a, \varphi_b \in \mathcal{E}(X, \theta; \phi)$ as well.

Proof After subtracting suitable constants from φ_0 and φ_1 and adding back the corresponding affine function of t, we may assume that $\varphi_0, \varphi_1 \le \phi$. It follows from Proposition 4.1.2 that $\varphi_t \le \phi$ for all $t \in (0, 1)$. In fact, we have the stronger estimate

$$\varphi_t \le t\varphi_1 + (1 - t)\varphi_0, \quad t \in (0, 1). \tag{4.6}$$

We first observe that when $\varphi_0, \varphi_1 \in \mathcal{E}(X, \theta; \phi)$, so is $\varphi_0 \wedge \varphi_1$, see Proposition 3.1.21. In particular, the constant subgeodesic $t \mapsto \varphi_0 \wedge \varphi_1$ is a candidate in the Perron envelope (4.5). So

$$\varphi_t \ge \varphi_0 \wedge \varphi_1, \quad t \in (0, 1). \tag{4.7}$$

By Proposition 4.1.3, $(\varphi_t)_{t\in(0,1)}$ is a subgeodesic.[2] It follows from Proposition 3.1.22 that $\varphi_t \in \mathcal{E}(X, \theta; \phi)$ for all $t \in (0, 1)$.

Next, we show that as $t \to 0+$, we have $\varphi_t \xrightarrow{L^1} \varphi_0$. The corresponding result at $t = 1$ is similar.

We first argue the special case where $\varphi_0 \le \varphi_1$. Take a constant $C > 0$ such that

$$\varphi_0 - C \le \varphi_1.$$

Then $(\varphi_0 - Ct)_{t\in(0,1)}$ is clearly a candidate in (4.5), see Example 4.1.1. Therefore, for all $t \in (0, 1)$,

$$\varphi_0 - Ct \le \varphi_t \le t\varphi_1 + (1 - t)\varphi_0. \tag{4.8}$$

[2] Be careful, here $t \in (0, 1)$ instead of $[0, 1]$.

It follows that $\varphi_t \xrightarrow{L^1} \varphi_0$ as $t \to 0+$.

Let us come back to the general case. By Corollary 4.1.1, $\{\varphi_t : t \in (0,1)\}$ is a relatively compact subset of $\mathrm{PSH}(X,\theta)$ with respect to the L^1-topology.

Let ψ be an L^1-cluster point of φ_t as $t \searrow 0$, and it suffices to show that $\psi = \varphi_0$.

For each $M \in \mathbb{N}$, we write

$$\varphi_0^M = \varphi_0 \wedge (\varphi_1 + M).$$

Observe that $\varphi_0^M \in \mathcal{E}(X,\theta;\phi)$ by Proposition 3.1.21. Let $(\varphi_t^M)_{t\in(0,1)}$ be the geodesic from φ_0^M to φ_1. Then it is clear that $\varphi_t^M \leq \varphi_t$ for all $t \in (0,1)$. Therefore,

$$\psi \geq \varphi_0 \wedge (\varphi_1 + M)$$

almost everywhere hence everywhere by Proposition 1.2.6. On the other hand, by (4.6), $\psi \leq \varphi_0$. So it suffices to show that

$$\varphi_0 \wedge (\varphi_1 + M) \xrightarrow{L^1} \varphi_0$$

as $M \to \infty$, which is shown in Proposition 3.1.24.

Now we have shown that $(\varphi_t)_{t\in[0,1]}$ is a subgeodesic.

Next, take $0 \leq a \leq b \leq 1$. We want to show that the restriction $(\varphi_t)_{t\in[a,b]}$ is the geodesic from φ_a to φ_b. We may assume that $a < b$. The argument is the standard *balayage* argument.

Let $(\psi_t)_{t\in(a,b)}$ be the geodesic from φ_a to φ_b. Since $(\varphi_t)_{t\in[a,b]}$ is a subgeodesic by Proposition 4.1.4, we have $\psi_t \geq \varphi_t$ for all $t \in (a,b)$.

We define

$$\eta_t = \begin{cases} \psi_t, & \text{if } t \in (a,b), \\ \varphi_t, & \text{if } t \in (0,1) \setminus (a,b). \end{cases}$$

Extending it by setting $\psi_a = \varphi_a$ and $\psi_b = \varphi_b$, the first part of the proof applied to the endpoints φ_a, φ_b shows that $(\psi_t)_{t\in[a,b]}$ is a subgeodesic from φ_a to φ_b. Since $\psi_t \geq \varphi_t$ for $t \in (a,b)$, the gluing lemma Lemma 4.1.1 shows that $(\eta_t)_{t\in[0,1]}$ is a subgeodesic from φ_0 to φ_1.

Therefore, for all $t \in (0,1)$, we have

$$\varphi_t \geq \eta_t.$$

In particular, for $t \in (a,b)$, we have

$$\varphi_t \geq \eta_t = \psi_t \geq \varphi_t.$$

In other words, $(\varphi_t)_{t\in(a,b)} = (\psi_t)_{t\in(a,b)}$ is the geodesic from φ_a to φ_b.

Finally, assume furthermore that $\varphi_0, \varphi_1 \in \mathcal{E}^1(X,\theta;\phi)$ (resp. $\mathcal{E}^\infty(X,\theta;\phi)$). Thanks to (4.7), Proposition 3.1.21 and Proposition 3.1.22, we find $\varphi_t \in \mathcal{E}^1(X,\theta;\phi)$ (resp. $\mathcal{E}^\infty(X,\theta;\phi)$) for all $t \in (0,1)$. □

Lemma 4.2.2 *Let $\varphi_0, \varphi_1 \in \mathrm{PSH}(X, \theta)$ with $\varphi_1 \leq \varphi_0$. Let $(\varphi_t)_{t\in(0,1)}$ be the geodesic from φ_0 to φ_1. Assume that $(\varphi_t)_{t\in[0,1]}$ is also a subgeodesic. Then φ_t is decreasing in $t \in [0, 1]$.*

Proof Fix $0 < r < s \leq 1$ and put $\delta = (s - r)/(1 - r)$. Then $(\varphi_{\delta+(1-\delta)t})_{t\in[0,1]}$ is a subgeodesic from φ_δ to φ_1. Since

$$\varphi_\delta \leq (1 - \delta)\varphi_0 + \delta\varphi_1 \leq \varphi_0,$$

we have $\varphi_{\delta+(1-\delta)t} \leq \varphi_t$ for all $t \in (0, 1)$ by the definition of geodesic. In particular, $\varphi_s = \varphi_{\delta+(1-\delta)r} \leq \varphi_r$.

It remains to prove that for any $t \in (0, 1)$, we have $\varphi_t \leq \varphi_0$, but this is an immediate consequence of Proposition 4.1.2:

$$\varphi_t \leq t\varphi_1 + (1 - t)\varphi_0 \leq \varphi_0.$$

Proposition 4.2.2 *Let $\varphi_1, \varphi_0 \in \mathcal{E}(X, \theta; \phi)$ with $\varphi_1 \leq \varphi_0$. Let $(\varphi_t)_{t\in(0,1)}$ be the geodesic from φ_0 to φ_1. Then*

$$s \sup_{\{\varphi_0\neq-\infty\}} (\varphi_1 - \varphi_0) = \sup_{\{\varphi_0\neq-\infty\}} (\varphi_s - \varphi_0) \tag{4.9}$$

for all $s \in [0, 1]$.

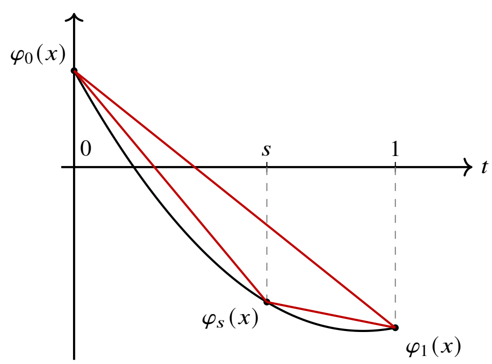

Fig. 4.1 Typical behaviour of $t \mapsto \varphi_t(x)$

Proof The notations in the proof are indicated in Fig. 4.1.[3]

We may assume that $s \in (0, 1)$ since there is nothing to prove when $s = 0$ or $s = 1$.

After replacing φ_t by $\varphi_t - C't$ for some large enough $C' > 0$, we may assume that $\varphi_1 \leq \varphi_0$. This procedure preserves the geodesic property by Example 4.2.1. Then φ_t is decreasing in t by Lemma 4.2.2.

[3] When dealing with convex functions, drawing a picture is the easiest way to keep track of the directions of inequalities.

Let

$$C = \sup_{\{\varphi_1 \neq -\infty\}} (\varphi_1 - \varphi_0) \leq 0. \tag{4.10}$$

Then by Proposition 1.2.6, we have

$$\varphi_1 \leq \varphi_0 + C.$$

So $(\varphi_1 - C(1-t))_{t\in(0,1)}$ is a candidate in (4.5) and hence

$$\varphi_1 - C(1-t) \leq \varphi_t, \quad t \in (0,1). \tag{4.11}$$

By Proposition 4.2.1, we have $\varphi_t \xrightarrow{L^1} \varphi_1$ as $t \to 1-$. Since φ_t is decreasing in $t \in (0,1)$, it follows that $\varphi_1 = \inf_{t\in(0,1)} \varphi_t$. Therefore, we can find a pluripolar set $Z \subseteq X$ such that $\varphi_t(x) \to \varphi_1(x) > -\infty$ as $t \to 1-$ for all $x \in X \setminus Z$.

Similarly, since $\varphi_0 = \sup_{t\in(0,1)}{}^* \varphi_t$, after enlarging Z, we may also guarantee that $\varphi_t(x) \longrightarrow \varphi_0(x) > -\infty$ as $t \to 0+$ for all $x \in X \setminus Z$ by Proposition 1.2.5.

For any such $x \in X \setminus Z$, the function $t \mapsto \varphi_t(x)$ is a real-valued continuous convex function on $[0,1]$. In particular, $t \mapsto \varphi_t(x)$ is absolutely continuous on $[0,1]$. Hence, for any $s \in [0,1)$, we have

$$\varphi_1(x) - \varphi_s(x) = \int_s^1 \frac{\mathrm{d}}{\mathrm{d}t}\varphi_t(x)\,\mathrm{d}t \leq (1-s) \lim_{t\to 1-} \frac{\varphi_1(x) - \varphi_t(x)}{1-t} \leq (1-s)C, \tag{4.12}$$

where the second inequality follows from (4.11).

Taking supremum in (4.12), we find that

$$\sup_{X\setminus Z} (\varphi_1 - \varphi_s) \leq (1-s) \sup_{x\in X\setminus Z} \lim_{t\to 1-} \frac{\varphi_1(x) - \varphi_t(x)}{1-t} \leq (1-s)C. \tag{4.13}$$

When $s = 0$, we deduce from Corollary 1.3.5 and (4.10) that

$$C = \sup_{\{\varphi_1 \neq -\infty\}} (\varphi_1 - \varphi_0) = \sup_{x\in X\setminus Z} \lim_{t\to 1-} \frac{\varphi_1(x) - \varphi_t(x)}{1-t}.$$

But this equality works equally well for the geodesic $(\varphi_{(1-s)t+s})_{t\in[0,1]}$. It follows that

$$\sup_{\{\varphi_1 \neq -\infty\}} (\varphi_1 - \varphi_s) = (1-s) \sup_{x\in X\setminus Z} \lim_{t\to 1-} \frac{\varphi_1(x) - \varphi_t(x)}{1-t} = (1-s)C.$$

Therefore, invoking Corollary 1.3.5 again, we deduce that all inequalities in (4.13) are in fact equalities. In other words,

$$\sup_{\{\varphi_1 \neq -\infty\}} (\varphi_1 - \varphi_0) = \sup_{x\in X\setminus Z} \lim_{t\to 1-} \frac{\varphi_1(x) - \varphi_t(x)}{1-t} = \sup_{\{\varphi_1 \neq -\infty\}} \frac{\varphi_1 - \varphi_s}{1-s}. \tag{4.14}$$

On the other hand, we have the trivial inequality when $s \in (0,1)$:

$$\sup_{\{\varphi_1\neq-\infty\}}(\varphi_1-\varphi_0)\leq s\sup_{\{\varphi_1\neq-\infty\}}\frac{\varphi_s-\varphi_0}{s}+(1-s)\sup_{\{\varphi_1\neq-\infty\}}\frac{\varphi_1-\varphi_s}{1-s}.$$

Together with (4.14), we find that for $s\in(0,1)$,

$$\sup_{\{\varphi_1\neq-\infty\}}(\varphi_1-\varphi_0)\leq\sup_{\{\varphi_1\neq-\infty\}}\frac{\varphi_s-\varphi_0}{s}.$$

The reverse inequality follows from the convexity, and hence for $s\in(0,1)$, we have

$$\sup_{\{\varphi_1\neq-\infty\}}\frac{\varphi_s-\varphi_0}{s}=\sup_{\{\varphi_1\neq-\infty\}}(\varphi_1-\varphi_0).$$

Using Corollary 1.3.5, we conclude (4.9). □

With an almost identical proof, we find

Proposition 4.2.3 *Let $\varphi_1,\varphi_0\in\mathcal{E}^\infty(X,\theta;\phi)$. Let $(\varphi_t)_{t\in(0,1)}$ be the geodesic from φ_0 to φ_1. Then*

$$t\inf_{\{\phi\neq-\infty\}}(\varphi_1-\varphi_0)=\inf_{\{\phi\neq-\infty\}}(\varphi_t-\varphi_0)$$

for all $t\in(0,1]$.

Definition 4.2.2 Let $\ell=(\ell_t)_{t\geq0}$ be a family in $\mathcal{E}(X,\theta;\phi)$. We say ℓ is a *geodesic ray* in $\mathcal{E}(X,\theta;\phi)$ emanating from ℓ_0 if for each $0\leq a\leq b$, the restriction $(\ell_t)_{t\in[a,b]}$ is a geodesic.

The set of geodesic rays in $\mathcal{E}(X,\theta;\phi)$ emanating from ϕ is denoted by $\mathcal{R}(X,\theta;\phi)$.

We say a geodesic ray $\ell\in\mathcal{R}(X,\theta;\phi)$ has *finite energy* if $\ell_t\in\mathcal{E}^1(X,\theta;\phi)$ for all $t>0$. The set of geodesic rays with finite energy is denoted by $\mathcal{R}^1(X,\theta;\phi)$.

We say a geodesic ray $\ell\in\mathcal{R}(X,\theta;\phi)$ is *bounded* if $\ell_t\in\mathcal{E}^\infty(X,\theta;\phi)$ for all $t\geq0$. The set of bounded geodesic rays is denoted by $\mathcal{R}^\infty(X,\theta;\phi)$.

Given $\ell,\ell'\in\mathcal{R}(X,\theta;\phi)$, we write $\ell\leq\ell'$ if $\ell_t\leq\ell'_t$ for each $t\geq0$.

When $\phi=V_\theta$, we usually omit it from the notations and write $\mathcal{R}(X,\theta)$, $\mathcal{R}^1(X,\theta)$ and $\mathcal{R}^\infty(X,\theta)$ respectively.

Proposition 4.2.4 *Let $\ell\in\mathcal{R}(X,\theta;\phi)$. Then there is a constant $C\in\mathbb{R}$ such that*

$$\sup_X\ell_t=Ct,\quad t\geq0.$$

Proof It follows from Proposition 4.2.2 that

$$\sup_{\{\phi\neq-\infty\}}(\ell_t-\phi)=t\sup_{\{\phi\neq-\infty\}}(\ell_1-\phi)$$

for all $t\geq0$.

It suffices to show that for any $t\geq0$,

$$\sup_{\{\phi\neq-\infty\}}(\ell_t-\phi)=\sup_X\ell_t.$$

This was already proved in Proposition 3.1.4. □

4.3 The metrics on the spaces of potentials and rays

Let X be a connected compact Kähler manifold of dimension n and θ be a smooth closed real $(1,1)$-form on X representing a big cohomology class. Fix a model potential $\phi \in \mathrm{PSH}(X,\theta)_{>0}$.

We first study a natural metric on $\mathcal{E}^1(X,\theta;\phi)$.

Definition 4.3.1 Let $\varphi,\psi \in \mathcal{E}^1(X,\theta;\phi)$, we define the *Darvas metric* d_1 as follows:

$$d_1(\varphi,\psi) = E^{\phi}_{\theta}(\varphi) + E^{\phi}_{\theta}(\psi) - 2E^{\phi}_{\theta}(\varphi \wedge \psi).$$

Note that by Proposition 3.1.21, $\varphi \wedge \psi \in \mathcal{E}^1(X,\theta;\phi)$.

Recall that E^{ϕ}_{θ} is defined in Definition 3.1.6.

In particular, if $\varphi \leq \psi$, we have

$$d_1(\varphi,\psi) = E^{\phi}_{\theta}(\psi) - E^{\phi}_{\theta}(\varphi). \tag{4.15}$$

We wish to show that d_1 is a complete metric. We first prove a contraction property:

Proposition 4.3.1 *Let $\varphi,\psi,\gamma \in \mathcal{E}^1(X,\theta;\phi)$. Then*

$$d_1(\varphi,\psi) \geq d_1(\varphi \wedge \gamma, \psi \wedge \gamma). \tag{4.16}$$

Proof **Step 1**. We first assume that $\varphi \geq \psi$. Then

$$\begin{aligned} d_1(\varphi,\psi) =& E^{\phi}_{\theta}(\varphi) - E^{\phi}_{\theta}(\psi) \\ \geq& E^{\phi}_{\theta}\Big((\varphi\wedge\gamma)\vee\psi\Big) - E^{\phi}_{\theta}(\psi) \qquad \text{by Lemma 3.1.3} \\ \geq& E^{\phi}_{\theta}(\varphi\wedge\gamma) - E^{\phi}_{\theta}(\psi\wedge\gamma) \qquad \text{by Corollary 3.1.6.} \end{aligned}$$

Step 2. We prove the general case.

By Step 1, we have

$$d_1(\varphi,\varphi\wedge\psi) \geq d_1(\varphi\wedge\gamma,\varphi\wedge\psi\wedge\gamma), \quad d_1(\psi,\varphi\wedge\psi) \geq d_1(\psi\wedge\gamma,\varphi\wedge\psi\wedge\gamma).$$

Adding the two inequalities together, we conclude (4.16). □

Lemma 4.3.1 *The function d_1 is a metric on $\mathcal{E}^1(X,\theta;\phi)$.*

Proof There are two facts to prove, as in the two steps below.

Step 1. Let $\varphi,\psi \in \mathcal{E}^1(X,\theta;\phi)$. Assume that $d_1(\varphi,\psi) = 0$. Then we will show that $\varphi = \psi$.

We first treat the case $\psi \leq \varphi$. Then $d_1(\varphi,\psi) = E_\theta^\phi(\varphi) - E_\theta^\phi(\psi) = 0$, and Proposition 3.1.18 gives

$$\int_X (\varphi - \psi)\, \theta_\psi^n = 0.$$

We conclude that $\psi = \varphi$ using the domination principle Theorem 2.4.5.

In general, put $\gamma = \varphi \wedge \psi$. Since

$$d_1(\varphi,\psi) = d_1(\varphi,\gamma) + d_1(\psi,\gamma),$$

both terms on the right vanish. Applying the comparable case to the pairs (φ,γ) and (ψ,γ) gives $\varphi = \gamma = \psi$.

Step 2. Let $\varphi,\psi,\gamma \in \mathcal{E}^1(X,\theta;\phi)$. We prove the triangle inequality:

$$d_1(\varphi,\psi) \leq d_1(\varphi,\gamma) + d_1(\psi,\gamma).$$

This can be translated to

$$E_\theta^\phi(\varphi\wedge\gamma) - E_\theta^\phi(\varphi\wedge\psi) \leq E_\theta^\phi(\gamma) - E_\theta^\phi(\psi\wedge\gamma).$$

We just have to compute using Proposition 4.3.1:

$$\begin{aligned} E_\theta^\phi(\gamma) - E_\theta^\phi(\psi\wedge\gamma) \geq& E_\theta^\phi(\varphi\wedge\gamma) - E_\theta^\phi(\varphi\wedge\psi\wedge\gamma) \\ \geq& E_\theta^\phi(\varphi\wedge\gamma) - E_\theta^\phi(\varphi\wedge\psi), \end{aligned}$$

where the second line follows from Lemma 3.1.3. □

We introduce an auxiliary functional.

Definition 4.3.2 Define $I_\theta \colon \mathcal{E}^1(X,\theta;\phi)\times\mathcal{E}^1(X,\theta;\phi) \to \mathbb{R}$ as follows:

$$I_\theta(\varphi,\psi) \coloneqq \int_X |\varphi-\psi| \left(\theta_\varphi^n + \theta_\psi^n\right). \tag{4.17}$$

We observe the following elementary equality.

Lemma 4.3.2 *Let $\varphi,\psi \in \mathcal{E}^1(X,\theta;\phi)$. Then*

$$I_\theta(\varphi,\psi) = I_\theta(\varphi\vee\psi,\varphi) + I_\theta(\varphi\vee\psi,\psi).$$

Proof By the plurifine locality of non-pluripolar products Proposition 2.2.1 (1), we can write

$$
\begin{aligned}
I_\theta(\varphi,\psi) &= \int_{\{\varphi<\psi\}} (\psi-\varphi)\left(\theta_\varphi^n+\theta_\psi^n\right) + \int_{\{\varphi>\psi\}} (\varphi-\psi)\left(\theta_\varphi^n+\theta_\psi^n\right),\\
I_\theta(\varphi\vee\psi,\varphi) &= \int_{\{\psi>\varphi\}} (\psi-\varphi)\left(\theta_\psi^n+\theta_\varphi^n\right),\\
I_\theta(\varphi\vee\psi,\psi) &= \int_{\{\psi<\varphi\}} (\varphi-\psi)\left(\theta_\psi^n+\theta_\varphi^n\right).
\end{aligned}
$$

The desired equality follows by comparing these terms. □

We have an interesting relation between d_1 and I_θ defined in Definition 4.3.2.

Theorem 4.3.1 *Let $\varphi,\psi\in\mathcal{E}^1(X,\theta;\phi)$. Then*

$$
\frac{1}{C_n} I_\theta(\varphi,\psi) \le d_1(\varphi,\psi) \le I_\theta(\varphi,\psi), \tag{4.18}
$$

where $C_n = 3\cdot 2^{n+2}(n+1)$.

Proof **Step 1**. We first prove the right-hand part of (4.18).

Thanks to Proposition 3.1.19 and Lemma 3.1.1, we have

$$
\begin{aligned}
E_\theta^\phi(\varphi) - E_\theta^\phi(\varphi\wedge\psi) &\le \int_X (\varphi-\varphi\wedge\psi)\,\theta_{\varphi\wedge\psi}^n\\
&\le \int_{\{\psi=\varphi\wedge\psi\}} (\varphi-\psi)\,\theta_\psi^n\\
&\le \int_X |\varphi-\psi|\,\theta_\psi^n.
\end{aligned}
$$

Similarly,

$$
E_\theta^\phi(\psi) - E_\theta^\phi(\varphi\wedge\psi) \le \int_X |\varphi-\psi|\,\theta_\varphi^n.
$$

Adding these inequalities up, we find

$$
d_1(\varphi,\psi) \le I_\theta(\varphi,\psi).
$$

Step 2. We prove the left-hand part of (4.18).

We claim that

$$
d_1\left(\varphi, \frac{\varphi+\psi}{2}\right) \le \frac{3(n+1)}{2} d_1(\varphi,\psi). \tag{4.19}
$$

For this purpose, we compute directly

$$
\begin{aligned}
& d_1\left(\varphi, \frac{\varphi+\psi}{2}\right) \\
=& d_1\left(\varphi, \varphi \wedge \frac{\varphi+\psi}{2}\right) + d_1\left(\frac{\varphi+\psi}{2}, \varphi \wedge \frac{\varphi+\psi}{2}\right) \\
\leq& d_1\left(\varphi, \varphi \wedge \psi\right) + d_1\left(\frac{\varphi+\psi}{2}, \varphi \wedge \psi\right) && \text{by Lemma 3.1.3} \\
\leq& \int_X (\varphi - \varphi \wedge \psi)\ \theta^n_{\varphi\wedge\psi} + \int_X \left(\frac{\varphi+\psi}{2} - \varphi \wedge \psi\right)\ \theta^n_{\varphi\wedge\psi} && \text{by Proposition 3.1.19} \\
=& \frac{3}{2}\int_X (\varphi - \varphi \wedge \psi)\ \theta^n_{\varphi\wedge\psi} + \frac{1}{2}\int_X (\psi - \varphi \wedge \psi)\ \theta^n_{\varphi\wedge\psi} \\
\leq& \frac{3(n+1)}{2} d_1\left(\varphi, \varphi \wedge \psi\right) + \frac{n+1}{2} d_1(\psi, \varphi \wedge \psi) && \text{by (3.35)} \\
\leq& \frac{3(n+1)}{2} d_1(\varphi, \psi),
\end{aligned}
$$

and (4.19) follows.

We now estimate the left-hand side of (4.19):

$$
\begin{aligned}
d_1\left(\varphi, \frac{\varphi+\psi}{2}\right) \geq& d_1\left(\varphi, \varphi \wedge \frac{\varphi+\psi}{2}\right) \\
\geq& \int_X \left(\varphi - \varphi \wedge \frac{\varphi+\psi}{2}\right)\ \theta^n_{\varphi} \\
\geq& \int_{\{\varphi>\psi\}} \left(\varphi - \varphi \wedge \frac{\varphi+\psi}{2}\right)\ \theta^n_{\varphi} \\
\geq& \int_{\{\varphi>\psi\}} \left(\varphi - \frac{\varphi+\psi}{2}\right)\ \theta^n_{\varphi} \\
=& \frac{1}{2}\int_{\{\varphi>\psi\}} (\varphi - \psi)\ \theta^n_{\varphi},
\end{aligned}
$$

as a consequence of Proposition 3.1.19.

Similarly,

$$
\begin{aligned}
d_1\left(\varphi, \frac{\varphi+\psi}{2}\right) \geq & d_1\left(\frac{\varphi+\psi}{2}, \varphi \wedge \frac{\varphi+\psi}{2}\right) \\
\geq & \int_X \left(\frac{\varphi+\psi}{2} - \varphi \wedge \frac{\varphi+\psi}{2}\right) \theta^n_{(\varphi+\psi)/2} \\
\geq & 2^{-n} \int_X \left(\frac{\varphi+\psi}{2} - \varphi \wedge \frac{\varphi+\psi}{2}\right) \theta^n_{\varphi} \\
\geq & 2^{-n} \int_{\{\varphi<\psi\}} \left(\frac{\varphi+\psi}{2} - \varphi \wedge \frac{\varphi+\psi}{2}\right) \theta^n_{\varphi} \\
\geq & 2^{-n} \int_{\{\varphi<\psi\}} \left(\frac{\varphi+\psi}{2} - \varphi\right) \theta^n_{\varphi} \\
= & 2^{-n-1} \int_{\{\varphi<\psi\}} (\psi-\varphi)\, \theta^n_{\varphi}.
\end{aligned}
$$

Adding these estimates up, we find

$$
2^{n+2} d_1\left(\varphi, \frac{\varphi+\psi}{2}\right) \geq \left(2^{n+1}+2\right) d_1\left(\varphi, \frac{\varphi+\psi}{2}\right) \geq \int_X |\varphi-\psi|\, \theta^n_{\varphi}.
$$

Therefore, combined with (4.19) we conclude

$$
3 \cdot 2^{n+1}(n+1) d_1(\varphi, \psi) \geq \int_X |\varphi-\psi|\, \theta^n_{\varphi}.
$$

By symmetry, we also get a similar expression after exchanging φ and ψ. Adding these inequalities together, we find

$$
3 \cdot 2^{n+2}(n+1) d_1(\varphi, \psi) \geq I_\theta(\varphi, \psi).
$$

The left-hand part of (4.18) then follows. □

Corollary 4.3.1 *Let $\varphi, \psi \in \mathcal{E}^1(X, \theta; \phi)$. Then*

$$
d_1(\varphi \vee \psi, \varphi) \leq C_n d_1(\varphi, \psi), \tag{4.20}
$$

where C_n is the constant in Theorem 4.3.1.

Proof By Theorem 4.3.1 and Lemma 4.3.2, we have

$$
d_1(\varphi \vee \psi, \varphi) \leq I_\theta(\varphi \vee \psi, \varphi) \leq I_\theta(\varphi, \psi) \leq C_n d_1(\varphi, \psi).
$$

This completes the proof. □

Lemma 4.3.3 *There is $A, B > 0$ so that for any $\varphi \in \mathcal{E}^1(X, \theta; \phi)$, we have*

$$
d_1(\varphi, \phi) \geq -\int_X \theta^n_{\phi} \cdot \sup_X \varphi \geq -A d_1(\varphi, \phi) - B. \tag{4.21}
$$

Proof When $\sup_X \varphi \leq 0$, the right-hand part of (4.21) is trivial. While as $\varphi \leq \phi$, we find

$$d_1(\phi, \varphi) = -E_\theta^\phi(\varphi) \geq -\int_X \theta_\phi^n \cdot \sup_{\{\phi \neq -\infty\}} (\varphi - \phi) = -\int_X \theta_\phi^n \cdot \sup_X \varphi,$$

where the last equality follows from Proposition 3.1.4.

We can therefore assume that $\sup_X \varphi > 0$. In this case, the left-hand part of (4.21) is trivial. Take a Kähler form $\omega \geq \theta$ on X. Then thanks to Theorem 3.1.1, we can find a constant $C > 0$ so that

$$\theta_\phi^n \leq C\omega^n.$$

Thanks to Proposition 1.5.1, there is a constant $C' > 0$, independent of the choice of φ, so that

$$\int_X \left(\phi - \varphi + \sup_X \varphi\right) \theta_\phi^n \leq C'.$$

We estimate

$$\begin{aligned} I_\theta(\phi, \varphi) &\geq \int_X |\varphi - \phi| \, \theta_\phi^n \\ &\geq \left(\sup_X \varphi\right) \int_X \theta_\phi^n - \int_X \left(\phi - \varphi + \sup_X \varphi\right) \theta_\phi^n \\ &\geq \left(\sup_X \varphi\right) \int_X \theta_\phi^n - C'. \end{aligned}$$

The right-hand part of (4.21) then follows from Theorem 4.3.1. □

Next we handle the completeness of d_1. The completeness can be proved in a more general framework.

Definition 4.3.3 Let E be a set. A *pre-rooftop structure* on E is a binary operator $\wedge : E \times E \to E$, satisfying the following axioms: For $x, y, z \in E$,

(1) $x \wedge y = y \wedge x$.
(2) $(x \wedge y) \wedge z = x \wedge (y \wedge z)$.
(3) $x \wedge x = x$.

We call $(E, \wedge)$ a *pre-rooftop space*.

A pre-rooftop structure $\wedge$ defines a partial order $\leq$ on E as follows:

$$x \leq y \quad \text{if and only if} \quad x \wedge y = x.$$

Here by abuse of notation, we use $\leq$ to denote the partial order.

In particular, it makes sense to talk about increasing and decreasing sequences in E.

Definition 4.3.4 Let (E, d) be a metric space. A *pre-rooftop structure* on (E, d) is a pre-rooftop structure $\wedge$ on E. We say $(E, d, \wedge)$ is a *pre-rooftop metric space*.

A *rooftop structure* on (E, d) is a pre-rooftop structure $\wedge$ on E such that

$$d(x \wedge z, y \wedge z) \leq d(x, y), \quad \forall x, y, z \in E. \tag{4.22}$$

We call $(E, d, \wedge)$ a *rooftop metric space*.

Lemma 4.3.4 *Let $(E, d, \wedge)$ be a rooftop metric space. Let $x, y, x', y' \in E$. Then*

$$d(x \wedge y, x' \wedge y') \leq d(x, x') + d(y, y'). \tag{4.23}$$

Proof We compute

$$d(x \wedge y, x' \wedge y') \leq d(x \wedge y, x \wedge y') + d(x \wedge y', x' \wedge y') \leq d(x, x') + d(y, y').$$

This completes the proof. □

Proposition 4.3.2 *Let $(E, d, \wedge)$ be a rooftop metric space. Then (E, d) is complete if and only if both of the following hold:*

(1) *Each increasing Cauchy sequence converges.*
(2) *Each decreasing Cauchy sequence converges.*

Proof The direct implication is trivial.

Conversely, assume that both conditions are true. Let $(x_j)_{j>0}$ be a Cauchy sequence in E. We want to prove that $(x_j)_j$ converges. By passing to a subsequence, we may assume that

$$d(x_j, x_{j+1}) \leq 2^{-j}.$$

For $k, j \geq 1$, let

$$y_j^k := x_k \wedge \cdots \wedge x_{k+j}.$$

Then $(y_j^k)_j$ is decreasing, and

$$d(y_j^k, y_{j+1}^k) \leq d(x_{k+j}, x_{k+j+1}) \leq 2^{-k-j}.$$

So $(y_j^k)_j$ is a decreasing Cauchy sequence. Define

$$y^k := \lim_{j\to\infty} y_j^k.$$

Then

$$d(y^k, y^{k+1}) = \lim_{j\to\infty} d(y_{j+1}^k, y_j^{k+1}) \leq d(x_k, x_{k+1}) \leq 2^{-k}.$$

So y^k is an increasing Cauchy sequence. Let

$$y := \lim_{k\to\infty} y^k.$$

For every $j \geq 0$, we have

$$d(y_j^k, x_k) \leq \sum_{r=k}^{k+j-1} d(x_r, x_{r+1}),$$

with the empty sum understood as 0. Indeed, this is trivial for $j = 0$, and the induction step follows from

$$y_j^k = x_k \wedge y_{j-1}^{k+1}$$

and

$$d(y_j^k, x_k) \leq d(y_{j-1}^{k+1}, x_k) \leq d(y_{j-1}^{k+1}, x_{k+1}) + d(x_{k+1}, x_k).$$

Letting $j \to \infty$, we get

$$d(y^k, x_k) \leq \sum_{r=k}^{\infty} d(x_r, x_{r+1}) \leq 2^{1-k}.$$

So $(x_k)_k$ converges to y. □

Theorem 4.3.2 *The metric space $(\mathcal{E}^1(X, \theta; \phi), d_1)$ is complete.*

It follows from the proof below that if $(\varphi_j)_{j>0}$ is a monotone sequence in $\mathcal{E}^1(X, \theta; \phi)$ with d_1-limit φ, then φ is the almost everywhere limit of $(\varphi_j)_{j>0}$. We will use this result without further explanation in the sequel.

Proof As we have seen in Lemma 4.3.1 and Proposition 4.3.1, the triple

$$\left(\mathcal{E}^1(X, \theta; \phi), d_1, \wedge\right)$$

is a rooftop metric space. Hence thanks to Proposition 4.3.2, it remains to show that each increasing or decreasing Cauchy sequence in $\mathcal{E}^1(X, \theta; \phi)$ converges.

We first consider an increasing sequence $(\varphi_i)_{i>0}$ in $\mathcal{E}^1(X, \theta; \phi)$. The Cauchy property simply means that

$$\sup_{i>0} E_\theta^\phi(\varphi_i) < \infty.$$

We claim that

$$\varphi \coloneqq \sup_{i>0}{}^* \varphi_i$$

is the d_1-limit of the sequence. We first observe that $(\sup_X \varphi_i)_{i>0}$ is bounded, as a consequence of Lemma 4.3.3. Therefore, Proposition 3.1.23 guarantees that $\varphi \in \mathcal{E}^1(X, \theta; \phi)$, and hence our assertion follows from Proposition 3.1.20.

Next we consider a decreasing sequence $(\varphi_i)_{i>0}$ in $\mathcal{E}^1(X, \theta; \phi)$. The Cauchy property simply means that

$$\inf_{i>0} E_\theta^\phi(\varphi_i) > -\infty.$$

We claim that

$$\varphi \coloneqq \inf_{i>0} \varphi_i$$

is the d_1-limit of the sequence. Equivalently, we need to show that $\varphi \in \mathcal{E}^1(X, \theta; \phi)$ and

$$E_\theta^\phi(\varphi) = \lim_{i\to\infty} E_\theta^\phi(\varphi_i). \tag{4.24}$$

We first observe that $(\sup_X \varphi_i)_{i>0}$ is bounded, as a consequence of Lemma 4.3.3. Therefore, $\varphi \in \mathrm{PSH}(X,\theta;\phi)$ and $\varphi_i \xrightarrow{L^1} \varphi$. We claim that $\varphi \in \mathcal{E}^1(X,\theta;\phi)$. For this purpose, we may assume that $\varphi_1 \le 0$. Since $(\varphi_i)_i$ is decreasing, $\varphi_i \le \varphi_1 \le 0$ for all $i > 0$. As $\varphi_i \le \phi$ and ϕ is model, each φ_i is a candidate for $P_\theta[\phi] = \phi$, so $\varphi_i \le \phi$. Thus, using the energy formula (3.26) and the fact that all mixed terms are non-positive, we get for each $i > 0$,

$$\frac{1}{n+1}\int_X (\varphi_i - \phi)\ \theta_{\varphi_i}^n \ge E_\theta^\phi(\varphi_i) \ge \inf_{j>0} E_\theta^\phi(\varphi_j) > -\infty.$$

We can then apply Lemma 3.1.9, with $\gamma = \phi$, to conclude that $\varphi \in \mathcal{E}^1(X,\theta;\phi)$. Our assertion (4.24) then follows from Proposition 3.1.16. We conclude the proof. □

Lemma 4.3.5 *Let $(\varphi_i)_{i>0}$ be a sequence in $\mathcal{E}^1(X,\theta;\phi)$ with d_1-limit $\varphi \in \mathcal{E}^1(X,\theta;\phi)$. Then after replacing $(\varphi_i)_{i>0}$ by a subsequence, we can find two sequences $(\psi_i)_{i>0}$ and $(\eta_i)_{i>0}$ in $\mathcal{E}^1(X,\theta;\phi)$ such that*

(1) *$(\psi_i)_{i>0}$ is decreasing with d_1 and pointwise limit φ;*
(2) *$(\eta_i)_{i>0}$ is increasing with d_1 and almost everywhere limit φ.*
(3) *$\eta_i \le \varphi_i \le \psi_i$ for all $i > 0$.*

Proof We first note that as a consequence of Lemma 4.3.3, $(\sup_X \varphi_i)_{i>0}$ is bounded. Let C_n be the constant in Theorem 4.3.1. After replacing $(\varphi_i)_{i>0}$ by a subsequence, we may assume that

$$d_1(\varphi_i, \varphi_{i+1}) \le \min\{C_n^{-2i}, 2^{-i}\}.$$

We first construct $(\psi_i)_{i>0}$. For this purpose, it suffices to define

$$\psi_i \coloneqq \sup_{j\ge i}{}^* \varphi_j$$

for each $i > 0$. It follows from Proposition 3.1.23 that $\psi_i \in \mathcal{E}^1(X,\theta;\phi)$ for each $i > 0$. For $k \ge 0$, set

$$\psi_i^k \coloneqq \varphi_i \vee \varphi_{i+1} \vee \cdots \vee \varphi_{i+k}.$$

Iterating Corollary 4.3.1 gives

$$d_1(\varphi_i, \psi_i^k) \le \sum_{a=i}^{i+k-1} C_n^{a+1-i} d_1(\varphi_a, \varphi_{a+1}) \le \sum_{a=i}^{i+k-1} C_n^{a+1-i} C_n^{-2a}.$$

Since ψ_i^k increases almost everywhere to ψ_i, letting $k \to \infty$ and using Proposition 3.1.20, we get

$$d_1(\varphi_i, \psi_i) \le \frac{C_n^{1-2i}}{1 - C_n^{-1}}.$$

Thus $\psi_i \to \varphi$ in d_1. Since $(\psi_i)_i$ is decreasing, the monotone case in the proof of Theorem 4.3.2 shows that ψ_i converges almost everywhere to φ, hence pointwise by Proposition 1.2.6.

Next we construct $(\eta_i)_{i>0}$. For each $i > 0$ and $k \geq 0$, we let

$$\eta_i^k \coloneqq \varphi_i \wedge \varphi_{i+1} \wedge \cdots \wedge \varphi_{i+k}.$$

Then

$$\begin{aligned} d_1(\varphi_i, \eta_i^k) \leq & \sum_{j=0}^{k-1} d_1\left(\eta_i^j, \eta_i^{j+1}\right) \\ \leq & \sum_{j=0}^{k-1} d_1\left(\varphi_{i+j}, \varphi_{i+j+1}\right) \quad \text{by Proposition 4.3.1} \\ \leq & \sum_{j=0}^{k-1} 2^{-i-j} \\ \leq & 2^{1-i}. \end{aligned}$$

Since $\eta_i^k \leq \varphi_i$, the preceding estimate gives

$$E_\theta^\phi(\eta_i^k) \geq E_\theta^\phi(\varphi_i) - 2^{1-i}, \quad k \geq 0.$$

The sequence $(\eta_i^k)_{k\geq 0}$ is decreasing, and the monotonicity of E_θ^ϕ shows that the sequence $\left(E_\theta^\phi(\eta_i^k)\right)_{k\geq 0}$ is decreasing and bounded from below, hence convergent. Thus, for $k \leq l$,

$$d_1(\eta_i^k, \eta_i^l) = E_\theta^\phi(\eta_i^k) - E_\theta^\phi(\eta_i^l),$$

and $(\eta_i^k)_{k\geq 0}$ is a decreasing Cauchy sequence. Therefore, the monotone case in the proof of Theorem 4.3.2 shows that

$$\eta_i \coloneqq \lim_{k\to\infty} \eta_i^k = \inf_{k\geq 0} \eta_i^k$$

is the d_1-limit of $(\eta_i^k)_{k\geq 0}$ and Proposition 3.1.20 shows that

$$d_1(\varphi_i, \eta_i) \leq 2^{1-i}.$$

By construction, $\eta_i = \inf_{j\geq i} \varphi_j$, hence $\eta_i \leq \varphi_i$ and $(\eta_i)_i$ is increasing. Therefore, φ is the d_1-limit of the increasing sequence $(\eta_i)_{i>0}$. As we have seen in the proof of Theorem 4.3.2, this implies that φ is also the almost everywhere limit of $(\eta_i)_{i>0}$. □

Theorem 4.3.3 *The functional $E_\theta^\phi\colon \mathcal{E}^1(X, \theta; \phi) \to \mathbb{R}$ is continuous.*

Proof Let $(\varphi_j)_{j>0}$ be a sequence in $\mathcal{E}^1(X, \theta; \phi)$ with d_1-limit $\varphi \in \mathcal{E}^1(X, \theta; \phi)$. We wish to show that

$$\lim_{j\to\infty} E_\theta^\phi(\varphi_j) = E_\theta^\phi(\varphi). \tag{4.25}$$

It suffices to prove that every subsequence has a further subsequence along which (4.25) holds. Thus, after replacing $(\varphi_j)_{j>0}$ by a subsequence, let $(\eta_j)_j$ and $(\psi_j)_j$ be the monotone sequences supplied by Lemma 4.3.5. From the construction, $\eta_j \le \varphi_j \le \psi_j$. By Proposition 3.1.20,

$$E^{\phi}_{\theta}(\eta_j) \to E^{\phi}_{\theta}(\varphi), \quad E^{\phi}_{\theta}(\psi_j) \to E^{\phi}_{\theta}(\varphi).$$

The monotonicity of E^{ϕ}_{θ} then squeezes $E^{\phi}_{\theta}(\varphi_j)$, proving (4.25). □

Next we recall two particular properties when $\phi = V_\theta$.[4]

Proposition 4.3.3 *Let $(\varphi_t)_{t\in[a,b]}$ be a geodesic in $\mathcal{E}^1(X,\theta)$. Then $t \mapsto E_\theta(\varphi_t)$ is a linear function of $t \in [a,b]$.*

See [DDNL18c, Theorem 3.12].

Proposition 4.3.4 *Let $(\varphi_0^i)_{i\in I}$, $(\varphi_1^i)_{i\in I}$ be two uniformly bounded from above increasing nets in $\mathcal{E}^\infty(X,\theta)$. Let $(\varphi_t^i)_{t\in(0,1)}$ be the geodesic from φ_0^i to φ_1^i for each $i \in I$. Then*

$$\left(\sup_{i\in I}{}^{*}\varphi_t^i\right)_{t\in(0,1)}$$

is the geodesic from $\sup_i{}^*\varphi_0^i$ to $\sup_i{}^*\varphi_1^i$.

Proof By Proposition 1.2.3 and Proposition 4.1.3, we may assume that I is countable. In this case, the assertion follows from [DDNL18c, Proposition 3.3] and Theorem 2.1.1. □

Proposition 4.3.5 *Let $(\varphi_t)_{t\in[0,1]}$, $(\psi_t)_{t\in[0,1]}$ be geodesics in $\mathcal{E}^1(X,\theta)$. Then the distance $d_1(\varphi_t,\psi_t)$ is a convex function of $t \in [0,1]$.*

Proof By definition of d_1 and Proposition 4.3.3, it suffices to show the concavity of

$$[0,1] \ni t \mapsto E_\theta(\varphi_t \wedge \psi_t).$$

Let $(\eta_t)_{t\in[0,1]}$ be the geodesic from $\varphi_0 \wedge \psi_0$ to $\varphi_1 \wedge \psi_1$. Then for each $t \in [0,1]$, we have $\eta_t \le \varphi_t \wedge \psi_t$. Hence, by the monotonicity of E_θ and Proposition 4.3.3,

$$E_\theta(\varphi_t \wedge \psi_t) \ge E_\theta(\eta_t) = (1-t)E_\theta(\varphi_0 \wedge \psi_0) + tE_\theta(\varphi_1 \wedge \psi_1)$$

for each $t \in [0,1]$. Applying the same argument to the restrictions of $(\varphi_t)_t$ and $(\psi_t)_t$ to any subinterval of $[0,1]$, which are again geodesics by Proposition 4.2.1, gives a similar inequality on every subinterval. Hence $t \mapsto E_\theta(\varphi_t \wedge \psi_t)$ is concave, and our assertion follows. □

In particular, we can introduce:

[4] We expect that these assertions hold even when $\phi \ne V_\theta$, but a proof in full generality is unknown to us.

Definition 4.3.5 Let $\ell, \ell' \in \mathcal{R}^1(X,\theta)$. We define

$$d_1(\ell,\ell') := \lim_{t\to\infty} \frac{1}{t} d_1(\ell_t,\ell'_t).$$

The limit exists in $[0,\infty]$ by Proposition 4.3.5.

Theorem 4.3.4 *The function d_1 defined in Definition 4.3.5 is a metric.*

One can actually show that $(\mathcal{R}^1(X,\theta), d_1)$ is a complete metric space. We do not need this fact in the sequel, so we omit the proof. See [DDNL21b, Theorem 2.14].

Proof Symmetry and the triangle inequality follow from the corresponding pointwise properties of d_1 on $\mathcal{E}^1(X,\theta)$.

We next show that $d_1(\ell,\ell') < \infty$ for any $\ell, \ell' \in \mathcal{R}^1(X,\theta)$. Let C_ℓ be the constant in Proposition 4.2.4, and set $\tilde{\ell}_t = \ell_t - C_\ell t$. Then $\tilde{\ell}$ is again a geodesic ray, $\sup_X \tilde{\ell}_t = 0$, and hence $\tilde{\ell}_t \leq V_\theta$. Thus, by Proposition 4.3.3,

$$d_1(\tilde{\ell}_t, V_\theta) = -E_\theta(\tilde{\ell}_t) = -tE_\theta(\tilde{\ell}_1).$$

Moreover, by (3.28), $d_1(\ell_t,\tilde{\ell}_t) = |C_\ell| t \int_X \theta^n_{V_\theta}$. Hence $d_1(\ell_t, V_\theta)$ grows at most linearly in t. The same argument applies to ℓ', and the triangle inequality gives $d_1(\ell,\ell') < \infty$.

It remains to prove separation: suppose that $\ell,\ell' \in \mathcal{R}^1(X,\theta)$ and $d_1(\ell,\ell') = 0$.

Fix $s > 0$, then it follows from Proposition 4.3.5 that

$$\frac{d_1(\ell_s,\ell'_s)}{s} \leq \lim_{t\to\infty} \frac{d_1(\ell_t,\ell'_t)}{t} = d_1(\ell,\ell') = 0.$$

Therefore, $\ell_s = \ell'_s$ and hence $\ell = \ell'$.

Definition 4.3.6 We define the *radial Monge–Ampère energy* $\mathbf{E}^\phi \colon \mathcal{R}(X,\theta;\phi) \to \mathbb{R} \cup \{\infty\}$ as follows:

$$\mathbf{E}^\phi(\ell) := \varlimsup_{t\to\infty} \frac{E^\phi_\theta(\ell_t)}{t}.$$

When $\phi = V_\theta$, we write $\mathbf{E}$ instead of $\mathbf{E}^{V_\theta}$.

Proposition 4.3.6 *Let $\ell,\ell' \in \mathcal{R}^1(X,\theta)$ and $\ell \leq \ell'$. Then*

$$d_1(\ell,\ell') = \mathbf{E}(\ell') - \mathbf{E}(\ell). \tag{4.26}$$

Proof By Proposition 4.3.3, the energy is linear along finite energy rays. Hence

$$E_\theta(\ell_t) = tE_\theta(\ell_1), \quad E_\theta(\ell'_t) = tE_\theta(\ell'_1).$$

Since $\ell_t \leq \ell'_t$ for all $t \geq 0$, (4.15) gives

$$\frac{1}{t} d_1(\ell_t,\ell'_t) = \frac{1}{t}\left(E_\theta(\ell'_t) - E_\theta(\ell_t)\right).$$

Letting $t \to \infty$ proves the assertion. □

Next we recall the $\vee$ operator at the level of geodesic rays.

Definition 4.3.7 Let $\ell, \ell' \in \mathcal{R}^\infty(X, \theta)$. We define $\ell \vee \ell'$ as the minimal ray in $\mathcal{R}^\infty(X, \theta)$ lying above both ℓ and ℓ'.

Proposition 4.3.7 *Given $\ell, \ell' \in \mathcal{R}^\infty(X, \theta)$. Then $\ell \vee \ell' \in \mathcal{R}^\infty(X, \theta)$ exists, and*

$$\mathbf{E}(\ell \vee \ell') = \lim_{t \to \infty} \frac{1}{t} E_\theta \left(\ell_t \vee \ell'_t\right). \tag{4.27}$$

Proof For each $t > 0$, let $(\ell''^t_s)_{s \in [0,t]}$ be the geodesic from V_θ to $\ell_t \vee \ell'_t$.

Step 1. We first show that for each fixed $s \geq 0$, ℓ''^t_s is increasing in $t \in [s, \infty)$.

To see this, fix $s \geq 0$ and choose $t' > t \geq s$. We need to show that

$$\ell''^{t'}_s \geq \ell''^t_s. \tag{4.28}$$

Since $(\ell''^{t'}_a)_{a \in [0,t]}$ is a geodesic, it suffices to show that $(\ell''^t_a)_{a \in [0,t]}$ is a candidate in the Perron envelope defining the former geodesic. In other words, in verifying (4.28), we may assume that either $s = 0$ or $s = t$. The case $s = 0$ is of course trivial. So it remains to prove the following:

$$\ell''^{t'}_t \geq \ell_t \vee \ell'_t.$$

By symmetry, it suffices to prove

$$\ell''^{t'}_t \geq \ell_t.$$

But since $(\ell_a)_{a \in [0,t']}$ is a candidate in the Perron envelope defining $\ell''^{t'}$, this inequality follows.

Step 2. Next, observe that for a fixed $s \geq 0$, we have

$$\sup_X \ell''^t_s \leq \frac{s}{t} \sup_X \ell''^t_t + \frac{t-s}{t} \sup_X \ell''^t_0 = \frac{s}{t} \left(\sup_X \ell_t\right) \vee \left(\sup_X \ell'_t\right)$$

for all $t > s$. The right-hand side is bounded from above by a constant independent of $t \geq s$ by Proposition 4.2.4. Let

$$(\ell \vee \ell')_s \coloneqq \sup_{t > s}{}^* \ell''^t_s. \tag{4.29}$$

Then Proposition 4.3.4 guarantees that $\ell \vee \ell' \in \mathcal{R}^\infty(X, \theta)$.

Step 3. We need to show that $\ell \vee \ell'$ defined in this way is indeed the minimal ray lying above ℓ and ℓ'.

First, by Step 1, we have

$$\ell''^t_s \geq \ell''^s_s \geq \ell_s$$

for any $t \geq s \geq 0$. Therefore,

$$(\ell \vee \ell')_s \geq \ell_s$$

for all $s \geq 0$. In other words, $\ell \vee \ell' \geq \ell$. Similarly, $\ell \vee \ell' \geq \ell'$.

Next, let $L \in \mathcal{R}^\infty(X,\theta)$ be a ray lying above both ℓ and ℓ'. Then we have

$$L_t \geq \ell_t \vee \ell'_t$$

for all $t \geq 0$. In particular,

$$L_s \geq \ell''^t_s$$

for all $t \geq s \geq 0$. It follows that

$$L_s \geq (\ell \vee \ell')_s$$

for all $s \geq 0$.

Step 4. It remains to argue (4.27). Since the energy is linear along bounded rays,

$$\mathbf{E}(\ell \vee \ell') = E_\theta\left((\ell \vee \ell')_1\right).$$

Moreover, ℓ''^t_1 increases to $(\ell \vee \ell')_1$ as $t \to \infty$. By Proposition 3.1.20,

$$E_\theta\left((\ell \vee \ell')_1\right) = \lim_{t\to\infty} E_\theta\left(\ell''^t_1\right).$$

Finally, by Proposition 4.3.3 applied to $(\ell''^t_s)_{s\in[0,t]}$,

$$E_\theta\left(\ell''^t_1\right) = \frac{1}{t} E_\theta\left(\ell_t \vee \ell'_t\right).$$

Combining these identities gives (4.27). □

Lemma 4.3.6 *For any $\ell, \ell' \in \mathcal{R}^\infty(X,\theta)$, we have*

$$d_1(\ell,\ell') \leq d_1(\ell,\ell\vee\ell') + d_1(\ell',\ell\vee\ell') \leq C_n d_1(\ell,\ell'), \tag{4.30}$$

where $C_n = 3(n+1)2^{n+2}$.

Proof The first inequality is trivial. As for the second, we estimate

$$\begin{aligned}
d_1(\ell,\ell\vee\ell') &= \mathbf{E}(\ell\vee\ell') - \mathbf{E}(\ell) && \text{by Proposition 4.3.6}\\
&= \lim_{t\to\infty} \frac{1}{t} E_\theta\left(\ell_t \vee \ell'_t\right) - \mathbf{E}(\ell) && \text{by (4.27)}\\
&= \lim_{t\to\infty} \frac{1}{t} d_1\left(\ell_t \vee \ell'_t, \ell_t\right) && \text{by (4.15).}
\end{aligned}$$

By symmetry, we find

$$d_1(\ell,\ell\vee\ell') + d_1(\ell',\ell\vee\ell') \leq \lim_{t\to\infty} \frac{1}{t}\Big(d_1\left(\ell_t\vee\ell'_t,\ell_t\right) + d_1\left(\ell_t\vee\ell'_t,\ell'_t\right)\Big).$$

By Theorem 4.3.1 and Lemma 4.3.2, for each $t > 0$,

$$d_1\left(\ell_t\vee\ell'_t,\ell_t\right) + d_1\left(\ell_t\vee\ell'_t,\ell'_t\right) \leq 3(n+1)2^{n+2} d_1(\ell_t,\ell'_t).$$

Now (4.30) follows. □

Example 4.3.1 Let $\varphi \in \mathrm{PSH}(X,\theta)$. Assume that $\varphi \leq 0$. For each $C > 0$, let $(\ell_t^{\varphi,C})_{t\in[0,C]}$ be the geodesic from V_θ to $(V_\theta - C) \vee \varphi$. For each $t \geq 0$, there is $\ell_t^{\varphi} \in \mathcal{E}^\infty(X,\theta)$ such that

$$\ell_t^{\varphi,C} \xrightarrow{d_1} \ell_t^{\varphi} \tag{4.31}$$

as $C \to \infty$. Then $\ell^\varphi \in \mathcal{R}^\infty(X,\theta)$ and

$$\mathbf{E}(\ell^\varphi) = \frac{1}{n+1}\sum_{j=0}^{n}\left(\int_X \theta_\varphi^j \wedge \theta_{V_\theta}^{n-j} - \int_X \theta_{V_\theta}^n\right). \tag{4.32}$$

Comparing the above approximants with cutoff parameters shifted by a bounded amount shows that $\ell^{\varphi+C} = \ell^\varphi$ for any $C \in \mathbb{R}$. Therefore, for a general $\varphi \in \mathrm{PSH}(X,\theta)$, we may simply define

$$\ell^\varphi := \ell^{\varphi - \sup_X \varphi}.$$

Then the conclusions of this example continue to hold.

Proof We first show that for each fixed $t \geq 0$, $\ell_t^{\varphi,C}$ is increasing in $C \geq t$.

To see this, choose $t \leq C_1 < C_2$. We need to show that

$$\ell_t^{\varphi,C_1} \leq \ell_t^{\varphi,C_2}.$$

Since both sides are geodesics for $t \in [0, C_1]$, it suffices to show that

$$(V_\theta - C_1) \vee \varphi \leq \ell_{C_1}^{\varphi,C_2}. \tag{4.33}$$

Now $((V_\theta - t) \vee \varphi)_{t\in[0,C_2]}$ is a subgeodesic from V_θ to $(V_\theta - C_2) \vee \varphi$ by Proposition 4.1.3.[5] It is therefore a candidate in the Perron envelope defining ℓ^{φ,C_2}, and evaluating at $t = C_1$ gives exactly (4.33).

From Proposition 4.1.2, we know that for any $C > t > 0$, we have

$$\ell_t^{\varphi,C} \leq \frac{t}{C}\Big((V_\theta - C) \vee \varphi\Big) + \frac{C-t}{C}\cdot V_\theta \leq 0,$$

while the linear subgeodesic $V_\theta - t$ is a candidate in the same Perron envelope, so $V_\theta - t \leq \ell_t^{\varphi,C}$. Therefore, by Proposition 1.2.2,

$$\ell_t^\varphi := \sup_{C>t}{}^{*} \ell_t^{\varphi,C} \in \mathcal{E}^\infty(X,\theta) \tag{4.34}$$

for all $t \geq 0$. Thanks to Proposition 3.1.20, we have

$$\ell_t^{\varphi,C} \xrightarrow{d_1} \ell_t^{\varphi}$$

[5] Here we need $\varphi \leq 0$.

as $C \to \infty$ for all $t \geq 0$. Applying Proposition 4.3.4 on each interval $[0, T]$ to the increasing family $(\ell_t^{\varphi,C})_{t\in[0,T]}$, $C \geq T$, shows that $\ell^\varphi \in \mathcal{R}^\infty(X, \theta)$.

It remains to compute the energy of ℓ^φ. We first fix $C \geq t > 0$ and compute using Proposition 4.3.3:

$$E_\theta\left(\ell_t^{\varphi,C}\right) = \frac{t}{C} E_\theta\Big((V_\theta - C) \vee \varphi\Big).$$

Letting $C \to \infty$ and applying Proposition 3.1.20, we find that

$$E_\theta(\ell_t^\varphi) = \lim_{C\to\infty} \frac{t}{C} E_\theta\Big((V_\theta - C) \vee \varphi\Big)$$

for any $t \geq 0$. It follows that

$$\mathbf{E}(\ell^\varphi) = \lim_{C\to\infty} \frac{1}{C} E_\theta\Big((V_\theta - C) \vee \varphi\Big).$$

Using the definition of E_θ, in order to obtain (4.32), it suffices to show that for each $j = 0, \ldots, n$, we have

$$\lim_{C\to\infty} \int_X \frac{(V_\theta - C) \vee \varphi - V_\theta}{C}\, \theta^j_{(V_\theta - C)\vee\varphi} \wedge \theta^{n-j}_{V_\theta} = \int_X \theta^j_\varphi \wedge \theta^{n-j}_{V_\theta} - \int_X \theta^n_{V_\theta}. \quad (4.35)$$

For this purpose, for each $C > 0$, we decompose X as $\{\varphi > V_\theta - C\}$ and $\{\varphi \leq V_\theta - C\}$. We have

$$\begin{aligned}&\int_{\{\varphi > V_\theta - C\}} \frac{(V_\theta - C) \vee \varphi - V_\theta}{C}\, \theta^j_{(V_\theta - C)\vee\varphi} \wedge \theta^{n-j}_{V_\theta}\\ =& \int_{\{\varphi > V_\theta - C\}} \frac{\varphi - V_\theta}{C}\, \theta^j_\varphi \wedge \theta^{n-j}_{V_\theta}.\end{aligned}$$

On the other hand,

$$\begin{aligned}&\int_{\{\varphi \leq V_\theta - C\}} \frac{(V_\theta - C) \vee \varphi - V_\theta}{C}\, \theta^j_{(V_\theta - C)\vee\varphi} \wedge \theta^{n-j}_{V_\theta}\\ =& -\int_{\{\varphi \leq V_\theta - C\}} \theta^j_{(V_\theta - C)\vee\varphi} \wedge \theta^{n-j}_{V_\theta}\\ =& -\int_X \theta^n_{V_\theta} + \int_{\{\varphi > V_\theta - C\}} \theta^j_\varphi \wedge \theta^{n-j}_{V_\theta}.\end{aligned}$$

Observe that for $C > 0$, the function $\mathbb{1}_{\{\varphi > V_\theta - C\}} C^{-1}(\varphi - V_\theta)$ is defined quasi-everywhere. Since $\varphi \leq V_\theta$, it takes values in $[-1, 0]$ outside a pluripolar set. When $C \to \infty$, these functions converge to 0 quasi-everywhere, so the first integral above converges to 0 by dominated convergence. Similarly, the indicators $\mathbb{1}_{\{\varphi > V_\theta - C\}}$ increase to 1 outside a pluripolar set, hence

$$\lim_{C\to\infty} \int_{\{\varphi > V_\theta - C\}} \theta^j_\varphi \wedge \theta^{n-j}_{V_\theta} = \int_X \theta^j_\varphi \wedge \theta^{n-j}_{V_\theta}.$$

Therefore, (4.35) follows. □

Chapter 5
Toric pluripotential theory on ample line bundles

There are two principal ways to formulate mathematical assertions (problems, conjectures, theorems, . . .): Russian and French. The Russian way is to choose the most simple and specific case (so that nobody could simplify the formulation preserving the main point). The French way is to generalize the statement as far as nobody could generalize it further.
— Vladimir Arnold[a]

[a] Vladimir Igorevich Arnold (1937–2010), who became a professor at l'Université Paris IX after the dissolution of USSR, was always sick of France (so am I!). In the public lecture entitled "Sur l'éducation mathématique" in 1997, he invented the famous joke "Combien font $2+3$?" to question the french education system.

This chapter develops the pluripotential theory of an ample toric line bundle L on a smooth projective toric variety X. Throughout, we work with a full-dimensional smooth (Delzant) lattice polytope $P \subseteq M_{\mathbb{R}}$, the associated fan Σ and toric variety $X = X_\Sigma$, and the toric ample line bundle $L = \mathcal{O}_X(D_P)$ defined by P.

The starting point is the well-known correspondence between toric plurisubharmonic metrics on L and proper convex functions on $N_{\mathbb{R}}$, made precise in Theorem 5.2.1. Under this dictionary, the basic operations of pluripotential theory correspond to elementary operations on convex functions and Lebesgue integrals, see Propositions 5.2.1, 5.2.3 and 5.2.5. In particular, non-pluripolar Monge–Ampère mass corresponds to the volume of an explicit convex body, the *Newton body* of φ (Definition 5.2.1).

The main technical input is Yao's theorem Theorem 5.2.2, which characterizes the characters lying in multiplier-ideal spaces in terms of the Newton body. Its asymptotic consequence Corollary 5.2.1 computes the leading order term of $h^0(X, L^k \otimes \mathcal{I}(k\varphi))$ by the volume of the Newton body. In the language of Chapter 7, this implies that all toric singularities are $\mathcal{I}$-good. This lies at the heart of the toric pluripotential theory on big line bundles developed in Chapter 12.

The chapter is organized as follows. Section 5.1 fixes notation and recalls the basic toric setup needed in the sequel. Section 5.2 then develops the analytic theory of toric psh functions and culminates in the proof of Theorem 5.2.2.

We assume the reader is familiar with basic toric geometry at the level of [CLS11]; the chapter can be safely skipped on a first reading if the reader's interest lies in the non-toric theory only. Some background on convex functions and convex bodies used throughout is recorded in Appendix A.

We remind the readers that throughout the whole book, convex functions are understood in the usual sense: the function $\mathbb{R} \ni x \mapsto x^2$ is *convex* instead of concave. This is different from the convention in [CLS11]. Please be careful when making comparisons.

5.1 Toric setup

Let T be a complex torus of dimension n[1] and let $T_c \subseteq T(\mathbb{C})$ denote the corresponding compact torus.

Write M for the character lattice of T, which is a free Abelian group of rank n. Similarly, let N be the cocharacter lattice of T, which is the dual lattice of M. Given $m \in M$, the corresponding character of T is denoted by χ^m. Write $M_\mathbb{R} = M \otimes_\mathbb{Z} \mathbb{R}$ and $N_\mathbb{R} = N \otimes_\mathbb{Z} \mathbb{R}$. The pairing between $M_\mathbb{R}$ and $N_\mathbb{R}$ is denoted by $\langle \bullet, \bullet \rangle$.

Let $P \subseteq M_\mathbb{R}$ be a full-dimensional *smooth*[2] lattice polytope[3].

Given any (closed) facet F of P, let $u_F \in N$ denote the unique ray generator (the first non-zero integral element) of the inward normal ray of F. Then P can be represented as

$$P = \{m \in M_\mathbb{R} : \langle m, u_F \rangle \geq -a_F \text{ for all facets } F \text{ of } P\} \tag{5.1}$$

for some uniquely determined integers a_F. The presentation is called the *facet presentation* of P.

Given any (closed) face Q of P, we let $\sigma_Q \subseteq N_\mathbb{R}$ be the closed convex cone generated by the u_F's, where F runs over all facets of P containing Q. When $Q = P$, σ_P is understood as $\{0\}$.

Let Σ be the *(inner) normal fan* of P. Namely,

$$\Sigma = \left\{\sigma_Q : Q \text{ is a face of } P\right\}.$$

The notation $\Sigma(1)$ denotes the set of rays in Σ. Note that $\Sigma(1)$ is in bijective correspondence with the set of facets of P. In fact, given any facet F of P, the cone σ_F is just the ray generated by u_F, namely, the inward normal ray of F.

For any $\rho \in \Sigma(1)$, let $u_\rho \in N$ denote the ray generator of ρ, namely the first non-zero element in $N \cap \rho$. If $\rho = \sigma_F$ for some facet F of P, then $u_\rho = u_F$.

Now the facet presentation (5.1) can be equivalently rewritten as

$$P = \left\{m \in M_\mathbb{R} : \langle m, u_\rho \rangle \geq -a_\rho \text{ for all } \rho \in \Sigma(1)\right\}.$$

Let $\mathrm{Supp}_P : N_\mathbb{R} \to \mathbb{R}$ denote the *support function* of P. Recall that the support function (Example A.1.2) of P is defined as

$$\mathrm{Supp}_P(n) = \max\left\{\langle m, n \rangle : m \in P\right\}.$$

Note that our support function differs from [CLS11, Proposition 4.2.14], where instead of a maximum, they took the minimum.

---

[1] Namely, an algebraic group defined over $\mathbb{C}$, which is isomorphic to $\mathbb{G}_\mathrm{m}^n$.

[2] Recall that *smooth* means that for every vertex $v \in P$, if we take the first lattice point w_E apart from v as one transverses each edge E of P containing v from v. Then $\{w_E - v\}_E$ forms a basis of M. See [CLS11, Definition 2.4.2]. We also say P is a *Delzant polytope* in this case.

[3] A *lattice polytope* in $M_\mathbb{R}$ is the convex hull of finitely many points in M.

Recall that the *characteristic function* $\chi_P\colon M_\mathbb{R} \to \{0,\infty\}$ of P is defined as in Example A.1.1:

$$\chi_P(m) := \begin{cases} 0, & m \in P; \\ \infty, & m \notin P. \end{cases}$$

Let $X = X_\Sigma$ be the smooth projective toric variety corresponding to Σ. See [CLS11, Theorem 3.1.5] for the construction of X and [CLS11, Theorem 3.1.19] for the smoothness of X. There is a canonical embedding $T \subseteq X$ as a dense Zariski open subset.

Let D be the Cartier divisor on X defined by P:

$$D = \sum_{\rho\in\Sigma(1)} a_\rho D_\rho,$$

where D_ρ is the toric prime divisor defined by ρ under the orbit-cone correspondence [CLS11, Theorem 3.2.6].

Let L be the toric line bundle induced by P, namely $L = \mathcal{O}_X(D)$. Since P is a smooth (Delzant) lattice polytope, L is very ample by [CLS11, Proposition 2.4.4]; in particular, L is ample.

We will choose the base e for the logarithm map

$$\mathbb{C}^* \to \mathbb{R}, \quad z \mapsto \log|z|^2.\text{[4]} \tag{5.2}$$

This choice is fixed throughout the book. Since we have a canonical identification $T(\mathbb{C}) \cong N \otimes_\mathbb{Z} \mathbb{C}^*$, the logarithm map then induces a tropicalization map after tensoring with N:

$$\mathrm{Trop}\colon T(\mathbb{C}) \to N_\mathbb{R}. \tag{5.3}$$

Before proceeding, it is always helpful to understand everything in our favorite example.

Example 5.1.1 We take $n = 1$ and $P = [0,1] \subseteq M_\mathbb{R} = \mathbb{R}$. In this case, the facet representation (5.1) becomes

$$P = \big\{m \in \mathbb{R} : \langle m, 1\rangle \geq 0, \langle m, -1\rangle \geq -1\big\},$$

with $u_{\{0\}} = 1$, $u_{\{1\}} = -1$, $a_{\{0\}} = 0$ and $a_{\{1\}} = 1$. The normal fan Σ is

$$\Sigma = \big\{(-\infty, 0], \{0\}, [0,\infty)\big\}.$$

See Fig. 5.1.

The corresponding toric variety is just $X = \mathbb{P}^1$. Under the orbit-cone correspondence, we have

$$D_{\{[0,\infty)\}} = [0], \quad D_{\{(-\infty,0]\}} = [\infty].$$

[4] Be careful when you compare with other references, some people prefer $\log|z|$, $-\log|z|$ or $-\log|z|^2$ instead.

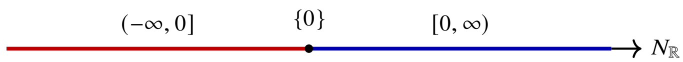

Fig. 5.1 The fan Σ of $\mathbb{P}^1$ in $N_{\mathbb{R}} = \mathbb{R}$, consisting of the two rays $(-\infty, 0]$, $[0, \infty)$ and the cone $\{0\}$.

The associated divisor $D = [\infty]$ and therefore,

$$L = \mathcal{O}_X(D) = \mathcal{O}_{\mathbb{P}^1}(1).$$

5.2 Toric plurisubharmonic functions

We continue to use the notations of Section 5.1.

Lemma 5.2.1 *Let $F\colon N_{\mathbb{R}} \to [-\infty, \infty]$ be a function. Then the following are equivalent:*

(1) *F is convex and takes values in $\mathbb{R}$, and*
(2) *$\mathrm{Trop}^* F$ is plurisubharmonic on $T(\mathbb{C})$.*

Proof We may choose an identification $N \cong \mathbb{Z}^n$ so that we have an identification $T(\mathbb{C}) \cong \mathbb{C}^{*n}$. Then Trop is identified with the map

$$\mathrm{Trop}\colon \mathbb{C}^{*n} \to \mathbb{R}^n, \quad (z_1, \dots, z_n) \mapsto \left(\log |z_1|^2, \dots, \log |z_n|^2\right).$$

(1) $\implies$ (2). By the standard convolution regularization of finite convex functions, let $F_k \in C^\infty(\mathbb{R}^n) \cap \mathrm{Conv}(\mathbb{R}^n)$ be a decreasing sequence with limit F. It follows from a straightforward computation that

$$\begin{aligned} \mathrm{dd^c}\, \mathrm{Trop}^* F_k(z_1, \dots, z_n) =& \frac{\mathrm{i}}{2\pi} \sum_{i,j=1}^{n} \partial_{ij} F_k \left(\log |z_1|^2, \dots, \log |z_n|^2\right) \\ & \times z_i^{-1} \overline{z_j}^{-1} \mathrm{d}z_i \wedge \mathrm{d}\overline{z_j}. \end{aligned} \tag{5.4}$$

So $\mathrm{Trop}^* F_k$ is plurisubharmonic. It follows from Proposition 1.2.2 that $\mathrm{Trop}^* F$ is plurisubharmonic.

(2) $\implies$ (1). It follows from Lemma 1.2.1 that F is finite. Moreover, by radial regularization, we may find a decreasing sequence φ_k of $(S^1)^n$-invariant smooth psh functions on $\mathbb{C}^{*n}$ with limit $\mathrm{Trop}^* F$. Write $\varphi_k = \mathrm{Trop}^* F_k$ for some function $F_k\colon \mathbb{R}^n \to \mathbb{R}$; it follows from (5.4) that F_k is convex for all k. Therefore, F is convex by Lemma A.1.2. □

Next we define a canonical Kähler form in $c_1(L)$.

Let $G_0\colon M_{\mathbb{R}} \to (-\infty, \infty]$ be defined as

$$G_0(m) := \begin{cases} \sum_{\rho\in\Sigma(1)} \left(\langle m,u_\rho\rangle + a_\rho\right)\log\left(\langle m,u_\rho\rangle + a_\rho\right)^{5}, & \text{if } m\in P,\\ \infty, & \text{otherwise.}\end{cases} \tag{5.5}$$

This is a closed proper convex function and $G_0 \sim \chi_P$, where $\sim$ is the relation defined in Definition A.1.8.

Let

$$F_0 = G_0^* \in \mathcal{E}^\infty(N_\mathbb{R}, P). \tag{5.6}$$

Here G_0^* is the Legendre transform of G_0, as recalled in Definition A.2.1. The set $\mathcal{E}^\infty(N_\mathbb{R}, P)$ is defined in Definition A.3.1.

By Guillemin's theorem [Gui94, CDG03], $\mathrm{dd^c}\,\mathrm{Trop}^* F_0$ can be extended to a unique Kähler form ω in $c_1(L)$. The Kähler form ω is clearly T_c-invariant.

For each $\rho\in\Sigma(1)$, we write

$$r_\rho(m) = \log\left(\langle m,u_\rho\rangle + a_\rho\right) + 1,\quad m\in P.$$

It follows from (5.5) that

$$\nabla G_0(m) = \sum_{\rho\in\Sigma(1)} r_\rho(m)u_\rho,\quad m\in \operatorname{Int} P. \tag{5.7}$$

Example 5.2.1 Let us move on with our favorite example Example 5.1.1. We continue to use the same notations. In this case,

$$G_0(m) = \begin{cases} m\log m + (1-m)\log(1-m), & \text{if } m\in[0,1],\\ \infty, & \text{otherwise.}\end{cases}$$

The Legendre transform is given[6] by

$$F_0(n) = \log\left(1+\mathrm{e}^n\right).$$

Composing with the tropicalization map, we find that

$$\omega|_{\mathbb{C}^*}(z) = \mathrm{dd^c}\log\left(1+|z|^2\right).$$

This is exactly the Fubini–Study metric as we have seen in Example 1.8.1.

We can now explain one subtlety: in our expression (5.5), there is no factor $1/2$ before the sum; the absence of the factor $1/2$ matches the choice $\mathrm{Trop}(z) = \log|z|^2$ rather than $\log|z|$ in our tropicalization map (5.2).

Let $\mathrm{PSH_{tor}}(X,\omega)$ denote the set of T_c-invariant ω-psh functions.

Theorem 5.2.1 *There are canonical bijections between the following three sets:*

[5] We understand that $0\log 0 = 0$ in this expression.

[6] This elementary computation fixes several normalization constants used later, so the formula is worth checking carefully.

(1) *The set of* $\varphi \in \mathrm{PSH}_{\mathrm{tor}}(X, \omega)$,
(2) *the set* $\mathcal{P}(N_{\mathbb{R}}, P)$ *in Definition A.3.1, namely, the set of convex functions* $F : N_{\mathbb{R}} \to \mathbb{R}$ *satisfying* $F \leq \mathrm{Supp}_P$, *and*
(3) *the set of closed proper convex functions* $G \in \mathrm{Conv}(M_{\mathbb{R}})$ *satisfying*

$$G|_{M_{\mathbb{R}} \setminus P} \equiv \infty.$$

For the notions of closedness and properness, we refer to Definition A.1.2 and Definition A.1.7.

Proof The bijection between (2) and (3) is the classical Legendre duality. Given F as in (2), we construct $G = F^*$ and *vice versa*, see Proposition A.2.5.

The map from (1) to (2) is given as follows: Given $\varphi \in \mathrm{PSH}_{\mathrm{tor}}(X, \omega)$, since φ is T_c-invariant, we can find $f : N_{\mathbb{R}} \to [-\infty, \infty)$ such that

$$\varphi|_{T(\mathbb{C})} = \mathrm{Trop}^* f. \tag{5.8}$$

We then define $F = f + F_0$. Then $\mathrm{Trop}^* F \in \mathrm{PSH}(T(\mathbb{C}))$. By Lemma 5.2.1, $F(n)$ is finite for any $n \in N_{\mathbb{R}}$ and F is convex. Moreover, $F \leq \mathrm{Supp}_P$ since φ is bounded from above on the compact space X and $F_0 \leq \mathrm{Supp}_P$.

Conversely, given a map $F \in \mathcal{P}(N_{\mathbb{R}}, P)$, then

$$\mathrm{Trop}^*(F - F_0) \in \mathrm{PSH}\left(T(\mathbb{C}), \omega|_{T(\mathbb{C})}\right).$$

Since $F \leq \mathrm{Supp}_P$ and $F_0 \sim \mathrm{Supp}_P$, the difference $F - F_0$ is bounded from above. Hence the corresponding local psh potentials are locally bounded from above near $X \setminus T(\mathbb{C})$. It follows from Theorem 1.2.1 that this function can be extended uniquely to an ω-psh function on X. The uniqueness of the extension guarantees its T_c-invariance.

The two maps are inverse to each other by construction. □

Given $\varphi \in \mathrm{PSH}_{\mathrm{tor}}(X, \omega)$, we will write F_φ and G_φ for the convex functions given by Theorem 5.2.1. From the proof, we have the following relations:

$$\varphi|_{T(\mathbb{C})} = \mathrm{Trop}^*\left(F_\varphi - F_0\right), \quad G_\varphi = F_\varphi^*. \tag{5.9}$$

Example 5.2.2 Let us take our favorite example Example 5.2.1 again. We will continue to use the same notations.

Recall that in Example 1.8.2 and Example 3.1.1, we constructed two S^1-invariant functions in $\mathrm{PSH}(X, \omega)$.

We begin with the function φ in Example 1.8.2. Recall that

$$\varphi(z) = \log \frac{|z|^2}{|z|^2 + 1}$$

for $z \in \mathbb{C}$. The function $f : \mathbb{R} \to \mathbb{R}$ in (5.8) is therefore

$$f(n) = \log \frac{\mathrm{e}^n}{1 + \mathrm{e}^n}.$$

Therefore, $F_\varphi : \mathbb{R} \to \mathbb{R}$ is
$$F_\varphi(n) = n.$$
Correspondingly, $G_\varphi : \mathbb{R} \to \mathbb{R}$ is
$$G_\varphi(m) = \begin{cases} 0, & \text{if } m = 1; \\ \infty, & \text{otherwise.} \end{cases}$$

Similarly, if ψ denotes the function in Example 3.1.1, then the function f in (5.8) is
$$f(n) = \begin{cases} -\log\left(\mathrm{e}^n + 1\right) + (-\log(-n)) \vee (n+2), & \text{if } n < -\log 2; \\ 2 + \log \frac{\mathrm{e}^n}{1+\mathrm{e}^n}, & \text{otherwise.} \end{cases}$$
Therefore,
$$F_\psi(n) = \begin{cases} (-\log(-n)) \vee (n+2), & \text{if } n < -\log 2; \\ 2 + n, & \text{otherwise.} \end{cases}$$
The Legendre transform is tricky to compute. Let λ be the large solution of $\log x = x-2$. So $\lambda \approx 3.146$. The smaller solution is around $0.159 < \log 2 \approx 0.693$. It might be helpful to have a look at the poorly drawn picture Fig. 5.2.

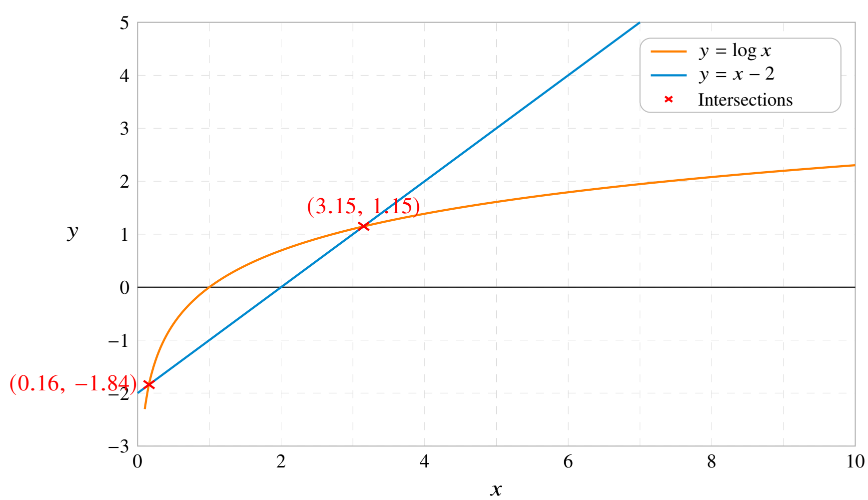

Fig. 5.2 The graphs of $\log x$ and $x - 2$.

It is immediate that $G_\psi(m) = \infty$ unless $m \in [0, 1]$. Let us assume that $m \in [0, 1]$. Then

$$\begin{aligned}G_\psi(m) &= \sup_{n\in\mathbb{R}} \left(mn - F_\psi(n)\right) \\ &= \sup_{n<-\log 2} (mn - (-\log(-n)) \vee (n+2)) \vee \sup_{n\geq -\log 2} (mn - n - 2) \\ &= \sup_{n>\log 2} (-mn + (\log n) \wedge (n-2)) \vee ((1-m)\log 2 - 2)\,.\end{aligned}$$

Let us focus on the first part, which can be decomposed further into

$$\begin{aligned}&\sup_{n>\log 2} (-mn + (\log n) \wedge (n-2)) \\ &= \sup_{n\in(\log 2,\lambda]} (n - 2 - mn) \vee \sup_{n>\lambda}(\log n - mn) \\ &= ((1-m)\lambda - 2) \vee \sup_{n>\lambda} (\log n - mn)\,.\end{aligned}$$

The latter part can be computed easily:

$$\sup_{n>\lambda} (\log n - mn) = \begin{cases} -\log m - 1, & \text{if } m \in \left(0, \lambda^{-1}\right]; \\ \log\lambda - m\lambda, & \text{if } m \in \left(\lambda^{-1}, 1\right]. \end{cases}$$

Putting everything together, we find

$$G_\psi(m) = \begin{cases} (-\log m - 1) \vee ((1-m)\lambda - 2)\,, & \text{if } m \in \left(0, \lambda^{-1}\right]; \\ (\log\lambda - m\lambda) \vee ((1-m)\lambda - 2)\,, & \text{if } m \in \left(\lambda^{-1}, 1\right]. \end{cases}$$

This can be further simplified, the final result is

$$G_\psi(m) = \begin{cases} -\log m - 1, & \text{if } m \in \left(0, \lambda^{-1}\right]; \\ (1-m)\lambda - 2, & \text{if } m \in \left(\lambda^{-1}, 1\right]; \\ \infty, & \text{otherwise.} \end{cases}$$

The graph of G_ψ on $(0, 1]$ is sketched in Fig. 5.3.

We observe a few elementary facts.

Proposition 5.2.1 *Given $\varphi, \psi \in \mathrm{PSH}_{\mathrm{tor}}(X, \omega)$. The following are equivalent:*

(1) $\varphi \leq \psi$,
(2) $F_\varphi \leq F_\psi$, *and*
(3) $G_\psi \leq G_\varphi$.

The same holds if we replace all $\leq$'s by $\preceq$.

Proof The equivalence between (1) and (2) follows from the definition (5.9). The equivalence between (2) and (3) follows from the definition of the Legendre transform. □

Similarly, we have

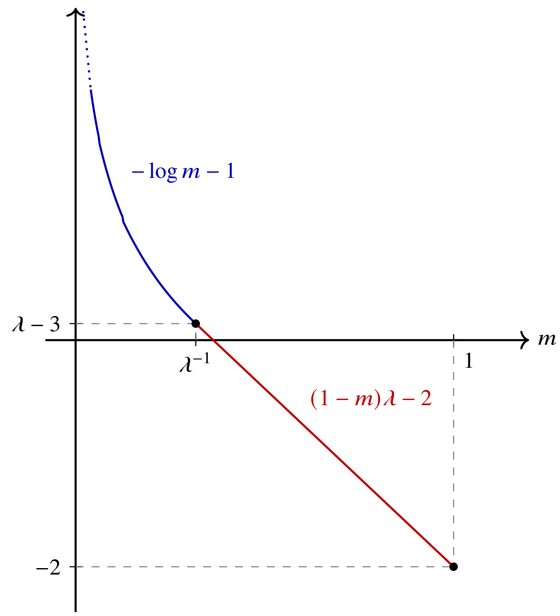

Fig. 5.3 The graph of G_ψ on $(0, 1]$.

Proposition 5.2.2 *Given $\varphi \in \mathrm{PSH}_{\mathrm{tor}}(X, \omega)$ and $C \in \mathbb{R}$. We have*

$$F_{\varphi+C} = F_\varphi + C, \quad G_{\varphi+C} = G_\varphi - C.$$

Proof The first equality follows from (5.9). The second is obtained by taking the Legendre transform. □

Proposition 5.2.3 *Given $\varphi, \psi \in \mathrm{PSH}_{\mathrm{tor}}(X, \omega)$ with $\varphi \wedge \psi \not\equiv -\infty$, then $\varphi \wedge \psi \in \mathrm{PSH}_{\mathrm{tor}}(X, \omega)$ and*

$$F_{\varphi \wedge \psi} = F_\varphi \wedge F_\psi, \quad G_{\varphi\wedge\psi} = G_\varphi \vee G_\psi.$$

The operators $\wedge$ and $\vee$ are defined in Definition A.1.5 and Definition A.1.6.

Proof By Proposition 3.1.1, we have $\varphi \wedge \psi \in \mathrm{PSH}(X, \omega)$. Since φ and ψ are T_c-invariant, the set of candidates in the definition of $\varphi \wedge \psi$ is preserved by the action of T_c. Hence $\varphi \wedge \psi$ is T_c-invariant as well. So $\varphi \wedge \psi$ is the biggest element in $\mathrm{PSH}_{\mathrm{tor}}(X, \omega)$ which is dominated by both φ and ψ. In view of Theorem 5.2.1 and Proposition 5.2.1, $G_{\varphi\wedge\psi}$ is the smallest closed proper convex function G on $M_\mathbb{R}$ dominating both G_φ and G_ψ, which is just $G_\varphi \vee G_\psi$.

The claim for F follows from Proposition A.2.3. □

Example 5.2.3 Now we can give an example of $\varphi, \psi \in \mathrm{PSH}_{\mathrm{tor}}(X, \omega)$ with $\varphi \wedge \psi \equiv -\infty$.

We take $P = [0, 1]$ so that $X = \mathbb{P}^1$ and ω is the Fubini–Study metric. Let $\varphi \in \mathrm{PSH}(X, \omega)$ be such that

$$\varphi(z) = \log \frac{|z|^2}{|z|^2+1}$$

for $z \in \mathbb{C}$. We have computed that G_φ in Example 5.2.2:

$$G_\varphi(m) = \begin{cases} 0, & \text{if } m = 1, \\ \infty, & \text{otherwise.} \end{cases}$$

Now we define $\psi \in \mathrm{PSH}_{\mathrm{tor}}(X, \omega)$ as the unique function such that

$$\psi(z) = \log \frac{1}{|z|^2+1}$$

for $z \in \mathbb{C}$. Then a similar computation shows that

$$G_\psi(m) = \begin{cases} 0, & \text{if } m = 0, \\ \infty, & \text{otherwise.} \end{cases}$$

Now we claim that $\varphi \wedge \psi \equiv -\infty$. Otherwise, we would have

$$G_{\varphi\wedge\psi} = G_\varphi \vee G_\psi \equiv \infty,$$

which is not proper.

Proposition 5.2.4 *Let $\{\varphi_i\}_{i\in I}$ be a non-empty family in $\mathrm{PSH}_{\mathrm{tor}}(X, \omega)$ uniformly bounded from above. Then ${\sup_{i\in I}}^* \varphi_i \in \mathrm{PSH}_{\mathrm{tor}}(X, \omega)$ and*

$$F_{{\sup_{i\in I}}^*\varphi_i} = \bigvee_{i\in I} F_{\varphi_i}, \quad G_{{\sup_{i\in I}}^*\varphi_i} = \mathrm{cl} \bigwedge_{i\in I} G_{\varphi_i}.$$

Moreover, if I is finite, then

$$G_{\max_{i\in I}\varphi_i} = \bigwedge_{i\in I} G_{\varphi_i}.$$

Similarly, if $\{\varphi_i\}_{i\in I}$ is a decreasing net in $\mathrm{PSH}_{\mathrm{tor}}(X, \omega)$ such that $\inf_{i\in I}\varphi_i \not\equiv -\infty$, then $\inf_{i\in I}\varphi_i \in \mathrm{PSH}_{\mathrm{tor}}(X, \omega)$ and

$$F_{\inf_{i\in I}\varphi_i} = \inf_{i\in I} F_{\varphi_i}, \quad G_{\inf_{i\in I}\varphi_i} = \bigvee_{i\in I} G_{\varphi_i}.$$

Recall that the closure cl is defined in Definition A.1.7.

Proof Thanks to Lemma A.1.2 and Proposition A.1.1, in both cases, the statement for F is clear. The corresponding statement for G is obtained via Proposition A.2.3. □

The complex Monge–Ampère operator is closely related to the real one:

Proposition 5.2.5 *Let $\varphi \in \mathrm{PSH}_{\mathrm{tor}}(X, \omega)$. Then*

$$\operatorname{Trop}_* \left(\omega|_{T(\mathbb{C})} + \mathrm{dd^c}\varphi|_{T(\mathbb{C})}\right)^n = \mathrm{MA}_{\mathbb{R}}(F_\varphi). \tag{5.10}$$

In particular,

$$\int_X \omega_\varphi^n = \int_{N_\mathbb{R}} \mathrm{MA}_{\mathbb{R}}(F_\varphi) = n! \operatorname{vol} \overline{\{G_\varphi < \infty\}}$$

and

$$\int_X \omega^n = n! \operatorname{vol} P.$$

Here the real Monge–Ampère operator is defined in Definition A.4.1. The normalization of the Lebesgue measure vol on $M_\mathbb{R}$ is such that the fundamental lattice cube has measure 1.[7]

Proof We only need to prove (5.10). By Proposition A.3.3, we can find a decreasing sequence of smooth convex functions F_j on $N_\mathbb{R}$ with limit F_φ. We write $F_j = F_{\varphi_j}$ for some $\varphi_j \in \mathrm{PSH}_{\mathrm{tor}}(X, \omega)$. By Theorem 2.1.1 and Theorem A.4.1, it suffices to establish (5.10) for the φ_j's. We may therefore reduce to the case where F_φ is smooth. We write $F = F_\varphi$ to simplify the notations. The notations $a_i = \log |z_i|^2$ will be used, where $i = 1, \dots, n$.

Next we fix an identification $N = \mathbb{Z}^n$. Fix a test function $f \in C^0_c(N_\mathbb{R})$, we need to show that

$$\int_{\mathbb{C}^{*n}} f(a_1, \dots, a_n)\, (\mathrm{dd^c} \operatorname{Trop}^* F(z_1, \dots, z_n))^n = \int_{\mathbb{R}^n} f\, \mathrm{MA}_{\mathbb{R}}(F).$$

Using Proposition A.4.1 and (5.4), this reduces to

$$\begin{aligned}\left(\frac{\mathrm{i}}{2\pi}\right)^n \int_{\mathbb{C}^{*n}} f(a_1, \dots, a_n) \left(\sum_{i,j=1}^n \partial_{i,j} F(a_1, \dots, a_n) z_i^{-1} \overline{z_j}^{-1} \mathrm{d}z_i \wedge \mathrm{d}\overline{z_j}\right)^n = \\ n! \int_{\mathbb{R}^n} f \det \nabla^2 F \,\mathrm{d}\operatorname{vol}.\end{aligned} \tag{5.11}$$

Expanding the bracket, we get

$$\begin{aligned}\left(\sum_{i,j=1}^n \partial_{i,j} F z_i^{-1} \overline{z_j}^{-1} \mathrm{d}z_i \wedge \mathrm{d}\overline{z_j}\right)^n = \sum_{i_1,\dots,i_n=1}^n \sum_{j_1,\dots,j_n=1}^n \partial_{i_1 j_1} F \cdots \partial_{i_n j_n} F \cdot \\ \mathrm{d}\log z_{i_1} \wedge \mathrm{d}\log \overline{z_{j_1}} \wedge \cdots \wedge \mathrm{d}\log z_{i_n} \wedge \mathrm{d}\log \overline{z_{j_n}},\end{aligned}$$

where $\mathrm{d}\log z_i = z_i^{-1}\mathrm{d}z_i$ and $\mathrm{d}\log \overline{z_i} = \overline{z_i}^{-1}\mathrm{d}\overline{z_i}$ are understood.

Using the apparent symmetry, the expression on the right-hand side becomes

[7] In some references like [GKZ08], the normalization is so that the fundamental lattice cube has measure $n!$. Be careful when making comparisons with these references.

$$\begin{aligned}
&\sum_{\sigma,\tau\in\mathfrak{S}_n}\prod_{k=1}^{n}\partial_{\sigma(k)\tau(k)}F\,\mathrm{d}\log z_{\sigma(1)}\wedge\mathrm{d}\log\overline{z_{\tau(1)}}\wedge\cdots\wedge\mathrm{d}\log z_{\sigma(n)}\wedge\mathrm{d}\log\overline{z_{\tau(n)}},\\
=&n!\sum_{\tau\in\mathfrak{S}_n}\prod_{k=1}^{n}\partial_{k\tau(k)}F\,\mathrm{d}\log z_1\wedge\mathrm{d}\log\overline{z_{\tau(1)}}\wedge\cdots\wedge\mathrm{d}\log z_n\wedge\mathrm{d}\log\overline{z_{\tau(n)}}\\
=&n!\sum_{\tau\in\mathfrak{S}_n}\operatorname{sgn}(\tau)\prod_{k=1}^{n}\partial_{k\tau(k)}F\,\mathrm{d}\log z_1\wedge\mathrm{d}\log\overline{z_1}\wedge\cdots\wedge\mathrm{d}\log z_n\wedge\mathrm{d}\log\overline{z_n}\\
=&n!\det\nabla^2F\mathrm{d}\log z_1\wedge\mathrm{d}\log\overline{z_1}\wedge\cdots\wedge\mathrm{d}\log z_n\wedge\mathrm{d}\log\overline{z_n},
\end{aligned}$$

where $\mathfrak{S}_n$ is the permutation group on $\{1,\dots,n\}$ and $\operatorname{sgn}(\tau)$ is the sign of τ.

Next, switch to polar coordinates for each z_i: Let $z_i = r_i\exp(\mathrm{i}\theta_i)$ and recall that $r_i = \exp(a_i/2)$. Then the left-hand side of (5.11) becomes

$$\begin{aligned}
&\frac{n!}{(2\pi)^n}\int_{\mathbb{R}^n\times[0,2\pi)^n} f\det\nabla^2F\,\mathrm{d}a_1\wedge\mathrm{d}\theta_1\wedge\cdots\wedge\mathrm{d}a_n\wedge\mathrm{d}\theta_n\\
=&n!\int_{\mathbb{R}^n} f\det\nabla^2F\,\mathrm{d}a_1\wedge\cdots\wedge\mathrm{d}a_n,
\end{aligned}$$

which is exactly what we have expected. □

Next we study the envelope operators developed in Chapter 3 in the toric setting.

Definition 5.2.1 Let $\varphi\in\mathrm{PSH}_{\mathrm{tor}}(X,\omega)$. We define its *Newton body* as

$$\Delta(\omega,\varphi) := \overline{\{G_\varphi<\infty\}}\subseteq P.$$

Note that $\Delta(\omega,\varphi)$ is a convex body.

By Proposition A.2.2, we have

$$\Delta(\omega,\varphi)=\overline{\nabla F_\varphi(N_{\mathbb{R}})}.$$

Example 5.2.4 By (5.5), we have

$$\Delta(\omega,0)=P.$$

In the case of Example 5.2.2, we have

$$\Delta(\omega,\varphi)=\{1\},\quad \Delta(\omega,\psi)=[0,1].$$

Observe that in the latter case,

$$\{G_\psi<\infty\}\subsetneq P.$$

Proposition 5.2.6 *Let* $\varphi\in\mathrm{PSH}_{\mathrm{tor}}(X,\omega)$. *Then* $P_\omega[\varphi]\in\mathrm{PSH}_{\mathrm{tor}}(X,\omega)$ *and*

$$G_{P_\omega[\varphi]}(x) = \begin{cases} G_0(x), & \text{if } x \in \Delta(\omega, \varphi); \\ \infty, & \text{otherwise.} \end{cases} \tag{5.12}$$

Proof By (3.4), we have

$$P_\omega[\varphi] = \sup_{C\in\mathbb{R}}{}^{*} \big((\varphi + C) \wedge 0\big).$$

It follows from Proposition 5.2.2, Proposition 5.2.3 and Proposition 5.2.4 that $P_\omega[\varphi] \in \mathrm{PSH}_{\mathrm{tor}}(X, \omega)$. Moreover, by the same propositions, we have

$$G_{P_\omega[\varphi]} = \operatorname{cl} \inf_{C\in\mathbb{R}} \big(G_0 \vee (G_\varphi - C)\big)\,.$$

We spell out the last closure. Set

$$K = \inf_{C\in\mathbb{R}} \big(G_0 \vee (G_\varphi - C)\big)\,.$$

If $x \in \operatorname{Dom} G_\varphi$, then $G_\varphi(x) - C \le G_0(x)$ for C large enough, hence $K(x) = G_0(x)$. If $x \notin \operatorname{Dom} G_\varphi$, then $K(x) = \infty$. Since $\Delta(\omega, \varphi) = \overline{\operatorname{Dom} G_\varphi}$ and G_0 is continuous on P, the lower semicontinuous regularization $\operatorname{cl} K$ is equal to G_0 on $\Delta(\omega, \varphi)$ and equal to ∞ outside $\Delta(\omega, \varphi)$. This is exactly (5.12).

Recall that $\mathrm{H}^0(X, L)$ can be identified with the vector space generated by χ^m for all $m \in P \cap M$, see [CLS11, Proposition 4.3.3]. In other words, a character χ^m of T gives a global section of L if and only if $m \in P$. This gives a beautiful characterization of the lattice points in P. The following theorem of Yi Yao gives an analogous characterization of the lattice points in the Newton body.

Theorem 5.2.2 (Yao) *Let $\varphi \in \mathrm{PSH}_{\mathrm{tor}}(X, \omega)$. Let $m \in M$.*

(1) *Suppose that $m \in \Delta(\omega, \varphi)$. Then $\chi^m \in \mathrm{H}^0(X, L \otimes \mathcal{I}(\varphi))$.*
(2) *There is a constant $C_0 > 0$ such that if there is $\rho \in \Sigma(1)$ with*

$$\langle m, -u_\rho\rangle - \mathrm{Supp}_{\Delta(\omega,\varphi)}(-u_\rho) > C_0, \tag{5.13}$$

then $\chi^m \notin \mathrm{H}^0(X, L \otimes \mathcal{I}(\varphi))$.
Moreover, the constant C_0 does not change if we replace P by a positive integer multiple of P.
(3) *More generally, there is a constant $C_0 > 0$ such that if $\sigma \in \Sigma$ is a cone of dimension p and $c_1, \ldots, c_p > 0$ are such that $\sum_{i=1}^p c_i = 1$, and*

$$\langle m, -u\rangle - \mathrm{Supp}_{\Delta(\omega,\varphi)}(-u) > C_0, \quad u = \sum_{i=1}^{p} c_i u_{\rho_i}, \tag{5.14}$$

where $\rho_1, \ldots, \rho_p$ are the rays in $\sigma(1)$, then $\chi^m \notin \mathrm{H}^0(X, L \otimes \mathcal{I}(\varphi))$.
Moreover, the constant C_0 does not change if we replace P by a positive integer multiple of P.

In (3), there are exactly p rays in $\sigma(1)$ since σ is smooth.

Proof It is convenient to use explicit coordinates. We will identify N with $\mathbb{Z}^n$ after choosing a basis. In this way, we get an identification $M = \mathbb{Z}^n$ and $T(\mathbb{C}) = \mathbb{C}^{*n}$. In this case, we have

$$\chi^m(z) = z^m$$

with the multi-index notation.

Observe that $\mathrm{H}^0(X, L \otimes \mathcal{I}(\varphi))$ is a T_c-invariant subspace of $\mathrm{H}^0(X, L)$. Since the T_c-weight spaces in $\mathrm{H}^0(X, L)$ are the lines $\mathbb{C}\chi^m$, it follows that $\mathrm{H}^0(X, L \otimes \mathcal{I}(\varphi))$ is the direct sum of suitable $\mathbb{C}\chi^m$'s. Due to Proposition 3.2.9, we may replace φ by $P_\omega[\varphi]$ and thanks to Proposition 5.2.6, we may assume that G_φ has the following form:

$$G_\varphi(x) = \begin{cases} G_0(x), & \text{if } x \in \Delta(\omega, \varphi); \\ \infty, & \text{otherwise.} \end{cases}$$

In particular, $F_\varphi \sim \mathrm{Supp}_{\Delta(\omega,\varphi)}$, so the difference $F_\varphi - \mathrm{Supp}_{\Delta(\omega,\varphi)}$ is bounded on $N_\mathbb{R}$. If $m \notin P$, then χ^m is not a section of L, so only the case $m \in M \cap P$ is relevant. For such an m, we need to know whether the following expression is finite or not:

$$\int_{\mathbb{C}^{*n}} |\chi^m|^2 \exp(-\mathrm{Trop}^* F_0 - \varphi)\, \omega^n. \tag{5.15}$$

On $T(\mathbb{C})$, we have $\mathrm{Trop}^* F_0 + \varphi = \mathrm{Trop}^* F_\varphi$. By Proposition 5.2.5, (5.15) is finite if and only if

$$\int_{\mathbb{R}^n} \exp\left(\langle m, n\rangle - F_\varphi(n)\right) \mathrm{MA}_\mathbb{R}(F_0)(n) < \infty.$$

Since $F_\varphi - \mathrm{Supp}_{\Delta(\omega,\varphi)}$ is bounded, this is equivalent to the finiteness of

$$\int_{\mathbb{R}^n} \exp\left(\langle m, n\rangle - \mathrm{Supp}_{\Delta(\omega,\varphi)}(n)\right) \mathrm{MA}_\mathbb{R}(F_0)(n).$$

By a change of variable, this integral is finite if and only if the following integral is:

$$\int_P \exp\left(\langle m, \nabla G_0(m')\rangle - \mathrm{Supp}_{\Delta(\omega,\varphi)}(\nabla G_0(m'))\right) \mathrm{d}m'. \tag{5.16}$$

Suppose that $m \in \Delta(\omega, \varphi)$. Then the integrand in (5.16) is bounded from above by 1, so (1) follows.

Next we consider (2). Put

$$H(n) = \langle m, n\rangle - \mathrm{Supp}_{\Delta(\omega,\varphi)}(n), \quad n \in N_\mathbb{R}.$$

This is a Lipschitz continuous positively homogeneous function on $N_\mathbb{R}$.

Suppose that m satisfies the assumptions of (2). Take $\rho \in \Sigma(1)$ so that (5.13) holds. The condition on $C_0 > 0$ will be clarified later on. Take an open subset U of P which satisfies the following two conditions:

- The intersection $U \cap Q$ has dimension $n-1$, where Q is the face of P defined by $\langle \bullet, u_\rho \rangle = -a_\rho$;
- U does not intersect other faces of P. □

See Fig. 5.4 for the visualization of U.

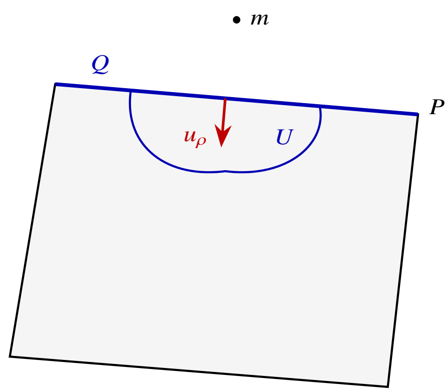

Fig. 5.4 The choice of U and the position of m.

Then by (5.7),

$$(\nabla G_0)\,|_U = -|r_\rho| u_\rho + O(1). \tag{5.17}$$

We claim that

$$\int_U \exp\left(\langle m, \nabla G_0(m')\rangle - \mathrm{Supp}_{\Delta(\omega,\varphi)}(\nabla G_0(m'))\right)\,\mathrm{d}m' = \infty.$$

In view of (5.13) and (5.17), and since H is Lipschitz, we have

$$H(\nabla G_0(m')) = |r_\rho(m')| H(-u_\rho) + O(1)$$

on U. After slightly shrinking U, this gives

$$H(\nabla G_0(m')) > \frac{1}{2} C_0 |r_\rho(m')|.$$

It suffices therefore to establish the following assertion:

$$\int_U \exp\left(-2^{-1} C_0 r_\rho(m')\right)\,\mathrm{d}m' = \infty.$$

Taking the definition of r_ρ into account, this is further equivalent to the following:

$$\int_U \left(\langle m', u_\rho \rangle + a_\rho\right)^{-2^{-1}C_0}\,\mathrm{d}m' = \infty.$$

This holds as long as $C_0 > 2$. This condition is independent of ρ. Furthermore, since replacing P by kP for some $k \in \mathbb{Z}_{>0}$ does not change the condition on C_0, we conclude the final assertion.

(3) This is similar to (2), but the notations get more complicated.

Let Q be the face of P corresponding to σ. Without loss of generality, we may assume that

$$c_1 \leq c_2 \leq \cdots \leq c_p.$$

We first take $\epsilon \in (0, 1/2)$ so that when $b_1, \ldots, b_p > 0$ satisfies

$$-\epsilon \leq \frac{b_j}{b_1} - \frac{c_j}{c_1} \leq \epsilon \quad \forall j = 2, \ldots, p,$$

we have

$$H\left(-\sum_{i=1}^{p} b_i u_{\rho_i}\right) > \frac{C_0}{2} \cdot \sum_{i=1}^{p} b_i.$$

This is possible thanks to (5.14).

Take a small open neighborhood V of Q in $M_{\mathbb{R}}$ with positive distances to all facets of P not containing Q. Consider the open subset U of $m' \in V \cap \operatorname{Int} P$ defined by the following conditions:

$$-\epsilon < \frac{-r_{\rho_j}(m')}{-r_{\rho_1}(m')} - \frac{c_j}{c_1} < \epsilon, \quad \forall j = 2, \ldots, p. \tag{5.18}$$

Then after possibly shrinking V so that $r_{\rho_i}(m') < 0$ for all $m' \in U$ and all $i = 1, \ldots, p$, we have

$$H\left(\sum_{i=1}^{p} r_{\rho_i}(m') u_{\rho_i}\right) > \frac{C_0}{2} \cdot \sum_{i=1}^{p} \left|r_{\rho_i}(m')\right|.$$

Thanks to (5.7), and again using the Lipschitz continuity of H, the bounded contribution of the facets not containing Q is negligible compared with $\sum_i |r_{\rho_i}(m')|$ after shrinking V. Thus for $m' \in U$,

$$H(\nabla G_0(m')) > \frac{C_0}{3} \cdot \sum_{i=1}^{p} \left|r_{\rho_i}(m')\right|.$$

Since $r_{\rho_i}(m') = \log\left(\langle m', u_{\rho_i}\rangle + a_{\rho_i}\right) + 1$, after shrinking V once more we may fix any $\delta \in (0, 1/3)$, say $\delta = 1/4$, so that

$$H(\nabla G_0(m')) > \delta C_0 \cdot \sum_{i=1}^{p} -\log\left(\langle m', u_{\rho_i}\rangle + a_{\rho_i}\right)$$

whenever $m' \in U$.

Now we claim that for C_0 large enough, the integral

$$\int_U \prod_{i=1}^{p} \left(\langle m', u_{\rho_i}\rangle + a_{\rho_i}\right)^{-\delta C_0} \, \mathrm{d}m' = \infty. \tag{5.19}$$

This will then conclude the proof as in (2).

Let

$$x_i = \langle m', u_{\rho_i}\rangle + a_{\rho_i}, \quad i = 1, \dots, p.$$

Since P is smooth, after adding affine coordinates along the face Q, the functions $x_1, \dots, x_p$ form part of a coordinate system in a neighborhood of Q, and the Lebesgue measure is comparable to $\mathrm{d}x_1 \cdots \mathrm{d}x_p$ times the Lebesgue measure in the remaining coordinates. Shrinking V and fixing the remaining coordinates in a small set of positive measure, it is therefore enough to find a region in the x-coordinates contained in U on which the corresponding p-dimensional integral diverges.

The inequalities in (5.18), together with $r_{\rho_i} = \log x_i + 1$, imply that after decreasing $\epsilon > 0$ and $\epsilon_0 > 0$ if necessary, the region $W \subseteq \mathbb{R}^p$ defined by

$$0 < x_1 < \epsilon_0, \quad x_1^{c_i/c_1+\epsilon} < x_i < x_1^{c_i/c_1-\epsilon}$$

for all $i = 2, \dots, p$ is contained in the projection of U. Consequently, for $\alpha = \delta C_0 > 1$, it is enough to prove the divergence of

$$\int_W \prod_{i=1}^{p} x_i^{-\alpha} \, \mathrm{d}x. \tag{5.20}$$

A straightforward computation shows that the integral is bounded from below by a positive multiple of

$$\int_0^{\epsilon_0} x_1^d \, \mathrm{d}x_1,$$

where

$$\begin{aligned} d &= -\alpha + (1-\alpha)\left(c_1^{-1} - 1 + \epsilon(p-1)\right) \\ &\leq -\alpha + (1-\alpha)(p-1) \\ &= -p\alpha + p - 1. \end{aligned}$$

When $\alpha > 1$, we have $d < -1$ and hence the integral (5.20) diverges. Thus it is enough to choose the same constant C_0 so that $C_0 > 2$ and $\delta C_0 > 1$. The estimates above depend only on logarithmic boundary coordinates, so this choice is unchanged after replacing P by kP.

Corollary 5.2.1 *Let* $\varphi \in \mathrm{PSH}_{\mathrm{tor}}(X, \omega)$. *Then*

$$\lim_{k\to\infty} \frac{n!}{k^n} h^0\left(X, L^k \otimes \mathcal{I}(k\varphi)\right) = n! \operatorname{vol} \Delta(\omega, \varphi).$$

Proof Applying Theorem 5.2.2 (1) to the line bundle L^k and the potential $k\varphi$, whose Newton body is $k\Delta(\omega, \varphi)$, we have

$$\#\left(k\Delta(\omega,\varphi)\cap M\right) \leq h^0\left(X, L^k\otimes\mathcal{I}(k\varphi)\right).$$

Here # denotes the number of elements. Therefore,

$$\lim_{k\to\infty} k^{-n}\#\left(k\Delta(\omega,\varphi)\cap M\right) \leq \varliminf_{k\to\infty} k^{-n}h^0\left(X, L^k\otimes\mathcal{I}(k\varphi)\right).$$

The left-hand side is just $\operatorname{vol}\Delta(\omega,\varphi)$, as follows from [KK12, Theorem 2]. So we conclude that

$$\varliminf_{k\to\infty}\frac{n!}{k^n}h^0\left(X, L^k\otimes\mathcal{I}(k\varphi)\right) \geq n!\operatorname{vol}\Delta(\omega,\varphi).$$

It remains to prove the reverse inequality.

We choose an identification $M\cong\mathbb{Z}^n$. In particular, there is a natural distance d on $M_{\mathbb{R}}\cong\mathbb{R}^n$. Given a convex body K and $\epsilon>0$, we let

$$K^\epsilon := \{x\in\mathbb{R}^n : \mathrm{d}(x,K)\leq\epsilon\}.$$

This is again a convex body, since it can be realized as the Minkowski sum of K with a closed ball of radius ϵ.

Thanks to Theorem 5.2.2 (3), and using the fact that the constant C_0 in Theorem 5.2.2 (3) is invariant under $P\mapsto kP$, we can find $C>0$ uniform in k such that for each $k\in\mathbb{Z}_{>0}$,

$$\left\{m\in M : \chi^m\in \mathrm{H}^0(X, L^k\otimes\mathcal{I}(k\varphi))\right\}\subseteq\left(k\Delta(\omega,\varphi)\right)^C.$$

Indeed, if a point m of kP is at distance at least C from $k\Delta(\omega,\varphi)$, the separating hyperplane theorem gives a unit direction $v\in N_{\mathbb{R}}$ for which

$$\langle m, v\rangle - \mathrm{Supp}_{k\Delta(\omega,\varphi)}(v)$$

is at least C. Since the fan Σ is finite and covers $N_{\mathbb{R}}$, after replacing v by a positive multiple we may write $v=-\lambda\sum_i c_i u_{\rho_i}$ with $\lambda>0$, with the ρ_i lying in a single cone of Σ, with $c_i>0$, and with $\sum_i c_i=1$. The possible values of λ are bounded from above independently of v, because Σ is finite. Choosing C large enough then makes (5.14) applicable. Fix $\epsilon>0$, then for large enough k, we have

$$\left(k\Delta(\omega,\varphi)\right)^C\subseteq k\left(\Delta(\omega,\varphi)^\epsilon\right).$$

It follows that

$$h^0\left(X, L^k\otimes\mathcal{I}(k\varphi)\right)\leq\#\left(k\left(\Delta(\omega,\varphi)^\epsilon\right)\cap M\right).$$

Applying [KK12, Theorem 2] again, we conclude that

$$\varlimsup_{k\to\infty}\frac{n!}{k^n}h^0\left(X, L^k\otimes\mathcal{I}(k\varphi)\right)\leq n!\operatorname{vol}\left(\Delta(\omega,\varphi)^\epsilon\right).$$

Letting $\epsilon \to 0+$ and applying Theorem C.1.2, we find

$$\varlimsup_{k\to\infty} \frac{n!}{k^n} h^0\left(X, L^k \otimes \mathcal{I}(k\varphi)\right) \leq n! \operatorname{vol} \Delta(\omega,\varphi).$$

Our assertion follows. □

Example 5.2.5 In general, in the setup of Theorem 5.2.2, there exists $m \in M \cap (P \setminus \Delta(\omega,\varphi))$ such that $\chi^m \in \mathrm{H}^0(X, L \otimes \mathcal{I}(\varphi))$.

As a concrete example, let us take $P = [0,1]$. Take φ so that $\Delta(\omega,\varphi) = [0,1/2]$. We claim that χ^1 is L^2-integrable.

It suffices to verify the convergence of (5.16). Recall that

$$\nabla G_0(m') = \log \frac{m'}{1-m'}, \quad m' \in [0,1],$$

while

$$\operatorname{Supp}_{[0,1/2]}(a) = \begin{cases} a/2, & \text{if } a > 0; \\ 0, & \text{otherwise.} \end{cases}$$

Therefore, (5.16) becomes

$$\int_0^{1/2} \frac{m'}{1-m'} \,\mathrm{d}m' + \int_{1/2}^1 \left(\frac{m'}{1-m'}\right)^{1/2} \mathrm{d}m' < \infty.$$

We interpret various classes of potentials studied in Section 3.1.3 in the toric setting.

Proposition 5.2.7 *Let $\varphi \in \mathrm{PSH}_{\mathrm{tor}}(X,\omega)$. Then the following are equivalent:*

(1) $\varphi \in \mathcal{E}^\infty(X,\omega)$;
(2) $F_\varphi \in \mathcal{E}^\infty(N_\mathbb{R}, P)$;
(3) $G_\varphi \sim G_0$.

The notation $\mathcal{E}^\infty(N_\mathbb{R}, P)$ is defined in Definition A.3.1.

Proof This follows immediately from Proposition 5.2.1. □

Proposition 5.2.8 *Let $\varphi \in \mathrm{PSH}_{\mathrm{tor}}(X,\omega)$. Then the following are equivalent:*

(1) $\varphi \in \mathcal{E}(X,\omega)$;
(2) $F_\varphi \in \mathcal{E}(N_\mathbb{R}, P)$;
(3) $\overline{\operatorname{Dom} G_\varphi} = P$.

The notation $\mathcal{E}(N_\mathbb{R}, P)$ is defined in Definition A.3.1.

Proof (1) $\iff$ (3). By Proposition 5.2.5

$$\int_X \omega_\varphi^n = \int_{T(\mathbb{C})} \left(\omega|_{T(\mathbb{C})} + \mathrm{dd^c}\varphi|_{T(\mathbb{C})}\right)^n = n! \operatorname{vol} \overline{\operatorname{Dom} G_\varphi}, \quad \int_X \omega^n = n! \operatorname{vol} P.$$

Therefore, (1) and (3) are equivalent.

(2) $\iff$ (3). This follows from Proposition A.2.2. □

Proposition 5.2.9 *Let* $\varphi \in \mathrm{PSH}_{\mathrm{tor}}(X,\omega)$. *Then*

$$E_\omega(\varphi) = n! \int_P \left(G_0 - G_\varphi\right) \,\mathrm{d}\,\mathrm{vol}\,.$$

Proof Both sides change by the same additive constant when we replace φ by $\varphi + A$, namely by

$$A \int_X \omega^n = A\, n! \,\mathrm{vol}\, P.$$

We may therefore assume that $\varphi \leq 0$. For $C > 0$, put

$$\varphi_C \coloneqq \varphi \vee (-C).$$

Then $\varphi_C \in \mathrm{PSH}_{\mathrm{tor}}(X,\omega) \cap L^\infty$ and $\varphi_C \searrow \varphi$ as $C \to \infty$. By Proposition 5.2.4, $G_{\varphi_C} \nearrow G_\varphi$. Applying the bounded case [BB13, Proposition 2.9] to φ_C, we get

$$E_\omega(\varphi_C) = n! \int_P \left(G_0 - G_{\varphi_C}\right) \,\mathrm{d}\,\mathrm{vol}\,.$$

Letting $C \to \infty$, the left-hand side converges to $E_\omega(\varphi)$ by Proposition 3.1.16. On the right-hand side, since $\varphi_C \leq 0$, we have $G_{\varphi_C} \geq G_0$, and the monotone convergence theorem applied to $G_{\varphi_C} - G_0 \nearrow G_\varphi - G_0$ gives the desired formula. □

Corollary 5.2.2 *Let* $\varphi \in \mathrm{PSH}_{\mathrm{tor}}(X,\omega)$. *Then the following are equivalent:*

(1) $\varphi \in \mathcal{E}^1(X,\omega)$;
(2) $F_\varphi \in \mathcal{E}^1(N_\mathbb{R}, P)$;
(3) $G_\varphi \in L^1(P)$.

The notation $\mathcal{E}^1(N_\mathbb{R}, P)$ is defined in Definition A.3.1.

Proof The equivalence between (1) and (3) follows from Proposition 5.2.9. The equivalence between (2) and (3) is the definition of $\mathcal{E}^1(N_\mathbb{R}, P)$, since $F_\varphi^* = G_\varphi$. □

Definition 5.2.2 We define

$$\begin{aligned}
\mathcal{E}^\infty_{\mathrm{tor}}(X,\omega) &= \mathcal{E}^\infty(X,\omega) \cap \mathrm{PSH}_{\mathrm{tor}}(X,\omega),\\
\mathcal{E}^1_{\mathrm{tor}}(X,\omega) &= \mathcal{E}^1(X,\omega) \cap \mathrm{PSH}_{\mathrm{tor}}(X,\omega),\\
\mathcal{E}_{\mathrm{tor}}(X,\omega) &= \mathcal{E}(X,\omega) \cap \mathrm{PSH}_{\mathrm{tor}}(X,\omega).
\end{aligned}$$

Corollary 5.2.3 *Let* $\varphi, \psi \in \mathcal{E}^1_{\mathrm{tor}}(X,\omega)$. *Then*

$$d_1(\varphi,\psi) = n! \int_P \left(2\left(G_\varphi \vee G_\psi\right) - G_\varphi - G_\psi\right) \,\mathrm{d}\,\mathrm{vol}\,.$$

Proof This follows from Proposition 5.2.9, Proposition 5.2.3 and Definition 4.3.1. □

Part II
The theory of $\mathcal{I}$-good singularities

This part is the technical core of the whole book. We develop the theory of $\mathcal{I}$-good singularities together with the analytic and algebraic tools surrounding them.

We briefly recall the four prerequisites from Part I on which this part rests: (i) the preorder $\preceq$ on $\mathrm{QPSH}(X)$ comparing potentials up to additive constants (Definition 1.5.2); (ii) the P-envelope $P_\theta[\bullet]$ and the $\mathcal{I}$-envelope $P_\theta[\bullet]_{\mathcal{I}}$ from Chapter 3, which extract the P- and $\mathcal{I}$-equivalence classes of a potential; (iii) the notion of *model potential* ($\varphi = P_\theta[\varphi]$) and the subspace $\mathrm{PSH}(X,\theta)_{>0}$ of θ-psh functions of positive non-pluripolar mass; (iv) the existence and basic properties of geodesic rays from Chapter 4.

The central thesis of this part is the following. In general, a single $\mathcal{I}$-equivalence class on $\mathrm{PSH}(X,\theta)_{>0}$ contains many P-equivalence classes, so that $\sim_P$ and $\sim_{\mathcal{I}}$ are genuinely distinct relations. The singularities arising naturally in geometric applications — analytic singularities, model potentials, full-mass potentials, toric potentials, elementary metrics — are, however, $\mathcal{I}$*-good*, and on such potentials the two relations agree. The asymptotic Riemann–Roch formula Theorem 7.4.1 captures the algebraic content of $\mathcal{I}$-goodness: for an $\mathcal{I}$-good metric, the asymptotics of multiplier-ideal cohomology recover the non-pluripolar volume of the underlying Chern current. In short, $\mathcal{I}$-good singularities are precisely those visible to algebraic geometry, and developing their theory is the goal of this part.

Chapter 6 sets up the general framework for comparing singularities: the P- and $\mathcal{I}$-preorders on $\mathrm{QPSH}(X)$ and the d_S-pseudometric on $\mathrm{PSH}(X,\theta)$, together with their continuity and convergence properties. The P-preorder is new, as are several basic properties of the d_S-pseudometric.

Chapter 7 introduces the notion of $\mathcal{I}$-good singularities (Definition 7.1.1) and proves the central equivalence theorem Theorem 7.1.1, giving three characterizations of $\mathcal{I}$-goodness: as the d_S-closure of analytic-singularity potentials, as the locus where non-pluripolar mass equals volume, and as the locus where the P- and $\mathcal{I}$-envelopes coincide. The chapter then develops the mixed volume theory of $\mathcal{I}$-good potentials and establishes the asymptotic Riemann–Roch formula Theorem 7.4.1.

Chapter 8 develops two key techniques for the inductive study of singularities: the *trace operator*, which gives a meaningful restriction of a quasi-psh function to a (possibly singular) subvariety modulo $\mathcal{I}$-equivalence, and the *analytic Bertini theorem*, controlling the behavior of a quasi-psh function along a generic member of a base-point-free linear system. The chapter culminates in a relative version of the asymptotic Riemann–Roch formula.

Chapter 9 develops the theory of *test curves* — concave decreasing families of model potentials — and proves the Ross–Witt Nyström correspondence Theorem 9.2.1, which identifies test curves modulo P-equivalence with geodesic rays via the Legendre transform. Our formulation is more general than all comparable results in the literature and is given in full detail. The parallel $\mathcal{I}$-model theory and the algebraic operations on test curves required in Chapter 13 are also developed here.

Chapter 10 constructs the *partial Okounkov bodies* $\Delta(L,\phi)$ associated with a psh metric ϕ on a big line bundle. These convex bodies extend the classical Newton–Okounkov body construction to the singular setting and provide a universal numerical invariant of the $\mathcal{I}$-equivalence class of ϕ. Both an algebraic and a transcendental theory are developed, and the construction yields nontrivial new results even in the toric setting.

Chapter 11 develops the theory of nef b-divisors and establishes the correspondence Theorem 11.1.3: after fixing a modified nef class α, positive currents with regular part in α and positive volume, modulo $\mathcal{I}$-equivalence, correspond to big and nef b-divisors with trace α on X. This makes precise the slogan that the $\mathcal{I}$-equivalence class of such a current is determined by cohomological data on the universe of all modifications of X.

These chapters are designed to be read linearly. However, once Chapters 6 to 8 have been digested (the comparison framework, the $\mathcal{I}$-good notion, and the trace and Bertini techniques), the reader may proceed to the application chapters in Part III and consult the remaining chapters of this part as needed: Chapter 10 is essential for Chapter 12, and Chapter 9 is essential for Chapter 13.

Chapter 6
Comparison of singularities

Algebra is the offer made by the devil to the mathematician. The devil says: "I will give you this powerful machine, it will answer any question you like. All you need to do is give me your soul: give up geometry and you will have this marvelous machine.
— Michael Atiyah[a]

[a] Sir Michael Francis Atiyah (1929–2019) wrote the influential *Introduction to commutative algebra* together with I. G. MacDonald, whose name is often omitted or misspelled.

A central theme of this book is that quasi-plurisubharmonic functions naturally come with several different notions of "equivalence of singularities". The finest is the preorder $\preceq$ of Definition 1.5.2, which compares potentials up to additive constants; this is also the first notion the reader has encountered. As we saw in Chapter 3, however, $\preceq$ is too rigid for many constructions: P-envelopes already collapse $\preceq$-classes into larger *P-equivalence* classes (Definition 3.1.2), and multiplier ideals collapse them further into *$\mathcal{I}$-equivalence* classes. This chapter promotes P- and $\mathcal{I}$-equivalence to genuine preorders on $\mathrm{QPSH}(X)$, identifies the relations between them, and introduces a pseudometric d_S that quantifies differences between singularities within the P-equivalence framework.

For orientation, the three preorders compared in this chapter and used pervasively in the rest of the book are:

relation	defined by	relative coarseness
$\varphi \preceq \psi$	$\varphi \leq \psi + C$ for some C (Definition 1.5.2)	finest
$\varphi \preceq_P \psi$	$P_\theta[\varphi] \leq P_\theta[\psi]$ (Definition 6.1.1)	coarser
$\varphi \preceq_{\mathcal{I}} \psi$	$\mathcal{I}(k\varphi) \subseteq \mathcal{I}(k\psi)$ for all $k \in \mathbb{Z}_{>0}$ (Definition 6.1.2)	coarsest

We use these three relations constantly throughout the rest of the book; the table fixes their relative strength at first encounter.

Section 6.1 introduces the P- and $\mathcal{I}$-preorders and develops their basic properties. The section culminates in Theorem 6.1.1, an existence theorem for geodesics with prescribed P-equivalence type.

Section 6.2 introduces the *d_S-pseudometric* on $\mathrm{PSH}(X, \theta)$ via the Darvas d_1-distance between the associated geodesic rays (Definition 6.2.1). We first characterize when $d_S(\varphi, \psi) = 0$ (it is precisely $\varphi \sim_P \psi$, see Proposition 6.2.2), and then establish the basic properties of d_S. The chapter closes with continuity statements (Theorems 6.2.3, 6.2.5 and 6.2.6) showing that the P- and $\mathcal{I}$-envelopes, the non-pluripolar volume, and equisingularity are all d_S-continuous; these continuity results are the technical engine driving applications in later chapters.

Results in this chapter will play a key role in the analytic side of the Ross–Witt Nyström correspondence Chapter 9 and the non-Archimedean theory Chapter 13.

6.1 The P and $\mathcal{I}$-preorders

Let X be a connected compact Kähler manifold of dimension n.

Recall that we defined a preorder on $\mathrm{QPSH}(X)$ in Definition 1.5.2 to compare the singularity types of quasi-plurisubharmonic functions. The problem with this preorder is that it is too fine. For our purposes, it is useful to consider rougher relations.

6.1.1 The definitions of the preorders

Recall that the P-envelope is defined in Definition 3.1.2.

Definition 6.1.1 For $\varphi, \psi \in \mathrm{QPSH}(X)$, we say φ is *P-more singular than* ψ and write $\varphi \preceq_P \psi$ if for some (equivalently, every, due to Lemma 6.1.1 below) closed smooth real $(1,1)$-form θ on X such that $\varphi, \psi \in \mathrm{PSH}(X,\theta)_{>0}$, we have

$$P_\theta[\varphi] \le P_\theta[\psi]. \tag{6.1}$$

If $\varphi \preceq_P \psi$ and $\psi \preceq_P \varphi$, we write $\varphi \sim_P \psi$ and say that φ and ψ have the same *P-singularity type*.

Recall that $\mathrm{PSH}(X,\theta)_{>0}$ was introduced in (2.8). Note that if $\varphi \preceq \psi$, then $\varphi \preceq_P \psi$. So the P-preorder is *coarser* than $\preceq$.

The P-equivalence relation can also be formulated using the language of relative full mass classes developed in Section 3.1.3. In fact, $\varphi, \psi \in \mathrm{PSH}(X,\theta)_{>0}$ satisfies $\varphi \sim_P \psi$ if and only if $\varphi, \psi \in \mathcal{E}(X,\theta;\phi)$ for a common model potential $\phi \in \mathrm{PSH}(X,\theta)_{>0}$.

The condition (6.1) is independent of the choice of θ:

Lemma 6.1.1 *Let $\varphi, \psi \in \mathrm{PSH}(X,\theta)_{>0}$. For any Kähler form ω on X, the following are equivalent:*

(1) $P_\theta[\varphi] \le P_\theta[\psi]$;
(2) $P_{\theta+\omega}[\varphi] \le P_{\theta+\omega}[\psi]$.

In particular, $\preceq_P$ defines a preorder on $\mathrm{QPSH}(X)$.

Proof (1) $\implies$ (2). Assume (1). Observe that

$$\varphi \le P_\theta[\varphi] \le P_{\theta+\omega}[\varphi].$$

Taking the $P_{\theta+\omega}$-envelope, we find

$$P_{\theta+\omega}[\varphi] \le P_{\theta+\omega}\Big[P_\theta[\varphi]\Big] \le P_{\theta+\omega}\Big[P_{\theta+\omega}[\varphi]\Big] = P_{\theta+\omega}[\varphi],$$

where the equality follows from Theorem 3.1.2. In particular,

$$P_{\theta+\omega}[\varphi] = P_{\theta+\omega}\Big[P_\theta[\varphi]\Big]. \tag{6.2}$$

A similar formula holds for ψ. So we see that (2) holds.

(2) $\implies$ (1). Assume (2). By (6.2), we may assume that φ and ψ are both model potentials in $\mathrm{PSH}(X,\theta)_{>0}$.

Observe that $\varphi \vee \psi \leq P_{\theta+\omega}[\psi]$. It follows that $P_{\theta+\omega}[\varphi \vee \psi] \leq P_{\theta+\omega}[\psi]$. The reverse inequality is trivial, so

$$P_{\theta+\omega}[\varphi \vee \psi] = P_{\theta+\omega}[\psi].$$

From the direction we have proved, for any $C \geq 1$,

$$P_{\theta+C\omega}[\varphi \vee \psi] = P_{\theta+C\omega}[\psi].$$

So by Proposition 3.1.3,

$$\int_X \left(\theta + C\omega + \mathrm{dd^c}(\varphi \vee \psi)\right)^n = \int_X \left(\theta + C\omega + \mathrm{dd^c}\psi\right)^n .$$

Since both sides are polynomials in C, the equality extends to $C = 0$, namely,

$$\int_X \theta_{\varphi\vee\psi}^n = \int_X \theta_\psi^n.$$

In particular, $\varphi \vee \psi \leq P_\theta[\psi] = \psi$ by (3.7). So (1) follows. □

As a consequence of Lemma 6.1.1, we can define the P-preorder at the level of currents. Given closed positive $(1,1)$-currents $T = \theta_\varphi$, $S = \theta'_\psi$, we write $T \preceq_P S$ (resp. $T \sim_P S$) if $\varphi \preceq_P \psi$ (resp. $\varphi \sim_P \psi$). This definition is independent of the decompositions of T and S.

As a first example of P-equivalence, we have:

Example 6.1.1 Let θ be a closed smooth real $(1,1)$-form on X and $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$. Then

$$\varphi \sim_P P_\theta[\varphi]^{1}.$$

This follows immediately from Theorem 3.1.2.

We give a very useful criterion of the P-equivalence in terms of the non-pluripolar masses.

Proposition 6.1.1 *Let θ be a closed smooth real $(1,1)$-form on X representing a big cohomology class. Let $\varphi, \psi \in \mathrm{PSH}(X,\theta)$ and $\varphi \leq \psi$. Then the following are equivalent:*

(1) *$\varphi \sim_P \psi$;*
(2) *for each $j = 0, \dots, n$, we have*

$$\int_X \theta_\varphi^j \wedge \theta_{V_\theta}^{n-j} = \int_X \theta_\psi^j \wedge \theta_{V_\theta}^{n-j}. \tag{6.3}$$

[1] It is not known to us whether the same holds when φ has vanishing mass.

Assume furthermore that $\varphi, \psi \in \mathrm{PSH}(X,\theta)_{>0}$, then these conditions are equivalent to the following:

(3) *We have*

$$\int_X \theta_\varphi^n = \int_X \theta_\psi^n.$$

Recall that V_θ is introduced in (2.9).

Proof We first prove the equivalence between (1) and (3) when $\varphi, \psi \in \mathrm{PSH}(X,\theta)_{>0}$.

(1) $\implies$ (3). Assume that $\varphi \sim_P \psi$. By Lemma 6.1.1, we have

$$P_\theta[\varphi] = P_\theta[\psi].$$

So (3) follows from Proposition 3.1.3.

(3) $\implies$ (1). It follows from Theorem 3.1.2 that $P_\theta[\varphi] = P_\theta[\psi]$, so (1) follows.

Let us come back to the general case with $\varphi, \psi \in \mathrm{PSH}(X,\theta)$.

(1) $\implies$ (2). Fix $j \in \{0, \dots, n\}$, we argue (6.3).

Take a Kähler form ω on X. By Lemma 6.1.1, for each $\epsilon > 0$, we have

$$P_{\theta+\epsilon\omega}[\varphi] = P_{\theta+\epsilon\omega}[\psi].$$

It follows from Proposition 3.1.3 that

$$\begin{aligned}\int_X (\theta + \epsilon\omega + \mathrm{dd^c}\psi)^j \wedge \theta_{V_\theta}^{n-j} &= \int_X \left(\theta + \epsilon\omega + \mathrm{dd^c} P_{\theta+\epsilon\omega}[\psi]\right)^j \wedge \theta_{V_\theta}^{n-j} \\ &= \int_X \left(\theta + \epsilon\omega + \mathrm{dd^c} P_{\theta+\epsilon\omega}[\varphi]\right)^j \wedge \theta_{V_\theta}^{n-j} \\ &= \int_X (\theta + \epsilon\omega + \mathrm{dd^c}\varphi)^j \wedge \theta_{V_\theta}^{n-j}.\end{aligned}$$

Since the two extremes are both polynomials in ϵ by the multi-linearity of the non-pluripolar products Proposition 2.2.1 (6), we conclude that the same holds when $\epsilon = 0$, that is, (6.3) holds.

(2) $\implies$ (1). Assume (6.3) holds for all $j = 0, \dots, n$. For each $t \in (0,1)$, we have

$$\int_X \theta^n_{t\varphi+(1-t)V_\theta} = \int_X \theta^n_{t\psi+(1-t)V_\theta}$$

by the binomial expansion. By the implication (3) $\implies$ (1), we have

$$t\varphi + (1-t)V_\theta \sim_P t\psi + (1-t)V_\theta$$

for each $t \in (0,1)$.

Fix a Kähler form ω on X. From the implication (1) $\implies$ (3), we have

$$\int_X (\theta+\omega)^n_{t\varphi+(1-t)V_\theta} = \int_X (\theta+\omega)^n_{t\psi+(1-t)V_\theta}.$$

Since both sides are polynomials in t by the multi-linearity of the non-pluripolar products Proposition 2.2.1 (6), the same holds when $t = 1$. From the implication (3) $\implies$ (1) again, we have $\varphi \sim_P \psi$. □

Next we introduce a different preorder.

Proposition 6.1.2 *Given $\varphi, \psi \in \mathrm{QPSH}(X)$, the following are equivalent:*

(1) *For any $k \in \mathbb{Z}_{>0}$, we have*
$$\mathcal{I}(k\varphi) \subseteq \mathcal{I}(k\psi);$$
(2) *for any $\lambda \in \mathbb{R}_{>0}$, we have*
$$\mathcal{I}(\lambda\varphi) \subseteq \mathcal{I}(\lambda\psi);$$
(3) *for any modification $\pi\colon Y \to X$ and any $y \in Y$, we have*
$$\nu(\pi^*\varphi, y) \geq \nu(\pi^*\psi, y);$$
(4) *for any proper bimeromorphic morphism $\pi\colon Y \to X$ from a Kähler manifold and any $y \in Y$, we have*
$$\nu(\pi^*\varphi, y) \geq \nu(\pi^*\psi, y);$$
(5) *for any prime divisor E over X, we have*
$$\nu(\varphi, E) \geq \nu(\psi, E).$$

Proof The proof is almost identical to that of Proposition 3.2.1. □

Definition 6.1.2 Let $\varphi, \psi \in \mathrm{QPSH}(X)$, we say φ is $\mathcal{I}$*-more singular than* ψ and write $\varphi \preceq_{\mathcal{I}} \psi$ if the equivalent conditions in Proposition 6.1.2 are satisfied.

It is clear that $\preceq_{\mathcal{I}}$ is a preorder on $\mathrm{QPSH}(X)$.

Note that $\varphi \preceq_{\mathcal{I}} \psi$ and $\psi \preceq_{\mathcal{I}} \varphi$ both hold if and only if $\varphi \sim_{\mathcal{I}} \psi$ in the sense of Definition 3.2.1.

Given closed positive $(1,1)$-currents $T = \theta_\varphi$, $S = \theta'_\psi$, we write $T \preceq_{\mathcal{I}} S$ (resp. $T \sim_{\mathcal{I}} S$) if $\varphi \preceq_{\mathcal{I}} \psi$ (resp. $\varphi \sim_{\mathcal{I}} \psi$). This definition is independent of the decompositions of T and S.

Lemma 6.1.2 *Let θ be a closed smooth real $(1,1)$-form on X representing a big cohomology class. Let $\varphi, \psi \in \mathrm{PSH}(X, \theta)_{>0}$. Then*
$$P_\theta[\varphi \vee \psi] = P_\theta\Big[P_\theta[\varphi] \vee P_\theta[\psi]\Big]. \tag{6.4}$$

Proof Since $\varphi \vee \psi \leq P_\theta[\varphi] \vee P_\theta[\psi]$, the $\leq$ direction of (6.4) follows. Conversely, it suffices to show that
$$P_\theta[\varphi \vee \psi] \geq P_\theta[\varphi] \vee P_\theta[\psi],$$
which follows from $\varphi \preceq \varphi \vee \psi$, $\psi \preceq \varphi \vee \psi$, and the monotonicity of the P-envelope with respect to $\preceq$. □

Lemma 6.1.3 *Let $\varphi, \psi \in \mathrm{QPSH}(X)$. Then the following are equivalent:*

(1) $\varphi \preceq_P \psi$ *(resp.* $\varphi \preceq_{\mathcal{I}} \psi$*);*
(2) $\varphi \vee \psi \sim_P \psi$ *(resp.* $\varphi \vee \psi \sim_{\mathcal{I}} \psi$*).*

Proof Take a closed real smooth $(1,1)$-form θ on X such that $\varphi, \psi \in \mathrm{PSH}(X,\theta)_{>0}$. We only prove the P case, the $\mathcal{I}$ case is similar.

(2) $\implies$ (1). Assume (2). By (2) and Example 6.1.1, $P_\theta[\varphi \vee \psi] = P_\theta[\psi] \sim_P \psi$. But $\varphi \leq P_\theta[\varphi \vee \psi]$, so (1) follows.

(1) $\implies$ (2). Assume (1). We may assume that φ, ψ are both model by Lemma 6.1.2. Then $\varphi \leq \psi$ and (2) follows. □

Corollary 6.1.1 *Let $\varphi, \psi \in \mathrm{QPSH}(X)$. Assume that $\varphi \preceq_P \psi$. Then $\varphi \preceq_{\mathcal{I}} \psi$.*

Therefore, the preorder $\preceq_{\mathcal{I}}$ is coarser than $\preceq_P$. Given $\varphi, \psi \in \mathrm{QPSH}(X)$, we therefore have the implications:

$$\varphi \preceq \psi \implies \varphi \preceq_P \psi \implies \varphi \preceq_{\mathcal{I}} \psi .$$

Proof Thanks to Lemma 6.1.3, we have $\varphi \vee \psi \sim_P \psi$. Take a closed smooth real $(1,1)$-form θ on X such that $\varphi, \psi \in \mathrm{PSH}(X,\theta)_{>0}$. Then by Proposition 3.2.9, we have

$$\psi \sim_{\mathcal{I}} P_\theta[\psi] = P_\theta\left[\varphi \vee \psi\right] \sim_{\mathcal{I}} \varphi \vee \psi .$$

Our assertion follows from Lemma 6.1.3. □

Next we give a few extra characterizations of the P-envelope.

Corollary 6.1.2 *Let θ be a closed smooth real $(1,1)$-form on X representing a big cohomology class. Assume that $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$. Then*

$$\begin{aligned} P_\theta[\varphi] &= \sup\left\{\psi \in \mathrm{PSH}(X,\theta) : \psi \leq 0, \psi \sim_P \varphi\right\} \\ &= \sup\left\{\psi \in \mathrm{PSH}(X,\theta) : \psi \leq 0, \psi \preceq_P \varphi\right\} . \end{aligned}$$

Just for comparison, let us recall a few other characterizations of the P-envelope for $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$:

$$\begin{aligned} P_\theta[\varphi] &= \sup{}^* \left\{\psi \in \mathrm{PSH}(X,\theta) : \psi \leq 0, \psi \preceq \varphi\right\} \\ &= \sup{}^* \left\{\psi \in \mathrm{PSH}(X,\theta) : \psi \leq 0, \psi \sim \varphi\right\} \\ &= \sup_{C \in \mathbb{Z}_{>0}}{}^* (\varphi + C) \wedge V_\theta \\ &= \sup\left\{\psi \in \mathrm{PSH}(X,\theta) : \psi \leq 0, \varphi \leq \psi, \int_X \theta_\varphi^n = \int_X \theta_\psi^n\right\} . \end{aligned}$$

Proof Note that $\psi \sim_P \varphi$ implies that $\psi \in \mathrm{PSH}(X, \theta)_{>0}$ by Proposition 3.1.3. We observe that

$$\begin{aligned}&\sup \{\psi \in \mathrm{PSH}(X, \theta) : \psi \leq 0, \psi \sim_P \varphi\}\\ =&\sup \{\psi \in \mathrm{PSH}(X, \theta) : \psi \leq 0, \varphi \leq \psi, \psi \sim_P \varphi\}\end{aligned}$$

since for any candidate ψ for the first supremum, $\psi \vee (\varphi - \sup_X \varphi)$ is larger than ψ, and is a candidate for the second supremum thanks to Lemma 6.1.3. So the first equality is a direct consequence of Proposition 6.1.1 and Theorem 3.1.2.

Next we prove the second equality. We only need to show that for any $\psi \in \mathrm{PSH}(X, \theta)$ with $\psi \leq 0$ and $\psi \preceq_P \varphi$, we have $\psi \leq P_\theta[\varphi]$.

By Lemma 6.1.3 and Example 6.1.1, we know that $P_\theta[\varphi] \vee \psi \sim_P \varphi$ and $P_\theta[\varphi] \vee \psi \leq 0$. It follows from the first equality that $\psi \leq P_\theta[\varphi]$. □

Similarly, we have a new characterization of the $\mathcal{I}$-envelope.

Corollary 6.1.3 *Let θ be a closed smooth real $(1,1)$-form on X representing a pseudoeffective cohomology class. Assume that $\varphi \in \mathrm{PSH}(X, \theta)$. Then*

$$P_\theta[\varphi]_{\mathcal{I}} = \sup \{\psi \in \mathrm{PSH}(X, \theta) : \psi \leq 0, \psi \preceq_{\mathcal{I}} \varphi\}.$$

Proof It suffices to show that for any $\psi \in \mathrm{PSH}(X, \theta)$ with $\psi \leq 0$ and $\psi \preceq_{\mathcal{I}} \varphi$, we have $\psi \leq P_\theta[\varphi]_{\mathcal{I}}$. By Lemma 6.1.3 and Proposition 3.2.6, we know that $P_\theta[\varphi]_{\mathcal{I}} \vee \psi \sim_{\mathcal{I}} \varphi$. Therefore, $P_\theta[\varphi]_{\mathcal{I}} \vee \psi$ is a candidate in the definition of $P_\theta[\varphi]_{\mathcal{I}}$ (cf. (3.53)), so

$$\psi \leq P_\theta[\varphi]_{\mathcal{I}} \vee \psi \leq P_\theta[\varphi]_{\mathcal{I}}.$$

We conclude the proof. □

Proposition 6.1.3 *Let θ be a closed real smooth $(1,1)$-form on X and $\varphi, \psi \in \mathrm{PSH}(X, \theta)$. Then the following are equivalent:*

(1) *$\varphi \preceq_{\mathcal{I}} \psi$;*
(2) *$P_\theta[\varphi]_{\mathcal{I}} \leq P_\theta[\psi]_{\mathcal{I}}$.*

Proof (1) $\implies$ (2). This follows immediately from Corollary 6.1.3.

(2) $\implies$ (1). It follows from Proposition 3.2.6 that

$$\varphi \sim_{\mathcal{I}} P_\theta[\varphi]_{\mathcal{I}} \leq P_\theta[\psi]_{\mathcal{I}} \sim_{\mathcal{I}} \psi.$$

We conclude (1). □

Example 6.1.2 Let us continue our example Example 3.1.1, where $X = \mathbb{P}^1$, ω is the Fubini–Study metric and $\varphi \in \mathrm{PSH}(X, \omega)$ has log-log singularity at 0. We have shown that $P_\omega[\varphi] = 0$ in (3.9), so $\varphi \sim_P 0$ and hence $\varphi \sim_{\mathcal{I}} 0$. In particular, P-equivalence is not equivalent to the equivalence of singularity types.

On the other hand, consider a potential $\psi \in \mathrm{PSH}(X, \omega)$ with log singularity at 0, as in Example 1.8.2. We know that $\nu(\psi, 0) = 1$ from the explicit expression (1.28). So $\psi \nsim_{\mathcal{I}} 0$ and hence $\psi \nsim_P 0$.

Moreover, $\psi \preceq_P \varphi$ and hence $\psi \preceq_{\mathcal{I}} \varphi$.

We give an example showing that P-equivalence is not equivalent to $\mathcal{I}$-equivalence.

Example 6.1.3 Let $X = \mathbb{P}^1$ and ω be the Fubini–Study metric. Let $K \subseteq \mathbb{P}^1$ be a polar Cantor set carrying an atom free probability measure μ supported on K (see [Car83, Page 31]). Write $\mu = \omega + \mathrm{dd^c}\varphi$ for some ω-subharmonic function φ. Since μ is atom free, we know that all Lelong numbers of φ are 0. On the other hand, φ has 0 non-pluripolar mass since K is pluripolar.

Then observe that $\varphi \sim_{\mathcal{I}} 0$ while $\varphi \nsim_P 0$.

For later use, we give the following definition.

Definition 6.1.3 Let L be a pseudo-effective line bundle on X. An *elementary metric* on L is a psh metric h on L such that there is a generalized Fubini–Study metric h' on L such that

$$\mathrm{dd^c}h \sim_P \mathrm{dd^c}h'.$$

The set of elementary metrics on L is denoted by $\mathrm{Ele}(L)$.

We also say $\mathrm{dd^c}h$ is elementary. If we have fixed a Hermitian metric h_0 on L, and if we represent h as $h_0 \exp(-\varphi)$, we also say the quasi-psh function φ is elementary.

Recall that the generalized Fubini–Study metrics are defined in Definition 1.8.7.

6.1.2 Properties of the preorders

Now we state a more natural version of the monotonicity theorem Theorem 2.4.3.

Proposition 6.1.4 *Let $\theta_1, \dots, \theta_n$ be closed real smooth $(1,1)$-forms on X. Let $\varphi_i, \psi_i \in \mathrm{PSH}(X, \theta_i)$ for $i = 1, \dots, n$. Assume that $\varphi_i \preceq_P \psi_i$ for each i. Then*

$$\int_X \theta_{1,\varphi_1} \wedge \dots \wedge \theta_{n,\varphi_n} \leq \int_X \theta_{1,\psi_1} \wedge \dots \wedge \theta_{n,\psi_n}.$$

In other words, the non-pluripolar mass is a decreasing quantity with respect to the P-singularity types of the potentials.

Proof Fix a Kähler form ω on X. For each $i = 1, \dots, n$, since $\varphi_i \preceq_P \psi_i$, we have

$$P_{\theta_i+\epsilon\omega}[\varphi_i] \leq P_{\theta_i+\epsilon\omega}[\psi_i]$$

for all $\epsilon > 0$. Therefore, by Proposition 3.1.3 and our previous monotonicity theorem Theorem 2.4.3, we have

$$\int_X (\theta_1 + \epsilon\omega)_{\varphi_1} \wedge \dots \wedge (\theta_n + \epsilon\omega)_{\varphi_n} \leq \int_X (\theta_1 + \epsilon\omega)_{\psi_1} \wedge \dots \wedge (\theta_n + \epsilon\omega)_{\psi_n}.$$

Letting $\epsilon \to 0+$ and applying the multi-linearity of the non-pluripolar products Proposition 2.2.1 (6), we find the desired inequality. ◻

Next we show that the P and $\mathcal{I}$-preorders are preserved by some natural operations.

Lemma 6.1.4 *Let $\pi\colon Y \to X$ be a proper bimeromorphic morphism from a Kähler manifold Y. Given two quasi-plurisubharmonic functions φ, ψ on X, then the following are equivalent:*

(1) $\varphi \preceq_P \psi$;
(2) $\pi^*\varphi \preceq_P \pi^*\psi$.

The same holds with $\mathcal{I}$ in place of P.

Proof In the P-case, this follows from Proposition 3.1.9, while in the $\mathcal{I}$-case, this follows from Proposition 3.2.5. □

Proposition 6.1.5 *Let $\varphi, \psi, \varphi', \psi' \in \mathrm{QPSH}(X)$. Assume that*

$$\varphi \preceq_P \psi, \quad \varphi' \preceq_P \psi'.$$

Then

$$\varphi + \varphi' \preceq_P \psi + \psi'.$$

The same holds with $\preceq_{\mathcal{I}}$ in place of $\preceq_P$.

Proof Take a Kähler form ω on X such that $\varphi, \psi, \varphi', \psi' \in \mathrm{PSH}(X, \omega)_{>0}$. The statement for $\preceq_{\mathcal{I}}$ is a simple consequence of Proposition 1.4.2. We only need to handle the case of $\preceq_P$.

Step 1. We first show that

$$P_\omega[\varphi] + P_\omega[\varphi'] \sim_P \varphi + \varphi'.$$

In fact, we clearly have

$$P_\omega[\varphi] + P_\omega[\varphi'] \geq \varphi + \varphi'.$$

So by Proposition 6.1.1, it suffices to show that they have the same mass. We compute

$$\begin{aligned}
&\int_X \left(2\omega + \mathrm{dd^c} P_\omega[\varphi] + \mathrm{dd^c} P_\omega[\varphi']\right)^n \\
=& \sum_{j=0}^n \binom{n}{j} \int_X \left(\omega + \mathrm{dd^c} P_\omega[\varphi]\right)^j \wedge \left(\omega + \mathrm{dd^c} P_\omega[\varphi']\right)^{n-j} \\
=& \sum_{j=0}^n \binom{n}{j} \int_X \omega_\varphi^j \wedge \omega_{\varphi'}^{n-j} \\
=& \int_X (2\omega + \mathrm{dd^c}\varphi + \mathrm{dd^c}\varphi')^n,
\end{aligned}$$

where we applied Proposition 3.1.3 to the mixed products on the third line.

Step 2. By Step 1, we may assume that $\varphi, \psi, \varphi', \psi'$ are all model potentials, after replacing them by their P_ω-envelopes. So $\varphi \leq \psi$ and $\varphi' \leq \psi'$. Our assertion follows. □

Proposition 6.1.6 *Let $(\varphi_i)_{i\in I}$, $(\psi_i)_{i\in I}$ be uniformly bounded from above non-empty families in* $\mathrm{QPSH}(X)$. *Assume that there exists a closed smooth real $(1,1)$-form θ such that $\varphi_i, \psi_i \in \mathrm{PSH}(X,\theta)$ and $\varphi_i \preceq_P \psi_i$ for all $i \in I$. Then*

$$\sup_{i\in I}{}^{*}\varphi_i \preceq_P \sup_{i\in I}{}^{*}\psi_i.$$

The same holds with $\preceq_{\mathcal{I}}$ in place of $\preceq_P$.

Proof By adding a Kähler form to θ, we may assume that $\varphi_i, \psi_i \in \mathrm{PSH}(X,\theta)_{>0}$ for all $i \in I$. The statement for $\preceq_{\mathcal{I}}$ is a simple consequence of Corollary 1.4.1, we only have to consider the statement for $\preceq_P$.

Step 1. We first handle the case where I is a directed set and $(\varphi_i)_{i\in I}$ and $(\psi_i)_{i\in I}$ are increasing nets.

In this case, our assertion follows simply from Proposition 3.1.13.

Step 2. We handle the case where I is finite. We may assume that $I = \{0, 1\}$. It suffices to show that

$$P_\theta[\varphi_0] \vee P_\theta[\varphi_1] \sim_P \varphi_0 \vee \varphi_1,$$

which follows from Lemma 6.1.2.

Step 3. The general case can be reduced to the two cases handled in Step 1 and Step 2. More precisely, by applying Proposition 1.2.3 to the two families and taking the union of the resulting countable subsets, we can find a countable subset $J \subseteq I$ such that

$$\sup_{j\in J}{}^{*}\varphi_j = \sup_{i\in I}{}^{*}\varphi_i, \quad \sup_{j\in J}{}^{*}\psi_j = \sup_{i\in I}{}^{*}\psi_i.$$

We may replace I by J and assume that I is countable. If I is finite, Step 2 applies directly. Otherwise, write $I = \mathbb{Z}_{>0}$ and set

$$\tilde{\varphi}_m = \varphi_1 \vee \cdots \vee \varphi_m, \quad \tilde{\psi}_m = \psi_1 \vee \cdots \vee \psi_m.$$

By Step 2, $\tilde{\varphi}_m \preceq_P \tilde{\psi}_m$ for every m. The two new sequences are increasing and have the same upper envelopes as the original families, so Step 1 applies.

We now refine Lemma 3.1.2 using the P-equivalence relation.

Proposition 6.1.7 *Let $\varphi, \psi, \varphi', \psi' \in \mathrm{PSH}(X,\theta)$ for some closed smooth real $(1,1)$-form θ on X. Assume that*

$$\varphi \sim_P \varphi', \quad \psi \sim_P \psi', \quad \varphi' \wedge \psi' \in \mathrm{PSH}(X,\theta)_{>0}.$$

Then

$$\varphi \wedge \psi \in \mathrm{PSH}(X,\theta)_{>0}, \quad \varphi \wedge \psi \sim_P \varphi' \wedge \psi'.$$

Proof We first observe that $\varphi, \varphi', \psi, \psi' \in \mathrm{PSH}(X,\theta)_{>0}$ by assumption. Let

$$\phi \coloneqq P_\theta[\varphi] = P_\theta[\varphi'], \quad \gamma = P_\theta[\psi] = P_\theta[\psi'].$$

Since $\varphi' \preceq \phi$ and $\psi' \preceq \gamma$, after adding a common constant to φ' and ψ', we may assume that $\varphi' \leq \phi$ and $\psi' \leq \gamma$. Then $\varphi' \wedge \psi' \leq \phi \wedge \gamma$, so the assumption

$\varphi' \wedge \psi' \in \mathrm{PSH}(X,\theta)_{>0}$ implies $\phi \wedge \gamma \in \mathrm{PSH}(X,\theta)_{>0}$ by monotonicity. It follows from Lemma 3.1.2 that

$$\int_X \theta^n_{\varphi'\wedge\psi'} = \int_X \theta^n_{\phi\wedge\gamma}.$$

Next, we apply Lemma 3.1.2 again to conclude that $\varphi \wedge \psi \in \mathrm{PSH}(X,\theta)$ and

$$\int_X \theta^n_{\varphi\wedge\psi} = \int_X \theta^n_{\phi\wedge\gamma} = \int_X \theta^n_{\varphi'\wedge\psi'} > 0.$$

Since $\varphi\wedge\psi \preceq \phi\wedge\gamma$ and the two potentials have the same positive mass, Theorem 3.1.2 gives $P_\theta[\varphi\wedge\psi] = \phi\wedge\gamma$. The same argument gives $P_\theta[\varphi'\wedge\psi'] = \phi\wedge\gamma$, hence $\varphi\wedge\psi \sim_P \varphi'\wedge\psi'$. □

We can formulate a generalization of Proposition 3.1.24 using the P-partial order.

Corollary 6.1.4 *Let $\varphi, \psi \in \mathrm{PSH}(X,\theta)_{>0}$. Assume that $\psi \preceq_P \varphi$. Then for any $C > 0$, $\psi \wedge (\varphi + C) \in \mathrm{PSH}(X,\theta)_{>0}$ and*

$$P_\theta[\varphi](\psi) = \sup_{C>0}{}^{*}\psi \wedge (\varphi + C) = \psi. \tag{6.5}$$

We refer to Definition 3.1.4 for the definition of $P_\theta[\varphi](\psi)$.

Proof Since $\psi \preceq_P \varphi$, by Proposition 6.1.7, we get

$$\psi \wedge (\varphi + C) \in \mathrm{PSH}(X,\theta)_{>0}$$

and $\psi \wedge (\varphi + C) \sim_P \psi$ for every $C > 0$.

Let

$$\eta \coloneqq \sup_{C>0}{}^{*}\psi \wedge (\varphi + C).$$

Then $\eta \sim_P \psi$. By Corollary 3.1.1, we have

$$\theta^n_\eta \leq \mathbb{1}_{\{\eta=\psi\}}\,\theta^n_\psi.$$

The domination principle Theorem 2.4.5 gives $\eta \geq \psi$. The reverse inequality is trivial, so $\eta = \psi$. □

The P-equivalence relation characterizes when a subgeodesic exists. See Definition 4.1.1 for the notion of subgeodesics.

Theorem 6.1.1 *Let $\varphi_0, \varphi_1 \in \mathrm{PSH}(X,\theta)_{>0}$. Then the following are equivalent:*

(1) *There is a subgeodesic from φ_0 to φ_1;*
(2) $\varphi_0 \sim_P \varphi_1$.

Proof (2) $\implies$ (1). This follows from Proposition 4.2.1.

(1) $\implies$ (2). Let $(\varphi_t)_{t\in(0,1)}$ be a subgeodesic from φ_0 to φ_1.

Step 1. We assume that $\varphi_0 \geq \varphi_1$.

We first replace the given subgeodesic by the geodesic from φ_0 to φ_1. Note that the geodesic is necessarily a subgeodesic as well by Lemma 4.2.1.

Observe that $t \mapsto \varphi_t$ is decreasing due to Lemma 4.2.2.

Let $\varphi_t = \varphi_1$ for all $t > 1$. Then by the gluing lemma Lemma 4.1.1, we find that $(\varphi_t)_{t\geq 0}$ is a subgeodesic ray.

Next, we consider the Legendre transform

$$\Gamma_\tau \coloneqq \inf_{t\geq 0}(\varphi_t - t\tau), \quad \tau \in \mathbb{R}.$$

It follows from Kiselman's principle Proposition 1.2.8 that $\Gamma_\tau \in \mathrm{PSH}(X,\theta) \cup \{-\infty\}$. Note that for $\tau > 0$, we clearly have $\Gamma_\tau \equiv -\infty$. On the other hand, for $\tau \leq 0$,

$$\Gamma_\tau = \inf_{t\in[0,1]}(\varphi_t - t\tau) \in \mathrm{PSH}(X,\theta).$$

By Legendre inversion (cf. Corollary A.2.1), for $t > 0$,

$$\varphi_t = \sup_{\tau<0}\left(\Gamma_\tau + t\tau\right).$$

Fix a Kähler form ω on X. It follows from Proposition 6.1.6 that for each $t > 0$,

$$\varphi_t \sim_P \sup_{\tau<0}{}^* P_{\theta+\omega}[\Gamma_\tau].$$

The right-hand side is independent of t. Here by adding ω, we no longer have to worry about the possibility where Γ_τ has vanishing mass.

Write

$$\varphi_0 = \sup_{t\in(0,1)}{}^*\varphi_t.$$

By Proposition 6.1.6 again, we find that

$$\varphi_0 \sim_P \sup_{\tau<0}{}^* P_{\theta+\omega}[\Gamma_\tau]$$

as well. So $\varphi_0 \sim_P \varphi_1$.

Step 2. We prove the general case.

Observe that $(\varphi_t \vee \varphi_1)_{t\in(0,1)}$ is a subgeodesic from $\varphi_0 \vee \varphi_1$ to φ_1. By Step 1,

$$\varphi_0 \vee \varphi_1 \sim_P \varphi_1.$$

Hence $\varphi_0 \leq_P \varphi_1$ by Lemma 6.1.3. The converse is proved similarly. Hence (2) follows.

Restating the (1) $\implies$ (2) direction in the theorem in terms of the complexification, we find the following interesting result.

Let $S = \{z \in \mathbb{C} : 0 < \operatorname{Re} z < 1\}$. We write $p_1 : X \times S \to X$ for the natural projection.

Corollary 6.1.5 *Let $\Phi \in \mathrm{PSH}(X \times S, p_1^*\theta)$. Assume that for any $c \in \mathbb{R}$, $x \in X$ and $z \in S$, we have*

$$\Phi(x,z) = \Phi(x, z + \mathrm{i}c).$$

Then $\int_X (\theta + \mathrm{dd^c}\Phi_z)^n$ *is independent of* $z \in S$*, where* $\Phi_z \in \mathrm{PSH}(X, \theta)$ *is given by* $\Phi_z(x) = \Phi(x, z)$.

Proof Fix a Kähler form ω on X. We may regard Φ as an element of $\mathrm{PSH}(X \times S, p_1^*(\theta+\omega))$. For any $0 < a < b < 1$, the restriction $(\Phi_t)_{t\in[a,b]}$ is then a subgeodesic from Φ_a to Φ_b in the class $\theta + \omega$. The slices have positive mass in this class, so Theorem 6.1.1 gives $\Phi_a \sim_P \Phi_b$. By Proposition 6.1.4, this directly implies

$$\int_X \theta^n_{\Phi_a} = \int_X \theta^n_{\Phi_b}.$$

Since Φ is invariant under imaginary translations, Φ_z depends only on $\operatorname{Re} z$, so the mass is independent of $z \in S$. □

6.2 The d_S-pseudometric

Let X be a connected compact Kähler manifold of dimension n and θ be a closed real smooth $(1, 1)$-form on X representing a big cohomology class. The goal of this section is to study a pseudometric on the space $\mathrm{PSH}(X, \theta)$.

6.2.1 The definition of the d_S-pseudometric

Recall that for any $\varphi \in \mathrm{PSH}(X, \theta)$, the geodesic ray $\ell^\varphi \in \mathcal{R}^1(X, \theta)$ is defined in Example 4.3.1.

Definition 6.2.1 For $\varphi, \psi \in \mathrm{PSH}(X, \theta)$, we define

$$d_S(\varphi, \psi) \coloneqq d_1(\ell^\varphi, \ell^\psi).$$

When we want to be more specific, we write $d_{S,\theta}$ instead of d_S.

The d_1 distance of geodesic rays is defined in Definition 4.3.5.

Proposition 6.2.1 *The function* d_S *defined in Definition 6.2.1 is a pseudometric on* $\mathrm{PSH}(X, \theta)$.

Proof This follows immediately from Theorem 4.3.4. □

When studying a pseudometric, the first thing is to understand when the distance between two elements vanishes.

We first prove a preparation:

Lemma 6.2.1 *Let* $\varphi, \psi \in \mathrm{PSH}(X, \theta)$*. Then*

$$d_S(\varphi, \psi) \le d_S(\varphi, \varphi \vee \psi) + d_S(\psi, \varphi \vee \psi) \le C_n d_S(\varphi, \psi),$$

where $C_n = 3(n + 1)2^{n+2}$.

We shall use the notations introduced in Example 4.3.1.

Proof We claim that

$$\ell^{\varphi} \vee \ell^{\psi} = \ell^{\varphi \vee \psi}. \tag{6.6}$$

Recall that $\vee$ is defined in Definition 4.3.7. Note that this assertion implies our desired inequality by Lemma 4.3.6.

In proving this assertion, we may assume that $\varphi, \psi \leq 0$ since

$$\ell^{\varphi + C} = \ell^{\varphi}, \quad \ell^{\psi + C} = \ell^{\psi}, \quad \ell^{(\varphi + C) \vee (\psi + C)} = \ell^{\varphi \vee \psi}$$

for any $C \in \mathbb{R}$.

In fact, it is clear that

$$\ell^{\varphi} \leq \ell^{\varphi \vee \psi}, \quad \ell^{\psi} \leq \ell^{\varphi \vee \psi},$$

so the $\leq$ direction in (6.6) holds.

Conversely, if $\ell' \in \mathcal{R}^1(X, \theta)$ and $\ell' \geq \ell^{\varphi} \vee \ell^{\psi}$, then for each $t \geq 0$,

$$\ell'_t \geq \Big((V_\theta - t) \vee \varphi \Big) \vee \Big((V_\theta - t) \vee \psi \Big) = (V_\theta - t) \vee (\varphi \vee \psi).$$

See (4.34) for the first inequality. Therefore, thanks to the definition of the Perron envelopes (cf. (4.5)), we have

$$\ell'_s \geq \ell_s^{\varphi \vee \psi, t}$$

for any $0 \leq s \leq t$. It follows from (4.34) that $\ell'_s \geq \ell_s^{\varphi \vee \psi}$ for any $s \geq 0$. $\square$

Proposition 6.2.2 *Let $\varphi, \psi \in \mathrm{PSH}(X, \theta)$. Then the following are equivalent:*

(1) $\varphi \sim_P \psi$*;*
(2) $d_S(\varphi, \psi) = 0$.

In particular, $d_S(\varphi, P_\theta[\varphi]) = 0$ for all $\varphi \in \mathrm{PSH}(X, \theta)_{>0}$.

Proof By Lemma 6.1.3, we have $\varphi \sim_P \psi$ if and only if $\varphi \sim_P \varphi \vee \psi$ and $\psi \sim_P \varphi \vee \psi$. By Lemma 6.2.1, $d_S(\varphi, \psi) = 0$ if and only if $d_S(\varphi, \varphi \vee \psi) = 0$ and $d_S(\psi, \varphi \vee \psi) = 0$. So it suffices to prove the assertion when $\varphi \leq \psi$. Assuming this, by Proposition 4.3.6, (2) holds if and only if

$$\mathbf{E}\left(\ell^{\varphi}\right) = \mathbf{E}\left(\ell^{\psi}\right),$$

where $\mathbf{E}$ is introduced in Definition 4.3.6. But by (4.32), this holds if and only if

$$\sum_{j=0}^{n} \int_X \theta_\varphi^j \wedge \theta_{V_\theta}^{n-j} = \sum_{j=0}^{n} \int_X \theta_\psi^j \wedge \theta_{V_\theta}^{n-j}.$$

Thanks to the monotonicity theorem Theorem 2.4.3, this holds if and only if for all $j = 0, \ldots, n$,

$$\int_X \theta_\varphi^j \wedge \theta_{V_\theta}^{n-j} = \int_X \theta_\psi^j \wedge \theta_{V_\theta}^{n-j},$$

which is equivalent to (1) by Proposition 6.1.1. $\square$

Lemma 6.2.2 *Suppose that $\varphi, \psi \in \mathrm{PSH}(X, \theta)$ and $\varphi \le_P \psi$. Then*

$$d_S(\varphi, \psi) = \frac{1}{n+1} \sum_{j=0}^{n} \left(\int_X \theta_\psi^j \wedge \theta_{V_\theta}^{n-j} - \int_X \theta_\varphi^j \wedge \theta_{V_\theta}^{n-j} \right).$$

Proof Set $\chi := \varphi \vee \psi$. By Lemma 6.1.3, we have $\chi \sim_P \psi$. Hence Proposition 6.2.2 gives $d_S(\chi, \psi) = 0$, and the pseudometric property gives

$$d_S(\varphi, \psi) = d_S(\varphi, \chi).$$

Since $\varphi \le \chi$, (4.32) and Proposition 4.3.6 give

$$d_S(\varphi, \chi) = \frac{1}{n+1} \sum_{j=0}^{n} \left(\int_X \theta_\chi^j \wedge \theta_{V_\theta}^{n-j} - \int_X \theta_\varphi^j \wedge \theta_{V_\theta}^{n-j} \right).$$

Finally, Proposition 6.1.1 applied to $\psi \le \chi$ gives

$$\int_X \theta_\chi^j \wedge \theta_{V_\theta}^{n-j} = \int_X \theta_\psi^j \wedge \theta_{V_\theta}^{n-j}$$

for all $j = 0, \ldots, n$. □

Corollary 6.2.1 *Suppose that $\varphi, \psi, \eta \in \mathrm{PSH}(X, \theta)$ and $\varphi \le_P \psi \le_P \eta$. Then*

$$d_S(\varphi, \eta) \ge d_S(\varphi, \psi), \quad d_S(\varphi, \eta) \ge d_S(\psi, \eta).$$

Proof This is an immediate consequence of Lemma 6.2.2 and the monotonicity theorem Proposition 6.1.4. □

Corollary 6.2.2 *For any $\varphi, \psi \in \mathrm{PSH}(X, \theta)$, we have*

$$\begin{aligned} d_S(\varphi, \psi) \le& \frac{1}{n+1} \sum_{j=0}^{n} \left(2 \int_X \theta_{\varphi \vee \psi}^j \wedge \theta_{V_\theta}^{n-j} - \int_X \theta_\varphi^j \wedge \theta_{V_\theta}^{n-j} - \int_X \theta_\psi^j \wedge \theta_{V_\theta}^{n-j} \right) \\ \le& C_n d_S(\varphi, \psi), \end{aligned} \tag{6.7}$$

where $C_n = 3(n+1)2^{n+2}$.

In particular, if $(\varphi_i)_{i \in I}$ is a net in $\mathrm{PSH}(X, \theta)$ with d_S-limit φ, then for each $j = 0, \ldots, n$,

$$\lim_{i \in I} \int_X \theta_{\varphi_i}^j \wedge \theta_{V_\theta}^{n-j} = \int_X \theta_\varphi^j \wedge \theta_{V_\theta}^{n-j} = \lim_{i \in I} \int_X \theta_{\varphi_i \vee \varphi}^j \wedge \theta_{V_\theta}^{n-j}.$$

Proof The estimates (6.7) follow from the combination of Lemma 6.2.2 and Lemma 6.2.1.

Suppose that $\varphi_i \xrightarrow{d_S} \varphi$. Then $\varphi_i \vee \varphi \xrightarrow{d_S} \varphi$ by Lemma 6.2.1. Therefore, Theorem 2.4.3 and Lemma 6.2.2 imply that

$$\lim_{i\in I}\int_X \theta^j_{\varphi_i\vee\varphi}\wedge\theta^{n-j}_{V_\theta}=\int_X \theta^j_{\varphi}\wedge\theta^{n-j}_{V_\theta}$$

for any $j=0,\dots,n$. The last assertion now follows from (6.7) and Theorem 2.4.3. □

Corollary 6.2.3 *Suppose that $(\varphi_i)_{i\in I}$ is an increasing net in $\mathrm{PSH}(X,\theta)$, uniformly bounded from above. Then*

$$\varphi_i \xrightarrow{d_S} \sup_{i\in I}{}^*\varphi_i.$$

If the φ_i's are all model potentials in $\mathrm{PSH}(X,\theta)_{>0}$, then so is $\sup_{i\in I}{}^*\varphi_i$, as we have seen in Proposition 3.1.13.

Proof Write $\varphi=\sup_{j\in I}{}^*\varphi_j$. Recall that by Proposition 1.2.2, $\varphi\in\mathrm{PSH}(X,\theta)$. By Lemma 6.2.2, it suffices to show that for each $k=0,\dots,n$, we have

$$\lim_{j\in I}\int_X \theta^k_{\varphi_j}\wedge\theta^{n-k}_{V_\theta}=\int_X \theta^k_{\varphi}\wedge\theta^{n-k}_{V_\theta}.$$

The latter follows from Corollary 2.4.1. □

Corollary 6.2.4 *Let $\varphi,\psi\in\mathrm{PSH}(X,\theta)$. Then*

$$\left|\int_X\theta^n_\varphi-\int_X\theta^n_\psi\right|\le D_n d_S(\varphi,\psi),$$

where $D_n=3(n+1)C_n$ with C_n being the same constant as in Lemma 6.2.1.

In other words, the non-pluripolar mass is uniformly continuous with respect to the d_S-pseudometric.

Proof We compute

$$\begin{aligned}\left|\int_X\theta^n_\varphi-\int_X\theta^n_\psi\right|\le&\left|2\int_X\theta^n_{\varphi\vee\psi}-\int_X\theta^n_\varphi-\int_X\theta^n_\psi\right|+2\left|\int_X\theta^n_{\varphi\vee\psi}-\int_X\theta^n_\varphi\right|\\ \le&(n+1)C_nd_S(\varphi,\psi)+2(n+1)d_S(\varphi,\varphi\vee\psi)\\ \le&(n+1)C_nd_S(\varphi,\psi)+2(n+1)C_nd_S(\varphi,\psi),\end{aligned}$$

where the first line is just the triangle inequality, the second line follows from Corollary 6.2.2 and the third line follows from Lemma 6.2.1. □

By contrast to the increasing case, for decreasing nets, the situation is different:

Corollary 6.2.5 *Suppose that $(\varphi_i)_{i\in I}$ is a decreasing net in $\mathrm{PSH}(X,\theta)$ such that $\varphi:=\inf_{i\in I}\varphi_i\not\equiv-\infty$. Then the following are equivalent:*

(1) *We have*

$$\varphi_i\xrightarrow{d_S}\varphi;$$

(2) *for each* $k = 0, \ldots, n$, *we have*

$$\lim_{j \in I} \int_X \theta_{\varphi_j}^k \wedge \theta_{V_\theta}^{n-k} = \int_X \theta_\varphi^k \wedge \theta_{V_\theta}^{n-k}. \tag{6.8}$$

If we assume furthermore that $\int_X \theta_\varphi^n > 0$, *then the above conditions are equivalent to the following:*

(3) *We have*

$$\lim_{j \in I} \int_X \theta_{\varphi_j}^n = \int_X \theta_\varphi^n.$$

When $\int_X \theta_\varphi^n > 0$ *and these assumptions are met, we also have*

$$P_\theta[\varphi] = \inf_{j \in I} P_\theta[\varphi_j]. \tag{6.9}$$

Proof Recall that by Proposition 1.2.2, $\varphi \in \mathrm{PSH}(X, \theta)$.

(1) $\iff$ (2). This follows immediately from Lemma 6.2.2.

Assume that $\int_X \theta_\varphi^n > 0$.

(2) $\implies$ (3). This is the case $k = n$ in (6.8).

(3) $\implies$ (2). Let $(b_j)_{j \in I}$ be a net converging to ∞ such that

$$b_j \in \left(1, \left(\frac{\int_X \theta_{\varphi_j}^n}{\int_X \theta_{\varphi_j}^n - \int_X \theta_\varphi^n}\right)^{1/n}\right).$$

This can be achieved for example by taking b_j as the midpoint of this interval. By Lemma 2.4.4, for each $j \in I$, we can find $\eta_j \in \mathrm{PSH}(X, \theta)$ such that

$$b_j^{-1} \eta_j + \left(1 - b_j^{-1}\right) \varphi_j \leq \varphi.$$

It follows from the monotonicity theorem Theorem 2.4.3 that for any $k = 0, \ldots, n$,

$$\int_X \theta_\varphi^k \wedge \theta_{V_\theta}^{n-k} \geq \left(1 - b_j^{-1}\right)^k \int_X \theta_{\varphi_j}^k \wedge \theta_{V_\theta}^{n-k}.$$

Taking the limit, we conclude the $\leq$ direction in (6.8). The $\geq$ direction follows from Theorem 2.4.3.

Finally, we argue (6.9). We may assume that $\varphi_j \leq 0$ for all $j \in I$ after passing to a tail. Let $\psi_j = P_\theta[\varphi_j] \geq \varphi_j$. It follows from Corollary 3.1.2 that ψ_j is a model potential. Let

$$\psi = \inf_{j \in I} \psi_j \geq \varphi.$$

By Proposition 3.1.11, ψ is a model potential. Since $\varphi \leq \psi \leq \psi_j$ for all $j \in I$, the monotonicity theorem Theorem 2.4.3 and Proposition 3.1.3 give

$$\int_X \theta_\varphi^n \leq \int_X \theta_\psi^n \leq \lim_{j \in I} \int_X \theta_{\psi_j}^n = \lim_{j \in I} \int_X \theta_{\varphi_j}^n = \int_X \theta_\varphi^n.$$

Hence ψ has the same mass as φ. As ψ is model and $\psi \geq \varphi$, Theorem 3.1.2 gives $\psi = P_\theta[\varphi]$. □

Having understood the increasing and decreasing cases, we shall handle general convergent sequences. In fact, since d_S is a pseudometric, the topology is completely determined by convergent sequences, so we do not need to consider nets in general.

Proposition 6.2.3 *Let $\varphi_j, \varphi \in \mathrm{PSH}(X,\theta)$ $(j \geq 1)$, $\varphi_j \xrightarrow{d_S} \varphi$. Assume that there is $\delta > 0$ such that*

$$\int_X \theta^n_{\varphi_j} \geq \delta$$

for all j and the φ_j's are all model potentials. Then up to replacing $(\varphi_j)_j$ by a subsequence, there is a decreasing sequence $(\psi_j)_j$ and an increasing sequence $(\eta_j)_j$ in $\mathrm{PSH}(X,\theta)$ such that

(1) $\psi_j \xrightarrow{d_S} \varphi$, $\eta_j \xrightarrow{d_S} \varphi$;
(2) $\psi_j \geq \varphi_j \geq \eta_j$ for all j.

In fact, for any $j \geq 1$, we will take

$$\eta_j = \inf_{k\in\mathbb{N}} \varphi_j \wedge \varphi_{j+1} \wedge \cdots \wedge \varphi_{j+k}, \quad \psi_j = \sup_{k\geq j}{}^* \varphi_k.$$

Proof First note that $\int_X \theta^n_\varphi \geq \delta$ as well, due to the d_S-continuity of non-pluripolar masses as proved in Corollary 6.2.4. Replacing φ by $P_\theta[\varphi]$, we may then assume that φ is a model potential in $\mathrm{PSH}(X,\theta)_{>0}$.

We are free to replace $(\varphi_j)_j$ by a subsequence. So we may assume that

$$d_S(\varphi_j, \varphi_{j+1}) \leq \frac{\delta 2^{-j-4}}{D_n C_n^{j+1}}, \quad d_S(\varphi, \varphi_j) \leq \frac{\delta 2^{-j-4}}{D_n}, \tag{6.10}$$

where C_n is the constant in Corollary 6.2.2, D_n is the constant in Corollary 6.2.4.

In particular, by Corollary 6.2.4,

$$\left| \int_X \theta^n_{\varphi_j} - \int_X \theta^n_\varphi \right| \leq \delta 2^{-j-4}. \tag{6.11}$$

Step 1. We handle the ψ_j's. For each $j \geq 1$ and $k \geq 1$, by Lemma 6.2.1 we have

$$\begin{aligned} d_S(\varphi_j, \varphi_j \vee \varphi_{j+1} \vee \cdots \vee \varphi_{j+k}) &\leq C_n\, d_S(\varphi_j, \varphi_{j+1} \vee \cdots \vee \varphi_{j+k}) \\ &\leq C_n\, d_S(\varphi_j, \varphi_{j+1}) \\ &\quad + C_n\, d_S(\varphi_{j+1}, \varphi_{j+1} \vee \cdots \vee \varphi_{j+k}). \end{aligned}$$

By iteration, we find

$$\begin{aligned} d_S(\varphi_j, \varphi_j \vee \varphi_{j+1} \vee \cdots \vee \varphi_{j+k}) &\leq \sum_{a=j}^{j+k-1} C_n^{a+1-j} d_S(\varphi_a, \varphi_{a+1}) \\ &\leq \frac{C_n^{-j}}{D_n} \sum_{a=j}^{j+k-1} \delta 2^{-a-4} \leq \frac{\delta 2^{-j-3}}{D_n}. \end{aligned}$$

Using Corollary 6.2.3, we have

$$\varphi_j \vee \varphi_{j+1} \vee \cdots \vee \varphi_{j+k} \xrightarrow{d_S} \psi_j$$

as $k \to \infty$. Hence

$$d_S(\varphi_j, \psi_j) \leq \frac{\delta 2^{-j-3}}{D_n}. \tag{6.12}$$

We conclude that $\psi_j \xrightarrow{d_S} \varphi$.

Moreover, we observe that

$$\varphi = \inf_{j \geq 1} P_\theta[\psi_j] \tag{6.13}$$

Indeed, set $\tilde{\psi}_j = P_\theta[\psi_j]$. Then $(\tilde{\psi}_j)_j$ is a decreasing sequence of model potentials, and $d_S(\tilde{\psi}_j, \psi_j) = 0$ by Proposition 6.2.2. Hence $\tilde{\psi}_j \xrightarrow{d_S} \varphi$. If $\chi = \inf_j \tilde{\psi}_j$, then Proposition 3.1.12 and Corollary 6.2.5 give $\tilde{\psi}_j \xrightarrow{d_S} \chi$. Thus $d_S(\chi, \varphi) = 0$. Since both χ and φ are model potentials, Proposition 6.2.2 gives $P_\theta[\chi] = P_\theta[\varphi]$, hence $\chi = \varphi$, proving (6.13).

Step 2. We consider the η_j's.

For each $j \geq 1$ and $k \geq 0$, we let

$$\eta_j^k \coloneqq \varphi_j \wedge \cdots \wedge \varphi_{j+k}.$$

Using (6.10), (6.12) and Corollary 6.2.4, we have

$$\left| \int_X \theta_{\psi_j}^n - \int_X \theta_\varphi^n \right| \leq \delta 2^{-j-2}$$

for all $j \geq 1$. Taking (6.11) into account, we have

$$\left| \int_X \theta_{\varphi_j}^n - \int_X \theta_{\psi_{j-1}}^n \right| \leq \delta 2^{-j} \tag{6.14}$$

for $j > 1$.

Step 2.1. We claim that for a fixed $j \geq 1$, for any $k \in \mathbb{N}$, we have $\eta_j^k \in \mathrm{PSH}(X, \theta)$ and

$$\int_X \theta_{\eta_j^k}^n \geq \int_X \theta_{\varphi_j}^n - \sum_{a=1}^{k} \delta 2^{-j-a}. \tag{6.15}$$

We argue by induction on $k \geq 0$. The case $k = 0$ is trivial. When $k > 0$, assume that the case $k - 1$ is known. Then

$$\begin{aligned}\int_X \theta^n_{\eta_j^{k-1}} + \int_X \theta^n_{\varphi_{j+k}} &\geq \int_X \theta^n_{\varphi_j} - \sum_{a=1}^{k-1} \delta 2^{-j-a} + \int_X \theta^n_{\psi_{j+k-1}} - \delta 2^{-j-k} \\ &\geq \int_X \theta^n_{\varphi_j} - \delta 2^{-j} + \int_X \theta^n_{\psi_{j+k-1}} > \int_X \theta^n_{\psi_{j+k-1}},\end{aligned}$$

where the first inequality follows from the inductive hypothesis and (6.14).

Observe that

$$\eta_j^{k-1} \vee \varphi_{j+k} \leq \psi_{j+k-1},$$

it follows from Proposition 3.1.5 that $\eta_j^k \in \operatorname{PSH}(X, \theta)$. By Theorem 3.1.3, we deduce that

$$\begin{aligned}\int_X \theta^n_{\eta_j^k} &\geq \int_X \theta^n_{\varphi_{j+k}} + \int_X \theta^n_{\eta_j^{k-1}} - \int_X \theta^n_{\psi_{j+k-1}} \\ &\geq \int_X \theta^n_{\varphi_j} - \sum_{a=1}^{k} \delta 2^{-j-a},\end{aligned}$$

where the second inequality follows from the inductive hypothesis and (6.14). Therefore, (6.15) follows.

Step 2.2. It follows by induction from Proposition 3.1.6 that for any $j \geq 1$, $k \geq 0$,

$$P_\theta\left[\eta_j^k\right] = \eta_j^k.$$

By Proposition 3.1.12, we have

$$\lim_{k\to\infty} \int_X \theta^n_{\eta_j^k} = \int_X \theta^n_{\eta_j}$$

for any $j \geq 1$. Letting $k \to \infty$ in (6.15), we find that

$$\int_X \theta^n_{\eta_j} \geq \int_X \theta^n_{\varphi_j} - \delta 2^{-j} > 0$$

for $j \geq 1$. Observe that we also have

$$\int_X \theta^n_{\eta_j} \leq \int_X \theta^n_{\varphi_j} \leq \int_X \theta^n_{\psi_j}$$

for $j \geq 1$ by Theorem 2.4.3. It follows from these inequalities and (6.11) that

$$\lim_{j\to\infty} \int_X \theta^n_{\eta_j} = \int_X \theta^n_{\varphi}.$$

The sequence $(\eta_j)_j$ is increasing. Hence Corollary 2.4.1 gives

$$\int_X \theta^n_\eta = \lim_{j\to\infty}\int_X \theta^n_{\eta_j} = \int_X \theta^n_\varphi,$$

where $\eta = \sup_{j\geq 1}{}^*\eta_j$. For fixed m, if $j \geq m$, then $\eta_j \leq \varphi_j \leq \psi_m$, while if $j < m$, then $\eta_j \leq \eta_m \leq \psi_m$. Thus η_j is a candidate in the definition of $P_\theta[\psi_m]$ for all j, m, and (6.13) gives $\eta \leq \varphi$. It follows from Proposition 6.1.1 that $\eta \sim_P \varphi$. By Corollary 6.2.3 and Proposition 6.2.2, we have $\eta_j \xrightarrow{d_S} \varphi$. □

Corollary 6.2.6 *Let $(\varphi_j)_{j\in I}$ be a Cauchy net (with respect to d_S) in* PSH(X,θ). *Assume that there is $\delta > 0$ such that $\int_X \theta^n_{\varphi_j} \geq \delta$ for all $j \in I$. Then $(\varphi_j)_{j\in I}$ converges with respect to d_S.*

In particular, if $(\varphi_j)_{j\in I}$ is a decreasing net such that $\int_X \theta^n_{\varphi_j} \geq \delta > 0$ for all $j \in I$, then $(\varphi_j)_{j\in I}$ converges with respect to d_S.

We can obviously relax the decreasing condition to the following: the P-singularity types of $(\varphi_j)_{j\in I}$ are decreasing.

Proof In a pseudometric space, sequential completeness implies completeness for Cauchy nets. So for the first assertion, we may assume that $(\varphi_j)_j$ is a Cauchy sequence.

We first treat the case of a decreasing sequence $(\gamma_j)_j$ with $\int_X \theta^n_{\gamma_j} \geq \delta$ for all j. Put $\tilde{\gamma}_j = P_\theta[\gamma_j]$. Then $(\tilde{\gamma}_j)_j$ is a decreasing sequence of model potentials, $d_S(\gamma_j, \tilde{\gamma}_j) = 0$, and

$$\int_X \theta^n_{\tilde{\gamma}_j} = \int_X \theta^n_{\gamma_j} \geq \delta$$

by Proposition 3.1.3. By Proposition 3.1.12, the decreasing limit $\tilde{\gamma} = \inf_j \tilde{\gamma}_j$ is model and

$$\lim_{j\to\infty}\int_X \theta^n_{\tilde{\gamma}_j} = \int_X \theta^n_{\tilde{\gamma}} \geq \delta.$$

Hence $\tilde{\gamma}_j \longrightarrow \tilde{\gamma}$ in d_S by Corollary 6.2.5, and therefore γ_j also converges in d_S.

Next let $(\varphi_j)_{j>0}$ be a d_S-Cauchy sequence with masses bounded from below by δ. Put

$$\tilde{\varphi}_j = P_\theta[\varphi_j].$$

By Corollary 3.1.2, Example 6.1.1, and Proposition 6.2.2, the $\tilde{\varphi}_j$'s are model potentials, $d_S(\varphi_j, \tilde{\varphi}_j) = 0$, and

$$\int_X \theta^n_{\tilde{\varphi}_j} = \int_X \theta^n_{\varphi_j} \geq \delta$$

by Proposition 3.1.3. Replacing φ_j by $\tilde{\varphi}_j$, we may therefore assume that the φ_j's are model potentials. Passing to a subsequence, we may also assume that

$$d_S(\varphi_j, \varphi_{j+1}) \leq C_n^{-2j},$$

where C_n is the constant in Corollary 6.2.2. Define

$$\psi_j = \sup_{k\geq j}{}^*\varphi_k$$

for each $j > 0$. As in the proof of Proposition 6.2.3 Step 1, especially (6.12), we know that

$$\lim_{j\to\infty} d_S(\varphi_j, \psi_j) = 0.$$

The sequence $(\psi_j)_j$ is decreasing and satisfies $\int_X \theta^n_{\psi_j} \geq \delta$ by Theorem 2.4.3. By the first paragraph, $(\psi_j)_j$ converges in d_S. Hence the chosen subsequence of $(\varphi_j)_j$ converges, and the original Cauchy sequence converges as well.

Finally, suppose that $(\varphi_j)_{j\in I}$ is a decreasing net with masses bounded from below by δ. For each $k = 0, \dots, n$, the net

$$\int_X \theta^k_{\varphi_j} \wedge \theta^{n-k}_{V_\theta}$$

is decreasing and bounded from below, hence Cauchy as a real-valued net. If $j' \geq j$, then $\varphi_{j'} \leq \varphi_j$, and Lemma 6.2.2 expresses $d_S(\varphi_{j'}, \varphi_j)$ as a finite sum of the corresponding differences of these mixed masses. For two sufficiently large indices, we choose a common upper bound in I and use the triangle inequality. It follows that $(\varphi_j)_{j\in I}$ is d_S-Cauchy. The first assertion gives its convergence. □

Lemma 6.2.3 *There is a constant $C > 0$ depending only on X and θ such that for any $\varphi \in \mathrm{PSH}(X, \theta)$ satisfying that θ_φ is a Kähler current, we have*

$$d_{S,\theta}\Big((1-\epsilon)\varphi, \varphi\Big) \leq C\epsilon$$

for small enough $\epsilon > 0$ such that $(1-\epsilon)\varphi \in \mathrm{PSH}(X, \theta)$.

Proof After adding a constant, we may assume that $\varphi \leq 0$. Since θ_φ is a Kähler current, $\theta_{(1-\epsilon)\varphi} \in \mathrm{PSH}(X, \theta)$ for all small enough $\epsilon > 0$. Fix such an ϵ. Then $\varphi \leq (1-\epsilon)\varphi$, so by Lemma 6.2.2, we can compute

$$\begin{aligned}
d_{S,\theta}\Big((1-\epsilon)\varphi, \varphi\Big) =& \frac{1}{n+1}\sum_{j=0}^{n}\left(\int_X \theta^j_{(1-\epsilon)\varphi} \wedge \theta^{n-j}_{V_\theta} - \int_X \theta^j_\varphi \wedge \theta^{n-j}_{V_\theta}\right)\\
=& \frac{1}{n+1}\sum_{j=0}^{n}\left(\int_X (1-\epsilon)^j \theta^j_\varphi \wedge \theta^{n-j}_{V_\theta} - \int_X \theta^j_\varphi \wedge \theta^{n-j}_{V_\theta}\right)\\
&+ \frac{1}{n+1}\sum_{j=0}^{n}\sum_{k=0}^{j-1}\binom{j}{k}(1-\epsilon)^k \epsilon^{j-k}\int_X \theta^{j-k} \wedge \theta^k_\varphi \wedge \theta^{n-j}_{V_\theta}.
\end{aligned}$$

Both terms are of the order of $O(\epsilon)$. □

6.2.2 Convergence theorems

Next we establish some important convergence theorems, allowing us to effectively manipulate the d_S-convergence.

Lemma 6.2.4 *Let $(\varphi_i)_{i\in I}$ be a net in* $\mathrm{PSH}(X,\theta)$ *and $\varphi \in \mathrm{PSH}(X,\theta)$. Assume that $\varphi_i \xrightarrow{d_S} \varphi$. Then for any $t \in (0,1]$,*

$$(1-t)\varphi_i + tV_\theta \xrightarrow{d_S} (1-t)\varphi + tV_\theta.$$

When $t=1$, the sum is understood as in Remark 2.4.2.

Proof Fix $t \in (0,1]$, we write

$$\varphi_{i,t} = (1-t)\varphi_i + tV_\theta, \quad \varphi_t = (1-t)\varphi + tV_\theta$$

for any $i \in I$.

By Corollary 6.2.2, it suffices to show that for each $j = 0, \dots, n$,

$$2\int_X \theta^j_{\varphi_{i,t}\vee\varphi_t} \wedge \theta^{n-j}_{V_\theta} - \int_X \theta^j_{\varphi_{i,t}} \wedge \theta^{n-j}_{V_\theta} - \int_X \theta^j_{\varphi_t} \wedge \theta^{n-j}_{V_\theta} \to 0. \tag{6.16}$$

Observe that

$$\varphi_{i,t} \vee \varphi_t = (1-t)(\varphi \vee \varphi_i) + tV_\theta.$$

So after binomial expansion, (6.16) follows from Corollary 6.2.2. □

Lemma 6.2.5 *Let $\varphi \in \mathrm{PSH}(X,\theta)$. For each $t \in (0,1)$, let $\varphi_t = (1-t)\varphi + tV_\theta$. Then*

$$\varphi_t \xrightarrow{d_S} \varphi$$

as $t \to 0+$.

Proof Since $\varphi \leq \varphi_t$, by Lemma 6.2.2, we need to show that for each $j = 1, \dots, n$, we have

$$\lim_{t\to 0+} \int_X \theta^j_{\varphi_t} \wedge \theta^{n-j}_{V_\theta} = \int_X \theta^j_{\varphi} \wedge \theta^{n-j}_{V_\theta}.$$

For this purpose, we compute

$$\begin{aligned}
&\int_X \theta^j_{\varphi_t} \wedge \theta^{n-j}_{V_\theta} - \int_X \theta^j_{\varphi} \wedge \theta^{n-j}_{V_\theta} \\
={}& \left((1-t)^j - 1\right) \int_X \theta^j_{\varphi} \wedge \theta^{n-j}_{V_\theta} \\
&+ \sum_{i=0}^{j-1} \binom{j}{i} (1-t)^i t^{j-i} \int_X \theta^i_{\varphi} \wedge \theta^{n-i}_{V_\theta}.
\end{aligned}$$

As $t \to 0+$, the right-hand side clearly tends to 0. □

The following convergence theorem lies at the heart of the whole theory.

Theorem 6.2.1 *Let $\theta_1, \dots, \theta_n$ be smooth closed real $(1,1)$-forms on X representing big cohomology classes. Suppose that $(\varphi_j^k)_{k\in I}$ are nets in $\mathrm{PSH}(X,\theta_j)$ and $\varphi_j \in \mathrm{PSH}(X,\theta_j)$ for $j = 1, \dots, n$. We assume that $\varphi_j^k \xrightarrow{d_S} \varphi_j$ for each $j = 1, \dots, n$. Then*

$$\lim_{k\in I} \int_X \theta_{1,\varphi_1^k} \wedge \cdots \wedge \theta_{n,\varphi_n^k} = \int_X \theta_{1,\varphi_1} \wedge \cdots \wedge \theta_{n,\varphi_n}. \tag{6.17}$$

Proof Since d_S is a pseudometric, in order to establish the continuity of mixed masses, it suffices to consider sequences instead of nets. So we may assume that $I = \mathbb{Z}_{>0}$ as ordered sets.

Step 1. We reduce to the case where φ_j^k, φ_j all have positive masses and there is a constant $\delta > 0$ such that for all $j = 1, \dots, n$ and $k \in I$,

$$\int_X \theta^n_{j,\varphi_j^k} > \delta.$$

Assuming in this case, the theorem has been proved, let us handle the general case. Take $t \in (0,1)$. By Lemma 6.2.4, we have

$$(1-t)\varphi_j^k + tV_{\theta_j} \xrightarrow{d_S} (1-t)\varphi_j + tV_{\theta_j}$$

as $k \to \infty$ for each $j = 1, \dots, n$. Note that for all $k \in I$,

$$\int_X \theta^n_{j,(1-t)\varphi_j^k + tV_{\theta_j}} \geq t^n \int_X \theta^n_{V_{\theta_j}} > 0.$$

Assume that we have proved the special case of the theorem, we have

$$\begin{aligned}
&\lim_{k\to\infty} \int_X \theta_{1,(1-t)\varphi_1^k + tV_{\theta_1}} \wedge \cdots \wedge \theta_{n,(1-t)\varphi_n^k + tV_{\theta_n}} \\
=& \int_X \theta_{1,(1-t)\varphi_1 + tV_{\theta_1}} \wedge \cdots \wedge \theta_{n,(1-t)\varphi_n + tV_{\theta_n}}.
\end{aligned}$$

More explicitly, for fixed k the left-hand side is the value at t of a polynomial $Q_k(t)$ of degree at most n, and the right-hand side is the value of a polynomial $Q(t)$ of degree at most n. Choose $n+1$ distinct numbers $t_0, \dots, t_n \in (0,1)$. Lagrange interpolation expresses $Q_k(0) - Q(0)$ as a fixed linear combination of the numbers $Q_k(t_a) - Q(t_a)$, which tend to 0. Hence the same formula holds at $t = 0$. From this, (6.17) follows.

Step 2. We now assume that the assumptions mentioned at the beginning of Step 1 are met. Note that we may assume that φ_j^k, φ_j are model potentials for all $j = 1, \dots, n$, $k > 0$ by Proposition 6.2.2 and Corollary 3.1.2.

We first handle the following monotone situation. Assume that, for each fixed i, the sequence $(\varphi_i^k)_k$ is either increasing or decreasing. Let $\tilde{\varphi}_i$ denote its pointwise limit, namely

$$\tilde{\varphi}_i = \inf_{k>0} \varphi_i^k$$

in the decreasing case and

$$\tilde{\varphi}_i = \sup_{k>0}{}^* \varphi_i^k$$

in the increasing case. We assume moreover that $\tilde{\varphi}_i \sim_P \varphi_i$ for every i. Since mixed non-pluripolar masses are invariant under replacing each entry by a P-equivalent potential, it suffices in this monotone situation to prove convergence to the mixed mass of the $\tilde{\varphi}_i$'s.

After reordering the indices, we may assume that there is $i_0 \in \{0, \dots, n\}$ such that $(\varphi_i^k)_k$ is decreasing for $i \leq i_0$ and increasing for $i > i_0$. Therefore, for each $k > 0$, using Theorem 2.4.3, we have

$$\int_X \theta_{1,\varphi_1^k} \wedge \cdots \wedge \theta_{n,\varphi_n^k} \geq \int_X \theta_{1,\tilde{\varphi}_1} \wedge \cdots \wedge \theta_{i_0,\tilde{\varphi}_{i_0}} \wedge \theta_{i_0+1,\varphi_{i_0+1}^k} \wedge \cdots \wedge \theta_{n,\varphi_n^k}.$$

Using Corollary 2.4.1, we therefore conclude that

$$\varliminf_{k\to\infty} \int_X \theta_{1,\varphi_1^k} \wedge \cdots \wedge \theta_{n,\varphi_n^k} \geq \int_X \theta_{1,\tilde{\varphi}_1} \wedge \cdots \wedge \theta_{n,\tilde{\varphi}_n}.$$

It remains to prove

$$\varlimsup_{k\to\infty} \int_X \theta_{1,\varphi_1^k} \wedge \cdots \wedge \theta_{n,\varphi_n^k} \leq \int_X \theta_{1,\tilde{\varphi}_1} \wedge \cdots \wedge \theta_{n,\tilde{\varphi}_n}. \tag{6.18}$$

By Theorem 2.4.3, for each $k > 0$, we have

$$\int_X \theta_{1,\varphi_1^k} \wedge \cdots \wedge \theta_{n,\varphi_n^k} \leq \int_X \theta_{1,\varphi_1^k} \wedge \cdots \wedge \theta_{i_0,\varphi_{i_0}^k} \wedge \theta_{i_0+1,\tilde{\varphi}_{i_0+1}} \wedge \cdots \wedge \theta_{n,\tilde{\varphi}_n}.$$

When proving (6.18), we may replace φ_j^k by $\tilde{\varphi}_j$ whenever $j > i_0$, $k > 0$. Thus, we are reduced to the case where for all i, $(\varphi_i^k)_k$ is decreasing.

By Proposition 3.1.12, the masses $\int_X \theta_{i,\varphi_i^k}^n$ decrease to $\int_X \theta_{i,\tilde{\varphi}_i}^n$. Hence, thanks to Lemma 2.4.4, for each $i = 1, \dots, n$, we may take an increasing sequence $(b_i^k)_k$ tending to ∞ satisfying

$$b_i^k \in \left(1, \left(\frac{\int_X \theta_{i,\varphi_i^k}^n}{\int_X \theta_{i,\varphi_i^k}^n - \int_X \theta_{i,\tilde{\varphi}_i}^n}\right)^{1/n}\right)$$

where the upper endpoint is understood as ∞ if the denominator is 0. For each such b_i^k, Lemma 2.4.4 gives $\rho_i^k \in \mathrm{PSH}(X, \theta_i)$ with

$$(b_i^k)^{-1}\rho_i^k + \left(1 - (b_i^k)^{-1}\right)\varphi_i^k \leq \tilde{\varphi}_i.$$

Then by Theorem 2.4.3 again,

$$\prod_{i=1}^{n}\left(1-(b_i^k)^{-1}\right)\int_X \theta_{1,\varphi_1^k}\wedge\cdots\wedge\theta_{n,\varphi_n^k}\leq\int_X \theta_{1,\tilde{\varphi}_1}\wedge\cdots\wedge\theta_{n,\tilde{\varphi}_n}.$$

Letting $k\to\infty$, we conclude (6.18). This proves the monotone situation.

We now return to the general case. It suffices to prove that any subsequence of

$$\int_X \theta_{1,\varphi_1^k}\wedge\cdots\wedge\theta_{n,\varphi_n^k}$$

has a further subsequence converging to the desired limit. Starting with an arbitrary subsequence, we pass to a further subsequence, still indexed by k, so that the construction in Proposition 6.2.3 applies simultaneously to the finitely many sequences $(\varphi_i^k)_k$, $i=1,\dots,n$.

Thus, after discarding finitely many terms and relabeling, for each i there are model potentials $\eta_i^k,\psi_i^k\in\mathrm{PSH}(X,\theta_i)$ such that $(\eta_i^k)_k$ is increasing, $(\psi_i^k)_k$ is decreasing, and

$$\eta_i^k\leq\varphi_i^k\leq\psi_i^k,\quad \eta_i^k\xrightarrow{d_S}\varphi_i,\quad \psi_i^k\xrightarrow{d_S}\varphi_i.$$

For the upper sequence, we take $\psi_i^k=P_{\theta_i}[\sup_{\ell\geq k}{}^*\varphi_i^\ell]$; as in (6.13), this gives $\inf_k\psi_i^k=\varphi_i$. For the lower sequence, the construction in Proposition 6.2.3 together with Proposition 3.1.11 shows that the η_i^k's are model after discarding finitely many terms, and the proof of Proposition 6.2.3 shows that their increasing limit is P-equivalent to φ_i.

By Theorem 2.4.3, for every k,

$$\int_X \theta_{1,\eta_1^k}\wedge\cdots\wedge\theta_{n,\eta_n^k}\leq\int_X \theta_{1,\varphi_1^k}\wedge\cdots\wedge\theta_{n,\varphi_n^k}\leq\int_X \theta_{1,\psi_1^k}\wedge\cdots\wedge\theta_{n,\psi_n^k}.$$

The monotone situation just proved applies to both the lower and upper sequences, and both limiting mixed masses are equal to

$$\int_X \theta_{1,\varphi_1}\wedge\cdots\wedge\theta_{n,\varphi_n}.$$

The squeeze theorem therefore gives the desired convergence along this further subsequence. This concludes the proof. □

Corollary 6.2.7 *Suppose that $(\varphi_i)_{i\in I}$ is a net in $\mathrm{PSH}(X,\theta)$ and $\varphi\in\mathrm{PSH}(X,\theta)$. Then the following are equivalent:*

(1) $\varphi_i\xrightarrow{d_S}\varphi$;

(2) $\varphi_i\vee\varphi\xrightarrow{d_S}\varphi$ *and*

$$\lim_{i\in I}\int_X\theta_{\varphi_i}^j\wedge\theta_{V_\theta}^{n-j}=\int_X\theta_{\varphi}^j\wedge\theta_{V_\theta}^{n-j}\tag{6.19}$$

for each $j=0,\dots,n$;

(3) *for each $j=0,\dots,n$,* (6.19) *holds and*

$$\lim_{i\in I}\int_X \theta^j_{\varphi_i\vee\varphi}\wedge\theta^{n-j}_{V_\theta}=\int_X \theta^j_{\varphi}\wedge\theta^{n-j}_{V_\theta}. \tag{6.20}$$

The corollary allows us to reduce a number of convergence problems related to d_S to the case $\varphi_i \geq \varphi$. This is the most handy way of establishing d_S-convergence in practice.

Proof The equivalence between (2) and (3) follows directly from Lemma 6.2.2.

(1) $\implies$ (2). That $\varphi_i\vee\varphi \xrightarrow{d_S} \varphi$ follows from Corollary 6.2.2. While (6.19) follows from Theorem 6.2.1.

(2) $\implies$ (1). By (6.7), we need to show that for each $j=0,\dots,n$, we have

$$2\int_X \theta^j_{\varphi_i\vee\varphi}\wedge\theta^{n-j}_{V_\theta}-\int_X \theta^j_{\varphi}\wedge\theta^{n-j}_{V_\theta}-\int_X \theta^j_{\varphi_i}\wedge\theta^{n-j}_{V_\theta}\to 0.$$

This follows from Theorem 6.2.1 and (6.19). □

Corollary 6.2.8 *Let $(\varphi_i)_{i\in I}$ be a net in $\mathrm{PSH}(X,\theta)$ and $\varphi\in\mathrm{PSH}(X,\theta)$. Let ω be a closed smooth positive $(1,1)$-form on X. Then the following are equivalent:*

(1) $\varphi_i \xrightarrow{d_{S,\theta}} \varphi$;

(2) $\varphi_i \xrightarrow{d_{S,\theta+\omega}} \varphi$.

In particular, there is no risk when we simply write $\varphi_i \xrightarrow{d_S} \varphi$. Similarly, consider two nets $(\varphi_i)_{i\in I}$, $(\psi_i)_{i\in I}$ in $\mathrm{PSH}(X,\theta)$, if both admit d_S-limits in $\mathrm{PSH}(X,\theta)$, then we can write $d_S(\varphi_i,\psi_i)\to 0$ without ambiguity.

Proof (1) $\implies$ (2). It suffices to show that for each $j=0,\dots,n$, we have

$$\begin{aligned}2\int_X (\theta+\omega)^j_{\varphi_i\vee\varphi}\wedge(\theta+\omega)^{n-j}_{V_{\theta+\omega}}-\int_X (\theta+\omega)^j_{\varphi_i}\wedge(\theta+\omega)^{n-j}_{V_{\theta+\omega}}\\ -\int_X (\theta+\omega)^j_{\varphi}\wedge(\theta+\omega)^{n-j}_{V_{\theta+\omega}}\to 0.\end{aligned}$$

Note that this quantity is a linear combination of terms of the following form:

$$\begin{aligned}2\int_X \theta^r_{\varphi_i\vee\varphi}\wedge\omega^{j-r}\wedge(\theta+\omega)^{n-j}_{V_{\theta+\omega}}-\int_X \theta^r_{\varphi_i}\wedge\omega^{j-r}\wedge(\theta+\omega)^{n-j}_{V_{\theta+\omega}}\\ -\int_X \theta^r_{\varphi}\wedge\omega^{j-r}\wedge(\theta+\omega)^{n-j}_{V_{\theta+\omega}},\end{aligned}$$

where $r=0,\dots,j$. By Theorem 6.2.1, it suffices to show that $\varphi\vee\varphi_i \xrightarrow{d_S} \varphi$. But this follows from Corollary 6.2.7.

(2) $\implies$ (1). From the direction we already proved, for each $C\geq 1$, we have

$$\varphi_i \xrightarrow{d_{S,\theta+C\omega}} \varphi.$$

By Theorem 6.2.1, it follows that

$$\lim_{i\in I}\int_X (\theta+C\omega)^j_{\varphi_i}\wedge\theta^{n-j}_{V_\theta}=\int_X(\theta+C\omega)^j_{\varphi}\wedge\theta^{n-j}_{V_\theta}$$

for all $j=0,\dots,n$. It follows that

$$\lim_{i\in I}\int_X \theta^j_{\varphi_i}\wedge\theta^{n-j}_{V_\theta}=\int_X\theta^j_{\varphi}\wedge\theta^{n-j}_{V_\theta}. \tag{6.21}$$

By Corollary 6.2.7, it remains to show that $\varphi_i\vee\varphi\xrightarrow{d_{S,\theta}}\varphi$. By Corollary 6.2.7 again, we know that $\varphi_i\vee\varphi\xrightarrow{d_{S,\theta+\omega}}\varphi$. So it suffices to apply (6.21) to $\varphi_i\vee\varphi$ instead of φ_i, and we conclude by Lemma 6.2.2. □

Corollary 6.2.9 *Let $\varphi_0,\varphi_1\in\mathrm{PSH}(X,\theta)$. Define $\varphi_t=t\varphi_1+(1-t)\varphi_0$ for $t\in(0,1)$. Then*

$$\varphi_t\xrightarrow{d_S}\varphi_0$$

as $t\to 0+$.

Proof First note that for each $j=0,\dots,n$,

$$\lim_{t\to 0+}\int_X \theta^j_{\varphi_t}\wedge\theta^{n-j}_{V_\theta}=\int_X\theta^j_{\varphi_0}\wedge\theta^{n-j}_{V_\theta}.$$

So thanks to Corollary 6.2.7, it remains to argue that for all $j=0,\dots,n$,

$$\lim_{t\to 0+}\int_X \theta^j_{\varphi_t\vee\varphi_0}\wedge\theta^{n-j}_{V_\theta}=\int_X\theta^j_{\varphi_0}\wedge\theta^{n-j}_{V_\theta}.$$

Observe that for $t\in(0,1)$, we have

$$\varphi_t\vee\varphi_0=t(\varphi_1\vee\varphi_0)+(1-t)\varphi_0,$$

so the desired limit follows by the same binomial expansion. □

We have the following sandwich criterion:

Corollary 6.2.10 *Let $(\varphi_i)_{i\in I},(\psi_i)_{i\in I},(\eta_i)_{i\in I}$ be three nets in $\mathrm{PSH}(X,\theta)$ and $\varphi\in\mathrm{PSH}(X,\theta)$. Assume that*

(1) *$\psi_i\preceq_P\varphi_i\preceq_P\eta_i$ for each $i\in I$;*
(2) *$\eta_i\xrightarrow{d_S}\varphi$, $\psi_i\xrightarrow{d_S}\varphi$.*

Then $\varphi_i\xrightarrow{d_S}\varphi$.

Proof By Corollary 6.2.8, we may replace θ by $\theta+\omega$, where ω is a Kähler form on X. In particular, we may assume that $\varphi_i,\psi_i,\eta_i\in\mathrm{PSH}(X,\theta)_{>0}$ for all $i\in I$. By Proposition 6.2.2, we may assume that φ_i,ψ_i,η_i are model potentials for all $i\in I$ and hence $\psi_i\le\varphi_i\le\eta_i$ for all $i\in I$.

It follows from Theorem 2.4.3 that for each $k = 0, \dots, n$, we have

$$\int_X \theta_{\psi_i}^k \wedge \theta_{V_\theta}^{n-k} \le \int_X \theta_{\varphi_i}^k \wedge \theta_{V_\theta}^{n-k} \le \int_X \theta_{\eta_i}^k \wedge \theta_{V_\theta}^{n-k}$$

for all $i \in I$. By Theorem 6.2.1, the limits, with respect to $i \in I$, of the two outer terms are $\int_X \theta_\varphi^k \wedge \theta_{V_\theta}^{n-k}$. It follows that

$$\lim_{i \in I} \int_X \theta_{\varphi_i}^k \wedge \theta_{V_\theta}^{n-k} = \int_X \theta_\varphi^k \wedge \theta_{V_\theta}^{n-k}. \tag{6.22}$$

By Corollary 6.2.7, it remains to prove that $\varphi_i \vee \varphi \xrightarrow{d_S} \varphi$. By Corollary 6.2.7 and Proposition 6.1.6, up to replacing ψ_i (resp. φ_i, η_i) by $\psi_i \vee \varphi$ (resp. $\varphi_i \vee \varphi, \eta_i \vee \varphi$), we may assume from the beginning that $\psi_i, \varphi_i, \eta_i \ge \varphi$. Now $\varphi_i \xrightarrow{d_S} \varphi$ by (6.22) and Lemma 6.2.2. □

Lemma 6.2.6 *Let $(\varphi_i)_{i \in I}$ be a net in $\mathrm{PSH}(X, \theta)$, and $\varphi \in \mathrm{PSH}(X, \theta)$. Fix $\lambda > 0$. Then the following are equivalent:*

(1) $\varphi_i \xrightarrow{d_{S,\theta}} \varphi$;

(2) $\lambda\varphi_i \xrightarrow{d_{S,\lambda\theta}} \lambda\varphi$.

Proof We have $V_{\lambda\theta} = \lambda V_\theta$. The middle expression in (6.7) for the pair $(\lambda\varphi_i, \lambda\varphi)$ in the class $\lambda\theta$ is therefore λ^n times the corresponding expression for (φ_i, φ) in the class θ. The equivalence follows from Corollary 6.2.2. □

Proposition 6.2.4 *Let $(\varphi_i)_{i \in I}$ (resp. $(\psi_i)_{i \in I}$) be a net in $\mathrm{PSH}(X, \theta)$ such that $\varphi_i \xrightarrow{d_S} \varphi \in \mathrm{PSH}(X, \theta)$ (resp. $\psi_i \xrightarrow{d_S} \psi \in \mathrm{PSH}(X, \theta)$). Then*

$$\varphi_i \vee \psi_i \xrightarrow{d_S} \varphi \vee \psi.$$

Proof Since d_S is a pseudometric, we may assume that both nets are actually sequences and $I = \mathbb{Z}_{>0}$. By Corollary 6.2.8, we may assume that the masses $\int_X \theta_\varphi^n > 0$, $\int_X \theta_\psi^n > 0$.

By Proposition 6.2.2 and Proposition 6.1.6, replacing all potentials by their P-envelopes changes neither the hypotheses nor the conclusion. Thus we may assume that the φ_j's, the ψ_j's, φ and ψ are all model. After discarding finitely many terms, the masses of the two sequences are bounded below by a positive constant. Applying Proposition 6.2.3 to both sequences and using the sandwich criterion Corollary 6.2.10, it is enough to treat the case where each of $(\varphi_j)_j$ and $(\psi_j)_j$ is monotone. In this reduced situation, $(\varphi_j)_j$ (resp. $(\psi_j)_j$) converges to φ (resp. ψ) almost everywhere.

We handle three cases separately.

Step 1. Assume that both sequences are increasing.

In this case, we have $\varphi_j \vee \psi_j \nearrow \varphi \vee \psi$ almost everywhere. Therefore, $\varphi_j \vee \psi_j \xrightarrow{d_S} \varphi \vee \psi$ by Corollary 6.2.3.

Step 2. Assume that one sequence, say $(\varphi_j)_j$ is increasing while the other is decreasing. Then we have

$$\varphi_j \vee \psi \le \varphi_j \vee \psi_j \le \varphi \vee \psi_j .$$

Thanks to Corollary 6.2.10, it suffices to show that both sides converge to $\varphi \vee \psi$ with respect to d_S. So we reduce to the case where both sequences are decreasing.

Step 3. Assume that both sequences are decreasing.

In this case, due to Corollary 6.2.5, it suffices to show that

$$\lim_{j\to\infty} \int_X \theta^n_{\varphi_j \vee \psi_j} = \int_X \theta^n_{\varphi\vee\psi}. \tag{6.23}$$

The $\geq$ direction follows from Theorem 2.4.3; it remains to argue the $\leq$ direction.

Thanks to Lemma 2.4.4, we may find a sequence $(\epsilon_j)_j$ in $(0,1)$ with limit 0 and a sequence $(\eta_j)_j$ in $\mathrm{PSH}(X,\theta)_{>0}$ such that

$$(1-\epsilon_j)\varphi_j + \epsilon_j \eta_j \le \varphi, \quad \eta_j \le \varphi_j .$$

It follows that for each $j \geq 1$, we have

$$(1-\epsilon_j)(\varphi_j \vee \psi_j) + \epsilon_j \eta_j \le \varphi \vee \psi_j .$$

Therefore by Theorem 2.4.3,

$$(1-\epsilon_j)^n \int_X \theta^n_{\varphi_j\vee\psi_j} \le \int_X \theta^n_{\varphi\vee\psi_j}.$$

Letting $j \to \infty$, we find that

$$\lim_{j\to\infty} \int_X \theta^n_{\varphi_j\vee\psi_j} \le \lim_{j\to\infty} \int_X \theta^n_{\varphi\vee\psi_j}.$$

Therefore, in order to prove (6.23), we may assume that one of the sequences is constant; let us say $\psi_j = \psi$ for all j. Repeating the same argument as before and constructing $(\epsilon_j)_j$, $(\eta_j)_j$ as above, we get

$$(1-\epsilon_j)^n \int_X \theta^n_{\varphi_j\vee\psi} \le \int_X \theta^n_{\varphi\vee\psi}.$$

Letting $j \to \infty$, we conclude (6.23). □

Proposition 6.2.5 *Let $(\varphi_i)_{i\in I}$, $(\psi_i)_{i\in I}$ be nets in $\mathrm{PSH}(X,\theta)$ such that $\varphi_i \xrightarrow{d_S} \varphi \in \mathrm{PSH}(X,\theta)$ and $\psi_i \xrightarrow{d_S} \psi \in \mathrm{PSH}(X,\theta)$. Assume that $\varphi_i \preceq_P \psi_i$ for all $i \in I$. Then $\varphi \preceq_P \psi$.*

Proof It follows from Proposition 6.2.4 that

$$\varphi_i \vee \psi_i \xrightarrow{d_S} \varphi \vee \psi .$$

By Lemma 6.1.3, we have $\varphi_i \vee \psi_i \sim_P \psi_i$ for all $i \in I$. In particular, by Proposition 6.2.2,

$$\varphi_i \vee \psi_i \xrightarrow{d_S} \psi.$$

By Proposition 6.2.2 again, $\varphi \vee \psi \sim_P \psi$ and hence $\varphi \preceq_P \psi$ by Lemma 6.1.3. □

Theorem 6.2.2 *Let θ_1, θ_2 be smooth real closed $(1,1)$-forms on X representing big cohomology classes. Suppose that $(\varphi_i)_{i\in I}$ (resp. $(\psi_i)_{i\in I}$) be a net in $\mathrm{PSH}(X,\theta_1)$ (resp. $\mathrm{PSH}(X,\theta_2)$) and $\varphi \in \mathrm{PSH}(X,\theta_1)$ (resp. $\psi \in \mathrm{PSH}(X,\theta_2)$). Consider the following three conditions:*

(1) $\varphi_i \xrightarrow{d_S} \varphi$;

(2) $\psi_i \xrightarrow{d_S} \psi$;

(3) $\varphi_i + \psi_i \xrightarrow{d_S} \varphi + \psi$.

Then any two of these conditions imply the third.

Proof By Corollary 6.2.8, we may assume that θ_1, θ_2 are both Kähler forms. We denote them by ω_1, ω_2 instead. Let $\omega = \omega_1 + \omega_2$.

(1)+(2) $\implies$ (3). It suffices to show that for each $r = 0, \dots, n$,

$$2\int_X \omega^r_{(\varphi_i+\psi_i)\vee(\varphi+\psi)} \wedge \omega^{n-r} - \int_X \omega^r_{\varphi_i+\psi_i} \wedge \omega^{n-r} - \int_X \omega^r_{\varphi+\psi} \wedge \omega^{n-r} \to 0.$$

Observe that for each $i \in I$,

$$(\varphi_i + \psi_i) \vee (\varphi + \psi) \leq (\varphi_i \vee \varphi) + (\psi_i \vee \psi).$$

The expression with the actual rooftop is non-negative and is bounded above by the same expression with this larger potential in place of $(\varphi_i + \psi_i) \vee (\varphi + \psi)$. Thus, it suffices to show that

$$2\int_X \omega^r_{(\varphi_i\vee\varphi)+(\psi_i\vee\psi)} \wedge \omega^{n-r} - \int_X \omega^r_{\varphi_i+\psi_i} \wedge \omega^{n-r} - \int_X \omega^r_{\varphi+\psi} \wedge \omega^{n-r} \to 0.$$

The left-hand side is a linear combination of

$$2\int_X \omega^a_{1,\varphi_i\vee\varphi} \wedge \omega^{r-a}_{2,\psi_i\vee\psi} \wedge \omega^{n-r} - \int_X \omega^a_{1,\varphi_i} \wedge \omega^{r-a}_{2,\psi_i} \wedge \omega^{n-r} - \int_X \omega^a_{1,\varphi} \wedge \omega^{r-a}_{2,\psi} \wedge \omega^{n-r}$$

with $a = 0, \dots, r$. Observe that $\varphi_i \vee \varphi \xrightarrow{d_S} \varphi$ and $\psi_i \vee \psi \xrightarrow{d_S} \psi$ by Corollary 6.2.2, each term tends to 0 by Theorem 6.2.1.

(1)+(3) $\implies$ (2). For each $C \geq 1$, from the direction we already proved together with Lemma 6.2.6,

$$C\varphi_i + \psi_i \xrightarrow{d_S} C\varphi + \psi.$$

By Theorem 6.2.1, for each $j = 0, \dots, n$,

$$\begin{aligned}&\lim_{i\in I}\int_X (C\omega_1+\omega_2+\mathrm{dd^c}(C\varphi_i+\psi_i))^j\wedge\omega_2^{n-j}\\ =&\int_X (C\omega_1+\omega_2+\mathrm{dd^c}(C\varphi+\psi))^j\wedge\omega_2^{n-j}.\end{aligned}$$

Both sides are polynomials in C. Comparing the constant terms gives

$$\lim_{i\in I}\int_X \omega_{2,\psi_i}^j\wedge\omega_2^{n-j}=\int_X \omega_{2,\psi}^j\wedge\omega_2^{n-j}. \tag{6.24}$$

Therefore, (2) follows if $\psi_i\geq\psi$ for each i by Lemma 6.2.2.

Next we prove the general case. By the direction that we already proved, we know that $\varphi_i+\psi\xrightarrow{d_S}\varphi+\psi$. By Proposition 6.2.4, we have

$$\varphi_i+(\psi_i\vee\psi)\xrightarrow{d_S}\varphi+\psi.$$

It follows from the special case above that $\psi_i\vee\psi\xrightarrow{d_S}\psi$. It follows from (6.24) and Corollary 6.2.7 that (2) holds.

(2)+(3) $\implies$ (1). This is similar.

Proposition 6.2.6 *Let $(\varphi_i)_{i\in I}$ be a net in $\mathrm{PSH}(X,\theta)$, decreasing with respect to the P-partial order. Assume that there is $\varphi\in\mathrm{PSH}(X,\theta)_{>0}$ with $\varphi_i\xrightarrow{d_S}\varphi$, then*

$$\inf_{i\in I}P_\theta[\varphi_i]=P_\theta[\varphi]. \tag{6.25}$$

Proof First observe that by our assumption, $(P_\theta[\varphi_i])_{i\in I}$ is a decreasing net. By the continuity of non-pluripolar masses Corollary 6.2.4, we have

$$\lim_{i\in I}\int_X\theta^n_{P_\theta[\varphi_i]}=\lim_{i\in I}\int_X\theta^n_{\varphi_i}=\int_X\theta^n_\varphi.$$

Therefore, replacing φ_i by $P_\theta[\varphi_i]$ and I by a tail, we may therefore assume that φ_i is model in $\mathrm{PSH}(X,\theta)_{>0}$ for each $i\in I$ and $(\varphi_i)_{i\in I}$ is decreasing. In this case, by Proposition 3.1.12, the potential

$$\psi\coloneqq\inf_{i\in I}\varphi_i$$

is model in $\mathrm{PSH}(X,\theta)_{>0}$ with

$$\int_X\theta^n_\psi=\lim_{i\in I}\int_X\theta^n_{\varphi_i}=\int_X\theta^n_\varphi.$$

Therefore, by Corollary 6.2.5, we find $\varphi_i\xrightarrow{d_S}\psi$. In particular, $\varphi\sim_P\psi$ by Proposition 6.2.2. Therefore,

$$\psi=P_\theta[\psi]=P_\theta[\varphi].$$

Therefore, (6.25) follows. □

Theorem 6.2.3 *The map*

$$P_\theta[\bullet]_{\mathcal{I}} : \mathrm{PSH}(X,\theta)_{>0} \to \mathrm{PSH}(X,\theta)_{>0}$$

is continuous with respect to d_S.

Proof Let $(\varphi_i)_{i\in\mathbb{Z}_{>0}}$ be a sequence in $\mathrm{PSH}(X,\theta)_{>0}$ such that $\varphi_i \xrightarrow{d_S} \varphi \in \mathrm{PSH}(X,\theta)_{>0}$. We want to show that

$$P_\theta[\varphi_i]_{\mathcal{I}} \xrightarrow{d_S} P_\theta[\varphi]_{\mathcal{I}}. \tag{6.26}$$

We may assume that the φ_i's and φ are all model potentials by Proposition 6.2.2.

By Proposition 6.2.3 and Corollary 6.2.10, we may assume that $(\varphi_i)_i$ is either increasing or decreasing. In the increasing case, we apply Proposition 3.2.14 and Corollary 6.2.3, while in the decreasing case, we apply Proposition 3.2.12, Proposition 3.1.12 and Corollary 6.2.5. □

We record the following result for later use.

Lemma 6.2.7 *Fix a Kähler form ω on X. As $\epsilon \to 0$ (from both sides), we have*

$$V_{\theta+\epsilon\omega} \xrightarrow{d_S} V_\theta. \tag{6.27}$$

Proof There are two assertions to prove, as detailed in the two steps.

Step 1. We first handle the case where $\epsilon \to 0+$.

In this case (6.27) means

$$V_{\theta+\epsilon\omega} \xrightarrow{d_{S,\theta+\omega}} V_\theta$$

as $\epsilon \to 0+$. So thanks to Corollary 6.2.5 and Proposition 3.1.12, it suffices to prove the following:

$$\inf_{\epsilon>0} P_{\theta+\omega}\left[V_{\theta+\epsilon\omega}\right] = P_{\theta+\omega}\left[V_\theta\right]. \tag{6.28}$$

First observe that

$$V_\theta = \inf_{\epsilon>0} V_{\theta+\epsilon\omega}.$$

In fact, the $\leq$ direction is trivial. As for the reverse inequality, it suffices to observe that the right-hand side lies in $\mathrm{PSH}(X,\theta)$. Therefore, due to Proposition 3.1.12,

$$\lim_{\epsilon\to 0+}\int_X (\theta+\epsilon\omega+\mathrm{dd^c}V_{\theta+\epsilon\omega})^n = \int_X \theta^n_{V_\theta}.$$

Therefore, for all $\epsilon>0$ small enough, we can find $\eta_\epsilon \in \mathrm{PSH}(X,\theta+\epsilon\omega)$ and $a_\epsilon > 0$ decreasing to 0 so that

$$(1-a_\epsilon)V_{\theta+\epsilon\omega} + a_\epsilon\eta_\epsilon \leq V_\theta.$$

Therefore, thanks to Proposition 3.1.10,

$$(1-a_\epsilon)P_{\theta+\omega}\left[V_{\theta+\epsilon\omega}\right]+a_\epsilon P_{\theta+\omega}\left[\eta_\epsilon\right]\leq P_{\theta+\omega}\left[V_\theta\right].$$

Since the envelopes $P_{\theta+\omega}[\eta_\epsilon]$ have supremum 0, they are uniformly bounded in L^1 by Proposition 1.5.1; hence $a_\epsilon P_{\theta+\omega}[\eta_\epsilon]\to 0$ in L^1. Letting $\epsilon\to 0$ and using Proposition 1.2.6, we conclude (6.28).

Step 2. We then handle the case where $\epsilon\to 0-$.

In this case, (6.27) simply means

$$V_{\theta-\epsilon\omega}\xrightarrow{d_{S,\theta}}V_\theta$$

as $\epsilon\to 0+$. But this follows from Corollary 6.2.3 if we can prove

$$\sup_{\epsilon>0}{}^* V_{\theta-\epsilon\omega}=V_\theta, \tag{6.29}$$

where we understand that ϵ is small enough so that $\theta-\epsilon\omega$ represents a big cohomology class. Since $\{\theta\}$ is big, we can find $\varphi\in\mathrm{PSH}(X,\theta)$ so that

$$\varphi\leq 0,\quad \theta_\varphi\geq\delta\omega$$

for some $\delta>0$. For a small $\epsilon>0$, we then have

$$\theta+\mathrm{dd^c}\left(\left(1-\frac{\epsilon}{\delta}\right)V_\theta+\frac{\epsilon}{\delta}\varphi\right)\geq\epsilon\omega.$$

Therefore,

$$\left(1-\frac{\epsilon}{\delta}\right)V_\theta+\frac{\epsilon}{\delta}\varphi\leq V_{\theta-\epsilon\omega}.$$

Letting $\epsilon\to 0+$, we then find

$$V_\theta\leq\sup_{\epsilon>0}{}^* V_{\theta-\epsilon\omega}.$$

The reverse inequality is trivial, hence (6.29) is established. □

6.2.3 Continuity of invariants

In this section, we prove the continuity of a few invariants of the singularities with respect to d_S.

Theorem 6.2.4 *Let $(\varphi_j)_{j\in I}$ be a net in $\mathrm{PSH}(X,\theta)$ and $\varphi_j\xrightarrow{d_S}\varphi\in\mathrm{PSH}(X,\theta)$. Then for any prime divisor E over X, we have*

$$\lim_{j\in I}\nu(\varphi_j,E)=\nu(\varphi,E). \tag{6.30}$$

Proof First observe that since d_S is a pseudometric, it suffices to prove (6.30) when $I = \mathbb{Z}_{>0}$ as partially ordered sets.

By Corollary 6.2.8, after adding a Kähler form to θ, we may assume that the masses of φ_j and of φ are bounded from below by a positive constant.

By Theorem 6.2.3, we may assume that φ_j and φ are both $\mathcal{I}$-model and hence model. When proving (6.30), we are free to pass to subsequences.

By Proposition 6.2.3 and a sandwich argument, we may assume that the sequence $(\varphi_j)_j$ is either increasing or decreasing. In the increasing case, the Lelong numbers $\nu(\varphi_j, E)$ decrease to a limit at least $\nu(\varphi, E)$, while L^1-convergence gives the opposite inequality by upper semicontinuity of Lelong numbers. In the decreasing case, (6.30) follows from Proposition 3.1.12. □

Theorem 6.2.5 *Let $(\varphi_j)_{j\in I}$ be a net in* $\mathrm{PSH}(X,\theta)$ *and $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$. Assume that $\varphi_j \xrightarrow{d_S} \varphi$. Then*

$$\operatorname{vol}\theta_{\varphi_j} \to \operatorname{vol}\theta_\varphi, \quad \int_X \theta_{\varphi_j}^n \to \int_X \theta_\varphi^n. \tag{6.31}$$

Recall the volume is defined in Definition 3.2.3. In fact, we do not have to assume the positivity of the mass of φ. The proof of the general statement is slightly more involved. See Corollary 7.3.1 below.

Proof The latter part of (6.31) is just a special case of Theorem 6.2.1. It remains to prove the former part.

We may therefore assume that $\int_X \theta_{\varphi_j}^n > 0$ for all $j \in I$ after passing to a tail. Then by Theorem 6.2.3, we have

$$P_\theta[\varphi_j]_{\mathcal{I}} \xrightarrow{d_S} P_\theta[\varphi]_{\mathcal{I}}.$$

Therefore, the first part of (6.31) follows again from Theorem 6.2.1. □

Next we show that d_S-convergent sequences have a sort of quasi-equisingular property (cf. (1.20)).

Theorem 6.2.6 *Let $\varphi_j, \varphi \in$* $\mathrm{PSH}(X,\theta)$ *($j \in \mathbb{Z}_{>0}$). Assume that $\varphi_j \xrightarrow{d_S} \varphi$. Then for each $\lambda' > \lambda > 0$, there is $j_0 > 0$ so that for $j \geq j_0$,*

$$\mathcal{I}(\lambda'\varphi_j) \subseteq \mathcal{I}(\lambda\varphi). \tag{6.32}$$

Proof Fix $\lambda' > \lambda > 0$, we want to find $j_0 > 0$ so that for $j \geq j_0$, (6.32) holds.

Step 1. We first assume that φ has analytic singularities.

Let $\pi\colon Y \to X$ be a log resolution of φ and let $E_1, \dots, E_N$ be all prime divisors in the polar locus of φ on Y. Recall that by Theorem 1.4.3, a local holomorphic function f lies in the right-hand side of (6.32) if and only if

$$\operatorname{ord}_{E_i}(f) > \lambda\nu(\varphi, E_i) - A_X(E_i) \tag{6.33}$$

whenever they make sense. Here A_X denotes the log discrepancy. Similarly, if f lies in the left-hand side of (6.32), then

$$\operatorname{ord}_{E_i}(f) > \lambda' \nu(\varphi_j, E_i) - A_X(E_i).$$

As Lelong numbers are continuous with respect to d_S by Theorem 6.2.4, we can find $j_0 > 0$ so that when $j \geq j_0$, $\lambda' \nu(\varphi_j, E_i) \geq \lambda \nu(\varphi, E_i)$ for all i. In particular, (6.33) follows.

Step 2. We handle the general case.

By Corollary 6.2.8, we are free to increase θ and assume that θ_φ is a Kähler current.

Take a quasi-equisingular approximation $(\psi_k)_k$ of φ in $\mathrm{PSH}(X, \theta)$. The existence is guaranteed by Theorem 1.6.2. Take $\lambda'' \in (\lambda, \lambda')$, then by definition, we can find $k > 0$ so that

$$\mathcal{I}(\lambda'' \psi_k) \subseteq \mathcal{I}(\lambda \varphi).$$

Observe that $\psi_k \geq \varphi$ for all $k > 0$. By Proposition 6.2.4, we then find that for any $k > 0$, we have $\varphi_j \vee \psi_k \xrightarrow{d_S} \varphi \vee \psi_k = \psi_k$ as $j \to \infty$. By Step 1, we can find $j_0 > 0$ so that for $j \geq j_0$,

$$\mathcal{I}\Big(\lambda'(\varphi_j \vee \psi_k)\Big) \subseteq \mathcal{I}(\lambda'' \psi_k).$$

It follows that for $j \geq j_0$,

$$\mathcal{I}(\lambda' \varphi_j) \subseteq \mathcal{I}(\lambda \varphi).$$

This completes the proof. □

Chapter 7
$\mathcal{I}$-good singularities

Le but de cette thèse est de munir son auteur du titre de Docteur.[a]
— Adrien Douady[b]*, at the beginning of his thesis*

[a] Similarly, the purpose of the current book is to make my complaints about France in the acknowledgments published.

[b] Adrien Douady (1935–2006) was a French mathematician known for his pioneering work in complex dynamics and fractal geometry. Along with John H. Hubbard, he proved important results about the Mandelbrot set and developed renormalization theory for polynomial mappings. He discovered the *Douady Rabbit*, a famous fractal Julia set.

Douady studied at École Normale Supérieure (the place where I began to hate France, thanks to Claude Viterbo) and taught at several French universities. He was also a member of the Bourbaki group.

Tragically, he died in a swimming accident in 2006.

The class of $\mathcal{I}$-good singularities is the central object of this book. Heuristically, $\mathcal{I}$-good potentials are those whose singularities are "visible to algebraic geometry": their non-pluripolar Monge–Ampère mass equals the volume defined by algebraic approximations, or equivalently the analytic and algebraic envelopes (P- and $\mathcal{I}$-envelopes, see Chapter 3) coincide. A great variety of potentials arising in practice — analytic singularities, model potentials, full-mass potentials, toric potentials, elementary metrics — turn out to be $\mathcal{I}$-good (see Examples 7.1.1, 7.1.2, 7.2.1 and 7.4.1), as do the potentials constructed from them. Many results in the rest of the book hold without restriction precisely because the relevant potentials are guaranteed to lie in this class.

The starting point is the equivalence theorem Theorem 7.1.1, which gives three characterizations of $\mathcal{I}$-goodness for $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$:

(1) φ is the d_S-limit of a sequence of analytic-singularity potentials (a closure-type condition);
(2) $\int_X \theta_\varphi^n = \operatorname{vol}\theta_\varphi$ (a mass-volume identity);
(3) $P_\theta[\varphi] = P_\theta[\varphi]_{\mathcal{I}}$ (a comparison of the P- and $\mathcal{I}$-envelopes).

Each of these conditions has its own use: (1) is the topological description that justifies calling $\mathcal{I}$-good potentials the "closure of the analytic class", (2) is what one wishes to use in volume computations, and (3) is what makes the algebraic side — multiplier ideals, asymptotic Riemann–Roch, base ideals — behave as expected.

The chapter is organized as follows. Section 7.1 establishes Theorem 7.1.1 and uses it to define $\mathcal{I}$-good potentials (Definition 7.1.1). It contains a stability property (Corollary 7.1.1), and gives a list of natural examples.

Section 7.2 develops further closure properties of the $\mathcal{I}$-good class: linearity (Proposition 7.2.1), behavior under suprema (Proposition 7.2.2), and the example of elementary metrics on a pseudo-effective line bundle.

Section 7.3 introduces the *mixed volume* of n-tuples of $\mathcal{I}$-good potentials (Definition 7.3.1) and studies its analytic properties: continuity along sequences (Theorem 7.3.1), a logarithmic Brunn–Minkowski-type inequality (Proposition 7.3.2), birational invariance (Proposition 7.3.3), and upper semicontinuity (Theorem 7.3.2).

Section 7.4 contains the main result of the chapter, the asymptotic Riemann–Roch formula Theorem 7.4.1: for a Hermitian pseudo-effective line bundle (L, h) on a connected compact Kähler manifold X of dimension n, we have

$$\lim_{k\to\infty} \frac{n!}{k^n} h^0\big(X, L^k \otimes \mathcal{I}(h^k)\big) = \operatorname{vol}\big(\mathrm{dd^c} h\big).$$

This is the algebraic content of $\mathcal{I}$-goodness: it is the asymptotic statement that the multiplier ideal cohomology of (L, h) recovers the non-pluripolar mass of the Chern current $\mathrm{dd^c} h$, and it serves as the foundation for the Okounkov body and partial Bergman kernel theories of Chapters 10 and 14.

7.1 The notion of $\mathcal{I}$-good singularities

Let X be a connected compact Kähler manifold of dimension n.

Theorem 7.1.1 *Let θ be a closed real smooth $(1,1)$-form on X representing a big cohomology class, and $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$. Then the following are equivalent:*

(1) *There exists a sequence $(\varphi_j)_{j>0}$ in $\mathrm{PSH}(X,\theta)$ with analytic singularities such that $\varphi_j \xrightarrow{d_S} \varphi$;*

(2) *we have*
$$\int_X \theta_\varphi^n = \operatorname{vol}\theta_\varphi; \tag{7.1}$$

(3) *we have*
$$P_\theta[\varphi] = P_\theta[\varphi]_{\mathcal{I}}. \tag{7.2}$$

In (1), we can in addition require that each θ_{φ_j} is a Kähler current.

Moreover, if θ_φ is a Kähler current, the sequence in (1) can be taken as any quasi-equisingular approximation of φ in $\mathrm{PSH}(X,\theta)$.

Since $(\mathrm{PSH}(X,\theta), d_S)$ is a pseudometric space, in (1) we can also replace the word *sequence* by *net*.

Recall that according to Corollary 3.2.1 and Proposition 3.2.9, one direction of (7.1) and (7.2) always holds:

$$\int_X \theta_\varphi^n \leq \operatorname{vol}\theta_\varphi, \quad P_\theta[\varphi] \leq P_\theta[\varphi]_{\mathcal{I}}.$$

Proof (1) $\implies$ (2). By Theorem 6.2.1, we have

$$\lim_{j\to\infty}\int_X \theta_{\varphi_j}^n = \int_X \theta_\varphi^n > 0.$$

We may therefore assume that $\int_X \theta_{\varphi_j}^n > 0$ for all $j \geq 1$. It follows from Proposition 3.2.10 that

$$\int_X \theta_{\varphi_j}^n = \operatorname{vol}\theta_{\varphi_j}$$

for any $j \geq 1$. Using Theorem 6.2.5, we conclude (7.1).

(2) $\iff$ (3). By definition,

$$\operatorname{vol}\theta_\varphi = \int_X \theta^n_{P_\theta[\varphi]_{\mathcal{I}}}.$$

Moreover, $\varphi - \sup_X \varphi \leq P_\theta[\varphi]_{\mathcal{I}}$. Thus, if (2) holds, Theorem 3.1.2 gives $P_\theta[\varphi]_{\mathcal{I}} \leq P_\theta[\varphi]$; the reverse inequality follows from Proposition 3.2.9. Conversely, if (3) holds, then (2) follows from Proposition 3.1.3.

(3) $\implies$ (1). Note that the condition in (1) characterizes the closure of analytic singularities in $\mathrm{PSH}(X,\theta)$.

Step 1. We first assume that θ_φ is a Kähler current. We will prove the following more general result in this case: Without assuming (3), $P_\theta[\varphi]_{\mathcal{I}}$ always lies in the closure of analytic singularities.

Let $(\varphi_j)_j$ be a quasi-equisingular approximation of φ in $\mathrm{PSH}(X,\theta)$. We will show that $\varphi_j \xrightarrow{d_S} P_\theta[\varphi]_{\mathcal{I}}$. Let

$$\psi = \inf_{j\in\mathbb{Z}_{>0}} P_\theta[\varphi_j].$$

Since $\varphi \leq \varphi_j$ for all $j > 0$, we have

$$\varphi - \sup_X \varphi \leq P_\theta[\varphi_j]$$

for all j, and hence

$$\varphi - \sup_X \varphi \leq \psi.$$

In particular, $\int_X \theta_\psi^n > 0$.

Now Proposition 6.2.2, Proposition 3.1.12 and Corollary 6.2.5 give

$$\varphi_j \xrightarrow{d_S} \psi.$$

Moreover, observe that ψ is $\mathcal{I}$-model by Proposition 3.2.12 and Proposition 3.2.10. Hence, if $\varphi \sim_{\mathcal{I}} \psi$, then

$$P_\theta[\varphi]_{\mathcal{I}} = P_\theta[\psi]_{\mathcal{I}} = \psi.$$

So it suffices to show that $\varphi \sim_{\mathcal{I}} \psi$. It only remains to argue that $\psi \preceq_{\mathcal{I}} \varphi$. For this purpose, take $\lambda > 0$. We need to show that

$$\mathcal{I}(\lambda\psi) \subseteq \mathcal{I}(\lambda\varphi).$$

By the strong openness Theorem 1.4.4, we may take $\lambda' > \lambda$ such that $\mathcal{I}(\lambda\psi) = \mathcal{I}(\lambda'\psi)$. It follows from the definition of the quasi-equisingular approximation that

$$\mathcal{I}(\lambda'\psi) \subseteq \mathcal{I}(\lambda'\varphi_j) \subseteq \mathcal{I}(\lambda\varphi)$$

for large enough j, where the first inclusion uses $\psi \leq P_\theta[\varphi_j]$ and $P_\theta[\varphi_j] \sim_{\mathcal{I}} \varphi_j$. Our assertion follows.

It follows from the proof that we may take φ_j so that θ_{φ_j} is a Kähler current for all $j \geq 1$.

Step 2. We handle the general case.

Assume (3) holds. By Lemma 2.4.5, we can find $\psi \in \mathrm{PSH}(X, \theta)$ so that θ_ψ is a Kähler current and $\psi \leq \varphi$. We let

$$\psi_j = (1 - j^{-1})\varphi + j^{-1}\psi$$

for each $j \in \mathbb{Z}_{>1}$. Then $(\psi_j)_j$ is an increasing sequence converging almost everywhere to φ. Then

$$P_\theta[\psi_j]_{\mathcal{I}} \xrightarrow{d_S} P_\theta[\varphi]_{\mathcal{I}} = P_\theta[\varphi]$$

by Proposition 3.2.14, Corollary 6.2.3. From Step 1, we know that each $P_\theta[\psi_j]_{\mathcal{I}}$ lies in the closure of analytic singularities, hence so is $P_\theta[\varphi] \sim_P \varphi$. Therefore, (1) follows. □

Definition 7.1.1 We say a potential $\varphi \in \mathrm{QPSH}(X)$ is $\mathcal{I}$*-good* if for some smooth closed real $(1,1)$-form θ on X such that $\varphi \in \mathrm{PSH}(X, \theta)_{>0}$, we have

$$P_\theta[\varphi] = P_\theta[\varphi]_{\mathcal{I}}. \tag{7.3}$$

Remark 7.1.1 In view of Theorem 7.1.1 and Corollary 3.2.1, the failure of $\mathcal{I}$-goodness of a given $\varphi \in \mathrm{PSH}(X, \theta)_{>0}$ can be characterized using the difference between the volume and the mass. We therefore introduce

$$\mathrm{Defect}(\theta_\varphi) := \operatorname{vol}\theta_\varphi - \int_X \theta_\varphi^n.$$

An immediate question is to verify that Definition 7.1.1 is independent of the choice of θ.

Lemma 7.1.1 *Let $\varphi \in \mathrm{PSH}(X, \theta)_{>0}$ for some smooth closed real $(1,1)$-form θ on X. Take a Kähler form ω on X. Then the following are equivalent:*

(1) $P_\theta[\varphi] = P_\theta[\varphi]_{\mathcal{I}}$;

(2) $P_{\theta+\omega}[\varphi] = P_{\theta+\omega}[\varphi]_{\mathcal{I}}$.

Proof (1) $\implies$ (2). By Theorem 7.1.1, we can find a sequence $(\varphi_j)_j$ in $\mathrm{PSH}(X,\theta)$ with analytic singularities such that $\varphi_j \xrightarrow{d_{S,\theta}} \varphi$. By Corollary 6.2.8, we have $\varphi_j \xrightarrow{d_{S,\theta+\omega}} \varphi$. Therefore, by Theorem 7.1.1 again, (2) holds.

(2) $\implies$ (1). Suppose that (1) fails, so that

$$\int_X (\theta + \mathrm{dd^c}\varphi)^n < \int_X (\theta + \mathrm{dd^c} P_\theta[\varphi]_{\mathcal{I}})^n.$$

Since $\varphi \le P_\theta[\varphi]_{\mathcal{I}}$, the monotonicity theorem Theorem 2.4.3 gives

$$\int_X \theta^i_\varphi \wedge \omega^{n-i} \le \int_X \theta^i_{P_\theta[\varphi]_{\mathcal{I}}} \wedge \omega^{n-i}$$

for all $i = 0, \dots, n$, and the inequality is strict for $i = n$. It follows that

$$\begin{aligned}
\int_X (\theta + \omega + \mathrm{dd^c}\varphi)^n &= \sum_{i=0}^n \binom{n}{i} \int_X \theta^i_\varphi \wedge \omega^{n-i} \\
&< \sum_{i=0}^n \binom{n}{i} \int_X \theta^i_{P_\theta[\varphi]_{\mathcal{I}}} \wedge \omega^{n-i} \\
&= \int_X (\theta + \omega + \mathrm{dd^c} P_\theta[\varphi]_{\mathcal{I}})^n \\
&\le \int_X (\theta + \omega + \mathrm{dd^c} P_{\theta+\omega}[\varphi]_{\mathcal{I}})^n.
\end{aligned}$$

So (2) fails as well. □

Corollary 7.1.1 *Let θ be a closed real smooth $(1,1)$-form on X representing a big cohomology class. Let $(\varphi_j)_{j\in I}$ be a net of $\mathcal{I}$-good potentials in $\mathrm{PSH}(X,\theta)$ such that $\varphi_j \xrightarrow{d_S} \varphi$. Then φ is $\mathcal{I}$-good.*

Note that we do not need to assume that $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$.

Proof By adding a Kähler form to θ, and using Lemma 7.1.1 and Corollary 6.2.8, we may assume that $\varphi_j, \varphi \in \mathrm{PSH}(X,\theta)_{>0}$ for all $j \in I$. It follows from Theorem 7.1.1 that

$$\int_X \theta^n_{\varphi_j} = \operatorname{vol}\theta_{\varphi_j}$$

for all $j \in I$. Taking limit with respect to j with the help of Theorem 6.2.5, we conclude that

$$\int_X \theta^n_\varphi = \operatorname{vol}\theta_\varphi.$$

Therefore, by Theorem 7.1.1 again, we find that φ is $\mathcal{I}$-good. □

Example 7.1.1 Assume that $\varphi \in \mathrm{QPSH}(X)$ has analytic singularities. Then φ is $\mathcal{I}$-good. This is proved in Proposition 3.2.10.

In particular, the potential in Example 1.8.2 is $\mathcal{I}$-good.

Example 7.1.2 Let θ be a closed real smooth $(1,1)$-form on X representing a big cohomology class, and let $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$[1] be an $\mathcal{I}$-model potential. Then φ is $\mathcal{I}$-good.

Example 7.1.3 Let θ be a closed real smooth $(1,1)$-form on X representing a big cohomology class, and let $\varphi \in \mathcal{E}(X,\theta)$. Then φ is $\mathcal{I}$-good. In fact, since $P_\theta[\varphi] = V_\theta$, we deduce that $P_\theta[\varphi]_{\mathcal{I}} = V_\theta$ as well.

In particular, the potential in Example 3.1.1 is $\mathcal{I}$-good.

A further class of examples of $\mathcal{I}$-good singularities will be given in Example 7.4.1 below.

There do exist, however, non-$\mathcal{I}$-good potentials.

Example 7.1.4 The potential in Example 6.1.3 is not $\mathcal{I}$-good. In fact, since all Lelong numbers of φ vanish, we know that $\varphi \sim_{\mathcal{I}} 0$, hence

$$P_{2\omega}[\varphi]_{\mathcal{I}} = 0.$$

On the other hand,

$$\int_X (2\omega + \mathrm{dd^c}\varphi)^1 = \int_X \omega < \int_X (2\omega),$$

where $(2\omega + \mathrm{dd^c}\varphi)^1$ is understood in the non-pluripolar sense.

Quasi-equisingular approximations and d_S-convergent sequences are related in the following manner:

Corollary 7.1.2 *Let $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$ and $(\epsilon_j)_j$ be a decreasing sequence in $\mathbb{R}_{\geq 0}$ with limit 0. Fix a Kähler form ω on X. Consider a decreasing sequence $(\varphi_j)_{j>0}$, where each $\varphi_j \in \mathrm{PSH}(X, \theta + \epsilon_j\omega)$ is a potential with analytic singularities. Assume that $\varphi = \inf_j \varphi_j$. Then the following are equivalent:*

(1) $\varphi_j \xrightarrow{d_S} P_\theta[\varphi]_{\mathcal{I}}$[2]*, and*
(2) $(\varphi_j)_j$ *is a quasi-equisingular approximation of φ.*

Proof By Corollary 6.2.8 and Example 7.1.2, we may replace θ by $\theta + C\omega$ for some large constant $C > 0$ and assume that $\varphi, \varphi_j \in \mathrm{PSH}(X, \theta - \omega)$ for all $j \geq 1$.

Here the limit in (1) is unchanged by this replacement. Indeed, $P_\theta[\varphi]_{\mathcal{I}}$ is $\mathcal{I}$-model, hence $\mathcal{I}$-good by Example 7.1.2. This implies

$$P_{\theta+C\omega}\left[P_\theta[\varphi]_{\mathcal{I}}\right] = P_{\theta+C\omega}\left[P_\theta[\varphi]_{\mathcal{I}}\right]_{\mathcal{I}} = P_{\theta+C\omega}[\varphi]_{\mathcal{I}}.$$

[1] It is not known to us whether the same holds when φ has vanishing mass.

[2] Just to be sure, this means $\varphi_j \xrightarrow{d_{S,\theta+\epsilon\omega}} P_\theta[\varphi]_{\mathcal{I}}$ for any $\epsilon > 0$. The choice of ϵ is irrelevant due to Corollary 6.2.8.

Thus the old and new targets are P-equivalent in the enlarged class, and Proposition 6.2.2 and Corollary 6.2.8 shows that condition (1) is unchanged.

(2) $\implies$ (1). This is already proved in the proof of Theorem 7.1.1.

(1) $\implies$ (2). This follows from Theorem 6.2.6. □

7.2 Properties of $\mathcal{I}$-good singularities

Let X be a connected compact Kähler manifold of dimension n.

We show that $\mathcal{I}$-goodness is preserved by a number of natural operations.

Proposition 7.2.1 *Let $\varphi, \psi \in \mathrm{QPSH}(X)$ be $\mathcal{I}$-good and $\lambda > 0$. Then the following potentials are all $\mathcal{I}$-good:*

(1) $\varphi + \psi$;
(2) $\varphi \vee \psi$;
(3) $\lambda\varphi$.

Proof Take a closed real smooth $(1,1)$-form θ on X such that $\varphi, \psi \in \mathrm{PSH}(X,\theta)_{>0}$. It follows from Theorem 7.1.1 that there are sequences $(\varphi_j)_j, (\psi_j)_j$ in $\mathrm{PSH}(X,\theta)$ with analytic singularities such that $\varphi_j \xrightarrow{d_S} \varphi$ and $\psi_j \xrightarrow{d_S} \psi$.

By Theorem 6.2.2, Proposition 6.2.4, we have

$$\varphi_j + \psi_j \xrightarrow{d_S} \varphi + \psi, \quad \varphi_j \vee \psi_j \xrightarrow{d_S} \varphi \vee \psi.$$

By Proposition 1.6.1, the potentials $\varphi_j + \psi_j$ and $\varphi_j \vee \psi_j$ have analytic singularities. So $\varphi + \psi$ and $\varphi \vee \psi$ are $\mathcal{I}$-good by Theorem 7.1.1.

On the other hand, Lemma 6.2.6 gives

$$\lambda\varphi_j \xrightarrow{d_S} \lambda\varphi.$$

Here the convergence is understood in the scaled class. So by Theorem 7.1.1, in order to show that $\lambda\varphi$ is $\mathcal{I}$-good, it suffices to show that $\lambda\varphi_j$ is $\mathcal{I}$-good. In other words, we may assume that φ has analytic singularities. We may further assume that $\varphi \le 0$.

In this case, take a decreasing sequence of rational numbers $\lambda_j \searrow \lambda$. Take a sufficiently large Kähler form ω on X so that $\lambda_j\varphi, \lambda\varphi \in \mathrm{PSH}(X,\omega)_{>0}$ for any j. Then observe that

$$\lambda\varphi = \sup_{j>0}{}^{*} \lambda_j\varphi.$$

Therefore, $\lambda_j\varphi \xrightarrow{d_S} \lambda\varphi$ by Corollary 6.2.3. Each $\lambda_j\varphi$ has analytic singularities, and hence is $\mathcal{I}$-good by Example 7.1.1. Therefore, $\lambda\varphi$ is $\mathcal{I}$-good by Theorem 7.1.1. □

Example 7.2.1 Let L be a pseudo-effective line bundle on X. Elementary metrics on L are defined in Definition 6.1.3. Let h be an elementary metric on L. Then $\mathrm{dd^c} h$ is $\mathcal{I}$-good.

This is a direct consequence of Proposition 7.2.1 and Example 7.1.1, and the fact that $\mathcal{I}$-goodness depends only on the P-singularity type.

Proposition 7.2.2 *Let $(\varphi_j)_{j\in I}$ be a non-empty family of $\mathcal{I}$-good potentials in $\mathrm{PSH}(X,\theta)$ for some closed real smooth $(1,1)$-form θ on X. Assume that this family is locally uniformly bounded from above. Then $\sup_{j\in I}{}^*\varphi_j$ is $\mathcal{I}$-good.*

Proof After adding a Kähler form to θ, we may assume that $\varphi_j \in \mathrm{PSH}(X,\theta)_{>0}$ for all $j \in I$.

When I is finite, this result follows from Proposition 7.2.1. When I is infinite, by Choquet's lemma Proposition 1.2.3, we may assume that $I = \mathbb{Z}_{>0}$. Replacing φ_j by $\varphi_1 \vee \cdots \vee \varphi_j$ and using Proposition 7.2.1, we may assume that the sequence $(\varphi_j)_j$ is increasing. In this case, as shown in Corollary 6.2.3,

$$\varphi_j \xrightarrow{d_S} \sup_{i\in\mathbb{Z}_{>0}}{}^*\varphi_i .$$

Therefore, $\sup_{i\in\mathbb{Z}_{>0}}{}^*\varphi_i$ is $\mathcal{I}$-good by Corollary 7.1.1. □

7.3 Mixed volumes

We first extend the notion of volume in Definition 3.2.3 to the mixed case. Let $\theta_1,\dots,\theta_n$ be smooth closed real $(1,1)$-forms on X representing pseudo-effective classes.

Definition 7.3.1 Let $\varphi_i \in \mathrm{PSH}(X,\theta_i)$ for $i = 1,\dots,n$. Write $T_i = \theta_i + \mathrm{dd^c}\varphi_i$ for each $i = 1,\dots,n$. We define the *mixed volume* $\mathrm{vol}(T_1,\dots,T_n)$ as follows:

(1) Suppose that $\mathrm{vol}\, T_i > 0$ for all $i = 1,\dots,n$. Then we let

$$\mathrm{vol}(T_1,\dots,T_n) = \int_X \left(\theta_1 + \mathrm{dd^c} P_{\theta_1}[\varphi_1]_{\mathcal{I}}\right) \wedge \cdots \wedge \left(\theta_n + \mathrm{dd^c} P_{\theta_n}[\varphi_n]_{\mathcal{I}}\right); \tag{7.4}$$

(2) in general, take a Kähler form ω on X, we define

$$\mathrm{vol}(T_1,\dots,T_n) = \lim_{\epsilon\to 0+} \mathrm{vol}(T_1+\epsilon\omega,\dots,T_n+\epsilon\omega). \tag{7.5}$$

Note that $\mathrm{vol}(T_1,\dots,T_n)$ does not depend on the choice of ω.

We first make a few observations: When $\mathrm{vol}\, T_i > 0$ for each $i = 1,\dots,n$, the definition (7.4) does not depend on how we represent T_i as $T_i = \theta_i + \mathrm{dd^c}\varphi_i$, this is a consequence of Theorem 2.4.3 and Proposition 3.2.4.

Next, when $\mathrm{vol}\, T_i > 0$ for each i, the definition (7.5) coincides with (7.4). In fact, in this case, for each i and each $\epsilon > 0$, we have

$$P_{\theta_i}[\varphi_i]_{\mathcal{I}} \sim_P P_{\theta_i+\epsilon\omega}\left[P_{\theta_i}[\varphi_i]_{\mathcal{I}}\right]_{\mathcal{I}} = P_{\theta_i+\epsilon\omega}[\varphi_i]_{\mathcal{I}}$$

as a consequence of Example 7.1.2. Hence using Proposition 6.1.4,

$$\begin{aligned}
&\lim_{\epsilon\to 0+} \operatorname{vol}(T_1+\epsilon\omega,\dots,T_n+\epsilon\omega)\\
=&\lim_{\epsilon\to 0+}\int_X \left(\theta_1+\epsilon\omega+\mathrm{dd^c}P_{\theta_1}[\varphi_1]_{\mathcal{I}}\right)\wedge\cdots\wedge\left(\theta_n+\epsilon\omega+\mathrm{dd^c}P_{\theta_n}[\varphi_n]_{\mathcal{I}}\right)\\
=&\operatorname{vol}(T_1,\dots,T_n).
\end{aligned}$$

Finally, for any closed positive $(1,1)$-current T on X, we have

$$\operatorname{vol}(T,\dots,T)=\operatorname{vol} T. \tag{7.6}$$

Write $T=\theta_\varphi$. In more concrete terms, we need to show that

$$\lim_{\epsilon\to 0+}\int_X (\theta+\epsilon\omega+\mathrm{dd^c}P_{\theta+\epsilon\omega}[\varphi]_{\mathcal{I}})^n=\int_X(\theta+\mathrm{dd^c}P_\theta[\varphi]_{\mathcal{I}})^n.$$

We may replace φ by $P_\theta[\varphi]_{\mathcal{I}}$ and assume that φ is $\mathcal{I}$-model in $\mathrm{PSH}(X,\theta)$. Then we claim that

$$\varphi=\inf_{\epsilon>0}P_{\theta+\epsilon\omega}[\varphi]_{\mathcal{I}}.$$

From this, our assertion follows from Proposition 3.1.12.

The $\leq$ direction is clear. For the converse, since φ is $\mathcal{I}$-model, it suffices to show that for each prime divisor E over X, we have

$$\nu(\varphi,E)\leq\nu\left(\inf_{\epsilon>0}P_{\theta+\epsilon\omega}[\varphi]_{\mathcal{I}},E\right).$$

We simply compute

$$\nu\left(\inf_{\epsilon>0}P_{\theta+\epsilon\omega}[\varphi]_{\mathcal{I}},E\right)\geq\sup_{\epsilon>0}\nu\left(P_{\theta+\epsilon\omega}[\varphi]_{\mathcal{I}},E\right)=\nu(\varphi,E).$$

Proposition 7.3.1

(1) *The mixed volume is symmetric in its n-variables.*
(2) *Consider closed positive $(1,1)$-currents $T_1,\dots,T_n,T_1'$ on X, then*

$$\operatorname{vol}(T_1+T_1',T_2,\dots,T_n)=\operatorname{vol}(T_1,T_2,\dots,T_n)+\operatorname{vol}(T_1',T_2,\dots,T_n). \tag{7.7}$$

(3) *Consider closed positive $(1,1)$-currents $T_1,\dots,T_n$ on X and $\lambda\geq 0$, then*

$$\operatorname{vol}(\lambda T_1,T_2,\dots,T_n)=\lambda\operatorname{vol}(T_1,T_2,\dots,T_n).$$

(4) *Suppose that $T_1,\dots,T_n,S_1,\dots,S_n$ are closed positive $(1,1)$-currents on X such that $T_i\preceq_{\mathcal{I}} S_i$ and $\{T_i\}=\{S_i\}$ for each $i=1,\dots,n$. Then*

$$\operatorname{vol}(T_1,\dots,T_n)\leq\operatorname{vol}(S_1,\dots,S_n). \tag{7.8}$$

(5) *Suppose that* $T_1, \dots, T_n$ *are closed positive* $(1,1)$*-currents on* X. *Then*

$$\operatorname{vol}(T_1, \dots, T_n) = \operatorname{vol}(\operatorname{Reg} T_1, \dots, \operatorname{Reg} T_n). \tag{7.9}$$

The notation Reg is defined in (1.24).

Proof (1) This follows from the symmetry of the mixed non-pluripolar product in (7.4) and then by the limiting definition (7.5).

(2) By definition of the mixed volume, we may assume that the relevant currents $T_1, \dots, T_n, T_1'$ are all Kähler currents. We write $T_i = \theta_i + \mathrm{dd^c}\varphi_i$ as before for each i and $T_1' = \theta_1' + \mathrm{dd^c}\varphi_1'$. The potentials $P_{\theta_1}[\varphi_1]_{\mathcal{I}}$ and $P_{\theta_1'}[\varphi_1']_{\mathcal{I}}$ are $\mathcal{I}$-model with positive masses, hence $\mathcal{I}$-good. By Proposition 7.2.1, their sum is $\mathcal{I}$-good. Moreover, it is $\mathcal{I}$-equivalent to $\varphi_1 + \varphi_1'$. Then thanks to Proposition 7.2.1,

$$P_{\theta_1}[\varphi_1]_{\mathcal{I}} + P_{\theta_1'}[\varphi_1']_{\mathcal{I}} \sim_P P_{\theta_1+\theta_1'}\left[P_{\theta_1}[\varphi_1]_{\mathcal{I}} + P_{\theta_1'}[\varphi_1']_{\mathcal{I}}\right]_{\mathcal{I}} = P_{\theta_1+\theta_1'}\left[\varphi_1 + \varphi_1'\right]_{\mathcal{I}}$$

Thus by Proposition 6.1.4, we have

$$\begin{aligned}
&\operatorname{vol}(T_1 + T_1', T_2, \dots, T_n)\\
={}& \int_X \left(\theta_1 + \theta_1' + \mathrm{dd^c} P_{\theta_1+\theta_1'}[\varphi_1 + \varphi_1']_{\mathcal{I}}\right) \wedge \left(\theta_2 + \mathrm{dd^c} P_{\theta_2}[\varphi_2]_{\mathcal{I}}\right) \wedge \cdots\\
&\wedge \left(\theta_n + \mathrm{dd^c} P_{\theta_n}[\varphi_n]_{\mathcal{I}}\right)\\
={}& \int_X \left(\theta_1 + \theta_1' + \mathrm{dd^c} P_{\theta_1}[\varphi_1]_{\mathcal{I}} + \mathrm{dd^c} P_{\theta_1'}[\varphi_1']_{\mathcal{I}}\right) \wedge \left(\theta_2 + \mathrm{dd^c} P_{\theta_2}[\varphi_2]_{\mathcal{I}}\right) \wedge \cdots\\
&\wedge \left(\theta_n + \mathrm{dd^c} P_{\theta_n}[\varphi_n]_{\mathcal{I}}\right)\\
={}& \operatorname{vol}(T_1, T_2, \dots, T_n) + \operatorname{vol}(T_1', T_2, \dots, T_n).
\end{aligned}$$

(3) This follows from (7.4) and then by the limiting definition (7.5).

(4) Thanks to the definition of the mixed volume, we may assume that $T_1, \dots, T_n$ and $S_1, \dots, S_n$ are all Kähler currents. In this case, our assertion follows from Proposition 6.1.4.

(5) Using (2) and (3), it suffices to establish the following: Suppose that

$$T = \sum_i c_i [E_i] \tag{7.10}$$

is a closed positive $(1,1)$-current on X, where $\{E_i\}$ is a countable collection of prime divisors on X and $c_i > 0$. Then

$$\operatorname{vol}(T, T_2, \dots, T_n) = 0 \tag{7.11}$$

for any closed positive $(1,1)$-currents $T_2, \dots, T_n$ on X.

Step 1. We first assume that (7.10) is a finite sum. Fix a Kähler form ω on X.

In this case, thanks to (2) and (3) again, we may assume that $T = [E]$ for some prime divisor E on X. Write $E = \theta_\varphi$, then φ has analytic singularities thanks to Proposition 1.8.1. Therefore, $P_{\theta+\epsilon\omega}[\varphi]_{\mathcal{I}} \sim \varphi$ for any $\epsilon > 0$ due to Proposition 3.2.10.

Therefore, writing $T_i = \theta_i + \mathrm{dd^c}\varphi_i$ for $i = 2, \dots, n$, and take $C > 0$ so that $C\omega + \theta_i$ are Kähler forms for each $i = 2, \dots, n$, then we have

$$\begin{aligned}
&\operatorname{vol}(T, T_2, \dots, T_n)\\
=&\lim_{\epsilon\to 0+}\int_X ([E]+\epsilon\omega)\wedge\left(\theta_2+\epsilon\omega+\mathrm{dd^c}P_{\theta_2+\epsilon\omega}[\varphi_2]_{\mathcal{I}}\right)\wedge\cdots\\
&\wedge\left(\theta_n+\epsilon\omega+\mathrm{dd^c}P_{\theta_n+\epsilon\omega}[\varphi_n]_{\mathcal{I}}\right)\\
=&\lim_{\epsilon\to 0+}\epsilon\int_X \omega\wedge\left(\theta_2+\epsilon\omega+\mathrm{dd^c}P_{\theta_2+\epsilon\omega}[\varphi_2]_{\mathcal{I}}\right)\wedge\cdots\\
&\wedge\left(\theta_n+\epsilon\omega+\mathrm{dd^c}P_{\theta_n+\epsilon\omega}[\varphi_n]_{\mathcal{I}}\right)\\
\leq&\lim_{\epsilon\to 0+}\epsilon\int_X \omega\wedge(\theta_2+C\omega)\wedge\cdots\wedge(\theta_n+C\omega)\\
=&0.
\end{aligned}$$

Step 2. We prove the case where $\{E_i\}$ is infinite. We may assume that i runs over $\mathbb{Z}_{>0}$.

Write $T_i = \theta_i + \mathrm{dd^c}\varphi_i$ for $i = 2, \dots, n$ as before. Fix a Kähler form ω on X so that $\theta_i + \omega$ is a Kähler form for each $i = 2, \dots, n$. Fix $\epsilon > 0$.

We can find $N_0 > 0$ so that for any $N \geq N_0$, the class of

$$\epsilon\omega + \sum_{i=N+1}^{\infty} c_i[E_i]$$

is Kähler. Take a Kähler form ω_N in this class. Then the currents

$$\sum_{i=1}^{N} c_i[E_i] + \omega_N, \quad \sum_{i=1}^{\infty} c_i[E_i] + \epsilon\omega$$

all lie in the same cohomology class. Moreover,

$$\sum_{i=1}^{\infty} c_i[E_i] + \epsilon\omega \preceq_{\mathcal{I}} \sum_{i=1}^{N} c_i[E_i] + \omega_N.$$

Therefore, by the monotonicity in (4), followed by the finite divisorial case of Step 1,

$$\begin{aligned}
\operatorname{vol}(T, T_2, \dots, T_n) &\leq \operatorname{vol}(T+\epsilon\omega, T_2, \dots, T_n)\\
&\leq \operatorname{vol}\left(\sum_{i=1}^{N} c_i[E_i] + \omega_N, T_2, \dots, T_n\right)\\
&= \operatorname{vol}(\omega_N, T_2, \dots, T_n)\\
&\leq \int_X \omega_N \wedge(\theta_2+\omega)\wedge\cdots\wedge(\theta_n+\omega).
\end{aligned}$$

Letting $N \to \infty$, and using $\{\omega_N\} \to \{\epsilon\omega\}$, we get

$$\operatorname{vol}(T, T_2, \ldots, T_n) \leq \epsilon \int_X \omega \wedge (\theta_2 + \omega) \wedge \cdots \wedge (\theta_n + \omega).$$

Since $\epsilon > 0$ is arbitrary, we conclude (7.11). □

Lemma 7.3.1 *Let ω be a Kähler form on X. Then there is a constant $C > 0$ depending only on $X, \omega, \{T_1\}, \ldots, \{T_n\}$ such that*

$$0 \leq \operatorname{vol}(T_1 + \epsilon\omega, \ldots, T_n + \epsilon\omega) - \operatorname{vol}(T_1, \ldots, T_n) \leq C\epsilon$$

for any $\epsilon \in [0, 1]$.

Proof By the multi-linearity established in Proposition 7.3.1, we can write

$$\operatorname{vol}(T_1 + \epsilon\omega, \ldots, T_n + \epsilon\omega) - \operatorname{vol}(T_1, \ldots, T_n)$$

as a linear combination of the mixed volumes between the T_i's and ω with coefficients ϵ^j for some $j \geq 1$. It suffices to show that

$$\operatorname{vol}(\omega, T_2, \ldots, T_n) \leq C,$$

where C depends only on $X, \omega, \{T_2\}, \ldots, \{T_n\}$. Represent T_i as $T_i = \theta_i + \mathrm{dd^c}\varphi_i$. Take a constant $D > 0$ so that $D\omega + \theta_i$ is a Kähler form for each $i = 2, \ldots, n$. Then

$$\operatorname{vol}(\omega, T_2, \ldots, T_n) \leq \operatorname{vol}(\omega, D\omega + \theta_2, \ldots, D\omega + \theta_n).$$

Our assertion follows. □

Next we show that Theorem 6.2.5 continues to hold even when φ has vanishing mass. We prove a slightly more general result:

Theorem 7.3.1 *Let $\theta_1, \ldots, \theta_n$ be smooth closed real $(1,1)$-forms representing big cohomology classes. Suppose that $(\varphi_j^k)_{k\in I}$ are nets in $\mathrm{PSH}(X, \theta_j)$ and $\varphi_j \in \mathrm{PSH}(X, \theta_j)$ for $j = 1, \ldots, n$. We assume that $\varphi_j^k \xrightarrow{d_S} \varphi_j$ for each $j = 1, \ldots, n$. Then*

$$\lim_{k\in I} \operatorname{vol}\left(\theta_{1,\varphi_1^k}, \ldots, \theta_{n,\varphi_n^k}\right) = \operatorname{vol}\left(\theta_{1,\varphi_1}, \ldots, \theta_{n,\varphi_n}\right). \tag{7.12}$$

Proof Fix a Kähler form ω, then for any $\epsilon > 0$, we have

$$\lim_{k\in I} \operatorname{vol}\left((\theta_1 + \epsilon\omega)_{\varphi_1^k}, \ldots, (\theta_n + \epsilon\omega)_{\varphi_n^k}\right) = \operatorname{vol}\left((\theta_1 + \epsilon\omega)_{\varphi_1}, \ldots, (\theta_n + \epsilon\omega)_{\varphi_n}\right)$$

as a consequence of Corollary 6.2.8, Theorem 6.2.1 and Theorem 6.2.3.

Now thanks to Lemma 7.3.1, there is $C > 0$ so that for each $k \in I$,

$$\begin{aligned}
&0 \leq \operatorname{vol}\left((\theta_1 + \epsilon\omega)_{\varphi_1^k}, \ldots, (\theta_n + \epsilon\omega)_{\varphi_n^k}\right) - \operatorname{vol}\left(\theta_{1,\varphi_1^k}, \ldots, \theta_{n,\varphi_n^k}\right) \leq C\epsilon, \\
&0 \leq \operatorname{vol}\left((\theta_1 + \epsilon\omega)_{\varphi_1}, \ldots, (\theta_n + \epsilon\omega)_{\varphi_n}\right) - \operatorname{vol}\left(\theta_{1,\varphi_1}, \ldots, \theta_{n,\varphi_n}\right) \leq C\epsilon.
\end{aligned}$$

Therefore, (7.12) follows. □

Corollary 7.3.1 *Let θ be a smooth closed real $(1,1)$-form representing a big cohomology class. Let $(\varphi_j)_{j\in I}$ be a net in $\operatorname{PSH}(X,\theta)$ and $\varphi \in \operatorname{PSH}(X,\theta)$. Assume that $\varphi_j \xrightarrow{d_S} \varphi$. Then*

$$\operatorname{vol}\theta_{\varphi_j} \to \operatorname{vol}\theta_\varphi, \quad \int_X \theta_{\varphi_j}^n \to \int_X \theta_\varphi^n. \tag{7.13}$$

Proof The first part of (7.13) is a special case of Theorem 7.3.1, while the second part of (7.13) is a special case of Theorem 6.2.1. □

The mixed volume has a log-concavity property:

Proposition 7.3.2 *Let $T_1,\dots,T_n$ be closed positive $(1,1)$-currents on X. Then*

$$\operatorname{vol}(T_1,\dots,T_n) \geq \prod_{i=1}^n (\operatorname{vol}T_i)^{1/n}.$$

Proof We may assume that $\operatorname{vol}T_i > 0$ for each $i=1,\dots,n$ since there is nothing to prove otherwise. In this case, we need to show that

$$\begin{aligned}&\int_X \left(\theta_1 + \mathrm{dd^c} P_{\theta_1}[\varphi_1]_{\mathcal{I}}\right) \wedge \cdots \wedge \left(\theta_n + \mathrm{dd^c} P_{\theta_n}[\varphi_n]_{\mathcal{I}}\right)\\ \geq &\prod_{i=1}^n \left(\int_X \left(\theta_i + \mathrm{dd^c}P_{\theta_i}[\varphi_i]_{\mathcal{I}}\right)^n\right)^{1/n}.\end{aligned}$$

This is a special case of Theorem 2.4.1. □

Next, we also study how the mixed volume behaves under bimeromorphic transforms.

Proposition 7.3.3 *Let $\pi\colon Y \to X$ be a proper bimeromorphic morphism from a Kähler manifold Y to X. Then*

$$\operatorname{vol}(\pi^*T_1,\dots,\pi^*T_n) = \operatorname{vol}(T_1,\dots,T_n).$$

Proof Using the definition of the mixed volume, we may easily reduce to the case where $\operatorname{vol}T_i > 0$ for each $i=1,\dots,n$. By Proposition 3.2.5, we know that if we write $T_i = \theta_i + \mathrm{dd^c}\varphi_i$, then

$$\pi^* P_{\theta_i}[\varphi_i]_{\mathcal{I}} = P_{\pi^*\theta_i}[\pi^*\varphi_i]_{\mathcal{I}}$$

for each $i=1,\dots,n$. In particular,

$$\operatorname{vol}\pi^*T_i = \operatorname{vol}T_i > 0.$$

Our assertion follows from the bimeromorphic invariance of the non-pluripolar product Proposition 2.4.2. □

As for pushforward, we also have a similar result. We need a preliminary result:

Lemma 7.3.2 *Let $\pi\colon Y \to X$ be a proper bimeromorphic morphism from a Kähler manifold Y. Then for any non-divisorial closed positive $(1,1)$-current T on Y, we have*

$$\pi^*\pi_*T = T + \sum_{i=1}^{N} c_i[E_i]$$

for finitely many π-exceptional divisors E_i and $c_i > 0$.

Proof Let E be the exceptional locus of π. Then

$$T = \mathbb{1}_{Y\setminus E}\pi^*\pi_*T.$$

Therefore,

$$\pi^*\pi_*T - T = \mathbb{1}_E\pi^*\pi_*T,$$

which has the stated form, due to the support theorems, see [Dem12b, Section 8]. □

Corollary 7.3.2 *Let $\pi\colon X \to Z$ be a proper bimeromorphic morphism from X to a Kähler manifold Z. Then*

$$\operatorname{vol}(T_1,\dots,T_n) = \operatorname{vol}(\pi_*T_1,\dots,\pi_*T_n). \tag{7.14}$$

Proof Observe that we may assume that $T_i = \operatorname{Reg} T_i$ for all $i = 1,\dots,n$. In fact, since the pushforward of the divisorial part of T_i is divisorial as well, hence by Proposition 7.3.1(5), they do not contribute to the volumes.

Now by Proposition 7.3.3, it remains to show that

$$\operatorname{vol}(T_1,\dots,T_n) = \operatorname{vol}(\pi^*\pi_*T_1,\dots,\pi^*\pi_*T_n).$$

By Lemma 7.3.2, the difference $\pi^*\pi_*T_i - T_i$ is divisorial, hence our desired equality follows from Proposition 7.3.1(5). □

A particular corollary of Corollary 7.3.2 will be useful later.

Corollary 7.3.3 *Let $\pi\colon X \to Z$ be a proper bimeromorphic morphism from X to a Kähler manifold Z. Assume that T is an $\mathcal{I}$-good closed positive $(1,1)$-current on X. Then so is π_*T.*

Proof We may assume that $\int_X T^n > 0$. Then by Corollary 7.3.2,

$$\operatorname{vol}\pi_*T = \operatorname{vol}T > 0$$

as well. Since T is $\mathcal{I}$-good, we have

$$\operatorname{vol}T = \int_X T^n.$$

But $\int_X T^n = \int_Z (\pi_*T)^n$, so

$$\operatorname{vol}\pi_*T = \int_Z (\pi_*T)^n > 0.$$

It follows that $\pi_* T$ is $\mathcal{I}$-good. □

Finally we establish a semicontinuity property of the mixed volumes.

Theorem 7.3.2 *Let $(\varphi_i^j)_{j\in J}$ $(i = 1,\dots,n)$ be nets in* $\mathrm{PSH}(X,\theta_i)$*, and $\varphi_i \in$* $\mathrm{PSH}(X,\theta_i)$ *$(i = 1,\dots,n)$. Assume that for each prime divisor E over X, we have*

$$\lim_{j\in J} \nu\left(\varphi_i^j, E\right) = \nu\left(\varphi_i, E\right), \quad i = 1,\dots,n.$$

Then

$$\varlimsup_{j\in J} \mathrm{vol}\left(\theta_1 + \mathrm{dd^c}\varphi_1^j, \dots, \theta_n + \mathrm{dd^c}\varphi_n^j\right) \leq \mathrm{vol}\left(\theta_1 + \mathrm{dd^c}\varphi_1, \dots, \theta_n + \mathrm{dd^c}\varphi_n\right).$$

Proof **Step 1**. We first assume that $\mathrm{vol}(\theta_i + \mathrm{dd^c}\varphi_i^j) > 0$ and $\mathrm{vol}(\theta_i + \mathrm{dd^c}\varphi_i) > 0$ for all $i = 1,\dots,n$ and $j \in J$.

Without loss of generality, we may assume that the φ_i^j's and the φ_i's are $\mathcal{I}$-model for all $i = 1,\dots,n$ and $j \in J$. Our assertion becomes

$$\varlimsup_{j\in J} \int_X \left(\theta_1 + \mathrm{dd^c}\varphi_1^j\right) \wedge \cdots \wedge \left(\theta_n + \mathrm{dd^c}\varphi_n^j\right) \leq \int_X \left(\theta_1 + \mathrm{dd^c}\varphi_1\right) \wedge \cdots \wedge \left(\theta_n + \mathrm{dd^c}\varphi_n\right). \tag{7.15}$$

For each $j \in J$, define

$$\psi_i^j \coloneqq \sup_{k\geq j}{}^* \varphi_i^k, \quad i = 1,\dots,n.$$

Observe that ψ_i^j is $\mathcal{I}$-good thanks to Proposition 7.2.2, since after the preceding reduction the tail family is bounded above by 0. It follows from Corollary 1.4.1 and our assumption that

$$\lim_{j\in J} \nu\left(\psi_i^j, E\right) = \nu\left(\varphi_i, E\right), \quad i = 1,\dots,n.$$

Moreover, for each $j \in J$, the same formula gives $\varphi_i \preceq_{\mathcal{I}} \psi_i^j$. Since φ_i is $\mathcal{I}$-model and ψ_i^j is $\mathcal{I}$-good, Proposition 6.1.3 gives

$$\varphi_i \leq P_{\theta_i}[\psi_i^j].$$

For each $i = 1,\dots,n$, we define

$$\psi_i = \inf_{j\in J} P_{\theta_i}[\psi_i^j].$$

In particular, $\psi_i \geq \varphi_i$, so $\int_X \theta_{i,\psi_i}^n > 0$. Since ψ_i^j is $\mathcal{I}$-good, $P_{\theta_i}[\psi_i^j]$ is both model and $\mathcal{I}$-model. Due to Proposition 3.2.13, ψ_i is $\mathcal{I}$-model. Thanks to Proposition 3.1.12, we know

$$\nu(\psi_i, E) = \nu(\varphi_i, E)$$

for any $i = 1, \ldots, n$ and any prime divisor E over X. In other words, $\psi_i \sim_{\mathcal{I}} \varphi_i$ for $i = 1, \ldots, n$. But both φ_i and ψ_i are $\mathcal{I}$-good, therefore,

$$\psi_i \sim_P \varphi_i, \quad i = 1, \ldots, n.$$

By Proposition 6.1.4, we have

$$\int_X (\theta_1 + \mathrm{dd^c}\psi_1) \wedge \cdots \wedge (\theta_n + \mathrm{dd^c}\psi_n) = \int_X (\theta_1 + \mathrm{dd^c}\varphi_1) \wedge \cdots \wedge (\theta_n + \mathrm{dd^c}\varphi_n).$$

Next by Proposition 6.1.4 again,

$$\begin{aligned}
&\varlimsup_{j\in J} \int_X \left(\theta_1 + \mathrm{dd^c}\varphi_1^j\right) \wedge \cdots \wedge \left(\theta_n + \mathrm{dd^c}\varphi_n^j\right) \\
\leq &\varlimsup_{j\in J} \int_X \left(\theta_1 + \mathrm{dd^c}\psi_1^j\right) \wedge \cdots \wedge \left(\theta_n + \mathrm{dd^c}\psi_n^j\right).
\end{aligned}$$

On the other hand, due to Proposition 3.1.12 and Corollary 6.2.5, for each $i = 1, \ldots, n$, we have

$$\psi_i^j \xrightarrow{d_S} \psi_i.$$

We conclude from Theorem 6.2.1 that

$$\varlimsup_{j\in J} \int_X \left(\theta_1 + \mathrm{dd^c}\psi_1^j\right) \wedge \cdots \wedge \left(\theta_n + \mathrm{dd^c}\psi_n^j\right) = \int_X (\theta_1 + \mathrm{dd^c}\psi_1) \wedge \cdots \wedge (\theta_n + \mathrm{dd^c}\psi_n).$$

Putting these equations together, (7.15) follows.

Step 2. Next we handle the general case.

Fix a Kähler form ω on X. For any $\epsilon \in (0, 1]$, from Step 1, we know that

$$\begin{aligned}
&\varlimsup_{j\in J} \operatorname{vol}\left(\theta_1 + \epsilon\omega + \mathrm{dd^c}\varphi_1^j, \ldots, \theta_n + \epsilon\omega + \mathrm{dd^c}\varphi_n^j\right) \\
\leq &\operatorname{vol}\left(\theta_1 + \epsilon\omega + \mathrm{dd^c}\varphi_1, \ldots, \theta_n + \epsilon\omega + \mathrm{dd^c}\varphi_n\right).
\end{aligned}$$

Using Lemma 7.3.1, we have

$$\begin{aligned}
&\varlimsup_{j\in J} \operatorname{vol}\left(\theta_1 + \mathrm{dd^c}\varphi_1^j, \ldots, \theta_n + \mathrm{dd^c}\varphi_n^j\right) \\
\leq &\varlimsup_{j\in J} \operatorname{vol}\left(\theta_1 + \epsilon\omega + \mathrm{dd^c}\varphi_1^j, \ldots, \theta_n + \epsilon\omega + \mathrm{dd^c}\varphi_n^j\right) \\
\leq &\operatorname{vol}\left(\theta_1 + \epsilon\omega + \mathrm{dd^c}\varphi_1, \ldots, \theta_n + \epsilon\omega + \mathrm{dd^c}\varphi_n\right) \\
\leq &\operatorname{vol}\left(\theta_1 + \mathrm{dd^c}\varphi_1, \ldots, \theta_n + \mathrm{dd^c}\varphi_n\right) + C\epsilon.
\end{aligned}$$

But since ϵ is arbitrary, our assertion follows. ◻

7.4 The volumes of Hermitian pseudo-effective line bundles

Let X be a connected compact Kähler manifold of dimension n.

Definition 7.4.1 A *Hermitian pseudo-effective line bundle* (L, h) on a complex manifold Y consists of a holomorphic line bundle L on Y together with a plurisubharmonic metric h on L.

Theorem 7.4.1 (Asymptotic Riemann–Roch formula) *Let (L, h) be a Hermitian pseudo-effective line bundle on X and T be a holomorphic line bundle on X. We have*

$$\lim_{k\to\infty} \frac{n!}{k^n} h^0\left(X, T \otimes L^k \otimes \mathcal{I}(h^k)\right) = \operatorname{vol}\left(\mathrm{dd^c} h\right). \tag{7.16}$$

In particular, the limit exists.

For the proof, let us fix a smooth Hermitian metric h_0 on L with $\theta = c_1(L, h_0)$. We identify h with $h_0 \exp(-\varphi)$ for some $\varphi \in \operatorname{PSH}(X, \theta)$. See Section 1.8 for the relevant notations.

Recall that when X admits a big line bundle, it is necessarily projective. See [MM07, Theorem 2.2.26].

We first handle the case where φ has analytic singularities.

Proposition 7.4.1 *Under the assumptions of Theorem 7.4.1, assume furthermore that φ has analytic singularities such that θ_φ is a Kähler current, then (7.16) holds.*

Proof **Step 1**. Reduce to the case of log singularities.

Let $\pi\colon Y \to X$ be a log resolution of φ. In this case, for each $k \in \mathbb{Z}_{>0}$, we have

$$h^0\Big(X, T \otimes L^k \otimes \mathcal{I}(k\varphi)\Big) = h^0\Big(Y, K_{Y/X} \otimes \pi^*T \otimes \pi^*L^k \otimes \mathcal{I}(k\pi^*\varphi)\Big).$$

By Proposition 3.2.5, we have

$$\operatorname{vol}(\mathrm{dd^c} h) = \operatorname{vol}(\mathrm{dd^c} \pi^* h).$$

Therefore, it suffices to argue (7.16) with $K_{Y/X} \otimes \pi^*T$, π^*L and π^*h in place of T, L and h.

Note that θ_φ is no longer a Kähler current, but it still has positive mass.

Step 2. Assume that φ has log singularities along an effective $\mathbb{Q}$-divisor D. We may assume that D is simple normal crossing by replacing X by a modification and D by the pull-back. We decompose D into irreducible components, say

$$D = \sum_{i=1}^{N} a_i D_i.$$

In this case, we can easily compute

$$\mathcal{I}(k\varphi) = \mathcal{O}_X\left(-\sum_{i=1}^{N}\lfloor ka_i\rfloor D_i\right)$$

for each $k \in \mathbb{Z}_{>0}$. Observe that $L - D$ is nef (see Lemma 1.6.1). Choose an integer $q > 0$ so that qD is integral. For each residue class $s = 0, \dots, q-1$, we have

$$\lfloor (mq+s)D\rfloor = mqD + \lfloor sD\rfloor.$$

Hence, on the residue class $k = mq + s$, the line bundle

$$T \otimes L^k \otimes \mathcal{O}_X\left(-\sum_{i=1}^{N}\lfloor ka_i\rfloor D_i\right)$$

is a fixed twist of $(q(L-D))^m$. Applying the asymptotic Riemann–Roch theorem [Laz04, Corollary 1.4.41] to the nef line bundle $q(L-D)$ and then letting $m \to \infty$ in each residue class, we conclude that

$$\lim_{k\to\infty} \frac{n!}{k^n} h^0\left(X, T \otimes L^k \otimes \mathcal{O}_X\left(-\sum_{i=1}^{N}\lfloor ka_i\rfloor D_i\right)\right) = (L-D)^n.$$

Observe that by Proposition 1.8.1,

$$\theta_\varphi = [D] + S,$$

where S is a closed positive $(1,1)$-current with bounded potential. Therefore,

$$(L-D)^n = \int_X S^n = \int_X \theta_\varphi^n.$$

By Example 7.1.1, we know that the right-hand side is exactly $\operatorname{vol}\theta_\varphi$. □

Proof (Proof of Theorem 7.4.1) **Step 1**. We first handle the case where θ_φ is a Kähler current. Fix a Kähler form $\omega \geq \theta$ on X such that $\theta_\varphi \geq 2\delta\omega$ for some $\delta \in (0,1)$.

Let $(\varphi_j)_j$ be a quasi-equisingular approximation of φ in $\mathrm{PSH}(X,\theta)$. We may assume that $\theta_{\varphi_j} \geq \delta\omega$ for all j. From Proposition 7.4.1, we know that for each $j \geq 1$,

$$\varlimsup_{k\to\infty} \frac{n!}{k^n} h^0\left(X, T\otimes L^k \otimes \mathcal{I}(k\varphi)\right) \leq \lim_{k\to\infty} \frac{n!}{k^n} h^0\left(X, T\otimes L^k \otimes \mathcal{I}(k\varphi_j)\right) = \operatorname{vol}\theta_{\varphi_j}.$$

It follows from Theorem 7.1.1 and Theorem 6.2.5 that the right-hand side converges to $\operatorname{vol}\theta_\varphi$ as $j \to \infty$. Therefore,

$$\varlimsup_{k\to\infty} \frac{n!}{k^n} h^0\left(X, T\otimes L^k \otimes \mathcal{I}(k\varphi)\right) \leq \operatorname{vol}\theta_\varphi.$$

Conversely, fix an integer $N > \delta^{-1}$. From Theorem 7.1.1 and Theorem 6.2.1, we know that

$$\lim_{j\to\infty}\int_X \theta^n_{\varphi_j} = \int_X \theta^n_{P_\theta[\varphi]_{\mathcal{I}}} > 0. \tag{7.17}$$

Therefore, by Lemma 2.4.4, we can find $j_0 > 0$ such that for $j \geq j_0$, there is $\psi \in \mathrm{PSH}(X,\theta)_{>0}$ (depending on j) with

$$(1-N^{-1})\varphi_j + N^{-1}\psi \leq P_\theta[\varphi]_{\mathcal{I}}. \tag{7.18}$$

For each $k > 0$, we write $k = k'N - r$, where $k' \in \mathbb{N}$ and $r \in \{0, 1, \dots, N-1\}$. Then we compute for $j \geq j_0$ and large enough k (to be specified shortly) that

$$\begin{aligned}
&h^0\left(X, T \otimes L^k \otimes \mathcal{I}(k\varphi)\right)\\
\geq &h^0\left(X, T \otimes L^{-r} \otimes L^{k'N} \otimes \mathcal{I}(k'N\varphi)\right)\\
\geq &h^0\left(X, T \otimes L^{-r} \otimes L^{k'N} \otimes \mathcal{I}\left(k'(\psi + (N-1)\varphi_j)\right)\right)\\
\geq &h^0\left(X, T \otimes L^{-r} \otimes L^{k'(N-1)} \otimes \mathcal{I}\left(k'N\varphi_j\right)\right),
\end{aligned}$$

where the third line follows from (7.18), the fourth line can be argued as follows: For large enough k, there is a non-zero section $s \in \mathrm{H}^0(X, L^{k'} \otimes \mathcal{I}(k'\psi))$ by Lemma 2.4.6. It follows from Lemma 1.6.3 that for large enough k,

$$\mathcal{I}\left(k'N\varphi_j\right) \subseteq \mathcal{I}_\infty\left(k'(N-1)\varphi_j\right).$$

It follows that multiplication by s gives an injective map

$$\begin{aligned}
&\mathrm{H}^0\left(X, T \otimes L^{-r} \otimes L^{k'(N-1)} \otimes \mathcal{I}\left(k'N\varphi_j\right)\right) \hookrightarrow\\
&\mathrm{H}^0\left(X, T \otimes L^{-r} \otimes L^{k'N} \otimes \mathcal{I}\left(k'\psi + k'(N-1)\varphi_j\right)\right).
\end{aligned}$$

Next observe that

$$(N-1)\theta + N\mathrm{dd^c}\varphi_j = N\theta_{\varphi_j} - \theta \geq N\delta\omega - \theta \geq (N\delta - 1)\omega.$$

Since $N > \delta^{-1}$, this is a Kähler current. Hence Proposition 7.4.1 is applicable. We let $k \to \infty$ to conclude that

$$\begin{aligned}
\varliminf_{k\to\infty}\frac{n!}{k^n}h^0(X, T\otimes L^k \otimes \mathcal{I}(k\varphi)) \geq& N^{-n}\int_X \left((N-1)\theta + N\mathrm{dd^c}\varphi_j\right)^n\\
=& \int_X \left((1-N^{-1})\theta + \mathrm{dd^c}\varphi_j\right)^n\\
\geq& \int_X \left(\theta + \mathrm{dd^c}\varphi_j\right)^n - CN^{-1},
\end{aligned}$$

where C is a constant independent of N and j. Letting $j \to \infty$ and then $N \to \infty$ and using (7.17), we find that

$$\varliminf_{k\to\infty} \frac{n!}{k^n} h^0\left(X, T\otimes L^k \otimes \mathcal{I}(k\varphi)\right) \geq \int_X \theta^n_{P_\theta[\varphi]_\mathcal{I}}.$$

Therefore, (7.16) follows.

Step 2. We handle the case where $\operatorname{vol}\theta_\varphi > 0$. We may assume that φ is $\mathcal{I}$-model.

First observe that L is big by Proposition 2.4.1. Hence X is projective. Take a very ample line bundle A on X and a Kähler form ω in $c_1(A)$. Take a Hermitian metric h_A on A with $\mathrm{dd^c} h_A = \omega$.

Fix $N \in \mathbb{Z}_{>0}$, we decompose any $k > 0$ as $k = k'N + r$ with $k' \in \mathbb{N}$ and $r \in \{0, 1, \dots, N-1\}$. Then

$$h^0\left(X, T\otimes L^k \otimes \mathcal{I}(k\varphi)\right) \leq h^0\left(X, T\otimes L^r \otimes L^{k'N} \otimes \mathcal{I}(k'N\varphi)\right).$$

Therefore,

$$\begin{aligned}
&\varlimsup_{k\to\infty} \frac{n!}{k^n} h^0\left(X, T\otimes L^k \otimes \mathcal{I}(k\varphi)\right)\\
\leq &\max_{r=0,\dots,N-1} \varlimsup_{k'\to\infty} \frac{n!}{k'^n N^n} h^0\left(X, T\otimes L^r \otimes L^{k'N} \otimes \mathcal{I}(k'N\varphi)\right)\\
\leq &\max_{r=0,\dots,N-1} \varlimsup_{k'\to\infty} \frac{n!}{k'^n N^n} h^0\left(X, T\otimes L^r \otimes L^{k'N} \otimes A^{k'} \otimes \mathcal{I}(k'N\varphi)\right)\\
= &\int_X \left(N^{-1}\omega + \theta + \mathrm{dd^c} P_{\theta+N^{-1}\omega}[\varphi]_\mathcal{I}\right)^n,
\end{aligned}$$

where the third line uses multiplication by a non-zero section of $A^{k'}$, and we have applied Step 1 to the Hermitian pseudo-effective line bundle $(L^N \otimes A, h^{\otimes N} \otimes h_A)$ on the fourth line. On the other hand, since φ is $\mathcal{I}$-good by Example 7.1.2, Lemma 7.1.1 gives

$$P_{\theta+N^{-1}\omega}[\varphi]_\mathcal{I} = P_{\theta+N^{-1}\omega}[\varphi].$$

It follows from Proposition 3.1.3 that

$$\varlimsup_{k\to\infty} \frac{n!}{k^n} h^0\left(X, T\otimes L^k \otimes \mathcal{I}(k\varphi)\right) \leq \int_X \left(N^{-1}\omega + \theta + \mathrm{dd^c}\varphi\right)^n.$$

Letting $N \to \infty$, we conclude

$$\varlimsup_{k\to\infty} \frac{n!}{k^n} h^0\left(X, T\otimes L^k \otimes \mathcal{I}(k\varphi)\right) \leq \int_X \theta^n_\varphi.$$

It remains to argue the reverse inequality.

Choose $\psi \in \mathrm{PSH}(X, \theta)$ such that θ_ψ is a Kähler current and $\psi \leq \varphi$. The existence of ψ is guaranteed by Lemma 2.4.5. Then for any $t \in (0, 1)$, we set

$$\varphi_t = (1-t)\varphi + t\psi.$$

It follows again from Step 1 that

$$\varliminf_{k\to\infty} \frac{n!}{k^n} h^0\left(X, T\otimes L^k \otimes \mathcal{I}(k\varphi)\right) \geq \varliminf_{k\to\infty} \frac{n!}{k^n} h^0\left(X, T\otimes L^k \otimes \mathcal{I}(k\varphi_t)\right) = \operatorname{vol}\theta_{\varphi_t}.$$

On the other hand, by Theorem 7.3.1 and Corollary 6.2.9,

$$\lim_{t\to 0+} \operatorname{vol}\theta_{\varphi_t} = \operatorname{vol}\theta_\varphi.$$

So we find

$$\varliminf_{k\to\infty} \frac{n!}{k^n} h^0\left(X, T\otimes L^k \otimes \mathcal{I}(k\varphi)\right) \geq \operatorname{vol}\theta_\varphi.$$

We conclude (7.16) in this case.

Step 3. We finally handle the case where $\operatorname{vol}\theta_\varphi = 0$. Replacing φ by $P_\theta[\varphi]_{\mathcal{I}}$, we may assume that φ is $\mathcal{I}$-model.

Assume that (7.16) fails. That is,

$$\varlimsup_{k\to\infty} \frac{n!}{k^n} h^0\left(X, T\otimes L^k\right) \geq \varlimsup_{k\to\infty} \frac{n!}{k^n} h^0\left(X, T\otimes L^k \otimes \mathcal{I}(k\varphi)\right) > 0,$$

then L is a big line bundle and hence X is projective.

Fix a very ample line bundle A on X and a Kähler form $\omega \in c_1(A)$. Take a decreasing sequence $(\epsilon_j)_j$ of rational numbers with limit 0 and a quasi-equisingular approximation $(\varphi_j)_j$ of φ with $\varphi_j \in \operatorname{PSH}(X, \theta + \epsilon_j\omega)_{>0}$.

We claim that as $j \to \infty$, the sequence $P_{\theta+\epsilon_j\omega}[\varphi_j]$ is decreasing with limit φ.

It is clear that this sequence is decreasing. Let ψ denote its limit for the moment. It is also clear that $\psi \geq \varphi$. Since φ is $\mathcal{I}$-model, it remains to show that $\psi \preceq_{\mathcal{I}} \varphi$. Fix $\lambda > 0$. By strong openness, choose $\lambda' > \lambda$ such that $\mathcal{I}(\lambda\psi) = \mathcal{I}(\lambda'\psi)$. For j large enough, the quasi-equisingularity of $(\varphi_j)_j$ gives

$$\mathcal{I}(\lambda'\psi) \subseteq \mathcal{I}(\lambda'\varphi_j) \subseteq \mathcal{I}(\lambda\varphi),$$

where the first inclusion uses $\psi \leq P_{\theta+\epsilon_j\omega}[\varphi_j]$ and $P_{\theta+\epsilon_j\omega}[\varphi_j] \sim_{\mathcal{I}} \varphi_j$. Thus $\psi \preceq_{\mathcal{I}} \varphi$, and the claim follows.

By our claim, Proposition 3.1.3 and Proposition 3.1.12, we find that

$$\lim_{j\to\infty} \int_X \left(\theta + \epsilon_j\omega + \mathrm{dd^c}\varphi_j\right)^n = \int_X \theta_\varphi^n = 0. \tag{7.19}$$

Fix $j > 0$, take an integer $N > 0$ so that $N\epsilon_j$ is an integer. Then we compute

$$
\begin{aligned}
&\overline{\lim_{k\to\infty}} \frac{n!}{k^n} h^0\left(X, T\otimes L^k\otimes \mathcal{I}(k\varphi)\right)\\
\leq &\overline{\lim_{k\to\infty}} \frac{n!}{k^n} h^0\left(X, T\otimes L^k\otimes \mathcal{I}(k\varphi_j)\right)\\
\leq &\max_{a=0,\dots,N-1} \overline{\lim_{k'\to\infty}} \frac{n!}{(k'N)^n} h^0\left(X, T\otimes L^a\otimes L^{Nk'}\otimes \mathcal{I}(Nk'\varphi_j)\right)\\
\leq &\max_{a=0,\dots,N-1} \overline{\lim_{k'\to\infty}} \frac{n!}{(k'N)^n} h^0\left(X, T\otimes L^a\otimes L^{Nk'}\otimes A^{k'N\epsilon_j}\otimes \mathcal{I}(Nk'\varphi_j)\right)\\
= &\frac{1}{N^n}\int_X \left(N\theta+\epsilon_j N\omega + N\mathrm{dd^c}\varphi_j\right)^n,
\end{aligned}
$$

where the third line follows by writing $k = Nk' + a$ as before, we applied Step 2 on the last line. Dividing by N^n, we find that

$$\overline{\lim_{k\to\infty}} \frac{n!}{k^n} h^0\left(X, T\otimes L^k\otimes \mathcal{I}(k\varphi)\right) \leq \int_X \left(\theta+\epsilon_j\omega+\mathrm{dd^c}\varphi_j\right)^n.$$

Since we know (7.19), letting $j \to \infty$, we conclude that

$$\overline{\lim_{k\to\infty}} \frac{n!}{k^n} h^0\left(X, T\otimes L^k\otimes \mathcal{I}(k\varphi)\right) = 0,$$

which is a contradiction. Hence (7.16) is established in full generality. □

Corollary 7.4.1 *Let L be a pseudo-effective line bundle on X, and let h be a smooth Hermitian metric on L with $\theta = c_1(L, h)$. Then we have*

$$\lim_{k\to\infty} \frac{n!}{k^n} h^0\left(X, L^k\right) = \int_X \theta^n_{V_\theta}. \tag{7.20}$$

Proof It suffices to observe that

$$\mathrm{H}^0\left(X, L^k\otimes \mathcal{I}\left(kV_\theta\right)\right) = \mathrm{H}^0\left(X, L^k\right) \tag{7.21}$$

for all $k > 0$, since for any holomorphic section s of L^k, the function $\log|s|^2_{h^k}$ is θ-psh thanks to Proposition 1.8.1 and therefore, we have

$$\frac{1}{k}\log|s|^2_{h^k} \leq V_\theta.$$

In particular, $s \in \mathrm{H}^0\left(X, L^k\otimes \mathcal{I}\left(kV_\theta\right)\right)$. □

This common quantity is the *volume* of L, usually denoted by $\operatorname{vol} L$. In view of Definition 3.2.4, we have

$$\operatorname{vol} L = \operatorname{vol} c_1(L). \tag{7.22}$$

Example 7.4.1 If X is a smooth projective toric variety and θ is invariant under the action of the compact torus, then any $\varphi \in \mathrm{PSH}_{\mathrm{tor}}(X, \theta)$ is $\mathcal{I}$-good.

Proof Thanks to Lemma 7.1.1, we may assume that $\theta \in c_1(L)$ for some toric invariant ample line bundle L. In this case, the result follows from Theorem 7.1.1, Theorem 7.4.1 and Theorem 5.2.2. □

Chapter 8
The trace operator

The difference between mathematicians and physicists is that after physicists prove a big result they think it is fantastic but after mathematicians prove a big result they think it is trivial.
— Lucien Szpiro[a]

[a] Lucien Szpiro (1941–2020) was a French mathematician known for his significant contributions to number theory and arithmetic geometry. His work often focused on problems related to Diophantine equations and the arithmetic of elliptic curves.

Szpiro is perhaps best known for Szpiro's Conjecture, which has deep connections to the famous abc conjecture in number theory, an important open problem with wide-ranging implications.

The goal of this chapter is to give a meaning to the *restriction* of a quasi-psh function $\varphi \in \mathrm{QPSH}(X)$ to a (possibly singular) subvariety $Y \subseteq X$. The naive answer $\varphi|_Y$ is unsatisfactory: it may be identically $-\infty$ even when $\nu(\varphi, Y) = 0$ (Example 8.1.1), and even when it is well-defined it does not interact well with quasi-equisingular approximations (Example 8.1.2). Following the philosophy of Chapter 7, we replace pointwise restriction by a restriction operation defined modulo $\mathcal{I}$-equivalence: the *trace operator* Tr_Y.

The trace operator is named after the Sobolev trace, and the analogy is exact:

Pluripotential theory	Real analysis
Quasi-psh functions	Functions
Quasi-psh functions with analytic singularities	Smooth functions
$\mathcal{I}$-good singularities	Measurable functions
$\mathcal{I}$-equivalence	Almost everywhere equality

In real analysis a reasonable function theory exists only for measurable functions and identifies functions modulo a.e. equality; in pluripotential theory the analogous workable class consists of the $\mathcal{I}$-good potentials, identified up to $\mathcal{I}$-equivalence. Quasi-psh functions with analytic singularities play the role of smooth functions: they admit an unambiguous pointwise restriction, and a general $\mathcal{I}$-good potential is a d_S-limit of such regular potentials, just as a measurable function is approximated almost everywhere by smooth ones. The Sobolev trace $\mathrm{Tr}\colon W^{1,p}(\Omega) \to L^p(\partial\Omega)$ takes a smooth function to its pointwise restriction and a Sobolev function to a class modulo a.e. equality; our trace Tr_Y behaves the same way relative to $\mathcal{I}$-equivalence.

The chapter is organized as follows.

Section 8.1 constructs the trace operator $\mathrm{Tr}_Y : \mathrm{QPSH}(X) \to \mathrm{QPSH}(\tilde{Y})/\!\sim_{\mathcal{I}}$ on the normalization $\tilde{Y}$ via quasi-equisingular approximations (Proposition 8.1.1 and Definitions 8.1.1 and 8.1.2). The construction rests on showing that d_S-limits of restrictions $\varphi_i|_{\tilde{Y}}$ depend only on φ, modulo $\mathcal{I}$-equivalence on $\tilde{Y}$.

Section 8.2 develops the basic functional properties of Tr_Y: linearity and behavior on decreasing limits (Propositions 8.2.1 and 8.2.2), together with the Ohsawa–Takegoshi-type extension result (Proposition 8.2.3 and Theorem 8.2.1) which is the analytic backbone of the trace.

Section 8.3 ties Tr_Y to the classical theory of *restricted volumes* of line bundles on Y: Proposition 8.3.1 and Theorem 8.3.1 show that the volume of $\mathrm{Tr}_Y\,\varphi$ on $\tilde{Y}$ recovers the restricted volume of the underlying line bundle.

Section 8.4 contains the relative version of the asymptotic Riemann–Roch formula Theorem 7.4.1: Theorems 8.4.1 and 8.4.2 compute the dimension of the cohomology of $L^k \otimes \mathcal{I}(h^k)$ restricted to Y in terms of the restricted volume of $\mathrm{dd^c}h$. This is the relative analogue of Theorem 7.4.1.

Section 8.5 establishes the *analytic Bertini theorem* Theorem 8.5.1: for a generic member of a base-point-free linear system, the trace recovers genuine pointwise restriction up to $\mathcal{I}$-equivalence (Corollaries 8.5.1 to 8.5.3). Together, the trace and the analytic Bertini theorem allow induction on $\dim X$ when studying singularities: the Bertini theorem handles the generic case, while Tr_Y handles the remaining (special) sections.

The trace operator and the analytic Bertini theorem are used pervasively in the rest of the book, in particular in the construction of partial Okounkov bodies in Chapter 10 (the inductive step uses successive trace operators to reduce the dimension).

8.1 The definition of the trace operator

Let X be a connected compact Kähler manifold and $Y \subseteq X$ be an irreducible analytic subset.

The trace operator gives a way to restrict a quasi-plurisubharmonic function on X to $\tilde{Y}$, the normalization of Y. It follows from [GK20, Proposition 3.5] that $\tilde{Y}$ is a normal Kähler space. We refer to Appendix B for the pluripotential theory on unibranch Kähler spaces.

For later applications, we need this generality even if initially we are only interested in the smooth case.

Before diving into the trace operator, let us try to understand what goes wrong with the naive restriction procedure: simply take $\varphi|_{\tilde{Y}}$. Both examples below are local, but can easily be globalized since the singularities are isolated.

Example 8.1.1 Consider the case where φ has log-log singularities as in Example 3.1.1. Locally we can take $\varphi(z) = -\log(-\log|z|^2)$. We have $\nu(\varphi, 0) = 0$ but $\varphi|_0 = -\infty$. So the naive restriction is not defined in this case.

Even if the naive restriction is defined, it does not behave well.

Example 8.1.2 Consider a psh function φ in two variables, say

$$\varphi(z, w) = \left(-\log(-\log|z|^2)\right) \vee \left(\log|w|^2\right).$$

Take an arbitrary quasi-equisingular approximation $(\varphi_j)_j$. Then each φ_j is locally bounded since $\nu(\varphi,(0,0)) = 0$. Let us consider the restrictions to $H = \{z = 0\}$. Then $\varphi_j|_H$ is still bounded, while

$$\varphi|_H(w) = \log |w|^2.$$

In other words, the restrictions $\varphi_j|_H$ fail to be a quasi-equisingular approximation of $\varphi|_H$.

Our trace operator gives an elegant way to solve both problems.

We first observe that if $\varphi \in \mathrm{QPSH}(X)$ has analytic singularities and $\nu(\varphi, Y) = 0$, then $\varphi|_Y \not\equiv -\infty$. This observation will be crucial in the sequel.

Proposition 8.1.1 *Let $\varphi \in \mathrm{QPSH}(X)$ be a function such that $\nu(\varphi, Y) = 0$. Let $(\varphi_i)_i$, $(\psi_i)_i$ be quasi-equisingular approximations of φ. Then*

$$\lim_{i\to\infty} d_S\left(\varphi_i|_{\tilde{Y}}, \psi_i|_{\tilde{Y}}\right) = 0. \tag{8.1}$$

Here in (8.1), d_S refers to any $d_{S,\theta}$, where θ is any closed smooth real $(1,1)$-form on $\tilde{Y}$ with $\varphi_i|_{\tilde{Y}}, \psi_i|_{\tilde{Y}} \in \mathrm{PSH}(\tilde{Y}, \theta)$. For different choices of θ, (8.1) remains equivalent, as explained in Corollary 6.2.8.

Proof After subtracting a common constant from φ, φ_i and ψ_i, we may assume that $\varphi_i \leq 0$ for all $i \geq 1$.

Take a Kähler form ω on X such that $\varphi_i, \psi_i \in \mathrm{PSH}(X, \omega/2)$ for all $i \geq 1$. By Corollary 6.2.8, we need to show that

$$\lim_{i\to\infty} d_{S,\omega|_{\tilde{Y}}}\left(\varphi_i|_{\tilde{Y}}, \psi_i|_{\tilde{Y}}\right) = 0.$$

Assume that this fails. Then, up to replacing the sequences by subsequences, we may assume that the following limit exists and

$$\lim_{i\to\infty} d_{S,\omega|_{\tilde{Y}}}\left(\varphi_i|_{\tilde{Y}}, \psi_i|_{\tilde{Y}}\right) > 0.$$

Take a Kähler form $\tilde{\omega}$ on $\tilde{Y}$, then

$$\lim_{i\to\infty} d_{S,\omega|_{\tilde{Y}}+\tilde{\omega}}\left(\varphi_i|_{\tilde{Y}}, \psi_i|_{\tilde{Y}}\right) > 0$$

by Corollary 6.2.8.

Since $\varphi_i \xrightarrow{d_S} P_\theta[\varphi]_{\mathcal{I}}$, $\psi_i \xrightarrow{d_S} P_\theta[\varphi]_{\mathcal{I}}$. Let

$$\chi_i \coloneqq \varphi_i \vee \psi_i.$$

By Proposition 6.2.4, we have $\chi_i \xrightarrow{d_S} P_\theta[\varphi]_{\mathcal{I}}$. Moreover, $(\chi_i)_i$ is again a quasi-equisingular approximation of the original potential φ as follows from Corollary 7.1.2.

The triangle inequality gives

$$d_S(\varphi_i|_{\tilde{Y}}, \psi_i|_{\tilde{Y}}) \leq d_S(\varphi_i|_{\tilde{Y}}, \chi_i|_{\tilde{Y}}) + d_S(\psi_i|_{\tilde{Y}}, \chi_i|_{\tilde{Y}}).$$

Thus, after passing to a subsequence and interchanging the two quasi-equisingular approximations if necessary, we may assume that

$$d_S(\varphi_i|_{\tilde{Y}}, \chi_i|_{\tilde{Y}}) \not\to 0.$$

Replacing $(\psi_i)_i$ by $(\chi_i)_i$, we may therefore assume that

$$\varphi_i \leq \psi_i$$

for all i, while preserving the contradiction assumption.

Take a decreasing sequence $(\epsilon_j)_j$ in $\mathbb{R}_{>0}$ with limit 0 such that $(1-\epsilon_j)\varphi_j \in \mathrm{PSH}(X,\omega)$. We first observe that

$$\lim_{i\to\infty} d_{S,\omega|_{\tilde{Y}}} \left(\varphi_i|_{\tilde{Y}}, (1-\epsilon_i)\varphi_i|_{\tilde{Y}}\right) = 0.$$

This is a consequence of Lemma 6.2.3. Hence, by Corollary 6.2.8, we find

$$\lim_{i\to\infty} d_{S,\omega|_{\tilde{Y}}+\tilde{\omega}} \left(\varphi_i|_{\tilde{Y}}, (1-\epsilon_i)\varphi_i|_{\tilde{Y}}\right) = 0.$$

Since $(\varphi_i)_i$ and $(\epsilon_i)_i$ are decreasing and $\varphi_i \leq 0$, the sequence $\left((1-\epsilon_i)\varphi_i|_{\tilde{Y}}\right)_i$ is decreasing. It has uniformly positive mass in the class $\omega|_{\tilde{Y}} + \tilde{\omega}$. Hence Corollary 6.2.6, together with the preceding estimate, gives a potential $\psi \in \mathrm{PSH}(\tilde{Y}, \omega|_{\tilde{Y}} + \tilde{\omega})$ such that

$$\varphi_i|_{\tilde{Y}} \xrightarrow{d_S} \psi.$$

Hence,

$$\lim_{i\to\infty} d_{S,\omega|_{\tilde{Y}}+\tilde{\omega}} \left(\psi, (1-\epsilon_i)\varphi_i|_{\tilde{Y}}\right) = 0.$$

Next by Proposition 1.6.3, we can find a subsequence $(\psi_{j_i})_{i\in\mathbb{Z}_{>0}}$ of $(\psi_j)_j$ such that for each $i \geq 1$,

$$\varphi_{j_i} \leq \psi_{j_i} \leq (1-\epsilon_i)\varphi_i.$$

Hence,

$$\varphi_{j_i}|_{\tilde{Y}} \leq \psi_{j_i}|_{\tilde{Y}} \leq (1-\epsilon_i)\varphi_i|_{\tilde{Y}}.$$

Therefore, by Corollary 6.2.1,

$$\begin{aligned}
\varlimsup_{i\to\infty} d_{S,\omega|_{\tilde{Y}}+\tilde{\omega}} \left(\varphi_{j_i}|_{\tilde{Y}}, \psi_{j_i}|_{\tilde{Y}}\right) &\leq \varlimsup_{i\to\infty} d_{S,\omega|_{\tilde{Y}}+\tilde{\omega}} \left(\varphi_{j_i}|_{\tilde{Y}}, (1-\epsilon_i)\varphi_i|_{\tilde{Y}}\right) \\
&= \varlimsup_{i\to\infty} d_{S,\omega|_{\tilde{Y}}+\tilde{\omega}} \left(\psi, (1-\epsilon_i)\varphi_i|_{\tilde{Y}}\right) \\
&=0,
\end{aligned}$$

which is a contradiction. ◻

Definition 8.1.1 Let $\varphi \in \mathrm{QPSH}(X)$ be a function such that $\nu(\varphi, Y) = 0$. We say a potential $\psi \in \mathrm{QPSH}(\tilde{Y})$ is a[1] *trace operator* of φ along Y if there is a quasi-equisingular approximation $(\varphi_j)_{j>0}$ of φ such that

$$\varphi_j|_{\tilde{Y}} \xrightarrow{d_S} \psi^{2}. \tag{8.2}$$

Indeed, choosing the background Kähler form on $\tilde{Y}$ sufficiently positive, the restricted quasi-equisingular approximation is a decreasing sequence with masses bounded below by a positive constant; hence it converges by Corollary 6.2.6. Observe that by Proposition 8.1.1, the condition (8.2) is independent of the choice of $(\varphi_j)_j$.

Later on in Theorem 12.3.2, we shall prove that the trace operator corresponds to the natural way of restricting convex bodies in the toric setting.

Proposition 8.1.2 *Let $\varphi \in \mathrm{QPSH}(X)$ such that $\nu(\varphi, Y) = 0$. Suppose that ψ and ψ' are trace operators of φ along Y. Then ψ and ψ' are $\mathcal{I}$-good and $\psi \sim_P \psi'$.*

Proof The restrictions of the analytic approximants have analytic singularities on $\tilde{Y}$, so the claim follows from Theorem 7.1.1. The fact that $\psi \sim_P \psi'$ follows from Proposition 8.1.1 and Proposition 6.2.2. □

Example 8.1.3 As a trivial example, when Y is just a single point, then $\mathrm{QPSH}(Y)$ is canonically identified with $\mathbb{R}$. Any constant $c \in \mathbb{R}$ is a trace operator of a function $\varphi \in \mathrm{QPSH}(X)$ satisfying $\nu(\varphi, Y) = 0$.

Definition 8.1.2 Let $\varphi \in \mathrm{QPSH}(X)$ such that $\nu(\varphi, Y) = 0$. We write $\mathrm{Tr}_Y(\varphi)$ for any trace operator of φ along Y.

Given a closed smooth real $(1,1)$-form θ on X. When $\mathrm{Tr}_Y(\varphi)$ can be chosen to lie in $\mathrm{PSH}(\tilde{Y}, \theta|_{\tilde{Y}})_{>0}$, we write

$$\mathrm{Tr}_Y^{\theta}(\varphi) \coloneqq P_{\theta|_{\tilde{Y}}}\left[\mathrm{Tr}_Y(\varphi)\right] = P_{\theta|_{\tilde{Y}}}\left[\mathrm{Tr}_Y(\varphi)\right]_{\mathcal{I}}.$$

The trace operator $\mathrm{Tr}_Y(\varphi)$ is therefore well-defined only up to P-equivalence by Proposition 8.1.2. Also observe that if $\varphi \in \mathrm{PSH}(X, \theta)$ for some smooth closed real $(1,1)$-form θ on X, then for any Kähler form ω on X, the trace operator $\mathrm{Tr}_Y^{\theta+\omega}(\varphi)$ is always defined. In particular, if θ_φ is a Kähler current, $\mathrm{Tr}_Y^{\theta}(\varphi)$ is always defined.

---

[1] Let us resume our analogy in the introduction of this chapter. In real analysis, the trace of a Sobolev function f is only defined up to almost-everywhere equality; we say a particular function on $\partial\Omega$ is *a* trace of f. Likewise here, we say ψ is *a* trace operator of φ rather than that the P-equivalence class of ψ is *the* trace operator of φ.

[2] To be more precise, what we mean is the following: We can find a closed smooth real $(1,1)$-form on X such that $\varphi \in \mathrm{PSH}(X, \theta)$. Then there is a Kähler form such that $\omega + \theta + \mathrm{dd^c}\varphi_j \geq 0$ for all $j \geq 1$. Take a Kähler form $\tilde{\omega}$ on $\tilde{Y}$ so that $\tilde{\omega} \geq (\theta + \omega)|_{\tilde{Y}}$ and that $\psi \in \mathrm{PSH}(\tilde{Y}, \tilde{\omega})$. Then our condition means that $\varphi_j|_{\tilde{Y}} \xrightarrow{d_{S,\tilde{\omega}}} \psi$. This condition is independent of the choices of θ, ω and $\tilde{\omega}$ by Corollary 6.2.8.

Remark 8.1.1 As in Remark 1.7.1, the trace operator could also be applied to closed positive $(1,1)$-currents on X. If $T \in \mathcal{Z}_+(X,\alpha)$ for some pseudo-effective class α on X (see Definition 1.7.3) and $\beta \in \mathrm{H}^{1,1}(\tilde{Y},\mathbb{R})$, then we write

$$\mathrm{Tr}_Y^{\beta}(T)$$

for any (if exists) closed positive $(1,1)$-current in β representing $\mathrm{Tr}_Y(T)$ when $\nu(T,Y)=0$.

Proposition 8.1.3 *Let $\varphi \in \mathrm{QPSH}(X)$ such that $\nu(\varphi,Y)=0$. Assume that $\varphi|_Y \not\equiv -\infty$. Then*

$$\varphi|_{\tilde{Y}} \preceq_P \mathrm{Tr}_Y(\varphi).$$

Proof Take a Kähler form ω such that ω_φ is a Kähler current. Let $(\varphi_j)_{j>0}$ be a quasi-equisingular approximation of φ in $\mathrm{PSH}(X,\omega)_{>0}$. We may assume that $\varphi_j \leq 0$ for all $j \geq 1$.

Then

$$\varphi_j|_{\tilde{Y}} \leq P_{\omega|_{\tilde{Y}}}\left[\varphi_j|_{\tilde{Y}}\right] \tag{8.3}$$

for all $j \geq 1$. In particular,

$$\varphi|_{\tilde{Y}} \leq \inf_{j\geq 1} P_{\omega|_{\tilde{Y}}}\left[\varphi_j|_{\tilde{Y}}\right].$$

Thanks to Corollary 6.2.5,

$$\mathrm{Tr}_Y(\varphi) \sim_P \inf_{j\geq 1} P_{\omega|_{\tilde{Y}}}[\varphi_j|_{\tilde{Y}}]. \tag{8.4}$$

We conclude our assertion. □

Example 8.1.4 Let $\varphi \in \mathrm{QPSH}(X)$ such that $\nu(\varphi,Y)=0$. Assume that φ has analytic singularities. Then

$$\mathrm{Tr}_Y(\varphi) \sim_P \varphi|_{\tilde{Y}}.$$

Example 8.1.5 Let $\varphi \in \mathrm{QPSH}(X)$. Take a closed real smooth $(1,1)$-form θ on X such that $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$, then

$$\mathrm{Tr}_X(\varphi) \sim_P P_\theta[\varphi]_{\mathcal{I}}, \quad \mathrm{Tr}_X^{\theta}(\varphi) = P_\theta[\varphi]_{\mathcal{I}}.$$

In particular, the trace operator can be regarded as a generalization of the $\mathcal{I}$-envelope.

Example 8.1.6 Assume that $\varphi \in \mathrm{PSH}(X,\theta)$ for some closed smooth real $(1,1)$-form θ on X, $\nu(\varphi,Y)=0$ and

$$\lim_{\epsilon\searrow 0}\int_{\tilde{Y}}\left(\theta|_{\tilde{Y}}+\epsilon\omega|_{\tilde{Y}}+\mathrm{dd^c}\,\mathrm{Tr}_Y^{\theta+\epsilon\omega}(\varphi)\right)^{\dim Y} > 0 \tag{8.5}$$

for any arbitrary choice of a Kähler form ω on X. Then it follows from Proposition 3.1.12 that $\mathrm{Tr}_Y^{\theta}(\varphi)$ is defined, and its mass is exactly the above limit.

As a consequence, we have the following formula:

$$\int_{\tilde{Y}} \left(\theta|_{\tilde{Y}} + \mathrm{dd^c}\,\mathrm{Tr}_Y^{\theta}(\varphi)\right)^{\dim Y} = \lim_{\epsilon \searrow 0} \int_{\tilde{Y}} \left(\theta|_{\tilde{Y}} + \epsilon\omega|_{\tilde{Y}} + \mathrm{dd^c}\,\mathrm{Tr}_Y^{\theta+\epsilon\omega}(\varphi)\right), \tag{8.6}$$

where the left-hand side is understood as 0 if $\mathrm{Tr}_Y^{\theta}(\varphi)$ is not defined.

Remark 8.1.2 The trace operator allows us to introduce the following extension of the moving Seshadri constant: Let $T \in \mathcal{Z}_+(X, \alpha)$ and $x \in X$, we define

$$\epsilon(T, x) := \inf_{V \ni x} \left(\frac{\mathrm{vol}\,\mathrm{Tr}_V^{\alpha|_{\tilde{V}}} T}{\mathrm{mult}_x V}\right)^{\frac{1}{\dim V}},$$

where $\mathrm{vol}\,\mathrm{Tr}_V^{\alpha|_{\tilde{V}}} T = 0$ if $\mathrm{Tr}_V^{\alpha|_{\tilde{V}}} T$ is not defined. Here V runs over all positive-dimensional closed irreducible analytic subsets of X containing x.

These moving Seshadri constants seem to be new. But since I do not have particularly good applications in mind, I will not study these objects in this book.

8.2 Properties of the trace operator

Let X be a connected compact Kähler manifold and $Y \subseteq X$ be an irreducible analytic subset.

We prove a few elementary properties of the trace operator.

Proposition 8.2.1 *Let $\varphi, \psi \in \mathrm{QPSH}(X)$, $\lambda > 0$. Assume that $\nu(\varphi, Y) = \nu(\psi, Y) = 0$. Then we have the following:*

(1) *Suppose that $\varphi \preceq_{\mathcal{I}} \psi$. Then $\mathrm{Tr}_Y(\varphi) \preceq_P \mathrm{Tr}_Y(\psi)$.*

(2) *We have*
$$\mathrm{Tr}_Y(\varphi + \psi) \sim_P \mathrm{Tr}_Y(\varphi) + \mathrm{Tr}_Y(\psi).$$

(3) *We have*
$$\mathrm{Tr}_Y(\lambda\varphi) \sim_P \lambda\,\mathrm{Tr}_Y(\varphi).$$

(4) *We have*
$$\mathrm{Tr}_Y(\varphi \vee \psi) \sim_P \mathrm{Tr}_Y(\varphi) \vee \mathrm{Tr}_Y(\psi).$$

As a consequence of (1), the P-equivalence class of $\mathrm{Tr}_Y(\varphi)$ depends only on the $\mathcal{I}$-equivalence class of φ.

Proof Take a closed smooth real $(1,1)$-form θ on X such that $\theta_\varphi, \theta_\psi$ are both Kähler currents. Let $(\varphi_j)_j$ and $(\psi_j)_j$ be quasi-equisingular approximations of φ and ψ in $\mathrm{PSH}(X, \theta)$ respectively. We may assume that $\varphi_j \leq 0$ and $\psi_j \leq 0$ for all $j \geq 1$.

(1) We choose the quasi-equisingular approximations compatibly. By Theorem 1.6.2, we may take φ_j and ψ_j with analytic singularities of types

$$\left(2^{-j}, \mathcal{I}(2^j\varphi)\right), \quad \left(2^{-j}, \mathcal{I}(2^j\psi)\right),$$

respectively. Since $\varphi \preceq_{\mathcal{I}} \psi$, we have

$$\mathcal{I}(2^j\varphi) \subseteq \mathcal{I}(2^j\psi)$$

for every j. Hence $\varphi_j \preceq \psi_j$. After changing φ_j by a constant, which does not affect its d_S-limit or its trace P-type, we may assume that

$$\varphi_j \leq \psi_j$$

for all j. Restricting to $\tilde{Y}$ and passing to the d_S-limits by Proposition 6.2.5, we obtain

$$\mathrm{Tr}_Y(\varphi) \preceq_P \mathrm{Tr}_Y(\psi).$$

(2) It follows from Theorem 6.2.2 that $\varphi_j + \psi_j \xrightarrow{d_S} P_\theta[\varphi]_{\mathcal{I}} + P_\theta[\psi]_{\mathcal{I}}$. However, by Proposition 3.2.11 and Proposition 7.2.1, we have

$$P_\theta[\varphi]_{\mathcal{I}} + P_\theta[\psi]_{\mathcal{I}} \sim_P P_{2\theta}[\varphi+\psi]_{\mathcal{I}}.$$

Therefore, by Proposition 6.2.2, Corollary 7.1.2 and Proposition 1.6.1, $(\varphi_j + \psi_j)_j$ is a quasi-equisingular approximation of $\varphi+\psi$. We conclude using Theorem 6.2.2.

(3) Let $(\lambda_j)_j$ be an increasing sequence of positive rational numbers with limit λ. Then $(\lambda_j\varphi_j)_j$ is a quasi-equisingular approximation of $\lambda\varphi$. Since $\lambda_j/\lambda \to 1$, Lemma 6.2.3 gives

$$d_S\left((\lambda_j/\lambda)\varphi_j|_{\tilde{Y}}, \varphi_j|_{\tilde{Y}}\right) \to 0.$$

After scaling the background class, Lemma 6.2.6 shows that $\lambda_j\varphi_j|_{\tilde{Y}}$ and $\lambda\varphi_j|_{\tilde{Y}}$ have the same d_S-limit. The assertion follows.

(4) By Proposition 6.2.4, we have

$$\varphi_j \vee \psi_j \xrightarrow{d_S} P_\theta[\varphi]_{\mathcal{I}} \vee P_\theta[\psi]_{\mathcal{I}}.$$

By Proposition 3.2.11 and Proposition 7.2.1, we have

$$P_\theta[\varphi]_{\mathcal{I}} \vee P_\theta[\psi]_{\mathcal{I}} \sim_P P_\theta[\varphi \vee \psi]_{\mathcal{I}}.$$

Therefore, our assertion follows exactly as in the proof of (2). □

The trace operator is continuous along d_S-convergent decreasing sequences.

Proposition 8.2.2 *Let $(\varphi_j)_{j\in I}$ be a decreasing net in* $\mathrm{QPSH}(X)$. *Assume that there exists a closed real smooth $(1,1)$-form θ such that $\varphi_j \in \mathrm{PSH}(X,\theta)$ for each $j \in I$. Assume that $\varphi_j \xrightarrow{d_S} \varphi \in \mathrm{QPSH}(X)$ and $\nu(\varphi, Y) = 0$. Then*

$$\mathrm{Tr}_Y(\varphi_j) \xrightarrow{d_S} \mathrm{Tr}_Y(\varphi).$$

In view of Corollary 7.1.2, the trace operator preserves the property of being a quasi-equisingular approximation, thereby solving the problem in Example 8.1.1.

Proof Since d_S is a pseudometric, it suffices to prove the sequential statement. Thus we assume that $(\varphi_j)_{j\geq 1}$ is a decreasing sequence and that

$$\varphi_j \xrightarrow{d_S} \varphi.$$

Take a Kähler form ω on X so that $\varphi_j, \varphi \in \mathrm{PSH}(X,\omega)_{>0}$. Set

$$\chi \coloneqq \inf_{j\geq 1} \varphi_j.$$

Then Corollary 6.2.5 gives

$$\varphi_j \xrightarrow{d_S} \chi.$$

Hence $d_S(\chi,\varphi) = 0$. By Proposition 6.2.2, we have $\chi \sim_P \varphi$, hence $\chi \sim_{\mathcal{I}} \varphi$ by Corollary 6.1.1. It follows from Proposition 8.2.1(1) that

$$\mathrm{Tr}_Y(\chi) \sim_P \mathrm{Tr}_Y(\varphi).$$

Therefore it suffices to prove that

$$\mathrm{Tr}_Y(\varphi_j) \xrightarrow{d_S} \mathrm{Tr}_Y(\chi).$$

Replacing φ by χ, we may assume from now on that

$$\varphi = \inf_{j\geq 1} \varphi_j.$$

After adding a Kähler form to ω, we may assume that ω_φ is a Kähler current.

For each $j \geq 1$, choose a quasi-equisingular approximation $(\varphi_j^k)_{k\geq 1}$ of φ_j in $\mathrm{PSH}(X,\omega)$. Using the standard construction in Theorem 1.6.2, we may choose these approximations compatibly in j, so that for each fixed k,

$$\varphi_{j+1}^k \leq \varphi_j^k$$

We shall choose an increasing sequence of integers k_j and put

$$\eta_j \coloneqq \varphi_j^{k_j}.$$

The sequence $(k_j)_j$ is chosen inductively: Choose $k_1 > 1$, for $j > 1$, we choose k_j large enough so that

(1) $k_j \geq k_{j-1}$;
(2) $d_{S,\omega}(\eta_j, P_\omega[\varphi_j]_{\mathcal{I}}) \leq 1/(j-1)$;
(3) $d_{S,\omega|_{\tilde{Y}}}(\eta_j|_{\tilde{Y}}, \mathrm{Tr}_Y^\omega(\varphi_j)) < 1/(j-1)$.

It follows from (1) that $(\eta_j)_j$ is decreasing in singularity types. By Theorem 6.2.3, we have

$$P_\omega[\varphi_j]_{\mathcal{I}} \xrightarrow{d_S} P_\omega[\varphi]_{\mathcal{I}}.$$

By (2) and Corollary 7.1.2, we have $\eta_j \xrightarrow{d_S} P_\omega[\varphi]_{\mathcal{I}}$. Observe that by Proposition 6.2.6,

$$P_\omega[\varphi]_{\mathcal{I}} = \inf_{j>0} P_\omega\left[\eta_j\right].$$

By Proposition 3.2.10, for each $j > 0$, η_j has analytic singularities. By Theorem 6.2.6, $(P_\omega[\eta_j])_j$ is a quasi-equisingular approximation of $P_\omega[\varphi]_{\mathcal{I}}$. Therefore, by definition of the trace operator and Proposition 8.2.1(1),

$$\eta_j|_{\tilde{Y}} \xrightarrow{d_S} \mathrm{Tr}_Y(\varphi).$$

It follows from (3) that

$$\mathrm{Tr}_Y(\varphi_j) \xrightarrow{d_S} \mathrm{Tr}_Y(\varphi).$$

We conclude the proof. □

Example 8.2.1 The trace operator is not continuous along increasing sequences. Let us consider the case $X = \mathbb{P}^2$ with coordinates (z_1, z_2) on $\mathbb{C}^2 \subseteq X$. Let ω_{FS} denote the Fubini–Study metric. The subvariety $Y \cong \mathbb{P}^1$ is defined by $z_2 = 0$. Consider an increasing sequence $(\varphi_j)_j$ in $\mathrm{PSH}(X, \omega_{\mathrm{FS}})$, whose potentials near $(0,0)$ are given by

$$\log|z_1|^2 \vee \left(j^{-1}\log|z_2|^2\right) + O(1).$$

The pointwise restriction of these potentials to Y are given locally by

$$\log|z_1|^2 + O(1).$$

On the other hand, locally

$$\log|z_1|^2 \vee \left(j^{-1}\log|z_2|^2\right) \to 0$$

almost everywhere as $j \to \infty$. So the trace operator is not continuous along the sequence $(\varphi_j)_j$.

Lemma 8.2.1 *Let $\pi\colon Z \to X$ be a proper bimeromorphic morphism, where Z is a connected Kähler manifold. Assume that W (resp. Y) is an irreducible analytic subset of Z (resp. X) such that the restriction $\Pi\colon W \to Y$ of π is defined and is bimeromorphic, so that we have the following commutative diagram*

$$\begin{tikzcd}
\tilde{W} \arrow[r] \arrow[d, "\tilde{\Pi}"] & W \arrow[r, hook] \arrow[d, "\Pi"] & Z \arrow[d, "\pi"] \\
\tilde{Y} \arrow[r] & Y \arrow[r, hook] & X.
\end{tikzcd}$$

Then for any $\varphi \in \mathrm{QPSH}(X)$ with $\nu(\varphi, Y) = 0$, we have

$$\tilde{\Pi}^* \operatorname{Tr}_Y(\varphi) \sim_P \operatorname{Tr}_W(\pi^*\varphi). \tag{8.7}$$

Proof We first observe that by Zariski's main theorem, $\nu(\pi^*\varphi, W) = 0$. So the right-hand side of (8.7) makes sense.

Step 1. Assume that φ has analytic singularities. It suffices to apply Example 8.1.4 to reformulate (8.7) as

$$\tilde{\Pi}^*(\varphi|_{\tilde{Y}}) \sim_P (\pi^*\varphi)|_{\tilde{W}}.$$

In fact, the strict equality holds, which is nothing but the functoriality of pullbacks.

Step 2. Next we handle the general case. Choose a smooth closed real $(1,1)$-form θ such that θ_φ is a Kähler current. Take a quasi-equisingular approximation $(\varphi_j)_j$ of φ in $\operatorname{PSH}(X,\theta)$. By Corollary 7.1.2, $(\pi^*\varphi_j)_j$ is a quasi-equisingular approximation of $\pi^*\varphi$. From Step 1, we know that for each j,

$$\tilde{\Pi}^* \operatorname{Tr}_Y(\varphi_j) \sim_P \operatorname{Tr}_W(\pi^*\varphi_j).$$

Letting $j \to \infty$, the left-hand side converges to $\tilde{\Pi}^* \operatorname{Tr}_Y(\varphi)$ by the definition of the trace operator, while the right-hand side converges to $\operatorname{Tr}_W(\pi^*\varphi)$, since $(\pi^*\varphi_j)_j$ is a quasi-equisingular approximation of $\pi^*\varphi$. This proves (8.7). □

Proposition 8.2.3 *Let $\varphi \in \operatorname{QPSH}(X)$ with $\nu(\varphi, Y) = 0$. Assume that Y is smooth. Then for any $\lambda > 0$, we have*

$$\mathcal{I}(\lambda \operatorname{Tr}_Y(\varphi)) \subseteq \operatorname{Res}_Y \mathcal{I}(\lambda\varphi). \tag{8.8}$$

See Definition 1.4.5 for the definition of Res_Y.

Proof Take a Kähler form ω on X such that ω_φ is a Kähler current.

Let $(\varphi_j)_j$ be a quasi-equisingular approximation of φ in $\operatorname{PSH}(X,\omega)$.

For each $j \geq 1$, we have

$$\operatorname{Tr}_Y(\varphi) \preceq_P \varphi_j|_Y.$$

Indeed, $\operatorname{Tr}_Y(\varphi)$ is P-equivalent to the decreasing d_S-limit of the restrictions $\varphi_j|_Y$, hence it is more singular than each $\varphi_j|_Y$.

For any $\lambda' > \lambda > 0$, we can find $j > 0$ so that

$$\mathcal{I}(\lambda'\varphi_j) \subseteq \mathcal{I}(\lambda\varphi).$$

By Corollary 6.1.1 and Theorem 1.4.5, we have

$$\mathcal{I}(\lambda' \operatorname{Tr}_Y(\varphi)) \subseteq \mathcal{I}(\lambda'\varphi_j|_Y) \subseteq \operatorname{Res}_Y \mathcal{I}(\lambda'\varphi_j) \subseteq \operatorname{Res}_Y \mathcal{I}(\lambda\varphi).$$

Thanks to Theorem 1.4.4, we conclude (8.8). □

Lastly, we turn our attention to global sections. For this we will need the following global Ohsawa–Takegoshi extension theorem for the trace operator:

Theorem 8.2.1 *Let L be a big line bundle on X and let θ be a closed real smooth $(1,1)$-form on X representing $c_1(L)$. Suppose that $\varphi \in \operatorname{PSH}(X,\theta)$ and θ_φ is a Kähler current.*

Assume that Y is smooth and $\nu(\varphi, Y) = 0$. Let T be a holomorphic line bundle on X. Then there exists k_0 such that for all $k \geq k_0$ and $s \in \mathrm{H}^0(Y, T|_Y \otimes L|_Y^k \otimes \mathcal{I}(k\,\mathrm{Tr}_Y^\theta(\varphi)))$, there exists an extension $\tilde{s} \in \mathrm{H}^0(X, T \otimes L^k \otimes \mathcal{I}(k\varphi))$.

It is of interest to know whether one can control the L^2-norm of $\tilde{s}$ in the above result.

Proof Fix a Kähler form ω on X and a Hermitian metric h_T on T. We may assume that $Y \neq X$ and that $\theta_\varphi \geq 3\delta\omega$ for some $\delta > 0$. Let $(\varphi_j)_j$ be the decreasing quasi-equisingular approximation of φ in $\mathrm{PSH}(X, \theta)$. We can assume that $\theta_{\varphi_j} \geq 2\delta\omega$ for all $j \geq 1$. Also, there exists $\epsilon_0 > 0$ such that $\theta_{(1+\epsilon)\varphi_j} \geq \delta\omega$ for any $\epsilon \in (0, \epsilon_0)$. Take $k_0 = k_0(\delta, h_T)$ as in Theorem 1.8.1.

We fix $k \geq k_0$ and $s \in \mathrm{H}^0(Y, T|_Y \otimes L|_Y^k \otimes \mathcal{I}(k\,\mathrm{Tr}_Y^\theta(\varphi)))$. By Theorem 1.4.4, there exists $\epsilon \in (0, \epsilon_0)$ such that $s \in \mathrm{H}^0(Y, T|_Y \otimes L|_Y^k \otimes \mathcal{I}(k(1+\epsilon)\,\mathrm{Tr}_Y^\theta(\varphi)))$.

Since $\mathrm{Tr}_Y^\theta(\varphi) \preceq_P \varphi_j|_Y$, Corollary 6.1.1 implies that $s \in \mathrm{H}^0(Y, T|_Y \otimes L|_Y^k \otimes \mathcal{I}(k(1+\epsilon)\varphi_j|_Y))$. By Theorem 1.8.1, there exists $\tilde{s}_j \in \mathrm{H}^0(X, T \otimes L^k \otimes \mathcal{I}(k(1+\epsilon)\varphi_j))$ such that $\tilde{s}_j|_Y = s$, for all j.

But by definition of quasi-equisingular approximation, we obtain that for high enough j the inclusion $\mathcal{I}(k(1+\epsilon)\varphi_j) \subseteq \mathcal{I}(k\varphi)$ holds. As a result, $\tilde{s}_j \in \mathrm{H}^0(X, T \otimes L^k \otimes \mathcal{I}(k\varphi))$ for high enough j, finishing the argument. □

8.3 Relation to the classical restricted volumes

Let X be a connected compact Kähler manifold of dimension n and Y be a connected submanifold of dimension m. Fix a big class $\alpha \in \mathrm{H}^{1,1}(X, \mathbb{R})$. Take a closed smooth real $(1,1)$-form $\theta \in \alpha$. Fix a Kähler form ω on X.

Recall that the notions of non-Kähler locus and non-nef locus was defined in Definition 1.7.6 and Definition 1.7.7.

When $Y \not\subseteq \mathrm{nK}(\alpha)$, Matsumura ([Mat13, Definition 1.4]) defines the restricted volume of $\{\theta\}$ to Y in the following manner:

$$\mathrm{vol}_{X|Y}(\alpha) \coloneqq \sup_{\varphi} \int_Y (\theta|_Y + \mathrm{dd^c}\varphi|_Y)^m\,, \tag{8.9}$$

where φ runs over elements in $\mathrm{PSH}(X, \theta)$ with analytic singularities such that $\varphi|_Y \not\equiv -\infty$. This definition is independent of the choice of θ.

In case $Y \not\subseteq \mathrm{nn}(\alpha)$, Collins–Tosatti [CT22] extend the above definition of restricted volume:

$$\mathrm{vol}_{X|Y}(\alpha) \coloneqq \lim_{\epsilon \searrow 0} \sup_{\varphi} \int_Y (\theta|_Y + \epsilon\omega|_Y + \mathrm{dd^c}\varphi|_Y)^m\,. \tag{8.10}$$

where φ runs over elements in $\mathrm{PSH}(X, \theta + \epsilon\omega)$ with analytic singularities such that $\varphi|_Y \not\equiv -\infty$. This definition is independent of the choice of θ.

These definitions extend the more classical definition in the algebraic setting due to [ELM+09].

Proposition 8.3.1 *Assume that* $Y \not\subseteq \mathrm{nK}(\alpha)$. *Then*

$$\mathrm{vol}_{X|Y}(\alpha) = \int_Y \left(\theta|_Y + \mathrm{dd^c}\,\mathrm{Tr}_Y^\theta(V_\theta)\right)^m = \int_Y (\theta|_Y + \mathrm{dd^c} V_\theta|_Y)^m. \tag{8.11}$$

Proof We start with the first equality of (8.11). Since $Y \not\subseteq \mathrm{nK}(\alpha)$, $V_\theta|_Y \not\equiv -\infty$ as a consequence of Lemma 2.4.1, hence also $\nu(V_\theta, Y) = 0$.

Take a quasi-equisingular approximation $(\varphi_j)_{j>0}$ of V_θ with $\varphi_j \in \mathrm{PSH}(X, \theta+\epsilon_j\omega)$. By Theorem 2.4.3, we have

$$\int_Y (\theta|_Y + \epsilon_j\omega|_Y + \mathrm{dd^c}\varphi_j|_Y)^m \geq \int_Y (\theta|_Y + \mathrm{dd^c}\chi|_Y)^m$$

for every admissible analytic-singularity potential χ in (8.9), after adding a constant to arrange $\chi \leq V_\theta$. Letting $j \to \infty$ and applying Example 8.1.6, we conclude that the $\geq$ direction in the first equality of (8.11) holds.

For the reverse direction, by definition, for any fixed $\epsilon > 0$, we have

$$\int_Y (\theta|_Y + \epsilon_j\omega|_Y + \mathrm{dd^c}\varphi_j|_Y)^m \leq \int_Y (\theta|_Y + \epsilon\omega|_Y + \mathrm{dd^c}\varphi_j|_Y)^m \leq \mathrm{vol}_{X|Y}(\{\theta\} + \epsilon\{\omega\})$$

for all large enough j. Letting $j \to \infty$ and $\epsilon \searrow 0$, using the continuity of $\mathrm{vol}_{X|Y}$ ([Mat13, Corollary 4.11]) together with Example 8.1.6, we conclude the first equality of (8.11).

Now we address the second equality. Due to Theorem 2.4.3, the defining formula (8.9), and the definition of V_θ, we obtain that

$$\mathrm{vol}_{X|Y}(\{\theta\}) \leq \int_Y (\theta|_Y + \mathrm{dd^c} V_\theta|_Y)^m.$$

The reverse equality now follows from the first equality of (8.11), Theorem 2.4.3 and the fact that $V_\theta|_Y \preceq_P \mathrm{Tr}_Y^\theta(V_\theta)$ as proved in Proposition 8.1.3. □

Theorem 8.3.1 *If* $Y \not\subseteq \mathrm{nn}(\alpha)$, *then*

$$\begin{aligned}
\mathrm{vol}_{X|Y}(\alpha) &= \lim_{\epsilon\to 0+} \int_Y (\theta|_Y + \epsilon\omega|_Y + \mathrm{dd^c} V_{\theta+\epsilon\omega}|_Y)^m \\
&= \lim_{\epsilon\to 0+} \int_Y \left(\theta|_Y + \epsilon\omega|_Y + \mathrm{dd^c}\,\mathrm{Tr}_Y^{\theta+\epsilon\omega}(V_{\theta+\epsilon\omega})\right)^m \\
&= \int_Y \left(\theta|_Y + \mathrm{dd^c}\,\mathrm{Tr}_Y^\theta(V_\theta)\right)^m,
\end{aligned} \tag{8.12}$$

where the last term is understood as 0 *if* $\mathrm{Tr}_Y^\theta(V_\theta)$ *is not defined.*

Proof Since $\nu(V_\theta, Y) = 0$, we have $Y \not\subseteq \mathrm{nK}(\alpha + \epsilon\{\omega\})$ for all $\epsilon > 0$. As a result, due to (8.10) and (8.11) only the last equality of (8.12) needs to be argued.

Thanks to Lemma 6.2.7, we have

$$V_{\theta+\epsilon\omega} \xrightarrow{d_S} V_\theta$$

as $\epsilon \to 0+$.

Therefore, using Proposition 8.2.2, we find

$$\operatorname{Tr}_Y\left(V_{\theta+\epsilon\omega}\right) \xrightarrow{d_S} \operatorname{Tr}_Y\left(V_\theta\right).$$

Therefore, thanks to Theorem 6.2.1, for any $\epsilon_0 > 0$, we have

$$\begin{aligned}\lim_{\epsilon\to 0+}\int_Y (\theta|_Y + \epsilon_0\omega|_Y)^m_{\operatorname{Tr}_Y^{\theta+\epsilon_0\omega}(V_{\theta+\epsilon\omega})} &= \int_Y (\theta|_Y + \epsilon_0\omega|_Y)^m_{\operatorname{Tr}_Y^{\theta+\epsilon_0\omega}(V_\theta)}\\ &\geq \int_Y (\theta|_Y)^m_{\operatorname{Tr}_Y^{\theta}(V_\theta)}.\end{aligned}$$

On the other hand,

$$\begin{aligned}\lim_{\epsilon_0\to 0+}\int_Y (\theta|_Y + \epsilon_0\omega|_Y)^m_{\operatorname{Tr}_Y^{\theta+\epsilon_0\omega}(V_\theta)} &= \lim_{\epsilon_0\to 0+}\int_Y (\theta|_Y + \epsilon_0\omega|_Y)^m_{\operatorname{Tr}_Y^{\theta}(V_\theta)}\\ &= \int_Y (\theta|_Y)^m_{\operatorname{Tr}_Y^{\theta}(V_\theta)}.\end{aligned}$$

These two equations together imply the existence of $(\epsilon_j)_{j>0}$ such that $0 < \epsilon_j < 1/j$ and

$$\lim_{j\to\infty}\int_Y \left(\theta|_Y + j^{-1}\omega|_Y\right)^m_{\operatorname{Tr}_Y^{\theta+j^{-1}\omega}(V_{\theta+\epsilon_j\omega})} = \int_Y (\theta|_Y)^m_{\operatorname{Tr}_Y^{\theta}(V_\theta)}. \tag{8.13}$$

Moreover, for each $j > 0$,

$$\begin{aligned}\int_Y \left(\theta|_Y + j^{-1}\omega|_Y\right)^m_{\operatorname{Tr}_Y^{\theta+j^{-1}\omega}(V_{\theta+\epsilon_j\omega})} &\geq \int_Y (\theta|_Y + \epsilon_j\omega|_Y)^m_{\operatorname{Tr}_Y^{\theta+\epsilon_j\omega}(V_{\theta+\epsilon_j\omega})}\\ &\geq \int_Y (\theta|_Y)^m_{\operatorname{Tr}_Y^{\theta}(V_\theta)}.\end{aligned} \tag{8.14}$$

Putting (8.13) and (8.14) together, it results that

$$\lim_{j\to\infty}\int_Y \left(\theta|_Y + \epsilon_j\omega|_Y\right)^m_{\operatorname{Tr}_Y^{\theta+\epsilon_j\omega}(V_{\theta+\epsilon_j\omega})} = \int_Y (\theta|_Y)^m_{\operatorname{Tr}_Y^{\theta}(V_\theta)}.$$

Finally, since $\int_Y \left(\theta|_Y + \epsilon\omega|_Y + \mathrm{dd^c}\operatorname{Tr}_Y^{\theta+\epsilon\omega}(V_{\theta+\epsilon\omega})\right)^m$ depends monotonically on $\epsilon > 0$, we conclude that

$$\lim_{\epsilon\to 0+}\int_Y \left(\theta|_Y + \epsilon\omega|_Y + \mathrm{dd^c}\operatorname{Tr}_Y^{\theta+\epsilon\omega}(V_{\theta+\epsilon\omega})\right)^m = \int_Y (\theta|_Y)^m_{\operatorname{Tr}_Y^{\theta}(V_\theta)}.$$

This completes the proof. □

8.4 Restricted volumes of line bundles

Let X be a connected projective manifold of dimension n and $Y \subseteq X$ be a connected submanifold of dimension m. Consider a big line bundle L on X, a Hermitian metric h_0 on L with $\theta = c_1(L, h_0)$. Let A be a very ample line bundle on X. Take a Hermitian metric h_A on A such that $\omega = \mathrm{dd^c} h_A$ is a Kähler form.

Using the trace operator, one could prove the following generalization of Theorem 7.4.1.

Theorem 8.4.1 *Let h be a singular plurisubharmonic metric on L with $\nu(\mathrm{dd^c} h, Y) = 0$. Assume that*

$$\lim_{\epsilon\to 0+} \int_Y \left(\mathrm{Tr}_Y^{c_1(L|_Y)+\epsilon\,\omega|_Y}\left(c_1(L,h)\right)\right)^m > 0. \tag{8.15}$$

Then for any holomorphic line bundle T on X we have

$$\int_Y \left(\mathrm{Tr}_Y^{c_1(L|_Y)}\left(c_1(L,h)\right)\right)^m = \lim_{k\to\infty} \frac{m!}{k^m} h^0\left(Y, T|_Y \otimes L|_Y^k \otimes \mathrm{Res}_Y\left(\mathcal{I}(h^k)\right)\right). \tag{8.16}$$

Recall that Res_Y is defined in Definition 1.4.5. Observe that by Example 8.1.6, (8.15) implies that $\mathrm{Tr}_Y^{c_1(L|_Y)}(c_1(L,h))$ is defined. So (8.16) is defined.

We will identify h with $\varphi \in \mathrm{PSH}(X, \theta)$ as in (1.27).

We only need to consider the case $Y \neq X$, since otherwise, the result is proved in Theorem 7.4.1. We will always assume $Y \neq X$ in the sequel.

Lemma 8.4.1 *Assume that φ has analytic singularities and θ_φ is a Kähler current. Suppose that $\varphi|_Y \not\equiv -\infty$. Then*

$$\int_Y (\theta|_Y + \mathrm{dd^c}\varphi|_Y)^m = \lim_{k\to\infty} \frac{m!}{k^m} \dim_{\mathbb{C}} \left\{ s|_Y : s \in \mathrm{H}^0(X, T\otimes L^k \otimes \mathcal{I}(k\varphi))\right\}. \tag{8.17}$$

Recall that $\mathcal{I}_\infty$ is defined in Definition 1.6.6.

Proof Suppose that $\epsilon \in (0,1)$ is small enough so that $(1-\epsilon)\varphi \in \mathrm{PSH}(X,\theta)$.

Using Theorem 7.4.1 we can start to write the following sequence of inequalities:

$$
\begin{aligned}
&\frac{1}{m!}\int_Y (\theta|_Y + \mathrm{dd^c}\varphi|_Y)^m \\
=& \lim_{k\to\infty} \frac{1}{k^m} h^0\left(Y, T|_Y \otimes L|_Y^k \otimes \mathcal{I}(k\varphi|_Y)\right)\\
\leq& \varliminf_{k\to\infty} \frac{1}{k^m} \dim\left\{ s|_Y : s\in \mathrm{H}^0\left(X, T\otimes L^k \otimes \mathcal{I}(k\varphi)\right)\right\} \quad \text{by Theorem 1.8.1}\\
\leq& \varlimsup_{k\to\infty} \frac{1}{k^m} \dim\left\{ s|_Y : s\in \mathrm{H}^0\left(X, T\otimes L^k \otimes \mathcal{I}(k\varphi)\right)\right\}\\
\leq& \varlimsup_{k\to\infty} \frac{1}{k^m} \dim\left\{ s|_Y : s\in \mathrm{H}^0\left(X, T\otimes L^k \otimes \mathcal{I}_\infty\left((1-\epsilon)k\varphi\right)\right)\right\} \quad \text{by Lemma 1.6.3}\\
\leq& \varlimsup_{k\to\infty} \frac{1}{k^m} h^0\left(Y, T|_Y \otimes L|_Y^k \otimes \mathcal{I}\left((1-\epsilon)k\varphi|_Y\right)\right)\\
=&\frac{1}{m!}\int_Y (\theta|_Y + (1-\epsilon)\mathrm{dd^c}\varphi|_Y)^m \quad \text{by Theorem 7.4.1.}
\end{aligned}
$$

Letting $\epsilon \to 0$, (8.17) follows from multi-linearity of the non-pluripolar product Proposition 2.2.1(6). □

Proposition 8.4.1 *In the setting of Theorem 8.4.1, assume that* $\mathrm{dd^c}h$ *is a Kähler current. Then* (8.16) *holds.*

Proof [3] Let $(\varphi_j)_j$ be a quasi-equisingular approximation of φ in $\mathrm{PSH}(X,\theta)$. After possibly replacing $(\varphi_j)_j$ by a subsequence, there exists $\epsilon_0 \in (0,1)\cap\mathbb{Q}$ such that the $\theta_{(1-\epsilon)\varphi_j}$'s are also Kähler currents for any $\epsilon\in(0,\epsilon_0)$.

We have the following inequalities:

[3] The original proof in [DX24a] is flawed. In general, the right-most term of (34) is only supported on Y, not necessarily eliminated by $\mathcal{I}_Y$.

$$
\begin{aligned}
&\frac{1}{m!}\int_Y \left(\theta|_Y + \mathrm{dd^c}\,\mathrm{Tr}_Y^\theta(\varphi)\right)^m \\
&= \lim_{k\to\infty} \frac{1}{k^m} h^0\left(Y, T|_Y \otimes L|_Y^k \otimes \mathcal{I}\left(k\,\mathrm{Tr}_Y^\theta(\varphi)\right)\right) \quad \text{by Theorem 7.4.1} \\
&\leq \varliminf_{k\to\infty} \frac{1}{k^m} h^0\left(Y, T|_Y \otimes L|_Y^k \otimes \mathrm{Res}_Y\left(\mathcal{I}(k\varphi)\right)\right) \quad \text{by Proposition 8.2.3} \\
&\leq \varlimsup_{k\to\infty} \frac{1}{k^m} h^0\left(Y, T|_Y \otimes L|_Y^k \otimes \mathrm{Res}_Y\left(\mathcal{I}(k\varphi)\right)\right) \\
&\leq \varlimsup_{k\to\infty} \frac{1}{k^m} h^0\left(Y, T|_Y \otimes L|_Y^k \otimes \mathrm{Res}_Y\,\mathcal{I}\left(k\varphi_j\right)\right) \\
&\leq \varlimsup_{k\to\infty} \frac{1}{k^m} h^0\left(Y, T|_Y \otimes L|_Y^k \otimes \mathrm{Res}_Y\,\mathcal{I}_\infty\left((1-\epsilon)k\varphi_j\right)\right) \quad \text{by Lemma 1.6.3} \\
&\leq \varlimsup_{k\to\infty} \frac{1}{k^m} h^0\left(Y, T|_Y \otimes L|_Y^k \otimes \mathcal{I}_\infty\left((1-\epsilon)k\varphi_j|_Y\right)\right) \\
&\leq \varlimsup_{k\to\infty} \frac{1}{k^m} h^0\left(Y, T|_Y \otimes L|_Y^k \otimes \mathcal{I}\left((1-\epsilon)k\varphi_j|_Y\right)\right) \\
&= \frac{1}{m!}\int_Y \left(\theta|_Y + (1-\epsilon)\mathrm{dd^c}\varphi_j|_Y\right)^m \quad \text{by Theorem 7.4.1.}
\end{aligned}
$$

Here we used the observation that for any $\psi \in \mathrm{QPSH}(X)$ with analytic singularities with $\psi|_Y \not\equiv -\infty$, we have

$$\mathrm{Res}_Y\,\mathcal{I}_\infty(\psi) \subseteq \mathcal{I}_\infty\left(\psi|_Y\right).$$

Letting $\epsilon \to 0+$, we conclude that

$$
\begin{aligned}
\frac{1}{m!}\int_Y \left(\theta|_Y + \mathrm{dd^c}\,\mathrm{Tr}_Y^\theta(\varphi)\right)^m &\leq \varliminf_{k\to\infty} \frac{1}{k^m} h^0\left(Y, T|_Y \otimes L|_Y^k \otimes \mathrm{Res}_Y\left(\mathcal{I}(k\varphi)\right)\right) \\
&\leq \varlimsup_{k\to\infty} \frac{1}{k^m} h^0\left(Y, T|_Y \otimes L|_Y^k \otimes \mathrm{Res}_Y\left(\mathcal{I}(k\varphi)\right)\right) \\
&\leq \frac{1}{m!}\int_Y \left(\theta|_Y + \mathrm{dd^c}\varphi_j|_Y\right)^m.
\end{aligned}
$$

Letting $j \to \infty$, we conclude the proof. □

Proof (Proof of Theorem 8.4.1) Using Proposition 8.2.3 and Theorem 7.4.1 we obtain that

$$
\begin{aligned}
\int_Y \left(\theta|_Y + \mathrm{dd^c}\,\mathrm{Tr}_Y^\theta(\varphi)\right)^m &= \lim_{k\to\infty} \frac{m!}{k^m} h^0\left(Y, T|_Y \otimes L|_Y^k \otimes \mathcal{I}\left(k\,\mathrm{Tr}_Y^\theta(\varphi)\right)\right) \\
&\leq \varliminf_{k\to\infty} \frac{m!}{k^m} h^0\left(Y, T|_Y \otimes L|_Y^k \otimes \mathrm{Res}_Y\left(\mathcal{I}(k\varphi)\right)\right).
\end{aligned}
$$

Now we address the other direction in (8.16). Let $\phi \in \mathrm{H}^0(X, A)$ be a section that does not vanish identically on Y. Such ϕ exists since A is very ample.

We fix $k_0 \in \mathbb{N}$. For any $k \geq 0$, we have $k = qk_0 + r$ with $q, r \in \mathbb{N}$ and $r \in \{0, \ldots, k_0 - 1\}$. Also, we have an injective linear map

$$\mathrm{H}^0\left(Y, T|_Y \otimes L|_Y^k \otimes \mathrm{Res}_Y \mathcal{I}(k\varphi)\right) \xrightarrow{\cdot\phi^{\otimes q}} \mathrm{H}^0\left(Y, T|_Y \otimes L|_Y^k \otimes A|_Y^q \otimes \mathrm{Res}_Y \mathcal{I}(k\varphi)\right).$$

Therefore,

$$\begin{aligned}
&\varlimsup_{k\to\infty} \frac{m!}{k^m} h^0\left(Y, T|_Y \otimes L|_Y^k \otimes \mathrm{Res}_Y \mathcal{I}(k\varphi)\right) \\
\leq &\varlimsup_{k\to\infty} \frac{m!}{k^m} h^0\left(Y, T|_Y \otimes L|_Y^k \otimes A|_Y^q \otimes \mathrm{Res}_Y \mathcal{I}(k\varphi)\right) \\
\leq &\max_{r=0,\ldots,k_0-1} \frac{1}{k_0^m} \varlimsup_{q\to\infty} \frac{m!}{q^m} h^0\left(Y, T|_Y \otimes L|_Y^{qk_0} \otimes A|_Y^q \otimes L|_Y^r \otimes \mathrm{Res}_Y \mathcal{I}((qk_0 + r)\varphi)\right) \\
\leq &\max_{r=0,\ldots,k_0-1} \frac{1}{k_0^m} \varlimsup_{q\to\infty} \frac{m!}{q^m} h^0\left(Y, T|_Y \otimes L|_Y^{qk_0} \otimes A|_Y^q \otimes L|_Y^r \otimes \mathrm{Res}_Y \mathcal{I}(k_0 q\varphi)\right) \\
= &\int_Y \left(\theta|_Y + k_0^{-1}\omega|_Y + \mathrm{dd^c}\, \mathrm{Tr}_Y^{\theta + k_0^{-1}\omega}(\varphi)\right)^m \\
= &\int_Y \left(\theta|_Y + k_0^{-1}\omega|_Y + \mathrm{dd^c}\, \mathrm{Tr}_Y^{\theta}(\varphi)\right)^m,
\end{aligned}$$

where the maximum is taken over the finitely many residue classes $k = qk_0 + r$, the following inequality uses $k_0 q \leq k$, the first equality follows by applying Proposition 8.4.1 for each r to the big line bundle $L^{k_0} \otimes A$, the Kähler current $k_0\theta_\varphi + \omega$, and twisting bundle $T \otimes L^r$, and the second equality follows from Proposition 3.1.3. Letting $k_0 \to \infty$, we conclude that

$$\varlimsup_{k\to\infty} \frac{m!}{k^m} h^0\left(Y, T|_Y \otimes L|_Y^k \otimes \mathrm{Res}_Y \mathcal{I}(k\varphi)\right) \leq \int_Y \left(\theta|_Y + \mathrm{dd^c}\, \mathrm{Tr}_Y^\theta(\varphi)\right)^m.$$

This completes the proof. □

Theorem 8.4.2 *Let $\varphi \in \mathrm{PSH}(X, \theta)$ such that $\nu(\varphi, Y) = 0$. Assume that θ_φ is a Kähler current. Then*

$$\int_Y \left(\theta|_Y + \mathrm{dd^c}\, \mathrm{Tr}_Y^\theta(\varphi)\right)^m = \lim_{k\to\infty} \frac{m!}{k^m} \dim_{\mathbb{C}} \left\{s|_Y : s \in \mathrm{H}^0\left(X, T \otimes L^k \otimes \mathcal{I}(k\varphi)\right)\right\}.$$

Proof This is a consequence of Theorem 7.4.1, Theorem 8.2.1 and Theorem 8.4.1:

$$
\begin{aligned}
\int_Y \left(\theta|_Y + \mathrm{dd^c}\, \mathrm{Tr}_Y^\theta(\varphi)\right)^m &= \lim_{k\to\infty} \frac{m!}{k^m} h^0\left(Y, T|_Y \otimes L|_Y^k \otimes \mathcal{I}\left(k\, \mathrm{Tr}_Y^\theta(\varphi)\right)\right) \\
&\leq \varliminf_{k\to\infty} \frac{m!}{k^m} \dim_{\mathbb{C}} \left\{ s|_Y : s \in \mathrm{H}^0\left(X, T \otimes L^k \otimes \mathcal{I}(k\varphi)\right)\right\} \\
&\leq \varlimsup_{k\to\infty} \frac{m!}{k^m} \dim_{\mathbb{C}} \left\{ s|_Y : s \in \mathrm{H}^0\left(X, T \otimes L^k \otimes \mathcal{I}(k\varphi)\right)\right\} \\
&\leq \lim_{k\to\infty} \frac{m!}{k^m} h^0\left(Y, T|_Y \otimes L|_Y^k \otimes \mathrm{Res}_Y\, \mathcal{I}(k\varphi)\right) \\
&= \int_Y \left(\theta|_Y + \mathrm{dd^c}\, \mathrm{Tr}_Y^\theta(\varphi)\right)^m .
\end{aligned}
$$

This completes the proof. □

Remark 8.4.1 One could also show that when (8.15) fails, the right-hand side of (8.16) is 0. See [DX24a].

8.5 Analytic Bertini theorems

Let X be a connected projective manifold of dimension $n \geq 1$.

The analytic Bertini theorem handles the restriction along a generic subvariety.

Theorem 8.5.1 *Let $\varphi \in \mathrm{QPSH}(X)$. Let $p\colon X \to \mathbb{P}^N$ be a morphism ($N \geq 1$). Define*

$$
\mathcal{G} := \left\{ H \in |\mathcal{O}_{\mathbb{P}^N}(1)| : H' := H \cap X \text{ is smooth and } \mathcal{I}(\varphi|_{H'}) = \mathrm{Res}_{H'}\big(\mathcal{I}(\varphi)\big)\right\}.
$$

Then $\mathcal{G} \subseteq |\mathcal{O}_{\mathbb{P}^N}(1)|$ is co-pluripolar.

Recall that co-pluripolar sets are defined in Definition 1.1.4. We adopt the convention that $\mathcal{I}(-\infty) = 0$.

Remark 8.5.1 Here and in the sequel, we slightly abuse the notation by writing $H \cap X$ for $p^{-1}H$, the scheme-theoretic inverse image of H. In other words, $H \cap X := H \times_{\mathbb{P}^N} X$.

By definition, any $H \in |\mathcal{O}_{\mathbb{P}^N}(1)|$ such that $p^{-1}H = \emptyset$ lies in $\mathcal{G}$.

Proof Take an ample line bundle L with a smooth Hermitian metric h such that $c_1(L,h) + \mathrm{dd^c}\varphi \geq 0$, where $c_1(L,h)$ is the first Chern form of (L,h), namely the curvature form of h. We introduce $\Lambda := |\mathcal{O}_{\mathbb{P}^N}(1)|$ to simplify our notations.

Step 1. We prove that the following set is co-pluripolar:

$$
\begin{aligned}
\mathcal{G}_L := \{ H \in \Lambda : H \cap X \text{ is smooth and } &\mathrm{H}^0\left(H \cap X, \omega_{H\cap X} \otimes L|_{H\cap X} \otimes \mathcal{I}(\varphi|_{H\cap X})\right) = \\
&\mathrm{H}^0\left(H \cap X, \omega_{H\cap X} \otimes L|_{H\cap X} \otimes \mathrm{Res}_{H\cap X}(\mathcal{I}(\varphi))\right)\} .
\end{aligned}
$$

Here $\omega_{H\cap X}$ denotes the dualizing sheaf of $H \cap X$.

Let $U \subseteq \Lambda \times X$ be the closed subvariety whose $\mathbb{C}$-points correspond to pairs $(H, x) \in \Lambda \times X$ with $p(x) \in H$. Let $\pi_1 : U \to \Lambda$ be the natural projection. We may assume that π_1 is surjective, as otherwise there is nothing to prove.

Observe that U is a local complete intersection scheme by *Krull's Hauptidealsatz* and *a fortiori* a Cohen–Macaulay scheme. It follows from miracle flatness [Mat89, Theorem 23.1] that the natural projection $\pi_2 \colon U \longrightarrow X$ is flat. As the fibers of π_2 over closed points of X are isomorphic to $\mathbb{P}^{N-1}$, it follows that π_2 is smooth. Thus, U is smooth as well. Moreover, observe that

$$\mathcal{I}(\pi_2^*\varphi) = \pi_2^*\mathcal{I}(\varphi) \tag{8.18}$$

by Proposition 1.4.5.

In the following, we will construct pluripolar sets $\Sigma_1 \subseteq \Sigma_2 \subseteq \Sigma_3 \subseteq \Sigma_4 \subsetneq \Lambda$ such that the behavior of π_1 is improved successively on the complement of Σ_i.

Step 1.1. The usual Bertini theorem shows that there is a proper Zariski closed set $\Sigma_1 \subsetneq \Lambda$ such that π_1 has smooth fibers outside Σ_1. Enlarging Σ_1, we may guarantee that π_1 and $\mathcal{I}(\pi_2^*\varphi)$ are both flat outside Σ_1. See [EGA IV$_2$, Théorème 6.9.1]. Then after further enlarging Σ_1 we may assume that H avoids all associated points of $\mathcal{O}_X/\mathcal{I}(\varphi)$, for all $H \in \Lambda \setminus \Sigma_1$. Let $\pi_{1,H}$ denote the fiber of π_1 at H and write $i_H : \pi_{1,H} \longrightarrow U$ for the inclusion morphism. We arrive at

$$\mathrm{Res}_{\pi_{1,H}}(\mathcal{I}(\pi_2^*\varphi)) = i_H^*\mathcal{I}(\pi_2^*\varphi)$$

for all $H \in \Lambda \setminus \Sigma_1$.[4]

Step 1.2. By Grauert's coherence theorem,

$$\mathcal{F}^i := R^i\pi_{1*}\left(\omega_{U/\Lambda} \otimes \pi_2^*L \otimes \mathcal{I}(\pi_2^*\varphi)\right)$$

is coherent for all i. Here $\omega_{U/\Lambda}$ denotes the relative dualizing sheaf of the morphism $U \longrightarrow \Lambda$. Thus, there is a proper Zariski closed set $\Sigma_2 \subsetneq \Lambda$ such that

(1) $\Sigma_2 \supseteq \Sigma_1$.
(2) The $\mathcal{F}^i$'s are locally free outside Σ_2.

We write $\mathcal{F} = \mathcal{F}^0$. By cohomology and base change [Har77, Theorem III.12.11], for any $H \in \Lambda \setminus \Sigma_2$, the fiber $\mathcal{F}|_H$ of $\mathcal{F}$ is given by

$$\mathcal{F}|_H = \mathrm{H}^0\left(\pi_{1,H}, \omega_{U/\Lambda}|_{\pi_{1,H}} \otimes \pi_2^*L|_{\pi_{1,H}} \otimes \mathrm{Res}_{\pi_{1,H}}(\mathcal{I}(\pi_2^*\varphi))\right).$$

Step 1.3. In order to proceed, we need to make use of the Hodge metric $h_{\mathcal{H}}$ on $\mathcal{F}$ defined in [HPS18]. We briefly recall its definition in our setting. By [HPS18, Section 22], we can find a proper Zariski closed set $\Sigma_3 \subsetneq \Lambda$ such that

(1) $\Sigma_3 \supseteq \Sigma_2$,
(2) π_1 is smooth outside Σ_3,
(3) both $\mathcal{F}$ and $\pi_{1*}\left(\omega_{U/\Lambda} \otimes \pi_2^*L\right)/\mathcal{F}$ are locally free outside Σ_3, and
(4) for each i,

$$R^i\pi_{1*}\left(\omega_{U/\Lambda} \otimes \pi_2^*L\right)$$

is locally free outside Σ_3.

[4] This subtle point was overlooked in the proof of [Xia22].

Then for any $H \in \Lambda \setminus \Sigma_3$,

$$\mathrm{H}^0(H \cap X, \omega_{H\cap X} \otimes L|_{H\cap X} \otimes \mathcal{I}(\varphi|_{H\cap X})) \subseteq \mathcal{F}|_H \subseteq \mathrm{H}^0(H \cap X, \omega_{H\cap X} \otimes L|_{H\cap X}).$$

See [HPS18, Lemma 22.1].

Now we can give the definition of the Hodge metric on $\Lambda\setminus\Sigma_3$. Given any $H \in \Lambda\setminus\Sigma_3$, any $\alpha \in \mathcal{F}|_H$, the Hodge metric is defined as

$$h_{\mathcal{H}}(\alpha, \alpha) := \int_{X\cap H} |\alpha|_h^2 \mathrm{e}^{-\varphi} \in [0, \infty].$$

Observe that $h_{\mathcal{H}}(\alpha, \alpha) < \infty$ if and only if $\alpha \in \mathrm{H}^0(H \cap X, \omega_{H\cap X} \otimes L|_{H\cap X} \otimes \mathcal{I}(\varphi|_{H\cap X}))$. Moreover, $h_{\mathcal{H}}(\alpha, \alpha) > 0$ if $\alpha \neq 0$. It is shown in [HPS18] (cf. [PT18, Theorem 3.3.5]) that $h_{\mathcal{H}}$ is indeed a singular Hermitian metric, and it extends to a positive metric on $\mathcal{F}$.

Step 1.4. The determinant $\det h_{\mathcal{H}}$ is singular at all $H \in \Lambda \setminus \Sigma_3$ such that

$$\mathrm{H}^0\Big(H \cap X, \omega_{H\cap X} \otimes L|_{H\cap X} \otimes \mathcal{I}(\varphi|_{H\cap X})\Big) \neq \mathcal{F}|_H.$$

As the map π_2 is smooth, we have $\pi_2^*\mathcal{I}(\varphi) = \mathcal{I}(\pi_2^*\varphi)$ by Proposition 1.4.5. Under the identification $\pi_{1,H} \cong H \cap X$, we have

$$\mathrm{Res}_{\pi_{1,H}}\Big(\pi_2^*\mathcal{I}(\varphi)\Big) \cong \mathrm{Res}_{H\cap X}\Big(\mathcal{I}(\varphi)\Big).$$

Thus, we have the following inclusions:

$$\begin{aligned}&\mathrm{H}^0\Big(H \cap X, \omega_{H\cap X} \otimes L|_{H\cap X} \otimes \mathcal{I}\,(\varphi|_{H\cap X})\Big)\\ \subseteq&\mathrm{H}^0\Big(H \cap X, \omega_{H\cap X} \otimes L|_{H\cap X} \otimes \mathrm{Res}_{H\cap X}\big(\mathcal{I}(\varphi)\big)\Big),\end{aligned}$$

the right-hand side being $\mathcal{F}|_H$.

Recall that the first inclusion follows from Theorem 1.4.5. Hence, $\det h_{\mathcal{H}}$ is singular at all $H \in |\mathcal{O}_{\mathbb{P}^N}(1)| \setminus \Sigma_3$ such that

$$\begin{aligned}&\mathrm{H}^0\Big(H \cap X, \omega_{H\cap X} \otimes L|_{H\cap X} \otimes \mathcal{I}\,(\varphi|_{H\cap X})\Big)\\ \neq&\mathrm{H}^0\Big(H \cap X, \omega_{H\cap X} \otimes L|_{H\cap X} \otimes \mathrm{Res}_{H\cap X}\big(\mathcal{I}(\varphi)\big)\Big).\end{aligned}$$

Let Σ_4 be the union of Σ_3 and the set of all such H. Since the Hodge metric $h_{\mathcal{H}}$ is positive ([PT18, Theorem 3.3.5] and [HPS18, Theorem 21.1]), its determinant $\det h_{\mathcal{H}}$ is also positive ([Rau15, Proposition 1.3] and [HPS18, Proposition 25.1]), it follows that Σ_4 is pluripolar. As a consequence, $\mathcal{G}_L$ is co-pluripolar.

Step 2.

Fix an ample invertible sheaf S on X. The same result holds with $L \otimes S^{\otimes a}$ in place of L. Thus, the set

$$A \coloneqq \bigcap_{a=0}^{\infty} \mathcal{G}_{L\otimes S^{\otimes a}}$$

is co-pluripolar. For each $H \in \Lambda$ such that $X \cap H$ is smooth and $\mathcal{I}(\varphi|_{X\cap H}) \neq \mathrm{Res}_{H\cap X}(\mathcal{I}(\varphi))$, let $\mathcal{K}$ be the following cokernel:

$$0 \to \mathcal{I}(\varphi|_{X\cap H}) \to \mathrm{Res}_{H\cap X}\big(\mathcal{I}(\varphi)\big) \to \mathcal{K} \to 0.$$

By Serre vanishing theorem, taking a large enough, we may guarantee that

$$\mathrm{H}^1\Big(X \cap H, \omega_{X\cap H} \otimes (L \otimes S^{\otimes a})|_{X\cap H} \otimes \mathcal{I}(\varphi|_{X\cap H})\Big) = 0$$

and

$$\mathrm{H}^0\Big(X \cap H, \omega_{X\cap H} \otimes (L \otimes S^{\otimes a})|_{X\cap H} \otimes \mathcal{K}\Big) \neq 0.$$

Then

$$\mathrm{H}^0\Big(X \cap H, \omega_{X\cap H} \otimes (L \otimes S^{\otimes a})|_{X\cap H} \otimes \mathcal{I}(\varphi|_{X\cap H})\Big) \neq$$
$$\mathrm{H}^0\Big(X \cap H, \omega_{X\cap H} \otimes (L \otimes S^{\otimes a})|_{X\cap H} \otimes \mathrm{Res}_{H\cap X}(\mathcal{I}(\varphi))\Big).$$

Thus, $H \notin A$. We conclude that $\mathcal{G}$ is co-pluripolar. □

Remark 8.5.2 More generally, the same technique implies the following general result: Let $f\colon X \to Y$ be a projective morphism between complex manifolds and (L, h) be a Hermitian pseudo-effective line bundle on X. Then for quasi-every[5] $y \in Y$, the fiber X_y is smooth and

$$\mathcal{I}(\lambda h|_{X_y}) = \mathrm{Res}_{X_y}(\mathcal{I}(\lambda h)).$$

In the sequel of this section, we fix a base-point free linear system Λ on X. For the results below that assert connectedness, we assume in addition that the set of connected smooth members of Λ is co-pluripolar in Λ.

Corollary 8.5.1 *Let $\varphi \in \mathrm{QPSH}(X)$. Then for quasi-every $H \in \Lambda$, we have $\varphi|_H \not\equiv -\infty$.*

Proof For quasi-every H, Theorem 8.5.1 gives

$$\mathcal{I}(\varphi|_H) = \mathrm{Res}_H\, \mathcal{I}(\varphi).$$

After removing the hyperplanes meeting the associated points of $\mathcal{O}_X/\mathcal{I}(\varphi)$ improperly, the right-hand side is nonzero. Since $\mathcal{I}(-\infty) = 0$, this implies $\varphi|_H \not\equiv -\infty$. □

Corollary 8.5.2 *Assume that $n \geq 2$. Let $\varphi \in \mathrm{QPSH}(X)$. Then quasi-every $H \in \Lambda$ is connected and smooth, satisfies $\nu(\varphi, H) = 0$ and we have*

$$\mathrm{Tr}_H(\varphi) \sim_{\mathcal{I}} \varphi|_H.$$

[5] That is, for all y outside a pluripolar subset of Y.

The connectedness assumption on H is used here because we developed most of the trace theory above for connected submanifolds.

Proof First observe that the set $\{x \in X : \nu(\varphi, x) > 0\}$ is a countable union of proper analytic subsets by Siu's semicontinuity theoremTheorem 1.4.1. It follows that a very general element in Λ is not contained in this set.

Fix an ample line bundle L so that there is a smooth Hermitian metric h_L such that $c_1(L, h_L) + \mathrm{dd^c}\varphi$ is a Kähler current. Thanks to Theorem 8.5.1 and the connectedness assumption on Λ, we can find a co-pluripolar set $\Lambda' \subseteq \Lambda$ such that each $H \in \Lambda'$ satisfies the following:

(1) H is smooth and connected;
(2) $\nu(\varphi, H) = 0$;
(3) $\mathcal{I}(k\varphi|_H) = \mathrm{Res}_H(\mathcal{I}(k\varphi))$ for all $k > 0$. □

It follows from Theorem 8.4.1 and Theorem 7.4.1 that

$$\int_H \left(c_1(L, h_L)|_H + \mathrm{dd^c}\,\mathrm{Tr}_H^{c_1(L,h_L)}(\varphi)\right)^{n-1} = \int_H \left(c_1(L, h_L)|_H + \mathrm{dd^c} P_{c_1(L,h_L)|_H}[\varphi|_H]_{\mathcal{I}}\right)^{n-1}.$$

Since $\varphi|_H \preceq_P \mathrm{Tr}_H(\varphi)$ by Proposition 8.1.3, we also have $\varphi|_H \preceq_{\mathcal{I}} \mathrm{Tr}_H(\varphi)$. Hence

$$P_{c_1(L,h_L)|_H}[\varphi|_H]_{\mathcal{I}} \le P_{c_1(L,h_L)|_H}[\mathrm{Tr}_H(\varphi)]_{\mathcal{I}}.$$

The trace is $\mathcal{I}$-good, so the right-hand side agrees with its P-envelope. The equality of the masses above and Theorem 3.1.2 then imply that the two sides have the same P-singularity type, hence

$$\mathrm{Tr}_H(\varphi) \sim_{\mathcal{I}} \varphi|_H.$$

Lemma 8.5.1 *Assume that $n \ge 2$. Let T be a closed positive $(1,1)$-current on X with $\int_X T^n > 0$. Then quasi-every $H \in \Lambda$ is connected and smooth, $T|_H$ is well-defined and satisfies*

$$\int_H T|_H^{n-1} > 0.$$

Proof Write $T = \theta_\varphi$ for some smooth closed real $(1,1)$-form θ on X and $\varphi \in \mathrm{PSH}(X, \theta)_{>0}$. Thanks to Lemma 2.4.5, we can find $\psi \in \mathrm{PSH}(X, \theta)$ such that θ_ψ is a Kähler current and $\psi \le \varphi$. By Corollary 8.5.1 and the connectedness assumption on Λ, we can find a co-pluripolar set $\Lambda' \subseteq \Lambda$ such that each $H \in \Lambda'$ satisfies:

(1) H is smooth and connected;
(2) the restriction $\psi|_H$ is not identically $-\infty$.

Since θ_ψ is a Kähler current, its restriction to such an H has positive non-pluripolar mass. As $\psi|_H \le \varphi|_H$, Theorem 2.4.3 gives

$$\int_H (\theta|_H + \mathrm{dd^c}\varphi|_H)^{n-1} \ge \int_H (\theta|_H + \mathrm{dd^c}\psi|_H)^{n-1} > 0.$$

We conclude the proof. □

Corollary 8.5.3 *Assume that $n \geq 2$. Let T be a closed positive $(1,1)$-current on X with $\operatorname{vol} T > 0$. Then quasi-every $H \in \Lambda$ is connected and smooth, and $\operatorname{Tr}_H^{[T]|_H}(T)$ is well-defined.*

Proof Write $T = \theta + \mathrm{dd}^{\mathrm{c}}\varphi$ and set $\psi = P_\theta[\varphi]_{\mathcal{I}}$. Then $\psi \sim_{\mathcal{I}} \varphi$ and

$$\int_X \theta_\psi^n = \operatorname{vol} T > 0.$$

By Lemma 8.5.1, for quasi-every $H \in \Lambda$, the hypersurface H is connected and smooth, $\psi|_H$ is well-defined, and

$$\int_H \left(\theta|_H + \mathrm{dd}^{\mathrm{c}}\psi|_H\right)^{n-1} > 0.$$

For such H, Corollary 8.5.2 gives $\operatorname{Tr}_H(\psi) \sim_{\mathcal{I}} \psi|_H$. Since $\varphi \sim_{\mathcal{I}} \psi$, Proposition 8.2.1 also gives $\operatorname{Tr}_H(\varphi) \sim_P \operatorname{Tr}_H(\psi)$. The assertion now follows from Example 8.1.6. □

Proposition 8.5.1 *Assume that $n \geq 2$. Let $\varphi, \psi \in \operatorname{QPSH}(X)$. Assume that $\varphi \preceq_P \psi$. Then quasi-every $H \in \Lambda$ is connected and smooth, and $\varphi|_H \preceq_P \psi|_H$.*

Proof Thanks to Lemma 6.1.3, we may replace φ by $\varphi \vee \psi$ and assume that $\varphi \sim_P \psi$. It suffices to show that $\varphi|_H \sim_P \psi|_H$ for quasi-every $H \in \Lambda$.

Take a smooth closed real $(1,1)$-form θ on X so that $\varphi, \psi \in \operatorname{PSH}(X, \theta)_{>0}$. It suffices to compare φ and ψ with $P_\theta[\varphi]$, so without loss of generality, we may assume that ψ is a model potential in $\operatorname{PSH}(X,\theta)_{>0}$. Up to adding a constant to φ, we may then assume that $\varphi \leq \psi$. It follows from Lemma 2.4.4 that we can find a sequence $(\eta_j)_j$ in $\operatorname{PSH}(X,\theta)_{>0}$ such that

$$j^{-1}\eta_j + \left(1 - j^{-1}\right)\psi \leq \varphi$$

for all $j \geq 2$. By Corollary 8.5.1, Lemma 8.5.1, we can find a co-pluripolar set $\Lambda' \subseteq \Lambda$ such that any $H \in \Lambda'$ satisfies:

(1) H is smooth and connected;
(2) $\eta_j|_H \in \operatorname{PSH}(H, \theta|_H)_{>0}$ for all $j \geq 2$ and $\psi|_H \in \operatorname{PSH}(H, \theta|_H)_{>0}$.

Therefore, taking Proposition 3.1.10 into account, we arrive at

$$j^{-1}P_{\theta|_H}\left[\eta_j|_H\right] + \left(1 - j^{-1}\right)P_{\theta|_H}\left[\psi|_H\right] \leq P_{\theta|_H}\left[\varphi|_H\right]$$

for all $j \geq 2$. The normalized envelopes $P_{\theta|_H}[\eta_j|_H]$ are uniformly bounded in L^1, hence

$$j^{-1}P_{\theta|_H}[\eta_j|_H] \xrightarrow{L^1} 0.$$

We conclude that

$$P_{\theta|_H}[\psi|_H] \leq P_{\theta|_H}[\varphi|_H]$$

and hence $\psi|_H \preceq_P \varphi|_H$. □

Lemma 8.5.2 *Assume that $n \geq 2$. Let θ be a closed smooth $(1,1)$-form on X representing a big cohomology class and $(\varphi_j)_j$ be a decreasing sequence in* $\mathrm{PSH}(X,\theta)$. *Assume that $\varphi \in \mathrm{PSH}(X,\theta)$ and $\varphi_j \xrightarrow{d_S} \varphi$. Furthermore, assume that $\varphi \leq \varphi_j$ for each $j > 0$. Then quasi-every $H \in \Lambda$ is connected and smooth, $\varphi_j|_H \not\equiv -\infty$ for all $j \geq 1$, $\varphi|_H \not\equiv -\infty$, and*

$$\varphi_j|_H \xrightarrow{d_S} \varphi|_H.$$

Proof By Corollary 6.2.8, we may assume that $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$. Using Lemma 2.4.4, we can find a decreasing sequence $(\epsilon_j)_j$ in $(0,1)$ with limit 0 and $\eta_j \in \mathrm{PSH}(X,\theta)_{>0}$ such that $\eta_j \leq \varphi_j$ and

$$\epsilon_j \eta_j + (1-\epsilon_j)\varphi_j \leq \varphi.$$

By Corollary 8.5.1, Lemma 8.5.1, we can find a co-pluripolar set $\Lambda' \subseteq \Lambda$ such that any $H \in \Lambda'$ satisfies:

(1) H is smooth and connected;
(2) $\eta_j|_H \in \mathrm{PSH}(H,\theta|_H)_{>0}$ for all $j \geq 1$ and $\varphi|_H \in \mathrm{PSH}(H,\theta|_H)_{>0}$.

Therefore, taking Proposition 3.1.10 into account, we arrive at

$$\epsilon_j P_{\theta|_H}\left[\eta_j|_H\right] + \left(1-\epsilon_j\right) P_{\theta|_H}\left[\varphi_j|_H\right] \leq P_{\theta|_H}\left[\varphi|_H\right].$$

Since the normalized envelopes $P_{\theta|_H}[\eta_j|_H]$ are uniformly bounded in L^1, we have $\epsilon_j P_{\theta|_H}[\eta_j|_H] \to 0$ in L^1. Letting $j \to \infty$, we get

$$\lim_{j\to\infty} P_{\theta|_H}\left[\varphi_j|_H\right] \leq P_{\theta|_H}\left[\varphi|_H\right].$$

The reverse inequality is trivial, so we get an equality here. By Theorem 2.4.3 and Proposition 3.1.12, we conclude that

$$\lim_{j\to\infty}\int_H \left(\theta|_H + \mathrm{dd^c}\varphi_j|_H\right)^{n-1} = \int_H \left(\theta|_H + \mathrm{dd^c}\varphi|_H\right)^{n-1}.$$

Therefore, using Corollary 6.2.5, we conclude that $\varphi_j|_H \xrightarrow{d_S} \varphi|_H$. □

Corollary 8.5.4 *Assume that $n \geq 2$. Let $\varphi \in \mathrm{QPSH}(X)$ be an $\mathcal{I}$-good potential. Then quasi-every $H \in \Lambda$ satisfies:*

(1) *H is connected and smooth;*
(2) *$\varphi|_H$ is $\mathcal{I}$-good;*
(3) *$\nu(\varphi, H) = 0$;*
(4) $\mathrm{Tr}_H\,\varphi \sim_P \varphi|_H$.

Furthermore, if θ is a closed smooth real $(1,1)$-form on X such that $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$, then we can further guarantee, on the same co-pluripolar set of $H \in \Lambda$, that $\mathrm{Tr}_H(\varphi)$ has a representative $\mathrm{Tr}_H(\varphi) \in \mathrm{PSH}(H,\theta|_H)_{>0}$.

Proof This is a consequence of Lemma 8.5.2, Theorem 7.1.1, Corollary 8.5.2 and Corollary 8.5.3. □

For later use, let us also prove a reverse Bertini theorem here.

Lemma 8.5.3 (Reverse Bertini theorem) *Let X be a complex manifold, $f\colon X \to \Delta^*$ be a projective surjective morphism to the punctured unit disk Δ^*. Let (L, h), (L, h') be Hermitian pseudo-effective line bundles on X with the same underlying line bundle. Assume that there is a biholomorphic S^1-action on (X, L) making f equivariant and such that h and h' are invariant under this action. Assume that for quasi-every $z \in \Delta^*$, X_z is smooth and $h|_{X_z} \sim_{\mathcal{I}} h|'_{X_z}$. Then $h \sim_{\mathcal{I}} h'$.*

Proof We need to show that $\mathcal{I}(kh) = \mathcal{I}(kh')$ for every positive integer k. Clearly, it suffices to prove the case $k = 1$. We will therefore prove $\mathcal{I}(h) = \mathcal{I}(h')$. First observe that it suffices to prove that

$$f_*(K_X \otimes L \otimes \mathcal{I}(h)) = f_*(K_X \otimes L \otimes \mathcal{I}(h')) \tag{8.19}$$

as subsheaves of $f_*(K_X \otimes L)$. In fact, suppose that (8.19) holds. Let H be a f-ample invertible sheaf on X. Then (8.19) also holds with $L \otimes H^m$ in place of L. It follows from Grauert–Remmert's version of Serre vanishing theorem [BS76, Theorem 2.1(A)] that $\mathcal{I}(h) = \mathcal{I}(h')$.

It remains to prove (8.19). Observe that both sides of (8.19) are locally free by [Mat22, Corollary 1.5]. We claim that it suffices to show that

$$f_*\left(K_X \otimes L \otimes \mathcal{I}(h)\right)_z = f_*\left(K_X \otimes L \otimes \mathcal{I}(h')\right)_z \tag{8.20}$$

for one $z \in \Delta^*$. In fact, this implies that the same holds outside a countable subset of Δ^*. But the set where (8.20) fails has to be S^1-invariant, it has to be empty.

In fact, we will prove (8.20) for quasi-every $z \in \Delta^*$. By cohomology and base change together with the analytic Bertini theorem Remark 8.5.2, for quasi-every $z \in \Delta^*$, we have

$$\begin{aligned} f_*\left(K_X \otimes L \otimes \mathcal{I}(h)\right)_z &= \mathrm{H}^0\left(X_z, K_X|_{X_z} \otimes L|_{X_z} \otimes \mathcal{I}(h|_{X_z})\right), \\ f_*\left(K_X \otimes L \otimes \mathcal{I}(h')\right)_z &= \mathrm{H}^0\left(X_z, K_X|_{X_z} \otimes L|_{X_z} \otimes \mathcal{I}(h'|_{X_z})\right). \end{aligned}$$

But we assumed that for quasi-every z, $h|_{X_z} \sim_{\mathcal{I}} h|'_{X_z}$, it follows that for quasi-every $z \in \Delta^*$, (8.20) holds. The proof is complete. □

Chapter 9
Test curves

Comment se fait-il que M. Gauss ait osé vous faire dire que la plupart de vos théorèmes lui étaient connus et qu'il en avait fait la découverte dès 1808. Cet excès d'impudence n'est pas croyable de la part d'un homme qui a assez de mérite personnel pour n'avoir pas besoin de s'approprier les découvertes des autres.
— Adrien-Marie Legendre[a], in a letter to Jacobi in 1827

[a] Adrien-Marie Legendre (1752–1833) was a French mathematician known for his foundational contributions to number theory, statistics, and mathematical analysis. Apart from his mathematical contributions, he also helped formalize the metric system during the French Revolution.

A geodesic ray $(\varphi_t)_{t\in[0,\infty)}$ in $\mathrm{PSH}(X,\theta)$ encodes the asymptotic behavior of singularities along a path of potentials, and is studied in Chapters 4 and 6. The *Legendre transform* provides an alternative parameterization: instead of recording the potential φ_t at each time t, one records, for each level τ, the model potential capturing the singularities of φ exceeding the linear rate τ. The resulting object, a concave decreasing curve $\tau \mapsto \Gamma_\tau$ of model potentials, is what we call a *test curve*. The aim of this chapter is to develop test curves as a self-contained theory and to establish their precise relation to geodesic rays, the *Ross–Witt Nyström correspondence*[1].

A test curve Γ is a concave decreasing family $(\Gamma_\tau)_{\tau<\Gamma_{\max}}$ of model potentials in $\mathrm{PSH}(X,\theta)$ whose "mass" $\int_X(\theta + \mathrm{dd}^c\Gamma_{-\infty})^n$ is positive (Definition 9.1.1). The condition that each Γ_τ is a model potential is essential: it is what makes test curves into a rigid combinatorial object, not just a concave curve in $\mathrm{PSH}(X,\theta)$. As with the analytic side, test curves admit relative versions parameterized by a model potential $\phi \in \mathrm{PSH}(X,\theta)_{>0}$ (Definition 9.1.2); these are the relevant objects for relative pluripotential theory.

The chapter is organized as follows.

Section 9.1 introduces test curves and develops their basic properties: independence of the auxiliary form θ (Proposition 9.1.2), behavior under bimeromorphic morphisms (Proposition 9.1.3), closedness in a suitable sense (Proposition 9.1.4), and a P-equivalence relation on test curves (Definition 9.1.3).

Section 9.2 establishes the Ross–Witt Nyström correspondence Theorem 9.2.1: the Legendre transform $\Gamma \mapsto \Gamma^*$ (Definition 9.2.1) is a bijection between test curves modulo P-equivalence and (suitable equivalence classes of) sublinear geodesic rays in $\mathrm{PSH}(X,\theta)$. Our formulation is more general than all comparable results in the literature. Two basic theorems due to He–Testorf–Wang (Proposition 9.2.2) and Hisamoto (Proposition 9.2.3) are recorded for later use.

[1] Witt and Nyström are both family names of a single person. Some Swedes have double family names. It should not be spelled as Witt-Nyström as some people do in the literature.

Section 9.3 develops the parallel theory for $\mathcal{I}$-model test curves (Definition 9.3.1) including the $\mathcal{I}$-envelope of test curves (Definition 9.3.2), the transition between the P- and $\mathcal{I}$-versions (Proposition 9.3.4), and the notion of *maximal geodesic rays* (Definition 9.3.3).

Section 9.4 defines algebraic operations on test curves in anticipation of the non-Archimedean pluripotential theory developed in Chapter 13, where test curves serve as the bridge between psh and non-Archimedean potentials.

We shall freely apply the convex-analytic results of Appendix A. Those results are stated for convex functions; when we apply them to concave functions, we always apply them to their negatives.

The Ross–Witt Nyström correspondence and its $\mathcal{I}$-model variant are the technical engines of Chapter 13, where they are used to identify certain test curves with non-Archimedean potentials.

9.1 The notion of test curves

Let X be a connected compact Kähler manifold of dimension n and θ be a smooth closed real $(1,1)$-form on X representing a big cohomology class.

Recall that the notion of model potentials is defined in Definition 3.1.3.

Definition 9.1.1 A *test curve* Γ in $\mathrm{PSH}(X,\theta)$ consists of a real number $\Gamma_{\max}$ together with a map $(-\infty,\Gamma_{\max}) \to \mathrm{PSH}(X,\theta)$ denoted by $\tau \mapsto \Gamma_\tau$ satisfying the following conditions:

(1) The map $\tau \mapsto \Gamma_\tau$ is concave and decreasing;
(2) $P_\theta[\Gamma_\tau] = \Gamma_\tau$;
(3) the potential
$$\Gamma_{-\infty} \coloneqq \sup_{\tau<\Gamma_{\max}}{}^{*}\Gamma_\tau \tag{9.1}$$
satisfies
$$\int_X (\theta + \mathrm{dd^c}\Gamma_{-\infty})^n > 0.$$

Let $\phi \in \mathrm{PSH}(X,\theta)_{>0}$ be a model potential. The set of test curves Γ with $\Gamma_{-\infty} = \phi$ is denoted by $\mathrm{TC}(X,\theta;\phi)$.

The union of all $\mathrm{TC}(X,\theta;\phi)$'s for various model potentials $\phi \in \mathrm{PSH}(X,\theta)_{>0}$ is denoted by $\mathrm{TC}(X,\theta)_{>0}$.

By (2), $\sup_X \Gamma_\tau = 0$ for each $\tau < \Gamma_{\max}$. So $\Gamma_{-\infty} \in \mathrm{PSH}(X,\theta)$ by Proposition 1.2.2. Moreover, $\Gamma_{-\infty}$ is a model potential by Proposition 3.1.13.

Remark 9.1.1 Sometimes it is convenient to extend Γ_τ to $\tau \geq \Gamma_{\max}$ as well. This can be done as follows: for $\tau > \Gamma_{\max}$, we set $\Gamma_\tau \equiv -\infty$. For $\tau = \Gamma_{\max}$, we set
$$\Gamma_\tau \coloneqq \inf_{\tau'<\Gamma_{\max}} \Gamma_{\tau'} \in \mathrm{PSH}(X,\theta).$$

his decreasing limit is not identically $-\infty$: indeed, the family $\{\Gamma_{\tau'} : \tau' < \Gamma_{\max}\}$ is normalized by $\sup_X \Gamma_{\tau'} = 0$, hence is relatively compact in L^1 by Proposition 1.5.1. Thus the pointwise infimum is a θ-psh function by Proposition 1.2.2.

Recall that according to our general principle, we speak of model potentials only when a potential has positive mass.

Lemma 9.1.1 *Assume that* $\Gamma \in \mathrm{TC}(X, \theta)_{>0}$. *Then for each* $\tau < \Gamma_{\max}$, *we have*

$$\int_X (\theta + \mathrm{dd^c}\Gamma_\tau)^n > 0. \tag{9.2}$$

In particular, Γ_τ *is a model potential in* $\mathrm{PSH}(X, \theta)_{>0}$.

The notations in the proof below are summarized in Fig. 9.1.

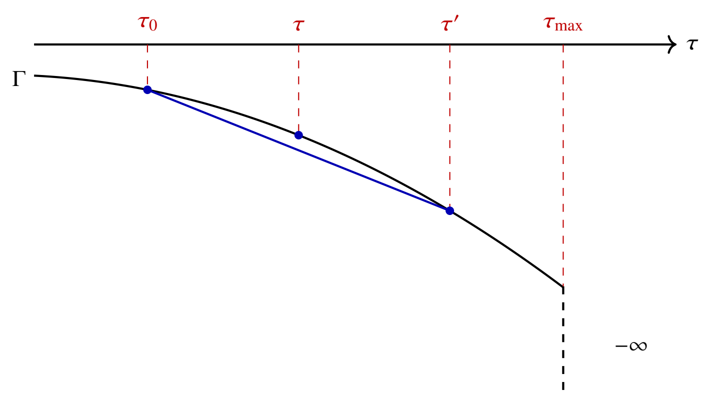

Fig. 9.1 The test curve Γ.

Proof Fix $\tau \in (-\infty, \Gamma_{\max})$.

By assumption, $\Gamma_{-\infty}$ has positive mass. By Corollary 2.4.1, we have

$$\int_X \theta^n_{\Gamma_{-\infty}} = \lim_{\tau \to -\infty} \int_X \theta^n_{\Gamma_\tau}.$$

In particular, for a sufficiently small $\tau_0 < \tau$, we have

$$\int_X \theta^n_{\Gamma_{\tau_0}} > 0.$$

Now take $\tau' \in (\tau, \Gamma_{\max})$ and $t \in (0,1)$ so that

$$\tau = (1-t)\tau' + t\tau_0.$$

From the concavity of Γ, we find that

$$\Gamma_\tau \geq (1-t)\Gamma_{\tau'} + t\Gamma_{\tau_0}.$$

By the monotonicity theorem Theorem 2.4.3 and the multi-linearity of the non-pluripolar products Proposition 2.2.1(6),

$$\int_X \theta^n_{\Gamma_\tau} \geq \int_X \theta^n_{(1-t)\Gamma_{\tau'}+t\Gamma_{\tau_0}} \geq t^n \int_X \theta^n_{\Gamma_{\tau_0}} > 0$$

and (9.2) follows. □

Proposition 9.1.1 *Let* $\Gamma \in \mathrm{TC}(X,\theta)_{>0}$. *Then the map*

$$[-\infty, \Gamma_{\max}) \to \mathbb{R}, \quad \tau \mapsto \log \int_X \theta^n_{\Gamma_\tau}$$

is continuous and concave on $(-\infty, \Gamma_{\max})$.

Proof The concavity of this function follows from Theorem 2.4.1 and Theorem 2.4.3. The continuity at $-\infty$ is a consequence of Corollary 2.4.1. The continuity on $(-\infty, \Gamma_{\max})$ follows from the fact that finite concave functions on open intervals are continuous. □

Definition 9.1.2 Let $\phi \in \mathrm{PSH}(X,\theta)_{>0}$ be a model potential.

A test curve $\Gamma \in \mathrm{TC}(X,\theta;\phi)$ is said to be *bounded* if for τ small enough, $\Gamma_\tau = \phi$. The subset of bounded test curves in $\mathrm{TC}(X,\theta;\phi)$ is denoted by $\mathrm{TC}^\infty(X,\theta;\phi)$. In this case, we write

$$\Gamma_{\min} := \sup\{\tau \in \mathbb{R} : \Gamma_\tau = \phi\}. \tag{9.3}$$

A test curve $\Gamma \in \mathrm{TC}(X,\theta;\phi)$ is said to have *finite energy* if

$$\mathbf{E}^\phi(\Gamma) := \Gamma_{\max} \int_X \theta^n_\phi + \int_{-\infty}^{\Gamma_{\max}} \left(\int_X \theta^n_{\Gamma_\tau} - \int_X \theta^n_\phi \right) \mathrm{d}\tau > -\infty. \tag{9.4}$$

When $\phi = V_\theta$, we write $\mathbf{E}$ instead of $\mathbf{E}^\phi$.

The subset of test curves with finite energy in $\mathrm{TC}(X,\theta;\phi)$ is denoted by $\mathrm{TC}^1(X,\theta;\phi)$.

Example 9.1.1 Given $\varphi \in \mathrm{PSH}(X,\theta)$, there is a canonically associated test curve $\Gamma^\varphi \in \mathrm{TC}^\infty(X,\theta;V_\theta)$: Set $\Gamma^\varphi_{\max} = 0$ and

$$\Gamma^\varphi_\tau = \begin{cases} V_\theta, & \text{if } \tau \leq -1; \\ P_\theta\left[(1+\tau)\varphi - \tau V_\theta\right], & \text{if } -1 < \tau < 0. \end{cases}$$

Note that Γ^φ is indeed a test curve, as follows from Proposition 3.1.10. Indeed, the concavity and monotonicity in τ follow from Proposition 3.1.10, and $\Gamma^\varphi_{-\infty} = V_\theta$ has positive mass.

We first observe that the notion of test curves does not really depend on the choice of θ within its cohomology class.

Proposition 9.1.2 *Let θ' be another smooth closed real $(1,1)$-form on X representing the same cohomology class as θ. Let $\phi \in \mathrm{PSH}(X,\theta)_{>0}$ be a model potential. Let $\phi' \in \mathrm{PSH}(X,\theta')_{>0}$ be the unique model potential satisfying $\phi \sim \phi'$.*

Then there is a canonical bijection

$$\mathrm{TC}(X,\theta;\phi) \xrightarrow{\sim} \mathrm{TC}(X,\theta';\phi').$$

This bijection induces the following bijections:

$$\mathrm{TC}^1(X,\theta;\phi) \xrightarrow{\sim} \mathrm{TC}^1(X,\theta';\phi'), \quad \mathrm{TC}^\infty(X,\theta;\phi) \xrightarrow{\sim} \mathrm{TC}^\infty(X,\theta';\phi').$$

These bijections satisfy the obvious cocycle conditions.

Proof Choose $g \in C^\infty(X)$ such that $\theta' = \theta + \mathrm{dd^c} g$. Given any $\Gamma \in \mathrm{TC}(X,\theta;\phi)$, we observe that $\Gamma' \colon (-\infty, \Gamma_{\max}) \to \mathrm{PSH}(X,\theta')$ defined as

$$\tau \mapsto P_{\theta'}[\Gamma_\tau - g]$$

lies in $\mathrm{TC}(X,\theta';\phi')$. In fact, the concavity and monotonicity of Γ' follow from Proposition 3.1.10. Moreover, applying Proposition 3.1.13 to the increasing net obtained as $\tau \to -\infty$ gives

$$\Gamma'_{-\infty} = P_{\theta'}[\Gamma_{-\infty} - g] = P_{\theta'}[\phi - g] = \phi'.$$

Thus $\Gamma' \in \mathrm{TC}(X,\theta';\phi')$.

Moreover, the choice of g is irrelevant since for any other choice of g, say g', we have

$$\Gamma_\tau - g \sim \Gamma_\tau - g'$$

for all $\tau < \Gamma_{\max}$. The inverse construction is obtained by replacing g with $-g$. The boundedness condition is clearly preserved. Finally, by Proposition 3.1.3, for every $\tau < \Gamma_{\max}$,

$$\int_X (\theta' + \mathrm{dd^c}\Gamma'_\tau)^n = \int_X (\theta + \mathrm{dd^c}\Gamma_\tau)^n,$$

and the same holds for ϕ and ϕ'. Hence the finite energy condition is preserved. The cocycle condition is immediate from the uniqueness of the model representative in a fixed singularity type. □

Proposition 9.1.3 *Let $\pi \colon Y \to X$ be a proper bimeromorphic morphism from a Kähler manifold Y. Then the pointwise pull-back induces a bijection*

$$\pi^* \colon \mathrm{TC}(X,\theta;\phi) \xrightarrow{\sim} \mathrm{TC}(Y,\pi^*\theta;\pi^*\phi).$$

Proof The bijectivity on the underlying potentials follows from Proposition 1.5.3. The compatibility with model potentials follows from Proposition 3.1.9. These two facts also preserve concavity, monotonicity and the value of $\Gamma_{-\infty}$, hence give the asserted bijection of test curves. □

Next we verify the closedness of a test curve as a family of concave functions, so that no pathology arises in the Legendre transforms considered below. The notion of closedness is recalled in Definition A.1.7.

Proposition 9.1.4 *Let Γ be a test curve in* $\mathrm{PSH}(X,\theta)$. *For each $x \in X$, the map $\mathbb{R} \ni \tau \mapsto \Gamma_\tau(x)$ is a closed concave function. Moreover, the map is proper as long as $\Gamma_{\Gamma_{\max}}(x) \neq -\infty$.*

Proof We prove closedness. Fix $x \in X$. Assume that $\Gamma_\tau(x) \neq -\infty$ for some $\tau \in \mathbb{R}$. We only need to argue the upper semicontinuity of $\tau \mapsto \Gamma_\tau(x)$. The upper semicontinuity is clear at $\tau \geq \Gamma_{\max}$, so we are reduced to prove the following:

$$\Gamma_\tau = \inf_{\tau' < \tau} \Gamma_{\tau'} \tag{9.5}$$

for any $\tau < \Gamma_{\max}$. Take $\tau'' \in (\tau, \Gamma_{\max})$. Outside the polar locus of $\Gamma_{\tau''}$, we know that (9.5) holds by continuity of real-valued concave functions. So (9.5) holds everywhere by Proposition 1.2.6.

The final assertion follows from properness of a finite closed concave function on its effective domain. □

Definition 9.1.3 Let $\Gamma \in \mathrm{TC}(X,\theta)_{>0}$ and ω be a smooth closed real positive $(1,1)$-form. Then we define $P_{\theta+\omega}[\Gamma] \in \mathrm{TC}(X,\theta+\omega)_{>0}$ as follows:

(1) Define
$$P_{\theta+\omega}[\Gamma]_{\max} = \Gamma_{\max};$$

(2) for each $\tau < \Gamma_{\max}$, define
$$P_{\theta+\omega}[\Gamma]_\tau = P_{\theta+\omega}[\Gamma_\tau].$$

It follows from Proposition 3.1.10 that the family $P_{\theta+\omega}[\Gamma]$ is concave and decreasing. Moreover, since ω is positive,

$$\int_X (\theta+\omega+\mathrm{dd^c}\Gamma_\tau)^n \geq \int_X (\theta+\mathrm{dd^c}\Gamma_\tau)^n > 0$$

for every $\tau < \Gamma_{\max}$, and hence $P_{\theta+\omega}[\Gamma]_\tau$ is a model potential by Corollary 3.1.2. Thus $P_{\theta+\omega}[\Gamma] \in \mathrm{TC}(X,\theta+\omega)_{>0}$.

Proposition 9.1.5 *Let $\Gamma \in \mathrm{TC}(X,\theta)_{>0}$ and ω be a closed real smooth positive $(1,1)$-form on X. Then*

$$P_{\theta+\omega}[\Gamma]_{-\infty} = P_{\theta+\omega}[\Gamma_{-\infty}].$$

Proof This follows from Proposition 3.1.13. □

9.2 Ross–Witt Nyström correspondence

Let X be a connected compact Kähler manifold of dimension n and θ be a smooth closed real $(1,1)$-form on X representing a big cohomology class. Fix a model potential $\phi \in \mathrm{PSH}(X,\theta)_{>0}$.

Proposition 9.1.4 allows us to talk about the Legendre transforms of test curves in the expected way.

The general definition of the Legendre transform Definition A.2.1 can be translated as follows:

Definition 9.2.1 Let $\Gamma \in \mathrm{TC}(X,\theta;\phi)$. We define its *Legendre transform* as the map $\Gamma^*\colon (0,\infty) \to \mathrm{PSH}(X,\theta)$ given by

$$\Gamma^*_t = \sup_{\tau\in\mathbb{R}} (t\tau + \Gamma_\tau)\,[2]. \tag{9.6}$$

Thanks to Remark 9.1.1, (9.6) can be equivalently written as

$$\Gamma^*_t = \sup_{\tau<\Gamma_{\max}} (t\tau + \Gamma_\tau) = \sup_{\tau\leq\Gamma_{\max}} (t\tau + \Gamma_\tau)\,.$$

It is sometimes handy to *define*

$$\Gamma^*_0 := \phi \tag{9.7}$$

at $t = 0$. But it is important to remember that, with this convention, (9.6) is not true at $t = 0$ in general.

Remark 9.2.1 Here we do not discuss the case $t < 0$ because its behavior is straightforward: take $x \in X$; if $\Gamma_\tau(x) = -\infty$ for all $\tau < \Gamma_{\max}$, then $\Gamma^*_t(x) = -\infty$; otherwise, $\Gamma^*_t(x) = \infty$.

The information about $t > 0$ suffices to characterize Γ.

Proposition 9.2.1 *Let* $\Gamma \in \mathrm{TC}(X,\theta;\phi)$. *Then*

$$\Gamma_\tau = \inf_{t>0} \left(\Gamma^*_t - t\tau\right) \tag{9.8}$$

for all $\tau \in \mathbb{R}$.

Due to our convention (9.7), in (9.8) we may equivalently take $t \geq 0$.

Proof Fix $x \in X$. We want to establish (9.8) at x. We distinguish two cases. First suppose that $\Gamma_\tau(x) = -\infty$ for all $\tau < \Gamma_{\max}$ and hence all $\tau \in \mathbb{R}$. In this case, we have $\Gamma^*_t(x) = -\infty$ for all $t > 0$. Therefore, (9.8) follows trivially.

Otherwise, by Remark 9.2.1, we know that $\Gamma^*_t(x) = \infty$ for all $t < 0$. The relative interior of the domain of $t \mapsto \Gamma^*_t(x)$ is contained in $(0,\infty)$. Therefore, (9.8) follows from Theorem A.2.1, Proposition 9.1.4. □

[2] There is no usc regularization in the preceding formula. This is not a typo.

In Definition 9.2.1, we made the nontrivial claim that $\Gamma^*_t \in \mathrm{PSH}(X,\theta)$ for all $t > 0$. Let us prove this.

Lemma 9.2.1 *Let $\Gamma \in \mathrm{TC}(X,\theta;\phi)$. Then $\Gamma^*_t \in \mathrm{PSH}(X,\theta)$ for all $t > 0$. In fact, Γ^* is upper semicontinuous as a function of $X \times (0,\infty)$.*

Proof We first observe that for each $x \in X$, we have

$$\Gamma^*_t(x) \leq t\Gamma_{\max} < \infty.$$

Let $R = \{a + ib \in \mathbb{C} : a > 0, b \in \mathbb{R}\}$. We consider

$$F\colon X \times R \to [-\infty,\infty), \quad (x, a+ib) \mapsto \Gamma^*_a(x).$$

Let $\pi\colon X \times R \to X$ be the natural projection. Observe that the upper semicontinuous regularization G of F is $\pi^*\theta$-psh by Proposition 1.2.2. It suffices to show that $F = G$. We let

$$E := \{(x,z) \in X \times R : F(x,z) < G(x,z)\}.$$

We want to argue that $E = \varnothing$. Since F is invariant under translations in the imaginary direction of R, the same is true for its upper semicontinuous regularization G. So E can be written as $B \times i\mathbb{R}$ for some set $B \subseteq X \times (0,\infty)$. Since E is a pluripolar set by Proposition 1.2.5, it has zero Lebesgue measure. Hence, B has zero Lebesgue measure. For each $x \in X$, write

$$B_x = \{t \in (0,\infty) : (x,t) \in B\}.$$

By Fubini's theorem, B_x has vanishing 1-dimensional Lebesgue measure for all $x \in X \setminus Z$, where $Z \subseteq X$ is a subset of measure 0. We may assume that $Z \supseteq \{\Gamma_{\Gamma_{\max}} = -\infty\}$ so that for $x \in X \setminus Z$, $\Gamma^*_t(x) \neq -\infty$ for all $t > 0$.

For any $x \in X \setminus Z$, both $t \mapsto F(x,t)$ and $G(x,t)$ are convex functions with values in $\mathbb{R}$ on $(0,\infty)$. They agree almost everywhere, hence everywhere by their continuity. It follows that for $x \in X \setminus Z$, we have $B_x = \varnothing$.

By Proposition 9.2.1, for any $x \in X$, we have

$$\Gamma_\tau(x) = \inf_{t>0}(F(x,t) - t\tau), \quad \tau < \Gamma_{\max}.$$

On the other hand, let

$$\chi_\tau(x) = \inf_{t>0}(G(x,t) - t\tau), \quad \tau < \Gamma_{\max}, \quad x \in X. \tag{9.9}$$

By Kiselman's principle Proposition 1.2.8, $\chi_\tau \in \mathrm{PSH}(X,\theta)$. But on $X \setminus Z$, we already know that $\Gamma_\tau = \chi_\tau$ for all $\tau < \Gamma_{\max}$. By Proposition 1.2.6,

$$\Gamma_\tau = \chi_\tau, \quad \tau < \Gamma_{\max}.$$

Now we conclude that $F(x,t) = G(x,t)$ by Corollary A.2.1. □

Corollary 9.2.1 *Let $\Gamma \in \mathrm{TC}(X,\theta;\phi)$. Then $\Gamma^*_t \in \mathcal{E}(X,\theta;\phi)$ for all $t > 0$.*

Proof Fix $t > 0$. We already know that $\Gamma_t^* \in \mathrm{PSH}(X, \theta)$ by Lemma 9.2.1. It suffices to show that

$$\Gamma_t^* \sim_P \phi.$$

From (9.6) and Proposition 6.1.6, we know that

$$\Gamma_t^* \sim_P \sup_{\tau < \Gamma_{\max}} {}^*\Gamma_\tau = \phi.$$

This completes the proof. □

Lemma 9.2.2 *Let $\Gamma \in \mathrm{TC}(X, \theta; \phi)$. Then*

$$\sup_X \Gamma_t^* = t\Gamma_{\max}$$

for all $t > 0$.

In particular, $t \mapsto \Gamma_t^* - t\Gamma_{\max}$ is a decreasing function in $t > 0$.

Proof Since $\Gamma_\tau \leq 0$ for all $\tau < \Gamma_{\max}$, we have

$$\sup_X \Gamma_t^* \leq t\Gamma_{\max}$$

for all $t > 0$. Conversely, for each $\tau < \Gamma_{\max}$,

$$\sup_X \Gamma_t^* \geq \sup_X (t\tau + \Gamma_\tau) = t\tau.$$

Letting $\tau \to \Gamma_{\max}$ gives the reverse inequality. The final assertion follows directly from the formula

$$\Gamma_t^* - t\Gamma_{\max} = \sup_{\tau < \Gamma_{\max}} \Big(\Gamma_\tau - t(\Gamma_{\max} - \tau)\Big).$$

This completes the proof. □

Lemma 9.2.3 *Given $\Gamma \in \mathrm{TC}(X, \theta; \phi)$, we have $\Gamma^* \in \mathcal{R}(X, \theta; \phi)$.*

See Definition 4.2.2 for the notation $\mathcal{R}(X, \theta; \phi)$.

Proof It follows from Lemma 9.2.1, (9.6) and Proposition 1.2.2 that Γ^* is a subgeodesic ray. By Corollary 9.2.1, for any $t > 0$, $\Gamma_t^* \in \mathcal{E}(X, \theta; \phi)$.

First observe that as $t \to 0+$, we have

$$\Gamma_t^* \xrightarrow{L^1} \phi. \tag{9.10}$$

By Lemma 9.2.2 and Proposition 1.5.1, it suffices to show each L^1-cluster point $\psi \in \mathrm{PSH}(X, \theta)$ of Γ_t^* as $t \to 0$ is equal to ϕ.

To see this, first observe that by (9.6), for any fixed $t > 0$,

$$\Gamma_t^* \leq t\Gamma_{\max} + \phi.$$

Therefore, $\psi \leq \phi$. On the other hand, for any fixed $\tau < \Gamma_{\max}$, by (9.6), we have

$$\Gamma^*_t \geq \Gamma_\tau + t\tau$$

for any $t > 0$. So $\psi \geq \Gamma_\tau$ almost everywhere and hence everywhere by Proposition 1.2.6. It follows that $\psi \geq \phi$. Therefore, $\psi = \phi$.

Assume that Γ^* is not a geodesic ray. Then we can find $0 \leq a < b$ such that $(\Gamma^*_t)_{t\in(a,b)}$ differs from the geodesic $(\eta_t)_{t\in(a,b)}$ from Γ^*_a to Γ^*_b. The existence of $(\eta_t)_t$ is guaranteed by Proposition 4.2.1. We consider the subgeodesic $(\ell_t)_{t>0}$ given by $\ell_t = \eta_t$ for $t \in (a,b)$ and $\ell_t = \Gamma^*_t$ otherwise. Note that ℓ is a subgeodesic due to the gluing lemma Lemma 4.1.1.

Consider the Legendre transform

$$\Gamma'_\tau = \inf_{t>0}(\ell_t - t\tau), \quad \tau \in \mathbb{R}.$$

Then $\Gamma'_\tau \geq \Gamma_\tau$ and $\Gamma'_\tau \in \mathrm{PSH}(X,\theta) \cup \{-\infty\}$ by Proposition 1.2.8 for all $\tau \in \mathbb{R}$.

We claim that

$$\Gamma'_\tau \leq \Gamma_\tau + (b-a)(\Gamma_{\max} - \tau), \quad \tau \in \mathbb{R}. \tag{9.11}$$

Observe that $\Gamma'_\tau \equiv -\infty$ when $\tau > \Gamma_{\max}$ by Lemma 9.2.2. Thus it suffices to consider $\tau \leq \Gamma_{\max}$. Put

$$C_\tau := (b-a)(\Gamma_{\max} - \tau) \geq 0.$$

For any $s \in [a,b]$, the monotonicity of $t \mapsto \Gamma^*_t - t\Gamma_{\max}$ from Lemma 9.2.2 gives

$$\Gamma^*_b - b\tau \leq \Gamma^*_s - s\tau + C_\tau.$$

Since $\ell_b = \Gamma^*_b$, we get

$$\Gamma'_\tau \leq \Gamma^*_b - b\tau \leq \Gamma^*_s - s\tau + C_\tau, \quad s \in [a,b].$$

For $s \notin [a,b]$, we have $\ell_s = \Gamma^*_s$, hence

$$\Gamma'_\tau \leq \Gamma^*_s - s\tau \leq \Gamma^*_s - s\tau + C_\tau.$$

Taking the infimum over all $s > 0$ gives

$$\Gamma'_\tau \leq \Gamma_\tau + C_\tau.$$

Therefore, (9.11) follows. In particular, for any $\tau < \Gamma_{\max}$, we have $\Gamma'_\tau \sim \Gamma_\tau$. Moreover, $\sup_X \ell_t \leq t\Gamma_{\max}$ for all $t > 0$: this is clear outside (a,b) from Lemma 9.2.2, and on (a,b) it follows from the convexity of subgeodesics applied to the geodesic $(\eta_t)_{t\in(a,b)}$. Hence, for $\tau < \Gamma_{\max}$,

$$\sup_X(\ell_t - t\tau) \leq t(\Gamma_{\max} - \tau).$$

Letting $t \to 0+$ gives $\Gamma'_\tau \leq 0$. It follows from the fact that Γ_τ is a model potential that $\Gamma_\tau = \Gamma'_\tau$ for all $\tau < \Gamma_{\max}$. Therefore, by Theorem A.2.1, we have $\Gamma^*_t = \ell_t$ for all $t > 0$, which is a contradiction. □

Given $\ell \in \mathcal{R}(X, \theta; \phi)$, define its Legendre transform

$$\ell^*_\tau \coloneqq \inf_{t>0}(\ell_t - t\tau), \quad \tau \in \mathbb{R}. \tag{9.12}$$

Lemma 9.2.4 *Given $\ell \in \mathcal{R}(X, \theta; \phi)$, then $\ell^* = (\ell^*_\tau)_{\tau < \sup_X \ell_1} \in \mathrm{TC}(X, \theta)$.*

Proof Note that it follows from Proposition 1.2.8 that $\ell^*_\tau \in \mathrm{PSH}(X, \theta) \cup \{-\infty\}$ for all $\tau \in \mathbb{R}$. It is clear that $\mathbb{R} \ni \tau \mapsto \ell^*_\tau$ is a decreasing and concave function.

By Proposition 4.2.4,

$$\sup_X \ell_t = t \sup_X \ell_1 \quad \forall t \geq 0.$$

Set $C = \sup_X \ell_1$. By Proposition 4.2.4, $\sup_X(\ell_t - Ct) = 0$ for all $t > 0$. For each x outside a pluripolar set, the function $t \mapsto \ell_t(x) - Ct$ is convex on $(0, \infty)$ and bounded above by 0, hence it is decreasing. By Proposition 1.2.6, the same monotonicity holds as an inequality of θ-psh functions. Thus $(\ell_t - Ct)_{t>0}$ is a decreasing net in $\mathrm{PSH}(X, \theta)$.

Its infimum is not identically $-\infty$, since the family is normalized by $\sup_X(\ell_t - Ct) = 0$ and is compact in L^1 by Proposition 1.5.1. Therefore

$$\ell^*_C = \inf_{t>0}(\ell_t - Ct) \in \mathrm{PSH}(X, \theta)$$

by Proposition 1.2.2.

On the other hand, for $\tau > \sup_X \ell_1$, the same argument shows that

$$\ell^*_\tau \equiv -\infty.$$

Therefore, $\ell^*_\tau \in \mathrm{PSH}(X, \theta)$ if and only if $\tau \leq \ell^*_{\max} \coloneqq \sup_X \ell_1$.

We claim that $(\ell^*_\tau)_{\tau < \ell^*_{\max}}$ is a test curve. We first observe that for $\tau < \ell^*_{\max}$, we have

$$\ell^*_\tau \leq \ell_1 - \tau \sim_P \phi.$$

Therefore,

$$\ell^*_\tau \leq_P \phi, \quad \forall \tau < \ell^*_{\max}. \tag{9.13}$$

Also observe that for any $\tau \leq \ell^*_{\max}$ and any $t > 0$, we have

$$\sup_X \ell^*_\tau \leq \sup_X \ell_t - t\tau = \ell^*_{\max} t - t\tau.$$

Letting $t \to 0+$, we find that for any $\tau \leq \ell^*_{\max}$, we have

$$\sup_X \ell^*_\tau \leq 0. \tag{9.14}$$

Fix $\tau < \ell^*_{\max}$, we want to argue that

$$P_\theta\left[\ell^*_\tau\right] = \ell^*_\tau. \tag{9.15}$$

First we claim that for any $C > 0$, we have

$$(\ell^*_\tau + C) \wedge \phi = (\ell^*_\tau + C) \wedge V_\theta. \tag{9.16}$$

The $\leq$ direction is trivial. We argue the reverse inequality, which reduces to

$$\phi \geq (\ell^*_\tau + C) \wedge V_\theta.$$

Since ϕ is model and $(\ell^*_\tau + C) \wedge V_\theta \leq 0$, it suffices to show that

$$\phi \geq_P (\ell^*_\tau + C) \wedge V_\theta,$$

which follows from (9.13). Therefore, (9.16) is established. Thanks to (9.14), we have the obvious inequality

$$(\ell^*_\tau + C) \wedge V_\theta \geq \ell^*_\tau$$

for any $C > 0$. Therefore, in order to prove (9.15), it remains to argue that for any $C > 0$,

$$(\ell^*_\tau + C) \wedge \phi \leq \ell^*_\tau. \tag{9.17}$$

For this purpose, let us consider the following geodesics: For any $M > 0$ and $t \in [0, 1]$, let

$$\ell^{1,M}_t = \ell_{tM} - tM\tau, \quad \ell^{2,M}_t = (\ell^*_\tau + C) \wedge \phi - Ct.$$

It is clear that at $t = 0, 1$, we have $\ell^{2,M}_t \leq \ell^{1,M}_t$. Hence, the same holds for all $t \in [0, 1]$. In particular, for any fixed $s \in (0, 1]$, we have

$$(\ell^*_\tau + C) \wedge \phi - Cs \leq \ell_{sM} - sM\tau$$

for all $M > 0$. Taking infimum with respect to $M > 0$, we find

$$(\ell^*_\tau + C) \wedge \phi - Cs \leq \ell^*_\tau.$$

Since $s \in (0, 1]$ is arbitrary, we conclude (9.17).

Finally, we check the condition at $-\infty$. Set

$$\psi := \sup_{\tau < \ell^*_{\max}}{}^* \ell^*_\tau.$$

The family $(\ell^*_\tau)_{\tau < \ell^*_{\max}}$ is locally uniformly bounded from above, so $\psi \in \mathrm{PSH}(X, \theta)$. Outside a pluripolar set, the function $t \mapsto \ell_t(x)$ is a finite closed convex function on $[0, \infty)$ with value $\phi(x)$ at $t = 0$. To justify the pointwise statement at $t = 0$, choose a sequence $q_j \in \mathbb{Q}_{>0}$ with $q_j \searrow 0$. By Corollary 1.2.1 and Proposition 1.2.5, after removing a pluripolar subset of X, we have $\ell_{q_j}(x) \to \phi(x)$. Together with the convexity of $t \mapsto \ell_t(x)$, this shows that $t \mapsto \ell_t(x)$ is closed at 0 and has value $\phi(x)$ there.

Applying the one-variable Legendre duality to this function gives

$$\psi(x) = \sup_{\tau < \ell^*_{\max}} \inf_{t>0} \left(\ell_t(x) - t\tau\right) = \ell_0(x) = \phi(x).$$

Hence $\psi = \phi$ by Proposition 1.2.6. In particular, $\ell^*_{-\infty} = \phi$ has positive mass. □

Theorem 9.2.1 *The Legendre transform in Definition 9.2.1 is a bijection*

$$\mathrm{TC}(X, \theta; \phi) \xrightarrow{\sim} \mathcal{R}(X, \theta; \phi). \tag{9.18}$$

Moreover, this bijection restricts to the following bijections:

$$\mathrm{TC}^1(X, \theta; \phi) \xrightarrow{\sim} \mathcal{R}^1(X, \theta; \phi), \quad \mathrm{TC}^\infty(X, \theta; \phi) \xrightarrow{\sim} \mathcal{R}^\infty(X, \theta; \phi). \tag{9.19}$$

For any $\Gamma \in \mathrm{TC}^1(X, \theta; \phi)$, *we have*

$$\mathbf{E}^\phi(\Gamma) = \mathbf{E}^\phi(\Gamma^*). \tag{9.20}$$

Recall that the two energy functionals in (9.20) are defined in (9.4) and Definition 4.3.6 respectively.

The correspondence (9.18) will be referred to as the *Ross–Witt Nyström correspondence*.

To appreciate this result, just consider the simple case where θ is a Kähler form and $\phi = 0$. In this case, elements in $\mathcal{R}^\infty(X, \theta)$ are rays of *bounded* potentials, while elements in $\mathrm{TC}^\infty(X, \theta)$ are rays of *singular* potentials. This result establishes a bridge between the pluripotential theory of regular potentials and that of singular potentials!

Proof **Step 1**. We first establish (9.18).

It follows from Lemma 9.2.3 that the forward map is well-defined. The inverse map is given by (9.12). We show that the inverse map is also well-defined. Given $\ell \in \mathcal{R}(X, \theta; \phi)$, we know from Lemma 9.2.4 that $\ell^* \in \mathrm{TC}(X, \theta)$. We need to show that $\ell^* \in \mathrm{TC}(X, \theta; \phi)$.

The equality $\ell^*_{-\infty} = \phi$ was proved in Lemma 9.2.4. Therefore, $\ell^* \in \mathrm{TC}(X, \theta; \phi)$ as expected.

The two operations are inverse to each other thanks to Corollary A.2.1. Hence, (9.18) is established.

Step 2. Next we consider the bounded situation. Namely, we want to establish the second half of (9.19).

Suppose that $\Gamma \in \mathrm{TC}^\infty(X, \theta; \phi)$. Take $\tau_0 \in \mathbb{R}$ so that $\Gamma_\tau = \phi$ for all $\tau \leq \tau_0$. It follows from (9.6) that

$$\Gamma^*_t \geq \phi + t\tau_0$$

for all $t > 0$. Therefore, $\Gamma^*_t \sim \phi$ for all $t > 0$ and hence $\Gamma^* \in \mathcal{R}^\infty(X, \theta; \phi)$.

Conversely, suppose that $\ell \in \mathcal{R}^\infty(X, \theta; \phi)$. Thanks to Proposition 4.2.3, there is a constant $C > 0$ such that

$$\ell_t \geq \phi - Ct.$$

Therefore, according to (9.12), we have

$$\ell^*_\tau \geq \inf_{t>0}\Big(\phi - (C+\tau)t\Big) = \phi$$

if $\tau \leq -C$. Therefore, $\ell^*_\tau = \phi$ for all $\tau \leq -C$.

Step 3. We establish (9.20) and the first half of (9.19).

Step 3.1. We reduce to the case where $\Gamma_{\max} = 0$.

Suppose that we define

$$\Gamma'_\tau = \Gamma_{\tau+\Gamma_{\max}}, \quad \forall \tau < 0.$$

Then $\Gamma' \in \mathrm{TC}(X,\theta;\phi)$ as well and for all $t > 0$,

$$\Gamma'^*_t = \sup_{\tau<0}\left(t\tau + \Gamma'_\tau\right) = \sup_{\tau<\Gamma_{\max}}\left(t\tau + \Gamma_\tau\right) - t\Gamma_{\max} = \Gamma^*_t - t\Gamma_{\max}.$$

Therefore,

$$\mathbf{E}^\phi\left(\Gamma'^*\right) = \mathbf{E}^\phi\left(\Gamma^*\right) - \Gamma_{\max}\int_X \theta^n_\phi.$$

by (3.28). Using (9.4), we also have

$$\begin{aligned}\mathbf{E}^\phi(\Gamma') &= \int_{-\infty}^0 \left(\int_X \theta^n_{\Gamma'_\tau} - \int_X \theta^n_\phi\right) \mathrm{d}\tau\\ &= \int_{-\infty}^{\Gamma_{\max}} \left(\int_X \theta^n_{\Gamma_\tau} - \int_X \theta^n_\phi\right) \mathrm{d}\tau\\ &= \mathbf{E}^\phi(\Gamma) - \Gamma_{\max}\int_X \theta^n_\phi.\end{aligned}$$

Therefore, it suffices to establish (9.20) for Γ' in place of Γ.

Step 3.2. We assume that $\Gamma_{\max} = 0$ and $\Gamma \in \mathrm{TC}^\infty(X,\theta;\phi)$. We prove (9.20).

For $N \in \mathbb{Z}_{>0}$, $M \in \mathbb{Z}$, we introduce the following:

$$\Gamma^{*,N,M}_t := \max_{\substack{k\in\mathbb{Z}\\ k\leq M}}\left(\Gamma_{k/2^N} + tk/2^N\right) \in \mathcal{E}^\infty(X,\theta;\phi), \quad t > 0.$$

We first claim that for all $t > 0$, $N \in \mathbb{Z}_{>0}$ and $M \in \mathbb{Z}_{<0}$,

$$\frac{t}{2^N}\int_X \theta^n_{\Gamma_{(M+1)/2^N}} \leq E^\phi_\theta\left(\Gamma^{*,N,M+1}_t\right) - E^\phi_\theta\left(\Gamma^{*,N,M}_t\right) \leq \frac{t}{2^N}\int_X \theta^n_{\Gamma_{M/2^N}}. \tag{9.21}$$

Assuming this, let us prove (9.20).

Fixing N, let $M = \lfloor 2^N\Gamma_{\min}\rfloor$. Recall that $\Gamma_{\min}$ is defined in (9.3). Then repeated applications of (9.21) yield

$$\sum_{j=M+1}^{0} \frac{t}{2^N} \int_X \theta^n_{\Gamma_{j/2^N}} \leq E^\phi_\theta\left(\Gamma_t^{*,N,0}\right) - E^\phi_\theta\left(\Gamma_t^{*,N,M}\right) \leq \sum_{j=M}^{-1} \frac{t}{2^N} \int_X \theta^n_{\Gamma_{j/2^N}}.$$

Since $M \leq 2^N \Gamma_{\min}$, we have

$$\Gamma_t^{*,N,M} = \phi + tM/2^N.$$

Using (3.28), we can continue to write

$$\sum_{j=M+1}^{0} \frac{t}{2^N} \left(\int_X \theta^n_{\Gamma_{j/2^N}} - \int_X \theta^n_\phi\right) \leq E^\phi_\theta\left(\Gamma_t^{*,N,0}\right) \leq \sum_{j=M}^{-1} \frac{t}{2^N} \left(\int_X \theta^n_{\Gamma_{j/2^N}} - \int_X \theta^n_\phi\right).$$

We now notice that we have Riemann sums on both the left and right of the above inequality. Moreover, as $N \to \infty$, the potentials $\Gamma_t^{*,N,0}$ increase to Γ_t^*, hence their energies converge to $E^\phi_\theta(\Gamma_t^*)$ by Proposition 3.1.20. Using Proposition 9.1.1, it is possible to let $N \to \infty$ and obtain

$$E^\phi_\theta(\Gamma_t^*) = t\mathbf{E}^\phi(\Gamma) \tag{9.22}$$

So (9.20) follows as desired.

It remains to argue (9.21). Fix $t > 0$, $N \in \mathbb{Z}_{>0}$ and $M \in \mathbb{Z}$. By Proposition 3.1.19,

$$\begin{aligned}\int_X \left(\Gamma_t^{*,N,M+1} - \Gamma_t^{*,N,M}\right) \theta^n_{\Gamma_t^{*,N,M+1}} \leq & E^\phi_\theta\left(\Gamma_t^{*,N,M+1}\right) - E^\phi_\theta\left(\Gamma_t^{*,N,M}\right) \\ & \leq \int_X \left(\Gamma_t^{*,N,M+1} - \Gamma_t^{*,N,M}\right) \theta^n_{\Gamma_t^{*,N,M}}.\end{aligned} \tag{9.23}$$

Clearly $\Gamma_t^{*,N,M+1} \geq \Gamma_t^{*,N,M}$. Moreover, since $\mathbb{R} \ni \tau \mapsto \Gamma_\tau + t\tau$ is concave, we notice that

$$U_t := \left\{\Gamma_t^{*,N,M+1} > \Gamma_t^{*,N,M}\right\} = \left\{\Gamma_{(M+1)/2^N} + 2^{-N}t > \Gamma_{M/2^N}\right\},$$

and on U_t we have

$$\Gamma_t^{*,N,M+1} = \Gamma_{(M+1)/2^N} + t(M+1)/2^N, \quad \Gamma_t^{*,N,M} = \Gamma_{M/2^N} + tM/2^N. \tag{9.24}$$

We also note that U_t is $\mathcal{F}$-open by Corollary 1.3.4. So from the lower bound in (9.23), we have

$$\begin{aligned}E^\phi_\theta\left(\Gamma_t^{*,N,M+1}\right) - E^\phi_\theta\left(\Gamma_t^{*,N,M}\right) &\geq \int_{U_t} \left(\Gamma_t^{*,N,M+1} - \Gamma_t^{*,N,M}\right) \theta^n_{\Gamma_t^{*,N,M+1}} \\ &= \int_{U_t} \left(\Gamma_{(M+1)/2^N} - \Gamma_{M/2^N} + t2^{-N}\right) \theta^n_{\Gamma_{(M+1)/2^N}} \\ &\geq \int_{\left\{\Gamma_{(M+1)/2^N} = 0\right\}} t2^{-N}\, \theta^n_{\Gamma_{(M+1)/2^N}},\end{aligned}$$

where on the second line, we applied (9.24) and Proposition 2.2.1, on the third line, we applied the fact that $\theta^n_{\Gamma_{(M+1)/2^N}}$ is supported on the set (cf. Theorem 3.1.1)

$$\left\{\Gamma_{(M+1)/2^N}=0\right\}\subseteq U_t\cap\left\{\Gamma_{M/2^N}=0\right\}.$$

When $(M+1)/2^N=0$, we use Proposition 3.1.11 to see that $P_\theta[\Gamma_0]=\Gamma_0$, since Γ_0 is the decreasing limit of the model potentials Γ_τ as $\tau\nearrow 0$. Thus Theorem 3.1.1 applies to this endpoint as well.

We have deduced the first inequality in (9.21). Next, we apply the upper bound part in (9.23) and compute similarly

$$\begin{aligned}
E^\phi_\theta(\Gamma^{*,N,M+1}_t)-E^\phi_\theta(\Gamma^{*,N,M}_t)&\leq\int_X\left(\Gamma^{*,N,M+1}_t-\Gamma^{*,N,M}_t\right)\theta^n_{\Gamma^{*,N,M}_t}\\
&=\int_{U_t}\left(\Gamma_{(M+1)/2^N}-\Gamma_{M/2^N}+t2^{-N}\right)\theta^n_{\Gamma_{M/2^N}}\\
&\leq\int_{\{\Gamma_{M/2^N}=0\}\cap U_t}\left(\Gamma_{(M+1)/2^N}+t2^{-N}\right)\theta^n_{\Gamma_{M/2^N}}\\
&\leq\int_{\{\Gamma_{M/2^N}=0\}\cap U_t}t2^{-N}\,\theta^n_{\Gamma_{M/2^N}}.
\end{aligned}$$

We conclude the latter half of (9.21).

Step 3.3. We assume that $\Gamma_{\max}=0$. Now $\Gamma\in\mathrm{TC}(X,\theta;\phi)$ only.

For each $\epsilon>0$, we introduce $\Gamma^\epsilon\in\mathrm{TC}^\infty(X,\theta;\phi)$ as follows:

(1) Let $\Gamma^\epsilon_{\max}=0$, and
(2) we set

$$\Gamma^\epsilon_\tau=\begin{cases}\phi, & \text{if } \tau\leq-\epsilon^{-1};\\ P_\theta\left[(1+\epsilon\tau)\Gamma_\tau-\epsilon\tau\phi\right], & \text{if } \tau\in\left(-\epsilon^{-1},0\right).\end{cases}$$

The concavity and monotonicity in τ follow from Proposition 3.1.10. The value at $-\infty$ is ϕ. Moreover, $\Gamma^\epsilon_\tau=\phi$ for $\tau\leq-\epsilon^{-1}$, so $\Gamma^\epsilon\in\mathrm{TC}^\infty(X,\theta;\phi)$.

It follows from Corollary 6.2.9 and Corollary 6.2.5 that for each $\tau<0$, the potentials Γ^ϵ_τ decrease to Γ_τ as $\epsilon\searrow 0$. Applying Proposition 3.1.12, we have

$$\lim_{\epsilon\to 0+}\int_X\left(\theta+\mathrm{dd^c}\Gamma^\epsilon_\tau\right)^n=\int_X(\theta+\mathrm{dd^c}\Gamma_\tau)^n$$

for all $\tau<0$. Hence, by the monotone convergence theorem and Step 3.2, we find

$$\mathbf{E}^\phi(\Gamma)=\lim_{\epsilon\to 0+}\mathbf{E}^\phi(\Gamma^\epsilon)=\lim_{\epsilon\to 0+}\mathbf{E}^\phi(\Gamma^{\epsilon *})=\lim_{\epsilon\to 0+}E^\phi_\theta(\Gamma^{\epsilon *}_1),\tag{9.25}$$

where the last equality follows from (9.22). Furthermore, according to Proposition A.2.3, we have

$$\Gamma^*_t=\inf_{\epsilon>0}\Gamma^{\epsilon *}_t$$

for all $t > 0$. Note that we do not have to take the closure of the right-hand side since it is automatically upper semicontinuous in t.

Now suppose that $\Gamma \in \mathrm{TC}^1(X, \theta; \phi)$. Then by (9.25), as $\epsilon \to 0+$, $(\Gamma_t^{\epsilon *})_\epsilon$ is a decreasing Cauchy net in $\mathcal{E}^1(X, \theta; \phi)$ and hence by Theorem 4.3.3 for each $t > 0$,

$$E_\theta^\phi\left(\Gamma_t^*\right) = \lim_{\epsilon \to 0+} E_\theta^\phi\left(\Gamma_t^{\epsilon,*}\right) = t\mathbf{E}^\phi(\Gamma) > -\infty,$$

where we have applied (9.22) and (9.25). Hence, $\Gamma^* \in \mathcal{R}^1(X, \theta; \phi)$. Moreover, (9.20) follows.

Conversely, suppose that $\Gamma^* \in \mathcal{R}^1(X, \theta; \phi)$. Then (9.25) implies that

$$\mathbf{E}^\phi(\Gamma) = \lim_{\epsilon \to 0+} E_\theta^\phi\left(\Gamma_1^{\epsilon *}\right) \geq E_\theta^\phi\left(\Gamma_1^*\right) > -\infty.$$

Hence, $\Gamma \in \mathrm{TC}^1(X, \theta; \phi)$. □

Remark 9.2.2 One could also consider geodesic rays emanating from another potential $\Phi \in \mathcal{E}(X, \theta; \phi)$. In this case, one can show that these geodesic rays are in bijection with Φ-twisted test curves: In Definition 9.1.1, we replace (2) by the following condition:

$$P_\theta\left[\Gamma_\tau\right](\Phi) = \Gamma_\tau.$$

Furthermore, we require that $\Gamma_{-\infty} = \Phi$.

The above results equally work in the twisted setting. The proofs are almost identical to the untwisted case.

As an immediate consequence of the proof, we have

Corollary 9.2.2 *Let $\ell \in \mathcal{R}^1(X, \theta; \phi)$. Then $[0, \infty) \ni t \mapsto E_\theta^\phi(\ell_t)$ is linear.*

Proof This follows from the same argument as that of (9.25). □

Corollary 9.2.3 *Let $\ell \in \mathcal{R}(X, \theta; \phi)$. Then $\sup_X \ell_t = \ell_{\max}^* t$ for any $t \geq 0$.*
In particular, $\ell_t - \ell_{\max}^* t$ is a decreasing function of $t \geq 0$.

Proof This follows from Lemma 9.2.2 and Theorem 9.2.1. □

Example 9.2.1 Let us see what the test curve in Example 9.1.1 corresponds to under the Ross–Witt Nyström correspondence. Fix $\varphi \in \mathrm{PSH}(X, \theta)$. We claim that

$$\ell^\varphi = \Gamma^{\varphi *}, \tag{9.26}$$

where ℓ^φ is as in Example 4.3.1. We may assume that $\varphi \leq 0$ since both sides are invariant after adding a constant to φ.

We first prove the easy direction $\ell^\varphi \geq \Gamma^{\varphi *}$, which is equivalent to $\ell^{\varphi *} \geq \Gamma^\varphi$. Since $\ell^{\varphi *}$ is a test curve, the latter is equivalent to

$$\ell_\tau^{\varphi *} \geq (1 + \tau)\varphi - \tau V_\theta$$

for all $\tau \in (-1, 0)$. By Legendre duality, this is equivalent to

$$\ell_t^{\varphi} \geq \sup_{\tau \in (-1,0)} \Big((1+\tau)\varphi - \tau V_\theta + t\tau \Big) = \varphi \vee (V_\theta - t) \tag{9.27}$$

for all $t > 0$.

Using the notations of Example 4.3.1, we find easily that

$$\ell_t^{\varphi, C} \geq \varphi \vee (V_\theta - t)$$

for any $C > 0$ and $t \in [0, C]$, since it holds at $t = 0$ and $t = C$. Letting $C \to \infty$, we find (9.27). Therefore, $\ell^{\varphi} \geq \Gamma^{\varphi *}$ follows.

In order to prove the equality in (9.26), it suffices to show that the two sides have the same energy, as a consequence of (4.26). So we compute

$$\begin{aligned}
\mathbf{E}\left(\Gamma^{\varphi *}\right) =& \mathbf{E}\left(\Gamma^{\varphi}\right) \\
=& \int_{-1}^{0} \left(\int_X \theta^n_{(1+\tau)\varphi - \tau V_\theta} - \int_X \theta^n_{V_\theta} \right) \mathrm{d}\tau \\
=& \sum_{j=0}^{n} \binom{n}{j} \int_X \theta^j_{\varphi} \wedge \theta^{n-j}_{V_\theta} \int_0^1 \tau^j (1-\tau)^{n-j} \,\mathrm{d}\tau - \int_X \theta^n_{V_\theta} \\
=& \sum_{j=0}^{n} \binom{n}{j} \frac{j!(n-j)!}{(n+1)!} \int_X \theta^j_{\varphi} \wedge \theta^{n-j}_{V_\theta} - \int_X \theta^n_{V_\theta} \\
=& \mathbf{E}(\ell^{\varphi}),
\end{aligned}$$

where we used the value of the β-function[3] on the fourth line, and the last line is just (4.32).

The multiplier ideal sheaves of a test curve can be characterized using the corresponding geodesic ray in a very simple manner.

Proposition 9.2.2 (He–Testorf–Wang) *Let $\ell \in \mathcal{R}(X, \theta; \phi)$. Given any $\tau < \ell^*_{\max}$ and $x \in X$, we have*

$$\mathcal{I}\left(\ell^*_\tau\right)_x = \left\{ f \in \mathcal{O}_{X,x} : |f|^2 \int_0^\infty \exp(-\ell_t + t\tau)\,\mathrm{d}t \text{ is integrable near } x \right\}. \tag{9.28}$$

Proof Fix $x \in X$, $\tau < \ell^*_{\max}$ and $f \in \mathcal{O}_{X,x}$. Fix a Kähler form ω on X.

Step 1. We first assume that f lies in the right-hand side of (9.28).

Given any $y \in X$, it follows from (9.12) that there is $t_0 > 0$ with

$$\ell^*_\tau(y) + 1 \geq \ell_{t_0}(y) - t_0\tau.$$

Observe that $t \mapsto \ell_t - t\ell^*_{\max}$ is decreasing in t by Corollary 9.2.3, it follows that for $t \in [t_0, t_0 + 1]$, we have

$$\ell^*_\tau(y) + 1 - t_0(\ell^*_{\max} - \tau) \geq \ell_{t_0}(y) - t_0\ell^*_{\max} \geq \ell_t(y) - t\ell^*_{\max}.$$

[3] Also known as Euler integral of the first kind.

Since $\tau < \ell^*_{\max}$ and $t_0 \geq t-1$, we deduce that

$$\ell^*_\tau(y) + 1 + \ell^*_{\max} - \tau \geq \ell_t(y) - t\tau, \quad t \in [t_0, t_0+1]. \tag{9.29}$$

Take a sufficiently small open neighborhood U of x such that

$$\int_U |f|^2 \int_0^\infty \exp(-\ell_t + t\tau)\, \mathrm{d}t\, \omega^n < \infty.$$

Applying (9.29), we deduce that

$$\int_U |f|^2 \exp\left(-\ell^*_\tau\right)\, \omega^n < \infty.$$

Therefore, $f \in \mathcal{I}(\ell^*_\tau)_x$.

Step 2. Assume that $f \in \mathcal{I}(\ell^*_\tau)_x$.

It follows from the strong openness theorem Theorem 1.4.4 that $f \in \mathcal{I}(\ell^*_{\tau+\epsilon})_x$ for some small enough $\epsilon > 0$ with $\tau + \epsilon < \ell^*_{\max}$. Take a sufficiently small open neighborhood U of x such that

$$\int_U |f|^2 \exp\left(-\ell^*_{\tau+\epsilon}\right)\, \omega^n < \infty.$$

We compute

$$\begin{aligned}\int_U |f|^2 \int_0^\infty \exp\left(-\ell_t + t\tau\right)\, \mathrm{d}t\, \omega^n \leq & \int_U |f|^2 \int_0^\infty \exp\left(-\ell^*_{\tau+\epsilon} - t\epsilon\right)\, \mathrm{d}t\, \omega^n \\ = & \frac{1}{\epsilon} \int_U |f|^2 \exp\left(-\ell^*_{\tau+\epsilon}\right)\, \omega^n \\ < & \infty.\end{aligned}$$

Therefore, f lies in the right-hand side of (9.28). □

The masses of potentials on a test curve can also be expressed in terms of the corresponding geodesic ray.

Proposition 9.2.3 (Hisamoto) *Let $\ell \in \mathcal{R}(X, \theta; \phi)$. Given any $\tau < \ell^*_{\max}$, we have*

$$\int_X \left(\theta + \mathrm{dd^c} \ell^*_\tau\right)^n = \int_{\{\dot{\ell}_0 \geq \tau\}} \theta^n_\phi. \tag{9.30}$$

Here $\dot{\ell}_0$ denotes the right-derivative of ℓ_t with respect to t at $t = 0$. It is well-defined quasi-everywhere, and hence the right-hand side of (9.30) makes sense.

Proof Fix $\tau < \ell^*_{\max}$. We first observe that it suffices to show that

$$\int_{\{\dot{\ell}_0 \geq \tau\}} \theta^n_\phi = \int_{\{\ell^*_\tau = \phi\}} \theta^n_\phi. \tag{9.31}$$

Assuming (9.31) for the moment. By Theorem 9.2.1, $\ell^* \in \mathrm{TC}(X,\theta;\phi)$. In particular, ℓ^*_τ is a model potential and

$$\ell^*_\tau \le \ell^*_{-\infty} = \phi.$$

Since ϕ is model as well, Proposition 3.1.7 gives

$$\theta^n_{\ell^*_\tau} = \mathbb{1}_{\{\ell^*_\tau = 0\}}\,\theta^n, \quad \theta^n_\phi = \mathbb{1}_{\{\phi=0\}}\,\theta^n.$$

Since $\ell^*_\tau \le \phi \le 0$, we have

$$\mathbb{1}_{\{\ell^*_\tau=\phi\}}\,\theta^n_\phi = \mathbb{1}_{\{\ell^*_\tau=\phi=0\}}\,\theta^n = \mathbb{1}_{\{\ell^*_\tau=0\}}\,\theta^n = \theta^n_{\ell^*_\tau}.$$

Therefore, (9.30) follows from (9.31).

In order to prove (9.31), it suffices to show that

$$\left\{\dot{\ell}_0 \ge \tau\right\} = \left\{\ell^*_\tau = \phi\right\} \quad \text{outside a pluripolar set.} \tag{9.32}$$

Take $x \in X$ so that $(\ell_t(x))_{t\ge 0}$ is finite and right-differentiable at $t = 0$. Note that this condition holds quasi-everywhere. Suppose that $\ell^*_\tau(x) = \phi(x)$. Then

$$\dot{\ell}_0(x) = \inf_{t>0} \frac{\ell_t(x) - \phi(x)}{t} \ge \inf_{t>0} \frac{\ell^*_\tau(x) + t\tau - \phi(x)}{t} = \tau.$$

Therefore, the $\supseteq$ direction in (9.32) follows. Conversely, suppose that $\dot{\ell}_0(x) \ge \tau$, then

$$\ell^*_\tau(x) = \inf_{t>0}\left(\ell_t(x) - t\tau\right) \ge \inf_{t>0}\left(\phi(x) + t\dot{\ell}_0(x) - t\tau\right) \ge \phi(x).$$

Together with $\ell^*_\tau \le \phi$, this gives $\ell^*_\tau(x) = \phi(x)$. Therefore, the $\subseteq$ direction in (9.32) follows as well. $\square$

Corollary 9.2.4 *Let $\ell \in \mathcal{R}^1(X,\theta;\phi)$. Then $\dot{\ell}_0 \in L^1(X,\theta^n_\phi)$ and*

$$\mathbf{E}^\phi(\ell) = \int_X \dot{\ell}_0\,\theta^n_\phi. \tag{9.33}$$

In particular, for every $t \ge 0$,

$$E^\phi_\theta(\ell_t) = t\int_X \dot{\ell}_0\,\theta^n_\phi.$$

Proof Set $\Gamma = \ell^*$. By Theorem 9.2.1, we have $\Gamma \in \mathrm{TC}^1(X,\theta;\phi)$ and

$$\mathbf{E}^\phi(\ell) = \mathbf{E}^\phi(\Gamma).$$

By Proposition 9.2.3, for every $\tau < \ell^*_{\max}$,

$$\int_X \theta^n_{\Gamma_\tau} = \int_{\{\dot{\ell}_0 \ge \tau\}} \theta^n_\phi.$$

Hence (9.4) gives

$$\mathbf{E}^{\phi}(\ell) = \ell^*_{\max} \int_X \theta^n_\phi - \int_{-\infty}^{\ell^*_{\max}} \int_{\{\dot{\ell}_0 < \tau\}} \theta^n_\phi \, \mathrm{d}\tau.$$

For $\tau > \ell^*_{\max}$, we have $\ell^*_\tau \equiv -\infty$ by Lemma 9.2.4. Then (9.31) shows that

$$\int_{\{\dot{\ell}_0 \geq \tau\}} \theta^n_\phi = 0.$$

The layer-cake formula therefore gives (9.33). The final assertion follows from Corollary 9.2.2, since $E^{\phi}_{\theta}(\ell_0) = E^{\phi}_{\theta}(\phi) = 0$. □

9.3 $\mathcal{I}$-model test curves

Let X be a connected compact Kähler manifold of dimension n and θ be a smooth closed real $(1,1)$-form on X representing a big cohomology class. Fix a model potential $\phi \in \mathrm{PSH}(X,\theta)_{>0}$.

Definition 9.3.1 A test curve $\Gamma \in \mathrm{TC}(X,\theta;\phi)$ is *$\mathcal{I}$-model* if for any $\tau < \Gamma_{\max}$, the potential Γ_τ is $\mathcal{I}$-model.

The subset of $\mathcal{I}$-model test curves in $\mathrm{TC}(X,\theta;\phi)$ is denoted by $\mathcal{E}^{\mathrm{NA}}(X,\theta;\phi)$. When $\phi = V_\theta$, we omit ϕ and write $\mathcal{E}^{\mathrm{NA}}(X,\theta)$ instead.

The union of the sets of $\mathcal{I}$-model test curves in $\mathrm{PSH}(X,\theta)$ for all model potentials $\phi \in \mathrm{PSH}(X,\theta)_{>0}$ is denoted by $\mathrm{PSH}^{\mathrm{NA}}(X,\theta)_{>0}$.

Note that $\Gamma_{\Gamma_{\max}}$ is automatically $\mathcal{I}$-model by Proposition 3.2.13.

Here we write NA with non-Archimedean in mind. The precise relation with non-Archimedean pluripotential theory will be clear in Chapter 13. This section and the next can be skipped on a first reading; the necessary results can be consulted when reading Chapter 13.

Proposition 9.3.1 *Let $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)_{>0}$. Then $\Gamma_{-\infty}$ is an $\mathcal{I}$-model potential.*

Proof This follows from Proposition 3.2.14. □

Proposition 9.3.2 *Let θ' be another smooth closed real $(1,1)$-form on X representing the same cohomology class as θ. Then there is a canonical bijection*

$$\mathrm{PSH}^{\mathrm{NA}}(X,\theta)_{>0} \xrightarrow{\sim} \mathrm{PSH}^{\mathrm{NA}}(X,\theta')_{>0}.$$

This bijection satisfies the obvious cocycle condition.

Proof Let $\theta' = \theta + \mathrm{dd^c} g$. By Proposition 9.1.2, the canonical map sends Γ to the test curve with levels

$$\Gamma'_\tau = P_{\theta'}[\Gamma_\tau - g].$$

If Γ_τ is $\mathcal{I}$-model, then it is $\mathcal{I}$-good by Example 7.1.2. Note that $\Gamma_\tau - g$ is $\mathcal{I}$-good since Γ_τ is. Hence

$$P_{\theta'}[\Gamma_\tau - g] = P_{\theta'}[\Gamma_\tau - g]_{\mathcal{I}},$$

so Γ'_τ is $\mathcal{I}$-model. The same argument applied to $-g$ gives the inverse map, and the cocycle condition follows from the one in Proposition 9.1.2. □

Proposition 9.3.3 *Let $\pi\colon Y \to X$ be a proper bimeromorphic morphism from a Kähler manifold Y. Then the pointwise pull-back induces a bijection*

$$\pi^*\colon \mathcal{E}^{\mathrm{NA}}(X,\theta;\phi) \xrightarrow{\sim} \mathcal{E}^{\mathrm{NA}}(Y,\pi^*\theta;\pi^*\phi).$$

Proof This is an immediate consequence of Proposition 9.1.3 and Proposition 3.2.5. □

Definition 9.3.2 Given $\Gamma \in \mathrm{TC}(X,\theta;\phi)$, we define its *$\mathcal{I}$-envelope* $P_\theta[\Gamma]_{\mathcal{I}}$ as the map

$$(-\infty,\Gamma_{\max}) \to \mathrm{PSH}(X,\theta), \quad \tau \mapsto P_\theta\left[\Gamma_\tau\right]_{\mathcal{I}}.$$

More generally, for any closed real smooth positive $(1,1)$-form ω on X, we define $P_{\theta+\omega}[\Gamma]_{\mathcal{I}}$ as the map

$$(-\infty,\Gamma_{\max}) \to \mathrm{PSH}(X,\theta+\omega), \quad \tau \mapsto P_{\theta+\omega}\left[\Gamma_\tau\right]_{\mathcal{I}}.$$

Proposition 9.3.4 *Let $\Gamma \in \mathrm{TC}(X,\theta;\phi)$. Then*

$$P_\theta[\Gamma]_{\mathcal{I}} \in \mathcal{E}^{\mathrm{NA}}(X,\theta;P_\theta[\phi]_{\mathcal{I}}).$$

More generally, for any closed real smooth positive $(1,1)$-form ω on X, we have

$$P_{\theta+\omega}[\Gamma]_{\mathcal{I}} \in \mathcal{E}^{\mathrm{NA}}(X,\theta+\omega;P_{\theta+\omega}[\phi]_{\mathcal{I}}).$$

Proof We prove the first assertion; the second is identical, replacing θ by $\theta+\omega$.

Set

$$\widetilde{\Gamma}_\tau \coloneqq P_\theta[\Gamma_\tau]_{\mathcal{I}}.$$

The family is decreasing by Lemma 3.2.1. If $\tau = (1-s)\tau_0 + s\tau_1$, then the concavity of Γ gives

$$\Gamma_\tau \geq (1-s)\Gamma_{\tau_0} + s\Gamma_{\tau_1}.$$

Hence, by Lemma 3.2.1 and Proposition 3.2.11,

$$\widetilde{\Gamma}_\tau \geq (1-s)\widetilde{\Gamma}_{\tau_0} + s\widetilde{\Gamma}_{\tau_1}.$$

Thus the family is concave. Finally,

$$\sup_{\tau<\Gamma_{\max}}{}^*\widetilde{\Gamma}_\tau = P_\theta[\phi]_{\mathcal{I}}$$

follows from Proposition 3.2.14. Since $P_\theta[\phi]_{\mathcal{I}} \geq \phi$, its mass is positive by the monotonicity theorem Theorem 2.4.3. This proves the assertion. □

Definition 9.3.3 Let $\phi \in \mathrm{PSH}(X, \theta)_{>0}$ be a model potential. A geodesic ray $\ell \in \mathcal{R}(X, \theta; \phi)$ is *maximal* if ℓ^* is $\mathcal{I}$-model, namely, if $\ell^* \in \mathcal{E}^{\mathrm{NA}}(X, \theta; \phi)$.

An important class of $\mathcal{I}$-model test curves is given by filtrations. We briefly recall the corresponding terminology.

Definition 9.3.4 Let L be a big line bundle. We write

$$R(X, L) = \bigoplus_{k=0}^{\infty} \mathrm{H}^0(X, L^k)$$

for the section ring[4] of L.

A *filtration* on $R(X, L)$ is a decreasing family of graded linear subspaces $(\mathcal{F}^\lambda)_{\lambda \in \mathbb{R}}$ of $R(X, L)$ with graded pieces

$$\mathcal{F}^\lambda = \bigoplus_{k=0}^{\infty} \mathcal{F}_k^\lambda,$$

such that the following conditions are satisfied:

- The filtration is left-continuous: For any $\lambda \in \mathbb{R}$, we have
$$\mathcal{F}^\lambda = \bigcap_{\lambda' < \lambda} \mathcal{F}^{\lambda'};$$
- the filtration is multiplicative: For any $\lambda, \lambda' \in \mathbb{R}$ and any $k, k' \in \mathbb{N}$, we have
$$\mathcal{F}_k^\lambda \cdot \mathcal{F}_{k'}^{\lambda'} \subseteq \mathcal{F}_{k+k'}^{\lambda+\lambda'};$$
- there is an integer $C > 0$ such that
$$\mathcal{F}_m^{Cm} = 0, \quad \mathcal{F}_m^{-Cm} = \mathrm{H}^0(X, L^m) \tag{9.34}$$
for all $m \in \mathbb{Z}_{>0}$.

Given a filtration $\mathcal{F}$ on $R(X, L)$, we define

$$\tau_k(\mathcal{F}) = \sup \left\{ \lambda \in \mathbb{R} : \mathcal{F}_k^\lambda \neq 0 \right\}.$$

By Fekete's lemma, we can introduce

$$\tau(\mathcal{F}) = \lim_{k \to \infty} \frac{1}{k} \tau_k(\mathcal{F}) = \sup_{k \in \mathbb{Z}_{>0}} \frac{1}{k} \tau_k(\mathcal{F}).$$

Note that $\tau(\mathcal{F})$ is bounded from above by the constant C in (9.34), hence finite.

[4] It is more natural to replace the direct sum by a disjoint union, leading to the notion of ringoids (*annénoïdes* in French), introduced by Ducros in [Duc21] in the context of Temkin's graded reduction of Berkovich germs.

Example 9.3.1 Let L be a big line bundle on X and $\mathcal{F}$ be a filtration on $R(X, L)$. Fix a smooth Hermitian metric h on L and write $\theta = c_1(L, h)$.

We introduce a few auxiliary functions. For each $k \in \mathbb{Z}_{>0}$, we introduce

$$\Gamma_\tau^{\mathcal{F},k} := \sup{}^* \left\{ \log |s|^2_{h^k} : s \in \mathcal{F}_k^{k\tau}, |s|^2_{h^k} \leq 1 \right\}.$$

When $k\tau \leq \tau_k(\mathcal{F})$, we know that $\mathcal{F}_k^{k\tau} \neq 0$. Moreover, Proposition 1.8.1 and Proposition 1.2.2 imply that

$$\Gamma_\tau^{\mathcal{F},k} \in \mathrm{PSH}(X, k\theta), \quad \tau \leq k^{-1}\tau_k(\mathcal{F}).$$

Observe that for $k, k' \in \mathbb{Z}_{>0}$, we have

$$\Gamma_\tau^{\mathcal{F},k+k'} \geq \Gamma_\tau^{\mathcal{F},k} + \Gamma_\tau^{\mathcal{F},k'}.$$

In particular, by Fekete's lemma,

$$\lim_{k \in \mathbb{Z}_{>0}} \frac{1}{k} \Gamma_\tau^{\mathcal{F},k} = \sup_{k \in \mathbb{Z}_{>0}} \frac{1}{k} \Gamma_\tau^{\mathcal{F},k} \tag{9.35}$$

exists for any $\tau < \tau(\mathcal{F})$.

We define $(\Gamma_\tau^{\mathcal{F}})_{\tau < \tau(\mathcal{F})}$ as follows:

$$\Gamma_\tau^{\mathcal{F}} := P_\theta \left[\sup_{k \in \mathbb{Z}_{>0}}{}^* \frac{1}{k} \Gamma_\tau^{\mathcal{F},k} \right]^{5}.$$

We claim that $\Gamma^{\mathcal{F}} \in \mathcal{E}^{\mathrm{NA}}(X, \theta)$ and is bounded.

It is clear that $(-\infty, \tau(\mathcal{F})) \ni \tau \mapsto \Gamma_\tau^{\mathcal{F}}$ is decreasing. We prove its concavity. By Proposition 3.1.10, it suffices to show that

$$(-\infty, \tau(\mathcal{F})) \ni \tau \mapsto \sup_{k \in \mathbb{Z}_{>0}}{}^* \frac{1}{k} \Gamma_\tau^{\mathcal{F},k}$$

is concave. In other words, we need to prove the following: Given $\tau_0 < \tau_1 < \tau(\mathcal{F})$ and $t \in (0, 1)$, we have

$$\sup_{k \in \mathbb{Z}_{>0}}{}^* \frac{1}{k} \Gamma_{t\tau_1 + (1-t)\tau_0}^{\mathcal{F},k} \geq t \sup_{k \in \mathbb{Z}_{>0}}{}^* \frac{1}{k} \Gamma_{\tau_1}^{\mathcal{F},k} + (1-t) \sup_{k \in \mathbb{Z}_{>0}}{}^* \frac{1}{k} \Gamma_{\tau_0}^{\mathcal{F},k}.$$

But thanks to Proposition 1.2.6 and Proposition 1.2.5, it suffices to show that

$$\sup_{k \in \mathbb{Z}_{>0}} \frac{1}{k} \Gamma_{t\tau_1 + (1-t)\tau_0}^{\mathcal{F},k} \geq t \sup_{k \in \mathbb{Z}_{>0}} \frac{1}{k} \Gamma_{\tau_1}^{\mathcal{F},k} + (1-t) \sup_{k \in \mathbb{Z}_{>0}} \frac{1}{k} \Gamma_{\tau_0}^{\mathcal{F},k}$$

[5] It is not clear if $P_\theta[\bullet]$ is necessary here. When L is ample, it is shown in [RWN14, Proposition 7.11] that it is not necessary. The proof in the reference relies on a Skoda division theorem [RWN14, Theorem 7.10], which is not known in the case of big line bundles.

for all $t \in (0,1)$. Take $s_i \in \mathcal{F}_{k_i}^{k_i\tau_i}$ for $i = 0, 1$ with $|s_i|^2_{h^{k_i}} \leq 1$, where $k_0, k_1 \in \mathbb{Z}_{>0}$. We need to prove that

$$\sup_{k\in\mathbb{Z}_{>0}} \frac{1}{k}\Gamma^{\mathcal{F},k}_{t\tau_1+(1-t)\tau_0} \geq \frac{1-t}{k_0}\log|s_0|^2_{h^{k_0}} + \frac{t}{k_1}\log|s_1|^2_{h^{k_1}}. \tag{9.36}$$

Approximate t by rational numbers from above, we may reduce to the case where $t \in \mathbb{Q}$. Write $t = p/q$ with $p, q \in \mathbb{Z}_{>0}$. Then

$$s := s_0^{k_1(q-p)} \otimes s_1^{k_0 p} \in \mathcal{F}_{k_0k_1q}^{k_0k_1\tau_0(q-p)+k_0k_1\tau_1 p},$$

and

$$\begin{aligned}
&\frac{1}{k_0k_1q}\log|s|^2_{h^{k_0k_1q}}\\
=&\frac{1}{k_0k_1q}\left(k_1(q-p)\log|s_0|^2 + k_0p\log|s_1|^2\right)\\
=&\frac{1-t}{k_0}\log|s_0|^2_{h^{k_0}} + \frac{t}{k_1}\log|s_1|^2_{h^{k_1}}.
\end{aligned}$$

So (9.36) follows.

Note that for each $k \in \mathbb{Z}_{>0}$, $\tau \leq k^{-1}\tau_k(\mathcal{F})$, we know that $k^{-1}\Gamma^{\mathcal{F},k}_\tau$ is $\mathcal{I}$-good by Proposition 7.2.2. The family is bounded above by 0, so it follows from the same proposition that the obstacle in the definition of $\Gamma^{\mathcal{F}}_\tau$ is $\mathcal{I}$-good. Hence the P- and $\mathcal{I}$-envelopes of this obstacle agree, so $\Gamma^{\mathcal{F}}_\tau$ is $\mathcal{I}$-model for each $\tau < \tau(\mathcal{F})$.

It remains to show that the test curve $\Gamma^{\mathcal{F}}$ is bounded and lies in $\mathcal{E}^{\mathrm{NA}}(X,\theta)$. Fix $\tau \leq -C$, where C is as in (9.34), we will show that

$$\Gamma^{\mathcal{F}}_\tau = V_\theta. \tag{9.37}$$

This follows from the Bergman kernel technique. Based on the theory developed so far, we can also give an elementary argument.

Fix $k \in \mathbb{Z}_{>0}$. Since $\tau \leq -C$, we have $\mathcal{F}_k^{k\tau} = \mathrm{H}^0(X, L^k)$. Let $s \in \mathrm{H}^0(X, L^k)$. After multiplying s by a constant, we may assume that $|s|^2_{h^k} \leq 1$. By the definition of $\Gamma^{\mathcal{F},k}_\tau$,

$$\Gamma^{\mathcal{F},k}_\tau \geq \log|s|^2_{h^k}.$$

Hence

$$s \in \mathrm{H}^0\left(X, L^k \otimes \mathcal{I}(\Gamma^{\mathcal{F},k}_\tau)\right).$$

Since

$$k\Gamma^{\mathcal{F}}_\tau \geq \Gamma^{\mathcal{F},k}_\tau,$$

we get

$$s \in \mathrm{H}^0\left(X, L^k \otimes \mathcal{I}(k\Gamma^{\mathcal{F}}_\tau)\right).$$

Thus

$$\mathrm{H}^0\left(X, L^k \otimes \mathcal{I}(k\Gamma^{\mathcal{F}}_\tau)\right) = \mathrm{H}^0(X, L^k)$$

for every $k > 0$. By Theorem 7.4.1,

$$\mathrm{vol}\left(\theta + \mathrm{dd^c}\Gamma_\tau^{\mathcal{F}}\right) = \mathrm{vol}\, L.$$

As $\Gamma_\tau^{\mathcal{F}}$ is $\mathcal{I}$-model, this volume is equal to $\int_X \left(\theta + \mathrm{dd^c}\Gamma_\tau^{\mathcal{F}}\right)^n$. Hence $\Gamma_\tau^{\mathcal{F}}$ has full Monge–Ampère mass. Since it is also model, it must be equal to V_θ. This proves (9.37).

Remark 9.3.1 There is an important special case of Example 9.3.1: Suppose that L is ample and $\mathcal{F}$ is the filtration induced by a smooth test configuration $(\mathcal{X}, \mathcal{L})$ of (X, L). Then the geodesic ray $\Gamma^{\mathcal{F}*}$ is exactly the Phong–Sturm geodesic ray associated with $(\mathcal{X}, \mathcal{L})$. See [RWN14, Section 9].

Remark 9.3.2 We deduce from Example 9.3.1 that the ray $\Gamma^{\mathcal{F}*}$ induced by a filtration $\mathcal{F}$ is maximal.

9.4 Operations on test curves

Let X be a connected compact Kähler manifold of dimension n and $\theta, \theta', \theta''$ be smooth closed real $(1,1)$-forms on X representing big cohomology classes.

In this section, we develop several general operations on test curves, anticipating the applications in non-Archimedean geometry in Chapter 13. One may read Chapter 13 first and consult this section when necessary.

Definition 9.4.1 Given $\Gamma \in \mathrm{TC}(X,\theta)_{>0}$, $\Gamma' \in \mathrm{TC}(X,\theta')_{>0}$, we say $\Gamma \leq \Gamma'$ if $\Gamma_{\max} \leq \Gamma'_{\max}$ and, for all $\tau < \Gamma_{\max}$, we have

$$\Gamma_\tau \leq_P \Gamma'_\tau. \tag{9.38}$$

Observe that (9.38) actually holds for all $\tau \in \mathbb{R}$ if $\theta = \theta'$. Here we use the convention that $-\infty \leq_P \varphi$ for any $\varphi \in \mathrm{QPSH}(X) \cup \{-\infty\}$. The relation $\leq$ defines a partial order on $\mathrm{TC}(X,\theta)_{>0}$.

Lemma 9.4.1 *Let $\Gamma, \Gamma' \in \mathrm{TC}(X,\theta)_{>0}$ and ω be a closed real smooth positive $(1,1)$-form on X. Then the following are equivalent:*

(1) $\Gamma \leq \Gamma'$;
(2) $P_{\theta+\omega}[\Gamma] \leq P_{\theta+\omega}[\Gamma']$.

Proof This follows from Example 6.1.1. □

Definition 9.4.2 Let $\Gamma \in \mathrm{TC}(X,\theta)_{>0}$ and $\Gamma' \in \mathrm{TC}(X,\theta')_{>0}$. Then we define $\Gamma + \Gamma' \in \mathrm{TC}(X, \theta+\theta')_{>0}$ as follows:

(1) We set

$$(\Gamma + \Gamma')_{\max} := \Gamma_{\max} + \Gamma'_{\max};$$

(2) for any $\tau < (\Gamma + \Gamma')_{\max}$, we define[6]

$$(\Gamma + \Gamma')_\tau := P_{\theta+\theta'}\left[\sup_{\delta\in\mathbb{R}} \left(\Gamma_\delta + \Gamma'_{\tau-\delta}\right)\right]. \tag{9.39}$$

Lemma 9.4.2 *Let $\Gamma \in \mathrm{TC}(X,\theta)_{>0}$ and $\Gamma' \in \mathrm{TC}(X,\theta')_{>0}$. Then for any $\tau < (\Gamma + \Gamma')_{\max}$, we have*

$$\sup_{\delta\in\mathbb{R}} \left(\Gamma_\delta + \Gamma'_{\tau-\delta}\right) \in \mathrm{PSH}(X, \theta + \theta').$$

This potential is $\mathcal{I}$-good if $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)_{>0}$ and $\Gamma' \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta')_{>0}$.

In particular, (9.39) in Definition 9.4.2 makes sense.

Proof Let

$$\eta_\tau = \sup_{\delta\in\mathbb{R}} \left(\Gamma_\delta + \Gamma'_{\tau-\delta}\right) = \sup_{\tau-\Gamma'_{\max}<\delta<\Gamma_{\max}} \left(\Gamma_\delta + \Gamma'_{\tau-\delta}\right)$$

for all $\tau \in \mathbb{R}$. Set

$$Z = \{x \in X : \Gamma_v(x) = -\infty \,\forall v \in \mathbb{R}\} \cup \left\{x \in X : \Gamma'_v(x) = -\infty \,\forall v \in \mathbb{R}\right\}.$$

It follows from Proposition A.2.4 that for any $x \in X \setminus Z$, we have

$$\eta_t^*(x) = \Gamma_t^*(x) + \Gamma_t'^*(x)$$

for all $t > 0$. The same trivially holds when $x \in Z$, so the equation holds everywhere. In particular, by Corollary A.2.1 and Proposition 1.2.8, we have

$$\eta_\tau = (\Gamma^* + \Gamma'^*)^*_\tau \in \mathrm{PSH}(X, \theta + \theta')$$

when $\tau < \Gamma_{\max} + \Gamma'_{\max}$.

Next, assume that Γ and Γ' are $\mathcal{I}$-model. We need to argue that so is $\Gamma + \Gamma'$. Fix $\tau < \Gamma_{\max} + \Gamma'_{\max}$. Then for each $\delta \in \mathbb{R}$ such that $\delta < \Gamma_{\max}$ and $\tau - \delta < \Gamma'_{\max}$, we know that $\Gamma_\delta \in \mathrm{PSH}(X,\theta)_{>0}$ and $\Gamma'_{\tau-\delta} \in \mathrm{PSH}(X,\theta')_{>0}$ by Lemma 9.1.1. It follows from Example 7.1.2 that Γ_δ and $\Gamma'_{\tau-\delta}$ are both $\mathcal{I}$-good, hence so is $\Gamma_\delta + \Gamma'_{\tau-\delta} \in \mathrm{PSH}(X, \theta+\theta')_{>0}$ by Proposition 7.2.1. These sums are bounded above by 0, so η_τ is $\mathcal{I}$-good by Proposition 7.2.2. Since $(\Gamma+\Gamma')_\tau = P_{\theta+\theta'}[\eta_\tau]$, we conclude that $\Gamma + \Gamma'$ is $\mathcal{I}$-model. □

Proposition 9.4.1 *Let $\Gamma \in \mathrm{TC}(X,\theta)_{>0}$ and $\Gamma' \in \mathrm{TC}(X,\theta')_{>0}$. Then $\Gamma + \Gamma' \in \mathrm{TC}(X, \theta+\theta')_{>0}$. Moreover,*

$$(\Gamma + \Gamma')_{-\infty} = P_{\theta+\theta'}\left[\Gamma_{-\infty} + \Gamma'_{-\infty}\right]. \tag{9.40}$$

When $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)_{>0}$ and $\Gamma' \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta')_{>0}$, we have $\Gamma + \Gamma' \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta+\theta')_{>0}$.

The operation + is commutative and associative.

---

[6] There is no usc regularization in the formula. It is not a typo.

Proof It follows immediately from Lemma 9.4.2 that $\Gamma+\Gamma' \in \mathrm{TC}(X,\theta+\theta')_{>0}$. It lies in $\mathrm{PSH}^{\mathrm{NA}}(X,\theta+\theta')_{>0}$ if $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)_{>0}$ and $\Gamma' \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta')_{>0}$.

We argue (9.40). By definition, for any small enough τ, we have

$$(\Gamma+\Gamma')_{-\infty} \geq (\Gamma+\Gamma')_{2\tau} \geq_P \Gamma_\tau + \Gamma'_\tau.$$

Letting $\tau \to -\infty$ and applying Proposition 6.2.5 and Theorem 6.2.2, we find that

$$(\Gamma+\Gamma')_{-\infty} \geq_P \Gamma_{-\infty} + \Gamma'_{-\infty}.$$

On the other hand, for each small enough τ, we have

$$(\Gamma+\Gamma')_\tau \sim_P \sup_{\delta\in\mathbb{R}} \left(\Gamma_\delta + \Gamma'_{\tau-\delta}\right) \leq_P \Gamma_{-\infty} + \Gamma'_{-\infty}$$

by Proposition 6.1.5 and Proposition 6.2.5. We apply Proposition 6.2.5 again, we conclude that

$$(\Gamma+\Gamma')_{-\infty} \leq_P \Gamma_{-\infty} + \Gamma'_{-\infty}.$$

So (9.40) follows.

Finally, let us show that $+$ is commutative and associative. Commutativity follows from the symmetry of the defining supremum. Let $\Gamma'' \in \mathrm{TC}(X,\theta'')_{>0}$. Then we want to show that

$$(\Gamma+\Gamma')+\Gamma'' = \Gamma + (\Gamma'+\Gamma'').$$

First observe that

$$((\Gamma+\Gamma')+\Gamma'')_{\max} = (\Gamma+(\Gamma'+\Gamma''))_{\max}\,.$$

Fix τ less than this common value. We compute that

$$\begin{aligned}
&((\Gamma+\Gamma')+\Gamma'')_\tau \\
=&P_{\theta+\theta'+\theta''}\left[\sup_{\delta_1\in\mathbb{R}}\left((\Gamma+\Gamma')_{\delta_1}+\Gamma''_{\tau-\delta_1}\right)\right] \\
\sim_P&\sup_{\delta_1\in\mathbb{R}}\left((\Gamma+\Gamma')_{\delta_1}+\Gamma''_{\tau-\delta_1}\right) \\
\sim_P&\sup_{\delta_1,\delta_2\in\mathbb{R}}\left(\Gamma_{\delta_2}+\Gamma'_{\delta_1-\delta_2}+\Gamma''_{\tau-\delta_1}\right),
\end{aligned}$$

where in the last line, we applied Proposition 6.2.5 and Proposition 6.1.5. Similarly, for $(\Gamma+(\Gamma'+\Gamma''))_\tau$, we get the same expression. The associativity follows. □

Lemma 9.4.3 *Let $\Gamma \in \mathrm{TC}(X,\theta)_{>0}$ and $\Gamma' \in \mathrm{TC}(X,\theta')_{>0}$. Then for any closed smooth positive $(1,1)$-forms ω and ω' on X, we have*

$$P_{\theta+\theta'+\omega+\omega'}[\Gamma+\Gamma'] = P_{\theta+\omega}[\Gamma] + P_{\theta'+\omega'}[\Gamma'].$$

Proof Observe that

$$\begin{aligned}P_{\theta+\theta'+\omega+\omega'}[\Gamma+\Gamma']_{\max} = &(P_{\theta+\omega}[\Gamma]+P_{\theta'+\omega'}[\Gamma'])_{\max}\\ =&\Gamma_{\max}+\Gamma'_{\max}.\end{aligned}$$

Take $\tau\in\mathbb{R}$ less than this common value, we need to verify that

$$(\Gamma+\Gamma')_\tau \sim_P \left(P_{\theta+\omega}[\Gamma]+P_{\theta'+\omega'}[\Gamma']\right)_\tau.$$

By definition, this means that

$$\sup_{\tau-\Gamma'_{\max}<\delta<\Gamma_{\max}}\left(\Gamma_\delta+\Gamma'_{\tau-\delta}\right)\sim_P \sup_{\tau-\Gamma'_{\max}<\delta<\Gamma_{\max}}\left(P_{\theta+\omega}[\Gamma_\delta]+P_{\theta'+\omega'}[\Gamma'_{\tau-\delta}]\right).$$

This is a consequence of Proposition 6.1.5 and Proposition 6.1.6. □

Definition 9.4.3 Let $\Gamma\in\mathrm{TC}(X,\theta)_{>0}$ and $C\in\mathbb{R}$, we define $\Gamma+C\in\mathrm{TC}(X,\theta)_{>0}$ as follows:

(1) We set

$$(\Gamma+C)_{\max}\coloneqq\Gamma_{\max}+C;$$

(2) for any $\tau<(\Gamma+C)_{\max}$, we set

$$(\Gamma+C)_\tau\coloneqq\Gamma_{\tau-C}.$$

It is obvious that if $\Gamma\in\mathrm{PSH}^{\mathrm{NA}}(X,\theta)_{>0}$, then so is $\Gamma+C$.

Proposition 9.4.2 *Let* $\Gamma\in\mathrm{TC}(X,\theta)_{>0}$, $\Gamma'\in\mathrm{TC}(X,\theta')_{>0}$ *and* $C,C'\in\mathbb{R}$. *Then*

(1) $(\Gamma+\Gamma')+C=\Gamma+(\Gamma'+C)=(\Gamma+C)+\Gamma'$*;*
(2) $\Gamma+(C+C')=(\Gamma+C)+C'$.

Proof (1) We first observe that

$$((\Gamma+\Gamma')+C)_{\max}=(\Gamma+(\Gamma'+C))_{\max}=((\Gamma+C)+\Gamma')_{\max}=\Gamma_{\max}+\Gamma'_{\max}+C.$$

Take any $\tau\in\mathbb{R}$ less than this common value. We compute

$$\begin{aligned}((\Gamma+\Gamma')+C)_\tau =&(\Gamma+\Gamma')_{\tau-C}=P_{\theta+\theta'}\left[\sup_{\delta\in\mathbb{R}}\left(\Gamma_\delta+\Gamma'_{\tau-C-\delta}\right)\right],\\ (\Gamma+(\Gamma'+C))_\tau =&P_{\theta+\theta'}\left[\sup_{\delta\in\mathbb{R}}\left(\Gamma_\delta+(\Gamma'+C)_{\tau-\delta}\right)\right]\\ =&P_{\theta+\theta'}\left[\sup_{\delta\in\mathbb{R}}\left(\Gamma_\delta+\Gamma'_{\tau-C-\delta}\right)\right],\\ ((\Gamma+C)+\Gamma')_\tau =&P_{\theta+\theta'}\left[\sup_{\delta\in\mathbb{R}}\left((\Gamma+C)_{C+\delta}+\Gamma'_{\tau-C-\delta}\right)\right]\\ =&P_{\theta+\theta'}\left[\sup_{\delta\in\mathbb{R}}\left(\Gamma_\delta+\Gamma'_{\tau-C-\delta}\right)\right].\end{aligned}$$

(2) Observe that

$$(\Gamma + (C + C'))_{\max} = ((\Gamma + C) + C')_{\max} = \Gamma_{\max} + C + C'.$$

For any $\tau \in \mathbb{R}$ less than this value, we have

$$(\Gamma + (C + C'))_\tau = \Gamma_{\tau - C - C'} = ((\Gamma + C) + C')_\tau .$$

This completes the proof. □

Definition 9.4.4 Let $\Gamma, \Gamma' \in \mathrm{TC}(X, \theta)_{>0}$. We define $\Gamma \vee \Gamma' \in \mathrm{TC}(X, \theta)_{>0}$ as follows:

(1) We set
$$(\Gamma \vee \Gamma')_{\max} := \Gamma_{\max} \vee \Gamma'_{\max};$$

(2) for any $\tau < (\Gamma \vee \Gamma')_{\max}$, we define

$$(\Gamma \vee \Gamma')_\tau := P_\theta \left[\mathrm{CE}\left(\rho \mapsto \Gamma_\rho \vee \Gamma'_\rho\right)_\tau \right]. \tag{9.41}$$

Recall that the upper concave envelope CE is defined in Definition A.1.4[7]. By construction, we have $\Gamma \vee \Gamma' \geq \Gamma$ and $\Gamma \vee \Gamma' \geq \Gamma'$.

On the right-hand side of (9.41), we first take the upper concave envelope $\mathrm{CE}\left(\rho \mapsto \Gamma_\rho \vee \Gamma'_\rho\right)$, this gives a family of functions indexed by $\rho \in \mathbb{R}$. Then the notation $\mathrm{CE}\left(\rho \mapsto \Gamma_\rho \vee \Gamma'_\rho\right)_\tau$ denotes the function obtained by evaluating this family at τ.

Lemma 9.4.4 *Let $\Gamma, \Gamma' \in \mathrm{TC}(X, \theta)_{>0}$. Then for any $\tau < \Gamma_{\max} \vee \Gamma'_{\max}$, we have*

$$\mathrm{CE}\left(\rho \mapsto \Gamma_\rho \vee \Gamma'_\rho\right)_\tau \in \mathrm{PSH}(X, \theta).$$

This potential is $\mathcal{I}$-good if $\Gamma, \Gamma' \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta)_{>0}$.

In particular, (9.41) in Definition 9.4.4 makes sense.

Proof To simplify the notation, we write

$$\psi_\tau = \mathrm{CE}\left(\rho \mapsto \Gamma_\rho \vee \Gamma'_\rho\right)_\tau$$

for all $\tau \in \mathbb{R}$. Thanks to Proposition A.2.3, for any fixed $x \in X$, we have

$$\psi^*_t(x) = \Gamma^*_t(x) \vee \Gamma'^*_t(x) \tag{9.42}$$

for all $t > 0$ as long as $\Gamma_\tau(x) \neq -\infty$ and $\Gamma'_\tau(x) \neq -\infty$ for some $\tau \in \mathbb{R}$. Otherwise, assume for instance that $\Gamma_\tau(x) = -\infty$ for all $\tau \in \mathbb{R}$; then by definition, $\psi_\tau(x) = \Gamma'_\tau(x)$

[7] In Definition A.1.4, we define the convex analogue, the *lower convex envelope*. The concave version is obtained by applying this definition to negatives.

for all $\tau \in \mathbb{R}$. Therefore, $\Gamma^*_t(x) = -\infty$ for all $t > 0$ and (9.42) continues to hold. The case where $\Gamma'_\tau(x) = -\infty$ for all τ is symmetric. Therefore, we have shown that

$$\psi^*_t = \Gamma^*_t \vee \Gamma'^*_t \in \mathrm{PSH}(X, \theta).$$

It follows from Proposition 4.1.3 that $(\psi^*_t)_{t \in [a,b]}$ is a subgeodesic for any $0 < a < b$.

Next we observe that $\psi_\bullet$ is closed by definition. So it follows from Proposition A.2.3 and Proposition 1.2.8 that

$$\psi_\tau = (\psi^*_\bullet)^*_\tau \in \mathrm{PSH}(X, \theta) \cup \{-\infty\}.$$

Due to Proposition 9.1.4 and Proposition A.1.2, there is a pluripolar set $Z \subsetneq X$ such that for $x \in X \setminus Z$, we have

$$\psi_\tau(x) = \sup \left\{ \lambda\Gamma_\rho(x) + (1-\lambda)\Gamma'_{\rho'}(x) : \lambda \in (0,1), \rho, \rho' \in \mathbb{R}, \lambda\rho + (1-\lambda)\rho' = \tau \right\}$$

for all $\tau < \Gamma_{\max} \vee \Gamma'_{\max}$. It follows from Proposition 1.2.6 that

$$\psi_\tau = \sup{}^* \left\{ \lambda\Gamma_\rho + (1-\lambda)\Gamma'_{\rho'} : \lambda \in (0,1), \rho, \rho' \in \mathbb{R}, \lambda\rho + (1-\lambda)\rho' = \tau \right\} \tag{9.43}$$

for all $\tau < \Gamma_{\max} \vee \Gamma'_{\max}$.

It follows from (9.43) that ψ_τ is $\mathcal{I}$-good if $\Gamma, \Gamma' \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta)_{>0}$, thanks to Proposition 7.2.1 and Proposition 7.2.2; the family in the upper envelope is bounded above by 0. □

Corollary 9.4.1 *Let $\Gamma, \Gamma' \in \mathrm{TC}(X, \theta)_{>0}$. Then $\Gamma \vee \Gamma' \in \mathrm{TC}(X, \theta)_{>0}$ and*

$$(\Gamma \vee \Gamma')_{-\infty} = P_\theta \left[\Gamma_{-\infty} \vee \Gamma'_{-\infty} \right]. \tag{9.44}$$

If $\Gamma, \Gamma' \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta)_{>0}$, then $\Gamma \vee \Gamma' \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta)_{>0}$.

For each $\Gamma'' \in \mathrm{TC}(X, \theta)_{>0}$ and each $\Gamma'' \geq \Gamma$ and $\Gamma'' \geq \Gamma'$, we have $\Gamma'' \geq \Gamma \vee \Gamma'$.

Moreover, the operation $\vee$ is associative and commutative.

In particular, given a finite family $\{\Gamma_i\}_{i \in I}$ in $\mathrm{TC}(X, \theta)_{>0}$, we can define

$$\bigvee_{i \in I} \Gamma_i$$

without ambiguity.

Proof It follows immediately from Lemma 9.4.4 that $\Gamma \vee \Gamma' \in \mathrm{TC}(X, \theta)_{>0}$, and it lies in $\mathrm{PSH}^{\mathrm{NA}}(X, \theta)_{>0}$ if $\Gamma, \Gamma' \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta)_{>0}$.

We argue (9.44). Set

$$\eta := \Gamma_{-\infty} \vee \Gamma'_{-\infty}.$$

Since $\Gamma \vee \Gamma' \geq \Gamma$ and $\Gamma \vee \Gamma' \geq \Gamma'$, we have $(\Gamma \vee \Gamma')_{-\infty} \geq \eta$. As $(\Gamma \vee \Gamma')_{-\infty}$ is a model potential, it follows that

$$(\Gamma \vee \Gamma')_{-\infty} \geq P_\theta[\eta].$$

Conversely, for each $\rho \in \mathbb{R}$ we have $\Gamma_\rho \vee \Gamma'_\rho \leq \eta$. Hence the constant curve with value η is a concave majorant of $\rho \mapsto \Gamma_\rho \vee \Gamma'_\rho$, and therefore

$$\mathrm{CE}\left(\rho \mapsto \Gamma_\rho \vee \Gamma'_\rho\right)_\tau \leq \eta, \quad \tau < \Gamma_{\max} \vee \Gamma'_{\max}.$$

Applying the monotonicity of the P-envelope and then letting $\tau \to -\infty$ gives

$$(\Gamma \vee \Gamma')_{-\infty} \leq P_\theta[\eta].$$

This proves (9.44).

Take Γ'' as in the statement of the corollary. First observe that

$$\Gamma''_{\max} \geq \Gamma_{\max} \vee \Gamma'_{\max} = (\Gamma \vee \Gamma')_{\max}.$$

Take $\tau < (\Gamma \vee \Gamma')_{\max}$, we argue that

$$\Gamma''_\tau \geq (\Gamma \vee \Gamma')_\tau.$$

By the concavity of Γ'', the potential Γ''_τ dominates the concave envelope

$$\mathrm{CE}\left(\rho \mapsto \Gamma_\rho \vee \Gamma'_\rho\right)_\tau.$$

Since Γ''_τ is model, the monotonicity of the P-envelope gives

$$\Gamma''_\tau \geq (\Gamma \vee \Gamma')_\tau.$$

Therefore,

$$\Gamma'' \geq \Gamma \vee \Gamma'.$$

The commutativity and associativity of $\vee$ are trivial. □

Lemma 9.4.5 *Let* $\Gamma, \Gamma' \in \mathrm{TC}(X,\theta)_{>0}$ *and* ω *be a closed smooth positive* $(1,1)$*-form on* X*. Then*

$$P_{\theta+\omega}[\Gamma \vee \Gamma'] = P_{\theta+\omega}[\Gamma] \vee P_{\theta+\omega}[\Gamma'].$$

Proof We first observe that

$$(P_{\theta+\omega}[\Gamma \vee \Gamma'])_{\max} = (P_{\theta+\omega}[\Gamma] \vee P_{\theta+\omega}[\Gamma'])_{\max} = \Gamma_{\max} \vee \Gamma'_{\max}.$$

Let $\tau \in \mathbb{R}$ be less than this common value. We need to show that

$$(\Gamma \vee \Gamma')_\tau \sim_P (P_{\theta+\omega}[\Gamma] \vee P_{\theta+\omega}[\Gamma'])_\tau.$$

We need the formula (9.43) proved in the proof of Lemma 9.4.4:

$$(\Gamma \vee \Gamma')_\tau \sim_P \sup{}^* \left\{\lambda\Gamma_\rho + (1-\lambda)\Gamma'_{\rho'} : \lambda \in (0,1), \rho, \rho' \in \mathbb{R}, \lambda\rho + (1-\lambda)\rho' = \tau\right\}.$$

A similar result holds with $P_{\theta+\omega}[\Gamma]$ and $P_{\theta+\omega}[\Gamma']$ in place of Γ and Γ'. So our assertion is a direct consequence of Proposition 6.1.5 and Proposition 6.1.6. □

Definition 9.4.5 Let $(\Gamma^i)_{i\in I}$ be an increasing net in $\mathrm{TC}(X,\theta)_{>0}$. Assume that

$$\sup_{i\in I}\Gamma^i_{\max} < \infty. \tag{9.45}$$

Then we define $\sup_{i\in I}{}^*\Gamma^i \in \mathrm{TC}(X,\theta)_{>0}$ as follows:

(1) We set
$$\left(\sup_{i\in I}{}^*\Gamma^i\right)_{\max} = \sup_{i\in I}\Gamma^i_{\max};$$
(2) for any $\tau < \sup_{i\in I}\Gamma^i_{\max}$, we let
$$\left(\sup_{i\in I}{}^*\Gamma^i\right)_{\tau} := \sup_{i\in I}{}^*\Gamma^i_{\tau}.$$

Proposition 9.4.3 *Let $(\Gamma^i)_{i\in I}$ be an increasing net in $\mathrm{TC}(X,\theta)_{>0}$ satisfying (9.45). Then $\sup_{i\in I}{}^*\Gamma^i$ as defined in Definition 9.4.5 lies in $\mathrm{TC}(X,\theta)_{>0}$. Moreover, if $\Gamma^i \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)_{>0}$ for all $i\in I$, then $\sup_{i\in I}{}^*\Gamma^i$ lies in $\mathrm{PSH}^{\mathrm{NA}}(X,\theta)_{>0}$ as well.*

Moreover, we have
$$\left(\sup_{i\in I}{}^*\Gamma^i\right)_{-\infty} = \sup_{i\in I}{}^*\Gamma^i_{-\infty}. \tag{9.46}$$

Proof Fix $\tau < \sup_i \Gamma^i_{\max}$. Choose i_0 such that $\tau < \Gamma^{i_0}_{\max}$. On the tail $i \geq i_0$, the potentials Γ^i_τ form an increasing net of model potentials of positive mass, uniformly bounded above by 0. Hence Proposition 3.1.13 shows that

$$\sup_{i\in I}{}^*\Gamma^i_\tau = \sup_{i\geq i_0}{}^*\Gamma^i_\tau$$

is model. Its mass is positive by monotonicity, since it dominates $\Gamma^{i_0}_\tau$. The decreasingness and concavity in τ follow from the directedness of the net and the corresponding properties of the curves Γ^i. Thus $\sup^*_i\Gamma^i$ is a test curve. If all Γ^i are $\mathcal{I}$-model, the same tail argument and Proposition 3.2.14 show that the supremum is $\mathcal{I}$-model.

It remains to argue (9.46). Fix any $i_0 \in I$. Since the curves are decreasing and are extended as in Remark 9.1.1, the value at $-\infty$ can be computed by taking the supremum over rational $\tau < \Gamma^{i_0}_{\max}$.

By Proposition 1.2.5, there is a pluripolar set $Z \subseteq X$ such that for any $x \in X \setminus Z$,

$$\left(\sup_{i\in I}{}^*\Gamma^i\right)_{-\infty}(x) = \sup_{\mathbb{Q}\ni\tau<\Gamma^{i_0}_{\max}}\left(\sup_{i\in I}{}^*\Gamma^i_\tau\right)(x) = \sup_{\mathbb{Q}\ni\tau<\Gamma^{i_0}_{\max},\, i\in I}\Gamma^i_\tau(x) = \sup_{i\in I}\Gamma^i_{-\infty}(x).$$

So they are equal everywhere by Proposition 1.2.6. □

Lemma 9.4.6 *Let $(\Gamma^i)_{i\in I}$ be an increasing net in $\mathrm{TC}(X,\theta)_{>0}$ satisfying (9.45). Assume that ω is a closed smooth positive $(1,1)$-form on X. Then*

$$P_{\theta+\omega}\left[\sup_{i\in I}{}^*\Gamma^i\right]=\sup_{i\in I}{}^*P_{\theta+\omega}\left[\Gamma^i\right].$$

Proof Observe that

$$\left(P_{\theta+\omega}\left[\sup_{i\in I}{}^*\Gamma^i\right]\right)_{\max}=\left(\sup_{i\in I}{}^*P_{\theta+\omega}\left[\Gamma^i\right]\right)_{\max}=\sup_{i\in I}\Gamma^i_{\max}.$$

Fix $\tau\in\mathbb{R}$ less than this common value.

It suffices to show that

$$\left(\sup_{i\in I}{}^*\Gamma^i\right)_\tau\sim_P\left(\sup_{i\in I}{}^*P_{\theta+\omega}\left[\Gamma^i\right]\right)_\tau.$$

This is an immediate consequence of Proposition 6.1.6. □

Definition 9.4.6 Let $(\Gamma^i)_{i\in I}$ be a non-empty family in $\mathrm{TC}(X,\theta)_{>0}$ satisfying (9.45). Then we define

$$\sup_{i\in I}{}^*\Gamma^i:=\sup_{J\in\mathrm{Fin}(I)}{}^*\left(\bigvee_{j\in J}\Gamma^j\right).\tag{9.47}$$

Recall that $\mathrm{Fin}(I)$ is the net of non-empty finite subsets of I, ordered by inclusion.

Observe that by Definition 9.4.4, we have

$$\sup_{J\in\mathrm{Fin}(I)}\left(\bigvee_{j\in J}\Gamma^j\right)_{\max}=\sup_{i\in I}\Gamma^i_{\max}<\infty.$$

So (9.47) makes sense. In particular,

$$\left(\sup_{i\in I}{}^*\Gamma^i\right)_{\max}=\sup_{i\in I}\Gamma^i_{\max}.\tag{9.48}$$

It is clear that Definition 9.4.6 extends both Definition 9.4.5 and Definition 9.4.4.

Proposition 9.4.4 *Let $(\Gamma^i)_{i\in I}$ be a non-empty family in $\mathrm{TC}(X,\theta)_{>0}$ satisfying (9.45). Then $\sup_{i\in I}{}^*\Gamma^i\in\mathrm{TC}(X,\theta)_{>0}$. Moreover, if $\Gamma^i\in\mathrm{PSH}^{\mathrm{NA}}(X,\theta)_{>0}$ for all $i\in I$, then so is $\sup_{i\in I}{}^*\Gamma^i$.*

Finally, we have

$$\left(\sup_{i\in I}{}^*\Gamma^i\right)_{-\infty}=P_\theta\left[\sup_{i\in I}{}^*\Gamma^i_{-\infty}\right].\tag{9.49}$$

Proof The first assertion and the second follow from Proposition 9.4.3 and Corollary 9.4.1.

It remains to argue (9.49). For this purpose, it suffices to show that

$$\left(\sup_{i\in I}{}^*\Gamma^i\right)_{-\infty}\sim_P\sup_{i\in I}{}^*\Gamma^i_{-\infty}.$$

For any $J \in \mathrm{Fin}(I)$, it follows from Corollary 9.4.1 and Proposition 6.1.6 that

$$\left(\bigvee_{j\in J} \Gamma^j\right)_{-\infty} \sim_P \bigvee_{j\in J} \Gamma^j_{-\infty}.$$

From this, applying Proposition 3.1.13, Proposition 6.1.6 and Proposition 9.4.3, we conclude our assertion. □

Lemma 9.4.7 *Let $(\Gamma^i)_{i\in I}$ be a non-empty family in $\mathrm{TC}(X,\theta)_{>0}$ satisfying (9.45). Assume that ω is a closed smooth positive $(1,1)$-form on X. Then*

$$P_{\theta+\omega}\left[\sup_{i\in I}{}^*\Gamma^i\right] = \sup_{i\in I}{}^* P_{\theta+\omega}\left[\Gamma^i\right].$$

Proof This is a direct consequence of Lemma 9.4.6 and Lemma 9.4.5. □

We prove a version of Choquet's lemma.

Proposition 9.4.5 *Let $(\Gamma^i)_{i\in I}$ be a non-empty family in $\mathrm{TC}(X,\theta)_{>0}$ satisfying (9.45). Then there is a countable subset $I' \subseteq I$ such that*

$$\sup_{i\in I}{}^*\Gamma^i = \sup_{i\in I'}{}^*\Gamma^i.$$

Proof We may assume that I is infinite.

Set

$$M := \sup_{i\in I} \Gamma^i_{\max}.$$

For each rational number $q < M$, apply Choquet's lemma Proposition 1.2.3 to the locally uniformly bounded above family

$$\left\{\left(\bigvee_{j\in J} \Gamma^j\right)_q : J \in \mathrm{Fin}(I)\right\} \subseteq \mathrm{PSH}(X,\theta).$$

We obtain a countable subfamily $\mathcal{J}_q \subseteq \mathrm{Fin}(I)$ such that

$$\sup_{J\in\mathrm{Fin}(I)}{}^* \left(\bigvee_{j\in J} \Gamma^j\right)_q = \sup_{J\in\mathcal{J}_q}{}^* \left(\bigvee_{j\in J} \Gamma^j\right)_q.$$

Let

$$I' := \bigcup_{q\in\mathbb{Q}, q<M} \bigcup_{J\in\mathcal{J}_q} J.$$

Then I' is countable and, for every rational $q < M$,

$$\left(\sup_{i\in I}{}^*\Gamma^i\right)_q = \left(\sup_{i\in I'}{}^*\Gamma^i\right)_q.$$

Moreover, $\sup_{i\in I'}\Gamma^i_{\max} = M$, since the last equality is non-trivial for every rational $q < M$. Let $\Gamma' = \sup_{i\in I'}{}^*\Gamma^i$. Then clearly, $\Gamma' \leq \Gamma$. We claim that they are actually equal.

Thanks to Proposition 6.1.1 and Lemma 9.1.1, it suffices to show that for any $\tau < \sup_{i\in I}{}^*\Gamma^i_{\max}$, we have

$$\int_X \left(\theta + \mathrm{dd^c}\Gamma'_\tau\right)^n = \int_X \left(\theta + \mathrm{dd^c}\Gamma_\tau\right)^n .$$

Since we know that this holds for τ lying in a dense subset, the same holds everywhere by Proposition 9.1.1. □

Proposition 9.4.6 *Let $(\Gamma^i)_{i\in I}$ be a non-empty family in $\mathrm{TC}(X,\theta)_{>0}$ satisfying (9.45). Let $C \in \mathbb{R}$. Then*

$$\sup_{i\in I}{}^*(\Gamma^i + C) = \sup_{i\in I}{}^*\Gamma^i + C.$$

Suppose that $(\Gamma'^i)_{i\in I}$ is another family in $\mathrm{TC}(X,\theta')_{>0}$ satisfying (9.45). Suppose that $\Gamma^i \leq \Gamma'^i$ for all $i \in I$. Then

$$\sup_{i\in I}{}^*\Gamma^i \leq \sup_{i\in I}{}^*\Gamma'^i.$$

Proof This is immediate by definition. □

Definition 9.4.7 Let $\Gamma \in \mathrm{TC}(X,\theta)_{>0}$ and $\lambda > 0$, we define $\lambda\Gamma \in \mathrm{TC}(X,\lambda\theta)_{>0}$ as follows:

(1) We set

$$(\lambda\Gamma)_{\max} = \lambda\Gamma_{\max};$$

(2) for any $\tau < (\lambda\Gamma)_{\max}$, we set

$$(\lambda\Gamma)_\tau = \lambda\Gamma_{\lambda^{-1}\tau}.$$

Proposition 9.4.7 *Let $\Gamma \in \mathrm{TC}(X,\theta)_{>0}$ and $\lambda > 0$. Then $\lambda\Gamma$ as defined in Definition 9.4.7 lies in $\mathrm{TC}(X,\lambda\theta)_{>0}$. Moreover, if $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)_{>0}$, then $\lambda\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X,\lambda\theta)_{>0}$.*

We have

$$(\lambda\Gamma)_{-\infty} = \lambda\Gamma_{-\infty}. \tag{9.50}$$

Proof This is immediate by definition. □

Proposition 9.4.8 *Let $\Gamma \in \mathrm{TC}(X,\theta)_{>0}$, $\Gamma' \in \mathrm{TC}(X,\theta')_{>0}$, $C \in \mathbb{R}$ and $\lambda, \lambda' > 0$, we have*

$$\begin{aligned}\lambda(\Gamma+\Gamma') &= \lambda\Gamma + \lambda\Gamma',\\ (\lambda\lambda')\Gamma &= \lambda(\lambda'\Gamma),\\ \lambda(\Gamma + C) &= \lambda\Gamma + \lambda C.\end{aligned}$$

Suppose that $(\Gamma^i)_{i\in I}$ is a non-empty family in $\mathrm{TC}(X,\theta)_{>0}$ satisfying (9.45). Then

$$\lambda \left(\sup_{i\in I}{}^{*} \Gamma^i \right) = \sup_{i\in I}{}^{*} (\lambda \Gamma^i).$$

Proof This is immediate by definition. □

Lemma 9.4.8 *Let* $\Gamma \in \mathrm{TC}(X,\theta)_{>0}$ *and* $\lambda > 0$. *Then for any closed smooth positive* $(1,1)$*-form* ω *on* X, *we have*

$$P_{\lambda\theta+\lambda\omega}[\lambda\Gamma] = \lambda P_{\theta+\omega}[\Gamma].$$

Proof This is clear by definition. □

Definition 9.4.8 Let $(\Gamma^i)_{i\in I}$ be a decreasing net in $\mathrm{TC}(X,\theta)_{>0}$. Assume that

$$\inf_{i\in I} \Gamma^i_{\max} > -\infty, \tag{9.51}$$

and

$$\inf_{i\in I} \int_X \left(\theta + \mathrm{dd^c}\Gamma^i_\tau\right)^n > 0, \quad \text{for some } \tau < \inf_{i\in I} \Gamma^i_{\max}. \tag{9.52}$$

Then we define $\inf_{i\in I} \Gamma^i \in \mathrm{TC}(X,\theta)_{>0}$ as follows:

(1) We set

$$\left(\inf_{i\in I} \Gamma^i \right)_{\max} = \inf_{i\in I} \Gamma^i_{\max};$$

(2) for any $\tau < \left(\inf_{i\in I} \Gamma^i\right)_{\max}$, we let

$$\left(\inf_{i\in I} \Gamma^i \right)_{\tau} \coloneqq \inf_{i\in I} \Gamma^i_{\tau}.$$

Proposition 9.4.9 *Let* $(\Gamma^i)_{i\in I}$ *be a decreasing net in* $\mathrm{TC}(X,\theta)_{>0}$ *satisfying* (9.51) *and* (9.52). *Then* $\inf_{i\in I} \Gamma^i \in \mathrm{TC}(X,\theta)_{>0}$.

Moreover, if $\Gamma^i \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)_{>0}$ *for all* $i \in I$, *then so is* $\inf_{i\in I} \Gamma^i$.

Proof The first assertion is an immediate consequence of Proposition 3.1.11, except for the positivity of the mass at $-\infty$. Choose τ_0 as in (9.52). Applying Proposition 3.1.12 gives

$$\int_X \left(\theta + \mathrm{dd^c} \inf_{i\in I} \Gamma^i_{\tau_0} \right)^n = \lim_{i\in I} \int_X \left(\theta + \mathrm{dd^c}\Gamma^i_{\tau_0} \right)^n > 0.$$

Hence the value at $-\infty$ has positive mass by monotonicity. The last assertion follows from Proposition 3.2.13. □

In general, it is not true that

$$\left(\inf_{i\in I} \Gamma^i \right)_{-\infty} = \inf_{i\in I} \Gamma^i_{-\infty}.$$

Lemma 9.4.9 *Let* $(\Gamma^i)_{i\in I}$ *be a decreasing net in* $\mathrm{TC}(X,\theta)_{>0}$ *satisfying* (9.51) *and* (9.52). *Assume that* ω *is a closed smooth positive* $(1,1)$*-form on* X. *Then*

$$P_{\theta+\omega}\left[\inf_{i\in I}\Gamma^i\right]=\inf_{i\in I}P_{\theta+\omega}\left[\Gamma^i\right].$$

Proof First observe that

$$\left(P_{\theta+\omega}\left[\inf_{i\in I}\Gamma^i\right]\right)_{\max}=\left(\inf_{i\in I}P_{\theta+\omega}\left[\Gamma^i\right]\right)_{\max}.$$

Let $\tau\in\mathbb{R}$ be less than this common value. Then we need to show the following:

$$P_{\theta+\omega}\left[\inf_{i\in I}\Gamma^i_\tau\right]\sim_P\inf_{i\in I}P_{\theta+\omega}\left[\Gamma^i_\tau\right]. \tag{9.53}$$

Applying Proposition 3.1.12 gives the mass convergence required in Corollary 6.2.5. Hence $\Gamma^i_\tau\xrightarrow{d_S}\inf_{j\in I}\Gamma^j_\tau$. Thanks to Corollary 6.2.8 and Corollary 6.2.5, we have

$$P_{\theta+\omega}\left[\Gamma^i_\tau\right]\xrightarrow{d_S}P_{\theta+\omega}\left[\inf_{j\in I}\Gamma^j_\tau\right],\quad P_{\theta+\omega}\left[\Gamma^i_\tau\right]\xrightarrow{d_S}\inf_{j\in I}P_{\theta+\omega}\left[\Gamma^j_\tau\right].$$

Hence, (9.53) follows from Proposition 6.2.2. □

Chapter 10
The theory of Okounkov bodies

It is very fortunate that, unlike people who dig for gold, mathematicians can freely share their precious treasures with everybody. Once you understand something really well, it feels great to explain it to all.
— Andrei Okounkov[a]

[a] Andrei Yuryevich Okounkov (1969–) is a Russian-American mathematician renowned for his contributions to representation theory. He was one of the key organizers of the ICM 2022 in St. Petersburg, which was unfortunately canceled under the indiscriminate discrimination against Russian citizens by the virtue signalers all over the western world after the war waged by the ruling class.

In this chapter, we apply our theory of singularities to the study of Okounkov bodies. We construct convex bodies $\Delta(L,\phi)$, the *partial Okounkov bodies*, attached to a big line bundle L on a smooth projective variety X together with a plurisubharmonic metric ϕ on L. These convex bodies provide a universal numerical invariant of the $\mathcal{I}$-equivalence class of ϕ, and reduce many problems in pluripotential theory to problems in convex geometry, which are often more tractable.

We develop two parallel constructions: an algebraic one in Section 10.3, where ϕ is encoded by the multiplier ideals $\mathcal{I}(k\phi)$, and a transcendental one in Section 10.4, where ϕ is treated directly as an analytic object via valuations on closed positive currents. The two constructions agree when both apply, but each has its own natural domain of definition and its own range of applications.

We assume some familiarity with the classical Okounkov bodies. Excellent references are the original papers of Lazarsfeld–Mustață [LM09] and Kaveh–Khovanskii [KK12]; we give a rapid review here.

Let X be an irreducible smooth projective variety of dimension n and L a big holomorphic line bundle on X. Given an admissible flag $X = Y_0 \supseteq Y_1 \supseteq \cdots \supseteq Y_n$ on X (Definition 10.2.1), one attaches to L a natural n-dimensional convex body $\Delta(L) \subseteq \mathbb{R}^n$, generalizing the Newton polytope construction in toric geometry (Definition 5.2.1). This construction was first considered by Okounkov [Oko96, Oko03] and extended by Lazarsfeld–Mustață [LM09] and Kaveh–Khovanskii [KK12]. The convex body $\Delta(L)$ is the *Okounkov body* of L with respect to the chosen flag. Concretely, the successive vanishing order along the flag defines a valuation $\nu\colon \mathbb{C}(X)^\times \to \mathbb{Z}^n$, and

$$\Gamma(L) \coloneqq \left\{(\nu(s),k) \in \mathbb{Z}^{n+1} : k \in \mathbb{N},\ s \in \mathrm{H}^0(X, L^k)^\times\right\}$$

is a semigroup whose closed convex cone, intersected with the slice $\{(x,1)\}$, is $\Delta(L)$. Lazarsfeld–Mustață [LM09] showed that $\Delta(L)$ depends only on the numerical class of L, and Jow [Jow10] showed that the Okounkov bodies for all flags determine this class, so $\Delta(L)$ may be regarded as a universal numerical invariant of L.

The aim of this chapter is to perform the analogous construction at the level of singularity types of psh metrics. Setting

$$\Gamma(L,\phi) := \left\{(\nu(s),k) \in \mathbb{Z}^{n+1} : k \in \mathbb{N},\ s \in \mathrm{H}^0(X, L^k \otimes \mathcal{I}(k\phi))^{\times}\right\},$$

one immediately encounters the obstruction that $\Gamma(L,\phi)$ is in general *not* a semigroup, so the Lazarsfeld–Mustață and Kaveh–Khovanskii constructions break down. The remedy is the notion of an *almost semigroup*: $\Gamma(L,\phi)$ does satisfy a near-semigroup property (Theorem 10.3.1), and the convex-body construction $\Gamma \mapsto \Delta(\Gamma)$ extends continuously from semigroups to almost semigroups (Theorem 10.1.2). The partial Okounkov body of (L,ϕ) is then

$$\Delta(L,\phi) := \Delta\big(\Gamma(L,\phi)\big),$$

and is a convex sub-body of $\Delta(L)$ — explaining the name *partial*.

The chapter is organized as follows.

Section 10.1 gives a self-contained introduction to almost semigroups: the basic definitions (Definition 10.1.2), the Hausdorff continuity theorem (Theorem 10.1.1), and the existence of a canonical convex-body extension (Theorem 10.1.2). Detailed proofs are deferred to Appendix C.

Section 10.2 sets up the geometric framework underlying both constructions. The algebraic side (Section 10.2.1) defines admissible flags on X (Definition 10.2.1) and recalls the associated $\mathbb{Z}^n$-valuation;; the transcendental side adapts these ideas to closed positive currents, defining a valuation on currents (Definition 10.2.3) and the lifting of flags through bimeromorphic resolutions (Definition 10.2.4 and Theorem 10.2.1).

Section 10.3 is the algebraic core of the chapter. After fixing the relevant spaces of sections, it recalls the classical Okounkov body Definition 10.3.4 and its volume formula (Corollary 10.3.1), then proves that $\Gamma(L,\phi)$ is an almost semigroup (Theorem 10.3.1), defines $\Delta(L,\phi)$, and computes its volume (Corollary 10.3.2). The remaining subsections develop the basic properties of $\Delta(L,\phi)$: continuity in ϕ (Theorem 10.3.2), concavity (Theorem 10.3.3), the Hausdorff convergence property (Theorem 10.3.4), and the recovery of Lelong numbers from the partial Okounkov body (Theorem 10.3.5).

Section 10.4 develops the transcendental theory. After comparison with the traditional approach, partial Okounkov bodies are constructed for arbitrary closed positive $(1,1)$-currents on a compact Kähler manifold, and a valuative characterization is established. The main result here is the equivalence of the algebraic and transcendental constructions on their common domain.

Section 10.5 relates partial Okounkov bodies to the test curves of Chapter 9: the family $(\Delta(L,\phi_\tau))_\tau$ defines an *Okounkov test curve*, providing a convex-geometric counterpart to the analytic Ross–Witt Nyström correspondence.

The universality of partial Okounkov bodies is captured by Corollary 10.3.3: the $\mathcal{I}$-equivalence class of ϕ is determined by the Okounkov bodies of (L,ϕ) for sufficiently many flags.

The results in this chapter will be applied in Chapter 12 to develop toric pluripotential theory on big line bundles.

10.1 Almost semigroups

We give a brief presentation of the theory of almost semigroups. The arguments are provided in Appendix C.

Fix an integer $n \geq 0$. Fix a closed convex cone $C \subseteq \mathbb{R}^n \times \mathbb{R}_{\geq 0}$ such that $C \cap \{x_{n+1} = 0\} = \{0\}$. Here x_{n+1} is the last coordinate of $\mathbb{R}^{n+1}$.

Write $\hat{\mathcal{S}}(C)$ for the set of subsets of $C \cap \mathbb{Z}^{n+1}$ and $\mathcal{S}(C)$ for the set of subsemigroups $S \subseteq C \cap \mathbb{Z}^{n+1}$. For each $k \in \mathbb{N}$ such that $S_k \neq \varnothing$ and $S \in \hat{\mathcal{S}}(C)$, we write

$$S_k \coloneqq \{x \in \mathbb{Z}^n : (x, k) \in S\}.$$

Note that S_k is a finite set by our assumption on C.

We introduce a pseudometric on $\hat{\mathcal{S}}(C)$ as follows:

$$d_{\mathrm{sg}}(S, S') \coloneqq \varlimsup_{k \to \infty} k^{-n} \left(|S_k| + |S'_k| - 2|(S \cap S')_k|\right). \tag{10.1}$$

Here $|\bullet|$ denotes the cardinality of a finite set. The quantity d_{sg} defined above is a pseudometric on $\hat{\mathcal{S}}(C)$. Given $S, S' \in \hat{\mathcal{S}}(C)$, we say S is equivalent to S' and write $S \sim S'$ if $d_{\mathrm{sg}}(S, S') = 0$. This is an equivalence relation.

The volume of $S \in \mathcal{S}(C)$ is defined as

$$\operatorname{vol} S \coloneqq \lim_{k \to \infty} (ka)^{-n} |S_{ka}| = \varlimsup_{k \to \infty} k^{-n} |S_k|,$$

where a is a sufficiently divisible positive integer. The existence of the limit and its independence from a both follow from the more precise result [KK12, Theorem 2].

We define $\overline{\mathcal{S}}(C)$ as the closure of $\mathcal{S}(C)$ in $\hat{\mathcal{S}}(C)$ with respect to the topology defined by the pseudometric d_{sg}. The function $\operatorname{vol}\colon \mathcal{S}(C) \to \mathbb{R}$ admits a unique 1-Lipschitz extension to

$$\operatorname{vol}\colon \overline{\mathcal{S}}(C) \to \mathbb{R}. \tag{10.2}$$

Given $S \in \hat{\mathcal{S}}(C)$, we will write $C(S) \subseteq C$ for the closed convex cone generated by $S \cup \{0\}$. Moreover, for each $k \in \mathbb{Z}_{>0}$ such that $S_k \neq \varnothing$, we define

$$\Delta_k(S) \coloneqq \operatorname{Conv}\left\{k^{-1}x \in \mathbb{R}^n : x \in S_k\right\} \subseteq \mathbb{R}^n.$$

Here Conv denotes the convex hull.

Definition 10.1.1 Let $\mathcal{S}'(C)$ be the subset of $\mathcal{S}(C)$ consisting of semigroups S such that S generates $\mathbb{Z}^{n+1}$ (as an Abelian group).

Note that for any $S \in \mathcal{S}'(C)$, the cone $C(S)$ has full dimension (i.e., the topological interior is non-empty). Given a full-dimensional subcone $C' \subseteq C$, we have $C' \cap \mathbb{Z}^{n+1} \in \mathcal{S}'(C)$.

We recall the following definition from [KK12].

Definition 10.1.2 Given $S \in \mathcal{S}'(C)$, its *Okounkov body* is defined by

$$\Delta(S) := \{x \in \mathbb{R}^n : (x, 1) \in C(S)\}.$$

Theorem 10.1.1 *For each $S \in \mathcal{S}'(C)$, we have*

$$\operatorname{vol} S = \lim_{k \to \infty} k^{-n} |S_k| = \operatorname{vol} \Delta(S) > 0. \tag{10.3}$$

Moreover, as $k \to \infty$,

$$\Delta_k(S) \xrightarrow{d_{\mathrm{Haus}}} \Delta(S). \tag{10.4}$$

Corollary 10.1.1 *Let $S, S' \in \mathcal{S}'(C)$. Assume that $\operatorname{vol}(S \cap S') > 0$. Then we have*

$$d_{\mathrm{sg}}(S, S') = \operatorname{vol}(S) + \operatorname{vol}(S') - 2 \operatorname{vol}(S \cap S').$$

Definition 10.1.3 We define $\overline{\mathcal{S}'(C)}_{>0}$ as the set of elements in the closure of $\mathcal{S}'(C)$ in $\hat{\mathcal{S}}(C)$ with positive volume. An element in $\overline{\mathcal{S}'(C)}_{>0}$ is called an *almost semigroup* in C.

Recall that the volume here is defined in (10.2).

Theorem 10.1.2 *The Okounkov body map $\Delta \colon \mathcal{S}'(C) \to \mathcal{K}_n$ as defined in Definition 10.1.2 admits a unique continuous extension*

$$\Delta \colon \overline{\mathcal{S}'(C)}_{>0} \to \mathcal{K}_n. \tag{10.5}$$

Moreover, for any $S \in \overline{\mathcal{S}'(C)}_{>0}$, we have

$$\operatorname{vol} S = \operatorname{vol} \Delta(S). \tag{10.6}$$

Corollary 10.1.2 *Suppose that $S, S' \in \overline{\mathcal{S}'(C)}_{>0}$ with $S \subseteq S'$. Then*

$$\Delta(S) \subseteq \Delta(S'). \tag{10.7}$$

10.2 Flags and valuations

10.2.1 The algebraic setting

Let X be an irreducible normal projective variety of dimension n.

Definition 10.2.1 An *admissible flag* $Y_\bullet$ on X is a flag of subvarieties

$$X = Y_0 \supseteq Y_1 \supseteq \cdots \supseteq Y_n$$

such that Y_i is irreducible of codimension i and is smooth at the point Y_n.

Given any admissible flag $Y_\bullet$, we can define a rank n valuation $\nu_{Y_\bullet} : \mathbb{C}(X)^\times \to \mathbb{Z}^n$. Here we consider $\mathbb{Z}^n$ as a totally ordered Abelian group with the lexicographic order. We sometimes write $\mathbb{Z}^n_{\mathrm{lex}}$ to emphasize this point.

If we identify the elements in $\mathbb{Z}^n$ with a row vector, the automorphism group $\mathrm{Aut}(\mathbb{Z}^n_{\mathrm{lex}})$ of $\mathbb{Z}^n_{\mathrm{lex}}$ is then identified with the subgroup of $\mathrm{GL}(n, \mathbb{Z})$ consisting of matrices of the form $\mathrm{I} + U$, where I is the identity matrix and U is a strictly upper triangular matrix with elements in $\mathbb{Z}$.

We recall the definition of $\nu_{Y_\bullet}$. Let $s^{(0)} = s \in \mathbb{C}(X)^\times$. For $i = 1, \ldots, n$, after localizing around Y_n, choose a local defining equation t_i of Y_i in Y_{i-1}, and set

$$\nu_i = \mathrm{ord}_{Y_i}(s^{(i-1)}), \qquad s^{(i)} = \left(s^{(i-1)} t_i^{-\nu_i} \right)|_{Y_i} \in \mathbb{C}(Y_i)^\times.$$

Then

$$\nu_{Y_\bullet}(s) = (\nu_1, \ldots, \nu_n) \in \mathbb{Z}^n.$$

It is easy to verify that $\nu_{Y_\bullet}$ is indeed a rank n valuation.

The same construction applies to a nonzero section $s \in \mathrm{H}^0(X, L)$ after choosing a local trivialization of L near Y_n; the result is independent of this choice, since units have valuation 0. We also write $\nu_{Y_\bullet}(D)$ for a non-zero effective Cartier divisor D, and extend the notation $\mathbb{Q}$-linearly to non-zero effective $\mathbb{Q}$-Cartier divisors.

Remark 10.2.1 Conversely, maximal-rank valuations of $\mathbb{C}(X)$ are closely related to flag valuations. More precisely, if a valuation has value group identified with $\mathbb{Z}^n_{\mathrm{lex}}$ and maximal rational rank n, then [CFK+17, Theorem 2.9] shows that it is equivalent[1] to, but not necessarily equal to, a valuation induced by an admissible flag on a modification of X.

10.2.2 The transcendental setting

Let X be a connected compact Kähler manifold of dimension n.

Definition 10.2.2 A *smooth flag* $Y_\bullet$ on X consists of a flag of connected closed submanifolds of X:

$$X = Y_0 \supseteq Y_1 \supseteq \cdots \supseteq Y_n,$$

where Y_i has dimension $n - i$.

In this section, we will fix a smooth flag $Y_\bullet$ on X.

[1] Two valuations ν, ν' with values in $\mathbb{Z}^n$ are equivalent if one can find a matrix G of the form $\mathrm{I} + N$, where N is strictly upper triangular with integral entries, such that $\nu' = \nu G$.

Definition 10.2.3 Let T be a closed positive $(1,1)$-current on X. We define the *valuation* of T along $Y_\bullet$ as

$$\nu_{Y_\bullet}(T) = \left(\nu_{Y_\bullet}(T)_1, \ldots, \nu_{Y_\bullet}(T)_n\right) \in \mathbb{R}^n_{\geq 0}$$

by induction on n. When $n = 0$, we define $\nu_{Y_\bullet}(T)$ as the unique point in $\mathbb{R}^0$. When $n \geq 1$, we define

$$\nu_{Y_\bullet}(T)_1 = \nu(T, Y_1);$$

Then for $i = 2, \ldots, n$, we define

$$\nu_{Y_\bullet}(T)_i = \nu_{Y_1 \supseteq \cdots \supseteq Y_n}\left(\mathrm{Tr}_{Y_1}(T - \nu(T, Y_1)[Y_1])\right)_{i-1}.$$

Proposition 10.2.1 *Let T be a closed positive $(1,1)$-current on X. Then $\nu_{Y_\bullet}(T) \in \mathbb{R}^n_{\geq 0}$ defined in Definition 10.2.3 is independent of the choices of the trace operators in the definition. Moreover, $\nu_{Y_\bullet}(T)$ depends only on the $\mathcal{I}$-equivalence class of T.*

Proof We will prove both statements at the same time by induction on $n \geq 0$. The cases $n = 0, 1$ are trivial.

Let us consider the case $n > 1$ and assume that the result is known in dimension $n - 1$. We first observe that $\nu_{Y_\bullet}(T)$ is independent of the choice of the trace operator: Different choices of $\mathrm{Tr}_{Y_1}(T - \nu(T, Y_1)[Y_1])$ are $\mathcal{I}$-equivalent by Proposition 8.1.2. Therefore, by induction, its valuation is well-defined.

Next, let T' be another closed positive $(1,1)$-current such that $T \sim_{\mathcal{I}} T'$. Using Proposition 3.2.1, we know that $\nu(T, Y_1) = \nu(T', Y_1)$. Therefore,

$$T - \nu(T, Y_1)[Y_1] \sim_{\mathcal{I}} T' - \nu(T', Y_1)[Y_1].$$

It follows by induction that

$$\nu_{Y_1 \supseteq \cdots \supseteq Y_n}\left(\mathrm{Tr}_{Y_1}(T - \nu(T, Y_1)[Y_1])\right) = \nu_{Y_1 \supseteq \cdots \supseteq Y_n}\left(\mathrm{Tr}_{Y_1}(T' - \nu(T', Y_1)[Y_1])\right).$$

This completes the proof. □

Example 10.2.1 When X is projective, and D is a non-zero effective Cartier divisor, we have

$$\nu_{Y_\bullet}([D]) = \nu_{Y_\bullet}(D),$$

where the right-hand side is defined in Section 10.2.1.

Proposition 10.2.2 *Let T, S be closed positive $(1,1)$-currents on X, $\lambda \in \mathbb{R}_{\geq 0}$. Then*

(1) *if $T \preceq_{\mathcal{I}} S$, we have*
$$\nu_{Y_\bullet}(T) \geq_{\mathrm{lex}} \nu_{Y_\bullet}(S). \tag{10.8}$$

(2) *We have the following additivity property:*
$$\nu_{Y_\bullet}(T + S) = \nu_{Y_\bullet}(T) + \nu_{Y_\bullet}(S), \quad \nu_{Y_\bullet}(\lambda T) = \lambda \nu_{Y_\bullet}(T). \tag{10.9}$$

Proof (1) We make an induction on $n \geq 0$. The case $n = 0, 1$ is trivial. Assume that $n \geq 2$ and the case $n-1$ is known. Observe that $\nu(T, Y_1) \geq \nu(S, Y_1)$, if the inequality is strict, we are done. So let us assume that $\nu(T, Y_1) = \nu(S, Y_1)$. By Proposition 8.2.1, we find that

$$\mathrm{Tr}_{Y_1}\Big(T - \nu(T, Y_1)[Y_1]\Big) \preceq_P \mathrm{Tr}_{Y_1}\Big(S - \nu(T, Y_1)[Y_1]\Big).$$

Hence the same comparison holds for $\preceq_{\mathcal{I}}$ by Corollary 6.1.1. By the inductive hypothesis, we conclude (10.8).

(2) We make an induction on $n \geq 0$. The cases $n = 0, 1$ are trivial. Assume that $n \geq 2$ and the case $n-1$ is known. By Proposition 1.4.2, we have

$$\nu(T + S, Y_1) = \nu(T, Y_1) + \nu(S, Y_1), \quad \nu(\lambda T, Y_1) = \lambda \nu(T, Y_1).$$

By Proposition 8.2.1, we have

$$\begin{aligned} \mathrm{Tr}_{Y_1}\Big(T + S - \nu(T+S, Y_1)[Y_1]\Big) &\sim_P \mathrm{Tr}_{Y_1}\Big(T - \nu(T, Y_1)[Y_1]\Big) \\ &\quad + \mathrm{Tr}_{Y_1}\Big(S - \nu(S, Y_1)[Y_1]\Big), \\ \mathrm{Tr}_{Y_1}\Big(\lambda T - \nu(\lambda T, Y_1)[Y_1]\Big) &\sim_P \lambda\, \mathrm{Tr}_{Y_1}\Big(T - \nu(T, Y_1)[Y_1]\Big). \end{aligned}$$

Since $\sim_P$ implies $\sim_{\mathcal{I}}$ by Corollary 6.1.1, the inductive hypothesis and the independence of the valuation from the trace representative conclude (10.9).

Assume that $n > 0$ for the remainder of this section.

Definition 10.2.4 Let $\pi\colon Z \to X$ be a proper bimeromorphic morphism where Z is a Kähler manifold. We say that a smooth flag $W_\bullet$ on Z is a *lifting* of $Y_\bullet$ to Z if $\pi(W_i) \subseteq Y_i$ and the induced map $W_i \to Y_i$ is bimeromorphic for each $i = 0, \ldots, n$.

In this case, we define $\mathrm{cor}(Y_\bullet, \pi) \in \mathrm{Aut}(\mathbb{Z}^n_{\mathrm{lex}})$ inductively as follows: When $n = 1$, we define $\mathrm{cor}(Y_\bullet, \pi) = [1]$; when $n > 1$, we set

$$\mathrm{cor}(Y_\bullet, \pi) \coloneqq \begin{bmatrix} 1 & \nu_{W_1 \supseteq \cdots \supseteq W_n}((\pi^*[Y_1] - [W_1])|_{W_1}) \\ 0 & \mathrm{cor}(Y_1 \supseteq \cdots \supseteq Y_n, \pi|_{W_1} : W_1 \to Y_1) \end{bmatrix}. \tag{10.10}$$

We observe that a lifting $W_\bullet$ of $Y_\bullet$ on Z is unique if it exists: For each $i = 0, \ldots, n-1$, the component W_{i+1} is necessarily the strict transform of Y_{i+1} with respect to the bimeromorphic morphism $W_i \to Y_i$. We shall also say that $(W_\bullet, \mathrm{cor}(Y_\bullet, \pi))$ is *the lifting* of $Y_\bullet$ to Z.

Proposition 10.2.3 *Let $\pi\colon Z \to X$ be a proper bimeromorphic morphism, where Z is a Kähler manifold. Let $W_\bullet$ be a lifting of $Y_\bullet$. Then for any closed positive $(1,1)$-current T on X, we have*

$$\nu_{W_\bullet}(\pi^* T) = \nu_{Y_\bullet}(T)\, \mathrm{cor}(Y_\bullet, \pi). \tag{10.11}$$

Proof We argue by induction on $n \geq 0$. The case $n = 0$ is trivial. In general, assume that $n \geq 1$ and the result is proved in dimension $n-1$.

For simplicity, we write $\nu = \nu_{Y_\bullet}$ and $\nu' = \nu_{W_\bullet}$. Let μ (resp. μ') be the valuation of currents defined by the truncated flag $Y_1 \supseteq \cdots \supseteq Y_n$ (resp. $W_1 \supseteq \cdots \supseteq W_n$). Then we need to show that

$$\begin{aligned} &\left[\nu'(\pi^*T)_1\ \mu'(\mathrm{Tr}_{W_1}(\pi^*T - \nu'(\pi^*T)_1[W_1]))\right]\\ =&\left[\nu(T)_1\ \mu(\mathrm{Tr}_{Y_1}(T - \nu(T)_1[Y_1]))\right]\mathrm{cor}(Y_\bullet, \pi). \end{aligned}\tag{10.12}$$

By Zariski's main theorem,

$$\nu'(\pi^*T)_1 = \nu(T)_1 =: c.$$

By the inductive hypothesis, we have

$$\mu'\Big(\Pi^*\,\mathrm{Tr}_{Y_1}(T - c[Y_1])\Big) = \mu\Big(\mathrm{Tr}_{Y_1}(T - c[Y_1])\Big)\,\mathrm{cor}\,(Y_1 \supseteq \cdots \supseteq Y_n, \Pi)\,, \tag{10.13}$$

where $\Pi\colon W_1 \longrightarrow Y_1$ is the restriction of π. Since $\pi^*[Y_1] = [W_1] + (\pi^*[Y_1] - [W_1])$, Lemma 8.2.1 and Proposition 8.2.1 give

$$\begin{aligned} \mathrm{Tr}_{W_1}\Big(\pi^*T - c[W_1]\Big) \sim_P{} & \mathrm{Tr}_{W_1}\Big(\pi^*(T - c[Y_1])\Big)\\ & + c\,\mathrm{Tr}_{W_1}\Big(\pi^*[Y_1] - [W_1]\Big)\\ \sim_P{} & \Pi^*\,\mathrm{Tr}_{Y_1}\Big(T - c[Y_1]\Big) + c\,\mathrm{Tr}_{W_1}\Big(\pi^*[Y_1] - [W_1]\Big). \end{aligned}$$

So

$$\mu'\Big(\mathrm{Tr}_{W_1}\,(\pi^*T - c[W_1])\Big) = \mu'\Big(\Pi^*\,\mathrm{Tr}_{Y_1}\,(T - c[Y_1])\Big) + c\mu'\Big(\mathrm{Tr}_{W_1}\,(\pi^*[Y_1] - [W_1])\Big).$$

Combining the above with (10.13), we see that (10.12) follows. □

Proposition 10.2.4 *Let $\pi\colon Z \to X$, $p\colon Z' \to Z$ be proper bimeromorphic morphisms with Z and Z' being Kähler manifolds. Assume that $Y_\bullet$ admits a lifting $W_\bullet$ (resp. $W'_\bullet$) to Z (resp. Z'). Then*

$$\mathrm{cor}(Y_\bullet, \pi\circ p) = \mathrm{cor}(Y_\bullet, \pi)\,\mathrm{cor}(W_\bullet, p). \tag{10.14}$$

Proof We let $\pi' = \pi \circ p$. Then $W'_\bullet$ is also the lifting of $W_\bullet$ with respect to p.

$$\begin{tikzcd} Z' \arrow[rr, "p"] \arrow[dr, "\pi'"'] & & Z \arrow[dl, "\pi"] \\ & X. & \end{tikzcd}$$

We argue by induction on $n \geq 1$. The case $n = 1$ is trivial. Assume that $n \geq 2$ and the case $n-1$ has been solved. Then by (10.10), the desired formula (10.14) can be reformulated as

$$\begin{bmatrix} 1 & \nu_{W_1'\supseteq\cdots\supseteq W_n'}((\pi'^*[Y_1]-[W_1'])|_{W_1'}) \\ 0 & \operatorname{cor}(Y_1\supseteq\cdots\supseteq Y_n, \pi'|_{W_1'}: W_1'\to Y_1)\end{bmatrix} = \begin{bmatrix} 1 & \nu_{W_1\supseteq\cdots\supseteq W_n}((\pi^*[Y_1]-[W_1])|_{W_1}) \\ 0 & \operatorname{cor}(Y_1\supseteq\cdots\supseteq Y_n, \pi|_{W_1}: W_1\to Y_1)\end{bmatrix} \cdot \begin{bmatrix} 1 & \nu_{W_1'\supseteq\cdots\supseteq W_n'}((p^*[W_1]-[W_1'])|_{W_1'}) \\ 0 & \operatorname{cor}(W_1\supseteq\cdots\supseteq W_n, p|_{W_1'}: W_1'\to W_1)\end{bmatrix}.$$

By the inductive hypothesis, this is equivalent to

$$\nu_{W_1'\supseteq\cdots\supseteq W_n'}\left((\pi'^*[Y_1]-[W_1'])|_{W_1'}\right) = \nu_{W_1'\supseteq\cdots\supseteq W_n'}\left((p^*[W_1]-[W_1'])|_{W_1'}\right) + \nu_{W_1\supseteq\cdots\supseteq W_n}\left((\pi^*[Y_1]-[W_1])|_{W_1}\right)\operatorname{cor}\left(W_1\supseteq\cdots\supseteq W_n, p|_{W_1'}: W_1'\to W_1\right),$$

which, thanks to Proposition 10.2.3, can be further rewritten as

$$\nu_{W_1'\supseteq\cdots\supseteq W_n'}\left((\pi'^*[Y_1]-[W_1'])|_{W_1'}\right) = \nu_{W_1'\supseteq\cdots\supseteq W_n'}\left((p^*[W_1]-[W_1'])|_{W_1'}\right) + \nu_{W_1'\supseteq\cdots\supseteq W_n'}\left(p|_{W_1'}^*\left((\pi^*[Y_1]-[W_1])|_{W_1}\right)\right).$$

Indeed, as divisors on W_1',

$$(\pi'^*[Y_1]-[W_1'])|_{W_1'} = (p^*[W_1]-[W_1'])|_{W_1'} + p|_{W_1'}^*((\pi^*[Y_1]-[W_1])|_{W_1}).$$

The desired equality is therefore exactly the additivity of the valuation, Proposition 10.2.2. □

Theorem 10.2.1 *Let $\pi\colon Z\to X$ be a proper bimeromorphic morphism from a reduced complex space Z. Then there is a modification $W\to X$ dominating $Z\to X$ such that $Y_\bullet$ admits a lifting to W.*

Recall that in this book, a modification means a finite composition of blowing-ups with smooth centers.

Proof By Hironaka's Chow lemma Theorem B.1.2, we may assume that π is a modification.

We begin by setting $W_0 = Z$. We will construct W_i inductively for each i. Assume that for $0\leq i<n$ a smooth partial flag $W_0\supseteq\cdots\supseteq W_i$ has been constructed on a modification $\pi_i: Z_i\to Z$ so that $\pi\circ\pi_i$ restricts to bimeromorphic morphisms $W_j\to Y_j$ for each $j=0,\ldots,i$.

By Zariski's main theorem, $W_i\to Y_i$ is an isomorphism outside a codimension 2 subset of Y_i. We let W_{i+1} be the strict transform of Y_{i+1} in W_i. The problem is that W_{i+1} is not necessarily smooth.

We will further modify Z_i and lift $W_1,\ldots,W_{i+1}$ in order to make the flag smooth. Take the embedded resolution of (W_j, W_{i+1}), say $W_j'\to W_j$ for each $j=0,\ldots,i$.

We have canonical embeddings $W'_i \hookrightarrow W'_{i-1} \hookrightarrow \cdots \hookrightarrow W'_0$ making the following diagram commutative:

$$\begin{tikzcd} W'_i \arrow[r, hook] \arrow[d] & W'_{i-1} \arrow[r, hook] \arrow[d] & \cdots \arrow[r, hook] \arrow[d, phantom, "\vdots"] & W'_0 \arrow[d] \\ W_i \arrow[r, hook] & W_{i-1} \arrow[r, hook] & \cdots \arrow[r, hook] & W_0 \end{tikzcd}$$

Let W'_{i+1} be the strict transform of W_{i+1} in W'_i. It suffices to define π_{i+1} as the morphism $W'_0 \to Z_i \to Z$ and replace $W_0 \supseteq \cdots \supseteq W_{i+1}$ by $W'_0 \supseteq \cdots \supseteq W'_{i+1}$. $\square$

We also include a more general version of Definition 10.2.3 for currents on bimeromorphic models.

Definition 10.2.5 Let $\pi\colon Z \to X$ be a proper bimeromorphic morphism from a normal compact Kähler space, and let S be a closed positive $(1,1)$-current on Z. Choose a proper bimeromorphic morphism $h\colon \widehat{Z} \to Z$ from a compact Kähler manifold such that $Y_\bullet$ admits a lifting $(W_\bullet, g)$ to $\widehat{Z}$ with respect to $\pi \circ h$, where

$$g = \operatorname{cor}(Y_\bullet, \pi \circ h).$$

We define

$$\nu_{Y_\bullet}(S) \coloneqq \nu_{W_\bullet}(h^*S)g^{-1} \in \mathbb{R}^n. \tag{10.15}$$

Note that it could happen that $\nu\nu_{Y_\bullet}(S) \notin \mathbb{R}^n_{\geq 0}$.

Lemma 10.2.1 *The definition in Definition 10.2.5 is independent of the choice of h. Moreover, if $p\colon Z' \to Z$ is a proper bimeromorphic morphism from a normal compact Kähler space, then, with respect to the model $\pi \circ p\colon Z' \to X$,*

$$\nu_{Y_\bullet}(p^*S) = \nu_{Y_\bullet}(S).$$

Proof We first prove independence of the choice of h. Let $h_i\colon \widehat{Z}_i \to Z$, $i = 1, 2$, be two choices, and let $(W^i_\bullet, g_i)$ be the corresponding liftings, with $g_i = \operatorname{cor}(Y_\bullet, \pi \circ h_i)$. Choose a common modification over Z, $q_i\colon U \to \widehat{Z}_i$ such that $Y_\bullet$ admits a lifting $W_\bullet$ to U. Then $W_\bullet$ is also the lifting of $W^i_\bullet$ with respect to q_i. Put

$$G_i = \operatorname{cor}(W^i_\bullet, q_i).$$

By Propositions 10.2.3 and 10.2.4,

$$\begin{aligned} \nu_{W_\bullet}(q_i^*h_i^*S)\operatorname{cor}(Y_\bullet, \pi \circ h_i \circ q_i)^{-1} &= \nu_{W^i_\bullet}(h_i^*S)G_i(g_iG_i)^{-1} \\ &= \nu_{W^i_\bullet}(h_i^*S)g_i^{-1}. \end{aligned}$$

Since $h_1 \circ q_1 = h_2 \circ q_2$, the left-hand side is independent of i. This proves that (10.15) is well-defined.

For the birational invariance, choose a proper bimeromorphic morphism $q\colon \widehat{Z} \to Z'$ from a compact Kähler manifold such that $Y_\bullet$ admits a lifting to $\widehat{Z}$ with respect to

$\pi \circ p \circ q$. Computing $\nu_{Y_\bullet}(p^*S)$ using q, and computing $\nu_{Y_\bullet}(S)$ using $p \circ q$, gives the same expression. □

Remark 10.2.2 Suppose that X is a normal projective variety. Consider a rank n (surjective) valuation $\nu\colon \mathbb{C}(X)^\times \to \mathbb{Z}^n$ and a closed positive $(1,1)$-current T on X. Then we can always define $\nu(T) \in \mathbb{R}^n$ as follows: Take a resolution $\pi\colon Y \to X$ such that there is a smooth flag $Y_\bullet$ on Y and $g \in \mathrm{Aut}(\mathbb{Z}^n_{\mathrm{lex}})$ such that

$$\nu = \nu_{Y_\bullet} g.$$

Then we define

$$\nu(T) \coloneqq \nu_{Y_\bullet}(\pi^* T) g.$$

This definition does not depend on the choice of π, by passing to a common resolution and applying Propositions 10.2.3 and 10.2.4.

10.3 Algebraic partial Okounkov bodies

Let X be a connected smooth complex projective variety of dimension n and (L, h) be a Hermitian big line bundle on X.

Let h_0 be a smooth Hermitian metric on L. Let $\theta = c_1(L, h_0)$. Then we can identify h with a function $\varphi \in \mathrm{PSH}(X, \theta)$. We will use interchangeably the notations (θ, φ) and (L, h). We assume that $\mathrm{vol}\,\theta_\varphi > 0$ in this section.

Fix a rank n valuation $\nu\colon \mathbb{C}(X)^\times \to \mathbb{Z}^n$, which without loss of generality can be assumed to be surjective.

We will adopt the notations of Section 10.1.

10.3.1 The spaces of sections

Definition 10.3.1 We will write

$$\begin{aligned}
\Gamma(\theta, \varphi) &\coloneqq \left\{ (\nu(s), k) : k \in \mathbb{N}, s \in \mathrm{H}^0(X, L^k \otimes \mathcal{I}(k\varphi))^\times \right\}, \\
\Delta_k(\theta, \varphi) &\coloneqq \mathrm{Conv} \left\{ k^{-1}\nu(s) : s \in \mathrm{H}^0(X, L^k \otimes \mathcal{I}(k\varphi))^\times \right\} \subseteq \mathbb{R}^n, \quad k \in \mathbb{Z}_{>0}.
\end{aligned}$$

When $\varphi = V_\theta$, we simply write $\Gamma(L)$ and $\Delta_k(L)$ instead.

Here and in the sequel, the cross notation means excluding 0. Here Conv denotes the convex hull. For large enough k, $\Delta_k(\theta, \varphi)$ is non-empty thanks to Theorem 7.4.1.

Definition 10.3.2 Assume that φ has analytic singularities. We define

$$\Gamma^\infty(\theta, \varphi) \coloneqq \left\{ (\nu(s), k) : k \in \mathbb{N}, s \in \mathrm{H}^0(X, L^k \otimes \mathcal{I}_\infty(k\varphi))^\times \right\}. \tag{10.16}$$

Recall that $\mathcal{I}_\infty$ is introduced in Definition 1.6.6.

For later use, we introduce a twisted version as well.

Definition 10.3.3 If T is a holomorphic line bundle on X, we introduce

$$\begin{aligned}\Delta_{k,T}(\theta,\varphi) :=& \mathrm{Conv}\left\{k^{-1}\nu(s) : s \in \mathrm{H}^0(X, T\otimes L^k \otimes \mathcal{I}(k\varphi))^\times\right\} \subseteq \mathbb{R}^n,\\ \Delta_{k,T}(L) :=& \mathrm{Conv}\left\{k^{-1}\nu(s) : s \in \mathrm{H}^0(X, T\otimes L^k)^\times\right\} \subseteq \mathbb{R}^n\end{aligned}$$

for all $k \in \mathbb{Z}_{>0}$.

10.3.2 Algebraic Okounkov bodies

Proposition 10.3.1 *There is a convex body $\Delta \in \mathcal{K}_n$ such that $\Gamma(L) \in \mathcal{S}'(\Delta)$.*

Recall that the notations $\mathcal{K}_n$ and $\mathcal{S}'(\Delta)$ are introduced in Appendix C.

Proof **Step 1**. We first show that there is $\Delta \in \mathcal{K}_n$ such that $\Delta_k(L) \subseteq \Delta$. For this purpose, using Remark 10.2.1, we may assume that ν is induced by an admissible flag $Y_\bullet$ on X. We may further assume that each Y_i is smooth.

Fix $s \in \mathrm{H}^0(X, L^k)^\times$ for some $k \in \mathbb{Z}_{>0}$. Assume that $s \neq 0$. We need to show that for each $i = 1, \dots, n$, $\nu(s)_i \leq Ck$ for some constant $C > 0$, independent of the choices of k and s.

Fix an ample divisor H on X. Take a large enough integer $b_1 > 0$ such that

$$(L - b_1 Y_1) \cdot H^{n-1} < 0.$$

Then $\nu(s)_1 \leq b_1 k$. Next take a large enough integer b_2 such that, for every $a \in [0, b_1]$,

$$\left((L - aY_1)|_{Y_1} - b_2 Y_2\right) \cdot H^{n-2} < 0.$$

After removing the first vanishing order of s along Y_1, the resulting section on Y_1 is a section of $k((L-aY_1)|_{Y_1})$ for some $a \in [0, b_1]$. Hence $\nu(s)_2 \leq b_2 k$. Continuing in this manner and choosing the constants uniformly over the compact ranges of the preceding normalized vanishing orders, we conclude that $\nu(s)_i/k$ is bounded for each i.

Step 2. Observe that $\Gamma(L)$ is clearly a semigroup. It remains to show that $\Gamma(L)$ generates $\mathbb{Z}^{n+1}$ as an Abelian group.

For this purpose, take two very ample divisors A and B so that $L = \mathcal{O}_X(A - B)$. After choosing A and B ample enough, we may guarantee that there exist sections $s_0 \in \mathrm{H}^0(X, A)$, $t_i \in \mathrm{H}^0(X, B)$ for $i = 0, \dots, n$ such that

$$\nu(s_0) = \nu(t_0) = 0$$

and $\nu(t_i)$ is the i-th unit vector $e_i \in \mathbb{R}^n$ for $i = 1, \dots, n$.

Since L is big, we can find $m_0 > 0$ such that for any $m \geq m_0$ we can find an effective divisor F_m on X linearly equivalent to $mL - B$. Let $f_m = \nu([F_m])$. Then we find that

$$(f_m, m), (f_m + e_1, m), \ldots, (f_m + e_n, m) \in \Gamma(L).$$

Since $(m+1)L$ is linearly equivalent to $A + F_m$, we have

$$(f_m, m+1) \in \Gamma(L).$$

It follows that $\Gamma(L)$ generates $\mathbb{Z}^{n+1}$. $\square$

Thanks to Proposition 10.3.1, we can introduce the next definition.

Definition 10.3.4 We define the *Okounkov body* of L with respect to the valuation ν as

$$\Delta_\nu(L) \coloneqq \Delta(\Gamma(L)).$$

When ν is induced by a smooth flag $Y_\bullet$ on X, we also write $\Delta_{Y_\bullet}(L)$ instead. The same convention applies to the partial Okounkov bodies studied below as well.

Proposition 10.3.2 *The Okounkov body $\Delta_\nu(L)$ depends only on the numerical class of L.*

See [LM09, Proposition 4.1] for the elegant proof.

Corollary 10.3.1 *We have*

$$\operatorname{vol} \Delta_\nu(L) = \frac{1}{n!} \operatorname{vol} L. \tag{10.17}$$

Proof This follows immediately from Proposition 10.3.1 and Theorem 10.1.1. $\square$

Proposition 10.3.3 *Assume that φ has analytic singularities and θ_φ is a Kähler current. Then we have*

$$\Gamma^\infty(\theta, \varphi) \in \mathcal{S}'(\Delta_\nu(L))$$

and

$$\operatorname{vol} \Gamma^\infty(\theta, \varphi) = \frac{1}{n!} \int_X \theta_\varphi^n.$$

Proof Replacing X by a modification, we may assume that φ has log singularities along an effective $\mathbb{Q}$-divisor D. See Theorem 1.6.1.

In this case,

$$\Gamma^\infty(\theta, \varphi) = \left\{ (\nu(s), k) : k \in \mathbb{N}, s \in \mathrm{H}^0\left(X, L^k \otimes \mathcal{O}_X(-\lceil kD \rceil)\right)^\times \right\}$$

Since $L - D$ is ample by Lemma 1.6.1, our assertion follows from the same argument as Proposition 10.3.1. $\square$

We first extend Theorem 10.1.1 to the twisted case.

Proposition 10.3.4 *For any holomorphic line bundle T on X, as $k \to \infty$*

$$\Delta_{k,T}(L) \xrightarrow{d_{\mathrm{Haus}}} \Delta_\nu(L).$$

Proof As L is big, we can take $k_0 \in \mathbb{Z}_{>0}$ so that

(1) $T^{-1} \otimes L^{k_0}$ admits a non-zero global holomorphic section s_0, and
(2) $T \otimes L^{k_0}$ admits a non-zero global holomorphic section s_1.

Then for $k \in \mathbb{Z}_{>k_0}$, we have injective linear maps

$$\mathrm{H}^0(X, L^{k-k_0}) \xrightarrow{\times s_1} \mathrm{H}^0(X, T \otimes L^k) \xrightarrow{\times s_0} \mathrm{H}^0(X, L^{k+k_0}).$$

It follows that

$$(k-k_0)\Delta_{k-k_0}(L) + \nu(s_1) \subseteq k\Delta_{k,T}(L) \subseteq (k+k_0)\Delta_{k+k_0}(L) - \nu(s_0).$$

Using Theorem 10.1.1, we conclude. □

Proposition 10.3.5 *Let L' be another big line bundle on X. Then*

$$\Delta_\nu(L) + \Delta_\nu(L') \subseteq \Delta_\nu(L \otimes L').$$

Proof Observe that for each $k \in \mathbb{N}$, we have

$$\Delta_k(L) + \Delta_k(L') \subseteq \Delta_k(L \otimes L').$$

So our assertion follows immediately from Theorem 10.1.1. □

Proposition 10.3.6 *For any $a \in \mathbb{Z}_{>0}$, we have*

$$\Delta_\nu(L^a) = a\Delta_\nu(L).$$

Proof This is an immediate consequence of Theorem 10.1.1. □

10.3.3 Construction of partial Okounkov bodies

Theorem 10.3.1 *We have*

$$\Gamma(\theta, \varphi) \in \overline{\mathcal{S}'(\Delta_\nu(L))}_{>0}.$$

We refer to Definition 10.1.3 for the definition of $\overline{\mathcal{S}'(\Delta_\nu(L))}_{>0}$.

This theorem allows us to give the following definition:

Definition 10.3.5 The *partial Okounkov body* of (L, h) is defined as

$$\Delta_\nu(L, h) = \Delta_\nu(\theta, \varphi) := \Delta\left(\Gamma(\theta, \varphi)\right). \tag{10.18}$$

When ν is induced by an admissible flag $Y_\bullet$ on X (see Definition 10.2.1), we also say that $\Delta_\nu(\theta,\varphi)$ is the *partial Okounkov body* of (L,h) or of (θ,φ) with respect to $Y_\bullet$. In this case, we also write $\Delta_{Y_\bullet}$ instead of Δ_ν.

Note that when h has minimal singularities, we have

$$\Delta_\nu(L,h)=\Delta_\nu(L).$$

So partial Okounkov bodies generalize Okounkov bodies.

Corollary 10.3.2 *We have*

$$\operatorname{vol}\Delta_\nu(\theta,\varphi)=\frac{1}{n!}\operatorname{vol}\theta_\varphi. \tag{10.19}$$

We will prove Theorem 10.3.1 and Corollary 10.3.2 at the same time. The proof relies on the pseudometric d_{sg} introduced in (10.1).

Proof **Step 1**. We first assume that φ has analytic singularities and θ_φ is a Kähler current.

We claim that

$$d_{\mathrm{sg}}(\Gamma^\infty(\theta,\varphi),\Gamma(\theta,\varphi))=0. \tag{10.20}$$

Observe that for each $\epsilon\in\mathbb{Q}_{>0}$, we have

$$\mathrm{H}^0\Big(X,L^k\otimes\mathcal{I}_\infty(k\varphi)\Big)\subseteq\mathrm{H}^0\Big(X,L^k\otimes\mathcal{I}(k\varphi)\Big)\subseteq\mathrm{H}^0\left(X,L^k\otimes\mathcal{I}_\infty(k(1-\epsilon)\varphi)\right)$$

for all large enough k. This is a consequence of Lemma 1.6.3. Therefore, it suffices to show that

$$\lim_{\mathbb{Q}\ni\epsilon\to 0+}\operatorname{vol}\Gamma^\infty(\theta,(1-\epsilon)\varphi)=\operatorname{vol}\Gamma^\infty(\theta,\varphi).$$

This follows from the explicit formula in Proposition 10.3.3.

Step 2. We next handle the case where θ_φ is a Kähler current.

Let $(\varphi_j)_j$ be a quasi-equisingular approximation of φ in $\mathrm{PSH}(X,\theta)$. Then $\varphi_j\xrightarrow{d_S}P_\theta[\varphi]_{\mathcal{I}}$ by Corollary 7.1.2.

In this case, it suffices to prove that

$$\Gamma(\theta,\varphi_j)\xrightarrow{d_{\mathrm{sg}}}\Gamma(\theta,\varphi). \tag{10.21}$$

In fact, by Theorem 7.4.1, we have

$$\begin{aligned}
&d_{\mathrm{sg}}\left(\Gamma(\theta,\varphi_j),\Gamma(\theta,\varphi)\right)\\
=&\varlimsup_{k\to\infty}k^{-n}\left(h^0\left(X,L^k\otimes\mathcal{I}(k\varphi_j)\right)-h^0\left(X,L^k\otimes\mathcal{I}(k\varphi)\right)\right)\\
=&\lim_{k\to\infty}k^{-n}h^0\left(X,L^k\otimes\mathcal{I}(k\varphi_j)\right)-\lim_{k\to\infty}k^{-n}h^0\left(X,L^k\otimes\mathcal{I}(k\varphi)\right)\\
=&\frac{1}{n!}\operatorname{vol}\theta_{\varphi_j}-\frac{1}{n!}\operatorname{vol}\theta_\varphi.
\end{aligned}$$

Letting $j \to \infty$, we conclude (10.21) by Theorem 6.2.5.

Step 3. Now we only assume that $\operatorname{vol}\theta_\varphi > 0$. We may replace φ with $P_\theta[\varphi]_{\mathcal{I}}$ and then assume that $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$.

Take a potential $\psi \in \mathrm{PSH}(X,\theta)$ such that $\psi \leq \varphi$ and θ_ψ is a Kähler current. The existence of ψ is proved in Lemma 2.4.5. For each $\epsilon \in (0,1)$, let $\varphi_\epsilon = (1-\epsilon)\varphi + \epsilon\psi$. It suffices to show that

$$\Gamma(\theta,\varphi_\epsilon) \xrightarrow{d_{\mathrm{sg}}} \Gamma(\theta,\varphi)$$

as $\epsilon \to 0+$. We compute using Theorem 7.4.1:

$$\begin{aligned}
&d_{\mathrm{sg}}\left(\Gamma(\theta,\varphi_\epsilon),\Gamma(\theta,\varphi)\right)\\
=&\varlimsup_{k\to\infty} k^{-n}\left(h^0\left(X, L^k\otimes\mathcal{I}(k\varphi)\right) - h^0\left(X, L^k\otimes\mathcal{I}(k\varphi_\epsilon)\right)\right)\\
=&\lim_{k\to\infty} k^{-n}h^0\left(X, L^k\otimes\mathcal{I}(k\varphi)\right) - \lim_{k\to\infty} k^{-n}h^0\left(X, L^k\otimes\mathcal{I}(k\varphi_\epsilon)\right)\\
=&\frac{1}{n!}\operatorname{vol}\theta_\varphi - \frac{1}{n!}\operatorname{vol}\theta_{\varphi_\epsilon}\\
\to&0
\end{aligned}$$

by Theorem 6.2.5, as $\epsilon \to 0+$. □

Remark 10.3.1 It follows from the proof that if φ has analytic singularities and θ_φ is a Kähler current, then (10.20) holds.

If we take a modification $\pi\colon Y \to X$ such that $\pi^*\varphi$ has log singularities along an effective $\mathbb{Q}$-divisor D on Y, then

$$\Delta_\nu(\theta,\varphi) = \Delta_\nu(\pi^*L - D) + \nu(D). \tag{10.22}$$

This is a very special case of Theorem 10.4.2.

10.3.4 Basic properties of partial Okounkov bodies

Proposition 10.3.7 *The partial Okounkov body $\Delta_\nu(L,h)$ depends only on $\mathrm{dd^c}h$, not on the explicit choices of L, h_0, h.*

Thanks to this result, given a closed positive $(1,1)$-current $T \in c_1(L)$ on X with $\int_X T^n > 0$, we can write

$$\Delta_\nu(T) \coloneqq \Delta_\nu(\theta,\varphi)$$

if $T = \theta + \mathrm{dd^c}\varphi$ for some $\varphi \in \mathrm{PSH}(X,\theta)$.

Proof There are two different claims to prove, as detailed in the two steps below.

Step 1. Let h_0' be another Hermitian metric on L. Set $\theta' = c_1(L,h_0')$. Write $\mathrm{dd^c}f = \theta - \theta'$. Let $\varphi' = \varphi + f \in \mathrm{PSH}(X,\theta')$. Then

$$\Delta_\nu(\theta,\varphi) = \Delta_\nu(\theta',\varphi'). \tag{10.23}$$

This is obvious since $\Gamma(\theta, \varphi) = \Gamma(\theta', \varphi')$.

Step 2. Let L' be another big line bundle on X. By Step 1, we may assume that the reference Hermitian metric h'_0 on L' is such that $c_1(L', h'_0) = \theta$.

Let h' be a plurisubharmonic metric on L' with $c_1(L, h) = c_1(L', h')$. Then

$$\Delta_\nu(L, h) = \Delta_\nu(L', h').$$

By the approximation step in the construction of $\Delta_\nu(L, h)$ and the continuity of the extension in Theorem 10.1.2, it is enough to prove the assertion when $c_1(L, h)$ has analytic singularities. After taking a birational resolution, it suffices to deal with the case where $c_1(L, h)$ has analytic singularities along an effective $\mathbb{Q}$-divisor D. By rescaling, we may also assume that D is a divisor. By Remark 10.3.1, we further reduce to the case where $c_1(L, h)$ is not singular.

In this case, the assertion is proved in Proposition 10.3.2. □

Proposition 10.3.8 *Let $\varphi, \psi \in \mathrm{PSH}(X, \theta)_{>0}$. Assume that $\varphi \preceq_{\mathcal{I}} \psi$. Then*

$$\Delta_\nu(\theta, \varphi) \subseteq \Delta_\nu(\theta, \psi). \tag{10.24}$$

In particular, we always have

$$\Delta_\nu(\theta, \varphi) \subseteq \Delta_\nu(L).$$

Proof This follows from Corollary 10.1.2. □

Theorem 10.3.2 *The Okounkov body map*

$$\Delta_\nu(\theta, \bullet) : (\mathrm{PSH}(X, \theta)_{>0}, d_S) \to (\mathcal{K}_n, d_{\mathrm{Haus}})$$

is continuous.

Proof Let $\varphi_j \to \varphi$ be a d_S-convergent sequence in $\mathrm{PSH}(X, \theta)_{>0}$. We want to show that

$$\Delta_\nu(\theta, \varphi_j) \xrightarrow{d_{\mathrm{Haus}}} \Delta_\nu(\theta, \varphi). \tag{10.25}$$

Replacing the potentials by their P-envelopes does not change their partial Okounkov bodies, since P-equivalent potentials are $\mathcal{I}$-equivalent, nor does it change their d_S-classes. Thus we may assume that all φ_j's and φ are model potentials.

By Proposition 6.2.3, after passing to a subsequence we can sandwich φ_j between a decreasing sequence and an increasing sequence, both converging to φ in d_S. Using Proposition 10.3.8 and Theorem C.1.1, it is enough to prove (10.25) when $(\varphi_j)_j$ is monotone. By Theorem 6.2.3, replacing φ_j and φ by their $\mathcal{I}$-envelopes preserves this reduction, so we may further assume that the φ_j's and φ are $\mathcal{I}$-model. In both cases, we claim that

$$\Gamma(\theta, \varphi_j) \xrightarrow{d_{\mathrm{sg}}} \Gamma(\theta, \varphi)$$

as $j \to \infty$. In fact, using Theorem 7.4.1, we can compute

$$\begin{aligned} d_{\mathrm{sg}}\left(\Gamma(\theta,\varphi_j),\Gamma(\theta,\varphi)\right) =& \varlimsup_{k\to\infty} k^{-n}\left|h^0\left(X,L^k\otimes\mathcal{I}(k\varphi_j)\right)-h^0\left(X,L^k\otimes\mathcal{I}(k\varphi)\right)\right| \\ =&\frac{1}{n!}\left|\operatorname{vol}\theta_{\varphi_j}-\operatorname{vol}\theta_\varphi\right|, \end{aligned}$$

which converges to 0 by Theorem 6.2.5. □

Proposition 10.3.9 *Let $\pi\colon Y\to X$ be a modification. Then*

$$\Delta_\nu(\pi^*L,\pi^*h)=\Delta_\nu(L,h).$$

Proof Thanks to Proposition 3.2.5, we may assume that φ is $\mathcal{I}$-model. By Theorem 7.1.1, we can find a sequence $(\varphi_j)_j$ with analytic singularities in $\mathrm{PSH}(X,\theta)$ such that $\varphi_j\xrightarrow{d_S}\varphi$. It is clear that $\pi^*\varphi_j\xrightarrow{d_S}\pi^*\varphi$. By Theorem 10.3.2, we may then reduce to the case where φ has analytic singularities. In this case, it suffices to apply Remark 10.3.1. □

Proposition 10.3.10 *Let (L',h') be another Hermitian big line bundle on X. Then*

$$\Delta_\nu(L,h)+\Delta_\nu(L',h')\subseteq\Delta_\nu(L\otimes L',h\otimes h').$$

Proof Take a Hermitian metric h_0' on L' and let $\theta'=c_1(L',h_0')$. We identify h' with $\varphi'\in\mathrm{PSH}(X,\theta')$. Then we need to show

$$\Delta_\nu(\theta,\varphi)+\Delta_\nu(\theta',\varphi')\subseteq\Delta_\nu(\theta+\theta',\varphi+\varphi'). \tag{10.26}$$

We observe that

$$P_\theta[\varphi]_{\mathcal{I}}+P_{\theta'}[\varphi']_{\mathcal{I}}\sim_{\mathcal{I}}\varphi+\varphi'.$$

Thus, after replacing φ and φ' by their $\mathcal{I}$-envelopes, in view of Proposition 10.3.8, we may assume that φ and φ' are $\mathcal{I}$-good.

By Theorem 7.1.1, we can find sequences $(\varphi_j)_j$ and $(\varphi_j')_j$ in $\mathrm{PSH}(X,\theta)_{>0}$ and $\mathrm{PSH}(X,\theta')_{>0}$ respectively such that

(1) φ_j and φ_j' both have analytic singularities for all $j\geq 1$, and
(2) $\varphi_j\xrightarrow{d_S}\varphi$, $\varphi_j'\xrightarrow{d_S}\varphi'$.

Then $\varphi_j+\varphi_j'\in\mathrm{PSH}(X,\theta+\theta')_{>0}$ and $\varphi_j+\varphi_j'\xrightarrow{d_S}\varphi+\varphi'$ by Theorem 6.2.2. Thus, by Theorem 10.3.2, we may assume that φ and φ' both have analytic singularities. Taking a birational resolution, we may further assume that they have log singularities. By Remark 10.3.1, we reduce to the case without singularities, in which case the result is just Proposition 10.3.5. □

Theorem 10.3.3 *Let $\varphi,\psi\in\mathrm{PSH}(X,\theta)_{>0}$. Then for any $t\in(0,1)$,*

$$\Delta_\nu\left(\theta,t\varphi+(1-t)\psi\right)\supseteq t\Delta_\nu(\theta,\varphi)+(1-t)\Delta_\nu(\theta,\psi). \tag{10.27}$$

Proof We may assume that t is rational as a consequence of Theorem 10.3.2. Similarly, as in the proof of Proposition 10.3.10, we may reduce to the case where both φ and ψ

have analytic singularities. In this case, let $N > 0$ be an integer such that Nt is an integer. Then for any $s \in \mathrm{H}^0(X, L^k \otimes \mathcal{I}_\infty(k\varphi))$ and $r \in \mathrm{H}^0(X, L^k \otimes \mathcal{I}_\infty(k\psi))$, we have

$$s^{tN} \otimes r^{N-tN} \in \mathrm{H}^0\left(X, L^{kN} \otimes \mathcal{I}_\infty\left(kN(t\varphi + (1-t)\psi)\right)\right).$$

By Theorem 10.1.1 and Remark 10.3.1, (10.27) follows. □

Proposition 10.3.11 *For any $a \in \mathbb{Z}_{>0}$,*

$$\Delta_\nu(a\theta, a\varphi) = a\Delta_\nu(\theta, \varphi).$$

Proof As in the proof of Proposition 10.3.10, we may assume that φ has log singularities. Using Remark 10.3.1, we reduce to the case without the singularities, which is proved in Proposition 10.3.6. □

In particular, if S is a closed positive $(1,1)$-current on X with $\int_X S^n > 0$ and such that

$$[S] \in \mathrm{NS}^1(X)_{\mathbb{Q}},$$

we can define

$$\Delta_\nu(S) \coloneqq a^{-1}\Delta_\nu(aS) \tag{10.28}$$

for a sufficiently divisible positive integer a. This definition is independent of the choice of a thanks to Proposition 10.3.11.

We also need the following perturbation. Let A be an ample line bundle on X. Fix a Hermitian metric h_A on A such that $\omega \coloneqq c_1(A, h_A)$ is a Kähler form on X.

Proposition 10.3.12 *As $\mathbb{Q}_{>0} \ni \delta \searrow 0$, the convex bodies $\Delta_\nu(\theta + \delta\omega + \mathrm{dd^c}\varphi)$ are decreasing and*

$$\Delta_\nu(\theta + \delta\omega + \mathrm{dd^c}\varphi) \xrightarrow{d_{\mathrm{Haus}}} \Delta_\nu(\theta_\varphi).$$

Proof Let $0 \leq \delta < \delta'$ be two rational numbers. Take $C \in \mathbb{N}_{>0}$ divisible enough, so that $C\delta$ and $C\delta'$ are both integers. Then by Proposition 10.3.10,

$$\Delta_\nu(C\theta + C\delta\omega + C\mathrm{dd^c}\varphi) \subseteq \Delta_\nu(C\theta + C\delta'\omega + C\mathrm{dd^c}\varphi).$$

It follows that

$$\Delta_\nu(\theta + \delta\omega + \mathrm{dd^c}\varphi) \subseteq \Delta_\nu(\theta + \delta'\omega + \mathrm{dd^c}\varphi).$$

On the other hand,

$$\operatorname{vol}\Delta_\nu(\theta + \delta\omega + \mathrm{dd^c}\varphi) = \frac{1}{n!}\operatorname{vol}(\theta + \delta\omega)_\varphi.$$

As $\delta \to 0+$, the right-hand side converges to

$$\operatorname{vol}\Delta_\nu(\theta, \varphi) = \frac{1}{n!}\operatorname{vol}\theta_\varphi,$$

thanks to Proposition 7.3.1. Our assertion therefore follows. □

10.3.5 The Hausdorff convergence property

Let T be a holomorphic line bundle on X. The goal of this section is to prove the following:

Theorem 10.3.4 *As $k \to \infty$, we have*

$$\Delta_{k,T}(\theta, \varphi) \xrightarrow{d_{\mathrm{Haus}}} \Delta_\nu(\theta, \varphi). \tag{10.29}$$

Recall that $\Delta_{k,T}(\theta, \varphi)$ is defined in Definition 10.3.3.

Although we are only interested in the untwisted case, the proof given below requires the twisted case.

We first observe that the sequence $\Delta_{k,T}(\theta, \varphi)$ is uniformly bounded: This follows easily from Proposition 10.3.4. So Blaschke's selection theorem Theorem C.1.1 is applicable. We will apply this observation without further comments.

Lemma 10.3.1 *Assume that φ has analytic singularities and θ_φ is a Kähler current. Then* (10.29) *holds.*

Proof Up to replacing X by a modification and twisting T accordingly, we may assume that φ has log singularities along an effective $\mathbb{Q}$-divisor D, see Proposition 10.3.9 and Theorem 1.6.1.

Take $\epsilon \in \mathbb{Q} \cap (0, 1)$. In this case, for large enough $k \in \mathbb{Z}_{>0}$ we have

$$\begin{aligned} \mathrm{H}^0\left(X, T \otimes L^k \otimes \mathcal{I}_\infty(k\varphi)\right) \subseteq & \mathrm{H}^0\left(X, T \otimes L^k \otimes \mathcal{I}(k\varphi)\right) \\ \subseteq & \mathrm{H}^0\left(X, T \otimes L^k \otimes \mathcal{I}_\infty(k(1-\epsilon)\varphi)\right). \end{aligned}$$

Take an integer $N \in \mathbb{Z}_{>0}$ so that ND is a divisor and $N\epsilon$ is an integer.

Let Δ' be the limit of a subsequence of $(\Delta_{k,T}(\theta, \varphi))_k$, say the sequence defined by the indices $k_1, k_2, \dots$. Thanks to Theorem C.1.1, it suffices to show that $\Delta' = \Delta_\nu(\theta, \varphi)$.

There exists $t \in \{0, 1, \dots, N-1\}$ such that $k_i \equiv t$ modulo N for infinitely many i, up to replacing $(k_i)_i$ by a subsequence, we may assume that $k_i \equiv t$ modulo N for all i. Write $k_i = Ng_i + t$. Then for large enough i, we have

$$\begin{aligned} \mathrm{H}^0\left(X, T \otimes L^{-N+t} \otimes L^{N(g_i+1)} \otimes \mathcal{I}_\infty\left(N(g_i+1)\varphi\right)\right) &\subseteq \mathrm{H}^0\left(X, T \otimes L^{k_i} \otimes \mathcal{I}(k_i\varphi)\right) \\ &\subseteq \mathrm{H}^0\left(X, T \otimes L^t \otimes L^{Ng_i} \otimes \mathcal{I}_\infty\left(g_iN(1-\epsilon)\varphi\right)\right). \end{aligned}$$

So

$$\begin{aligned} (g_i+1)\Delta_{g_i+1, T\otimes L^{-N+t}}(NL - ND) + N(g_i+1)\nu(D) &\subseteq (Ng_i + t)\Delta_{k_i,T}(\theta, \varphi) \\ &\subseteq g_i\Delta_{g_i, T\otimes L^t}(NL - N(1-\epsilon)D) + Ng_i(1-\epsilon)\nu(D). \end{aligned}$$

Letting $i \to \infty$, by Proposition 10.3.4,

$$\Delta_\nu(L - D) + \nu(D) \subseteq \Delta' \subseteq \Delta_\nu(L - (1-\epsilon)D) + (1-\epsilon)\nu(D).$$

Letting $\epsilon \to 0+$, using the continuity of classical Okounkov bodies in the big cone recalled in Theorem 10.4.1, we find that

$$\Delta' = \Delta_\nu(L - D) + \nu(D) = \Delta_\nu(\theta, \varphi),$$

where we applied Remark 10.3.1 as well. Our assertion follows. □

Lemma 10.3.2 *Assume that θ_φ is a Kähler current. Then as $\mathbb{Q} \ni \beta \to 0+$, we have*

$$\Delta_\nu\Big((1-\beta)\theta, \varphi\Big) \xrightarrow{d_{\mathrm{Haus}}} \Delta_\nu(\theta, \varphi).$$

Here and in the sequel, $\Delta_\nu((1-\beta)\theta, \varphi) = \Delta_\nu((1-\beta)\theta + \mathrm{dd^c}\varphi)$ is defined in (10.28).

Proof By Proposition 10.3.10, we have

$$\Delta_\nu\Big((1-\beta)\theta, \varphi\Big) + \beta\Delta_\nu(L) \subseteq \Delta_\nu(\theta, \varphi).$$

In particular, if Δ' is the Hausdorff limit of a subnet of $(\Delta((1-\beta)\theta, \varphi))_\beta$, then $\Delta' \subseteq \Delta_\nu(\theta, \varphi)$. But

$$\begin{aligned}
\operatorname{vol}\Delta' =& \lim_{\beta\to 0+} \operatorname{vol}\Delta_\nu((1-\beta)\theta, \varphi)\\
=& \frac{1}{n!}\lim_{\beta\to 0+}\int_X \Big((1-\beta)\theta + \mathrm{dd^c}P_{(1-\beta)\theta}[\varphi]_{\mathcal{I}}\Big)^n\\
=& \frac{1}{n!}\int_X (\theta + \mathrm{dd^c}P_\theta[\varphi]_{\mathcal{I}})^n.
\end{aligned}$$

Since we have not developed the theory of nef b-divisors yet, we give a direct proof as well. Take a Kähler form ω so that $\theta_\varphi \geq \omega$. Let $\psi = P_{\theta-\omega}[\varphi]_{\mathcal{I}}$. Then $\varphi \sim_{\mathcal{I}} \psi$. In order to establish the last equality, we may replace φ by ψ and hence assume that φ is $\mathcal{I}$-good. For small enough $\beta > 0$, the form $(1-\beta)\theta - (\theta - \omega) = \omega - \beta\theta$ is Kähler, so this remains true in the class $(1-\beta)\{\theta\}$ by Lemma 7.1.1. Thus the equality of masses in an $\mathcal{I}$-equivalence class reduces the desired equality to

$$\lim_{\beta\to 0+}\int_X ((1-\beta)\theta + \mathrm{dd^c}\varphi)^n = \int_X \theta_\varphi^n,$$

which is obvious.

It follows that $\Delta' = \Delta_\nu(\theta, \varphi)$. We conclude by Theorem C.1.1. □

Proof (Proof of Theorem 10.3.4) Fix a very ample line bundle H on X and a Kähler form $\omega \in c_1(H)$ such that $\omega \geq \theta$.

Step 1. We first handle the case where θ_φ is a Kähler current, say $\theta_\varphi \geq 2\delta\omega$ for some $\delta \in (0, 1)$. Take a quasi-equisingular approximation $(\varphi_j)_j$ of φ in $\mathrm{PSH}(X, \theta)$. We may assume that $\theta_{\varphi_j} \geq \delta\omega$ for all $j \geq 1$. After adding constants, we also assume that $\varphi \leq 0$ and $\varphi_j \leq 0$ for all j.

Let Δ' be a limit of a subsequence of $(\Delta_{k,T}(\theta,\varphi))_k$. Let us say the indices of the subsequence are $k_1 < k_2 < \cdots$. By Theorem C.1.1, it suffices to show that $\Delta' = \Delta_\nu(\theta,\varphi)$.

Observe that for each $j \geq 1$, we have $\Delta' \subseteq \Delta_\nu(\theta,\varphi_j)$ by Lemma 10.3.1. Letting $j \to \infty$, we find $\Delta' \subseteq \Delta_\nu(\theta,\varphi)$ as a consequence of Theorem 10.3.2. Therefore, it suffices to prove that

$$\operatorname{vol}\Delta' \geq \operatorname{vol}\Delta_\nu(\theta,\varphi).$$

Fix an integer $N > \delta^{-1}$. Observe that for any $j \geq 1$, we have $\varphi_j \in \mathrm{PSH}(X,(1-N^{-1})\theta)$. Similarly, $\varphi \in \mathrm{PSH}(X,(1-N^{-1})\theta)$. By Lemma 10.3.2, it suffices to argue that

$$\operatorname{vol}\Delta' \geq \operatorname{vol}\Delta_\nu\left((1-N^{-1})\theta,\varphi\right). \tag{10.30}$$

Step 1.1. We first reduce to the case where $N|k_i$ for all i.

We are free to replace $(k_i)_i$ by a subsequence, so we may assume that $k_i \equiv a$ modulo N for all $i \geq 1$, where $a \in \{0,1,\dots,N-1\}$. We write $k_i = g_iN + a$. Observe that for each $i \geq 1$,

$$\mathrm{H}^0\left(X, T\otimes L^{k_i}\otimes \mathcal{I}(k_i\varphi)\right) \supseteq \mathrm{H}^0\left(X, T\otimes L^{-N+a}\otimes L^{g_iN+N}\otimes \mathcal{I}\left((g_iN+N)\varphi\right)\right).$$

Up to replacing T by $T\otimes L^{-N+a}$, we may therefore assume that $a = 0$, so that $k_i = g_iN$.

Step 1.2. Write $k_i = g_iN$ for all i. We prove (10.30).

Since $\varphi_j \in \mathrm{PSH}(X,(1-N^{-1})\theta)$, we have $\frac{N}{N-1}\varphi_j \in \mathrm{PSH}(X,\theta)$. Applying Lemma 2.4.4 to $P_\theta[\varphi]_{\mathcal{I}}$ and $\frac{N}{N-1}\varphi_j$, we can find $j' \in \mathbb{Z}_{>0}$ such that for all $j \geq j'$, there is $\psi_j \in \mathrm{PSH}(X,\theta)_{>0}$ satisfying

$$P_\theta[\varphi]_{\mathcal{I}} \geq \varphi_j + N^{-1}\psi_j.$$

Fix $j \geq j'$. It suffices to show that

$$\Delta_\nu\left((1-N^{-1})\theta,\varphi_j\right) + \nu' \subseteq \Delta' \tag{10.31}$$

for some $\nu' \in \mathbb{R}^n$. In fact, if this is true, we have

$$\operatorname{vol}\Delta' \geq \operatorname{vol}\Delta_\nu\left((1-N^{-1})\theta,\varphi_j\right).$$

Letting $j \to \infty$ and applying Corollary 6.2.8 and Theorem 10.3.2, we conclude (10.30).

It remains to prove (10.31). As in the proof of Theorem 7.4.1, there is $g_0 > 0$ such that for any $g \geq g_0$, we can find a non-zero section $s_g \in \mathrm{H}^0(X, L^g\otimes\mathcal{I}(g\psi_j))$ such that we get an injective linear map

$$\mathrm{H}^0\left(X, T\otimes L^{g(N-1)}\otimes\mathcal{I}(gN\varphi_j)\right) \xrightarrow{\times s_g} \mathrm{H}^0\left(X, T\otimes L^{gN}\otimes\mathcal{I}(gN\varphi)\right).$$

In particular, when $g = g_i$ for some i large enough, we then find

$$\Delta_{g_i,T}\left((N-1)\theta, N\varphi_j\right) + (g_i)^{-1}\nu(s_{g_i}) \subseteq N\Delta_{k_i,T}(\theta,\varphi).$$

We observe that the $(g_i)^{-1}\nu(s_{g_i})$'s are bounded as both convex bodies appearing in this equation are bounded when i varies. After passing to a subsequence, we may assume that $(g_i)^{-1}\nu(s_{g_i})$ converges. Then Lemma 10.3.1, followed by division by N, gives a vector $\nu' \in \mathbb{R}^n$ such that (10.31) holds.

Step 2. Next we handle the general case.

Let Δ' be the Hausdorff limit of a subsequence of $(\Delta_{k,T}(\theta,\varphi))_k$, say the subsequence with indices $k_1 < k_2 < \cdots$. By Theorem C.1.1, it suffices to prove that $\Delta' = \Delta_\nu(\theta,\varphi)$.

Take $\psi \in \mathrm{PSH}(X,\theta)$ such that θ_ψ is a Kähler current and $\psi \leq \varphi$. The existence of ψ follows from Lemma 2.4.5.

Then for any $\epsilon \in \mathbb{Q} \cap (0,1)$,

$$\Delta_{k,T}(\theta,\varphi) \supseteq \Delta_{k,T}\left(\theta, (1-\epsilon)\varphi + \epsilon\psi\right)$$

for all $k \geq 1$. It follows from Step 1 that

$$\Delta' \supseteq \Delta_\nu\left(\theta, (1-\epsilon)\varphi + \epsilon\psi\right).$$

Letting $\epsilon \to 0$ and applying Theorem 10.3.2, we have $\Delta' \supseteq \Delta_\nu(\theta,\varphi)$. It remains to establish that

$$\operatorname{vol}\Delta' \leq \operatorname{vol}\Delta_\nu(\theta,\varphi). \tag{10.32}$$

Fix an integer $N > 0$, it suffices to argue that

$$\operatorname{vol}\Delta' \leq \frac{1}{n!}\int_X \left(N^{-1}\omega + \theta + \mathrm{dd^c} P_{N^{-1}\omega+\theta}[\varphi]_{\mathcal{I}}\right)^n. \tag{10.33}$$

Assuming this, letting $N \to \infty$, we conclude (10.32), thanks to Proposition 7.3.1.

After adding a constant to φ, which does not change the multiplier ideals or the current $\theta + \mathrm{dd^c}\varphi$, we may assume that $\varphi \leq 0$.

Step 2.1. We first reduce to the case $N|k_i$ for all i.

For this purpose, we are free to replace $k_1 < k_2 < \cdots$ by a subsequence. We may then assume that $k_i \equiv a$ modulo N for all $i \geq 1$ for some $a \in \{0,1,\ldots,N-1\}$. We write $k_i = g_i N + a$. Observe that

$$\mathrm{H}^0\left(X, T\otimes L^{k_i}\otimes \mathcal{I}(k_i\varphi)\right) \subseteq \mathrm{H}^0\left(X, T\otimes L^a \otimes L^{g_iN}\otimes\mathcal{I}(g_iN\varphi)\right).$$

Up to replacing T by $T\otimes L^a$, we may assume that $a = 0$.

Step 2.2. We write $k_i = g_iN$ for all i.

Take a non-zero section $s \in \mathrm{H}^0(X,H)$.

We have an injective linear map

$$\mathrm{H}^0\left(X, T\otimes L^{gN}\otimes\mathcal{I}(gN\varphi)\right) \xrightarrow{\times s^g} \mathrm{H}^0\left(X, T\otimes H^g\otimes L^{gN}\otimes\mathcal{I}(gN\varphi)\right)$$

for each $g \geq 1$. In particular, for each $i \geq 1$,

$$k_i \Delta_{k_i,T}(\theta, \varphi) + g_i \nu(s) \subseteq g_i \Delta_{g_i,T}(\omega + N\theta, N\varphi).$$

Letting $i \to \infty$, by Step 1, we have

$$N\Delta' + \nu(s) \subseteq \Delta_\nu(\omega + N\theta, N\varphi).$$

So

$$\operatorname{vol}\Delta' \leq \operatorname{vol}\Delta_\nu\left(N^{-1}\omega + \theta, \varphi\right) = \frac{1}{n!}\int_X \left(N^{-1}\omega + \theta + \mathrm{dd^c} P_{N^{-1}\omega+\theta}[\varphi]_{\mathcal{I}}\right)^n.$$

This completes the proof. □

10.3.6 Recover Lelong numbers from partial Okounkov bodies

Theorem 10.3.5 *Let E be a prime divisor on X. Let $Y_\bullet$ be an admissible flag with $E = Y_1$. Then*

$$\nu(\varphi, E) = \min_{x \in \Delta_{Y_\bullet}(\theta,\varphi)} x_1. \tag{10.34}$$

Here x_1 denotes the first component of x.

The picture in Fig. 10.1 illustrates the formula: the left supporting hyperplane of $\Delta_{Y_\bullet}(\theta, \varphi)$ is given by $x_1 = \nu(\varphi, Y_1)$.

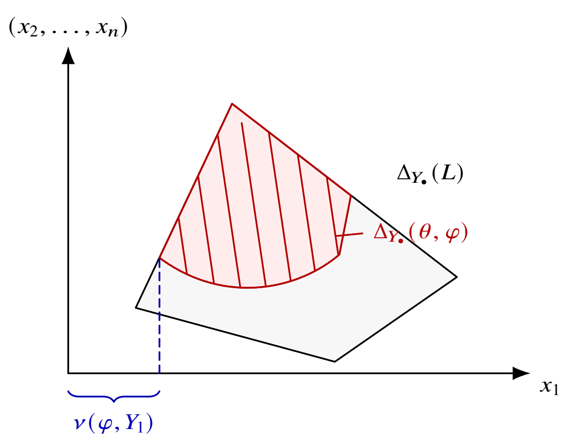

Fig. 10.1 Partial Okounkov body and Lelong number

Proof Replacing φ by $P_\theta[\varphi]_{\mathcal{I}}$, we may assume that φ is $\mathcal{I}$-good.

Step 1. We first reduce to the case where φ has analytic singularities.

By Theorem 7.1.1, we can find a sequence $(\varphi_j)_j$ in $\mathrm{PSH}(X,\theta)_{>0}$ with analytic singularities such that $\varphi_j \xrightarrow{d_S} \varphi$. It follows from Theorem 10.3.2 that

$$\Delta_{Y_\bullet}(\theta,\varphi_j) \xrightarrow{d_{\mathrm{Haus}}} \Delta_{Y_\bullet}(\theta,\varphi).$$

Therefore,

$$\lim_{j\to\infty} \min_{x\in\Delta_{Y_\bullet}(\theta,\varphi_j)} x_1 = \min_{x\in\Delta_{Y_\bullet}(\theta,\varphi)} x_1.$$

In view of Theorem 6.2.4, it suffices to prove (10.34) with φ_j in place of φ.

Step 2. Assume that φ has analytic singularities. In view of Proposition 10.3.9 and Theorem 1.6.1, after replacing X by a modification, we may assume that φ has log singularities along an effective $\mathbb{Q}$-divisor F.

Perturbing L by an ample $\mathbb{Q}$-line bundle by Proposition 10.3.12, we may assume that θ_φ is a Kähler current. Therefore, $L-F$ is ample by Lemma 1.6.1. Finally, by rescaling, we may assume that F is a divisor and L is a line bundle.

By Theorem 10.3.4, we know that

$$\min_{x\in\Delta_{Y_\bullet}(\theta,\varphi)} x_1 = \lim_{k\to\infty} \min_{x\in\Delta_k(\theta,\varphi)} x_1.$$

By definition,

$$\min_{x\in\Delta_k(\theta,\varphi)} x_1 = k^{-1}\operatorname{ord}_E \mathrm{H}^0\left(X, L^k\otimes \mathcal{I}(k\varphi)\right).$$

In view of Proposition 1.4.4, it remains to show that

$$\lim_{k\to\infty} k^{-1}\operatorname{ord}_E \mathrm{H}^0\left(X, L^k\otimes \mathcal{I}(k\varphi)\right) = \lim_{k\to\infty} k^{-1}\operatorname{ord}_E \mathcal{I}(k\varphi). \tag{10.35}$$

The $\geq$ direction is trivial, we prove the converse. Observe that

$$\mathrm{H}^0\left(X, L^k\otimes \mathcal{I}(k\varphi)\right) = \mathrm{H}^0\left(X, L^k\otimes \mathcal{O}_X(-kF)\right), \quad \mathcal{I}(k\varphi)=\mathcal{O}(-kF).$$

As $L-F$ is ample, for large enough k, we have

$$\operatorname{ord}_E \mathrm{H}^0\left(X, L^k\otimes \mathcal{O}_X(-kF)\right) = \operatorname{ord}_E(kF).$$

Thus, (10.35) follows. □

Corollary 10.3.3 *Let $\varphi,\psi\in \mathrm{PSH}(X,\theta)_{>0}$. If*

$$\Delta_{W_\bullet}(\pi^*\theta,\pi^*\varphi)\subseteq \Delta_{W_\bullet}(\pi^*\theta,\pi^*\psi)$$

for all modifications $\pi\colon Y\to X$ and all admissible flags $W_\bullet$ on Y, then $\varphi\preceq_{\mathcal{I}}\psi$.

Proof By Theorem 10.3.5, the assumed inclusions imply that $\nu(\pi^*\varphi, E) \geq \nu(\pi^*\psi, E)$ for every prime divisor E on every modification π. Hence $\varphi \preceq_{\mathcal{I}} \psi$ by Proposition 6.1.2. □

Corollary 10.3.4 *Let E be a prime divisor over X. Then*

$$\nu(V_\theta, E) = \lim_{k\to\infty} \frac{1}{k} \operatorname{ord}_E \mathrm{H}^0(X, L^k). \tag{10.36}$$

Proof After replacing X by a modification, we may assume that E is a prime divisor on X. Note that since X is projective, we can always find an admissible flag with E being the first component.

Then (10.36) follows from Theorem 10.3.5 and the fact that $\Delta_{Y_\bullet}(\theta, V_\theta) = \Delta_{Y_\bullet}(L)$ for any admissible flag $Y_\bullet$ on X. □

10.4 Transcendental partial Okounkov bodies

Let X be a connected compact Kähler manifold of dimension $n > 0$. Fix a smooth flag $Y_\bullet$ on X. We will extend the theory of partial Okounkov bodies in the previous section to the transcendental setting.

10.4.1 The traditional approach to the Okounkov body problem

Definition 10.4.1 Let α be a big cohomology class on X. We define the *Okounkov body* of α with respect to the flag $Y_\bullet$ as

$$\Delta_{Y_\bullet}(\alpha) := \overline{\left\{\nu_{Y_\bullet}(S) : S \in \mathcal{Z}_+(X, \alpha), S \text{ has gentle analytic singularities}\right\}}. \tag{10.37}$$

See Definition 1.6.5 for the definition of gentle analytic singularities.

The results of [DRWN+26] can be summarized as follows:

Theorem 10.4.1 *For any big cohomology class α on X, the set $\Delta_{Y_\bullet}(\alpha) \subseteq \mathbb{R}^n$ is a convex body satisfying the following properties:*

(1) *We have*

$$\operatorname{vol} \Delta_{Y_\bullet}(\alpha) = \frac{1}{n!} \operatorname{vol} \alpha;$$

(2) *given another big cohomology class α' on X, we have*

$$\Delta_{Y_\bullet}(\alpha) + \Delta_{Y_\bullet}(\alpha') \subseteq \Delta_{Y_\bullet}(\alpha + \alpha');$$

(3) *let $\pi\colon Y \to X$ be a proper bimeromorphic morphism where Y is a Kähler manifold. Assume that $(W_\bullet, g)$ is the lifting of $Y_\bullet$ to Y. Then*

$$\Delta_{W_\bullet}(\pi^*\alpha) = \Delta_{Y_\bullet}(\alpha)g;$$

(4) *the map $\alpha \mapsto \Delta_{Y_\bullet}(\alpha)$ is continuous in the big cone with respect to the Hausdorff metric;*

(5) *for any small enough $t > 0$, we have*

$$\left\{y \in \mathbb{R}^{n-1} : (t, y) \in \Delta_{Y_\bullet}(\alpha)\right\} = \Delta_{Y_1 \supseteq \cdots \supseteq Y_n}\Big((\alpha - t[Y_1])|_{Y_1}\Big).$$

See Definition 10.2.4 for the notion of lifting. The proof requires some techniques not covered in the current book; this theorem may be used as a black box.

10.4.2 Definitions of partial Okounkov bodies

Let θ be a closed real smooth $(1,1)$-form on X representing a big cohomology class α.

Let $T = \theta_\varphi \in \mathcal{Z}_+(X, \alpha)$. We shall define a convex body $\Delta_{Y_\bullet}(T) \subseteq \mathbb{R}^n$, which is also written as $\Delta_{Y_\bullet}(\theta, \varphi)$. This convex body is called the *partial Okounkov body* of T with respect to the flag $Y_\bullet$.

10.4.2.1 The case of analytic singularities

Definition 10.4.2 When T is a Kähler current with analytic singularities, we take a modification $\pi\colon Y \to X$ so that

(1)
$$\pi^*T = [D] + R, \tag{10.38}$$
where D is an effective $\mathbb{Q}$-divisor on Y and R is a closed positive $(1,1)$-current with bounded potential, and

(2) the lifting $(Z_\bullet, g)$ of $Y_\bullet$ to Y exists.

Define
$$\Delta_{Y_\bullet}(T) \coloneqq \Delta_{Z_\bullet}([R])g^{-1} + \nu_{Z_\bullet}([D])g^{-1}.$$

The existence of π is guaranteed by Theorem 1.6.1 and Theorem 10.2.1.

Lemma 10.4.1 *The convex body $\Delta_{Y_\bullet}(T)$ defined in Definition 10.4.2 is independent of the choice of π.*

Proof Take another map $\pi'\colon Y' \to X$ with the same properties. We want to show that π and π' define the same $\Delta_{Y_\bullet}(T)$. We may assume that π' dominates π through $p\colon Y' \to Y$, so that we have a commutative diagram

$$\begin{tikzcd} Y' \arrow[rr, "p"] \arrow[dr, "\pi'"'] & & Y \arrow[dl, "\pi"] \\ & X. & \end{tikzcd}$$

We take D and R as in (10.38). Then

$$\pi'^*T = [p^*D] + p^*R.$$

Write $(Z_\bullet, g)$ and $(Z'_\bullet, g')$ for the liftings of $Y_\bullet$ to Y and Y' respectively. We need to prove that

$$\Delta_{Z_\bullet}([R])g^{-1} + \nu_{Z_\bullet}([D])g^{-1} = \Delta_{Z'_\bullet}([p^*R])g'^{-1} + \nu_{Z'_\bullet}([p^*D])g'^{-1}.$$

This follows from Theorem 10.4.1, Proposition 10.2.3 and Proposition 10.2.4. □

The above proof also describes the bimeromorphic behavior of $\Delta_{Y_\bullet}(T)$:

Lemma 10.4.2 *Let $T \in \mathcal{Z}_+(X, \alpha)$ be a Kähler current with analytic singularities. Let $\pi\colon Y \to X$ be a proper bimeromorphic morphism and $(W_\bullet, g)$ be the lifting of $Y_\bullet$ to Y. Then*

$$\Delta_{W_\bullet}(\pi^*T) = \Delta_{Y_\bullet}(T)g.$$

Lemma 10.4.3 *Assume that $T, S \in \mathcal{Z}_+(X, \alpha)$ are two Kähler currents with analytic singularities and $T \preceq S$. Then*

$$\Delta_{Y_\bullet}(T) \subseteq \Delta_{Y_\bullet}(S) \subseteq \Delta_{Y_\bullet}(\alpha).$$

Moreover,

$$\operatorname{vol} \Delta_{Y_\bullet}(T) = \frac{1}{n!}\int_X T^n. \tag{10.39}$$

Proof We first show that

$$\Delta_{Y_\bullet}(T) \subseteq \Delta_{Y_\bullet}(S).$$

Using Lemma 10.4.2, we may assume that T and S have log singularities along effective $\mathbb{Q}$-divisors E and F respectively. By assumption, $E \geq F$. Replacing T and S by $T - [F]$ and $S - [F]$ respectively, we may assume that $F = 0$.

In this case, we need to show that

$$\Delta_{Y_\bullet}(\alpha) \supseteq \Delta_{Y_\bullet}(\alpha - [E]) + \nu_{Y_\bullet}([E]),$$

which is obvious.

Next we prove that

$$\Delta_{Y_\bullet}(T) \subseteq \Delta_{Y_\bullet}(\alpha).$$

By Lemma 10.4.2 and Theorem 10.4.1 again, we may assume that T has log singularities. We take D and R as in (10.38). We need to show that

$$\Delta_{Y_\bullet}(\alpha - [D]) + \nu_{Y_\bullet}([D]) \subseteq \Delta_{Y_\bullet}(\alpha),$$

which is again obvious.

Finally, (10.39) follows immediately from Theorem 10.4.1. □

10.4.2.2 The case of Kähler currents

Definition 10.4.3 Let $T \in \mathcal{Z}_+(X,\alpha)$ be a Kähler current. Take a quasi-equisingular approximation $(T_j)_j$ of T in $\mathcal{Z}_+(X,\alpha)$. Then we define

$$\Delta_{Y_\bullet}(T) := \bigcap_{j=1}^{\infty} \Delta_{Y_\bullet}(T_j).$$

Lemma 10.4.4 *The convex body $\Delta_{Y_\bullet}(T)$ in Definition 10.4.3 is independent of the choices of the T_j's.*

In particular, if T also has analytic singularities, then the $\Delta_{Y_\bullet}(T)$'s defined in Definition 10.4.3 and in Definition 10.4.2 coincide.

Proof Let $(S_j)_j$ be another quasi-equisingular approximation of T in $\mathcal{Z}_+(X,\alpha)$. By Proposition 1.6.3, for any small rational $\epsilon > 0$, $j > 0$, we can find $k > 0$ so that

$$S_k \preceq (1-\epsilon)T_j.$$

It is more convenient to use the language of θ-psh functions at this point. Let ψ_k (resp. φ_k) denote the potentials in $\mathrm{PSH}(X,\theta)$ corresponding to S_k (resp. T_k) for each $k \geq 1$. Note that ψ_k and φ_k are unique up to additive constants.

By Lemma 10.4.3,

$$\bigcap_{k=1}^{\infty} \Delta_{Y_\bullet}(\theta,\psi_k) \subseteq \Delta_{Y_\bullet}(\theta,(1-\epsilon)\varphi_j).$$

On the other hand, observe that

$$\bigcap_{\epsilon\in\mathbb{Q}_{>0}\text{ small enough}} \Delta_{Y_\bullet}(\theta,(1-\epsilon)\varphi_j) = \Delta_{Y_\bullet}(\theta,\varphi_j).$$

In fact, the $\supseteq$ direction follows from Lemma 10.4.3, so it suffices to show that the two sides have the same volume, which follows from (10.39).

It follows that

$$\bigcap_{k=1}^{\infty} \Delta_{Y_\bullet}(\theta,\psi_k) \subseteq \bigcap_{j=1}^{\infty} \Delta_{Y_\bullet}(\theta,\varphi_j).$$

The other inclusion follows by symmetry. □

The same argument shows that

Corollary 10.4.1 *Suppose that $T, S \in \mathcal{Z}_+(X,\alpha)$ are two Kähler currents satisfying $T \preceq_{\mathcal{I}} S$. Then*

$$\Delta_{Y_\bullet}(T) \subseteq \Delta_{Y_\bullet}(S) \subseteq \Delta_{Y_\bullet}(\alpha).$$

Proposition 10.4.1 *Let $T \in \mathcal{Z}_+(X, \alpha)$ be a Kähler current. Then*

$$\operatorname{vol} \Delta_{Y_\bullet}(T) = \frac{1}{n!} \operatorname{vol} T. \tag{10.40}$$

Proof Take a quasi-equisingular approximation $(T_j)_j$ of T in $\mathcal{Z}_+(X, \alpha)$. Note that $\Delta_{Y_\bullet}(T_j)$ is decreasing in j, as follows from Lemma 10.4.3. Our assertion follows from (10.39) and Theorem 6.2.5. □

Lemma 10.4.5 *Let $T \in \mathcal{Z}_+(X, \alpha)$ be a Kähler current and ω be a Kähler form on X. Then*

$$\Delta_{Y_\bullet}(T) \subseteq \Delta_{Y_\bullet}(T + \omega). \tag{10.41}$$

Moreover,

$$\Delta_{Y_\bullet}(T) = \bigcap_{\epsilon > 0} \Delta_{Y_\bullet}(T + \epsilon\omega). \tag{10.42}$$

Proof We first prove (10.41). Taking quasi-equisingular approximations, we reduce immediately to the case where T has analytic singularities. By Lemma 10.4.2, we may assume that T has log singularities. Take D and R as in (10.38). By definition again, it suffices to show that

$$\Delta_{Y_\bullet}([R]) \subseteq \Delta_{Y_\bullet}([R] + [\omega]),$$

which is clear by definition.

Next we prove (10.42). Thanks to (10.41), it remains to prove that both sides have the same volume:

$$\lim_{\epsilon \to 0+} \operatorname{vol}(T + \epsilon\omega) = \operatorname{vol} T.$$

This is proved in Proposition 7.3.1. □

10.4.2.3 The general case

Definition 10.4.4 Let $T \in \mathcal{Z}_+(X, \alpha)$. Take a Kähler form ω on X, we define

$$\Delta_{Y_\bullet}(T) = \bigcap_{j=1}^{\infty} \Delta_{Y_\bullet}(T + j^{-1}\omega). \tag{10.43}$$

The same definition makes sense when α is only pseudo-effective.

This definition is clearly independent of the choice of ω by Lemma 10.4.5. Moreover, it extends Definition 10.4.3 and Definition 10.4.2 as a result of Lemma 10.4.5.

The main properties of $\Delta_{Y_\bullet}(T)$ are summarized as follows:

Theorem 10.4.2 *The convex bodies $\Delta_{Y_\bullet}(T)$ satisfy the following properties:*

(1) *Suppose that* $T \in \mathcal{Z}_+(X,\alpha)_{>0}$. *We have*

$$\operatorname{vol}\Delta_{Y_\bullet}(T) = \frac{1}{n!}\operatorname{vol}T. \tag{10.44}$$

(2) *For* $T, S \in \mathcal{Z}_+(X,\alpha)$ *satisfying* $T \preceq_{\mathcal{I}} S$, *we have*

$$\Delta_{Y_\bullet}(T) \subseteq \Delta_{Y_\bullet}(S) \subseteq \Delta_{Y_\bullet}(\alpha).$$

(3) *For any current* $T \in \mathcal{Z}_+(X,\alpha)$ *with minimal singularities, we have*

$$\Delta_{Y_\bullet}(T) = \Delta_{Y_\bullet}(\alpha).$$

(4) *The map* $\mathcal{Z}_+(X,\alpha)_{>0} \to \mathcal{K}_n$ *given by* $T \mapsto \Delta_{Y_\bullet}(T)$ *is continuous, where we endow the* d_S*-pseudometric on* $\mathcal{Z}_+(X,\alpha)_{>0}$ *and the Hausdorff topology on* $\mathcal{K}_n$.

(5) *Let* $\pi\colon Y \to X$ *be a proper bimeromorphic morphism, where* Y *is a Kähler manifold. Assume that the lifting* $(W_\bullet, g)$ *of* $Y_\bullet$ *to* Y *exists. Then for any* $T \in \mathcal{Z}_+(X,\alpha)_{>0}$, *we have*

$$\Delta_{W_\bullet}(\pi^*T) = \Delta_{Y_\bullet}(T)g.$$

(6) *For* $T, S \in \mathcal{Z}_+(X,\alpha)$, *we have*

$$\Delta_{Y_\bullet}(T) + \Delta_{Y_\bullet}(S) \subseteq \Delta_{Y_\bullet}(T+S). \tag{10.45}$$

Proof (1) By (10.43) and (10.40), for any Kähler form ω on X,

$$\operatorname{vol}\Delta_{Y_\bullet}(T) = \lim_{j\to\infty}\operatorname{vol}\Delta_{Y_\bullet}(T + j^{-1}\omega) = \frac{1}{n!}\lim_{j\to\infty}\operatorname{vol}(T + j^{-1}\omega).$$

The right-hand side is computed in Proposition 7.3.1. Hence, (10.44) follows.

(2) Fix a Kähler form ω on X. By Corollary 10.4.1, for each $j \geq 1$,

$$\Delta_{Y_\bullet}(T + j^{-1}\omega) \subseteq \Delta_{Y_\bullet}(S + j^{-1}\omega) \subseteq \Delta_{Y_\bullet}(\alpha + j^{-1}[\omega]).$$

It remains to show that

$$\Delta_{Y_\bullet}(\alpha) = \bigcap_{j=1}^{\infty}\Delta_{Y_\bullet}(\alpha + j^{-1}[\omega]).$$

The $\subseteq$ direction is clear. Comparing the volumes using Theorem 10.4.1, we conclude that equality holds.

(3) This follows from (1) and (2).

(4) Let $(T_j)_j$ be a sequence in $\mathcal{Z}_+(X,\alpha)_{>0}$ converging to $T \in \mathcal{Z}_+(X,\alpha)_{>0}$ with respect to d_S. We want to show that $\Delta_{Y_\bullet}(T_j) \xrightarrow{d_{\mathrm{Haus}}} \Delta_{Y_\bullet}(T)$. By Proposition 6.2.3 and (2), we may assume that the singularity type of T_j is either increasing or decreasing. In either monotone case, (2) gives monotonicity of the convex bodies, while (1) and Corollary 6.2.4 give convergence of their volumes to $\operatorname{vol}\Delta_{Y_\bullet}(T)$. Every Hausdorff

cluster point is therefore a monotone limit with the same volume as $\Delta_{Y_\bullet}(T)$, hence is equal to $\Delta_{Y_\bullet}(T)$.

(5) We may assume that T is $\mathcal{I}$-good. It follows from (4) and Theorem 7.1.1 that we may reduce to the case where T has analytic singularities. Our assertion follows from Lemma 10.4.2.

(6) By (10.43), in order to prove (10.45), we may assume that T and S are both Kähler currents. Take quasi-equisingular approximations $(T_j)_j$ and $(S_j)_j$ of T and S respectively. By Theorem 6.2.2, $T_j + S_j \xrightarrow{d_S} T + S$. By (4), we may therefore assume that T and S have analytic singularities. Replacing X by a suitable modification, we may assume that T and S both have log singularities, say

$$T = [D] + R, \quad S = [D'] + R',$$

where D and D' are $\mathbb{Q}$-divisors on X and R and R' are closed positive $(1,1)$-currents with bounded potentials. We need to show that

$$\Delta_{Y_\bullet}([R]) + \Delta_{Y_\bullet}([R']) + \nu_{Y_\bullet}([D]) + \nu_{Y_\bullet}([D']) \subseteq \Delta_{Y_\bullet}([R+R']) + \nu_{Y_\bullet}([D+D']).$$

By Proposition 10.2.2, this is equivalent to

$$\Delta_{Y_\bullet}([R]) + \Delta_{Y_\bullet}([R']) \subseteq \Delta_{Y_\bullet}([R+R']),$$

which is already proved in Theorem 10.4.1. □

Corollary 10.4.2 *Assume that L is a big line bundle on X and h is a plurisubharmonic metric on L with positive volume. Then*

$$\Delta_{Y_\bullet}(\mathrm{dd^c} h) = \Delta_{Y_\bullet}(L, h). \tag{10.46}$$

Similarly, the definition (10.28) is compatible with the definition in Definition 10.4.4.

Proof We may assume that $\mathrm{dd^c} h$ has positive mass and is $\mathcal{I}$-good. By the d_S-continuity of both sides of (10.46) as proved in Theorem 10.4.2 and Theorem 10.3.2, together with Theorem 7.1.1, we may assume that $\mathrm{dd^c} h$ has analytic singularities.

In this case, using the birational invariance of both sides of (10.46) as proved in Proposition 10.3.9 and Theorem 10.4.2, we may assume that $\mathrm{dd^c} h$ has log singularities. Finally, after all these reductions, the equality (10.46) holds by construction. □

10.4.3 The valuative characterization

In this section, we will characterize the partial Okounkov bodies using valuations of currents.

Lemma 10.4.6 *Let β be a big and nef class on X. Then*

$$\left\{ y \in \mathbb{R}^{n-1} : (0, y) \in \Delta_{Y_\bullet}(\beta) \right\} = \Delta_{Y_1 \supseteq \cdots \supseteq Y_n}(\beta|_{Y_1}). \tag{10.47}$$

Here the right-hand side is understood in the sense of Definition 10.4.4 when $\beta|_{Y_1}$ is not big.

Proof **Step 1**. We first reduce to the case where β is a Kähler class.

Take a Kähler class α on X. It follows from the volume formula in Theorem 10.4.1 that

$$\Delta_{Y_\bullet}(\beta) = \bigcap_{\epsilon>0} \Delta_{Y_\bullet}(\beta+\epsilon\alpha).$$

Since $\beta|_{Y_1}$ is nef, hence pseudo-effective, Definition 10.4.4 similarly gives

$$\Delta_{Y_1\supseteq\cdots\supseteq Y_n}(\beta|_{Y_1}) = \bigcap_{\epsilon>0} \Delta_{Y_1\supseteq\cdots\supseteq Y_n}(\beta|_{Y_1}+\epsilon\alpha|_{Y_1}).$$

So it suffices to prove (10.47) with $\beta+\epsilon\alpha$ in place of β.

Step 2. Assume that β is a Kähler class. The $\supseteq$ direction in (10.47) follows from the extension theorem Theorem 1.6.3. To prove the other direction, recall that by Theorem 10.4.1, for $t>0$ small enough, we have

$$\left\{y\in\mathbb{R}^{n-1} : (t,y)\in\Delta_{Y_\bullet}(\beta)\right\} = \Delta_{Y_1\supseteq\cdots\supseteq Y_n}\left((\beta-t[Y_1])|_{Y_1}\right).$$

As $t\to 0+$, the right-hand side converges to $\Delta_{Y_1\supseteq\cdots\supseteq Y_n}(\beta|_{Y_1})$ with respect to the Hausdorff metric as a consequence of Theorem 10.4.1, while the left-hand side converges to

$$\left\{y\in\mathbb{R}^{n-1} : (0,y)\in\Delta_{Y_\bullet}(\beta)\right\}$$

by Lemma C.1.2. We conclude our assertion. □

Lemma 10.4.7 *Let $T\in\mathcal{Z}_+(X,\alpha)$ be a Kähler current. Assume that $\nu(T,Y_1)=0$. Then*

$$\left\{y\in\mathbb{R}^{n-1} : (0,y)\in\Delta_{Y_\bullet}(T)\right\} = \Delta_{Y_1\supseteq\cdots\supseteq Y_n}\left(\mathrm{Tr}_{Y_1}^{\alpha|_{Y_1}}(T)\right). \tag{10.48}$$

More generally, if $T\in\mathcal{Z}_+(X,\alpha)$ and $\nu(T,Y_1)=0$, suppose in addition that $\mathrm{Tr}_{Y_1}^{\alpha|_{Y_1}}(T)$ is defined, then (10.48) still holds.

See Remark 8.1.1 for the definition of $\mathrm{Tr}_{Y_1}^{\alpha|_{Y_1}}(T)$. Note that $\Delta_{Y_1\supseteq\cdots\supseteq Y_n}\left(\mathrm{Tr}_{Y_1}^{\alpha|_{Y_1}}(T)\right)$ is independent of the choice of the representative $\mathrm{Tr}_{Y_1}^{\alpha|_{Y_1}}(T)$.

Remark 10.4.1 More generally, the same argument shows the following result: Let $k=0,\dots,n$ and $T\in\mathcal{Z}_+(X,\alpha)$ such that $\nu(T,Y_k)=0$. Assume that $\mathrm{Tr}_{Y_k}^{\alpha|_{Y_k}}(T)$ is defined. Then

$$\left\{y\in\mathbb{R}^{n-k} : (0,\dots,0,y)\in\Delta_{Y_\bullet}(T)\right\} = \Delta_{Y_k\supseteq\cdots\supseteq Y_n}\left(\mathrm{Tr}_{Y_k}^{\alpha|_{Y_k}}(T)\right). \tag{10.49}$$

Also note that this result extends [Jow10, Theorem 3.4] and hence gives simpler proofs of [Jow10, Theorem A, Theorem B].

Proof Let ω be a Kähler form on X. The last assertion follows from the first by perturbing θ to $\theta + \epsilon\omega$.

Step 1. We first handle the case where T has analytic singularities. Let $\pi\colon Z \to X$ be a modification such that

(1) $Y_\bullet$ admits a lifting $(W_\bullet, g)$, and
(2) $\pi^*T = [D] + R$, where D is an effective $\mathbb{Q}$-divisor on Z and R is closed positive $(1,1)$-current with bounded potential.

This is possible by Theorem 1.6.1 and Theorem 10.2.1.

By Lemma 8.2.1,

$$\Pi^* \operatorname{Tr}_{Y_1}(T) \sim_P \operatorname{Tr}_{W_1}(\pi^*T),$$

where $\Pi\colon W_1 \to Y_1$ is the restriction of π. It follows from Theorem 10.4.2 that

$$\begin{aligned}\Delta_{W_1\supseteq\cdots\supseteq W_n}(\operatorname{Tr}_{W_1}(\pi^*T)) =&\Delta_{Y_1\supseteq\cdots\supseteq Y_n}(\operatorname{Tr}_{Y_1}(T))\operatorname{cor}(Y_1\supseteq\cdots\supseteq Y_n,\Pi),\\ \Delta_{W_\bullet}(\pi^*T) =&\Delta_{Y_\bullet}(T)g.\end{aligned}$$

Taking (10.10) into account, we find that it suffices to show that

$$\left\{y\in\mathbb{R}^{n-1} : (0,y)\in\Delta_{W_\bullet}(\pi^*T)\right\} = \Delta_{W_1\supseteq\cdots\supseteq W_n}(\operatorname{Tr}_{W_1}(\pi^*T)).$$

We may assume that π is the identity map. Since $\nu(T,Y_1)=0$, the divisor D does not contain Y_1. Hence we have

$$T = [D]+R,\quad T|_{Y_1} = [D]|_{Y_1} + R|_{Y_1}.$$

Note that $[D]|_{Y_1}$ is the current of integration along an effective $\mathbb{Q}$-divisor on Y_1.

In particular,

$$\begin{aligned}\Delta_{Y_\bullet}(T) =&\Delta_{Y_\bullet}([R]) + \nu_{Y_\bullet}([D]),\\ \Delta_{Y_1\supseteq\cdots\supseteq Y_n}(T|_{Y_1}) =&\Delta_{Y_1\supseteq\cdots\supseteq Y_n}([R]|_{Y_1}) + \nu_{Y_1\supseteq\cdots\supseteq Y_n}([D]|_{Y_1}).\end{aligned}$$

The first coordinate of $\nu_{Y_\bullet}([D])$ is 0, and the remaining coordinates are exactly $\nu_{Y_1\supseteq\cdots\supseteq Y_n}([D]|_{Y_1})$. Thus it suffices to show that

$$\left\{y\in\mathbb{R}^{n-1} : (0,y)\in\Delta_{Y_\bullet}([R])\right\} = \Delta_{Y_1\supseteq\cdots\supseteq Y_n}([R]|_{Y_1}).$$

Since R has bounded potential, its cohomology class is nef; it is big because T is a Kähler current. Thus the last identity is exactly Lemma 10.4.6.

Step 2. Next we consider the case where T is a Kähler current. Take a quasi-equisingular approximation $(T_j)_j$ of T in $\mathcal{Z}_+(X,\alpha)$. From Step 1, we know that for large $j\geq 1$,

$$\left\{y\in\mathbb{R}^{n-1} : (0,y)\in\Delta_{Y_\bullet}(T_j)\right\} = \Delta_{Y_1\supseteq\cdots\supseteq Y_n}(\operatorname{Tr}_{Y_1}(T_j)).$$

Letting $j\to\infty$ and applying Theorem 10.4.2 and Proposition 8.2.2, we conclude (10.48). $\square$

Theorem 10.4.3 *Assume that $T \in \mathcal{Z}_+(X,\alpha)_{>0}$ is a Kähler current. We have*

$$\min_{\text{lex}} \Delta_{Y_\bullet}(T) = \nu_{Y_\bullet}(T). \tag{10.50}$$

Here the minimum is with respect to the lexicographic order.

Proof We argue by induction on $n \geq 0$. The case $n = 0$ is of course trivial. Let us assume that $n > 0$ and the case $n-1$ has been proved.

We first observe that by Theorem 10.4.2,

$$\Delta_{Y_\bullet}(T - \nu(T,Y_1)[Y_1]) + (\nu(T,Y_1), 0, \dots, 0) \subseteq \Delta_{Y_\bullet}(T).$$

Comparing the volumes of both sides using Theorem 10.4.2 and Proposition 7.3.1, we find that equality holds:

$$\Delta_{Y_\bullet}(T - \nu(T,Y_1)[Y_1]) + (\nu(T,Y_1), 0, \dots, 0) = \Delta_{Y_\bullet}(T).$$

Replacing T by $T - \nu(T,Y_1)[Y_1]$, we may therefore assume that $\nu(T,Y_1) = 0$. It suffices to apply Lemma 10.4.7 and the inductive hypothesis. □

Corollary 10.4.3 *For any $T \in \mathcal{Z}_+(X,\alpha)$,*

$$\nu_{Y_\bullet}(T) \in \Delta_{Y_\bullet}(T) \subseteq \Delta_{Y_\bullet}(\alpha).$$

Proof When T is a Kähler current, this follows from Theorem 10.4.3.

In general, by definition, $\nu_{Y_\bullet}(T) = \nu_{Y_\bullet}(T + \omega)$ for any Kähler form ω on X. It follows that

$$\nu_{Y_\bullet}(T) \in \Delta_{Y_\bullet}(T + \omega)$$

for any Kähler form ω. It follows that $\nu_{Y_\bullet}(T) \in \Delta_{Y_\bullet}(T)$. □

Theorem 10.4.4 *For any $T \in \mathcal{Z}_+(X,\alpha)_{>0}$,*

$$\Delta_{Y_\bullet}(T) = \overline{\left\{\nu_{Y_\bullet}(S) : S \in \mathcal{Z}_+(X,\alpha), S \leq_{\mathcal{I}} T\right\}}. \tag{10.51}$$

In particular, [2]

$$\Delta_{Y_\bullet}(\alpha) = \overline{\left\{\nu_{Y_\bullet}(S) : S \in \mathcal{Z}_+(X,\alpha)\right\}}. \tag{10.52}$$

Remark 10.4.2 We expect that the closure operation in (10.51) is not necessary. This problem is closely related to the Dirichlet problem of the trace operator, see Page 449 for more details.

[2] According to Ya Deng, the definition (10.52) of $\Delta_{Y_\bullet}(\alpha)$ was what Demailly originally proposed for Deng's thesis. Due to the lack of the techniques of the trace operators, Deng had to work with analytic singularities instead. As a consequence, the transcendental analogue of Proposition 10.3.9 is not obvious. This is one of the two main technical difficulties of Theorem 10.4.1. This problem also led me to finally develop the theory of trace operators, a notion that had been on the author's mind for several years.

Proof The $\supseteq$ direction in (10.51) follows from Corollary 10.4.3 and Theorem 10.4.2(2).

Let us write

$$D_{Y_\bullet}(T) = \left\{\nu_{Y_\bullet}(S) : S \in \mathcal{Z}_+(X,\alpha), S \preceq_{\mathcal{I}} T\right\}$$

for the time being.

Step 1. Assume that T has analytic singularities. We have

$$\begin{aligned}\Delta_{Y_\bullet}(T) \supseteq & \overline{D_{Y_\bullet}(T)} \\ \supseteq & \overline{\left\{\nu_{Y_\bullet}(S) : \mathcal{Z}_+(X,\alpha) \ni S \text{ has gentle analytic singularities}, S \preceq T\right\}}.\end{aligned}$$

It follows easily from Theorem 10.4.1 that the volume of the right-hand side is equal to the volume of $\Delta_{Y_\bullet}(T)$, so (10.51) holds.

Step 2. Assume that T is a Kähler current. Take a quasi-equisingular approximation $T_j \in \mathcal{Z}_+(X,\alpha)$ of T. Next we use the language of psh functions. Let $\varphi_j, \varphi \in \mathrm{PSH}(X,\theta)$ be the potentials corresponding to T_j, T for each $j \geq 1$.

Fix an integer $N > 0$. For large enough $j \geq 1$, we can find $\psi_j \in \mathrm{PSH}(X,\theta)_{>0}$ such that

$$P_\theta[\varphi]_{\mathcal{I}} \geq (1-N^{-1})\varphi_j + N^{-1}\psi_j.$$

The existence of ψ_j follows from Lemma 2.4.4. It follows that

$$\begin{aligned}D_{Y_\bullet}(T) \supseteq & D_{Y_\bullet}\left(\theta + \mathrm{dd^c}\left((1-N^{-1})\varphi_j + N^{-1}\psi_j\right)\right) \\ \supseteq & (1-N^{-1})D_{Y_\bullet}(T_j) + N^{-1}D_{Y_\bullet}(\theta + \mathrm{dd^c}\psi_j).\end{aligned}$$

We claim that

$$\overline{D_{Y_\bullet}(T)} \supseteq \bigcap_{j=1}^{\infty} \overline{D_{Y_\bullet}(T_j)}.$$

Let x be a point in the right-hand side. Fix $N > 0$. For all large j, choose

$$y_j \in D_{Y_\bullet}(T_j), \quad z_j \in D_{Y_\bullet}(\theta + \mathrm{dd^c}\psi_j)$$

with $y_j \to x$. By the preceding inclusion,

$$(1-N^{-1})y_j + N^{-1}z_j \in D_{Y_\bullet}(T).$$

The points x, y_j, z_j all lie in the compact set $\Delta_{Y_\bullet}(\alpha)$ after taking closures, by Corollary 10.4.3. Hence, letting $j \to \infty$ and then $N \to \infty$, we get $x \in \overline{D_{Y_\bullet}(T)}$, proving the claim. Therefore, by Step 1, we conclude that

$$\Delta_{Y_\bullet}(T) = \bigcap_{j=1}^{\infty} \overline{\Delta_{Y_\bullet}(T_j)} = \bigcap_{j=1}^{\infty} \overline{D_{Y_\bullet}(T_j)} \subseteq \overline{D_{Y_\bullet}(T)}.$$

The reverse direction is already known.

Step 3. Finally, consider the general case. Take a Kähler current $T' \in \mathcal{Z}_+(X, \alpha)$ more singular than T (the existence of which is given by Lemma 2.4.5). For each $\epsilon \in (0, 1)$, we know that

$$\Delta_{Y_\bullet}((1-\epsilon)T + \epsilon T') = \overline{D_{Y_\bullet}((1-\epsilon)T + \epsilon T')} \subseteq \overline{D_{Y_\bullet}(T)}.$$

Letting $\epsilon \to 0+$ and using the d_S-continuity in Theorem 10.4.2(4), we find that

$$\Delta_{Y_\bullet}(T) \subseteq \overline{D_{Y_\bullet}(T)}.$$

As the other inclusion is already known, we conclude. □

Corollary 10.4.4 *Assume that* $T \in \mathcal{Z}_+(X, \alpha)_{>0}$. *We have*

$$\min_{\text{lex}} \Delta_{Y_\bullet}(T) = \nu_{Y_\bullet}(T). \tag{10.53}$$

Proof By Theorem 10.4.4, it is clear that

$$\min_{\text{lex}} \Delta_{Y_\bullet}(T) \leq_{\text{lex}} \nu_{Y_\bullet}(T).$$

On the other hand, we clearly have

$$\Delta_{Y_\bullet}(T) \subseteq \Delta_{Y_\bullet}(T + \omega)$$

for any Kähler form ω on X. It follows that

$$\min_{\text{lex}} \Delta_{Y_\bullet}(T) \geq_{\text{lex}} \min_{\text{lex}} \Delta_{Y_\bullet}(T + \omega).$$

By Theorem 10.4.3, the right-hand side is just $\nu_{Y_\bullet}(T + \omega) = \nu_{Y_\bullet}(T)$. We conclude the proof. □

10.5 Okounkov test curves

Fix $n \in \mathbb{N}$. Let $\Delta, \Delta' \subseteq \mathbb{R}^n$ be convex bodies with positive volumes. The standard Lebesgue measure on $\mathbb{R}^n$ is denoted by vol.

Recall that $\mathcal{K}_n$ denotes the set of convex bodies in $\mathbb{R}^n$ and d_{Haus} denotes the Hausdorff metric. We refer to Appendix C for the basic properties of these objects.

We study the notion of Okounkov test curves in this section; this leads to the definition of the Duistermaat–Heckman measure of a non-Archimedean metric in Section 13.3 below. This section can be skipped on a first reading.

Definition 10.5.1 An *Okounkov test curve* relative to Δ consists of

(1) a number $\Delta_{\max} \in \mathbb{R}$ and

(2) an assignment $(-\infty, \Delta_{\max}) \ni \tau \mapsto \Delta_\tau \in \mathcal{K}_n$ satisfying

a. the assignment $\tau \mapsto \Delta_\tau$ is decreasing and concave[3];

b. we have $\Delta_\tau \xrightarrow{d_{\text{Haus}}} \Delta$ as $\tau \to -\infty$.

The set of Okounkov test curves relative to Δ is denoted by $\mathrm{TC}(\Delta)$.

An Okounkov test curve $\Delta_\bullet$ relative to Δ is *bounded* if $\Delta_\tau = \Delta$ when τ is small enough. The subset of bounded Okounkov test curves is denoted by $\mathrm{TC}^\infty(\Delta)$.

An Okounkov test curve $\Delta_\bullet$ relative to Δ is said to have *finite energy* if

$$\mathbf{E}(\Delta_\bullet) := n!\Delta_{\max} \operatorname{vol} \Delta + n! \int_{-\infty}^{\Delta_{\max}} (\operatorname{vol} \Delta_\tau - \operatorname{vol} \Delta)\, \mathrm{d}\tau > -\infty. \tag{10.54}$$

The subset of Okounkov test curves with finite energy is denoted by $\mathrm{TC}^1(\Delta)$.

Given $\Delta_\bullet \in \mathrm{TC}(\Delta)$ and $\Delta'_\bullet \in \mathrm{TC}(\Delta')$, we say $\Delta_\bullet \leq \Delta'_\bullet$ if $\Delta_{\max} \leq \Delta'_{\max}$ and for any $\tau < \Delta_{\max}$, we have $\Delta_\tau \subseteq \Delta'_\tau$.

Sometimes it is convenient to introduce

$$\Delta_{\Delta_{\max}} = \bigcap_{\tau < \Delta_{\max}} \Delta_\tau \in \mathcal{K}_n. \tag{10.55}$$

We shall always make this extension in the sequel when we talk about $\Delta_{\Delta_{\max}}$. Observe that $(-\infty, \Delta_{\max}] \ni \tau \mapsto \Delta_\tau$ is still concave.

Proposition 10.5.1 *Any Okounkov test curve $(\Delta_\tau)_{\tau<\Delta_{\max}}$ relative to Δ is continuous in τ. Moreover,* $\operatorname{vol} \Delta_\tau > 0$ *for all $\tau < \Delta_{\max}$.*

Proof We first claim that $\operatorname{vol} \Delta_{\tau'} > 0$ for all $\tau' < \Delta_{\max}$. By Condition (2b) in Definition 10.5.1 and Theorem C.1.2, we know that $\operatorname{vol} \Delta_{\tau''} > 0$ when τ'' is small enough. Fix one such τ''. We may assume that $\tau'' \leq \tau'$ since otherwise there is nothing to prove. Next take $\tau''' \in (\tau', \Delta_{\max})$. Take $t \in (0, 1)$ such that $\tau' = t\tau''' + (1 - t)\tau''$. It follows that

$$\operatorname{vol} \Delta_{\tau'} \geq \operatorname{vol} (t\Delta_{\tau'''} + (1 - t)\Delta_{\tau''}) \geq (1 - t)^n \operatorname{vol} \Delta_{\tau''} > 0.$$

Next we claim that $\operatorname{vol} \Delta_\tau$ is continuous for $\tau < \Delta_{\max}$. In fact, it follows from Theorem C.1.4 that $(-\infty, \Delta_{\max}) \ni \tau \mapsto \log \operatorname{vol} \Delta_\tau$ is concave, but we have already known that it is finite, hence the continuity follows.

Next we show that

$$\Delta_\tau = \bigcap_{\tau' < \tau} \Delta_{\tau'}.$$

The $\subseteq$ direction is obvious. By the continuity of the volume, both sides have the same volume and the volume is positive, we therefore obtain the equality.

Similarly, we have

[3] Here concavity refers to the concavity with respect to the Minkowski sum.

$$\Delta_\tau = \overline{\bigcup_{\tau'>\tau} \Delta_{\tau'}}.$$

The continuity of Δ_τ at $\tau < \Delta_{\max}$ is proved. □

Definition 10.5.2 A *test function* on Δ is a function $G\colon \Delta \to [-\infty, \infty)$ such that

(1) G is concave,
(2) G is finite on $\operatorname{Int}\Delta$, and
(3) G is upper semicontinuous.

A test function G is *bounded* if G is bounded from below.
A test function G has *finite energy* if

$$\mathbf{E}(G) \coloneqq n! \int_\Delta G \,\mathrm{d}\lambda > -\infty. \tag{10.56}$$

Definition 10.5.3 Let $\Delta_\bullet \in \mathrm{TC}(\Delta)$. We define its *Legendre transform* as

$$G[\Delta_\bullet]\colon \Delta \to [-\infty, \infty), \quad a \mapsto \sup\{\tau < \Delta_{\max} : a \in \Delta_\tau\}.$$

Given a test function $G\colon \Delta \to [-\infty, \infty)$, we define its *inverse Legendre transform* $\Delta[G]_\bullet$ as the Okounkov test curve relative to Δ defined as follows:

(1) $\Delta[G]_{\max} = \sup_\Delta G$, and
(2) for each $\tau < \sup_\Delta G$, we set

$$\Delta[G]_\tau = \{x \in \Delta : G \geq \tau\}.$$

We observe that

$$G[\Delta_\bullet](a) = \max\{\tau \leq \Delta_{\max} : a \in \Delta_\tau\}, \text{ if } G[\Delta_\bullet](a) > -\infty. \tag{10.57}$$

Lemma 10.5.1 *Let $\Delta_\bullet \in \mathrm{TC}(\Delta)$. Then $G[\Delta_\bullet]$ defined in Definition 10.5.3 is a test function.*

Similarly, if $G\colon \Delta \to [-\infty, \infty)$ is a test function, then $\Delta[G]_\bullet$ is an Okounkov test curve.

Proof First suppose that $\Delta_\bullet \in \mathrm{TC}(\Delta)$. We want to verify that $G[\Delta_\bullet]$ satisfies the conditions in Definition 10.5.2.

We first verify the concavity. Take $a, b \in \Delta$. We want to prove that for any $t \in (0, 1)$,

$$G[\Delta_\bullet](ta + (1-t)b) \geq tG[\Delta_\bullet](a) + (1-t)G[\Delta_\bullet](b). \tag{10.58}$$

There is nothing to prove if $G[\Delta_\bullet](a)$ or $G[\Delta_\bullet](b)$ is $-\infty$. So we assume that both are finite. In this case, by (10.57),

$$a \in \Delta_{G[\Delta_\bullet](a)}, \quad b \in \Delta_{G[\Delta_\bullet](b)}.$$

Thus,

$$ta + (1-t)b \in t\Delta_{G[\Delta_\bullet](a)} + (1-t)\Delta_{G[\Delta_\bullet](b)} \subseteq \Delta_{tG[\Delta_\bullet](a)+(1-t)G[\Delta_\bullet](b)}.$$

We deduce that

$$G[\Delta_\bullet](ta + (1-t)b) \geq tG[\Delta_\bullet](a) + (1-t)G[\Delta_\bullet](b).$$

Therefore, (10.58) follows.

It is clear that $G[\Delta_\bullet]$ is finite on the interior of Δ. It remains to argue that $G[\Delta_\bullet]$ is upper semicontinuous.

Let $(a_i)_{i\geq 1}$ be a sequence in Δ with limit $a \in \Delta$. Define $\tau_i = G[\Delta_\bullet](a_i)$. Let $\tau = \overline{\lim}_i \tau_i$. We need to show that

$$G[\Delta_\bullet](a) \geq \tau. \tag{10.59}$$

There is nothing to prove if $\tau = -\infty$. We assume that this is not the case. Passing to a subsequence, we may assume that $\tau_i \to \tau$. In particular, we can assume that $\tau_i \neq -\infty$ for all $i \geq 1$. It follows from (10.57) that $a_i \in \Delta_{\tau_i}$ for all $i \geq 1$. If $\tau < \Delta_{\max}$, then $\Delta_{\tau_i} \xrightarrow{d_{\mathrm{Haus}}} \Delta_\tau$, so Theorem C.1.3 implies that $a \in \Delta_\tau$. If $\tau = \Delta_{\max}$, then the same conclusion follows from the definition $\Delta_{\Delta_{\max}} = \bigcap_{\tau' < \Delta_{\max}} \Delta_{\tau'}$. Thus, (10.59) follows.

Conversely, suppose that $G \colon \Delta \to [-\infty, \infty)$ is a test function. We argue that $\Delta[G]_\bullet$ is an Okounkov test curve. We verify the conditions in Definition 10.5.1.

Firstly, for each $\tau < \sup_\Delta G$, the set $\Delta[G]_\tau$ is a convex body as G is concave and usc. Moreover, $\Delta[G]_\tau$ is clearly decreasing in τ.

Secondly, for each $a \in \Delta$, we can write $a = \lim_i a_i$ with $a_i \in \operatorname{Int} \Delta$. By assumption, G is finite at a_i. Thus,

$$a \in \overline{\{G > -\infty\}} = \overline{\bigcup_{\tau < \sup_\Delta G} \Delta[G]_\tau}.$$

By Theorem C.1.3, $\Delta[G]_\tau \xrightarrow{d_{\mathrm{Haus}}} \Delta$ as $\tau \to -\infty$.

Thirdly, $\Delta[G]$ is concave. To see, take $\tau, \tau' < \Delta_{\max}$, we need to prove that for any $t \in (0,1)$,

$$\Delta[G]_{t\tau+(1-t)\tau'} \supseteq t\Delta[G]_\tau + (1-t)\Delta[G]_{\tau'}. \tag{10.60}$$

Let $a \in \Delta[G]_\tau$ and $b \in \Delta[G]_{\tau'}$. We have $G(a) \geq \tau$ and $G(b) \geq \tau'$. As G is concave, we have $G(ta + (1-t)b) \geq t\tau + (1-t)\tau'$. Thus,

$$ta + (1-t)b \in \Delta[G]_{t\tau+(1-t)\tau'}$$

and (10.60) follows. □

Theorem 10.5.1 *The Legendre transform and inverse Legendre transform are inverse to each other, defining a bijection between* $\mathrm{TC}(\Delta)$ *and the set of test functions on* Δ*.*

Under this bijection, $\mathrm{TC}^1(\Delta)$ *corresponds to test functions on* Δ *with finite energy and* $\mathrm{TC}^\infty(\Delta)$ *corresponds to bounded test functions on* Δ*.*

Proof Thanks to Lemma 10.5.1, in order to prove the first assertion, it only remains to see that the Legendre transform and the inverse Legendre transform are inverse to each other, which is immediate by definition.

It is obvious that $\mathrm{TC}^\infty(\Delta)$ corresponds to bounded test functions. Moreover, a direct computation shows that if $\Delta_\bullet \in \mathrm{TC}(\Delta)$, then

$$\mathbf{E}(\Delta_\bullet) = \mathbf{E}(G[\Delta_\bullet]),$$

concluding the $\mathrm{TC}^1(\Delta)$ case. □

Proposition 10.5.2 *Let $(\Delta^i)_{i\in I}$ be a decreasing net in $\mathcal{K}_n$ such that $\Delta \subseteq \bigcap_{i\in I} \Delta^i$. Consider a decreasing net $(\Delta^i_\bullet)_{i\in I}$ with $\Delta^i_\bullet \in \mathrm{TC}(\Delta^i)$ for all $i \in I$ such that there is $\Delta_\bullet \in \mathrm{TC}(\Delta)$ satisfying the following properties:*

(1) $\Delta_{\max} = \lim_{i\in I} \Delta^i_{\max}$;

(2) *for any $\tau < \Delta_{\max}$, we have $\Delta^i_\tau \xrightarrow{d_{\mathrm{Haus}}} \Delta_\tau$.*

Then for any $a \in \Delta$, we have

$$\lim_{i\in I} G[\Delta^i_\bullet](a) = G[\Delta_\bullet](a). \tag{10.61}$$

Note that in general,

$$\Delta \subsetneq \bigcap_{i\in I} \Delta^i.$$

Proof Fix $a \in \Delta$. It follows immediately from the definition of G that the net $(G[\Delta^i_\bullet](a))_{i\in I}$ is decreasing and the $\geq$ direction in (10.61) holds. Let us prove the reverse inequality. Let τ denote the left-hand side of (10.61) for the moment. By definition, for any $\epsilon > 0$ and any $i \in I$, we have $a \in \Delta^i_{\tau-\epsilon}$. It follows that

$$a \in \Delta_{\tau-\epsilon}.$$

Therefore,

$$\tau \leq G[\Delta_\bullet](a).$$

This completes the proof. □

Similarly, for increasing nets, we have:

Proposition 10.5.3 *Let $(\Delta^i)_{i\in I}$ be an increasing net in $\mathcal{K}_n$ with Hausdorff limit Δ such that $\operatorname{vol}\Delta^i > 0$ for all $i \in I$. Consider an increasing net $(\Delta^i_\bullet)_{i\in I}$ with $\Delta^i_\bullet \in \mathrm{TC}(\Delta^i)$ for all $i \in I$. Let $\Delta_{\max} = \lim_{i\in I} \Delta^i_{\max}$. For any $\tau < \Delta_{\max}$, let Δ_τ be the Hausdorff limit of Δ^i_τ. Then $\Delta_\bullet \in \mathrm{TC}(\Delta)$ and*

$$\lim_{i\in I} G[\Delta^i_\bullet](a) = G[\Delta_\bullet](a) \tag{10.62}$$

for any $a \in \operatorname{Int}\Delta$.

Proof It is obvious that $\Delta_\bullet \in \mathrm{TC}(\Delta)$.

Fix $a \in \operatorname{Int}\Delta$. Since $\Delta^i \xrightarrow{d_{\mathrm{Haus}}} \Delta$, we have $a \in \Delta^i$ for all large enough i. By definition, the net $(G[\Delta^i_\bullet](a))_{i\in I}$ is increasing and the $\leq$ direction in (10.62) holds. Let us write $\tau = G[\Delta_\bullet](a)$ for the time being. By definition of G, for any $\epsilon > 0$, we have

$$a \in \Delta_{\tau-\epsilon/2}.$$

The concavity of $\Delta_\bullet$ guarantees that

$$a \in \operatorname{Int}\Delta_{\tau-\epsilon}.$$

Since $\Delta^i_{\tau-\epsilon} \xrightarrow{d_{\mathrm{Haus}}} \Delta_{\tau-\epsilon}$ and $a \in \operatorname{Int}\Delta_{\tau-\epsilon}$, it follows that for all large enough $i \in I$,

$$a \in \Delta^i_{\tau-\epsilon}.$$

Therefore,

$$\tau - \epsilon \leq G[\Delta^i_\bullet](a)$$

for all large enough i. Taking the limit with respect to i and then with respect to ϵ, we conclude the desired inequality. □

Definition 10.5.4 Let $\Delta_\bullet$ be an Okounkov test curve relative to Δ. We define the *Duistermaat–Heckman measure* $\mathrm{DH}(\Delta_\bullet)$ as

$$\mathrm{DH}(\Delta_\bullet) := G[\Delta_\bullet]_*(\mathrm{vol}).$$

It is a Radon measure on $\mathbb{R}$.

In other words, $\mathrm{DH}(\Delta_\bullet)$ is the distribution of the random variable $G[\Delta_\bullet]$.

Proposition 10.5.4 *Let $\Delta_\bullet \in \mathrm{TC}(\Delta)$. Let $m \in \mathbb{Z}_{>0}$. Then the following extended moment identity holds:*

$$\int_{\mathbb{R}} x^m \,\mathrm{DH}(\Delta_\bullet)(x) = \Delta_{\max}^m \operatorname{vol}\Delta + m\int_{-\infty}^{\Delta_{\max}} \tau^{m-1}(\operatorname{vol}\Delta_\tau - \operatorname{vol}\Delta)\,\mathrm{d}\tau \qquad (10.63)$$

and

$$\int_{\mathbb{R}} \mathrm{DH}(\Delta_\bullet) = \operatorname{vol}\Delta. \qquad (10.64)$$

Proof In fact, (10.64) follows immediately from the definition. Let $G = G[\Delta_\bullet]$. We first assume that G is bounded from below. Then (10.63) follows from a straightforward computation:

$$
\begin{aligned}
&\int_{\mathbb{R}} x^m \,\mathrm{DH}(\Delta_\bullet)(x)\\
=&\int_\Delta G[\Delta_\bullet](a)^m \,\mathrm{d}\,\mathrm{vol}(a)\\
=&\int_\Delta \left(\Delta_{\max}^m - \int_{G[\Delta_\bullet](a)}^{\Delta_{\max}} m\tau^{m-1}\,\mathrm{d}\tau\right) \mathrm{d}\,\mathrm{vol}(a)\\
=&\Delta_{\max}^m \operatorname{vol}\Delta - m\int_{\mathbb{R}}\int_\Delta \mathbb{1}_{[G[\Delta_\bullet](a),\Delta_{\max}]}(\tau)\tau^{m-1}\,\mathrm{d}\,\mathrm{vol}(a)\,\mathrm{d}\tau\\
=&\Delta_{\max}^m \operatorname{vol}\Delta - m\int_{-\infty}^{\Delta_{\max}}\int_{\Delta\setminus\Delta_\tau} \tau^{m-1}\,\mathrm{d}\,\mathrm{vol}(a)\,\mathrm{d}\tau\\
=&\Delta_{\max}^m \operatorname{vol}\Delta - m\int_{-\infty}^{\Delta_{\max}} \tau^{m-1}\left(\operatorname{vol}\Delta - \operatorname{vol}\Delta_\tau\right)\,\mathrm{d}\tau.
\end{aligned}
$$

In general, set $G_A = G \vee (-A)$. The same pointwise identity applied to G_A gives

$$
\int_\Delta G_A^m \,\mathrm{d}\,\mathrm{vol} = \Delta_{\max}^m \operatorname{vol}\Delta + m\int_{-A}^{\Delta_{\max}} \tau^{m-1}(\operatorname{vol}\Delta_\tau - \operatorname{vol}\Delta)\,\mathrm{d}\tau.
$$

Letting $A \to \infty$ gives (10.63): if m is even, this follows from monotone convergence applied to $G_A^m \nearrow G^m$; if m is odd, apply monotone convergence to $\Delta_{\max}^m - G_A^m \nearrow \Delta_{\max}^m - G^m$. Thus both sides of (10.63) are interpreted in the corresponding extended sense. □

Lemma 10.5.2 *Let $(\Delta^i)_{i\geq 1}$ be a decreasing sequence in $\mathcal{K}_n$ with limit Δ. Suppose that $(\Delta^i_\bullet)_{i\geq 1}$ is a decreasing sequence with $\Delta^i_\bullet \in \mathrm{TC}(\Delta^i)$. Suppose that there is $\Delta_\bullet \in \mathrm{TC}(\Delta)$ such that*

(1) $\Delta_{\max} = \lim_{i\to\infty} \Delta^i_{\max}$;

(2) *for any $\tau < \Delta_{\max}$, we have $\Delta^i_\tau \xrightarrow{d_{\mathrm{Haus}}} \Delta_\tau$.*

Then $\mathrm{DH}(\Delta^i_\bullet) \rightharpoonup \mathrm{DH}(\Delta_\bullet)$.

Proof Let $f \in C_b(\mathbb{R})$. It follows from Proposition 10.5.2 that

$$
G[\Delta^i_\bullet] \to G[\Delta_\bullet]
$$

pointwise on Δ. Hence the dominated convergence theorem gives

$$
\int_\Delta f(G[\Delta^i_\bullet])\,\mathrm{d}\lambda \to \int_\Delta f(G[\Delta_\bullet])\,\mathrm{d}\lambda.
$$

Since $\Delta^i \xrightarrow{d_{\mathrm{Haus}}} \Delta$, Theorem C.1.2 gives $\operatorname{vol}(\Delta^i\setminus\Delta) \to 0$. The contribution of $\Delta^i\setminus\Delta$ is bounded by $\|f\|_\infty \operatorname{vol}(\Delta^i\setminus\Delta)$, so

$$
\int_{\Delta^i} f(G[\Delta^i_\bullet])\,\mathrm{d}\lambda \to \int_\Delta f(G[\Delta_\bullet])\,\mathrm{d}\lambda.
$$

This is the weak convergence. □

Similarly, we have

Lemma 10.5.3 *Let $(\Delta^i)_{i\geq 1}$ be an increasing sequence in $\mathcal{K}_n$ with Hausdorff limit Δ such that $\operatorname{vol}\Delta^i > 0$ for all $i \geq 1$. Consider an increasing sequence $(\Delta^i_\bullet)_{i\geq 1}$ with $\Delta^i_\bullet \in \mathrm{TC}(\Delta^i)$ for all $i \geq 1$. Let $\Delta_\bullet \in \mathrm{TC}(\Delta)$ be defined as*

(1) $\Delta_{\max} = \lim_{i\to\infty} \Delta^i_{\max}$;
(2) *for any $\tau < \Delta_{\max}$, Δ_τ is the Hausdorff limit of Δ^i_τ.*

Then we have

$$\mathrm{DH}(\Delta^i_\bullet) \rightharpoonup \mathrm{DH}(\Delta_\bullet).$$

Proof Let $f \in C_b(\mathbb{R})$. It follows from Proposition 10.5.3 that

$$G[\Delta^i_\bullet] \to G[\Delta_\bullet]$$

almost everywhere on Δ. Extending $f(G[\Delta^i_\bullet])$ by 0 outside Δ^i, we also have $\mathbb{1}_{\Delta^i} \to \mathbb{1}_\Delta$ almost everywhere on Δ, since the boundary of the convex body Δ has Lebesgue measure zero. The dominated convergence theorem gives

$$\int_{\Delta^i} f(G[\Delta^i_\bullet])\,\mathrm{d}\lambda \to \int_\Delta f(G[\Delta_\bullet])\,\mathrm{d}\lambda.$$

This is the weak convergence. □

The main source of Okounkov test curves is the following:

Theorem 10.5.2 *Let X be a connected compact Kähler manifold and θ be a closed smooth real $(1,1)$-form on X representing a big cohomology class α. Let $Y_\bullet$ be a smooth flag on X and $\Gamma \in \mathrm{TC}(X,\theta)_{>0}$. Then the map*

$$(-\infty, \Gamma_{\max}) \ni \tau \mapsto \Delta_{Y_\bullet}(\theta,\Gamma)_\tau \coloneqq \Delta_{Y_\bullet}(\theta,\Gamma_\tau)$$

defines an Okounkov test curve relative to $\Delta_{Y_\bullet}(\theta,\Gamma_{-\infty})$.

If furthermore $\Gamma \in \mathrm{TC}^1(X,\theta;\Gamma_{-\infty})$ (resp. $\mathrm{TC}^\infty(X,\theta;\Gamma_{-\infty})$), then we have $\Delta_{Y_\bullet}(\theta,\Gamma) \in \mathrm{TC}^1(\Delta_{Y_\bullet}(\theta,\Gamma_{-\infty}))$ (resp. $\mathrm{TC}^\infty(\Delta_{Y_\bullet}(\theta,\Gamma_{-\infty}))$).

See Definition 9.1.1 and Definition 9.1.2 for the relevant definitions.

Proof Consider $\Gamma \in \mathrm{TC}(X,\theta)_{>0}$. We need to verify that $\Delta_{Y_\bullet}(\theta,\Gamma)$ is an Okounkov test curve relative to $\Delta_{Y_\bullet}(\theta,\Gamma_{-\infty})$.

First observe that $\tau \mapsto \Delta_{Y_\bullet}(\theta,\Gamma_\tau)$ is concave and decreasing for $\tau < \Gamma_{\max}$. This is a direct consequence of Theorem 10.4.4.

Next we show that as $\tau \to -\infty$, we have

$$\Delta_{Y_\bullet}(\theta,\Gamma_\tau) \xrightarrow{d_{\mathrm{Haus}}} \Delta_{Y_\bullet}(\theta,\Gamma_{-\infty}).$$

By Theorem 10.4.2(2), the bodies $\Delta_{Y_\bullet}(\theta,\Gamma_\tau)$ are contained in $\Delta_{Y_\bullet}(\theta,\Gamma_{-\infty})$ and increase as $\tau \to -\infty$. Thus it suffices to compute their volumes:

$$\lim_{\tau\to-\infty} \operatorname{vol} \Delta_{Y_\bullet}(\theta,\Gamma_\tau) = \frac{1}{n!}\lim_{\tau\to-\infty} \operatorname{vol}(\theta+\mathrm{dd^c}\Gamma_\tau) = \frac{1}{n!}\operatorname{vol}(\theta+\mathrm{dd^c}\Gamma_{-\infty})$$
$$= \operatorname{vol}\Delta_{Y_\bullet}(\theta,\Gamma_{-\infty}),$$

where we applied Theorem 10.4.2 and Theorem 6.2.5.

When $\Gamma \in \mathrm{TC}^\infty(X,\theta;\Gamma_{-\infty})$, it is clear that $\Delta_{Y_\bullet}(\theta,\Gamma) \in \mathrm{TC}^\infty(\Delta_{Y_\bullet}(\theta,\Gamma_{-\infty}))$.

When $\Gamma \in \mathrm{TC}^1(X,\theta;\Gamma_{-\infty})$, by Theorem 10.4.2(1), (9.4) and (10.54), we have

$$\mathbf{E}^{\Gamma_{-\infty}}(\Gamma) = \mathbf{E}(\Delta_{Y_\bullet}(\theta,\Gamma)).$$

So $\Delta_{Y_\bullet}(\theta,\Gamma) \in \mathrm{TC}^1(\Delta_{Y_\bullet}(\theta,\Gamma_{-\infty}))$. □

Remark 10.5.1 As a special case of this construction, suppose that Γ is the test curve induced by a test configuration as in Example 9.3.1 and Remark 9.3.1, then for any $\tau < \Gamma_{\max}$, $\Delta_{Y_\bullet}(\theta,\Gamma_\tau)$ is the Okounkov body of a graded linear series

$$\bigoplus_{k=0}^{\infty} \mathcal{F}_k^{k\tau},$$

where $\mathcal{F}$ is the filtration induced by the test configuration. See [Xia25b, Theorem 5.28] for the details. In particular, in this case, our theory of partial Okounkov bodies recovers the Okounkov bodies of the filtered linear series in the sense of [BC11].

Chapter 11
The theory of b-divisors

The mathematician's patterns, like the painter's or the poet's must be beautiful; the ideas, like the colors or the words must fit together in a harmonious way. Beauty is the first test: There is no permanent place in this world for ugly mathematics.
— Godfrey Harold Hardy[a]

[a] Godfrey Harold Hardy (1877–1947) was a British mathematician famous for his work in number theory and mathematical analysis. Apart from his research, Hardy was a strong advocate for pure mathematics and believed that mathematics should be pursued for its own beauty, not just for practical use.

He remained lifelong unmarried and dedicated much of his life entirely to mathematics, fitting into the common stereotype of a mathematician.

A central theme of this book is the principle that the algebraic shadow of a psh singularity should be controlled by cohomological data on all modifications of the underlying manifold. The cleanest manifestation of this principle is the theory of *nef b-divisors*: cohomological objects on the universe of all modifications $Y \to X$ which precisely capture the $\mathcal{I}$-equivalence information of closed positive $(1,1)$-currents in a fixed modified nef class. The aim of this chapter is to develop this theory and to state the precise correspondence as Theorem 11.1.3: after fixing a modified nef class α, the assignment "current $\mapsto$ associated nef b-divisor" gives a bijection between $\mathcal{G}(\alpha)/\sim_{\mathcal{I}}$ and the big and nef b-divisors whose trace on X is α.

The notion of b-divisor goes back to Shokurov: a b-divisor on X records, for every modification $\pi: Y \to X$, a Weil divisor or a cohomology class $\mathbb{D}_\pi \in \mathrm{H}^{1,1}(Y, \mathbb{R})$, the various $\mathbb{D}_\pi$'s being required to be compatible under push-forward. Such an object is therefore intrinsically birational, lives on the universe of all modifications simultaneously, and forgets nothing about pull-backs to higher models. The class of *nef b-divisors* is a natural subclass playing in b-divisor theory the role of nef classes in classical algebraic geometry, and admits a satisfying intersection theory.

The main analytic input is that closed positive $(1,1)$-currents naturally produce nef b-divisors. Given T on X, we define $\mathbb{D}(T)_\pi := \{\mathrm{Reg}\,\pi^*T\} \in \mathrm{H}^{1,1}(Y, \mathbb{R})$ for every modification $\pi: Y \to X$. Theorem 11.1.1 shows that this assignment factors through $\mathcal{I}$-equivalence and lands in the nef b-divisor cone. Theorems 11.1.2 and 11.1.3 then upgrade this map to a bijection in the big case, with $\preceq_{\mathcal{I}}$ on currents corresponding to a preorder $\leq$ on nef b-divisors (Corollary 11.1.6). The continuity of this correspondence in the d_S-pseudometric is recorded in Proposition 11.1.2.

The chapter is organized as follows.

Section 11.1 introduces the basic notions: b-divisors and Cartier b-divisors (Definition 11.1.4), nef b-divisors (Definition 11.1.5), and the volume of a b-divisor (Definition 11.1.2). Its second subsection Section 11.1.2 attaches a nef b-divisor to each closed positive $(1,1)$-current (Theorem 11.1.1) and proves the main correspondence theorem Theorem 11.1.3, together with several corollaries on degeneracy (Corollaries 11.1.2 and 11.1.4), $\mathcal{I}$-equivalence (Corollary 11.1.3), and order-preservation (Corollary 11.1.6).

Section 11.2 develops the intersection theory of nef b-divisors. The starting point is the extension of the classical Dang–Favre intersection product to the b-divisor setting (Definition 11.2.1); we then establish its basic properties.

We use the convention of denoting the current of integration along a prime divisor D by $[D]$ and the associated cohomology class by $\{D\}$, to avoid confusion between analytic and cohomological objects.

11.1 The notion of b-divisors

The b-divisors defined in this section are sometimes called b-divisor classes. We always omit the word *classes* to save space.

Let X be a connected compact Kähler manifold of dimension n.

Let us recall the following elementary result regarding how cohomology behaves under blow-up.

Proposition 11.1.1 *Let $\pi\colon Y \to X$ be a blow-up with* connected *smooth center of codimension at least* 2 *with exceptional divisor E. Then there is a natural identification*

$$\mathrm{H}^{1,1}(Y,\mathbb{R}) = \mathrm{H}^{1,1}(X,\mathbb{R}) \oplus \mathbb{R}\{E\}. \tag{11.1}$$

See [RYY19] for a much more general result.

11.1.1 Nef b-divisors

Definition 11.1.1 A *(Weil) b-divisor* $\mathbb{D}$ over X is an assignment $(\mathbb{D}_\pi)_{\pi\colon Y\to X}$, where $\pi\colon Y \to X$ runs over all modifications of X such that

(1) $\mathbb{D}_\pi \in \mathrm{H}^{1,1}(Y,\mathbb{R})$;
(2) The classes are compatible under push-forwards: If $\pi'\colon Z \to X$ and $\pi\colon Y \to X$ are both modifications of X and π' dominates π through $g\colon Z \to Y$, then $g_*\mathbb{D}_{\pi'} = \mathbb{D}_\pi$.

We also write $\mathbb{D}_Y = \mathbb{D}_\pi$ if there is no risk of confusion.

Given two Weil b-divisors $\mathbb{D}$ and $\mathbb{D}'$ over X, we say $\mathbb{D} \leq \mathbb{D}'$ if for each modification π of X, we have $\mathbb{D}_\pi \leq \mathbb{D}'_\pi$. Recall that by definition, this means the class $\mathbb{D}'_\pi - \mathbb{D}_\pi$ is pseudo-effective.

The class $\mathbb{D}_X$ is called the *root* of $\mathbb{D}$. The set of Weil b-divisors over X has the natural structure of a real vector space.

Definition 11.1.2 The *volume* of a Weil b-divisor $\mathbb{D}$ over X is

$$\operatorname{vol}\mathbb{D} := \lim_{\pi : Y \to X} \operatorname{vol}\mathbb{D}_Y .$$

The right-hand side is a decreasing net due to Proposition 3.2.8, hence the limit always exists.

We say $\mathbb{D}$ is *big* if $\operatorname{vol}\mathbb{D} > 0$.

Lemma 11.1.1 *Let $(\mathbb{D}_i)_{i\in I}$ be a net of b-divisors converging to $\mathbb{D}$. Then*

$$\varlimsup_{i\in I} \operatorname{vol}\mathbb{D}_i \leq \operatorname{vol}\mathbb{D}. \tag{11.2}$$

If the net is decreasing, then

$$\lim_{i\in I} \operatorname{vol}\mathbb{D}_i = \operatorname{vol}\mathbb{D}.$$

Here we say $(\mathbb{D}_i)_{i\in I}$ converges to $\mathbb{D}$ if for any modification $\pi : Y \to X$, we have $\mathbb{D}_{i,Y} \to \mathbb{D}_Y$ with respect to the Euclidean topology.

Proof Let $\pi : Y \to X$ be a modification. Then

$$\operatorname{vol}\mathbb{D}_Y = \lim_{i\in I} \operatorname{vol}\mathbb{D}_{i,Y} \geq \varlimsup_{i\in I} \operatorname{vol}\mathbb{D}_i .$$

The inequality (11.2) follows. As for the decreasing case, it suffices to observe that both sides of (11.2) can be written as

$$\inf_i \inf_{\pi : Y \to X} \operatorname{vol}\mathbb{D}_{i,Y} .$$

This completes the proof. □

Example 11.1.1 In general, equality need not hold in (11.2). Let $X = \mathbb{P}^2$, and let $H = c_1(\mathcal{O}_{\mathbb{P}^2}(1))$. Choose a sequence of points $p_i \in X$ converging to the generic point in the following sense: for every algebraic curve $C \subseteq X$, the set of indices i with $p_i \in C$ is finite.

Let

$$\pi_i : X_i \to X$$

be the blow-up of X at p_i, with exceptional divisor E_i. Let $\mathbb{D}_i$ be the Cartier b-divisor realized on X_i by the nef class

$$\pi_i^* H - \{E_i\},$$

and let $\mathbb{D}$ be the Cartier b-divisor realized on X by H. Then $\mathbb{D}_i \to \mathbb{D}$. Indeed, for any fixed modification $\rho : Y \to X$, the morphism ρ is an isomorphism over p_i for all large enough i, hence the trace of $\mathbb{D}_i$ on Y is $\rho^* H$ for all large enough i.

On the other hand,

$$\operatorname{vol}\mathbb{D}_i = \left(\pi_i^* H - \{E_i\}\right)^2 = 0$$

for all i, whereas

$$\operatorname{vol}\mathbb{D} = H^2 = 1.$$

Thus the inequality in (11.2) is strict for this sequence.

Definition 11.1.3 A *Cartier b-divisor* $\mathbb{D}$ over X is a Weil b-divisor $\mathbb{D}$ over X such that there exists a modification $\pi\colon Y \to X$ and a class $\alpha_Y \in \mathrm{H}^{1,1}(Y,\mathbb{R})$ so that for each $\pi'\colon Z \to X$ dominating π, the class $\mathbb{D}_Z$ is the pull-back of α_Y. Any such (π, α_Y) is called a *realization* of $\mathbb{D}$.

By abuse of language, we also say (Y, α_Y) is a realization of $\mathbb{D}$. The realization is not unique in general.

Definition 11.1.4 A Cartier b-divisor $\mathbb{D}$ over X is *nef* if there exists a realization $(\pi\colon Y \to X, \alpha_Y)$ of $\mathbb{D}$ such that α_Y is nef.

Definition 11.1.5 A Weil b-divisor $\mathbb{D}$ over X is *nef* if it is the limit of a net of nef Cartier b-divisors $(\mathbb{D}_i)_i$ over X.

In other words, for each modification $\pi\colon Y \to X$, we have $\mathbb{D}_{i,Y} \to \mathbb{D}_Y$.

Note that thanks to Proposition 1.7.1, each $\mathbb{D}_Y$ is necessarily modified nef, but it is not nef in general. The notion of modified nef classes is defined in Definition 1.7.9.

A priori, for a Cartier b-divisor, nefness could mean either the notion defined by Definition 11.1.4 or the one defined by Definition 11.1.5. We will show in Corollary 11.1.5 that they are actually equivalent. Before that, by a nef Cartier b-divisor, we always mean nefness in the sense of Definition 11.1.4.

Our definition Definition 11.1.5 amounts to defining the set of Weil b-divisors as the closure of the set of Cartier b-divisors in $\varprojlim_\pi \mathrm{H}^{1,1}(Y,\mathbb{R})$ with respect to the projective limit topology. In particular, the limit of a converging net of nef b-divisors is still nef.

11.1.2 The b-divisors of currents

Let T be a closed positive $(1,1)$-current on X.

Given any modification $\pi\colon Y \to X$, we define

$$\mathbb{D}(T)_Y \coloneqq \{\operatorname{Reg}\pi^*T\} \in \mathrm{H}^{1,1}(Y,\mathbb{R}). \tag{11.3}$$

We observe that if T' is another closed positive $(1,1)$-current on X and $\lambda \geq 0$, then

$$\mathbb{D}(T+T') = \mathbb{D}(T) + \mathbb{D}(T'), \quad \mathbb{D}(\lambda T) = \lambda\mathbb{D}(T).$$

We shall use these identities implicitly in the sequel.

Note that when T has analytic singularities, $\mathbb{D}(T)$ is Cartier.

Theorem 11.1.1 *Let T be a closed positive $(1,1)$-current on X. Then $\mathbb{D}(T)$ is nef. Moreover,*

$$\operatorname{vol} T = \operatorname{vol}\mathbb{D}(T). \tag{11.4}$$

Proof Let ω be a Kähler form on X.

Step 1. Reduce to the case where T is a Kähler current.

Note that $\mathbb{D}(\omega)$ is the Cartier b-divisor realized by $(X, \{\omega\})$. The b-divisors $\mathbb{D}(T+\epsilon\omega) = \mathbb{D}(T) + \epsilon\mathbb{D}(\omega)$ decrease to $\mathbb{D}(T)$ as $\epsilon \searrow 0$. Hence, by Lemma 11.1.1 and Proposition 7.3.1, it suffices to prove our assertion with $T + \epsilon\omega$ in place of T.

Step 2. We prove the assertion under the additional assumption that T has analytic singularities.

Let $\pi\colon Y \to X$ be a modification so that

$$\pi^*T = [D] + R,$$

where D is an effective $\mathbb{Q}$-divisor on Y and R is a closed positive $(1,1)$-current with locally bounded potentials. See Theorem 1.6.1 for the existence of π. Then $\mathbb{D}(T)$ is the nef Cartier b-divisor realized by $(\pi, \{R\})$. Note that (11.4) is immediate in this case.

Step 3. We prove the assertion for a general Kähler current T. Next, take a closed smooth real $(1,1)$-form θ in the cohomology class of T and write $T = \theta_\varphi$ for some $\varphi \in \mathrm{PSH}(X, \theta)$. Since T is a Kähler current, we choose the quasi-equisingular approximation $(\varphi_j)_j$ of φ so that, after discarding finitely many terms, each $\theta + \mathrm{dd^c}\varphi_j$ is a Kähler current. This is possible by Demailly regularization, in the form used in the proof of Theorem 7.1.1. We claim that

$$\mathbb{D}(\theta + \mathrm{dd^c}\varphi_j) \to \mathbb{D}(\theta + \mathrm{dd^c}\varphi). \tag{11.5}$$

By definition of this convergence, we need to establish the following: for every modification $\pi\colon Y \to X$, we have

$$\left\{\mathrm{Reg}\left(\pi^*(\theta + \mathrm{dd^c}\varphi_j)\right)\right\} \to \left\{\mathrm{Reg}\left(\pi^*(\theta + \mathrm{dd^c}\varphi)\right)\right\}.$$

This follows immediately from Theorem 6.2.4 if $\mathrm{Sing}(\pi^*T)$ has only finitely many components.

Fix a Kähler form ω_Y on Y. We want to show that for any $\epsilon > 0$, we can find $j_0 > 0$ so that when $j \geq j_0$,

$$\left\{\mathrm{Reg}\left(\pi^*(\theta + \mathrm{dd^c}\varphi_j)\right)\right\} \leq \left\{\mathrm{Reg}\left(\pi^*(\theta + \mathrm{dd^c}\varphi)\right)\right\} + \epsilon\{\omega_Y\}. \tag{11.6}$$

Write the divisorial part of $\pi^*\theta + \mathrm{dd^c}\pi^*\varphi_j$ and $\pi^*\theta + \mathrm{dd^c}\pi^*\varphi$ as

$$\sum_{i=1}^\infty a_i^j[E_i], \quad \sum_{i=1}^\infty a_i[E_i].$$

Then $a_i^j \leq a_i$.

We can find an integer $N > 0$ large enough so that, in the pseudo-effective order of cohomology classes,

$$\sum_{i=N+1}^\infty a_i\{E_i\} \leq \frac{\epsilon}{2}\{\omega_Y\}.$$

By Theorem 6.2.4, we can take j_0 large enough so that for $j > j_0$,

$$\sum_{i=1}^{N}(a_i - a_i^j)\{E_i\} \leq \frac{\epsilon}{2}\{\omega_Y\}.$$

Then (11.6) follows, and (11.5) is established.

As a consequence, $\mathbb{D}(T)$ is nef and the volume can be computed using Lemma 11.1.1:

$$\operatorname{vol}\mathbb{D}(T) = \lim_{j\to\infty}\operatorname{vol}\mathbb{D}\left(\theta + \mathrm{dd^c}\varphi_j\right) = \lim_{j\to\infty}\operatorname{vol}\left(\theta + \mathrm{dd^c}\varphi_j\right) = \operatorname{vol}T.$$

Hence, (11.4) follows. □

Conversely, we want to realize nef b-divisors as $\mathbb{D}(T)$. We first prove a continuity result.

Proposition 11.1.2 *Let θ be a closed real smooth $(1,1)$-form on X, let $(\varphi_i)_{i\in I}$ be a net in $\mathrm{PSH}(X,\theta)$, and let $\varphi \in \mathrm{PSH}(X,\theta)$. If $\varphi_i \xrightarrow{d_S} \varphi$, then*

$$\mathbb{D}\left(\theta_{\varphi_i}\right) \to \mathbb{D}\left(\theta_{\varphi}\right). \tag{11.7}$$

Proof Fix a modification $\pi\colon Y \to X$. Fix a Kähler form ω on X. It suffices to establish the following:

$$\mathbb{D}\left(\theta_{\varphi_i}\right)_Y \to \mathbb{D}\left(\theta_{\varphi}\right)_Y. \tag{11.8}$$

As a map from the pseudometric space $\mathrm{PSH}(X,\theta)$ to the finite-dimensional Euclidean space $\mathrm{H}^{1,1}(Y,\mathbb{R})$, the continuity of $\mathbb{D}(\bullet)_Y$ can be tested on sequences. Thus, without loss of generality, we may assume that $(\varphi_i)_i$ is a sequence and $I = \mathbb{Z}_{>0}$. It is enough to prove that every subsequence has a further subsequence for which (11.9) holds.

Since $\pi^*\varphi_i \xrightarrow{d_S} \pi^*\varphi$, when proving (11.8), we may assume without loss of generality that π is the identity map on X. Therefore, we are reduced to the following assertion:

$$\left\{\operatorname{Reg}\theta_{\varphi_i}\right\} \to \left\{\operatorname{Reg}\theta_{\varphi}\right\}. \tag{11.9}$$

After adding a Kähler form to θ, we may also assume that θ_φ is a Kähler current. By Theorem 6.2.3, replacing all potentials by their $\mathcal{I}$-envelopes does not change the divisorial parts and preserves d_S-convergence. The masses are then bounded from below along a tail, so after passing to a subsequence Proposition 6.2.3 supplies increasing and decreasing envelopes $\eta_i \leq \varphi_i \leq \psi_i$ which both converge to φ in d_S. Since

$$\mathbb{D}(\theta_{\eta_i})_X \leq \mathbb{D}(\theta_{\varphi_i})_X \leq \mathbb{D}(\theta_{\psi_i})_X,$$

it is enough to treat the case where $(\varphi_i)_i$ is monotone.

Write the divisorial parts of θ_{φ_i} and θ_φ as

$$\sum_{m=1}^{\infty} a_m^i[D_m], \quad \sum_{m=1}^{\infty} a_m[D_m].$$

By Theorem 6.2.4, $a_m^i \to a_m$ for each fixed m. If $(\varphi_i)_i$ is decreasing, then $a_m^i \le a_m$ for all i, m. The finite-tail argument in the proof of (11.6) then gives

$$\{\operatorname{Reg}\theta_\varphi\} \le \{\operatorname{Reg}\theta_{\varphi_i}\} \le \{\operatorname{Reg}\theta_\varphi\} + \epsilon\{\omega\}$$

for all sufficiently large i.

If $(\varphi_i)_i$ is increasing, then $a_m \le a_m^i \le a_m^1$. We first choose N so that the tails of both $\sum_m a_m[D_m]$ and $\sum_m a_m^1[D_m]$ are at most $(\epsilon/3)\{\omega\}$ in the pseudo-effective order. The convergence of the finitely many coefficients a_m^i, $m \le N$, then gives

$$\{\operatorname{Reg}\theta_{\varphi_i}\} \le \{\operatorname{Reg}\theta_\varphi\} \le \{\operatorname{Reg}\theta_{\varphi_i}\} + \epsilon\{\omega\}$$

for all sufficiently large i. This proves (11.9). □

Lemma 11.1.2 *Let $\pi\colon X \to Z$ be a proper bimeromorphic morphism from X to a Kähler manifold Z. Consider non-divisorial closed positive $(1,1)$ currents T, S on X in the same cohomology class. If $T \le_{\mathcal{I}} S$, then $\pi_* T \le_{\mathcal{I}} \pi_* S$.*

Proof By Hironaka's Chow lemma Theorem B.1.2 and Lemma 6.1.4, we may assume that π is a modification.

By Lemma 7.3.2,

$$\pi^*\pi_* T = T + F_T\,, \quad \pi^*\pi_* S = S + F_S,$$

where F_T and F_S are effective π-exceptional divisors. Since $\{T\} = \{S\}$, the exceptional divisors F_T and F_S have the same cohomology class. Factoring π into blow-ups and using Proposition 11.1.1 at each step, the classes of π-exceptional prime divisors are linearly independent in the relative cohomology; hence $F_T = F_S$. It follows that

$$\pi^*\pi_* T = T + F_T \le_{\mathcal{I}} S + F_S = \pi^*\pi_* S.$$

Our assertion then follows from Lemma 6.1.4. □

Theorem 11.1.2 *Each big and nef b-divisor $\mathbb{D}$ over X can be realized as $\mathbb{D}(T)$ for some $T \in \mathbb{D}_X$. Furthermore, we may always assume that T is $\mathcal{I}$-good.*

Note that T is not unique. The current T is necessarily non-divisorial.

Proof Fix a big and nef b-divisor $\mathbb{D}$ over X.

For each $\pi\colon Y \to X$, we take a current with minimal singularities T_Y in $\mathbb{D}_Y$. We claim that $\mathbb{D}(\pi_* T_Y)$ coincides with $\mathbb{D}$ up to the level of Y: For any modification $\pi'\colon Z \to X$ dominated by π through a morphism $g\colon Y \to Z$, we have

$$\mathbb{D}_Z = \mathbb{D}(\pi_* T_Y)_Z.$$

The notations are summarized in the following commutative diagram:

$$\begin{tikzcd} Y \arrow[rr, "g"] \arrow[dr, "\pi"'] & & Z \arrow[dl, "\pi'"] \\ & X. & \end{tikzcd} \tag{11.10}$$

After unfolding the definitions, this means

$$\mathrm{Reg}(\pi'^{*}\pi_{*}T_Y) \in \mathbb{D}_Z.$$

Note that

$$\mathrm{Reg}(\pi'^{*}\pi_{*}T_Y) = \mathrm{Reg}(\pi'^{*}\pi'_{*}g_{*}T_Y).$$

Due to Proposition 1.7.1 and Proposition 3.2.8, we know that $\mathbb{D}_Y$ is modified nef and big. In particular, T_Y is non-divisorial, hence so is g_*T_Y by Lemma 1.7.2. It follows from Lemma 7.3.2 that

$$\mathrm{Reg}(\pi'^{*}\pi'_{*}g_{*}T_Y) = \mathrm{Reg}(g_*T_Y) = g_*T_Y \in \mathbb{D}_Z.$$

Note that

$$\mathrm{vol}\, T_Y \geq \mathrm{vol}\, \mathbb{D} > 0. \tag{11.11}$$

Next we claim that the P-singularity types of the net $(\pi_*T_Y)_Y$ form a decreasing net.

To see this, let us fix a diagram as (11.10). We need to show that

$$\pi_*T_Y \preceq_P \pi'_*T_Z.$$

Since T_Z has minimal singularities, we have $g_*T_Y \preceq_{\mathcal{I}} T_Z$. In particular, Lemma 11.1.2 guarantees that $\pi_*T_Y \preceq_{\mathcal{I}} \pi'_*T_Z$. But by Corollary 7.3.3, both π_*T_Y and π'_*T_Z are $\mathcal{I}$-good, so there is no difference between the P-preorder and the $\mathcal{I}$-preorder in this case. Our assertion follows.

Next observe that the net $(\pi_*T_Y)_Y$ has a d_S-limit as a consequence of (11.11) and Corollary 6.2.6. Take a closed positive $(1,1)$-current $T \in \mathbb{D}_X$ such that

$$\pi_*T_Y \xrightarrow{d_S} T.$$

It follows from Proposition 11.1.2 that

$$\mathbb{D}(\pi_*T_Y) \to \mathbb{D}(T).$$

Therefore, we conclude that

$$\mathbb{D}(T) = \mathbb{D}.$$

Thanks to Theorem 11.1.1, $\mathrm{vol}\, T > 0$. Write $T = \theta + \mathrm{dd^c}\varphi$ for some $\varphi \in \mathrm{PSH}(X,\theta)$, then

$$T' := \theta + \mathrm{dd^c} P_\theta[\varphi]_{\mathcal{I}}$$

is $\mathcal{I}$-good, non-divisorial and $\mathbb{D}(T') = \mathbb{D}(T)$. □

Corollary 11.1.1 *Let $\mathbb{D}$ be a b-divisor over X. Then the following are equivalent:*

(1) $\mathbb{D}$ *is nef;*
(2) *for each modification $\pi\colon Y \to X$, the class $\mathbb{D}_Y$ is modified nef.*

Proof (1) $\implies$ (2). Suppose that (1) holds. Fix a modification $\pi\colon Y \to X$. Then we need to show that $\mathbb{D}_Y$ is modified nef. Since modified nefness is a closed condition, after approximating $\mathbb{D}$ by nef Cartier b-divisors, we may assume that $\mathbb{D}$ itself is a nef Cartier b-divisor. We can then find a modification $\pi'\colon Z \to X$ dominating π so that $\mathbb{D}$ is realized by a nef class α on Z. Then $\mathbb{D}_Y$ is nothing but the pushforward of α, and hence modified nef thanks to Proposition 1.7.1.

(2) $\implies$ (1). Suppose that (2) holds. We need to show that $\mathbb{D}$ is nef. Fix a Kähler form ω on X. It suffices to show that $\mathbb{D} + \epsilon\mathbb{D}(\omega)$ is nef for each $\epsilon > 0$. After replacing $\mathbb{D}$ by the latter, we may further assume that $\mathbb{D}_Y$ is big for each modification $\pi\colon Y \to X$, and $\operatorname{vol}\mathbb{D}_Y$ has a uniform positive lower bound. In this case, the argument of Theorem 11.1.2 shows that $\mathbb{D} = \mathbb{D}(T)$ for some closed positive $(1,1)$-current in $\mathbb{D}_X$. Therefore, $\mathbb{D}$ is nef, thanks to Theorem 11.1.1. □

Let α be a modified nef class on X. We write $\mathcal{G}(\alpha)$ for the set of closed positive $(1,1)$-currents T on X with $T = \operatorname{Reg} T \in \alpha$ and $\operatorname{vol} T > 0$.

Theorem 11.1.3 *There is a natural bijection from $\mathcal{G}(\alpha)/\sim_{\mathcal{I}}$ to the set of big and nef b-divisors $\mathbb{D}$ over X with $\mathbb{D}_X = \alpha$.*

Proof Given $T \in \mathcal{G}(\alpha)$, we associate the b-divisor $\mathbb{D}(T)$. It is big and nef due to Theorem 11.1.1. This map clearly descends to $\mathcal{G}(\alpha)/\sim_{\mathcal{I}}$.

This map is surjective by Theorem 11.1.2. Now we show that it is injective. Let $T, T' \in \mathcal{G}(\alpha)$. Assume that $\mathbb{D}(T) = \mathbb{D}(T')$, we want to show that $T \sim_{\mathcal{I}} T'$.

Let E be a prime divisor over X, it suffices to show that

$$\nu(T, E) = \nu(T', E). \tag{11.12}$$

We may assume that E is not a prime divisor on X, as otherwise both sides vanish.

Choose a sequence of blow-ups with smooth *connected* centers

$$Y \coloneqq X_k \to X_{k-1} \to \cdots \to X_0 \coloneqq X$$

so that E is a prime divisor on Y, exceptional with respect to $X_k \to X_{k-1}$. Denote the composition by $\pi\colon Y \to X$. Thanks to Proposition 11.1.1,

$$\mathrm{H}^{1,1}(X_k, \mathbb{R}) = \mathrm{H}^{1,1}(X_{k-1}, \mathbb{R}) \oplus \mathbb{R}\{E_k\},$$

where $E_k = E$ is the exceptional divisor of $X_k \to X_{k-1}$.

By induction,

$$\mathrm{H}^{1,1}(Y, \mathbb{R}) = \mathrm{H}^{1,1}(X, \mathbb{R}) \oplus \bigoplus_{i=1}^{k} \mathbb{R}\{E_i\},$$

where E_i is the exceptional divisor of $X_i \to X_{i-1}$. Now by Lemma 7.3.2,

$$\operatorname{Reg} \pi^* T = \pi^* T - \sum_{i=1}^{k} \nu(T, E_i)[E_i]. \tag{11.13}$$

In particular, the cohomology class of $\operatorname{Reg} \pi^* T$ determines $\nu(T, E)$. Hence, (11.12) follows. □

Corollary 11.1.2 *The set of nef b-divisors over X with root α can be naturally identified with*

$$\varprojlim_{\omega} \left(\mathcal{G}(\alpha + \omega)/\sim_{\mathcal{I}} \right),$$

where ω runs over the directed set of Kähler forms on X (with respect to the preorder of reverse domination), and given two Kähler forms $\omega \leq \omega'$ the transition map

$$\mathcal{G}(\alpha + \omega)/\sim_{\mathcal{I}} \to \mathcal{G}(\alpha + \omega')/\sim_{\mathcal{I}}$$

is induced by the map $\mathcal{G}(\alpha + \omega) \to \mathcal{G}(\alpha + \omega')$ sending T to $T + \omega' - \omega$.

It is tempting to extend these results to more general currents, not necessarily non-divisorial ones. For this purpose, we introduce the following definition:

Definition 11.1.6 An *augmented nef b-divisor* over X is a pair $(\mathbb{D}, D)$, where

(1) $\mathbb{D}$ is a nef b-divisor over X;
(2) D is a formal sum

$$D = \sum_E c_E E, \quad c_E \in \mathbb{R}_{\geq 0},$$

where E runs over the set of prime divisors on X,

such that the following condition is satisfied:

$$\sum_E c_E \{E\}$$

is convergent as a sum in $\mathrm{H}^{1,1}(X, \mathbb{R})$.

The *cohomology class* of $(\mathbb{D}, D)$ is defined as

$$\mathbb{D}_X + \sum_E c_E \{E\} \in \mathrm{H}^{1,1}(X, \mathbb{R}).$$

The *volume* of $(\mathbb{D}, D)$ is defined as

$$\operatorname{vol}(\mathbb{D}, D) = \operatorname{vol} \mathbb{D}.$$

Recall that we defined $\mathcal{Z}_+(X, \alpha)$ as the set of closed positive $(1,1)$-currents $T \in \alpha$ in Definition 1.7.3. We introduce a further notation here:

$$\mathcal{Z}_+(X, \alpha)_{>0} := \{ T \in \mathcal{Z}_+(X, \alpha) : \operatorname{vol} T > 0 \}.$$

Corollary 11.1.3 *There is a canonical bijection between the following sets:*

(1) *The set* $\mathcal{Z}_+(X,\alpha)_{>0}/\sim_{\mathcal{I}}$*;*
(2) *the set of augmented nef b-divisors over* X *with positive volume with cohomology class* α.

More precisely, given a current $T \in \mathcal{Z}_+(X,\alpha)_{>0}$, we associate $(\mathbb{D}, D)$ as follows:

$$\mathbb{D} = \mathbb{D}(T), \quad D = \sum_E \nu(T,E)E.$$

Proof This is a direct consequence of Theorem 11.1.3 and Siu's decomposition Lemma 1.7.1. □

In fact, Corollary 11.1.3 can also be reformulated in an elementary manner, without referring to b-divisors at all. Suppose that we are given a big cohomology class α on X and a non-negative real number c_E for each prime divisor E over X. A natural question is: When is there a closed positive $(1,1)$-current $T \in \alpha$ with positive volume such that $\nu(T,E) = c_E$ for every E? Then Corollary 11.1.3 says that T exists if and only if the following two conditions hold:

(1)
$$\lim_{\pi\colon Y\to X} \operatorname{vol}\left(\pi^*\alpha - \sum_{E\subseteq Y} c_E\{E\}\right) > 0[1],$$
where π runs over all modifications of X;
(2) for each modification $\pi\colon Y \to X$, the class $\pi^*\alpha - \sum_{E\subseteq Y} c_E\{E\}$ is modified nef.

Similarly, we have the following generalization of Corollary 11.1.2.

Corollary 11.1.4 *There is a canonical bijection between the following two sets:*

(1) *The set of*
$$\varprojlim_{\omega} \left(\mathcal{Z}_+(X,\alpha+\omega)_{>0}/\sim_{\mathcal{I}}\right),$$
where ω *runs over the directed set of Kähler forms on* X*;*
(2) *the set of augmented nef b-divisors over* X *with cohomology class* α.

Corollary 11.1.5 *Let* $\mathbb{D}$ *be a Cartier b-divisor over* X*. Then* $\mathbb{D}$ *is nef in the sense of Definition 11.1.4 if and only if it is nef in the sense of Definition 11.1.5.*

Proof We only handle the nontrivial implication. Assume that $\mathbb{D}$ is nef in the sense of Definition 11.1.5. We want to show that $\mathbb{D}$ is nef in the sense of Definition 11.1.4. After replacing X by a realization of $\mathbb{D}$, we may assume that $\mathbb{D}$ is realized by (X,α) for some cohomology class $\alpha \in \mathrm{H}^{1,1}(X,\mathbb{R})$.

We first assume that $\mathbb{D}$ is big. Choose a non-divisorial closed positive $(1,1)$-current T on X such that $\mathbb{D} = \mathbb{D}(T)$. Now $\mathbb{D} = \mathbb{D}(T)$ means that for each modification

[1] In particular, implicitly, the sum $\sum_{E\subseteq Y} c_E\{E\}$ converges in $\mathrm{H}^{1,1}(Y,\mathbb{R})$.

$\pi\colon Y \to X$, the current π^*T is non-divisorial. In particular, T has vanishing generic Lelong number along each prime divisor over X, see (11.13). Equivalently, T has vanishing Lelong number everywhere. By Demailly's regularization theorem, a pseudo-effective class carrying a positive current with vanishing Lelong number at every point is nef. Hence $\alpha = \{T\}$ is nef.

In the general case, fix a Kähler form ω on the realization of $\mathbb{D}$. For every $\epsilon > 0$, the Cartier b-divisor $\mathbb{D} + \epsilon\mathbb{D}(\omega)$ is big and nef in the sense of Definition 11.1.5, hence is nef in the sense of Definition 11.1.4 by the big case. Thus $\alpha + \epsilon\{\omega\}$ is nef for all $\epsilon > 0$, and the closedness of the nef cone shows that α is nef. □

Corollary 11.1.6 *Let T and T' be non-divisorial closed positive $(1,1)$-currents on X. Suppose that $\{T\} = \{T'\}$. Then the following are equivalent:*

(1) $\mathbb{D}(T) \leq \mathbb{D}(T')$*;*
(2) $T \preceq_{\mathcal{I}} T'$*.*

Proof Let E be a prime divisor over X. Choose a sequence of blow-ups with smooth connected centers as in the proof of Theorem 11.1.3, so that E appears as one of the exceptional prime divisors on the resulting model $\pi\colon Y \to X$. By (11.13), the difference

$$\mathbb{D}(T')_Y - \mathbb{D}(T)_Y$$

is represented in the relative cohomology by

$$\sum_i \left(\nu(T, E_i) - \nu(T', E_i)\right)\{E_i\}.$$

Using Proposition 11.1.1 inductively, or equivalently the negativity lemma for exceptional classes, pseudo-effectivity of this exceptional class is equivalent to the non-negativity of all coefficients. Hence $\mathbb{D}(T) \leq \mathbb{D}(T')$ is equivalent to

$$\nu(T, E) \geq \nu(T', E)$$

for every prime divisor E over X, which is exactly $T \preceq_{\mathcal{I}} T'$ by Proposition 6.1.2. □

Corollary 11.1.7 *Let $\mathbb{D}$ be a nef b-divisor over X. Then there is a decreasing sequence of nef and big Cartier b-divisors $\mathbb{D}_i$ over X with limit $\mathbb{D}$.*

Proof Take a Kähler form ω on X. By Theorem 11.1.2, for each $i > 0$, we can find a non-divisorial Kähler current $T_i \in \mathbb{D}_X + i^{-1}\{\omega\}$ such that

$$\mathbb{D}(T_i) = \mathbb{D} + i^{-1}\mathbb{D}(\omega).$$

We observe that T_i and T_{i+1} have the same $\mathcal{I}$-singularity type. Indeed, applying Corollary 11.1.6 to T_i and $T_{i+1} + (i^{-1} - (i+1)^{-1})\omega$ shows that these two currents are $\mathcal{I}$-equivalent; adding the smooth form does not change the singularity type. Let $(T_i^j)_j$ be quasi-equisingular approximations of T_i such that

(1) T_i^j is a Kähler current in $\mathbb{D}_X + i^{-1}\{\omega\}$ for $j \geq j_0(i)$, and

(2) for each fixed j, the singularity type of T_i^j is independent of i.

Note that (2) is possible by using the Bergman kernel construction of the quasi-equisingular approximations, since the T_i's have the same multiplier ideals.

For fixed j, the b-divisor

$$\mathbb{B}_j \coloneqq \mathbb{D}(T_i^j) - i^{-1}\mathbb{D}(\omega)$$

is independent of i. Since the quasi-equisingular approximations are decreasing in j, the sequence $(\mathbb{B}_j)_j$ is decreasing, and Proposition 11.1.2 gives $\mathbb{B}_j \to \mathbb{D}$. It suffices to take $\mathbb{D}_i = \mathbb{B}_{j_i} + i^{-1}\mathbb{D}(\omega) = \mathbb{D}(T_i^{j_i})$, where j_i is chosen to be a sufficiently fast increasing sequence of positive integers with $j_i \geq j_0(i)$. □

11.2 Properties of the intersection product

Let X be a connected compact Kähler manifold of dimension n.

Definition 11.2.1 Let $\mathbb{D}_1, \ldots, \mathbb{D}_n$ be big and nef b-divisors over X. Then we define their intersection as

$$(\mathbb{D}_1, \ldots, \mathbb{D}_n) \coloneqq \operatorname{vol}(T_1, \ldots, T_n),$$

where $T_1, \ldots, T_n$ are closed positive $(1,1)$-currents in $\mathbb{D}_{1,X}, \ldots, \mathbb{D}_{n,X}$ respectively such that $\mathbb{D}(T_i) = \mathbb{D}_i$.

In general, if the $\mathbb{D}_i$'s are only nef, we define

$$(\mathbb{D}_1, \ldots, \mathbb{D}_n) \coloneqq \lim_{\epsilon \to 0+} (\mathbb{D}_1 + \epsilon\mathbb{D}(\omega), \ldots, \mathbb{D}_n + \epsilon\mathbb{D}(\omega)),$$

where ω is a Kähler form on X.

The definition makes sense thanks to Theorem 11.1.2. It does not depend on the choices of $T_1, \ldots, T_n$ since they are uniquely defined up to $\mathcal{I}$-equivalence, as proved in Theorem 11.1.3.

When $\mathbb{D}_1, \ldots, \mathbb{D}_n$ are big and nef, the two definitions coincide as follows from Lemma 11.2.1 below.

We first note that even when the T_i's have vanishing volumes, the two intersection products still agree.

Proposition 11.2.1 *Let $T_1, \ldots, T_n$ be closed positive $(1,1)$-currents on X. Then*

$$(\mathbb{D}(T_1), \ldots, \mathbb{D}(T_n)) = \operatorname{vol}(T_1, \ldots, T_n).$$

This is a trivial consequence of the definitions.

Proposition 11.2.2

(1) *The product in Definition 11.2.1 is symmetric in its n-variable.*

(2) *Let* $\mathbb{D}_1, \ldots, \mathbb{D}_n, \mathbb{D}'_1$ *be nef b-divisors over* X. *Then*

$$\left(\mathbb{D}_1 + \mathbb{D}'_1, \ldots, \mathbb{D}_n\right) = \left(\mathbb{D}_1, \ldots, \mathbb{D}_n\right) + \left(\mathbb{D}'_1, \ldots, \mathbb{D}_n\right).$$

(3) *Let* $\mathbb{D}_1, \ldots, \mathbb{D}_n$ *be nef b-divisors over* X *and* $\lambda \geq 0$. *Then*

$$\left(\lambda\mathbb{D}_1, \ldots, \mathbb{D}_n\right) = \lambda\left(\mathbb{D}_1, \ldots, \mathbb{D}_n\right).$$

Proof This follows immediately from Proposition 7.3.1. □

Proposition 11.2.3 *The product in Definition 11.2.1 is monotonically increasing in each variable.*

Proof Let $\mathbb{D}_1, \ldots, \mathbb{D}_n$ and $\mathbb{D}'$ be nef b-divisors over X so that $\mathbb{D}_1 \leq \mathbb{D}'$. We want to show that

$$\left(\mathbb{D}_1, \ldots, \mathbb{D}_n\right) \leq \left(\mathbb{D}', \mathbb{D}_2, \ldots, \mathbb{D}_n\right).$$

We can easily reduce to the case where $\mathbb{D}_1, \ldots, \mathbb{D}_n$ and $\mathbb{D}'$ are all big. In this case, take $\mathcal{I}$-good non-divisorial closed positive $(1,1)$-currents $T_1, \ldots, T_n$ and T' so that $\mathbb{D}(T_i) = \mathbb{D}_i$ for all $i = 1, \ldots, n$ and $\mathbb{D}(T') = \mathbb{D}'$. Furthermore, we may assume that the T_i's and T' are Kähler currents by the perturbation argument.

Let $(T_i^j)_j$ be a quasi-equisingular approximation of T_i for $i = 2, \ldots, n$. It follows from Theorem 6.2.1 that

$$\int_X T_1 \wedge \cdots \wedge T_n = \lim_{j\to\infty} \int_X T_1 \wedge T_2^j \wedge \cdots \wedge T_n^j.$$

It suffices to show that for all $j \geq 1$,

$$\int_X T_1 \wedge T_2^j \wedge \cdots \wedge T_n^j \leq \int_X T' \wedge T_2^j \wedge \cdots \wedge T_n^j.$$

Therefore, we have reduced to the case where $T_2, \ldots, T_n$ have analytic singularities. After a resolution, we may assume that they have log singularities along $\mathbb{Q}$-divisors. By Proposition 7.3.1(5), we can further reduce to the case where $T_2, \ldots, T_n$ have bounded local potentials. Perturbing $T_2, \ldots, T_n$ by a Kähler form, we may further assume that $\{T_2\}, \ldots, \{T_n\}$ are Kähler classes. By Proposition 7.3.1(4), we finally reduce to the case where $T_2, \ldots, T_n$ are Kähler forms. In this case, the assertion follows from the positivity of $T' - T_1$. □

Lemma 11.2.1 *Let* ω *be a Kähler form on* X. *Fix a compact set* $K \subseteq \mathrm{H}^{1,1}(X, \mathbb{R})$. *Let* $\mathbb{D}_1, \ldots, \mathbb{D}_n$ *be nef b-divisors over* X *such that* $\mathbb{D}_{i,X} \in K$ *for each* $i = 1, \ldots, n$. *Then there is a constant* C *depending only on* $X, K, \{\omega\}$ *such that for any* $\epsilon \in [0,1]$, *we have*

$$0 \leq \left(\mathbb{D}_1 + \epsilon\mathbb{D}(\omega), \ldots, \mathbb{D}_n + \epsilon\mathbb{D}(\omega)\right) - \left(\mathbb{D}_1, \ldots, \mathbb{D}_n\right) \leq C\epsilon.$$

Proof By Proposition 11.2.3, the left-hand side is non-negative. For the upper bound, expand the first term multilinearly using Proposition 11.2.2. Every non-zero error term contains at least one factor $\epsilon\mathbb{D}(\omega)$, and the remaining factors have roots in the compact set $K + \{\omega\}$. The corresponding intersection numbers are uniformly bounded by monotonicity and the cohomological mixed intersections with a fixed large Kähler class depending only on K and $\{\omega\}$. Summing these finitely many terms gives the required estimate. □

We first make a consistency check.

Proposition 11.2.4 *Suppose that $\mathbb{D}$ is a nef b-divisor over X. Then*

$$(\mathbb{D}, \ldots, \mathbb{D}) = \operatorname{vol} \mathbb{D}.$$

Proof Using Lemma 11.2.1 and Lemma 11.1.1, we may easily reduce to the case where $\mathbb{D}$ is nef and big. In this case, take a non-divisorial closed positive $(1,1)$-current T in $\mathbb{D}_X$ such that $\mathbb{D}(T) = \mathbb{D}$. Then we need to show that

$$\operatorname{vol} \mathbb{D} = \operatorname{vol} T,$$

which is proved in Theorem 11.1.1. □

Proposition 11.2.5 *Let $\mathbb{D}_1, \ldots, \mathbb{D}_n$ be nef b-divisors over X. Then*

$$(\mathbb{D}_1, \ldots, \mathbb{D}_n) \geq \prod_{i=1}^{n} (\operatorname{vol} \mathbb{D}_i)^{1/n}.$$

Proof We may assume that $\operatorname{vol} \mathbb{D}_i > 0$ for each $i = 1, \ldots, n$ since there is nothing to prove otherwise. In this case, our assertion follows from Proposition 7.3.2. □

Proposition 11.2.6 *The product in Definition 11.2.1 is upper semicontinuous in the following sense. Suppose that $(\mathbb{D}_i^j)_{j \in J}$ are nets of nef b-divisors over X with limits $\mathbb{D}_i$ for each $i = 1, \ldots, n$. Then*

$$\varlimsup_{j \in J} \left(\mathbb{D}_1^j, \ldots, \mathbb{D}_n^j \right) \leq (\mathbb{D}_1, \ldots, \mathbb{D}_n).$$

Proof **Step 1**. We first assume that the $\mathbb{D}_i^j$'s and the $\mathbb{D}_i$'s are all big.

Take $\mathcal{I}$-good non-divisorial closed positive $(1,1)$-currents T_i^j and T_i so that $\mathbb{D}(T_i^j) = \mathbb{D}_i^j$ and $\mathbb{D}(T_i) = \mathbb{D}_i$. Note that by our assumption and the proof of Theorem 11.1.3, for any prime divisor E over X, we have

$$\lim_{j \in J} \nu(T_i^j, E) = \nu(T_i, E).$$

So our assertion follows from Theorem 7.3.2.

Step 2. Next we handle the general case.

Take a Kähler form ω on X. Then by Lemma 11.2.1, for any $\epsilon \in (0, 1]$, we have

$$\begin{aligned}
\overline{\lim_{j\in J}}\left(\mathbb{D}_1^j,\dots,\mathbb{D}_n^j\right) &\leq \overline{\lim_{j\in J}}\left(\mathbb{D}_1^j+\epsilon\mathbb{D}(\omega),\dots,\mathbb{D}_n^j+\epsilon\mathbb{D}(\omega)\right)\\
&\leq (\mathbb{D}_1+\epsilon\mathbb{D}(\omega),\dots,\mathbb{D}_n+\epsilon\mathbb{D}(\omega))\\
&\leq (\mathbb{D}_1,\dots,\mathbb{D}_n)+C\epsilon.
\end{aligned}$$

Since ϵ is arbitrary, our assertion follows. □

Proposition 11.2.7 *The product in Definition 11.2.1 is continuous along decreasing nets in each variable. In other words, if $(\mathbb{D}_i^j)_{j\in J}$ ($i=1,\dots,n$) are decreasing nets of nef b-divisors over X with limits $\mathbb{D}_i$. Then*

$$\lim_{j\in J}\left(\mathbb{D}_1^j,\dots,\mathbb{D}_n^j\right)=(\mathbb{D}_1,\dots,\mathbb{D}_n)\,.$$

Proof This is a straightforward consequence of Proposition 11.2.3 and Proposition 11.2.6. □

Remark 11.2.1 The algebraic nef b-divisors are introduced by Dang–Favre [DF22], as their intersection theory. It is a straightforward application of the results proved in this section that our transcendental intersection theory coincides with theirs in the algebraic setting.

Part III
Applications

The three chapters of this part each apply the singularity theory of Part II to a substantial question in pluripotential theory. They are logically independent of each other; each draws on a different subset of the machinery developed earlier. We summarize the dependencies here so that a reader interested in only one application can navigate to it directly.

Chapter 12 develops pluripotential theory on a *big* toric line bundle, generalizing the classical ample-toric dictionary of Chapter 5. The construction relies in an essential way on the partial Okounkov body machinery of Chapter 10 and on the trace operator of Chapter 8.

Chapter 13 develops a non-Archimedean pluripotential theory on a compact Kähler manifold X, parallel to and compatible with the algebraic theory of Boucksom–Jonsson. Non-Archimedean potentials are defined as inverse systems of $\mathcal{I}$-model test curves (Chapter 9 and Section 9.3), with the operations from Section 9.4 providing the usual algebraic structure (sums, maxima, scalings, etc.). The chapter also constructs the Duistermaat–Heckman measure of a non-Archimedean metric and culminates in the comparison theorem Theorem 13.4.3, which identifies our space of non-Archimedean potentials with the Boucksom–Jonsson space in the projective setting.

Chapter 14 proves the convergence of *partial Bergman kernels*: for an arbitrary θ-psh weight, a closed non-pluripolar weighted subset $(K, v) \subseteq X$, and a Bernstein–Markov measure on K, the normalized partial Bergman measure converges weakly to the partial equilibrium measure (Theorem 14.3.1). This generalizes the celebrated theorem of Berman–Boucksom–Witt Nyström [BBWN11] to arbitrary singular weights and to relative weighted subsets; the asymptotic Riemann–Roch formula of Chapter 7 is the technical engine driving the proof.

The three chapters draw on different external prerequisites. Chapter 12 requires basic toric geometry; [CLS11] and [Ful93] are excellent references. Chapter 13 is best read in tandem with the Boucksom–Jonsson theory; the article [BJ22a] is the canonical source. Chapter 14 relies on familiarity with [BBWN11]. None of these prerequisites is logically required, but each will substantially ease the reading.

Chapter 12
Toric pluripotential theory on big line bundles

C'est l'harmonie des diverses parties, leur symétrie, leur heureux balancement; c'est en un mot tout ce qui y met de l'ordre, tout ce qui leur donne de l'unité, ce qui nous permet par conséquent d'y voir clair et d'en comprendre l'ensemble en même temps que les détails.
— *Henri Poincaré*[a], L'avenir des mathématiques

[a] Henri Poincaré (1854–1912) was a French mathematician, physicist, and philosopher of science. He is considered one of the greatest mathematicians of all time and a pioneer of several modern mathematical fields. He also played a key role in the development of special relativity, and was one of the first to understand the deep connection between mathematics and physics.

In this chapter, we develop the toric pluripotential theory on a *big* toric line bundle $L = \mathcal{O}_X(D)$ on a smooth projective toric variety X. The ample case was treated in Chapter 5; the principal new feature in the big setting is that the polytope $P_D \subseteq M_{\mathbb{R}}$ associated with D is in general only rational (not a lattice polytope) and the faces of P_D no longer correspond to toric-invariant subvarieties of X. This rules out the direct use of the ample-case dictionary and forces the use of the partial Okounkov body machinery from Chapter 10.

The development proceeds in three stages. In the first stage (Section 12.1), we set up the big-toric framework: the polytope P_D, the Cartier data $\{m_\sigma\}_{\sigma\in\Sigma}$, the augmented base locus, and the base-locus computation Proposition 12.1.1 that drives the partial Okounkov body construction in this setting (see also Corollary 12.1.1).

The second stage (Section 12.2) computes the partial Okounkov body of a toric psh metric on a big toric line bundle. After recalling the relevant Newton-body construction (Proposition 12.2.1) and comparing it with the classical algebraic Okounkov body (Proposition 12.2.2), we prove the central computation Theorem 12.2.1: for a toric θ-psh potential φ with θ_φ a Kähler current, the partial Okounkov body $\Delta(L, \varphi)$ is, modulo a unimodular transformation, the Newton body of φ. This gives the partial Okounkov body machinery a fully explicit description in the toric setting.

The pluripotential-theoretic consequences are then derived. Two non-trivial new applications are emphasized:

- Corollary 12.2.2, an explicit formula for the Lelong numbers of a toric psh metric on a big line bundle along a toric-invariant prime divisor in terms of the Newton body. This is new even in the ample regime;
- Theorem 12.2.2, an explicit control of the potential with minimal singularities on L.

The third stage (Section 12.3) develops the full pluripotential theory in the big toric setting. The toric–convex dictionary (Theorem 12.3.1 and Corollary 12.3.3) extends the ample-case identification of toric θ-psh potentials with proper convex functions on $N_{\mathbb{R}}$. The basic operations — pointwise sums (Proposition 12.3.1), rooftop (Proposition 12.3.2), suprema and decreasing limits (Proposition 12.3.3) — all admit clean toric expressions, and the P-equivalence classes are characterized in Corollary 12.3.2. We then describe the toric subgeodesics and geodesics (Proposition 12.3.5 and Corollaries 12.3.4 to 12.3.6) and the resulting toric Ross–Witt Nyström correspondence (Corollary 12.3.7), the toric trace operator (Theorem 12.3.2) with its volume identity (Corollary 12.3.8), and the toric mixed-volume formula (Theorem 12.3.3). The augmented base locus formula (Corollary 12.3.9) brings the chapter back to the algebraic geometry of D.

Throughout, we use the toric setup of Chapter 5, and freely apply the partial Okounkov body machinery of Chapter 10 as well as the trace operator of Chapter 8.

12.1 Toric setup

Let T be a complex torus of dimension n with character lattice M and cocharacter lattice N. Some basic terminology is recalled in Section 5.1. Recall that T_c is the compact torus contained in $T(\mathbb{C})$.

Consider a rational polyhedral fan Σ in $N_{\mathbb{R}}$ corresponding to an n-dimensional smooth projective toric variety X.

Let

$$D = \sum_{\rho \in \Sigma(1)} a_\rho D_\rho$$

be a T-invariant big divisor on X. Let $P_D \subseteq M_{\mathbb{R}}$ be the following polytope[1]

$$P_D = \left\{ m \in M_{\mathbb{R}} : \langle m, u_\rho \rangle \geq -a_\rho \quad \forall \rho \in \Sigma(1) \right\}. \tag{12.1}$$

Since we have assumed that D is big, P_D is n-dimensional.

Let $L = \mathcal{O}_X(D)$. Note that replacing D by a linearly equivalent divisor amounts to replacing P_D by an integral translation.

Recall that for each $\rho \in \Sigma(1)$, u_ρ denotes the ray generator of ρ. Let $\{m_\sigma\}_{\sigma \in \Sigma}$ denote the *Cartier data* associated with D. In other words, for each $\sigma \in \Sigma$, $m_\sigma \in M$ satisfies that

$$\langle m_\sigma, u_\rho \rangle = -a_\rho, \quad \forall \rho \in \sigma(1).$$

The element $m_\sigma \in M$ is well-defined modulo

$$M(\sigma) \coloneqq \sigma^\perp \cap M, \tag{12.2}$$

where

$$\sigma^\perp \coloneqq \{ m \in M_{\mathbb{R}} : \langle m, u \rangle = 0 \quad \forall u \in \sigma \}.$$

[1] Note that P_D is not necessarily a lattice polytope, see [CLS11, Example 10.5.4]. In fact, any rational polytope with positive volume can be realized in this manner.

Moreover, if τ is a face of σ, then

$$m_\sigma \equiv m_\tau \mod M(\tau). \tag{12.3}$$

See [CLS11, Theorem 4.2.8]. In particular, for an n-dimensional $\sigma \in \Sigma$, the element m_σ is uniquely determined. In general, for $\sigma \in \Sigma(n)$, one need not have $m_\sigma \in P_D$. In fact, $m_\sigma \in P_D$ for all $\sigma \in \Sigma(n)$ if and only if D is base-point free. See [CLS11, Theorem 6.1.7].

Note that for any n-dimensional cone σ in Σ and any $\rho \in \sigma(1)$, we have

$$\langle m - m_\sigma, u_\rho \rangle \geq 0, \quad \forall m \in P_D, \tag{12.4}$$

as a consequence of (12.1) and the choice of m_σ.

Recall that

$$D|_{U_\sigma} = \operatorname{div}(\chi^{-m_\sigma})|_{U_\sigma} \tag{12.5}$$

for all $\sigma \in \Sigma$, where U_σ is the affine subvariety of X corresponding to σ. See [CLS11, Proposition 4.1.2].

Next consider a T-invariant irreducible subvariety $Y \subseteq X$. Since X is smooth, so is Y. Let σ be the cone in Σ corresponding to Y. We observe that σ corresponds to a face Q_σ of P_D:

$$Q_\sigma = \left\{ m \in P_D : \langle m, u_\rho \rangle = -a_\rho \quad \forall \rho \in \sigma(1) \right\}. \tag{12.6}$$

The dimension of σ is not necessarily equal to the codimension of Q as we will see in Example 12.1.2.

We have the following characterization of the base locus of D. We are not aware of a direct reference; the result is likely folklore.

Proposition 12.1.1 *The base locus* $\mathrm{Bs}(D)$ *(with the reduced complex structure) of* D *is a toric-invariant (possibly reducible) subvariety given by the union of* $V(\tau)$*, where* τ *runs over elements in* Σ *satisfying the following condition:*

$$a_\rho + \langle m, u_\rho \rangle > 0 \quad \text{for some } \rho \in \tau(1) \tag{12.7}$$

for each $m \in M \cap P_D$.

Here $V(\tau)$ denotes the toric subvariety of X corresponding to τ. In other words,

$$V(\tau) = \bigcap_{\rho \in \tau(1)} D_\rho.$$

Proof Recall that

$$\mathrm{H}^0(X, L) \cong \bigoplus_{m \in M \cap P_D} \mathbb{C}\chi^m.$$

See [CLS11, Proposition 4.3.3] for example. So we only need to understand the common zeros of $D + \operatorname{div} \chi^m$ for all $m \in M \cap P_D$. But we know that

$$D + \operatorname{div} \chi^m = \sum_{\rho \in \Sigma(1)} \left(a_\rho + \langle m, u_\rho \rangle\right) D_\rho.$$

Our assertion follows. □

Corollary 12.1.1 *The stable base locus (with the reduced complex structure) of D is a toric-invariant (possibly reducible) subvariety given by the union of the $V(\tau)$'s, where τ runs over elements in Σ satisfying (12.7) for each $m \in P_D$.*

Geometrically, the condition means $Q_\tau = \varnothing$.

Proof It follows from Proposition 12.1.1 that the stable base locus of D is given by the union of the $V(\tau)$'s, where τ runs over elements of Σ satisfying (12.7) for all $m \in M_{\mathbb{Q}} \cap P_D$. That is,

$$M_{\mathbb{Q}} \cap \left\{ m \in P_D : a_\rho + \langle m, u_\rho \rangle = 0 \quad \forall \rho \in \tau(1) \right\} = \varnothing.$$

Since the latter part is a rational polytope, this statement is equivalent to

$$\left\{ m \in P_D : a_\rho + \langle m, u_\rho \rangle = 0 \quad \forall \rho \in \tau(1) \right\} = \varnothing. \tag{12.8}$$

Our first assertion follows. □

We will keep two examples in mind.

Example 12.1.1 In this case, Σ is the fan in Fig. 12.1 consisting of three 2-dimensional cones σ_1, σ_2 and σ_3; three 1-dimensional cones σ_4, σ_5 and σ_6; one 0-dimensional cone σ_0.

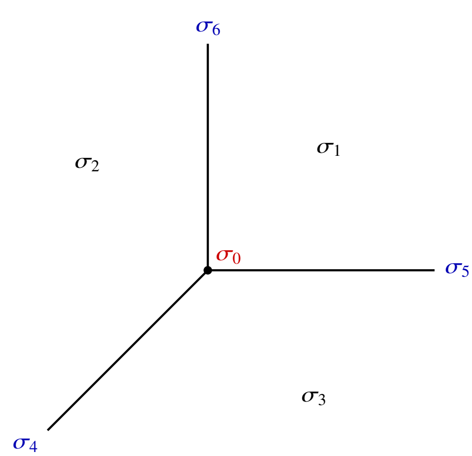

Fig. 12.1 The fan Σ of $\mathbb{P}^2$. The three rays $\sigma_4, \sigma_5, \sigma_6 \in \Sigma(1)$ subdivide $N_{\mathbb{R}}$ into three full-dimensional cones $\sigma_1, \sigma_2, \sigma_3 \in \Sigma(2)$; the origin is the unique cone of dimension zero.

The fan Σ is just the fan of $X = \mathbb{P}^2$. Under the orbit-cone correspondence, we have

$$
\begin{aligned}
D_{\sigma_1} =& \{[1:0:0]\}, \quad D_{\sigma_2} = \{[0:1:0]\}, \quad D_{\sigma_3} = \{[0:0:1]\},\\
D_{\sigma_4} =& \{[0:X_1:X_2] : X_1X_2 \neq 0\}, \quad D_{\sigma_5} = \{[X_0:0:X_2] : X_0X_2 \neq 0\},\\
D_{\sigma_6} =& \{[X_0:X_1:0] : X_0X_1 \neq 0\}, \quad D_{\sigma_0} = \mathbb{P}^2.
\end{aligned}
$$

In particular, $\Sigma(1) = \{\sigma_4, \sigma_5, \sigma_6\}$. We shall take

$$D = D_{\sigma_4}.$$

In other words,

$$a_{\sigma_5} = a_{\sigma_6} = 0, \quad a_{\sigma_4} = 1.$$

Note that the ray generators are given by

$$u_{\sigma_4} = (-1,-1), \quad u_{\sigma_5} = (1,0), \quad u_{\sigma_6} = (0,1).$$

It follows that

$$P_D = \{m = (m_1, m_2) \in \mathbb{R}^2 : m_1 + m_2 \leq 1, m_1 \geq 0, m_2 \geq 0\}.$$

Therefore, P_D is just the polytope in Fig. 12.2. In this case, the Cartier data for

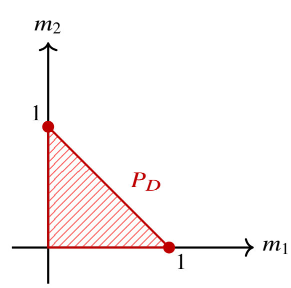

Fig. 12.2 The polytope P_D.

2-dimensional cones are given as follows:

$$m_{\sigma_1} = (0,0), \quad m_{\sigma_2} = (1,0), \quad m_{\sigma_3} = (0,1);$$

while the remaining Cartier data are determined by (12.3).

In this case, $L = \mathcal{O}_X(D) = \mathcal{O}_{\mathbb{P}^2}(1)$. Hence the line bundle L is ample.

We also observe that

$$
\begin{aligned}
Q_{\sigma_1} =& \{(0,0)\}, \quad Q_{\sigma_2} = \{(1,0)\}, \quad Q_{\sigma_3} = \{(0,1)\},\\
Q_{\sigma_4} =& \{(m_1,m_2) : m_1 \geq 0, m_2 \geq 0, m_1 + m_2 = 1\},\\
Q_{\sigma_5} =& \{0\} \times [0,1], \quad Q_{\sigma_6} = [0,1] \times \{0\},\\
Q_{\sigma_0} =& P_D.
\end{aligned}
$$

Next we give a non-ample example.

Example 12.1.2 Let Σ be the fan shown in Fig. 12.3. Comparing with our previous

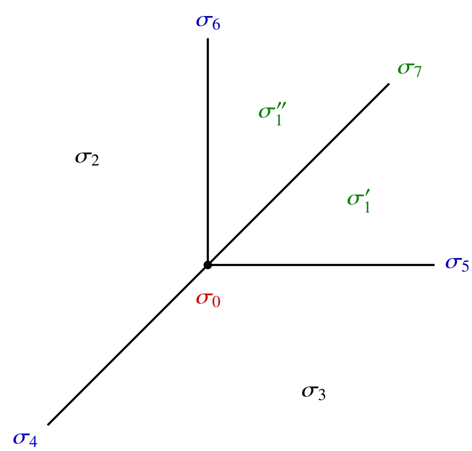

Fig. 12.3 The fan of $\mathrm{Bl}_0\mathbb{P}^2$. Adding the ray σ_7 generated by $(1, 1)$ subdivides the cone σ_1 into two full-dimensional cones σ_1', σ_1''.

example Fig. 12.1, we have divided σ_1 from the middle, giving rise to two additional 2-dimensional cones σ_1' and σ_1'', and one additional 1-dimensional cone σ_7.

The corresponding $X = \mathrm{Bl}_0\mathbb{P}^2$ is just the blow-up of $\mathbb{P}^2$ at the origin 0 and hence $L = \pi^*\mathcal{O}_{\mathbb{P}^2}(1)$. Let $\pi \colon X \longrightarrow \mathbb{P}^2$ denote the blow-up morphism. Let

$$D = D_{\sigma_4}.$$

Then D is the pull-back of the divisor D in Example 12.1.1. Note that D is not ample, since it has degree 0 on the exceptional divisor.

In this case, we have

$$\Sigma(1) = \{\sigma_4, \sigma_5, \sigma_6, \sigma_7\},$$

and D_{σ_7} is just the exceptional divisor.

The corresponding ray generators are

$$u_{\sigma_4} = (-1, -1), \quad u_{\sigma_5} = (1, 0), \quad u_{\sigma_6} = (0, 1), \quad u_{\sigma_7} = (1, 1),$$

while

$$m_{\sigma_1'} = m_{\sigma_1''} = (0, 0), \quad m_{\sigma_2} = (1, 0), \quad m_{\sigma_3} = (0, 1).$$

Therefore, P_D is the same as in Fig. 12.2.

We also observe that

$$
\begin{aligned}
Q_{\sigma_1'} =&\{(0,0)\}, \quad Q_{\sigma_1''} = \{(0,0)\}, \quad Q_{\sigma_7} = \{(0,0)\} \\
Q_{\sigma_2} =&\{(1,0)\}, \quad Q_{\sigma_3} = \{(0,1)\}, \\
Q_{\sigma_4} =&\{(m_1,m_2) : m_1 \geq 0, m_2 \geq 0, m_1+m_2 = 1\}, \\
Q_{\sigma_5} =&\{0\} \times [0,1], \quad Q_{\sigma_6} = [0,1] \times \{0\}, \\
Q_{\sigma_0} =&P_D.
\end{aligned}
$$

12.2 Toric partial Okounkov bodies

We continue to use the notations in Section 12.1.

In order to study the toric-invariant singular plurisubharmonic metrics on L, we need to fix a reference toric-invariant smooth Hermitian metric, so that the psh metrics can be identified with quasi-psh functions. Unlike the ample case studied in Chapter 5, in the case of big line bundles, there does not seem to be a natural choice of a smooth Hermitian metric on L similar to Guillemin's metric.

We shall fix a T_c-invariant Hermitian metric h on L so that $\theta = c_1(L,h)$.

The first observation is the following:

Lemma 12.2.1 *There is a smooth function $F_\theta \colon N_{\mathbb{R}} \to \mathbb{R}$ such that*

$$\theta = \mathrm{dd^c}\, \mathrm{Trop}^* F_\theta \quad \text{on } T(\mathbb{C}).$$

Proof **Step 1**. We first prove the existence of F_θ for a specific choice of h.

Write D as the difference of two toric-invariant ample divisors, say $D_1 - D_2$. Let h_1 and h_2 be the associated Guillemin's metrics on D_1 and D_2 respectively. We define $h = h_1 \otimes h_2^{-1}$.

In this case, our assertion follows since it holds in the case of Guillemin's metrics.

Step 2. In general, fix h_0 as in Step 1. Then the general h can be written as $h_0 \exp(-g)$ for some smooth function g on X. Since both metrics are T_c-invariant, so is g, and on the open torus we may write $g = \mathrm{Trop}^* r$ for some smooth function r on $N_{\mathbb{R}}$. Hence it suffices to modify F_θ in Step 1 by r to conclude. □

Note that F_θ is well-defined up to a linear term.

Next, we make an additional requirement on F_θ to fix the linear term. Let s_D be a rational section of L corresponding to D. Then s_D is well-defined up to a non-zero multiple. By Lelong–Poincaré formula Proposition 1.8.1, we have

$$\mathrm{dd^c}\left(\mathrm{Trop}^* F_\theta + \log |s_D|_h^2\right) = 0$$

on $T(\mathbb{C})$. Therefore, $\mathrm{Trop}^* F_\theta + \log |s_D|_h^2$ is the tropicalization of a linear function. Hence, after adding a linear function to F_θ, we can guarantee that

$$\mathrm{Trop}^* F_\theta + \log |s_D|_h^2 = 0 \tag{12.9}$$

from now on. Note that a different choice of s_D means adding a constant to F_θ.

Summarizing the situation, we have chosen the following data in addition to the fan Σ so far: A divisor D, a Hermitian metric h on L and a rational section s_D of L subject to various conditions.

12.2.1 Newton bodies

Let $\mathrm{PSH}_{\mathrm{tor}}(X,\theta)$ be the set of T_c-invariant functions in $\mathrm{PSH}(X,\theta)$.

Definition 12.2.1 A function $\varphi \in \mathrm{PSH}_{\mathrm{tor}}(X,\theta)$ can be written as

$$\varphi|_{T(\mathbb{C})} = \mathrm{Trop}^* f$$

for some unique function $f\colon N_{\mathbb{R}} \to [-\infty,\infty)$. Then we define $F_\varphi\colon N_{\mathbb{R}} \to \mathbb{R}$ as follows:

$$F_\varphi = F_\theta + f. \tag{12.10}$$

Observe that F_φ is a convex function and takes finite values by Lemma 5.2.1. In particular, f is also real-valued. Once D and h are fixed, F_φ is well-defined up to a constant since F_θ is.

Definition 12.2.2 Let $\varphi \in \mathrm{PSH}_{\mathrm{tor}}(X,\theta)$, we define its *Newton body* as

$$\Delta(\theta,\varphi) \coloneqq \overline{\nabla F_\varphi(N_{\mathbb{R}})} \subseteq M_{\mathbb{R}}.$$

Note that $\Delta(\theta,\varphi)$ is independent of the choice of s_D.

The Newton body $\Delta(\theta,\varphi)$ depends on the choice of D, not only on the associated line bundle L: A different choice of D inducing the same line bundle corresponds to a translation of $\Delta(\theta,\varphi)$. We will see in a while (Theorem 12.2.1) that once D is fixed $\Delta(\theta,\varphi)$ depends only on the current θ_φ. Hence, the choice of h is irrelevant.

For the moment, it is not clear if $\Delta(\theta,\varphi) \subseteq P_D$ or not. We shall prove this result using the theory of partial Okounkov bodies in the next section.

Proposition 12.2.1 *Let $\varphi \in \mathrm{PSH}_{\mathrm{tor}}(X,\theta)$. Then*

$$\mathrm{Trop}_* \left(\theta|_{T(\mathbb{C})} + \mathrm{dd^c}\varphi|_{T(\mathbb{C})}\right)^n = \mathrm{MA}_{\mathbb{R}}\left(F_\varphi\right). \tag{12.11}$$

In particular,

$$\int_X \theta_\varphi^n = n!\,\mathrm{vol}\,\Delta(\theta,\varphi) \tag{12.12}$$

Proof Let F_0 be a smooth convex function on $N_{\mathbb{R}}$ such that $\mathrm{dd^c}\,\mathrm{Trop}^* F_0$ can be extended to a Kähler form on X. For example, Guillemin's construction (5.5) with respect to a suitable Delzant polytope gives such an example.

Write ω for a Kähler form on X extending $\mathrm{dd^c}\,\mathrm{Trop}^* F_0$. Then for any large enough $C > 0$, $\theta + C\omega$ is a Kähler form. So we conclude from Proposition 5.2.5 that

$$\mathrm{Trop}_* \left((\theta + C\omega)|_{T(\mathbb{C})} + \mathrm{dd^c}\varphi|_{T(\mathbb{C})}\right)^n = \mathrm{MA}_{\mathbb{R}}(F_\varphi + CF_0).$$

Since both sides are polynomials in C, we conclude that the same holds for $C = 0$. Therefore, (12.11) follows.

(12.12) is a direct consequence of (12.11). □

12.2.2 Partial Okounkov bodies

There are some canonical choices of smooth flags in the toric setting.

Since X is smooth and projective, we can choose a full-dimensional cone σ in Σ with rays $\rho_1, \dots, \rho_n \in \sigma(1)$ such that $u_{\rho_1}, \dots, u_{\rho_n}$ form a basis of N. Define

$$Y_i = D_{\rho_1} \cap \cdots \cap D_{\rho_i}, \quad i = 1, \dots, n.$$

Then $Y_\bullet$ is a smooth flag on X. Let

$$\Phi\colon M \to \mathbb{Z}^n, \quad m \mapsto \left(\langle m - m_\sigma, u_{\rho_1}\rangle, \dots, \langle m - m_\sigma, u_{\rho_n}\rangle\right). \tag{12.13}$$

Then Φ is an isomorphism of lattices. It induces an $\mathbb{Z}$-affine isomorphism

$$\Phi_{\mathbb{R}}\colon M_{\mathbb{R}} \to \mathbb{R}^n.$$

Proposition 12.2.2 *We have*

$$k^{-1}\nu_{Y_\bullet}\left(\mathrm{H}^0(X, L^k)^\times\right) = \Phi_{\mathbb{R}}\left(P_D \cap k^{-1}M\right) \tag{12.14}$$

for any $k \in \mathbb{Z}_{>0}$. In particular,

$$\Delta_{Y_\bullet}(L) = \Phi_{\mathbb{R}}(P_D). \tag{12.15}$$

Recall that $\Delta_{Y_\bullet}(L) \subseteq \mathbb{R}^n$ is the Okounkov body defined in Definition 10.3.4.

Proof We first reduce to the case where $D|_{U_\sigma} = 0$. In fact, replacing D by $D + \mathrm{div}\,\chi^{m_\sigma}$ would result in changing P_D to $P_D - m_\sigma$. So in view of (12.5), we may assume that $D|_{U_\sigma} = 0$ and hence $m_\sigma = 0$.

Fix $k \in \mathbb{Z}_{>0}$. Let $s \in \mathrm{H}^0(X, L^k)$ be a non-zero toric eigen-section, say χ^m for some $m \in kP_D \cap M$. The zero-divisor of s on U_σ is given by

$$\sum_{i=1}^n \langle m, u_{\rho_i}\rangle D_{\rho_i},$$

see [CLS11, Proposition 4.1.2]. Therefore,

$$\nu_{Y_\bullet}(s) = \Big(\langle m, u_{\rho_1}\rangle, \dots, \langle m, u_{\rho_n}\rangle\Big) = \Phi(m).$$

Conversely, every section of L^k is a finite linear combination of such characters. In the coordinates on U_σ determined by $u_{\rho_1}, \dots, u_{\rho_n}$, the Okounkov valuation of a non-zero linear combination is the lexicographically first exponent among the characters with non-zero coefficient. Hence arbitrary sections give no valuations beyond the right-hand side of (12.14), and (12.14) follows. □

Example 12.2.1 Let us continue the example of $\mathbb{P}^2$ in Example 12.1.1. We use the same notations. Take σ_1 as our reference cone, and $\rho_1 = \sigma_5$, $\rho_2 = \sigma_6$. Then

$$Y_1 = \Big\{[X_0 : 0 : X_2] : X_0 X_2 \neq 0\Big\}, \quad Y_2 = \Big\{[X_0 : 0 : 0] : X_0 \neq 0\Big\}.$$

The map Φ is given by

$$\Phi(m_1, m_2) = (m_1, m_2).$$

In this case, we see easily

$$\Delta_{Y_\bullet}(\mathcal{O}_{\mathbb{P}^2}(1)) = P_D$$

is the polytope in Fig. 12.2.

Example 12.2.2 Let us continue the example of $\mathrm{Bl}_0\mathbb{P}^2$ in Example 12.1.2. This time, let us take σ_1' as our reference cone and $\rho_1 = \sigma_5$, $\rho_2 = \sigma_7$. Then Y_1 is just the strict transform of the line $\{[X_0 : 0 : X_2] : X_0 X_2 \neq 0\}$ in $\mathbb{P}^2$, while Y_2 is the point $Y_1 \cap E$, where E is the exceptional divisor.

In this case, the map Φ is given by

$$\Phi(m_1, m_2) = (m_1, m_1 + m_2).$$

We find that

$$\Delta_{Y_\bullet}\Big(\mathrm{Bl}_0\mathbb{P}^2, \pi^*\mathcal{O}_{\mathbb{P}^2}(1)\Big)$$

is the polytope in Fig. 12.4.

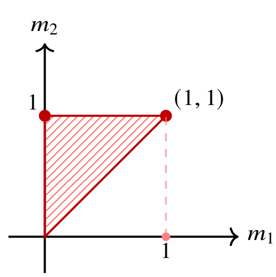

Fig. 12.4 The Okounkov body $\Delta_{Y_\bullet}(\mathrm{Bl}_0\mathbb{P}^2, \pi^*\mathcal{O}_{\mathbb{P}^2}(1))$.

Note that it differs from the polytope in Example 12.2.1.[2]

Theorem 12.2.1 *Let $\varphi \in \mathrm{PSH}_{\mathrm{tor}}(X,\theta)_{>0}$. Then*

$$\Phi_{\mathbb{R}}\left(\Delta(\theta,\varphi)\right) = \Delta_{Y_\bullet}(\theta,\varphi). \tag{12.16}$$

In particular,

$$n!\,\mathrm{vol}\,\Delta(\theta,\varphi) = \mathrm{vol}\,\theta_\varphi. \tag{12.17}$$

In particular, once D is fixed, the Newton body $\Delta(\theta,\varphi)$ depends only on the current θ_φ, not on the specific choices of h, φ and s_D. It makes sense to write

$$\Delta\left(\theta_\varphi\right) = \Delta(\theta,\varphi).$$

A generalization of (12.17) to mixed volumes will be proved in Theorem 12.3.3.

Proof We first reduce to the case where $D|_{U_\sigma} = 0$. In fact, changing D to $D + \mathrm{div}\,\chi^{m_\sigma}$ would result in changing F_θ to $F_\theta - m_\sigma$. Hence, F_φ changes to $F_\varphi - m_\sigma$. Therefore, $\Delta(\theta,\varphi)$ becomes $\Delta(\theta,\varphi) - m_\sigma$. Taking (12.5) into consideration, we may assume that $m_\sigma = 0$.

Step 1. We first reduce to the case where θ_φ is a Kähler current.

By Lemma 2.4.5, we can find $\psi \in \mathrm{PSH}(X,\theta)$ such that $\psi \leq \varphi$ and θ_ψ is a Kähler current. Taking the average along T_c, we may assume that ψ is T_c-invariant.

For each $t \in (0,1)$, we let

$$\varphi_t = (1-t)\psi + t\varphi.$$

Suppose that Kähler current case is known. Then we get

$$\Phi_{\mathbb{R}}\left(\Delta(\theta,\varphi_t)\right) = \Delta_{Y_\bullet}(\theta,\varphi_t)$$

for any $t \in (0,1)$. It follows from Theorem A.4.2 that

$$\Phi_{\mathbb{R}}\left(\Delta(\theta,\varphi)\right) \supseteq \Phi_{\mathbb{R}}\left(\Delta(\theta,\varphi_t)\right) = \Delta_{Y_\bullet}(\theta,\varphi_t)$$

for any $t \in (0,1)$. Thanks to Theorem 10.3.2, we have

$$\Phi_{\mathbb{R}}\left(\Delta(\theta,\varphi)\right) \supseteq \Delta_{Y_\bullet}(\theta,\varphi).$$

Comparing the volumes of both sides using Proposition 12.2.1, (10.19), and the $\mathcal{I}$-goodness of toric singularities Example 7.4.1, we find that

[2] Although these examples are almost trivial, they did confuse me a lot at the beginning of 2023, when Kewei Zhang, Tamás Darvas and I were collaborating on [DRWN+26]. At that time, Kewei himself already proved the main theorem for a generic flag. I realized that some simple birational geometry would suffice to prove the same result for general flags. I persuaded myself and Kewei that the Okounkov bodies are always birationally invariant, and deduced some apparently wrong conclusions. I got no clue for a couple of weeks, then one day, on the noisy metro line 7 of Paris, I got nothing to do, so I said to myself: Why not compute the simplest toric examples? Then after a few minutes, all of a sudden, the whole picture became completely clear.

$$n!\operatorname{vol}\Phi_{\mathbb{R}}\left(\Delta(\theta,\varphi)\right)=\int_X\theta_\varphi^n=\operatorname{vol}\theta_\varphi=n!\operatorname{vol}\Delta_{Y_\bullet}(\theta,\varphi).$$

In particular, we conclude (12.16).

Step 2. We handle the case where θ_φ is a Kähler current.

Let $(\varphi_j)_j$ be a quasi-equisingular approximation of φ in $\mathrm{PSH}(X,\theta)$. Since θ_φ is a Kähler current, we may take the approximation so that each θ_{φ_j} is a Kähler current as well.

We may assume that φ_j is T_c-invariant for each $j\geq 1$ from the construction of [Dem12a, Theorem 13.21].

Now assume that the result is known for each φ_j. Then

$$\Phi_{\mathbb{R}}\left(\Delta(\theta,\varphi_j)\right)=\Delta_{Y_\bullet}(\theta,\varphi_j).$$

In particular,

$$\Phi_{\mathbb{R}}\left(\Delta(\theta,\varphi)\right)\subseteq\Delta_{Y_\bullet}(\theta,\varphi_j)$$

for each $j\geq 1$. It follows from Theorem 10.3.2 that

$$\Phi_{\mathbb{R}}\left(\Delta(\theta,\varphi)\right)\subseteq\Delta_{Y_\bullet}(\theta,\varphi).$$

Comparing the volumes of both sides using Proposition 12.2.1, (10.19), and Example 7.4.1, we conclude (12.16).

Step 3. It remains to handle the case where φ has analytic singularities and θ_φ is a Kähler current. In fact, we may assume that for some $q\in\mathbb{Z}_{>0}$, φ has the form

$$\varphi=\frac{1}{q}\log\sum_{i=1}^a|s_i|^2_{h^q}+O(1),$$

where $s_1,\dots,s_a\in\mathrm{H}^0(X,L^q)$ are toric eigen-sections. This follows from the proof of Step 2 and the construction of [Dem12a, Theorem 13.21].

Let $m_1,\dots,m_a\in qP_D\cap M$ be the lattice points corresponding to $s_1,\dots,s_a$. Observe that

$$\begin{aligned}\Delta(\theta,\varphi)=&\overline{\nabla F_\varphi(N_{\mathbb{R}})}\\=&\left\{m\in M_{\mathbb{R}}:\ F_\varphi(n)-\langle m,n\rangle\text{ is bounded from below}\right\}\\=&\left\{m\in M_{\mathbb{R}}:\ \frac{1}{q}\log\textstyle\sum_{i=1}^a\mathrm{e}^{\langle m_i,n\rangle}-\langle m,n\rangle\text{ is bounded below in }n\right\}\\=&q^{-1}\mathrm{Conv}\{m_1,\dots,m_a\},\end{aligned}$$

where we have applied (12.9) on the second line and Lemma A.5.2 on the third line. In particular, by the same separation argument as in Lemma A.5.1, for any $k\in\mathbb{Z}_{>0}$ and any $m\in k\Delta(\theta,\varphi)\cap M$, we have

$$|\chi^m|^2\mathrm{e}^{-k\varphi}$$

is bounded from above on $T(\mathbb{C})$. In other words, the section s of L^k corresponding to m satisfies

$$s \in \mathrm{H}^0\left(X, L^k \otimes \mathcal{I}_\infty(k\varphi)\right).$$

Therefore,

$$\nu_{Y_\bullet}(s) = \Phi(m) \in k\Delta_k(\theta, \varphi),$$

where Δ_k is defined in Section 10.3. Hence,

$$\Phi\left(k\Delta(\theta, \varphi) \cap M\right) \subseteq k\Delta_k(\theta, \varphi).$$

Letting $k \to \infty$ and applying Theorem 10.3.4, we find that

$$\Phi_{\mathbb{R}}\left(\Delta(\theta, \varphi)\right) \subseteq \Delta_{Y_\bullet}(\theta, \varphi).$$

Comparing the volumes of both sides using Proposition 12.2.1 and (10.19), we conclude that the equality holds and (12.16) follows. □

The following two consequences are both due to Yi Yao.

Corollary 12.2.1 *Let E be a T-invariant prime divisor on X corresponding to a ray $\rho \in \Sigma(1)$. Then for any $\varphi \in \mathrm{PSH}_{\mathrm{tor}}(X, \theta)_{>0}$, we have*

$$\nu(\varphi, E) = \inf\left\{\langle m - m_\rho, u_\rho\rangle : m \in \Delta(\theta, \varphi)\right\}.$$

Proof This follows immediately from Theorem 12.2.1 and Theorem 10.3.5. In fact, since X is projective and smooth, there is always a T-invariant smooth flag $Y_\bullet$ with $Y_1 = E$. □

This result seems new even in the ample setting. Intuitively, after taking Theorem 12.2.2 into consideration as well, in the ample case the generic Lelong number $\nu(\varphi, E)$ is the rescaled "distance" from $\Delta(\theta, \varphi)$ to the facet of P_D corresponding to E.[3]

Corollary 12.2.2 *Let $Y \subseteq X$ be a T-invariant subvariety corresponding to a cone σ in Σ, and let $\varphi \in \mathrm{PSH}_{\mathrm{tor}}(X, \theta)_{>0}$. Then the following are equivalent:*

(1) *$\nu(\varphi, Y) = 0$;*
(2) *there is a point $m \in \Delta(\theta, \varphi)$ such that $\langle m - m_\rho, u_\rho\rangle = 0$ for any $\rho \in \sigma(1)$;*
(3) *we have*

$$\Delta(\theta, \varphi) \cap Q_\sigma \neq \emptyset.$$

Recall that Q_σ is defined in (12.6).

Proof (2) $\Longleftrightarrow$ (3). This follows from the definition of Q_σ in (12.6).

(1) $\Longleftrightarrow$ (2). Let $\rho_1, \ldots, \rho_r$ be the rays of σ. Up to replacing D by a translation, we may assume that $m_\sigma = 0$. Hence, we may take $m_{\rho_i} = 0$ for all i.

Let $\pi\colon Z \to X$ be the blow-up of X along Y. See [CLS11, Page 132] for the basic properties of the toric blow-up. Take the divisor $\pi^* D$ on Z. We choose the pull-back

[3] Be cautious! In the big setting, in general, the condition $\langle m - m_\rho, u_\rho\rangle = 0$ does not necessarily define a facet of P_D. Hence the intuition fails.

metric $\pi^* h$ on $\pi^* L$. Then $F_{\pi^*\theta}$ can be taken as $\pi^* F_\theta$ by (12.9). It follows that $\Delta(\theta, \varphi) = \Delta(\pi^*\theta, \pi^*\varphi)$. On the other hand, the ray corresponding to the exceptional divisor E is generated by $u_{\rho_1} + \cdots + u_{\rho_r}$. Since X is smooth, this vector is primitive.

Recall that the support function of $\pi^* D$ is the same as the support function of D, see [CLS11, Proposition 6.2.7]. In particular, we can take the Cartier datum $m_\rho = m_\sigma \mod M(\rho)$, where ρ is the ray corresponding to E.

It follows from Corollary 12.2.1 and Lemma 1.4.1 that

$$\nu(\varphi, Y) = \nu(\pi^*\varphi, E) = \inf\left\{\langle m - m_\sigma, u_{\rho_1} + \cdots + u_{\rho_r}\rangle : m \in \Delta(\theta, \varphi)\right\}. \tag{12.18}$$

Our assertion follows in view of (12.4). □

It follows from (12.18) that

$$\nu(\varphi, Y) \geq \sum_{i=1}^{r} \nu(\varphi, E_i),$$

where the E_i's are the prime divisors corresponding to the rays of σ. This inequality seems to be new as well.

The following consequence of Theorem 12.2.1 is the key to the development of the toric pluripotential theory.

Theorem 12.2.2 *We have*

$$F_{V_\theta} \in \mathcal{E}^\infty(N_\mathbb{R}, P_D).^{4} \tag{12.19}$$

In particular,

$$\int_X \theta_{V_\theta}^n = n! \operatorname{vol} P_D. \tag{12.20}$$

Recall that $\mathcal{E}^\infty$ is defined in Definition A.3.1. The equation (12.19) says

$$F_{V_\theta} - \operatorname{Supp}_{P_D} \quad \text{is bounded.} \tag{12.21}$$

In particular,

$$\Delta(\theta, V_\theta) = P_D \tag{12.22}$$

and hence the Newton bodies $\Delta(\theta, \varphi)$ are all contained in P_D.

Proof Take $\varphi = V_\theta$ in Theorem 12.2.1, we find

$$\Phi_\mathbb{R}\left(\Delta(\theta, V_\theta)\right) = \Delta_{Y_\bullet}(\theta, V_\theta) = \Delta_{Y_\bullet}(L) = \Phi_\mathbb{R}(P_D),$$

where we applied Proposition 12.2.2 in the last equality. Therefore, (12.22) follows. In particular, (12.20) follows from Proposition 12.2.1.

Next we prove (12.21). Take a smooth function $\chi \colon N_\mathbb{R} \to [0, 1]$ so that

(1) $\operatorname{Supp} \chi$ is compact;

[4] Initially I was only able to show $F_{V_\theta} \in \mathcal{E}(N_\mathbb{R}, P_D)$, the strengthened version was suggested by Robert Berman.

(2) $\chi \equiv 1$ on a neighbourhood of 0.

Fix a T_c-invariant volume form μ on $T(\mathbb{C})$. Then $(\mathrm{Trop}^* \chi)\mu$ can be regarded as a T_c-invariant smooth non-negative measure on X. Take a suitable normalizing constant $C > 0$, there is a unique solution

$$\theta_\varphi^n = C\,(\mathrm{Trop}^*\chi)\,\mu, \quad \sup_X \varphi = 0, \quad \varphi \in \mathcal{E}^\infty(X,\theta). \tag{12.23}$$

This is proved in [BEGZ10]. The uniqueness of φ further guarantees its T_c-invariance. Since φ has minimal singularities, $F_\varphi \sim F_{V_\theta}$. Next, using Proposition 12.2.1, the Monge–Ampère equation (12.23) implies

$$\mathrm{MA}_{\mathbb{R}}\left(F_\varphi\right) = C\chi\,\mathrm{Trop}_*\,\mu.$$

It follows from the regularity theorem [BB13, Theorem 2.19] and (12.22) that $F_\varphi \sim \mathrm{Supp}_{P_D}$. Hence, (12.21) follows. □

In particular, thanks to Corollary 12.2.1,

$$\nu(V_\theta, E) = \min\left\{\langle m - m_\rho, u_\rho\rangle : m \in P_D\right\}. \tag{12.24}$$

As an interesting consequence of (12.24), we have a geometric description of the divisorial Zariski decomposition of [Bou02b] in the toric setting: The negative part of D is given by

$$\sum_{\rho\in\Sigma(1)} \min\left\{\langle m - m_\rho, u_\rho\rangle : m \in P_D\right\} D_\rho.$$

A prime divisor D_ρ appears in the negative part if and only if the corresponding condition

$$\langle m, u_\rho\rangle \geq -a_\rho$$

is redundant in defining P_D by (12.1). The modified nef part of D is the $\mathbb{Q}$-divisor obtained after removing these redundancies.

12.3 The pluripotential theory

We continue to use the notations in Section 12.1.

Theorem 12.3.1 *There are canonical bijections*[5] *between the following sets:*

(1) *The set of* $\varphi \in \mathrm{PSH}_{\mathrm{tor}}(X,\theta)$*;*
(2) *the set of* $F \in \mathcal{P}(N_{\mathbb{R}}, P_D)$*, and*

[5] In the earlier version of this book, I required additional conditions in (2) and (3), namely $F \leq F_{V_\theta}$ and $G \geq G_{V_\theta}$ respectively. These conditions can be removed, since G_{V_θ} is always bounded, as suggested by Robert Berman. See Theorem 12.2.2.

(3) *the set of closed proper convex functions* $G \in \mathrm{Conv}(M_{\mathbb{R}})$ *satisfying*

$$G|_{M_{\mathbb{R}} \setminus P_D} \equiv \infty.$$

The set $\mathcal{P}(N_{\mathbb{R}}, P_D)$ is defined in Definition A.3.1. As before, we write F_φ, G_φ for the functions determined by this construction.

Proof The proof is similar to that of Theorem 5.2.1, but due to its importance, we give the details. Again, the correspondence between (2) and (3) follows easily from Proposition A.2.5.

Given φ, we can construct F_φ in (2) as explained earlier in (12.10). Conversely, suppose that $F \in \mathcal{P}(N_{\mathbb{R}}, P_D)$, then $F \leq F_{V_\theta}$ by Theorem 12.2.2. Then

$$\mathrm{Trop}^* (F - F_\theta) \in \mathrm{PSH}\left(T(\mathbb{C}), \theta|_{T(\mathbb{C})}\right)$$

by Lemma 5.2.1. Since $F \leq F_{V_\theta}$, we see that $\mathrm{Trop}^*(F - F_\theta)$ is bounded from above. It follows that Grauert–Remmert's extension theorem Theorem 1.2.1 is applicable, and this function extends to a unique θ-psh function φ. The uniqueness of the extension guarantees that $\varphi \in \mathrm{PSH}_{\mathrm{tor}}(X, \theta)$.

The two maps are inverse to each other by construction. □

We fix a model potential $\phi \in \mathrm{PSH}_{\mathrm{tor}}(X, \theta)_{>0}$ with Newton body $\Delta(\theta, \phi)$.

A similar argument guarantees the following:

Corollary 12.3.1 *There is a canonical bijection between the following sets:*

(1) *The set of* $\varphi \in \mathrm{PSH}_{\mathrm{tor}}(X, \theta; \phi)$*,*
(2) *the set of* $F \in \mathcal{P}(N_{\mathbb{R}}, \Delta(\theta, \phi))$*, and*
(3) *the set of closed proper convex functions* $G \in \mathrm{Conv}(M_{\mathbb{R}})$ *satisfying*

$$G|_{M_{\mathbb{R}} \setminus \Delta(\theta, \phi)} = \infty.$$

Moreover, under these correspondences, we have the following bijections:

(1) *The set* $\mathcal{E}_{\mathrm{tor}}(X, \theta; \phi)$*,*
(2) *the set of* $F \in \mathcal{E}(N_{\mathbb{R}}, \Delta(\theta, \phi))$*, and*
(3) *the set of closed proper convex functions* $G \in \mathrm{Conv}(M_{\mathbb{R}})$ *satisfying*

$$\mathrm{Int}\{G < \infty\} = \Delta(\theta, \phi).$$

Here the notations are defined as follows:

$$\begin{aligned} \mathrm{PSH}_{\mathrm{tor}}(X, \theta; \phi) &:= \{\varphi \in \mathrm{PSH}_{\mathrm{tor}}(X, \theta) : \varphi \leq \phi\}, \\ \mathcal{E}_{\mathrm{tor}}(X, \theta; \phi) &:= \mathcal{E}(X, \theta; \phi) \cap \mathrm{PSH}_{\mathrm{tor}}(X, \theta). \end{aligned}$$

The proofs of the following results are similar to the ample case studied in Chapter 5. We omit the details.

Proposition 12.3.1 *Given* $\varphi \in \mathrm{PSH}_{\mathrm{tor}}(X, \theta)$ *and* $C \in \mathbb{R}$*. We have*

$$F_{\varphi + C} = F_\varphi + C, \quad G_{\varphi + C} = G_\varphi - C.$$

Proposition 12.3.2 *Given $\varphi, \psi \in \mathrm{PSH}_{\mathrm{tor}}(X,\theta)$, assume that $\varphi \wedge \psi \not\equiv -\infty$, then $\varphi \wedge \psi \in \mathrm{PSH}_{\mathrm{tor}}(X,\theta)$ and*

$$F_{\varphi\wedge\psi} = F_\varphi \wedge F_\psi, \quad G_{\varphi\wedge\psi} = G_\varphi \vee G_\psi .$$

Proposition 12.3.3 *Let $(\varphi_i)_{i\in I}$ be a family in $\mathrm{PSH}_{\mathrm{tor}}(X,\theta)$ uniformly bounded from above. Then $\sup_{i\in I}{}^*\varphi_i \in \mathrm{PSH}_{\mathrm{tor}}(X,\theta)$ and*

$$F_{\sup_{i\in I}{}^*\varphi_i} = \bigvee_{i\in I} F_{\varphi_i}, \quad G_{\sup_{i\in I}{}^*\varphi_i} = \mathrm{cl} \bigwedge_{i\in I} G_{\varphi_i} .$$

Moreover, if I is finite, then

$$G_{\bigvee_{i\in I}\varphi_i} = \bigwedge_{i\in I} G_{\varphi_i} .$$

Similarly, if $\{\varphi_i\}_{i\in I}$ is a decreasing net in $\mathrm{PSH}_{\mathrm{tor}}(X,\theta)$ such that $\inf_{i\in I}\varphi_i \not\equiv -\infty$, then $\inf_{i\in I}\varphi_i \in \mathrm{PSH}_{\mathrm{tor}}(X,\theta)$ and

$$F_{\inf_{i\in I}\varphi_i} = \bigwedge_{i\in I} F_{\varphi_i}, \quad G_{\inf_{i\in I}\varphi_i} = \bigvee_{i\in I} G_{\varphi_i} .$$

Proposition 12.3.4 *Let $\varphi \in \mathrm{PSH}_{\mathrm{tor}}(X,\theta)$. Then $P_\theta[\varphi] \in \mathrm{PSH}_{\mathrm{tor}}(X,\theta)$ and*

$$G_{P_\theta[\varphi]}(x) = \begin{cases} G_{V_\theta}(x), & \text{if } x \in \Delta(\theta,\varphi); \\ \infty, & \text{otherwise.} \end{cases} \tag{12.25}$$

Proof By (3.4),

$$P_\theta[\varphi] = \sup_{C\in\mathbb{R}}{}^*\big((\varphi + C) \wedge V_\theta\big).$$

Hence $P_\theta[\varphi] \in \mathrm{PSH}_{\mathrm{tor}}(X,\theta)$ by Propositions 12.3.1 to 12.3.3. The same results give

$$G_{P_\theta[\varphi]} = \mathrm{cl} \inf_{C\in\mathbb{R}} \left(G_{V_\theta} \vee (G_\varphi - C)\right) .$$

Put

$$K = \inf_{C\in\mathbb{R}} \left(G_{V_\theta} \vee (G_\varphi - C)\right) .$$

If $x \in \mathrm{Dom}\, G_\varphi$, then $G_\varphi(x) - C \leq G_{V_\theta}(x)$ for all large enough C, so $K(x) = G_{V_\theta}(x)$. If $x \notin \mathrm{Dom}\, G_\varphi$, then $K(x) = \infty$. By Proposition A.2.2, we have $\Delta(\theta,\varphi) = \overline{\mathrm{Dom}\, G_\varphi}$. Moreover, Theorem 12.2.2 gives $F_{V_\theta} \sim \mathrm{Supp}_{P_D}$, hence $G_{V_\theta} \sim \chi_{P_D}$ by Legendre duality. In particular, G_{V_θ} is finite and continuous on P_D. Thus the lower semicontinuous regularization of K is equal to G_{V_θ} on $\Delta(\theta,\varphi)$ and equal to ∞ outside $\Delta(\theta,\varphi)$. ◻

As a consequence, we have

Corollary 12.3.2 *Let $\varphi, \psi \in \mathrm{PSH}_{\mathrm{tor}}(X,\theta)$. Then the following are equivalent:*

(1) *$\varphi \preceq_P \psi$;*

(2) $\varphi \preceq_{\mathcal{I}} \psi$;
(3) $\Delta(\theta, \varphi) \subseteq \Delta(\theta, \psi)$.

Proof (1) $\iff$ (2). This follows from the $\mathcal{I}$-goodness of φ and ψ, see Example 7.4.1.

(1) $\iff$ (3). When $\varphi, \psi \in \mathrm{PSH}_{\mathrm{tor}}(X, \theta)_{>0}$, this follows from Proposition 12.3.4.

In general, fix an ample toric-invariant divisor H on X and a toric-invariant Hermitian metric h_H on $\mathcal{O}_X(H)$ such that $\omega_H := \mathrm{dd^c} h_H$ is a Kähler form. We perturb L to $L + m^{-1}H$ for $m \in \mathbb{Z}_{>0}$, then thanks to Lemma A.3.1, we find that

$$\Delta(\theta + m^{-1}\omega_H, \varphi) := m^{-1}\Delta(m\theta + \omega_H, m\varphi) = \Delta(\theta, \varphi) + m^{-1}\Delta\left(\mathcal{O}_X(H)\right).$$

Then since $\varphi, \psi \in \mathrm{PSH}_{\mathrm{tor}}(X, \theta + m^{-1}\omega_H)_{>0}$, we know that $\varphi \preceq_P \psi$ if and only if

$$\Delta(\theta, \varphi) + m^{-1}\Delta\left(\mathcal{O}_X(H)\right) \subseteq \Delta(\theta, \psi) + m^{-1}\Delta\left(\mathcal{O}_X(H)\right)$$

for any $m \in \mathbb{Z}_{>0}$. The latter condition is equivalent to $\Delta(\theta, \varphi) \subseteq \Delta(\theta, \psi)$. $\square$

Corollary 12.3.3 *There is a canonical bijection*

$$\mathrm{PSH}_{\mathrm{tor}}(X, \theta)/\sim_P = \mathrm{PSH}_{\mathrm{tor}}(X, \theta)/\sim_{\mathcal{I}} \to \{K \in \mathcal{K}_n : K \subseteq P_D\}.$$

Here $\mathcal{K}_n$ denotes the set of convex bodies in $\mathbb{R}^n$, as in Appendix C.1.

More precisely, given $\varphi \in \mathrm{PSH}_{\mathrm{tor}}(X, \theta)$, the corresponding convex body is $\Delta(\theta, \varphi)$.

Proof Thanks to Corollary 12.3.2, we have the equality

$$\mathrm{PSH}_{\mathrm{tor}}(X, \theta)/\sim_P = \mathrm{PSH}_{\mathrm{tor}}(X, \theta)/\sim_{\mathcal{I}},$$

and the map

$$\mathrm{PSH}_{\mathrm{tor}}(X, \theta)/\sim_{\mathcal{I}} \to \{K \in \mathcal{K}_n : K \subseteq P_D\}$$

is well-defined and injective.

In order to see the surjectivity, take $K \in \mathcal{K}_n$, $K \subseteq P_D$. Let χ_K be the characteristic function of K as defined in Example A.1.1. Since χ_K satisfies Condition (3) of Theorem 12.3.1, we can find $\varphi \in \mathrm{PSH}_{\mathrm{tor}}(X, \theta)$ so that $\chi_K = G_\varphi$. Then $\Delta(\theta, \varphi) = K$ due to Proposition A.2.2. $\square$

Next we handle subgeodesics.

Proposition 12.3.5 *Let $\varphi_0, \varphi_1 \in \mathrm{PSH}_{\mathrm{tor}}(X, \theta)$. There is a canonical bijection between the following sets:*

(1) *The set of T_c-invariant subgeodesics from φ_0 to φ_1;*
(2) *the set of convex functions $F : N_{\mathbb{R}} \times (0, 1) \to \mathbb{R}$ such that for each $r \in (0, 1)$, the function*

$$F_r : N_{\mathbb{R}} \to \mathbb{R}, \quad n \mapsto F(n, r) \tag{12.26}$$

satisfies $F_r \to F_{\varphi_1}$ (resp. $F_r \to F_{\varphi_0}$) everywhere as $r \to 1-$ (resp. $r \to 0+$).

Proof We begin with a subgeodesic $(\varphi_t)_{t\in(0,1)}$ from φ_0 to φ_1. Then we define $F\colon N_{\mathbb{R}}\times(0,1)\to\mathbb{R}$ as follows:

$$F(n,t)=F_{\varphi_t}(n).$$

Define F_t as in (12.26), we have

$$\operatorname{Trop}^* F_t-\operatorname{Trop}^* F_\theta=\varphi_t,\quad t\in(0,1).$$

By definition, as $t\to 0+$, $\varphi_t\to\varphi_0$ almost everywhere. By Fubini's theorem, $F_t\to F_{\varphi_0}$ almost everywhere, hence everywhere by Theorem A.1.2. Similarly, $F_t\to F_{\varphi_1}$ everywhere as $t\to 1-$.

Next we show that F is convex. Let $p_1\colon X\times S\to X$ be the projection, where

$$S:=\{z\in\mathbb{C}:\mathrm{e}^{-1}<|z|^2<1\}.$$

Since $(\varphi_t)_{t\in(0,1)}$ is a subgeodesic, its complexification Φ is $p_1^*\theta$-psh. Recall that Φ is defined as

$$\Phi(x,z)=\operatorname{Trop}^*\left(F_{-\log|z|^2}-F_\theta\right)(x). \tag{12.27}$$

In particular, $\Psi\colon T(\mathbb{C})\times S\to\mathbb{R}$ defined by

$$\Psi(x,z):=\Phi(x,z)+\operatorname{Trop}^* F_\theta(x)=\operatorname{Trop}^* F_{-\log|z|^2}(x)$$

is plurisubharmonic and $T_c\times S^1$-invariant. Fix a small enough $\epsilon>0$. We can find a decreasing sequence of $T_c\times S^1$-invariant plurisubharmonic functions Ψ_i on $T(\mathbb{C})\times S_\epsilon$ converging to Ψ everywhere, where

$$S_\epsilon:=\{z\in\mathbb{C}:\mathrm{e}^{-1+\epsilon}<|z|^2<\mathrm{e}^{-\epsilon}\}.$$

Let us write

$$\Psi_i(x,z)=\operatorname{Trop}^* F_{i,-\log|z|^2}(x)$$

for some $F_i\colon N_{\mathbb{R}}\times(\epsilon,1-\epsilon)\to\mathbb{R}$.

The same computation as in Lemma 5.2.1 shows that F_i is convex. It follows that F, as the decreasing limit of F_i, is also convex on $N_{\mathbb{R}}\times(\epsilon,1-\epsilon)$. Since $\epsilon>0$ is arbitrary, we conclude that F is convex on $N_{\mathbb{R}}\times(0,1)$.

Conversely, suppose that we are given F in (2). We define $\Phi\colon T(\mathbb{C})\times S\to\mathbb{R}$ using (12.27). The arguments in the previous part can be reversed to show that Φ is $p_1^*\theta|_{T(\mathbb{C})\times S}$-psh.

By our assumption, for each $t\in(0,1)$, we have

$$F_t\le tF_{\varphi_1}+(1-t)F_{\varphi_0}\le F_{V_\theta}+C \tag{12.28}$$

for some constant $C\in\mathbb{R}$ independent of the choice of t. Therefore, Φ is bounded from above and hence, by Theorem 1.2.1, admits a unique $p_1^*\theta$-psh extension to $X\times S$, which we still denote by Φ. We let

$$\varphi_t(x) = \Phi(x, \mathrm{e}^{-t/2})$$

for all $t \in (0,1)$ and $x \in X$. We claim that (φ_t) is a subgeodesic from φ_0 to φ_1.

For this purpose, we only need to show that $(\varphi_t)_{t\in(0,1)}$ has the correct boundary value. But from our assumption in (2), we know that as $t \to 0+$ (resp. $t \to 1-$), $\varphi_t \to \varphi_0$ (resp. $\varphi_t \to \varphi_1$) almost everywhere. In particular, $\sup_X \varphi_t \geq -C'$ for some large constant $C' > 0$ independent of $t \in (0,1)$. Therefore, together with (12.28), we deduce from Proposition 1.5.1 that $\{\varphi_t\}_{t\in(0,1)}$ is a relatively compact family with respect to the L^1-topology. We need to show that each cluster point ψ as $t \to 0+$ is equal to φ_0. But we already know that $\psi = \varphi_0$ almost everywhere. Hence we deduce $\psi = \varphi_0$ from Proposition 1.2.6. As $t \to 0+$, we have $\varphi_t \xrightarrow{L^1} \varphi_0$. Similarly, as $t \to 1-$, we have $\varphi_t \xrightarrow{L^1} \varphi_1$.

The two constructions are inverse to each other by construction.

Corollary 12.3.4 *Let $\varphi_0, \varphi_1 \in \mathrm{PSH}_{\mathrm{tor}}(X,\theta)$. Then there is a canonical bijection between the following sets:*

(1) *The set of T_c-invariant subgeodesics from ψ_0 to ψ_1, where $\psi_0, \psi_1 \in \mathrm{PSH}_{\mathrm{tor}}(X,\theta)$ and $\psi_0 \leq \varphi_0$, $\psi_1 \leq \varphi_1$;*
(2) *the set of closed proper convex functions Ψ on $M_{\mathbb{R}} \times \mathbb{R}$ such that there is a closed proper convex function $G \in \mathrm{Conv}(M_{\mathbb{R}})$ satisfying*

$$G(m) + (s \vee 0) \geq \Psi(m,s) \geq G_{\varphi_0}(m) \vee \left(G_{\varphi_1}(m) + s\right) \tag{12.29}$$

for all $m \in M_{\mathbb{R}}$ and $s \in \mathbb{R}$.

Proof Let us begin with a subgeodesic $(\psi_t)_{t\in(0,1)}$ as in (1). Let F be the convex function as in Proposition 12.3.5. We extend F to a function $F\colon N_{\mathbb{R}} \times \mathbb{R} \to \mathbb{R}$ as follows: For any $n \in N_{\mathbb{R}}$, we define

$$F(n,t) = \begin{cases} F_{\psi_t}(n), & \text{if } 0 < t < 1, \\ F_{\psi_0}(n), & \text{if } t = 0, \\ F_{\psi_1}(n), & \text{if } t = 1, \\ \infty, & \text{if } t > 1 \text{ or } t < 0. \end{cases}$$

Then F is a proper closed convex function on $N_{\mathbb{R}} \times \mathbb{R}$. Let Ψ be the Legendre transform of F. Then Ψ is a proper closed convex function on $M_{\mathbb{R}} \times \mathbb{R}$ by Theorem A.2.1. By (A.2), for any $m \in M_{\mathbb{R}}$ and $s \in \mathbb{R}$, we have

$$\begin{aligned} \Psi(m,s) &= \sup_{n\in N_{\mathbb{R}}, t\in[0,1]} \left(\langle m,n\rangle + ts - F(n,t)\right) \\ &= \sup_{t\in[0,1]} \left(ts + F_t^*(m)\right). \end{aligned}$$

Therefore, the latter half of (12.29) follows. Next recall that

$$\eta := \inf_{t\in(0,1)} \psi_t \in \mathrm{PSH}_{\mathrm{tor}}(X,\theta),$$

as follows from Proposition 4.1.2. Therefore,

$$\begin{aligned}\Psi(m,s) =& \sup_{n\in N_{\mathbb{R}},t\in[0,1]} (\langle m,n\rangle + ts - F(n,t))\\ \leq& \sup_{n\in N_{\mathbb{R}},t\in[0,1]} \big(\langle m,n\rangle + ts - F_\eta(n)\big)\\ =& \sup_{t\in[0,1]} ts + G_\eta(m)\\ =&(s\vee 0) + G_\eta(m).\end{aligned}$$

Conversely, let us begin with a function Ψ as in (2). Let F be the Legendre transform of Ψ. We first observe that $F(n,t) = \infty$ for all $n \in N_{\mathbb{R}}$ and $t \notin [0,1]$. Indeed, the upper bound $\Psi \leq G + (s\vee 0)$ gives

$$F(n,t) \geq G^*(n) + \sup_{s\in\mathbb{R}} (ts - (s\vee 0)),$$

and the last supremum is infinite exactly when $t \notin [0,1]$.

For $t \in [0,1]$, the lower bound $\Psi \geq G_{\varphi_0} \vee (G_{\varphi_1} + s)$ implies

$$\Psi(m,s) \geq (1-t)G_{\varphi_0}(m) + tG_{\varphi_1}(m) + ts.$$

Consequently,

$$F(n,t) \leq \big((1-t)G_{\varphi_0} + tG_{\varphi_1}\big)^*(n) \leq (1-t)F_{\varphi_0}(n) + tF_{\varphi_1}(n)$$

for all $t \in [0,1]$ and $n \in N_{\mathbb{R}}$. On the other hand, the upper bound $\Psi \leq G + (s\vee 0)$ gives

$$\begin{aligned}F(n,t) =& \sup_{m\in M_{\mathbb{R}},s\in\mathbb{R}} (\langle m,n\rangle + ts - \Psi(m,s))\\ \geq& \sup_{m\in M_{\mathbb{R}},s\in\mathbb{R}} (\langle m,n\rangle + ts - G(m) - (s\vee 0))\\ =&G^*(n) + \sup_{s\in\mathbb{R}} (ts - (s\vee 0))\\ =&\begin{cases} G^*(n), & \text{if } t \in [0,1],\\ \infty, & \text{otherwise.}\end{cases}\end{aligned}$$

The convexity of F gives one-sided boundary limits

$$F_0(n) := \lim_{t\to 0+} F(n,t), \quad F_1(n) := \lim_{t\to 1-} F(n,t).$$

The estimates above give $F_0 \leq F_{\varphi_0}$ and $F_1 \leq F_{\varphi_1}$, and also show that $F_0, F_1 \in \mathcal{P}(N_{\mathbb{R}}, P_D)$. Let $\psi_0, \psi_1 \in \mathrm{PSH}_{\mathrm{tor}}(X,\theta)$ be the corresponding potentials given by Theorem 12.3.1. Then $\psi_0 \leq \varphi_0$ and $\psi_1 \leq \varphi_1$. Applying Proposition 12.3.5 to the restriction of F to $N_{\mathbb{R}} \times (0,1)$, we get a subgeodesic $(\psi_t)_{t\in(0,1)}$ from ψ_0 to ψ_1. Thus $(\psi_t)_{t\in(0,1)}$ satisfies (1). The two operations are inverse to each other by construction. □

As an immediate corollary,

Corollary 12.3.5 *Let $\varphi_0, \varphi_1 \in \mathrm{PSH}_{\mathrm{tor}}(X,\theta) \cap \mathrm{PSH}(X,\theta)_{>0}$. Then the following are equivalent:*

(1) *$\varphi_0 \sim_P \varphi_1$;*
(2) *there is a subgeodesic from φ_0 to φ_1.*

If these conditions are satisfied, let $(\varphi_t)_{t\in(0,1)}$ be the geodesic from φ_0 to φ_1. Then $\varphi_t \in \mathrm{PSH}_{\mathrm{tor}}(X,\theta)$ for all $t \in (0,1)$ and

$$G_{\varphi_t} = (1-t)G_{\varphi_0} + tG_{\varphi_1}. \tag{12.30}$$

Proof The equivalence between (1) and (2) follows from a general fact Theorem 6.1.1. In the toric case, (2) $\implies$ (1) can also be argued more directly.

Assume that these conditions are satisfied. Let $(\varphi_t)_{t\in(0,1)}$ be the geodesic from φ_0 to φ_1. Since the geodesic is the upper envelope of toric subgeodesics, $\varphi_t \in \mathrm{PSH}_{\mathrm{tor}}(X,\theta)$ for all $t \in (0,1)$. Let Ψ' be the proper convex function on $M_{\mathbb{R}} \times \mathbb{R}$ defined by Corollary 12.3.4. Then Ψ' is the minimum of all Ψ satisfying (12.29). We claim that

$$\Psi'(m,s) = G_{\varphi_0}(m) \vee \left(G_{\varphi_1}(m) + s\right). \tag{12.31}$$

It suffices to show that the right-hand side is proper, namely, $G_{\varphi_0} \vee G_{\varphi_1}$ is not identically ∞. But recall that by Proposition 4.1.2, we have $\varphi_0 \wedge \varphi_1 \in \mathrm{PSH}(X,\theta)$. Therefore, by Proposition 12.3.2,

$$G_{\varphi_0} \vee G_{\varphi_1} = G_{\varphi_0 \wedge \varphi_1} \not\equiv \infty.$$

In particular, (12.31) follows.

Now by construction,

$$G_{\varphi_t}(m) = \inf_{s\in\mathbb{R}} \left(\Psi'(m,s) - st\right) = (1-t)G_{\varphi_0}(m) + tG_{\varphi_1}(m)$$

for all $t \in (0,1)$. So (12.30) follows. □

Let $\phi \in \mathrm{PSH}_{\mathrm{tor}}(X,\theta) \cap \mathrm{PSH}(X,\theta)_{>0}$ be a model potential. We write

$$\mathcal{R}_{\mathrm{tor}}(X,\theta;\phi) \coloneqq \left\{\ell \in \mathcal{R}(X,\theta;\phi) : \ell_t \in \mathrm{PSH}_{\mathrm{tor}}(X,\theta) \quad \forall t \geq 0\right\}.$$

Recall that $\mathcal{R}(X,\theta;\phi)$ is defined in Definition 4.2.2.

Corollary 12.3.6 *There is a canonical bijection between the following sets:*

(1) *The set of $\ell \in \mathcal{R}_{\mathrm{tor}}(X,\theta;\phi)$;*
(2) *The set of proper closed convex functions g on $M_{\mathbb{R}}$ with $\overline{\mathrm{Dom}\, g} = \Delta(\theta,\phi)$.*

Moreover, given $\ell \in \mathcal{R}_{\mathrm{tor}}(X, \theta; \phi)$, then

$$G_{\ell_t} = G_\phi + tg, \quad \forall t > 0. \tag{12.32}$$

Proof First observe that given g as in (2), (12.32) indeed induces a geodesic ray in $\mathcal{R}_{\mathrm{tor}}(X, \theta; \phi)$ thanks to Corollary 12.3.5. Therefore, we have a map from (2) to (1).

Conversely, given $\ell \in \mathcal{R}_{\mathrm{tor}}(X, \theta; \phi)$, it follows from Corollary 12.3.5 that G_{ℓ_t} is linear in $t \geq 0$ after restriction to $\mathrm{Int}\{G_\phi < \infty\}$. Let

$$g'(m) = \begin{cases} G_{\ell_1}(m) - G_\phi(m), & m \in \mathrm{Int}\{G_\phi < \infty\}; \\ \infty, & m \in M_{\mathbb{R}} \setminus \mathrm{Int}\{G_\phi < \infty\}. \end{cases}$$

Note that for $m \in \mathrm{Int}\{G_\phi < \infty\}$, we actually have

$$g'(m) = \lim_{t \to \infty} t^{-1} G_{\ell_t}(m).$$

Hence, g' is a proper convex function on $M_{\mathbb{R}}$. Define $g = \mathrm{cl}\, g'$. Then g is a proper closed convex function on $M_{\mathbb{R}}$, and $\overline{\mathrm{Dom}\, g} = \Delta(\theta, \phi)$. We have therefore a map from (1) to (2).

It remains to argue that this map is the converse of the preceding map from (2) to (1). The non-trivial point is to verify (12.32) holds for the g we just constructed. Fix $t > 0$, we need to show that

$$G_{\ell_t} = G_\phi + tg.$$

This holds on $\mathrm{Int}\{G_\phi < \infty\}$ by Corollary 12.3.5 and the definition of g, but then it holds everywhere thanks to Proposition A.1.4. Our assertion follows. □

In particular, we can make the corresponding test curve more explicit.

Corollary 12.3.7 *Let $\ell \in \mathcal{R}_{\mathrm{tor}}(X, \theta; \phi)$ and g be the function as in Corollary 12.3.6. Then*

$$\ell^*_{\max} = -\inf_{M_{\mathbb{R}}} g, \tag{12.33}$$

and for each $\tau < -\inf_{M_{\mathbb{R}}} g$, the function ℓ^*_τ is toric-invariant, and

$$G_{\ell^*_\tau}(m) = \begin{cases} G_\phi(m), & \text{if } g(m) \leq -\tau; \\ \infty, & \text{otherwise.} \end{cases}$$

In particular, for such τ,

$$\Delta\left(\theta, \ell^*_\tau\right) = \{g \leq -\tau\}.$$

In other words,

$$\left(\Delta\left(\theta, \ell^*_\tau\right)\right)_{\tau < -\inf_{M_{\mathbb{R}}} g}$$

is the inverse Legendre transform of $-g$ using the terminology of Definition 10.5.3.

As a consequence, in the toric setting, the Ross–Witt Nyström correspondence Theorem 9.2.1 reduces essentially to Theorem 10.5.1.

Proof Fix $\tau \in \mathbb{R}$, then
$$\ell^*_\tau = \inf_{t>0} (\ell_t - t\tau)$$
is clearly toric-invariant. Therefore,
$$F_{\ell^*_\tau} = \inf_{t>0} \left(F_{\ell_t} - t\tau\right).$$
Fix $m \in M_\mathbb{R}$, we compute
$$\begin{aligned}
G_{\ell^*_\tau}(m) &= \sup_{n\in N_\mathbb{R}} \left(\langle m,n\rangle - F_{\ell^*_\tau}(n)\right)\\
&= \sup_{t>0}\sup_{n\in N_\mathbb{R}} \left(\langle m,n\rangle - F_{\ell_t}(n) + t\tau\right)\\
&= \sup_{t>0} \left(G_{\ell_t}(m) + t\tau\right)\\
&= G_\phi(m) + \sup_{t>0} t\,(g(m)+\tau)\\
&= \begin{cases} G_\phi(m), & \text{if } g(m)+\tau \leq 0;\\ \infty, & \text{otherwise.}\end{cases}
\end{aligned}$$
Our assertions follow. □

Next we consider the trace operator studied in Chapter 8. We wish to understand the trace operator in the toric situation. For this purpose, we will need to fix a T-invariant subvariety $Y \subseteq X$. Let σ be the corresponding cone in Σ, and let $Q_\sigma \subseteq P_D$ be the subset defined in (12.6). The cocharacter lattice of Y is given by
$$N(\sigma) \coloneqq N/N \cap \langle\sigma\rangle,$$
where $\langle\sigma\rangle$ is the linear span of σ. See [CLS11, (3.2.6)]. In particular, we have a canonical identification of the character lattice $M(\sigma)$ of Y:
$$M(\sigma) = \sigma^\perp \cap M,$$
which is compatible with our previous notation (12.2). Let $i_\sigma\colon M(\sigma) \to M$ be the inclusion map. Let T_Y be the torus of Y. Then we have a natural surjection $q_T : T \to T_Y$. In particular, the tropicalization map
$$\mathrm{Trop}\colon T(\mathbb{C}) \to N_\mathbb{R}$$
descends to the tropicalization map of Y:
$$\mathrm{Trop}_Y : T_Y(\mathbb{C}) \to N(\sigma)_\mathbb{R}.$$
Choose a representative $m_\sigma \in M$ of the Cartier datum along σ. Then $m - m_\sigma \in M(\sigma)_\mathbb{R}$ for every $m \in Q_\sigma$. We let

$$D_Y = \sum_{\substack{\rho \in \Sigma(1) \\ \rho \npreceq \sigma}} \left(a_\rho + \langle m_\sigma, u_\rho \rangle\right) D_\rho|_Y ,$$

where $\rho \npreceq \sigma$ means that ρ is not a face of σ. Then $O_Y(D_Y) = L|_Y$. Equivalently, D_Y is the divisor associated to the rational section

$$s_{D_Y} := (\chi^{m_\sigma} s_D)|_Y$$

of $L|_Y$. Write P_{D_Y} for the polytope of D_Y. With this choice, $Q_\sigma - m_\sigma$ is naturally contained in P_{D_Y}.

Theorem 12.3.2 *With the above choice of representative m_σ, let $\varphi \in \mathrm{PSH}_{\mathrm{tor}}(X,\theta)_{>0}$ satisfy $\nu(\varphi, Y) = 0$. If $\mathrm{Tr}^\theta_Y(\varphi)$ is defined and $\mathrm{vol}(\theta|_Y, \mathrm{Tr}^\theta_Y(\varphi)) > 0$[6], then*

$$\Delta\left(\theta|_Y, \mathrm{Tr}^\theta_Y(\varphi)\right) = (\Delta(\theta,\varphi) \cap Q_\sigma) - m_\sigma$$

as subsets of $M(\sigma)_\mathbb{R}$.

Observe that the condition $\nu(\varphi, Y) = 0$ means exactly that $\Delta(\theta,\varphi) \cap Q_\sigma \neq \varnothing$ by Corollary 12.2.2.

Since Y itself is a smooth toric variety, the preceding constructions of X all apply to Y. We briefly summarize the situation in Table 12.1.

Notions for X	Notions for Y
N	$N(\sigma)$
M	$M(\sigma)$
Σ	$\mathrm{Star}(\sigma)$
D	D_Y
L	$L\|_Y$
h	$h\|_Y$
θ	$\theta\|_Y$
Trop	Trop_Y
P_D	P_{D_Y}
s_D	s_{D_Y}

Table 12.1 The correspondence between X and Y

Recall that $\mathrm{Star}(\sigma)$ is the fan in $N(\sigma)_\mathbb{R}$ consisting of $\bar{\tau}$ for all faces $\tau \in \Sigma$ containing σ, where $\bar{\tau}$ is the image of τ in $N(\sigma)_\mathbb{R}$. See [CLS11, Proposition 3.2.7].

Proof The idea of the proof is that since Lemma 10.4.7 and Remark 10.4.1 describe how partial Okounkov bodies behave under restriction, and Theorem 12.2.1 compares partial Okounkov bodies with Newton bodies, we can deduce the behavior of Newton bodies under restriction as well.

[6] Note that $\mathrm{Tr}^\theta_Y(\varphi) \in \mathrm{PSH}_{\mathrm{tor}}(Y, \theta|_Y)$.

First we note that by our assumption, $L|_Y$ is a big line bundle. In particular, if we set $r = \dim \sigma$, then $\dim Y = n - r$.

We also note that $\mathrm{Tr}_Y^\theta(\varphi)$ is $T_{Y,c}$-invariant. Indeed, for any $g \in T_c$, let g_Y be the induced automorphism of Y. Then $g_Y^* \mathrm{Tr}_Y(\varphi)$ is again a trace of $g^*\varphi = \varphi$ along Y. Hence $g_Y^* \mathrm{Tr}_Y(\varphi) \sim_P \mathrm{Tr}_Y(\varphi)$ by Proposition 8.1.2. Since $\theta|_Y$ is $T_{Y,c}$-invariant and the P-envelope depends only on the P-equivalence class, we get $g_Y^* \mathrm{Tr}_Y^\theta(\varphi) = \mathrm{Tr}_Y^\theta(\varphi)$. As the image of T_c in $\mathrm{Aut}(Y)$ is $T_{Y,c}$, the assertion follows.

For this purpose, let σ^0 be an n-dimensional cone of Σ containing σ. The image $\overline{\sigma^0}$ in $N(\sigma)$ is then an $(n-r)$-dimensional cone of $\mathrm{Star}(\sigma)$. We shall use these cones as the reference cones while defining the partial Okounkov bodies.

We list the rays in $\sigma^0(1)$ as follows:

$$\rho_1, \dots, \rho_n, \tag{12.34}$$

where $\rho_1, \dots, \rho_r \in \sigma(1)$ and hence $\rho_{r+1}, \dots, \rho_n \notin \sigma(1)$. In particular, the images

$$\overline{\rho_{r+1}}, \dots, \overline{\rho_n} \tag{12.35}$$

of the latter give a list of $\overline{\sigma^0}(1)$.

We construct the flag $Y_\bullet$ on X using the rays (12.34) and the flag $Z_\bullet$ on Y using the rays (12.35). Note that

$$Z_i = Y_{r+i},$$

where $i = 1, \dots, n-r$.

Next we compute the Cartier data associated with $\overline{\sigma^0}$ for the divisor D_Y. By definition, $m_{\overline{\sigma^0}} \in M(\sigma)$ is the unique element satisfying

$$\langle m_{\overline{\sigma^0}}, u_{\overline{\rho_j}} \rangle = -a_{\rho_j} - \langle m_\sigma, u_{\rho_j} \rangle$$

for all $j = r+1, \dots, n$. Since $m_{\sigma^0} - m_\sigma$ vanishes on σ, it lies in $M(\sigma)$, and the preceding formula gives

$$m_{\overline{\sigma^0}} = m_{\sigma^0} - m_\sigma.$$

Let $\Phi \colon M \to \mathbb{Z}^n$ and $\Psi \colon M(\sigma) \to \mathbb{Z}^{n-r}$ be defined as

$$\begin{aligned}
\Phi(m) &= \big(\langle m - m_{\sigma^0}, u_{\rho_1}\rangle, \dots, \langle m - m_{\sigma^0}, u_{\rho_n}\rangle\big) \\
&= \big(\langle m, u_{\rho_1}\rangle, \dots, \langle m, u_{\rho_n}\rangle\big) + (a_{\rho_1}, \dots, a_{\rho_n}), \\
\Psi(m) &= \Big(\langle m - m_{\overline{\sigma^0}}, u_{\overline{\rho_{r+1}}}\rangle, \dots, \langle m - m_{\overline{\sigma^0}}, u_{\overline{\rho_n}}\rangle\Big)
\end{aligned}$$

Observe that for $i = r+1, \dots, n$, we have

$$u_{\overline{\rho_i}} = u_{\rho_i} \mod N \cap \langle \sigma \rangle,$$

so

$$\begin{aligned}
\Psi(m) = \;&\big(\langle m, u_{\rho_{r+1}}\rangle, \dots, \langle m, u_{\rho_n}\rangle\big) \\
&+ \big(a_{\rho_{r+1}} + \langle m_\sigma, u_{\rho_{r+1}}\rangle, \dots, a_{\rho_n} + \langle m_\sigma, u_{\rho_n}\rangle\big)
\end{aligned}$$

for $m \in M(\sigma)$. Therefore, we have a commutative diagram

$$\begin{tikzcd} m_\sigma + M(\sigma)_{\mathbb{R}} \arrow[r, "\Phi_{\mathbb{R}}"] \arrow[d, "m \mapsto m - m_\sigma"'] & \mathbb{R}^n \arrow[d, "L"] \\ M(\sigma)_{\mathbb{R}} \arrow[r, "\Psi_{\mathbb{R}}"] & \mathbb{R}^{n-r}, \end{tikzcd}$$

where $L\colon \mathbb{R}^n \to \mathbb{R}^{n-r}$ is the map

$$(b_1, \ldots, b_n) \mapsto (b_{r+1}, \ldots, b_n).$$

By Theorem 12.2.1, we have

$$\Phi_{\mathbb{R}}\Big(\Delta(\theta,\varphi)\Big) = \Delta_{Y_\bullet}(\theta,\varphi), \quad \Psi_{\mathbb{R}}\Big(\Delta(\theta|_Y, \mathrm{Tr}_Y^\theta(\varphi))\Big) = \Delta_{Z_\bullet}(\theta|_Y, \mathrm{Tr}_Y^\theta(\varphi)).$$

While by Lemma 10.4.7 and Remark 10.4.1,

$$\begin{aligned} \Delta_{Z_\bullet}\Big(\theta|_Y, \mathrm{Tr}_Y^\theta(\varphi)\Big) =& L\Big(\Delta_{Y_\bullet}(\theta,\varphi) \cap (\{0\}^r \times \mathbb{R}^{n-r})\Big) \\ =& L\Big(\Phi_{\mathbb{R}}(\Delta(\theta,\varphi)) \cap (\{0\}^r \times \mathbb{R}^{n-r})\Big) \\ =& L \circ \Phi_{\mathbb{R}}\Big(\Delta(\theta,\varphi) \cap Q_\sigma\Big) \\ =& \Psi_{\mathbb{R}}\Big((\Delta(\theta,\varphi) \cap Q_\sigma) - m_\sigma\Big). \end{aligned}$$

In the third line, we used $\Delta(\theta,\varphi) \subseteq P_D$ from Theorem 12.2.2, the identity (12.6), and the fact that $m_{\sigma^0} - m_\sigma$ vanishes on σ. Since $\Psi_{\mathbb{R}}$ is an affine isomorphism, our assertion follows. □

Corollary 12.3.8 *For any $\varphi \in \mathrm{PSH}_{\mathrm{tor}}(X,\theta)$ with $\nu(\varphi, Y) = 0$, we have*

$$\mathrm{vol}\left(\theta|_Y + \mathrm{dd^c}\,\mathrm{Tr}_Y^\theta(\varphi)\right) = (\dim Y)!\, \mathrm{vol}_{\dim Y}\left(\Delta(\theta,\varphi) \cap Q_\sigma\right). \tag{12.36}$$

The left-hand side of (12.36) is understood as 0 if $\mathrm{Tr}_Y^\theta(\varphi)$ is not defined. On the right-hand side, we translate $\Delta(\theta,\varphi) \cap Q_\sigma$ by $-m_\sigma$ and compute the $\dim Y$-dimensional Lebesgue measure on $M(\sigma)_{\mathbb{R}}$, normalized so that a fundamental parallelepiped for $M(\sigma)$ has volume 1.

Proof Take a toric-invariant ample line bundle H on X and a toric-invariant Kähler metric $\omega \in c_1(H)$. For $\epsilon \in \mathbb{Q}_{>0}$, write $Q_{\sigma,\epsilon}$ for the face corresponding to σ for the divisor $D + \epsilon H$, with the same convention on the Cartier datum along σ. Since $\theta_\varphi + \epsilon\omega$ is a toric Kähler current and $\nu(\varphi, Y) = 0$, its trace along Y is defined and has positive volume. Hence Theorem 12.3.2 gives

$$\mathrm{vol}\left(\theta|_Y + \epsilon\omega|_Y + \mathrm{dd^c}\,\mathrm{Tr}_Y^{\theta+\epsilon\omega}(\varphi)\right) = (\dim Y)!\,\mathrm{vol}_{\dim Y}\Big(\Delta(\theta+\epsilon\omega,\varphi) \cap Q_{\sigma,\epsilon}\Big).$$

Moreover, by Lemma A.3.1,

$$\Delta(\theta + \epsilon\omega, \varphi) \cap Q_{\sigma,\epsilon} = (\Delta(\theta, \varphi) \cap Q_\sigma) + \epsilon Q_\sigma^H,$$

where Q_σ^H denotes the corresponding face of the polytope of H. Indeed, the defining inequalities along the rays of σ are non-negative on both summands, so equality for their sum forces equality on each summand. Combining this equation with Example 8.1.6 and letting $\epsilon \to 0+$, we find (12.36). □

As a corollary, we obtain an elegant characterization of the augmented base locus in the toric setting.

Corollary 12.3.9 *The augmented base locus (with the reduced complex structure) of D is a toric-invariant (possibly reducible) subvariety given by the union of the $V(\tau)$'s, where τ runs over the elements in Σ such that*

$$\dim Q_\tau < n - \dim \tau\,^7.$$

Intuitively, we should think of $n - \dim \tau$ as the *expected* dimension of Q_τ. This corollary says that Q_τ fails to attain the expected dimension if and only if it corresponds to a subvariety in the augmented base locus.

With the help of some knowledge of convex geometry, we can also deduce this corollary from Corollary 12.1.1.

Proof Apply Corollary 12.3.8 to $\varphi = V_\theta$. Together with Nakamaye's theorem [ELM+09, Theorem 5.7] and Theorem 8.3.1, this says that $V(\tau)$ lies in the augmented base locus exactly when the $\dim V(\tau)$-dimensional volume of Q_τ is zero, which is equivalent to $\dim Q_\tau < n - \dim \tau$. □

Example 12.3.1 Let us consider the example Example 12.1.2 again, we consider the subvariety corresponding to σ_7. We have

$$Q_{\sigma_7} = \left\{ m \in P_D : \langle m, (1,1) \rangle = 0 \right\}.$$

The situation is explained in Fig. 12.5. We find that $\dim Q_{\sigma_7} = 0$, but the expected dimension is $2 - 1 = 1$. So the corresponding divisor, namely the exceptional divisor on $\mathrm{Bl}_0\mathbb{P}^2$ is in the non-Kähler locus. Similarly, we can verify that the non-Kähler locus consists only of this divisor.

Finally, we compute the mixed volumes of Newton bodies.

Theorem 12.3.3 *Take T-invariant big divisors $D_1, \ldots, D_n$ on X. Consider T_c-invariant closed positive $(1,1)$-currents T_i in $c_1(\mathcal{O}_X(D_i))$ for $i = 1, \ldots, n$. Then*

$$n!\,\mathrm{vol}\Big(\Delta(T_1), \ldots, \Delta(T_n)\Big) = \mathrm{vol}\,(T_1, \ldots, T_n)\,. \tag{12.37}$$

The mixed volume of currents is studied in Section 7.3.

---

[7] Here we understand that $\dim \varnothing = -\infty$.

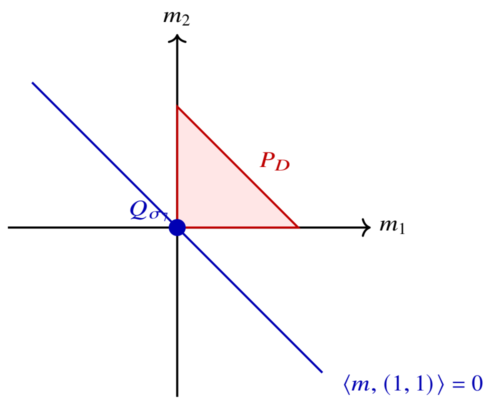

Fig. 12.5 The face $Q_{\sigma_7} = \{m \in P_D : \langle m, (1,1)\rangle = 0\}$ is the single point $\{(0,0)\}$, of dimension 0 rather than the expected 1.

Proof It suffices to prove the following assertion:

$$\Delta(T_1 + T_2) = \Delta(T_1) + \Delta(T_2).^{8} \tag{12.38}$$

In fact, assuming this, by induction, for each tuple of non-negative integers $C_1, \ldots, C_n$, we have

$$\Delta\left(\sum_{i=1}^{n} C_i T_i\right) = \sum_{i=1}^{n} C_i \Delta(T_i).$$

Comparing the volumes, we have

$$\operatorname{vol} \Delta\left(\sum_{i=1}^{n} C_i T_i\right) = \sum_{i_1,\ldots,i_n=1}^{n} C_{i_1} \cdots C_{i_n} \operatorname{vol}\left(\Delta\left(T_{i_1}\right), \ldots, \Delta\left(T_{i_n}\right)\right).$$

On the other hand, by Proposition 12.2.1, the $\mathcal{I}$-goodness of toric singularities Example 7.4.1, and Proposition 7.3.1, we have

$$n! \operatorname{vol} \Delta\left(\sum_{i=1}^{n} C_i T_i\right) = \operatorname{vol}\left(\sum_{i=1}^{n} C_i T_i\right) = \sum_{i_1,\ldots,i_n=1}^{n} C_{i_1} \cdots C_{i_n} \operatorname{vol}\left(T_{i_1}, \ldots, T_{i_n}\right).$$

Comparing the coefficients of the above two expressions, we conclude (12.37).

It only remains to establish (12.38). For this purpose, we fix smooth T_c-invariant Hermitian metrics h_1 and h_2 on $\mathcal{O}_X(D_1)$ and $\mathcal{O}_X(D_2)$ respectively. Correspondingly, take $h_1 \otimes h_2$ as the reference Hermitian metric on $\mathcal{O}_X(D_1 + D_2)$. Write

$$\theta_1 = \mathrm{dd^c} h_1, \quad \theta_2 = \mathrm{dd^c} h_2.$$

[8] This is slightly problematic when $n = 1$, since T_2 is not defined *stricto sensu*. We take D_2, T_2 satisfying the same assumptions as D_1, T_1 in this case.

It follows from the definition that $F_{\theta_1} + F_{\theta_2}$ is a candidate for $F_{\theta_1+\theta_2}$; see Lemma 12.2.1 and (12.9). Take $\varphi_i \in \mathrm{PSH}_{\mathrm{tor}}(X, \theta_i)$ with $T_i = \theta_i + \mathrm{dd^c}\varphi_i$ for $i = 1, 2$. Then $F_{\varphi_1} + F_{\varphi_2}$ is a candidate for $F_{\varphi_1+\varphi_2}$; see (12.10). Therefore, (12.38) follows from the equality of closed gradient images proved in Lemma A.3.1. □

Chapter 13
Non-Archimedean pluripotential theory

A good theorem lasts forever. Once proved, it will always stay proved, and other mathematicians are free to use it and build on it as they please, sometimes to great effect.
— John Tate[a]

[a] John Torrence Tate Jr. (1925–2019), the grandfather of Dustin Clausen, was one of the greatest minds in the whole history of America. However, his aversion to publishing papers arguably impeded the progress of the development of mathematics to some extent. For example, his foundational work on rigid non-Archimedean geometry was written in 1962, but was not available to the public until 1971.

This chapter develops a non-Archimedean pluripotential theory on a compact Kähler manifold X, parallel to and compatible with the Boucksom–Jonsson theory in the algebraic setting [BJ22a]. Our approach is grounded in the theory of $\mathcal{I}$-good singularities (Chapter 7) and test curves (Chapter 9): non-Archimedean metrics on a cohomology class $[\theta]$ are defined as suitable inverse systems of $\mathcal{I}$-model test curves, and the construction is shown to extend the algebraic Boucksom–Jonsson framework in the projective case.

The motivating observation is the following. The Ross–Witt Nyström correspondence Theorem 9.2.1 identifies the geodesic rays in $\mathrm{PSH}(X,\theta)$ with P-equivalence classes of concave curves of model potentials. Restricting to the $\mathcal{I}$-equivalence side, one expects a similar correspondence between $\mathcal{I}$-equivalence classes of rays and $\mathcal{I}$-model test curves. The latter form the basic "building blocks" of a non-Archimedean potential theory.

The chapter is organized as follows.

Section 13.1 introduces the space of non-Archimedean potentials. We define

$$\mathrm{PSH}^{\mathrm{NA}}(X,\theta) \coloneqq \varprojlim_{\omega\in\mathrm{K\ddot{a}h}(X)} \mathrm{PSH}^{\mathrm{NA}}(X,\theta+\omega)_{>0},$$

the inverse limit being taken along the $\mathcal{I}$-projection transition maps $P_{\theta+\omega}[\,\cdot\,]_{\mathcal{I}}$ from Proposition 9.3.4, and prove the basic properties.

Section 13.2 develops the basic operations on non-Archimedean potentials: sums (Definition 13.2.2), additive constants and translation invariance (Propositions 13.2.6 and 13.2.7), maxima and Lorentzian-type combinations (Definition 13.2.4), monotonicity of the non-Archimedean volume (Proposition 13.2.2), and a Choquet-type result (Proposition 13.2.11). These operations are the non-Archimedean counterparts of the operations on test curves developed in Section 9.4.

Section 13.3 constructs the Duistermaat–Heckman measure $\mathrm{DH}(\Gamma)$ of a non-Archimedean metric $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta)$. The construction extends many known constructions in the literature.

Section 13.4 establishes the comparison with the Boucksom–Jonsson algebraic theory [BJ22a]. After recalling their construction in the projective setting, we construct the analytification of X together with the analytification of a non-Archimedean metric in our sense, and define a class of *piecewise linear* non-Archimedean metrics (Definition 13.4.5) which are dense in the space of all non-Archimedean potentials (Corollary 13.4.1). The comparison theorem Theorem 13.4.3 then identifies our $\mathrm{PSH}^{\mathrm{NA}}(X, \theta)$ with the Boucksom–Jonsson space of psh non-Archimedean metrics on projective manifolds, in a manner compatible with all volume, mass and trace structures (Theorem 13.4.5).

A closely related theory has been developed independently by Mesquita-Piccione [MP25], where a Berkovich-like analytification of a compact Kähler manifold is constructed; we refer to that paper for the comparison with our approach.

This chapter relies heavily on the theory of $\mathcal{I}$-model test curves from Section 9.3 and on the operations on test curves from Section 9.4. It is in principle independent of Chapters 10 to 12, and the reader interested only in the comparison theorem may proceed directly to Section 13.4 after reading Section 13.1.

13.1 The definition of non-Archimedean metrics

Let X be a connected compact Kähler manifold of dimension n. Let $\mathrm{K\ddot{a}h}(X)$ be the set of Kähler forms on X with the partial order given as follows: We say $\omega \preceq \omega'$ if $\omega \geq \omega'$. Note that the partially ordered set $\mathrm{K\ddot{a}h}(X)$ is a directed set.

Let θ be a closed smooth real $(1,1)$-form.

Definition 13.1.1 We define

$$\mathrm{PSH}^{\mathrm{NA}}(X, \theta) = \varprojlim_{\omega \in \mathrm{K\ddot{a}h}(X)} \mathrm{PSH}^{\mathrm{NA}}(X, \theta + \omega)_{>0}$$ [1]

in the category of sets, where the transition maps are given as follows: Suppose that $\omega, \omega' \in \mathrm{K\ddot{a}h}$ and $\omega \geq \omega'$. Then the transition map is defined in Proposition 9.3.4:

$$P_{\theta+\omega}[\bullet]_{\mathcal{I}} : \mathrm{PSH}^{\mathrm{NA}}(X, \theta + \omega')_{>0} \to \mathrm{PSH}^{\mathrm{NA}}(X, \theta + \omega)_{>0}. \tag{13.1}$$

Note that the transition maps are compatible:

$$P_{\theta+\omega''}[P_{\theta+\omega'}[\varphi]_{\mathcal{I}}]_{\mathcal{I}} = P_{\theta+\omega''}[\varphi]_{\mathcal{I}}, \quad \forall \varphi \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta + \omega)_{>0}$$

[1] The cumbersome projective limit can be avoided if instead of relying on the language of quasi-plurisubharmonic functions, we use that of augmented nef b-divisors developed in Definition 11.1.6 instead. But given the fact that these two formulations are completely equivalent to each other by Corollary 11.1.4, we just stick to the slightly more traditional language here.

for $\omega \leq \omega' \leq \omega''$, since both sides are $\mathcal{I}$-model and are $\mathcal{I}$-equivalent to φ.

Recall that $\mathrm{PSH}^{\mathrm{NA}}(X, \theta)_{>0}$ is defined in Definition 9.3.1.

In general, when we denote an element in $\mathrm{PSH}^{\mathrm{NA}}(X, \theta)$ by Γ, its component in $\mathrm{PSH}^{\mathrm{NA}}(X, \theta + \omega)_{>0}$ ($\omega \in \mathrm{K\ddot{a}h}(X)$) will be written either as $\Gamma^{\theta+\omega}$ or as $P_{\theta+\omega}[\Gamma]_{\mathcal{I}}$.

Note that $\Gamma^{\theta+\omega}_{\max}$ is independent of the choice of $\omega \in \mathrm{K\ddot{a}h}(X)$. We denote this common value by $\Gamma_{\max}$.

Remark 13.1.1 Thanks to Proposition 9.3.2, for any other θ' representing $[\theta]$, we have a canonical bijection

$$\mathrm{PSH}^{\mathrm{NA}}(X, \theta) \xrightarrow{\sim} \mathrm{PSH}^{\mathrm{NA}}(X, \theta').$$

Moreover, these bijections satisfy the cocycle condition. If we view the set of closed real smooth $(1,1)$-forms representing $[\theta]$ as a category with a unique morphism between any two objects, then we can define

$$\mathrm{PSH}^{\mathrm{NA}}(X, [\theta]) = \varprojlim_{\theta} \mathrm{PSH}^{\mathrm{NA}}(X, \theta).$$

This definition is independent of the choice of the explicit representative of the cohomology class $[\theta]$.

However, given the fact that our notation is already quite heavy, we stick to the set $\mathrm{PSH}^{\mathrm{NA}}(X, \theta)$. The constructions below are independent of the choice of θ within its cohomology class.

Proposition 13.1.1 *Let $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta)$. Then given $\omega, \omega' \in \mathrm{K\ddot{a}h}(X)$ with $\omega \geq \omega'$, we have*

$$P_{\theta+\omega}\left[\Gamma^{\theta+\omega'}_{-\infty}\right] = P_{\theta+\omega}\left[\Gamma^{\theta+\omega'}_{-\infty}\right]_{\mathcal{I}} = \Gamma^{\theta+\omega}_{-\infty}.$$

Proof Since for any $\tau < \Gamma_{\max}$, the potential $\Gamma^{\theta+\omega'}_{\tau}$ is $\mathcal{I}$-good by Example 7.1.2, it follows that

$$P_{\theta+\omega}\left[\Gamma^{\theta+\omega'}_{\tau}\right] = P_{\theta+\omega}\left[\Gamma^{\theta+\omega'}_{\tau}\right]_{\mathcal{I}} = \Gamma^{\theta+\omega}_{\tau}$$

for all $\tau < \Gamma_{\max}$. Our assertion follows from Proposition 3.1.13 and Proposition 3.2.14. □

Proposition 13.1.2 *There is a natural injective map*

$$\mathrm{PSH}^{\mathrm{NA}}(X, \theta)_{>0} \hookrightarrow \mathrm{PSH}^{\mathrm{NA}}(X, \theta), \quad \Gamma \mapsto \left(P_{\theta+\omega}[\Gamma]_{\mathcal{I}}\right)_{\omega \in \mathrm{K\ddot{a}h}(X)}.$$

In the sequel, we do not distinguish an element of $\mathrm{PSH}^{\mathrm{NA}}(X, \theta)_{>0}$ from its image in $\mathrm{PSH}^{\mathrm{NA}}(X, \theta)$. Note that given $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta)_{>0}$, the value of $\Gamma_{\max}$ does not depend on whether we view it as an element of $\mathrm{PSH}^{\mathrm{NA}}(X, \theta)_{>0}$ or of $\mathrm{PSH}^{\mathrm{NA}}(X, \theta)$.

Proof The map is well-defined by the compatibility of the transition maps. It suffices to argue its injectivity. Suppose that $\Gamma, \Gamma' \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta)_{>0}$ and

$$P_{\theta+\omega}[\Gamma]_{\mathcal{I}} = P_{\theta+\omega}[\Gamma']_{\mathcal{I}}$$

for some Kähler form ω on X. Then $\Gamma_{\max} = \Gamma'_{\max}$ and for any $\tau < \Gamma_{\max}$, we have

$$\Gamma_\tau \sim_{\mathcal{I}} \Gamma'_\tau$$

by Proposition 6.1.3. It follows again from Proposition 6.1.3 that

$$\Gamma_\tau = \Gamma'_\tau.$$

This completes the proof. □

Definition 13.1.2 Let $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta)$. We define its *volume* as follows:

$$\operatorname{vol} \Gamma := \lim_{\omega \in \mathrm{K\ddot{a}h}(X)} \int_X \left(\theta + \omega + \mathrm{dd^c} \Gamma_{-\infty}^{\theta+\omega}\right)^n \in [0, \infty).$$

Observe that the net is decreasing, so the limit exists.

Proposition 13.1.3 *Let* $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta)_{>0}$. *Then*

$$\operatorname{vol} \Gamma = \int_X (\theta + \mathrm{dd^c} \Gamma_{-\infty})^n .$$

Proof This follows from Proposition 3.1.12, Corollary 3.1.3 and Proposition 9.1.5. □

Lemma 13.1.1 *Let* $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta)$, *let* ω *be a Kähler form on* X, *and let* $0 < t \leq 1$. *Then for every* $\tau < \Gamma_{\max}$, *we have*

$$\begin{aligned} &\int_X \left(\theta + t\omega + \mathrm{dd^c} \Gamma_{-\infty}^{\theta+t\omega}\right)^n - \int_X \left(\theta + t\omega + \mathrm{dd^c} \Gamma_{\tau}^{\theta+t\omega}\right)^n \\ \leq &\int_X \left(\theta + \omega + \mathrm{dd^c} \Gamma_{-\infty}^{\theta+\omega}\right)^n - \int_X \left(\theta + \omega + \mathrm{dd^c} \Gamma_{\tau}^{\theta+\omega}\right)^n . \end{aligned}$$

Proof For $\sigma \in \{-\infty, \tau\}$, put

$$S_t^\sigma = \theta + t\omega + \mathrm{dd^c} \Gamma_\sigma^{\theta+t\omega}.$$

By the transition compatibility and Propositions 3.1.3 and 13.1.1 and Example 7.1.2, projecting $\Gamma_\sigma^{\theta+t\omega}$ to the class $\theta + \omega$ does not change the total mass computed in the larger class. Thus

$$\int_X \left(\theta + \omega + \mathrm{dd^c} \Gamma_{\sigma}^{\theta+\omega}\right)^n = \int_X \left(S_t^\sigma + (1-t)\omega\right)^n .$$

Expanding the right-hand side and subtracting the two identities for $\sigma = -\infty$ and $\sigma = \tau$, we get

$$
\begin{aligned}
&\int_X \left(\theta+\omega+\mathrm{dd^c}\Gamma^{\theta+\omega}_{-\infty}\right)^n - \int_X \left(\theta+\omega+\mathrm{dd^c}\Gamma^{\theta+\omega}_{\tau}\right)^n \\
&= \sum_{a=0}^{n} \binom{n}{a} (1-t)^{n-a} \left(\int_X (S_t^{-\infty})^a \wedge \omega^{n-a} - \int_X (S_t^{\tau})^a \wedge \omega^{n-a} \right).
\end{aligned}
$$

Since $\Gamma^{\theta+t\omega}_{-\infty} \geq \Gamma^{\theta+t\omega}_{\tau}$, each summand is non-negative by the monotonicity theorem for mixed non-pluripolar products. Keeping only the term $a=n$ gives the desired inequality. □

Lemma 13.1.2 *The image of the canonical injection*

$$
\mathrm{PSH}^{\mathrm{NA}}(X,\theta)_{>0} \hookrightarrow \mathrm{PSH}^{\mathrm{NA}}(X,\theta)
$$

is given by the set of $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)$ *with positive volume.*

Proof By Proposition 13.1.3, it is clear that the image of an element in $\mathrm{PSH}^{\mathrm{NA}}(X,\theta)_{>0}$ has positive volume.

Conversely, take $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)$ with positive volume. We want to construct $\Gamma' \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)_{>0}$ representing Γ.

Fix a Kähler form ω on X. Define

$$
\Gamma'_\tau := \lim_{k\to\infty} \Gamma^{\theta+k^{-1}\omega}_{\tau}, \qquad \tau < \Gamma_{\max}. \tag{13.2}
$$

We claim that it suffices to show

$$
\varliminf_{k\to\infty} \int_X \left(\theta+k^{-1}\omega+\mathrm{dd^c}\Gamma^{\theta+k^{-1}\omega}_{\tau}\right)^n > 0 \tag{13.3}
$$

for some $\tau < \Gamma_{\max}$. If this holds, then the argument of Lemma 9.1.1 implies that the same holds for all $\tau < \Gamma_{\max}$. Then Proposition 3.1.12 guarantees that $\Gamma' \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)_{>0}$ and represents Γ.

It remains to argue (13.3). Let $\epsilon = \operatorname{vol}\Gamma > 0$. Take $\tau < \Gamma_{\max}$ so that

$$
\int_X \left(\theta+\omega+\mathrm{dd^c}\Gamma^{\theta+\omega}_{-\infty}\right)^n - \int_X \left(\theta+\omega+\mathrm{dd^c}\Gamma^{\theta+\omega}_{\tau}\right)^n < \epsilon/2.
$$

Therefore, by Lemma 13.1.1, for any $k \geq 1$,

$$
\int_X \left(\theta+k^{-1}\omega+\mathrm{dd^c}\Gamma^{\theta+k^{-1}\omega}_{-\infty}\right)^n - \int_X \left(\theta+k^{-1}\omega+\mathrm{dd^c}\Gamma^{\theta+k^{-1}\omega}_{\tau}\right)^n < \epsilon/2.
$$

By the definition of $\operatorname{vol}\Gamma$, the first integral in the preceding line tends to ϵ as $k \to \infty$. Hence

$$
\varliminf_{k\to\infty} \int_X \left(\theta+k^{-1}\omega+\mathrm{dd^c}\Gamma^{\theta+k^{-1}\omega}_{\tau}\right)^n \geq \epsilon/2 > 0,
$$

which proves (13.3). □

Example 13.1.1 Given $\varphi \in \mathrm{PSH}(X, \theta)$, we can define an associated non-Archimedean metric $\Gamma^{\varphi} \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta)$ as follows:

(1) $\Gamma^{\varphi}_{\max} = 0$;
(2) for any $\omega \in \mathrm{K\ddot{a}h}(X)$ and any $\tau < 0$, we set

$$\Gamma^{\varphi, \theta+\omega}_{\tau} = P_{\theta+\omega}[\varphi]_{\mathcal{I}}.$$

Such non-Archimedean metrics are called *homogeneous* non-Archimedean metrics.

Observe that

$$\operatorname{vol}\Gamma^{\varphi} = \operatorname{vol}\theta_{\varphi}.$$

See Proposition 7.3.1 and the footnote there.

Definition 13.1.3 Let ω be a closed real smooth positive $(1,1)$-form on X. We define the map

$$P_{\theta+\omega}[\bullet]_{\mathcal{I}} : \mathrm{PSH}^{\mathrm{NA}}(X, \theta) \to \mathrm{PSH}^{\mathrm{NA}}(X, \theta + \omega)$$

as follows: Given $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta)$, we define $P_{\theta+\omega}[\Gamma]_{\mathcal{I}}$ as the element such that for any $\omega' \in \mathrm{K\ddot{a}h}(X)$, we have

$$P_{\theta+\omega}[\Gamma]^{\theta+\omega+\omega'}_{\mathcal{I}} = \Gamma^{\theta+\omega+\omega'}.$$

It is straightforward to check that under the identification of Proposition 13.1.2, the map $P_{\theta+\omega}[\bullet]_{\mathcal{I}}$ extends the map (13.1).

Proposition 13.1.4 *The maps $P_{\theta+\omega}[\bullet]_{\mathcal{I}}$ in Definition 13.1.3 together induce a bijection*

$$\mathrm{PSH}^{\mathrm{NA}}(X, \theta) \xrightarrow{\sim} \varprojlim_{\omega \in \mathrm{K\ddot{a}h}(X)} \mathrm{PSH}^{\mathrm{NA}}(X, \theta + \omega). \tag{13.4}$$

Proof It is a tautology that the maps $P_{\theta+\omega}[\bullet]_{\mathcal{I}}$ in Definition 13.1.3 are compatible with the transition maps. So the map (13.4) is well-defined. It is injective by the same argument as Proposition 13.1.2. We argue the surjectivity.

By unfolding the definitions, an object in the target of (13.4) is an assignment: With each $\omega \in \mathrm{K\ddot{a}h}(X)$, we associate a family $(\Gamma^{\omega,\omega'})_{\omega' \in \mathrm{K\ddot{a}h}(X)}$ satisfying:

(1) $\Gamma^{\omega,\omega'} \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta + \omega + \omega')_{>0}$ for each $\omega, \omega' \in \mathrm{K\ddot{a}h}(X)$;
(2) for each $\omega, \omega', \omega'' \in \mathrm{K\ddot{a}h}(X)$ satisfying $\omega'' \geq \omega'$, we have

$$P_{\theta+\omega+\omega''}\left[\Gamma^{\omega,\omega'}\right]_{\mathcal{I}} = \Gamma^{\omega,\omega''};$$

(3) for each $\omega, \omega', \omega'' \in \mathrm{K\ddot{a}h}(X)$ satisfying $\omega \leq \omega'$, we have

$$P_{\theta+\omega'+\omega''}\left[\Gamma^{\omega,\omega''}\right]_{\mathcal{I}} = \Gamma^{\omega',\omega''}.$$

The preimage of such an object is given by the family $(\Gamma^{\theta+\omega})_{\omega \in \mathrm{K\ddot{a}h}(X)}$ given by

$$\Gamma^{\theta+\omega} = \Gamma^{\omega/2,\omega/2}.$$

The fact that the image of Γ is as expected follows directly from the definitions. □

With an almost identical argument involving Proposition 3.1.12, we get

Proposition 13.1.5 *The maps $P_{\theta+\omega}[\bullet]_{\mathcal{I}}$ in Definition 13.1.3 and the injective maps Proposition 13.1.2 together induce bijections*

$$\mathrm{PSH}^{\mathrm{NA}}(X,\theta) \xrightarrow{\sim} \varprojlim_{\omega} \mathrm{PSH}^{\mathrm{NA}}(X,\theta+\omega)_{>0} \xrightarrow{\sim} \varprojlim_{\omega} \mathrm{PSH}^{\mathrm{NA}}(X,\theta+\omega), \tag{13.5}$$

where ω runs over either the partially ordered set of all smooth closed real positive $(1,1)$-forms with positive volume[2] *on X or* $\mathrm{K\ddot{a}h}(X)$.

Corollary 13.1.1 *Let $\pi\colon Y \to X$ be a proper bimeromorphic morphism from a compact Kähler manifold Y. Then π^* induces a bijection*

$$\mathrm{PSH}^{\mathrm{NA}}(X,\theta) \xrightarrow{\sim} \mathrm{PSH}^{\mathrm{NA}}(Y,\pi^*\theta).$$

Proof This follows immediately from Proposition 13.1.5. □

It is immediate to verify that π^* in Corollary 13.1.1 extends the map Proposition 9.3.3.

13.2 Operations on non-Archimedean metrics

Let X be a connected compact Kähler manifold of dimension n and $\theta, \theta', \theta''$ be closed real smooth $(1,1)$-forms on X representing big cohomology classes.

This section relies heavily on Section 9.4. We shall use the notions introduced in that section without further explanations.

Definition 13.2.1 Let $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)$, $\Gamma' \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta')$. We say $\Gamma \leq \Gamma'$ if for some $\omega \in \mathrm{K\ddot{a}h}(X)$, we have

$$\Gamma^{\theta+\omega} \leq \Gamma'^{\theta'+\omega}.$$

This notion is independent of the choice of ω thanks to Lemma 9.4.1.

Moreover, we have the following:

Proposition 13.2.1 *Let $\Gamma, \Gamma' \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)$ and ω be a closed smooth positive $(1,1)$-form on X. Then the following are equivalent:*

(1) $\Gamma \leq \Gamma'$;
(2) $P_{\theta+\omega}[\Gamma]_{\mathcal{I}} \leq P_{\theta+\omega}[\Gamma']_{\mathcal{I}}$.

Proof This follows immediately from Lemma 9.4.1. □

Observe that this definition coincides with the corresponding definition in Definition 9.4.1 when $\Gamma, \Gamma' \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)_{>0}$.

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[2] This partially ordered set is not directed.

Proposition 13.2.2 *Let $\Gamma, \Gamma' \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)$. Assume that $\Gamma \leq \Gamma'$. Then*

$$\operatorname{vol}\Gamma \leq \operatorname{vol}\Gamma'.$$

Proof It suffices to show that for any Kähler form ω on X, we have

$$\operatorname{vol}\Gamma^{\theta+\omega}_{-\infty} \leq \operatorname{vol}\Gamma'^{\theta+\omega}_{-\infty},$$

which is an immediate consequence of Theorem 2.4.3. □

Definition 13.2.2 Let $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)$ and $\Gamma' \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta')$. Then we define $\Gamma+\Gamma' \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta+\theta')$ as the unique element such that for any $\omega \in \mathrm{Käh}(X)$, we have

$$(\Gamma+\Gamma')^{\theta+\theta'+2\omega} = \Gamma^{\theta+\omega} + \Gamma'^{\theta'+\omega}.$$

This definition yields an element in $\mathrm{PSH}^{\mathrm{NA}}(X,\theta+\theta')$ by Lemma 9.4.3 and it extends the definition in Definition 9.4.2 by Lemma 9.4.3 as well.

Proposition 13.2.3 *Let $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)$ and $\Gamma' \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta')$. Suppose that ω, ω' are two smooth closed positive $(1,1)$-forms on X. Then*

$$P_{\theta+\omega+\theta'+\omega'}[\Gamma+\Gamma']_{\mathcal{I}} = P_{\theta+\omega}[\Gamma]_{\mathcal{I}} + P_{\theta'+\omega'}[\Gamma']_{\mathcal{I}}.$$

Proof This is a direct consequence of Lemma 9.4.3. □

Proposition 13.2.4 *The operation $+$ is commutative and associative: For any $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)$, $\Gamma' \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta')$ and $\Gamma'' \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta'')$, we have*

$$\Gamma+\Gamma' = \Gamma'+\Gamma, \quad (\Gamma+\Gamma')+\Gamma'' = \Gamma+(\Gamma'+\Gamma'').$$

Proof This is a direct consequence of Proposition 9.4.1. □

Definition 13.2.3 Let $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)$ and $C \in \mathbb{R}$. We define $\Gamma+C \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)$ as the unique element such that for any $\omega \in \mathrm{Käh}(X)$, we have

$$(\Gamma+C)^{\theta+\omega} = \Gamma^{\theta+\omega} + C.$$

It is obvious from Definition 9.4.3 that $\Gamma + C \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)$. It is also obvious that this definition extends Definition 9.4.3.

Proposition 13.2.5 *Let $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)$ and $C \in \mathbb{R}$. Suppose that ω is a smooth closed positive $(1,1)$-form on X. Then*

$$P_{\theta+\omega}[\Gamma]_{\mathcal{I}} + C = P_{\theta+\omega}[\Gamma+C]_{\mathcal{I}}.$$

Proof This is clear by definition. □

Proposition 13.2.6 *Let $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)$, $\Gamma' \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta')$ and $C, C' \in \mathbb{R}$. Then*

(1) $(\Gamma+\Gamma')+C = \Gamma+(\Gamma'+C) = (\Gamma+C)+\Gamma'$;

(2) $\Gamma + (C + C') = (\Gamma + C) + C'$.

Proof This is a direct consequence of Proposition 9.4.2. □

Proposition 13.2.7 *Let* $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta)$ *and* $C \in \mathbb{R}$. *Then*

$$\operatorname{vol} \Gamma = \operatorname{vol}(\Gamma + C).$$

Proof It suffices to show that for each Kähler form ω on X,

$$\operatorname{vol} \Gamma^{\theta+\omega}_{-\infty} = \operatorname{vol}(\Gamma + C)^{\theta+\omega}_{-\infty},$$

which is obvious. □

Definition 13.2.4 Let $\Gamma, \Gamma' \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta)$, we define $\Gamma \vee \Gamma' \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta)$ as the unique element such that for any $\omega \in \mathrm{Käh}(X)$, we have

$$(\Gamma \vee \Gamma')^{\theta+\omega} = \Gamma^{\theta+\omega} \vee \Gamma'^{\theta+\omega}.$$

It follows from Lemma 9.4.5 that $\Gamma \vee \Gamma' \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta)$ and this definition extends the corresponding definition in Definition 9.4.4.

Proposition 13.2.8 *Let* $\Gamma, \Gamma' \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta)$ *and* ω *be a closed smooth positive* $(1, 1)$*-form on* X. *Then*

$$P_{\theta+\omega}[\Gamma \vee \Gamma']_{\mathcal{I}} = P_{\theta+\omega}[\Gamma]_{\mathcal{I}} \vee P_{\theta+\omega}[\Gamma']_{\mathcal{I}}.$$

Proof This is a direct consequence of Lemma 9.4.5. □

Proposition 13.2.9 *The operation* $\vee$ *is commutative and associative.*

In particular, given a finite non-empty family $(\Gamma^i)_{i \in I}$ in $\mathrm{PSH}^{\mathrm{NA}}(X, \theta)$, we then define $\bigvee_{i \in I} \Gamma^i$ in the obvious way.

Proof This is a direct consequence of Corollary 9.4.1. □

Definition 13.2.5 Let $(\Gamma^i)_{i \in I}$ be a non-empty family in $\mathrm{PSH}^{\mathrm{NA}}(X, \theta)$. Assume that

$$\sup_{i \in I} \Gamma^i_{\max} < \infty. \tag{13.6}$$

Then we define $\sup^*_{i \in I} \Gamma^i \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta)$ as the unique element such that for any $\omega \in \mathrm{Käh}(X)$, we have

$$\left(\sup_{i \in I}{}^* \Gamma^i \right)^{\theta+\omega} = \sup_{i \in I}{}^* \Gamma^{i, \theta+\omega}.$$

It follows immediately from Lemma 9.4.7 that $\sup^*_{i \in I} \Gamma^i \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta)$ and this definition extends Definition 9.4.6. Moreover, this definition clearly extends Definition 13.2.4 as well.

Proposition 13.2.10 *Let* $(\Gamma^i)_{i\in I}$ *be a non-empty family in* $\mathrm{PSH}^{\mathrm{NA}}(X,\theta)$ *satisfying* (13.6). *Assume that* ω *is a closed smooth positive* $(1,1)$*-form on* X. *Then*

$$P_{\theta+\omega}\left[\sup_{i\in I}{}^*\Gamma^i\right]_{\mathcal{I}} = \sup_{i\in I}{}^* P_{\theta+\omega}\left[\Gamma^i\right]_{\mathcal{I}}.$$

Proof This is a direct consequence of Lemma 9.4.7. □

We also have a non-Archimedean version of Choquet's lemma.

Proposition 13.2.11 *Let* $(\Gamma^i)_{i\in I}$ *be a non-empty family in* $\mathrm{PSH}^{\mathrm{NA}}(X,\theta)$ *satisfying* (13.6). *Then there exists a countable subfamily* $I'\subseteq I$ *such that*

$$\sup_{i\in I}{}^*\Gamma^i = \sup_{i\in I'}{}^*\Gamma^i.$$

Proof Fix a Kähler form ω_0 on X. For each $m\geq 1$, Proposition 9.4.5 gives a countable subfamily $I_m\subseteq I$ such that

$$\sup_{i\in I}{}^* P_{\theta+m^{-1}\omega_0}[\Gamma^i]_{\mathcal{I}} = \sup_{i\in I_m}{}^* P_{\theta+m^{-1}\omega_0}[\Gamma^i]_{\mathcal{I}}.$$

Set $I'=\bigcup_{m\geq 1} I_m$. This is countable. We claim that it works.

Let $\omega'\in \mathrm{K\ddot{a}h}(X)$. Choose m so large that $\omega'\geq m^{-1}\omega_0$. By Proposition 13.2.10 and the compatibility of the transition maps,

$$\begin{aligned}
\sup_{i\in I}{}^* P_{\theta+\omega'}[\Gamma^i]_{\mathcal{I}} =& P_{\theta+\omega'}\left[\sup_{i\in I}{}^* P_{\theta+m^{-1}\omega_0}[\Gamma^i]_{\mathcal{I}}\right]_{\mathcal{I}}\\
=& P_{\theta+\omega'}\left[\sup_{i\in I'}{}^* P_{\theta+m^{-1}\omega_0}[\Gamma^i]_{\mathcal{I}}\right]_{\mathcal{I}}\\
=& \sup_{i\in I'}{}^* P_{\theta+\omega'}[\Gamma^i]_{\mathcal{I}}.
\end{aligned}$$

Thus the two non-Archimedean metrics have the same component in every class $\theta+\omega'$, and the assertion follows. □

Proposition 13.2.12 *Let* $(\Gamma^i)_{i\in I}$ *be a non-empty family in* $\mathrm{PSH}^{\mathrm{NA}}(X,\theta)$ *satisfying* (13.6). *Let* $C\in\mathbb{R}$. *Then*

$$\sup_{i\in I}{}^*(\Gamma^i + C) = \sup_{i\in I}{}^*\Gamma^i + C.$$

Suppose that $(\Gamma'^i)_{i\in I}$ *is another family in* $\mathrm{PSH}^{\mathrm{NA}}(X,\theta')$ *satisfying* (13.6). *Suppose that* $\Gamma^i\leq\Gamma'^i$ *for all* $i\in I$. *Then*

$$\sup_{i\in I}{}^*\Gamma^i \leq \sup_{i\in I}{}^*\Gamma'^i.$$

Proof This is an immediate consequence of Proposition 9.4.6. □

Proposition 13.2.13 *Let* $(\Gamma^i)_{i\in I}$ *be an increasing net in* $\mathrm{PSH}^{\mathrm{NA}}(X,\theta)$ *satisfying* (13.6). *Then*

$$\operatorname{vol}\left(\sup_{i\in I}{}^*\Gamma^i\right) = \lim_{i\in I} \operatorname{vol}\Gamma^i. \tag{13.7}$$

Proof The $\geq$ direction in (13.7) is a direct consequence of Proposition 13.2.2. It remains to prove the reverse inequality.

Note that (13.7) holds when $\operatorname{vol}\Gamma^i > 0$ for each $i \in I$, as a consequence of Proposition 9.4.3, Corollary 6.2.3 and Theorem 6.2.5.

In particular, for each Kähler form ω on X, we have

$$\operatorname{vol}\left(\sup_{i\in I}{}^*\Gamma^{i,\theta+\omega}\right) = \lim_{i\in I} \operatorname{vol}\Gamma^{i,\theta+\omega}.$$

For our purpose, we need to show that for any $\epsilon > 0$, we can find ω so that

$$\sup_{i\in I} \operatorname{vol}\Gamma^{i,\theta+\omega} < \sup_{i\in I} \operatorname{vol}\Gamma^i + \epsilon.$$

We shall show that it is possible to choose ω so that the stronger statement holds:

$$\operatorname{vol}\Gamma^{i,\theta+\omega} < \operatorname{vol}\Gamma^i + \epsilon, \quad \forall i \in I.$$

Equivalently, we need to choose ω so that for any Kähler form ω' on X dominated by ω, we have

$$\operatorname{vol}\Gamma^{i,\theta+\omega}_{-\infty} < \operatorname{vol}\Gamma^{i,\theta+\omega'}_{-\infty} + \epsilon/2, \quad \forall i \in I.$$

Fix a Kähler form ω_0 on X and a Kähler form Ω such that $\Omega \geq \theta + \omega_0$. It is enough to take $\omega = \delta\omega_0$ for some $\delta \in (0,1)$. If ω' is dominated by ω, then $\theta + \omega' \leq \Omega$. Set

$$T_i \coloneqq \theta + \omega' + \mathrm{dd^c}\Gamma^{i,\theta+\omega'}_{-\infty}.$$

Using Proposition 9.1.5 to compute the first volume with the same singularity type, we get

$$\begin{aligned}
\operatorname{vol}\Gamma^{i,\theta+\omega}_{-\infty} - \operatorname{vol}\Gamma^{i,\theta+\omega'}_{-\infty} &= \int_X (T_i + \omega - \omega')^n - \int_X T_i^n \\
&= \sum_{a=0}^{n-1} \binom{n}{a} \int_X T_i^a \wedge (\omega - \omega')^{n-a} \\
&\leq \sum_{a=0}^{n-1} \binom{n}{a} \int_X \Omega^a \wedge \omega^{n-a}.
\end{aligned}$$

In the last inequality we used the monotonicity of mixed non-pluripolar products, together with $0 \leq \omega - \omega' \leq \omega$ and $\theta + \omega' \leq \Omega$. We can choose ω so that the right-hand side is less than $\epsilon/2$. Our assertion then follows. □

Definition 13.2.6 Let $(\Gamma^i)_{i\in I}$ be a decreasing net in $\mathrm{PSH}^{\mathrm{NA}}(X,\theta)$. Assume that

$$\inf_{i\in I} \Gamma^i_{\max} > -\infty, \tag{13.8}$$

then we define $\inf_{i\in I}\Gamma^i \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)$ as the unique element such that for each $\omega\in \mathrm{Käh}(X)$, we have

$$\left(\inf_{i\in I}\Gamma^i\right)^{\theta+\omega} = \inf_{i\in I}\Gamma^{i,\theta+\omega}. \tag{13.9}$$

We observe that

$$\left(\inf_{i\in I}\Gamma^i\right)^{\theta+\omega} \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta+\omega)_{>0}.$$

This follows from Proposition 9.4.9. Moreover, by Lemma 9.4.9, we have $\inf_{i\in I}\Gamma^i \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)$, and this definition extends Definition 9.4.8.

In general,

$$\mathrm{vol}\left(\inf_{i\in I}\Gamma^i\right) \leq \lim_{i\in I}\mathrm{vol}\,\Gamma^i$$

as a consequence of Proposition 13.2.2. But the reverse inequality fails in general.

Proposition 13.2.14 *Let $(\Gamma^i)_{i\in I}$ be a decreasing net in $\mathrm{PSH}^{\mathrm{NA}}(X,\theta)$ satisfying (13.8). Assume that ω is a closed smooth positive $(1,1)$-form on X. Then*

$$P_{\theta+\omega}\left[\inf_{i\in I}\Gamma^i\right]_{\mathcal{I}} = \inf_{i\in I}P_{\theta+\omega}\left[\Gamma^i\right]_{\mathcal{I}}.$$

Proof This follows from Lemma 9.4.9. □

Proposition 13.2.15 *Let $(\Gamma^i)_{i\in I}$ be a decreasing net in $\mathrm{PSH}^{\mathrm{NA}}(X,\theta)$ satisfying (13.8). Let $C\in\mathbb{R}$. Then*

$$\inf_{i\in I}(\Gamma^i + C) = \inf_{i\in I}\Gamma^i + C.$$

Suppose that $(\Gamma'^i)_{i\in I}$ is another decreasing net in $\mathrm{PSH}^{\mathrm{NA}}(X,\theta')$ satisfying (13.8). Suppose that $\Gamma^i\leq\Gamma'^i$ for all $i\in I$. Then

$$\inf_{i\in I}\Gamma^i \leq \inf_{i\in I}\Gamma'^i.$$

Proof This is clear by definition. □

Definition 13.2.7 Let $\Gamma\in\mathrm{PSH}^{\mathrm{NA}}(X,\theta)$ and $\lambda\in\mathbb{R}_{>0}$. Then we define $\lambda\Gamma\in\mathrm{PSH}^{\mathrm{NA}}(X,\lambda\theta)$ as the unique element such that for any $\omega\in\mathrm{Käh}(X)$, we have

$$(\lambda\Gamma)^{\lambda\theta+\omega} = \lambda\Gamma^{\theta+\lambda^{-1}\omega}.$$

It follows immediately from Lemma 9.4.8 that $\lambda\Gamma\in\mathrm{PSH}^{\mathrm{NA}}(X,\lambda\theta)$ and this definition extends Definition 9.4.7.

Proposition 13.2.16 *Let $\Gamma\in\mathrm{PSH}^{\mathrm{NA}}(X,\theta)$ and $\lambda\in\mathbb{R}_{>0}$. Then for any closed smooth positive $(1,1)$-form ω on X, we have*

$$P_{\lambda\theta+\omega}[\lambda\Gamma]_{\mathcal{I}} = \lambda P_{\theta+\lambda^{-1}\omega}[\Gamma]_{\mathcal{I}}.$$

Proof This follows immediately from Lemma 9.4.8. □

Proposition 13.2.17 *Let $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)$, $\Gamma' \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta')$, $C \in \mathbb{R}$ and $\lambda, \lambda' > 0$, we have*

$$\begin{aligned}\lambda(\Gamma+\Gamma') =&\lambda\Gamma+\lambda\Gamma',\\ (\lambda\lambda')\Gamma =&\lambda(\lambda'\Gamma),\\ \lambda(\Gamma+C) =&\lambda\Gamma+\lambda C.\end{aligned}$$

Suppose that $(\Gamma^i)_{i\in I}$ is a non-empty family in $\mathrm{PSH}^{\mathrm{NA}}(X,\theta)$ satisfying (13.6). Then

$$\lambda\left(\sup_{i\in I}{}^*\Gamma^i\right) = \sup_{i\in I}{}^*(\lambda\Gamma^i).$$

If $(\Gamma^i)_{i\in I}$ is a decreasing net in $\mathrm{PSH}^{\mathrm{NA}}(X,\theta)$ satisfying (13.8), then

$$\lambda\left(\inf_{i\in I}\Gamma^i\right) = \inf_{i\in I}(\lambda\Gamma^i).$$

Proof Everything except the last assertion follows from Proposition 9.4.8. The last assertion follows directly from the definition of the trace operator. □

Proposition 13.2.18 *Let $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)$ and $\lambda \in \mathbb{R}_{>0}$. Then*

$$\mathrm{vol}\,(\lambda\Gamma) = \lambda^n \,\mathrm{vol}\,\Gamma.$$

Proof This is clear by definition. □

Definition 13.2.8 Let $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)$. Let $Y \subseteq X$ be an irreducible analytic subset. We say that the trace operator of Γ along Y is *well-defined* if

$$\nu\left(\Gamma_\tau^{\theta+\omega}, Y\right) = 0$$

for all sufficiently negative τ and every $\omega \in \mathrm{Käh}(X)$. We define

$$(\mathrm{Tr}_Y(\Gamma))_{\max} \coloneqq \sup\left\{\tau < \Gamma_{\max} : \nu\left(\Gamma_\tau^{\theta+\omega}, Y\right) = 0\right\}.$$

Here ω is any Kähler form on X; this number is independent of the choice of ω by the compatibility of the transition maps.

In this case, we define $\mathrm{Tr}_Y(\Gamma) \in \mathrm{PSH}^{\mathrm{NA}}(\tilde{Y}, \theta|_{\tilde{Y}})$[3] as the unique element such that for any $\omega \in \mathrm{Käh}(\tilde{Y})$, the component

$$\mathrm{Tr}_Y(\Gamma)^{\theta|_{\tilde{Y}}+\omega} \in \mathrm{PSH}^{\mathrm{NA}}(\tilde{Y}, \theta|_{\tilde{Y}}+\omega)_{>0} \tag{13.10}$$

is defined as follows:

(1) We let

[3] Here $\tilde{Y} \to Y$ is the normalization of Y.

$$\left(\mathrm{Tr}_Y(\Gamma)^{\theta|_{\tilde{Y}}+\omega}\right)_{\max} = (\mathrm{Tr}_Y(\Gamma))_{\max}\,; \tag{13.11}$$

(2) for each $\tau \in \mathbb{R}$ less than $(\mathrm{Tr}_Y(\Gamma))_{\max}$, we define

$$\mathrm{Tr}_Y(\Gamma)_\tau^{\theta|_{\tilde{Y}}+\omega} := P_{\theta|_{\tilde{Y}}+\omega}\left[\mathrm{Tr}_Y^{\theta+\tilde{\omega}}\left(\Gamma_\tau^{\theta+\tilde{\omega}}\right)\right]_{\mathcal{I}},$$

where $\tilde{\omega}$ is an arbitrary Kähler form on X such that $\omega \geq \tilde{\omega}|_{\tilde{Y}}$.

It follows from [GK20, Proposition 3.5] that $\tilde{Y}$ is a normal Kähler space and hence $\tilde{\omega}$ exists. We observe that the choice of the trace operator $\mathrm{Tr}_Y^{\theta+\tilde{\omega}}\left(\Gamma_\tau^{\theta+\tilde{\omega}}\right)$ is irrelevant since two different choices are $\mathcal{I}$-equivalent. Moreover, (13.10) holds as a consequence of Proposition 8.1.2 and Proposition 8.2.1. It is therefore clear that $\mathrm{Tr}_Y(\Gamma) \in \mathrm{PSH}^{\mathrm{NA}}(\tilde{Y}, \theta|_{\tilde{Y}})$.

Proposition 13.2.19 *Let $\pi\colon Y \to X$ be a proper bimeromorphic morphism from a compact Kähler manifold Y. Then all definitions in this section are invariant under pullback to Y.*

The meaning is clear in most cases. In the case of the trace operator, this means the following: Suppose that $Z \subseteq X$ is an analytic subset and $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta)$ has non-trivial restriction to Z. Suppose that Z is not contained in the non-isomorphism locus of π so that the strict transform W of Z is defined. If we write $\Pi\colon W \to Z$ for the restriction of π and $\tilde{\Pi}\colon \tilde{W} \to \tilde{Z}$ the strict transform of Π, then we have

$$\tilde{\Pi}^* \,\mathrm{Tr}_Z(\Gamma) = \mathrm{Tr}_W(\pi^*\Gamma).$$

The relevant notations are summarized in the following diagram:

$$\begin{tikzcd} \tilde{W} \arrow[r] \arrow[d, "\tilde{\Pi}"] & W \arrow[r] \arrow[d, "\Pi"] & Y \arrow[d, "\pi"] \\ \tilde{Z} \arrow[r] & Z \arrow[r] & X. \end{tikzcd}$$

Proof We only prove the assertion for the trace operator, as the other proofs are similar.

We shall use the notations above. Observe that for any closed positive smooth $(1,1)$-form ω on X with positive mass, we have

$$\left(\tilde{\Pi}^* \,\mathrm{Tr}_Z(\Gamma)\right)_{\max} = (\mathrm{Tr}_Z(\Gamma))_{\max} = \sup\left\{\tau < \Gamma_{\max} : \nu\left(\Gamma_\tau^{\theta+\omega}, Z\right) = 0\right\},$$

and

$$\begin{aligned}(\mathrm{Tr}_W(\pi^*\Gamma))_{\max} &= \sup\left\{\tau < \Gamma_{\max} : \nu\left((\pi^*\Gamma)_\tau^{\pi^*\theta+\pi^*\omega}, W\right) = 0\right\}\\ &= \sup\left\{\tau < \Gamma_{\max} : \nu\left(\pi^*\Gamma_\tau^{\theta+\omega}, W\right) = 0\right\}\\ &= \sup\left\{\tau < \Gamma_{\max} : \nu\left(\Gamma_\tau^{\theta+\omega}, Z\right) = 0\right\}.\end{aligned}$$

Here we implicitly used Proposition 13.1.5. Therefore,

$$\left(\tilde{\Pi}^* \operatorname{Tr}_Z(\Gamma)\right)_{\max} = \left(\operatorname{Tr}_W(\pi^*\Gamma)\right)_{\max}.$$

Let $\tau \in \mathbb{R}$ be less than this common value. Take a Kähler form $\omega_{\tilde{Z}}$ (resp. $\omega_{\tilde{W}}$) on $\tilde{Z}$ (resp. $\tilde{W}$). Take a Kähler form ω_Y on Y (resp. ω_X on X) such that

$$\omega_{\tilde{Z}} \geq \omega_X|_{\tilde{Z}}, \quad \omega_{\tilde{W}} \geq \omega_Y|_{\tilde{W}}, \quad \omega_Y \geq \pi^*\omega_X.$$

We want to show that

$$\left(\tilde{\Pi}^* \operatorname{Tr}_Z(\Gamma)\right)_\tau^{\theta|_{\tilde{Z}}+\omega_{\tilde{Z}}} \sim_P \left(\operatorname{Tr}_W(\pi^*\Gamma)\right)_\tau^{(\pi^*\theta)|_{\tilde{W}}+\omega_{\tilde{W}}}.$$

Unfolding the definitions, we reduce to

$$\tilde{\Pi}^* \operatorname{Tr}_Z^{\theta+\omega_X}\left(\Gamma_\tau^{\theta+\omega_X}\right) \sim_P \operatorname{Tr}_W^{\pi^*\theta+\omega_Y}\left((\pi^*\Gamma)_\tau^{\pi^*\theta+\omega_Y}\right).$$

Using Proposition 8.2.1, this is equivalent to

$$\tilde{\Pi}^* \operatorname{Tr}_Z\left(\Gamma_\tau^{\theta+\omega_X}\right) \sim_P \operatorname{Tr}_W\left((\pi^*\Gamma)_\tau^{\pi^*\theta+\pi^*\omega_X}\right).$$

This is a consequence of Lemma 8.2.1. □

13.3 Duistermaat–Heckman measures

Let X be a connected compact Kähler manifold of dimension n and θ be a closed real smooth $(1,1)$-form on X representing a big cohomology class.

Definition 13.3.1 Assume that X admits a smooth flag $Y_\bullet$. Let $\Gamma \in \operatorname{PSH}^{\mathrm{NA}}(X,\theta)_{>0}$. The *Duistermaat–Heckman measure* $\operatorname{DH}(\Gamma)$ of Γ is defined as

$$\operatorname{DH}(\Gamma) := n! \cdot \operatorname{DH}\left(\Delta_{Y_\bullet}(\theta,\Gamma)\right).$$

Recall that $\Delta_{Y_\bullet}(\theta,\Gamma) \in \operatorname{TC}(\Delta_{Y_\bullet}(\theta,\Gamma_{-\infty}))$ is the Okounkov test curve defined in Theorem 10.5.2. See Definition 10.5.4 for the definition of the Duistermaat–Heckman measure of an Okounkov test curve.

Theorem 13.3.1 *Assume that X admits a smooth flag $Y_\bullet$. The Duistermaat–Heckman measure $\operatorname{DH}(\Gamma)$ of $\Gamma \in \operatorname{PSH}^{\mathrm{NA}}(X,\theta)_{>0}$ in Definition 13.3.1 is independent of the choice of the smooth flag $Y_\bullet$. Furthermore, for any $m \in \mathbb{Z}_{>0}$, the following extended moment identity holds:*

$$\int_{\mathbb{R}} x^m \operatorname{DH}(\Gamma)(x) = \Gamma_{\max}^m \operatorname{vol}\Gamma + m\int_{-\infty}^{\Gamma_{\max}} \tau^{m-1}\left(\operatorname{vol}(\theta+\mathrm{dd^c}\Gamma_\tau) - \operatorname{vol}\Gamma\right) \mathrm{d}\tau, \tag{13.12}$$

and

$$\int_{\mathbb{R}} \mathrm{DH}(\Gamma) = \operatorname{vol}\Gamma. \tag{13.13}$$

Proof We observe that the moments of the random variable $G[\Delta_{Y_\bullet}(\theta,\Gamma)]$ as computed in Proposition 10.5.4 are independent of the choice of the flag: In fact, they are given by (13.12) and (13.13) thanks to Theorem 10.4.2(1).

Assume first that Γ is bounded. Since the Duistermaat–Heckman measure has bounded support in this case (cf. Theorem 10.5.1), we conclude that $\mathrm{DH}(\Gamma)$ is uniquely determined.

In general, we may assume that $\Gamma_{\max} = 0$. For each $\epsilon > 0$, we define $\Gamma^\epsilon \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)_{>0}$ as follows:

(1) Let $\Gamma^\epsilon_{\max} = 0$, and
(2) we set

$$\Gamma^\epsilon_\tau = \begin{cases} \Gamma_{-\infty}, & \text{if } \tau \le -\epsilon^{-1}, \\ P_\theta\left[(1+\epsilon\tau)\Gamma_\tau - \epsilon\tau\Gamma_{-\infty}\right], & \text{if } \tau \in \left(-\epsilon^{-1}, 0\right). \end{cases}$$

Then it follows from the argument of Theorem 9.2.1 Step 3.3 that $\Delta_{Y_\bullet}(\Gamma)_\tau$ is the decreasing limit of $\Delta_{Y_\bullet}(\Gamma^\epsilon)_\tau$ for any $\tau < \Gamma_{\max}$ as $\epsilon \to 0+$. So $\mathrm{DH}(\Gamma^\epsilon) \rightharpoonup \mathrm{DH}(\Gamma)$ by Lemma 10.5.2. It follows that $\mathrm{DH}(\Gamma)$ is independent of the choice of the flag. □

More generally, when X does not admit a smooth flag, we may choose a modification $\pi\colon Y \to X$ so that Y admits a flag. We define

$$\mathrm{DH}(\Gamma) \coloneqq \mathrm{DH}(\pi^*\Gamma). \tag{13.14}$$

It follows from Theorem 10.4.2(5) that this measure is independent of the choice of π.

Proposition 13.3.1 *Let*

$$(\Gamma^i)_{i\ge 1} \subseteq \mathrm{PSH}^{\mathrm{NA}}(X,\theta)_{>0}, \quad \Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)_{>0}.$$

Assume one of the following conditions holds:

(1) *The sequence $(\Gamma^i)_{i\ge1}$ is decreasing and $\Gamma = \inf_{i\ge1}\Gamma^i$. Assume that*

$$\operatorname{vol}\Gamma = \lim_{i\to\infty} \operatorname{vol}\Gamma^i. \tag{13.15}$$

(2) *The sequence $(\Gamma^i)_{i\ge1}$ is increasing and $\Gamma = \sup_{i\ge1}{}^*\Gamma^i$.*

Then

$$\mathrm{DH}(\Gamma^i) \rightharpoonup \mathrm{DH}(\Gamma). \tag{13.16}$$

Proof We may assume that X admits a smooth flag $Y_\bullet$.

Assume (1). Note that (13.15) implies that

$$\Gamma_{-\infty} = \inf_{i\ge1} \Gamma^i_{-\infty}.$$

We want to derive (13.16) from Lemma 10.5.2. It boils down to prove the following: For any $\tau < \Gamma_{\max}$, we have

$$\Delta_{Y_\bullet}(\theta, \Gamma^i_\tau) \xrightarrow{d_{\mathrm{Haus}}} \Delta_{Y_\bullet}(\theta, \Gamma_\tau).$$

This follows immediately from Theorem 10.4.2(1) and Proposition 3.1.12.

The proof under the assumption (2) is similar. We only need to apply Lemma 10.5.3 instead of Lemma 10.5.2. □

Remark 13.3.1 When $[\theta]$ is a Hodge class and Γ is induced by a test configuration as in Example 9.3.1 and Remark 9.3.1, our Duistermaat–Heckman measure coincides with the more traditional definition of [BHJ17, Section 3.2]. This is explained in [Xia25b, Remark 7.17].

13.4 Comparison with Boucksom–Jonsson's theory

13.4.1 A brief recap of Boucksom–Jonsson's theory

In this section, we briefly recall the non-Archimedean global pluripotential theory à la Boucksom–Jonsson [BJ22a]. Our presentation is selective, so the original paper remains the best companion for this section.

13.4.1.1 Valuations

Let X be an irreducible reduced variety over $\mathbb{C}$ of dimension n. We recall the Berkovich analytification X^{An} of X with respect to the trivial valuation on $\mathbb{C}$.

Definition 13.4.1 A (real-valued) *valuation* on X (or a *valuation* of $\mathbb{C}(X)$) is a map $v\colon \mathbb{C}(X) \to (-\infty, \infty]$ satisfying the following conditions:

(1) For $f \in \mathbb{C}(X)$, $v(f) = \infty$ if and only if $f = 0$;
(2) For $f, g \in \mathbb{C}(X)$, $v(fg) = v(f) + v(g)$;
(3) For $f, g \in \mathbb{C}(X)$, $v(f + g) \geq v(f) \wedge v(g)$.

The set of valuations on X is denoted by X^{val}. The center of a valuation v is the scheme-theoretic point $c = c(v)$ of X such that $v \geq 0$ on $\mathcal{O}_{X,c}$ and $v > 0$ on the maximal ideal $\mathfrak{m}_{X,c}$ of $\mathcal{O}_{X,c}$. The center is unique if it exists, and it exists if X is proper.

In the remainder of this section, we assume that X is projective.

As a set, X^{An} is the set of *semi-valuations* on X, in other words, real-valued valuations v on irreducible reduced subvarieties Y of X that are trivial on $\mathbb{C}$. We call Y the *support* of the semi-valuation v. In other words,

$$X^{\mathrm{An}} = \coprod_{Y} Y^{\mathrm{val}}.$$

We will write $v_{\mathrm{triv}} \in X^{\mathrm{An}}$ for the trivial valuation on X: $v_{\mathrm{triv}}(f) = 0$ for any $f \in \mathbb{C}(X)^{\times}$.

We endow X^{An} with the coarsest topology such that

(1) for any Zariski open subset $U \subseteq X$, the subset U^{An} of X^{An} consisting of semi-valuations whose supports meet U is open;
(2) for each Zariski open subset $U \subseteq X$ and each $f \in \mathrm{H}^0(U, \mathcal{O}_X)$ (here $\mathcal{O}_X$ is the sheaf of regular functions), the map $|f|\colon U^{\mathrm{An}} \to \mathbb{R}$ sending v to $\exp(-v(f))$ is continuous.

See [Ber93] for more details.

We will be most interested in divisorial valuations. Recall that a *divisorial valuation* on X is a valuation of the form $t \operatorname{ord}_E$, where $t \in \mathbb{Q}_{>0}$ and E is a prime divisor over X. The set of divisorial valuations on X is denoted by X^{div}. When $\mathbb{Q}_{>0}$ is replaced by $\mathbb{R}_{>0}$, we can similarly define a space $X^{\mathrm{div}}_{\mathbb{R}}$.

Given any coherent ideal $\mathfrak{a}$ on X and any $v \in X^{\mathrm{An}}$, we define

$$v(\mathfrak{a}) := \min\{v(f) : f \in \mathfrak{a}_{c(v)}\} \in [0, \infty], \tag{13.17}$$

where $c(v)$ is the center of the valuation v on X.

Given any valuation v on X, the Gauss extension of v is a valuation $\sigma(v)$ on $X \times \mathbb{A}^1$:

$$\sigma(v)\left(\sum_i f_i t^i\right) := \min_i (v(f_i) + i). \tag{13.18}$$

Here t is the standard coordinate on $\mathbb{A}^1 = \operatorname{Spec} \mathbb{C}[t]$. The key property is that when v is a divisorial valuation, then so is $\sigma(v)$. See [BHJ17, Lemma 4.2].

13.4.1.2 Non-Archimedean plurisubharmonic functions

Let X be an irreducible complex projective variety of dimension n and L be a holomorphic pseudo-effective $\mathbb{Q}$-line bundle on X. Through the GAGA morphism $X^{\mathrm{An}} \to X$ of ringed spaces, L can be pulled-back to an analytic line bundle L^{An} on X. The purpose of this section is to study the psh metrics on L^{An}. We will follow the approach of [BJ22a], which avoids the direct treatment of L^{An} itself.

Following [BJ22a, Definition 2.18], we define $\mathcal{H}^{\mathrm{gf}}_{\mathbb{Q}}(L^{\mathrm{An}})$, the set of *(rational) generically finite Fubini–Study functions* $\phi\colon X^{\mathrm{An}} \to [-\infty, \infty)$ that are of the following form:

$$\phi = \frac{1}{m} \max_j \{\log |s_j| + \lambda_j\}. \tag{13.19}$$

Here $m \in \mathbb{Z}_{>0}$ is an integer such that L^m is a line bundle, the s_j's are a finite collection of nonzero sections in $\mathrm{H}^0(X, L^m)$, and $\lambda_j \in \mathbb{Q}$. We follow the convention of Boucksom–Jonsson by writing $\log |s_j|(v) = -v(s_j)$.

Definition 13.4.2 ([BJ22a, Definition 4.1]) A *plurisubharmonic metric* (or *psh metric* for short) on L^{An} is a function $\phi\colon X^{\mathrm{An}} \to [-\infty, \infty)$ that is not identically $-\infty$, and is the pointwise limit of a decreasing net $(\phi_i)_{i\in I}$, where $\phi_i \in \mathcal{H}^{\mathrm{gf}}_{\mathbb{Q}}(L_i^{\mathrm{An}})$ for some $\mathbb{Q}$-line bundles L_i on X satisfying $c_1(L_i) \to c_1(L)$ in $\mathrm{NS}^1(X)_{\mathbb{R}}$.

The set of psh metrics on L^{An} is denoted by $\mathrm{PSH}(L^{\mathrm{An}})$. We endow $\mathrm{PSH}(L^{\mathrm{An}})$ with the topology of pointwise convergence on X^{div}. This topology is Hausdorff as functions in $\mathrm{PSH}(L^{\mathrm{An}})$ are completely determined by their restriction on X^{div}:

Theorem 13.4.1 ([BJ22a, Theorem 4.22]) *Let $\phi \in \mathrm{PSH}(L^{\mathrm{An}})$ and $\psi : X^{\mathrm{An}} \to [-\infty, \infty)$ be an usc function. Assume that $\phi \leq \psi$ on X^{div}. Then the same holds on X^{An}.*

Proposition 13.4.1 ([BJ22a, Theorem 4.7]) *Let $\phi, \phi' \in \mathrm{PSH}(L^{\mathrm{An}})$. Then so is their pointwise maximum $\phi \vee \phi'$.*

Proposition 13.4.2 *Let H be an ample line bundle on X. Consider $\phi \in \mathrm{PSH}((L+H)^{\mathrm{An}})$. Assume that for each $m \in \mathbb{Z}_{>0}$, we have $\phi \in \mathrm{PSH}((L+m^{-1}H)^{\mathrm{An}})$. Then $\phi \in \mathrm{PSH}(L^{\mathrm{An}})$.*

This is a special case of [BJ22a, (PSH2)] on Page 45.

Next we note that we may use sequences instead of nets in the definition of $\mathrm{PSH}(L^{\mathrm{An}})$:

Theorem 13.4.2 ([BJ22a, Corollary 12.18]) *Let S be an ample line bundle on X. Let $\phi \in \mathrm{PSH}(L^{\mathrm{An}})$. Then there is a sequence of rational numbers $\varepsilon_i \searrow 0$ and a decreasing sequence $\phi_i \in \mathcal{H}^{\mathrm{gf}}_{\mathbb{Q}}((L+\varepsilon_i S)^{\mathrm{An}})$ such that ϕ is the pointwise limit of ϕ_i, as $i \to \infty$.*

The space $\mathrm{PSH}(L^{\mathrm{An}})$ inherits most of the expected properties of (Archimedean) psh functions ([BJ22a, Theorem 4.7]). However, the following compactness result is not known:

Conjecture 13.4.1 ([BJ22a, §5]) Assume that X is unibranch. Then every bounded from above increasing net of elements in $\mathrm{PSH}(L^{\mathrm{An}})$ converges in $\mathrm{PSH}(L^{\mathrm{An}})$.

This prediction is equivalent to the so-called envelope conjecture [BJ22a, Conjecture 5.14]: the regularized supremum of a bounded from above family of functions in $\mathrm{PSH}(L^{\mathrm{An}})$ lies in $\mathrm{PSH}(L^{\mathrm{An}})$. See [BJ22a, Theorem 5.11] for the proof of the equivalence. This conjecture is proved when X is smooth and L is nef in [BJ22a]. More recently, in [BJ22b], Boucksom–Jonsson further established the case when X is smooth and L is pseudo-effective.

13.4.2 The analytifications

Let X be a connected projective manifold of dimension n. Let θ be a closed smooth real $(1,1)$-form on X representing a pseudo-effective cohomology class.

13.4.2.1 The transcendental setting

Definition 13.4.3 For $\varphi \in \mathrm{PSH}(X,\theta)$, we define the *analytification* $\varphi^{\mathrm{An}}\colon X^{\mathrm{An}} \to [-\infty, 0]$ as follows:

$$\varphi^{\mathrm{An}}(v) := -v(\varphi) = -\lim_{k\to\infty} \frac{1}{k} v\left(\mathcal{I}(k\varphi)\right). \tag{13.20}$$

By Theorem 1.4.2 and Fekete's lemma, the limit in (13.20) exists.

Note that we can also write

$$\varphi^{\mathrm{An}}(v) = \inf_{k\in\mathbb{Z}_{>0}} -2^{-k} v\left(\mathcal{I}(2^k\varphi)\right). \tag{13.21}$$

When $v = t\,\mathrm{ord}_E$ for some prime divisor E over X, $\varphi^{\mathrm{An}}(v) = -t\nu(\varphi, E)$ by Proposition 1.4.4.

Definition 13.4.4 Let $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)$. We define the *analytification* $\Gamma^{\mathrm{An}}\colon X^{\mathrm{div}} \to [-\infty, \infty)$ of Γ as follows: For any $\omega \in \mathrm{K\ddot{a}h}(X)$, we define

$$\Gamma^{\mathrm{An}}(v) := \sup_{\tau\le\Gamma_{\max}} \left(\Gamma_\tau^{\theta+\omega,\mathrm{An}}(v) + \tau\right). \tag{13.22}$$

Clearly, (13.22) is independent of the choice of ω.

Note that (13.22) can be equivalently written as

$$\Gamma^{\mathrm{An}}(v) = \sup_{\tau\le\Gamma_{\max}} \left(\Gamma_\tau^{\theta+\omega,\mathrm{An}}(v) + \tau\right) = \sup_{\tau\in\mathbb{R}} \left(\Gamma_\tau^{\theta+\omega,\mathrm{An}}(v) + \tau\right)$$

with $(-\infty)^{\mathrm{An}}(v) = -\infty$ understood.

Proposition 13.4.3 *Let* $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)_{>0}$ *with* $\Gamma_{\max} \le 0$. *Let* Ψ *be the complexification of* Γ^*. *Then*

$$\Gamma^{\mathrm{An}}(v) = -\sigma(v)(\Psi) \quad \forall v \in X^{\mathrm{div}}. \tag{13.23}$$

See Definition 4.1.2 for the definition of the complexification $\Psi \in \mathrm{QPSH}(X\times\Delta)$. Note that since $\Gamma_{\max} \le 0$, by Corollary 9.2.3 and Theorem 1.2.1, Ψ extends uniquely to a quasi-psh function on $X \times \Delta$.

Proof Recall that

$$\Psi(x,\delta) = \sup_{\tau\le\Gamma_{\max}} \left(\Gamma_\tau(x) - \log|\delta|^2\tau\right) \quad \text{for } x \in X, \delta \in \Delta^*.$$

By (13.18), we have $\sigma(v)(\log|\delta|^2) = 1$ and $\sigma(v)(\Gamma_\tau) = v(\Gamma_\tau)$ for all $\tau \leq \Gamma_{\max}$. So we have

$$\sigma(v)(\Gamma_\tau(x) - \log|\delta|^2\tau) = v(\Gamma_\tau) - \tau.$$

Lastly, since $\sigma(v)$ is a divisorial valuation on $X \times \Delta$, by Corollary 1.4.1, we conclude (13.23). ◻

Definition 13.4.5 Let $N \in \mathbb{N}$, and $A_0, \dots, A_N$ be a finite collection of elements in $\mathrm{PSH}(X, \theta)$, and $\tau_0 > \tau_1 > \cdots > \tau_N$ be finitely many real numbers. Then the *piecewise linear curve* $A = (A_\tau)_{\tau\in\mathbb{R}}$ in $\mathrm{PSH}(X, \theta) \cup \{-\infty\}$ associated with these data is the affine interpolation of these data:

(1) $A_{\tau_i} = A_i$ for $i = 0, \dots, N$;
(2) $A_\tau = A_{\tau_N}$ for $\tau \leq \tau_N$;
(3) for $t \in (0, 1)$ and $i = 0, \dots, N - 1$, we have

$$A_{(1-t)\tau_i+t\tau_{i+1}} = (1 - t)A_{\tau_i} + tA_{\tau_{i+1}};$$

(4) $A_\tau \equiv -\infty$ for $\tau > \tau_0$.

The *analytification* of A is the function $A^{\mathrm{An}}\colon X^{\mathrm{An}} \to [-\infty, \infty)$ defined as follows:

$$A^{\mathrm{An}}(v) \coloneqq \sup_{\tau\leq\tau_0} (A^{\mathrm{An}}_\tau(v) + \tau) = \max_{i=0,\dots,N} \left(A^{\mathrm{An}}_{\tau_i}(v) + \tau_i\right) \quad \forall v \in X^{\mathrm{An}}. \tag{13.24}$$

We also say $A = (A_\tau)_{\tau\leq\tau_0}$ is a piecewise linear curve in $\mathrm{PSH}(X, \theta)$.

Remark 13.4.1 Note that $\tau \mapsto A_\tau$ is upper semicontinuous, but not necessarily concave. Let $(A'_\tau)_{\tau\in\mathbb{R}}$ be the upper concave envelope of $\tau \mapsto A_\tau$. Then it can be inductively constructed as follows:

(1) For $\tau \in (\tau_0, \infty)$, we let $A'_\tau \equiv -\infty$;
(2) we set $A'_{\tau_0} = A_{\tau_0}$;
(3) define inductively for $j = 0, \dots, N - 1$ the following: For $\tau \in [\tau_{j+1}, \tau_j)$, we set

$$A'_\tau = \max_{i=j+1,\dots,N} \left(\frac{\tau_j - \tau}{\tau_j - \tau_i} A_{\tau_i} + \frac{\tau - \tau_i}{\tau_j - \tau_i} A'_{\tau_j}\right) \vee A'_{\tau_j};$$

(4) for $\tau \in (-\infty, \tau_N)$, we set $A'_\tau = A_{\tau_N}$.

This construction is just a reformulation of the general formula Proposition A.1.2.

In particular, $A'_\tau \in \mathrm{PSH}(X, \theta)$ for all $\tau \leq \tau_0$.

Note that A' is not necessarily piecewise linear.

Lemma 13.4.1 *Let A be a piecewise linear curve in* $\mathrm{PSH}(X, \theta)$. *Let $(A'_\tau)_{\tau\in\mathbb{R}}$ be the upper concave envelope of $\tau \mapsto A_\tau$. Define $\tilde{A} \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta)$ by*

$$\tilde{A}^{\theta+\omega}_\tau \coloneqq P_{\theta+\omega}[A'_\tau]_{\mathcal{I}}, \quad \omega \in \mathrm{K\ddot{a}h}(X), \quad \tau < \tau_0.$$

Moreover,

$$A^{\mathrm{An}} = \tilde{A}^{\mathrm{An}} \quad \text{on } X^{\mathrm{div}}. \tag{13.25}$$

Here τ_0 is as in Definition 13.4.5.

Proof We continue to use the notations in Definition 13.4.5. The fact that $\tilde{A} \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)$ follows from Remark 13.4.1 and the cocycle property of the $\mathcal{I}$-projection. In order to prove (13.25), we fix $v \in X^{\mathrm{div}}$ and $\omega \in \mathrm{K\ddot{a}h}(X)$. By Remark 13.4.1,

$$\tau \mapsto \left(P_{\theta+\omega}[A'_\tau]_{\mathcal{I}}\right)^{\mathrm{An}}(v) = \left(A'_\tau\right)^{\mathrm{An}}(v)$$

is just the upper concave envelope of

$$\tau \mapsto A_\tau^{\mathrm{An}}(v).$$

Therefore, (13.25) follows. □

13.4.2.2 The algebraic setting

Let L be a $\mathbb{Q}$-line bundle on X and h be a Hermitian metric on L with $\theta = c_1(L,h)$.

Lemma 13.4.2 *For any $\varphi \in \mathrm{PSH}(X,\theta)$, we have $\varphi^{\mathrm{An}} \in \mathrm{PSH}(L^{\mathrm{An}})$.*

Proof After replacing L with a sufficiently high power, we may assume that L is a line bundle. Take a very ample line bundle H on X. By Siu's uniform global generation theorem [Siu98], [Dem12a, Theorem 6.27] there exists $b > 0$ large enough so that $H^b \otimes L^k \otimes \mathcal{I}(k\varphi)$ is globally generated for all $k > 0$. Let $\{s_i\}_i$ be a finite set of global sections that generate the sheaf $H^b \otimes L^k \otimes \mathcal{I}(k\varphi)$. Then

$$v(\mathcal{I}(k\varphi)) = \min_i v(s_i).$$

It follows that $v \mapsto -k^{-1}v\left(\mathcal{I}(k\varphi)\right)$ lies in $\mathcal{H}^{\mathrm{gf}}_{\mathbb{Q}}((L+\frac{b}{k}H)^{\mathrm{An}})$. Using (13.21), we conclude that $\varphi^{\mathrm{An}} \in \mathrm{PSH}(L^{\mathrm{An}})$. □

Lemma 13.4.3 *Let Γ be a piecewise linear curve in $\mathrm{PSH}(X,\theta)$. Then $\Gamma^{\mathrm{An}} \in \mathrm{PSH}(L^{\mathrm{An}})$.*

Proof The result follows from (13.24), Proposition 13.4.1 and Lemma 13.4.2. □

Lemma 13.4.4 *Let R be a commutative $\mathbb{C}$-algebra of finite type and I be an ideal of $R[t]$. If for any $a \in S^1$, $a^*I \subseteq I$, then I is stable under the $\mathbb{C}^*$-action. Moreover, there are ideals $I_0 \subseteq I_1 \subseteq \cdots \subseteq I_m$ in R so that*

$$I = I_0 + I_1 t + \cdots + I_m(t^m). \tag{13.26}$$

Proof It suffices to argue that I can be expanded as in (13.26). To see this, assume that $a \in I$. We can write $a = a_0 + a_1 t + \cdots + a_m t^m$ with $a_i \in R$. Then our assumption implies that $\sum_i a_i \rho^i t^i \in I$ as well for all $\rho \in S^1$. So by the Lagrange interpolation formula, $a_i t^i \in I$ for all i. Therefore, we can write I as $I_0 + I_1 t + I_2 t^2 + \cdots$ for some ideals $I_0 \subseteq I_1 \subseteq \cdots$ in R. But as R is noetherian, there is $m \geq 0$ so that $I_{m'} = I_m$ for $m' > m$. (13.26) follows. □

Lemma 13.4.5 *Let X be a complex projective variety and $p : X \times \mathbb{C} \to X$ be the natural projection. Assume that $\mathcal{I}$ is an analytic coherent ideal sheaf on $X \times \mathbb{C}$. Assume that $\mathcal{I}|_{X\times\mathbb{C}^*} = p^*\mathcal{J}$ for some coherent ideal sheaf $\mathcal{J}$ on X. Then $\mathcal{I}$ is the analytification of an algebraic coherent ideal sheaf.*

Proof Let $q : X \times (\mathbb{P}^1 \setminus \{0\}) \to X$ be the natural projection. As $\mathbb{C}^* \subset \mathbb{P}^1 \setminus \{0\}$ we can glue $q^*\mathcal{J}$ with $\mathcal{I}$ to get an analytic coherent ideal sheaf on $X \times \mathbb{P}^1$. By the GAGA principle, this ideal sheaf is necessarily algebraic, hence so is its restriction to $X \times \mathbb{C}$. □

Next we point out a version of Siu's uniform global generation lemma [Siu88] that we will need in the proof of our next theorem:

Lemma 13.4.6 *Let L be a big line bundle on X such that $c_1(L) = \{\theta\}$ and $\Phi \in$ $\mathrm{PSH}(X \times \Delta, p_1^*\theta)$, where Δ is the unit disk. Let G be an ample line bundle on X. Then there exists $k > 0$, depending only on X and G, such that $p_1^*(G^k \otimes L^m) \otimes \mathcal{I}(m\Phi)$ is globally generated for all $m \in \mathbb{N}$.*

Proof The argument for this is exactly the same as the one in [BBJ21, Lemma 5.6] with Nadel's vanishing theorem replaced by the family version proved by Matsumura in [Mat22, Theorem 1.7]. □

Proposition 13.4.4 *Let $\phi \in \mathrm{PSH}(X, \theta)_{>0}$ be a model potential and $\ell \in \mathcal{R}(X, \theta; \phi)$ with $\sup_X \ell_1 \leq 0$. Let Φ be the complexification of ℓ. Then the function*

$$v \mapsto -\sigma(v)(\Phi) \quad \text{for } v \in X^{\mathrm{div}}$$

admits a unique extension to an element of $\mathrm{PSH}(L^{\mathrm{An}})$.

Proof We may assume that L is a line bundle. Observe that the extension is unique if it exists by Theorem 13.4.1.

Let $p_1 \colon X \times \mathbb{C} \to X$ be the projection. For each $t > 0$, we have $\ell_t \in \mathcal{E}(X, \theta; \phi)$, hence $\ell_t \sim_P \phi$ by Theorem 3.1.2 and $\ell_t \sim_{\mathcal{I}} \phi$ by Corollary 6.1.1. Thanks to Proposition 1.4.5 and Lemma 8.5.3, for each $m \in \mathbb{Z}_{>0}$, we have

$$\mathcal{I}(m\Phi)|_{X\times\Delta^*} = p_1^*\mathcal{I}(m\phi)|_{X\times\Delta^*}.$$

In particular, $\mathcal{I}(m\Phi)$ admits an S^1-invariant coherent extension to $X \times \mathbb{C}$, obtained by gluing it with $p_1^*\mathcal{I}(m\phi)$ over $X \times \mathbb{C}^*$.

From Lemma 13.4.4 and Lemma 13.4.5, we get that

$$\mathcal{I}(m\Phi) = \mathfrak{a}_0 + \mathfrak{a}_1 t + \cdots + \mathfrak{a}_{N-1}t^{N-1} + \mathfrak{a}_N(t^N)\,, \tag{13.27}$$

where the $\mathfrak{a}_i$'s are coherent ideal sheaves on X.

Using Lemma 13.4.6, there exists an ample line bundle T over X such that $p_1^*(T \otimes L^m) \otimes \mathcal{I}(m\Phi)$ is globally generated, which is equivalent to $T \otimes L^m \otimes \mathfrak{a}_i$ being globally generated for all i.[4]

[4] In contrast with the case where ϕ is bounded, explored in [BBJ21], $\mathfrak{a}_N \neq \mathcal{O}_X$ in general.

We define

$$\varphi_m(v) := -\frac{1}{m}\sigma(v)(\mathcal{I}(m\Phi)) = -\frac{1}{m}\min_i(v(\mathfrak{a}_i) + i), \quad v \in X^{\mathrm{div}}.$$

From the right-hand side of the formula, φ_m can be extended to an element in $\mathcal{H}^{\mathrm{gf}}_{\mathbb{Q}}((L + m^{-1}T)^{\mathrm{An}})$, which we denote by the same symbol.

For $v \in X^{\mathrm{div}}$,

$$-\sigma(v)(\Phi) = \lim_{m\to\infty} -\frac{1}{2^m}\sigma(v)(\mathcal{I}(2^m\Phi)) = \lim_{m\to\infty}\varphi_{2^m}(v)$$

and the right-hand side defines an element in $\mathrm{PSH}(L^{\mathrm{An}})$ by definition, since $\{\varphi_{2^m}\}_m$ is decreasing by the subadditivity of multiplier ideals Theorem 1.4.2. □

Corollary 13.4.1 *Let $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta)$. Then Γ^{An} defined in Definition 13.4.4 admits a unique extension to $\mathrm{PSH}(L^{\mathrm{An}})$.*

The extension will be denoted by the same notation Γ^{An}.

Proof Observe that the extension is unique if it exists by Theorem 13.4.1. We may assume that $\Gamma_{\max} = 0$ without loss of generality.

When $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta)_{>0}$, our assertion follows from Proposition 13.4.4 and Proposition 13.4.3.

In general, fix an ample line bundle H on X and a Kähler form $\omega \in c_1(H)$. Then we know that

$$\Gamma^{\mathrm{An}} = \left(\Gamma^{m^{-1}\omega}\right)^{\mathrm{An}} \in \mathrm{PSH}((L + m^{-1}H)^{\mathrm{An}})$$

for any $m \in \mathbb{Z}_{>0}$. Therefore, $\Gamma^{\mathrm{An}} \in \mathrm{PSH}(L^{\mathrm{An}})$ by Proposition 13.4.2. □

13.4.3 The comparison theorem

Let X be a connected projective manifold of dimension n. Let L be a pseudo-effective $\mathbb{Q}$-line bundle on X and h be a Hermitian metric on L with $\theta = c_1(L, h)$.

Thanks to Corollary 13.4.1, we already have a map

$$\mathrm{PSH}^{\mathrm{NA}}(X, \theta) \to \mathrm{PSH}(L^{\mathrm{An}}), \quad \Gamma \mapsto \Gamma^{\mathrm{An}}. \tag{13.28}$$

We observe that for $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta)$ and a Kähler form ω on X, we have

$$\left(P_{\theta+\omega}[\Gamma]_{\mathcal{I}}\right)^{\mathrm{An}} = \Gamma^{\mathrm{An}}. \tag{13.29}$$

Also observe that

$$\Gamma_{\max} = \Gamma^{\mathrm{An}}(v_{\mathrm{triv}}), \tag{13.30}$$

Lemma 13.4.7 *The map (13.28) is order preserving. Moreover, suppose that $\Gamma, \Gamma' \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta)$ satisfies that $\Gamma^{\mathrm{An}} \leq \Gamma'^{\mathrm{An}}$, then $\Gamma \leq \Gamma'$.*

In particular, the map (13.28) *is injective.*

Proof The map (13.28) is order preserving by definition. Let us take $\Gamma, \Gamma' \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta)$ with $\Gamma^{\mathrm{An}} \leq \Gamma'^{\mathrm{An}}$. Fix a Kähler form ω on X.

Let $v \in X^{\mathrm{div}}$ and $t \in \mathbb{Q}_{>0}$. Then, using (13.22) and the homogeneity of divisorial valuations, we notice that

$$t\Gamma^{\mathrm{An}}\left(t^{-1}v\right) = \sup_{\tau \in \mathbb{R}} \left(\left(\Gamma_\tau^{\theta+\omega}\right)^{\mathrm{An}}(v) + t\tau\right). \tag{13.31}$$

A similar equality holds for Γ'. Therefore, by Corollary A.2.1, we have

$$\left(\Gamma_\tau^{\theta+\omega}\right)^{\mathrm{An}} \leq \left(\Gamma'^{\theta+\omega}_\tau\right)^{\mathrm{An}}$$

for all $\tau \in \mathbb{R}$. It follows that

$$\Gamma_\tau^{\theta+\omega} \leq_{\mathcal{I}} \Gamma'^{\theta+\omega}_\tau$$

by Proposition 6.1.2. Since both sides are $\mathcal{I}$-model, Proposition 6.1.3 gives

$$\Gamma_\tau^{\theta+\omega} \leq \Gamma'^{\theta+\omega}_\tau$$

for all $\tau \in \mathbb{R}$. Hence $\Gamma \leq \Gamma'$ by the definition of the order. □

Lemma 13.4.8 *Let $\phi \in \mathcal{H}^{\mathrm{gf}}_{\mathbb{Q}}(L^{\mathrm{An}})$. Then there is a piecewise linear curve A in* $\mathrm{PSH}(X, \theta)$ *with $\phi = A^{\mathrm{An}}$. In particular, ϕ is in the image of* (13.28).

Note that from the proof below, the test curve Γ corresponding to ϕ satisfies the following: For any $\tau \leq \Gamma_{\max}$, Γ_τ is elementary. See Definition 6.1.3 for the definition of elementary metrics.

Proof Let us write

$$\phi = \frac{1}{m} \max_{i=1,\dots,M} \left(\log|s_i| + \lambda_i\right), \tag{13.32}$$

where $m \in \mathbb{Z}_{>0}$, $s_1, \dots, s_M$ are a finite number of sections of L^m and $\lambda_1, \dots, \lambda_M \in \mathbb{Q}$.

Write I_τ for the set of i such that $m^{-1}\lambda_i = \tau$. We denote the finitely many τ so that I_τ is non-empty as $\tau_0 > \cdots > \tau_N$. For each $i = 0, \dots, N$, we write

$$A_{\tau_i} = \frac{1}{m} \max_{j \in I_{\tau_i}} \log|s_j|^2_{h^m}.$$

We define A as the piecewise linear curve associated with the A_{τ_i}'s and the τ_i's. For every $v \in X^{\mathrm{An}}$,

$$A^{\mathrm{An}}(v) = \max_{i=0,\dots,N} \left(A^{\mathrm{An}}_{\tau_i}(v) + \tau_i\right) = \frac{1}{m} \max_{j=1,\dots,M} \left(\log|s_j|(v) + \lambda_j\right) = \phi(v),$$

where we used the convention $\log|s_j|(v) = -v(s_j)$.

The final assertion follows from Lemma 13.4.1 and Theorem 13.4.1. □

Proposition 13.4.5 *Let* $(\Gamma_i)_{i\in I}$ *be a decreasing net in* $\mathrm{PSH}^{\mathrm{NA}}(X,\theta)$. *Assume that* (13.8) *is satisfied. Then*

$$\left(\inf_{i\in I}\Gamma_i\right)^{\mathrm{An}}=\inf_{i\in I}\Gamma_i^{\mathrm{An}}.$$

Proof Take a Kähler form ω on X. We need to show that

$$\left(\inf_{i\in I}P_{\theta+\omega}[\Gamma_i]_{\mathcal{I}}\right)^{\mathrm{An}}=\inf_{i\in I}\Gamma_i^{\mathrm{An}}.$$

Therefore, after replacing θ by $\theta+\omega$, we may assume that $\Gamma_i\in\mathrm{PSH}^{\mathrm{NA}}(X,\theta)_{>0}$ for all $i\in I$ and $\inf_{i\in I}\Gamma_i\in\mathrm{PSH}^{\mathrm{NA}}(X,\theta)_{>0}$. Fix $v\in X^{\mathrm{div}}$. By Theorem 13.4.1, it suffices to prove that

$$\sup_{\tau\in\mathbb{R}}\left(\left(\inf_{i\in I}\Gamma_{i,\tau}\right)^{\mathrm{An}}(v)+\tau\right)=\inf_{i\in I}\sup_{\tau\in\mathbb{R}}\left(\Gamma_{i,\tau}^{\mathrm{An}}(v)+\tau\right).\tag{13.33}$$

As in the proof of Lemma 9.4.9, the decreasing net $\Gamma_{i,\tau}$ converges to $\inf_{i\in I}\Gamma_{i,\tau}$ in d_S. Hence Theorem 6.2.4 gives

$$\left(\inf_{i\in I}\Gamma_{i,\tau}\right)^{\mathrm{An}}(v)=\inf_{i\in I}\Gamma_{i,\tau}^{\mathrm{An}}(v),$$

so (13.33) is a consequence of Proposition A.2.3. □

Theorem 13.4.3 *The map* (13.28) *is an order preserving bijection.*

Proof The map (13.28) is an order preserving injection by Lemma 13.4.7. It remains to prove that it is surjective. Let $\phi\in\mathrm{PSH}(L^{\mathrm{An}})$. We want to construct $\Gamma\in\mathrm{PSH}^{\mathrm{NA}}(X,\theta)$ with $\Gamma^{\mathrm{An}}=\phi$.

Let H be an ample line bundle and $(\epsilon_i)_i$ be a decreasing sequence of rational numbers with limit 0, $\phi_i\in\mathcal{H}^{\mathrm{gf}}_{\mathbb{Q}}((L+\epsilon_iH)^{\mathrm{An}})$ such that

$$\phi=\inf_{i>0}\phi_i.$$

The existence of these data is guaranteed by Theorem 13.4.2. Fix a Kähler form $\omega\in c_1(H)$.

Thanks to Lemma 13.4.8, we can find $\Gamma^i\in\mathrm{PSH}^{\mathrm{NA}}(X,\theta+\epsilon_i\omega)$ with $(\Gamma^i)^{\mathrm{An}}=\phi_i$. It follows from Lemma 13.4.7 that

$$\Gamma^i\geq P_{\theta+\epsilon_i\omega}\left[\Gamma^{i+1}\right]_{\mathcal{I}}\geq\Gamma^{i+1}.$$

Therefore, for any $\omega'\in\mathrm{K\ddot{a}h}(X)$, after discarding finitely many i so that $\omega'\geq\epsilon_i\omega$, the sequence $(P_{\theta+\omega'}\left[\Gamma^i\right]_{\mathcal{I}})_i$ is decreasing. This does not change its infimum. We let

$$\Gamma^{\theta+\omega'}=\inf_{i\gg1}P_{\theta+\omega'}\left[\Gamma^i\right]_{\mathcal{I}}\in\mathrm{PSH}^{\mathrm{NA}}(X,\theta+\omega').$$

Note that the infimum is defined thanks to (13.30). It follows from Proposition 13.4.5 that

$$\left(\Gamma^{\theta+\omega'}\right)^{\mathrm{An}} = \phi.$$

From this, it is clear that for $\omega', \omega'' \in \mathrm{Käh}(X)$ with $\omega' \leq \omega''$, we have

$$P_{\theta+\omega''}\left[\Gamma^{\theta+\omega'}\right]_{\mathcal{I}} = \Gamma^{\theta+\omega''}.$$

It follows that $(\Gamma^{\theta+\omega'})_{\omega'\in\mathrm{Käh}(X)}$ defines an element Γ in $\mathrm{PSH}^{\mathrm{NA}}(X,\theta)$ and $\Gamma^{\mathrm{An}} = \phi$.□

Theorem 13.4.4 *Under the bijection Theorem 13.4.3, the operations on* $\mathrm{PSH}^{\mathrm{NA}}(X,\theta)$ *defined in Section 13.2 all correspond to the corresponding operations on* $\mathrm{PSH}(L^{\mathrm{An}})$ *in Boucksom–Jonsson's theory.*

The meaning is immediate for all operations except for the trace operator, and the proofs are elementary, as we have seen in Proposition 13.4.5 for the infimum operator. We shall only restate and prove the case of trace operators; the remaining arguments are recorded in [Xia25a].

Theorem 13.4.5 *Let $Y \subseteq X$ be an irreducible analytic subset. Consider an element* $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)$ *with well-defined restriction to Y. Then*

$$\mathrm{Tr}_Y(\Gamma)^{\mathrm{An}}|_{Y^{\mathrm{div}}} = \Gamma^{\mathrm{An}}|_{Y^{\mathrm{div}}}. \tag{13.34}$$

Observe that there is a canonical identification $Y^{\mathrm{div}} = \tilde{Y}^{\mathrm{div}}$. Recall that a generalized Fubini–Study metric is defined in Definition 1.8.7.

Proof We first reduce to the positive mass case. Choose a Kähler form ω_0 on X and set $\widehat{\Gamma} = P_{\theta+\omega_0}[\Gamma]_{\mathcal{I}}$. Replacing θ by $\theta+\omega_0$ and Γ by $\widehat{\Gamma}$ does not change Γ^{An}, by (13.29); it also only replaces $\mathrm{Tr}_Y(\Gamma)$ by the corresponding $\mathcal{I}$-projection on $\tilde{Y}$, whose analytification is unchanged. Hence we may assume that $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)_{>0}$. Let $\phi = \Gamma^{\mathrm{An}} \in \mathrm{PSH}(L^{\mathrm{An}})$. By Lemma 13.4.9, $\phi(\nu_{Y,\mathrm{triv}}) \neq -\infty$.

Take an ample line bundle S on X and a Kähler form ω in $c_1(S)$. Write ϕ as the decreasing limit of a sequence $(\phi^i)_i$ of elements in $\mathcal{H}^{\mathrm{gf}}_{\mathbb{Q}}((L+i^{-1}S)^{\mathrm{An}})$ as in Theorem 13.4.2.

Take $\Gamma^i \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta+i^{-1}\omega)$ such that $\Gamma^{i,\mathrm{An}} = \phi^i$. Note that by Lemma 13.1.2, $\Gamma^i \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta+i^{-1}\omega)_{>0}$.

Write $\widehat{\Gamma}^i := P_{\theta+\omega}[\Gamma^i]_{\mathcal{I}}$ and $\widehat{\Gamma} := P_{\theta+\omega}[\Gamma]_{\mathcal{I}}$. Applying Proposition 13.4.5 to the decreasing sequence $(\widehat{\Gamma}^i)_i$ and then using the injectivity in Lemma 13.4.7, we get $\inf_i \widehat{\Gamma}^i = \widehat{\Gamma}$. Hence, by the componentwise definition of this infimum, for any $\tau < \Gamma_{\max}$, we have

$$\inf_{i\in\mathbb{N}} \widehat{\Gamma}^i_\tau = \widehat{\Gamma}_\tau.$$

In particular, $\widehat{\Gamma}^i_\tau \xrightarrow{d_{S,\theta+\omega}} \widehat{\Gamma}_\tau$ for all $\tau < \Gamma_{\max}$.

By Lemma 13.4.9 again, each $\widehat{\Gamma}^i$ has non-trivial restriction to Y. By Proposition 8.2.2, for any Kähler form ω' on $\tilde{Y}$ satisfying $\omega' \geq \omega|_{\tilde{Y}}$ we have

$$\mathrm{Tr}_Y\left(\widehat{\Gamma}^i_\tau\right) \xrightarrow{d_S} \mathrm{Tr}_Y\left(\widehat{\Gamma}_\tau\right)$$

for any $\tau < (\mathrm{Tr}_Y(\Gamma))_{\max}$. Thanks to Theorem 6.2.4, for every $c\,\mathrm{ord}_F \in Y^{\mathrm{div}}$ and every such τ,

$$\mathrm{Tr}_Y\left(\widehat{\Gamma}_\tau\right)^{\mathrm{An}}(c\,\mathrm{ord}_F) = \inf_{i\geq 1}\mathrm{Tr}_Y\left(\widehat{\Gamma}^i_\tau\right)^{\mathrm{An}}(c\,\mathrm{ord}_F).$$

Applying Proposition A.2.3 to the Legendre transforms in τ, and using again that $\mathcal{I}$-projection does not change analytification, we get

$$\mathrm{Tr}_Y(\Gamma)^{\mathrm{An}}(c\,\mathrm{ord}_F) = \inf_{i\geq 1}\mathrm{Tr}_Y(\Gamma^i)^{\mathrm{An}}(c\,\mathrm{ord}_F).$$

Since also $\Gamma^{\mathrm{An}} = \inf_i \Gamma^{i,\mathrm{An}}$, it suffices to prove (13.34) with Γ^i in place of Γ.

In other words, we have reduced to the case where $\phi \in \mathcal{H}^{\mathrm{gf}}_{\mathbb{Q}}(L^{\mathrm{An}})$ and L is big.

Let $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X,\theta)_{>0}$ with $\Gamma^{\mathrm{An}} = \phi$. By Lemma 13.4.8, we can find a concave curve $(\Gamma'_\tau)_{\tau\leq\Gamma_{\max}}$ such that Γ'_τ is a generalized Fubini–Study metric for each $\tau \leq \Gamma_{\max}$ and such that

$$\Gamma_\tau = P_\theta[\Gamma'_\tau]_{\mathcal{I}} = P_\theta[\Gamma'_\tau].$$

Here the second equality follows from Example 7.1.1, since generalized Fubini–Study metrics have analytic singularities. It follows that for any $c\,\mathrm{ord}_F \in \tilde{Y}^{\mathrm{div}}$,

$$\begin{aligned}
\phi|_{Y^{\mathrm{An}}}(c\,\mathrm{ord}_F) &= \sup_{\tau<\Gamma_{\max}}\left(\Gamma'^{\mathrm{An}}_\tau(c\,\mathrm{ord}_F)+\tau\right)\\
&= \sup_{\tau<\Gamma_{\max}}\left((\Gamma'_\tau|_{\tilde{Y}})^{\mathrm{An}}(c\,\mathrm{ord}_F)+\tau\right)\\
&= \sup_{\tau<\Gamma_{\max}}\left(\mathrm{Tr}_Y(\Gamma'_\tau)^{\mathrm{An}}(c\,\mathrm{ord}_F)+\tau\right)\\
&= \sup_{\tau<\Gamma_{\max}}\left(\mathrm{Tr}_Y(\Gamma_\tau)^{\mathrm{An}}(c\,\mathrm{ord}_F)+\tau\right)\\
&= \mathrm{Tr}_Y(\Gamma)^{\mathrm{An}}(c\,\mathrm{ord}_F).
\end{aligned}$$

The third equality follows from Example 8.1.4, since Γ'_τ has analytic singularities. The fourth equality follows from $\Gamma_\tau = P_\theta[\Gamma'_\tau]$, Proposition 8.2.1 and Corollary 6.1.1. It remains to justify the second line. Namely, we want to show that for any generalized Fubini–Study metric φ whose restriction to $\tilde{Y}$ is not identically $-\infty$, we have

$$\varphi^{\mathrm{An}}(c\,\mathrm{ord}_F) = (\varphi|_{\tilde{Y}})^{\mathrm{An}}(c\,\mathrm{ord}_F). \tag{13.35}$$

We can immediately reduce to the case where φ is a Fubini–Study metric, and then to the case

$$\varphi = \log|s|^2_{h_0},$$

where s is a holomorphic section of L, not vanishing identically on Y, in which case (13.35) is obvious. ◻

Lemma 13.4.9 *Let* $\Gamma \in \mathrm{PSH}^{\mathrm{NA}}(X, \theta)$ *and* $Y \subseteq X$ *be an irreducible analytic subset. Then the following are equivalent:*

(1) $\Gamma^{\mathrm{An}}(v_{Y,\mathrm{triv}}) \neq -\infty$*;*
(2) $\Gamma^{\mathrm{An}}|_{Y^{\mathrm{An}}} \not\equiv -\infty$*;*
(3) Γ *has a well-defined restriction to* Y.

Here $v_{Y,\mathrm{triv}}$ denotes the trivial valuation of $\mathbb{C}(Y)$.

Proof The equivalence between (1) and (2) is a simple consequence of the maximum principle [BJ22a, Lemma 1.4(i)].

To see the equivalence between (1) and (3), it suffices to observe that for any $\varphi \in \mathrm{PSH}(X, \theta)$,

$$\varphi^{\mathrm{An}}(v_{Y,\mathrm{triv}}) = \begin{cases} -\infty, & \text{if } \nu(\varphi, Y) > 0; \\ 0, & \text{if } \nu(\varphi, Y) = 0. \end{cases}$$

This completes the proof. ◻

Chapter 14
Partial Bergman kernels

I speak twelve languages: English is the bestest.
— Stefan Bergman[a]

[a] Stefan Bergman (1895–1977), bearing a very Scandinavian name, was a Polish-American mathematician best known for his work in complex analysis, especially in several complex variables. He introduced the Bergman kernel, a fundamental concept in complex analysis that has influenced many areas of mathematics and theoretical physics.

Bergman was born in Poland (then part of the Russian Empire), and studied in Berlin. He fled Europe during World War II and eventually settled in the United States.

The classical Bergman kernel of a Hermitian line bundle L on a compact Kähler manifold X encodes, via its asymptotic expansion as the tensor power L^k goes to infinity, deep geometric information about the underlying metric. The aim of this chapter is to extend this idea to the *partial* Bergman kernel, defined by restricting attention to sections cut out by an ideal, typically the multiplier ideal of an auxiliary psh weight on a closed subset $K \subseteq X$.

The principal new result is Theorem 14.3.1: for a θ-psh potential $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$ and a closed non-pluripolar weighted subset (K,v), together with any Bernstein–Markov measure $\mu \in \mathrm{BM}(K,v)$, the normalized partial Bergman measure on X converges weakly to the partial equilibrium measure of (K,v,φ) as $k \to \infty$. This generalizes the celebrated result of Berman–Boucksom–Witt Nyström [BBWN11] (the case $\varphi = V_\theta$) to the relative setting; the assumptions are essentially the mildest under which the statement makes sense. We strongly recommend reading [BBWN11] before starting this chapter.

Section 14.1 introduces the *relative P-envelope* $P_{\theta,K}[\varphi](v)$ together with the *relative $\mathcal{I}$-envelope* $P_{\theta,K}[\varphi]_{\mathcal{I}}(v)$, in which the constraint $\eta \leq 0$ defining the usual P- and $\mathcal{I}$-envelopes is replaced by a constraint of the form $\eta|_K \leq v$. We establish their basic properties, including monotonicity, mass-preservation, and the relation to the existing envelope operators of Chapter 3.

Section 14.2 develops the equilibrium theory associated with a *weighted subset* (K,v) (Definition 14.2.1). We show that the Monge–Ampère measure $\theta^n_{P_{\theta,K}[\varphi](v)}$ is supported on the contact locus $\{P_{\theta,K}[\varphi](v)|_K = v\} \subseteq K$ (Lemma 14.2.1), thereby justifying the name *partial equilibrium measure*. We then introduce the *partial equilibrium energy* $\mathcal{E}^{\varphi}_{\theta,K}(v)$ (Definition 14.2.2) and prove its differentiability (Proposition 14.2.1), which will be the key analytic input for the quantization result of the next section.

Section 14.3 develops the quantization. We first introduce *Bernstein–Markov measures* $\mu \in \mathrm{BM}(K,v)$ (Definition 14.3.2), the natural class of measures on K

relative to which the Bernstein–Markov inequality holds. We then define the partial Bergman kernel and prove the main convergence theorem Theorem 14.3.1, first in the smooth case, then for potentials with analytic singularities, and finally in full generality via d_S-approximations.

14.1 Partial envelopes

Let X be a connected compact Kähler manifold of dimension n and $K \subseteq X$ be a closed non-pluripolar set. Let θ be a smooth closed real $(1,1)$-form on X representing a pseudo-effective cohomology class.

Definition 14.1.1 Given a function $f\colon K \to [-\infty, \infty]$ and $\varphi \in \mathrm{PSH}(X,\theta)$, we introduce the *relative P-envelope* of φ (with respect to K, f, θ) as

$$P_{\theta,K}[\varphi](f) \coloneqq \sup{}^*\left\{\eta \in \mathrm{PSH}(X,\theta) : \eta|_K \le f \text{ quasi-everywhere and } \eta \le \varphi\right\}. \tag{14.1}$$

Similarly, we define the *relative $\mathcal{I}$-envelope* of φ (with respect to K, f, θ) as

$$P_{\theta,K}[\varphi]_{\mathcal{I}}(f) \coloneqq \sup{}^*\left\{\eta \in \mathrm{PSH}(X,\theta) : \eta|_K \le f \text{ quasi-everywhere and } \eta \le_{\mathcal{I}} \varphi\right\}. \tag{14.2}$$

By $\eta|_K \le f$ quasi-everywhere, we mean that there is a pluripolar set $S \subseteq X$ so that $\eta|_{K\setminus S} \le f|_{K\setminus S}$.

Observe that when f is bounded from below, neither envelope is identically $-\infty$.

The envelopes contain essentially all envelope operators that we have studied in this book:

Example 14.1.1 When $K = X$ and $f = 0$, we have

$$P_{\theta,X}[\varphi](0) = P_\theta[\varphi], \quad P_{\theta,X}[\varphi]_{\mathcal{I}}(0) = P_\theta[\varphi]_{\mathcal{I}}.$$

When $K = X$, we have

$$P_{\theta,X}[\varphi](f) = P_\theta[\varphi](f).$$

These definitions reduce to the envelopes studied in Chapter 3. On the other hand, when $\varphi = V_\theta$, we have

$$\begin{aligned} P_{\theta,K}[V_\theta](f) &= P_{\theta,K}[V_\theta]_{\mathcal{I}}(f) \\ &= \sup{}^*\left\{\eta \in \mathrm{PSH}(X,\theta) : \eta|_K \le f \text{ quasi-everywhere}\right\}. \end{aligned}$$

In an even more special case where $\varphi = V_\theta$ and $K = X$, we find that

$$P_{\theta,X}[V_\theta](f) = \sup{}^*\left\{\eta \in \mathrm{PSH}(X,\theta) : \eta \le f \text{ quasi-everywhere}\right\}.$$

This is nothing but $P_\theta(f)$ as we studied in Definition 2.4.1.

Proposition 14.1.1 *Let* $f, g\colon K \to [-\infty, \infty]$ *be functions and* $\varphi, \psi \in \mathrm{PSH}(X, \theta)$. *Then we have the following:*

(1) *We have*
$$P_{\theta,K}[\varphi](f) \leq P_{\theta,K}[\varphi]_{\mathcal{I}}(f);$$

(2) *the envelopes are quasi-everywhere dominated by* f *on* K*:*
$$\begin{aligned} P_{\theta,K}[\varphi](f)\big|_K &\leq f \text{ quasi-everywhere}, \\ P_{\theta,K}[\varphi]_{\mathcal{I}}(f)\big|_K &\leq f \text{ quasi-everywhere}; \end{aligned}$$

(3) *suppose that* $\varphi \preceq_{\mathcal{I}} \psi$*, then*
$$P_{\theta,K}[\varphi]_{\mathcal{I}}(f) \leq P_{\theta,K}[\psi]_{\mathcal{I}}(f);$$

(4) *if* $f \leq g$*, then*
$$P_{\theta,K}[\varphi](f) \leq P_{\theta,K}[\varphi](g), \quad P_{\theta,K}[\varphi]_{\mathcal{I}}(f) \leq P_{\theta,K}[\varphi]_{\mathcal{I}}(g);$$

(5) *for any* $t \in (0, 1)$*, we have*
$$\begin{aligned} P_{\theta,K}\left[t\varphi + (1-t)\psi\right](f) &\geq t P_{\theta,K}[\varphi](f) + (1-t) P_{\theta,K}[\psi](f), \\ P_{\theta,K}\left[t\varphi + (1-t)\psi\right]_{\mathcal{I}}(f) &\geq t P_{\theta,K}[\varphi]_{\mathcal{I}}(f) + (1-t) P_{\theta,K}[\psi]_{\mathcal{I}}(f). \end{aligned}$$

Proof (1) This follows from $\eta \leq \varphi \Rightarrow \eta \preceq_{\mathcal{I}} \varphi$.

(2) This follows from Proposition 1.2.5.

(3) This follows by inclusion of the defining candidate sets.

(4) This follows by inclusion of the defining candidate sets.

(5) If $\eta, \eta' \in \mathrm{PSH}(X, \theta)$ are such that
$$\begin{aligned} &\eta|_K \leq f \text{ quasi-everywhere}, \quad \eta'|_K \leq f \text{ quasi-everywhere}, \\ &\eta \leq \varphi, \quad \eta' \leq \psi, \end{aligned}$$
then for $t \in (0, 1)$,
$$(t\eta + (1-t)\eta')|_K \leq f \text{ quasi-everywhere}, \quad t\eta + (1-t)\eta' \leq t\varphi + (1-t)\psi.$$
Therefore,
$$t\eta + (1-t)\eta' \leq P_{\theta,K}\left[t\varphi + (1-t)\psi\right](f).$$
Since η and η' are arbitrary, the first inequality follows. The second follows from the same argument. □

Similar to the usual envelopes studied in Chapter 3, the P-partial envelopes are well-behaved only when φ has positive mass.

Lemma 14.1.1 *Suppose that* $f\colon K \to (-\infty, \infty]$ *is bounded from below, and* $\varphi \in \mathrm{PSH}(X, \theta)_{>0}$. *Then*

$$P_{\theta,K}[\varphi](f) = \sup{}^*\Big\{\eta \in \mathrm{PSH}(X,\theta)_{>0} : \eta|_K \leq f \text{ quasi-everywhere and } \eta \leq \varphi\Big\}. \tag{14.3}$$

In other words, in this situation, in the definition (14.1), we may restrict to only those η with positive mass.

Proof By our assumption, we can find $C > 0$ so that $\varphi|_K - C \leq f$. Then if η is a candidate in (14.1), $\eta \vee (\varphi - C)$ is also a candidate, and $\eta \vee (\varphi - C) \in \mathrm{PSH}(X,\theta)_{>0}$. Therefore, $\eta \vee (\varphi - C)$ is also a candidate for the right-hand side of (14.3). The equality (14.3) follows. □

Note that in the most important cases, we can actually replace the condition $\eta \leq f$ quasi-everywhere in (14.1) by everywhere.

Lemma 14.1.2 *Let* $f\colon K \to (-\infty, \infty]$ *be bounded from below, and* $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$. *Then*

$$\begin{aligned} P_{\theta,K}[\varphi](f) &= \sup{}^*\Big\{\eta \in \mathrm{PSH}(X,\theta) : \eta|_K \leq f \text{ and } \eta \leq \varphi\Big\} \\ &= \sup{}^*\Big\{\eta \in \mathrm{PSH}(X,\theta)_{>0} : \eta|_K \leq f \text{ and } \eta \leq \varphi\Big\}. \end{aligned} \tag{14.4}$$

Proof Both $\geq$ directions in (14.4) are trivial.

Conversely, let η be a candidate in (14.3). Let $S \subseteq X$ be a pluripolar set so that $\eta|_{K\setminus S} \leq f|_{K\setminus S}$.

We now claim that we can choose $\chi \in \mathrm{PSH}(X,\theta)_{>0}$ with $\chi|_S \equiv -\infty$ and $\chi \leq \eta$.

Assume this for the moment. For any $\delta \in (0,1)$, we have

$$(1-\delta)\eta|_K + \delta\chi|_K \leq f, \quad (1-\delta)\eta + \delta\chi \leq \varphi.$$

Hence, $(1-\delta)\eta + \delta\chi$ is dominated by the right-hand side of (14.4). Letting $\delta \to 0+$, we conclude that η is dominated by the right-hand side of (14.4) and hence, (14.4) follows.

It remains to construct χ. Since φ has positive mass, the class $[\theta]$ is big. Fix a Kähler form ω_0 on X and choose $\rho \in \mathrm{PSH}(X,\theta)$ with $\theta_\rho \geq 2\epsilon_0\omega_0$ for some $\epsilon_0 > 0$. By Theorem 1.1.5, there is a quasi-psh function q such that $q|_S \equiv -\infty$. Choose $A > 0$ so that $A\omega_0 + \mathrm{dd^c} q \geq 0$. For $a > 0$ small enough, $\rho + aq \in \mathrm{PSH}(X,\theta)$ and still has pole $-\infty$ on S. Replacing ρ by $\rho + aq$, we may therefore assume that $\rho|_S \equiv -\infty$. Since $\int_X \theta_\eta^n > 0$, fixing such a ρ and taking $\epsilon \in (0,1)$ small enough, we have

$$\int_X \theta^n_{\epsilon\rho + (1-\epsilon)V_\theta} + \int_X \theta^n_\eta > \int_X \theta^n_{V_\theta}.$$

Thus, by Proposition 3.1.5 and Theorem 3.1.3, we have

$$\left(\epsilon\rho + (1-\epsilon)V_\theta\right) \wedge \eta \in \mathrm{PSH}(X,\theta)_{>0}.$$

It suffices to define this function as χ. □

Corollary 14.1.1 *Suppose that* $f\colon K \to \mathbb{R}$ *is bounded, and* $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$. *Then there is a constant* $C = C(\sup_K f, K, X) > 0$ *such that*

$$P_{\theta,K}[\varphi](f) \le C, \quad P_{\theta,K}[\varphi]_{\mathcal{I}}(f) \le C.$$

Proof Since K is non-pluripolar, there is $C = C(\sup_K f, K, X) > 0$ so that for any $\eta \in \mathrm{PSH}(X,\theta)$ with $\eta|_K \le f$ quasi-everywhere, we have $\eta \le C$, see Remark 1.5.2. The assertion for $P_{\theta,K}[\varphi](f)$ follows from Lemma 14.1.2, and the assertion for $P_{\theta,K}[\varphi]_{\mathcal{I}}(f)$ follows from Proposition 14.1.1. □

Proposition 14.1.2 *Let* $f\colon K \to (-\infty,\infty]$ *be a function bounded from below and* $\varphi,\psi \in \mathrm{PSH}(X,\theta)_{>0}$. *Suppose that* $\varphi \preceq_P \psi$. *Then*

$$P_{\theta,K}[\varphi](f) \le P_{\theta,K}[\psi](f).$$

Proof Let $\eta \in \mathrm{PSH}(X,\theta)_{>0}$ be such that $\eta|_K \le f$ quasi-everywhere and $\eta \le \varphi$. Then $\eta \preceq_P \psi$, and thanks to Corollary 6.1.4 for any $C > 0$, we have $\eta \wedge (\psi + C) \in \mathrm{PSH}(X,\theta)_{>0}$. But then $\eta \wedge (\psi + C)$ is a candidate in the definition of $P_{\theta,K}[\psi](f)$, and we find

$$\eta \wedge (\psi + C) \le P_{\theta,K}[\psi](f).$$

By Corollary 6.1.4 again, we get

$$\eta \le P_{\theta,K}[\psi](f).$$

Our assertion follows now from Lemma 14.1.1. □

Corollary 14.1.2 *Let* $f\colon K \to (-\infty,\infty]$ *be a function bounded from below and* $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$. *Then*

$$\begin{aligned} P_{\theta,K}[\varphi](f) &= \sup{}^*\left\{\eta \in \mathrm{PSH}(X,\theta) : \eta|_K \le f \text{ quasi-everywhere and } \eta \preceq_P \varphi\right\} \\ &= \sup{}^*\left\{\eta \in \mathrm{PSH}(X,\theta)_{>0} : \eta|_K \le f \text{ quasi-everywhere and } \eta \preceq_P \varphi\right\} \\ &= \sup{}^*\left\{\eta \in \mathrm{PSH}(X,\theta) : \eta|_K \le f \text{ and } \eta \preceq_P \varphi\right\} \\ &= \sup{}^*\left\{\eta \in \mathrm{PSH}(X,\theta)_{>0} : \eta|_K \le f \text{ and } \eta \preceq_P \varphi\right\}. \end{aligned}$$

Proof We first apply Proposition 14.1.2,

$$\begin{aligned} P_{\theta,K}[\varphi](f) =& P_{\theta,K}\left[P_\theta[\varphi]\right](f) \\ &= \sup{}^*\left\{\eta \in \mathrm{PSH}(X,\theta) : \eta|_K \le f \text{ quasi-everywhere and } \eta \preceq_P \varphi\right\}. \end{aligned}$$

Next observe that by Lemma 14.1.1, the latter can also be written as

$$\sup{}^*\Big\{\eta \in \mathrm{PSH}(X,\theta)_{>0} : \eta|_K \leq f \text{ quasi-everywhere and } \eta \preceq_P \varphi\Big\}.$$

Next we apply Lemma 14.1.2; the remaining statements follow. □

Corollary 14.1.3 *Let $f\colon K \to (-\infty,\infty]$ be a function bounded from below and $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$ be $\mathcal{I}$-good. Then*

$$P_{\theta,K}[\varphi]_{\mathcal{I}}(f) = P_{\theta,K}[\varphi](f).$$

Proof In fact, since φ is $\mathcal{I}$-good, a potential $\eta \in \mathrm{PSH}(X,\theta)$ satisfies $\eta \preceq_P \varphi$ if and only if $\eta \preceq_{\mathcal{I}} \varphi$. Therefore, our assertion follows from Corollary 14.1.2 and (14.2).□

Next, we make the following observations about the singularity types of our envelopes:

Lemma 14.1.3 *Let $f\colon K \to \mathbb{R}$ be a bounded function, and $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$. We have*

$$P_{\theta,K}[\varphi](f) \sim P_\theta[\varphi], \quad P_{\theta,K}[\varphi]_{\mathcal{I}}(f) \sim P_\theta[\varphi]_{\mathcal{I}}. \tag{14.5}$$

Proof We first prove the P-part of (14.5).

If $\eta \in \mathrm{PSH}(X,\theta)$ with $\eta \leq \varphi$ and $\eta \leq 0$, then $\eta + \inf_K f$ is a candidate in the definition (14.1), and hence,

$$\eta + \inf_K f \leq P_{\theta,K}[\varphi](f).$$

It follows that

$$P_\theta[\varphi] + \inf_K f \leq P_{\theta,K}[\varphi](f).$$

Conversely, let $\eta \in \mathrm{PSH}(X,\theta)$ be such that

$$\eta|_K \leq f, \quad \eta \leq \varphi.$$

Since K is non-pluripolar, by Remark 1.5.2 we can find $C = C(\sup_K f, K, X) > 0$ so that $\eta \leq C$. Hence $\eta - C$ is a candidate in the definition of $P_\theta[\varphi]$, and hence

$$\eta - C \leq P_\theta[\varphi].$$

By Lemma 14.1.2, this implies that

$$P_{\theta,K}[\varphi](f) \leq P_\theta[\varphi] + C.$$

The proof of the $\mathcal{I}$-part is similar, but we spell it out since this comparison will be used below. If $\eta \in \mathrm{PSH}(X,\theta)$ satisfies $\eta \leq 0$ and $\eta \sim_{\mathcal{I}} \varphi$, then $\eta + \inf_K f$ is a candidate in (14.2), because bounded perturbations preserve the relation $\preceq_{\mathcal{I}}$. Hence

$$P_\theta[\varphi]_{\mathcal{I}} + \inf_K f \leq P_{\theta,K}[\varphi]_{\mathcal{I}}(f).$$

Conversely, let $\eta \in \mathrm{PSH}(X,\theta)$ be a candidate in (14.2). Since K is non-pluripolar and f is bounded, the same compactness estimate gives a constant $C > 0$, independent of η, such that $\eta \le C$. Then $\eta - C \le 0$ and $\eta - C \le_{\mathcal{I}} \varphi$. By Corollary 6.1.3,

$$\eta - C \le P_\theta[\varphi]_{\mathcal{I}}.$$

Taking the upper envelope over all such η gives

$$P_{\theta,K}[\varphi]_{\mathcal{I}}(f) \le P_\theta[\varphi]_{\mathcal{I}} + C.$$

This completes the proof. □

Corollary 14.1.4 *Let $f\colon K \to \mathbb{R}$ be a bounded function, and $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$. Then*

$$P_{\theta,K}[\varphi](f) = P_{\theta,X}[\varphi]\left(P_{\theta,K}[V_\theta](f)\right). \tag{14.6}$$

Proof Take any $\eta \in \mathrm{PSH}(X,\theta)$ with $\eta \le \varphi$ and $\eta|_K \le f$ quasi-everywhere, then

$$\eta \le P_{\theta,K}[\varphi](f) \le P_{\theta,K}[V_\theta](f).$$

Therefore,

$$\eta \le P_{\theta,X}[\varphi]\left(P_{\theta,K}[V_\theta](f)\right).$$

So we conclude the $\le$ direction in (14.6).

For the reverse direction, take $\eta \in \mathrm{PSH}(X,\theta)_{>0}$ such that

$$\eta \le \varphi, \quad \eta \le P_{\theta,K}[V_\theta](f) \quad \text{quasi-everywhere.}$$

Then $\eta|_K \le f$ quasi-everywhere by Proposition 14.1.1. Hence

$$\eta \le P_{\theta,K}[\varphi](f).$$

By Corollary 14.1.2, the $\ge$ direction in (14.6) follows. □

14.2 Partial equilibrium measures

Let X be a connected compact Kähler manifold of dimension n, and θ be a smooth closed real $(1,1)$-form on X representing a pseudo-effective cohomology class.

Definition 14.2.1 A *weighted subset* of X is a pair (K,v) consisting of a closed non-pluripolar subset $K \subseteq X$ and a function $v \in C^0(K)$.

The next result motivates our terminology to call the measures $\theta^n_{P_{\theta,K}[\varphi](f)}$ the *partial equilibrium measures* of our context:

Lemma 14.2.1 *Let (K,v) be a weighted subset of X, and $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$. Then*

$$\int_{X\setminus K}\theta^n_{P_{\theta,K}[\varphi](\nu)}=0,\quad \int_K \mathbb{1}_{\{P_{\theta,K}[\varphi](\nu)|_K\neq\nu\}}\theta^n_{P_{\theta,K}[\varphi](\nu)}=0. \tag{14.7}$$

More precisely, we have

$$\theta^n_{P_{\theta,K}[\varphi](\nu)}\leq \mathbb{1}_{\{P_{\theta,K}[\varphi](\nu)|_K=P_{\theta,K}[V_\theta](\nu)|_K=\nu\}}\,\theta^n_{P_{\theta,K}[V_\theta](\nu)}. \tag{14.8}$$

Here $\{P_{\theta,K}[\varphi](\nu)|_K\neq\nu\}$ and $\{P_{\theta,K}[\varphi](\nu)|_K=P_{\theta,K}[V_\theta](\nu)|_K=\nu\}$ are understood as consisting of points of K satisfying these constraints.

Proof **Step 1**. We first prove (14.7) when $\varphi=V_\theta$.

Let $S\subseteq X$ be a proper analytic subset, such that V_θ is locally bounded on $X\setminus S$. Such S exists by Lemma 2.4.1. Fix holomorphic balls $B\Subset B'\Subset X\setminus S$.

By Proposition 1.2.3 and Lemma 14.1.2, we can take an increasing sequence $(\eta_j)_j$ in $\mathrm{PSH}(X,\theta)$ such that

$$\eta_j\nearrow P_{\theta,K}[V_\theta](\nu)\text{ a.e.},\quad \eta_j|_K\leq\nu\text{ for all }j\geq 1.$$

Note that $P_{\theta,K}[V_\theta](\nu)\sim V_\theta$ by Lemma 14.1.3. Take $C>0$ so that

$$\left(P_{\theta,K}[V_\theta](\nu)\right)\big|_K-C\leq\nu.$$

Then we may replace η_j by $\eta_j\vee(P_{\theta,K}[V_\theta](\nu)-C)$ and assume that $\eta_j\sim V_\theta$ for each j. In particular, η_j is locally bounded on $X\setminus S$.

For each $j\geq 1$, let $\gamma_j\in\mathrm{PSH}(X,\theta)$ be the balayage of η_j on B, defined in Proposition 1.5.4. Note that $(\gamma_j)_j$ is increasing due to Proposition 1.2.10.

For the first equality in (14.7), we may assume that $B\Subset X\setminus(S\cup K)$. It suffices to show that $\theta^n_{P_{\theta,K}[V_\theta](\nu)}$ does not charge B.

Since B lies outside K by assumption, we also have $\gamma_j|_K=\eta_j|_K\leq\nu$ for all $j\geq 1$. In particular, $\gamma_j\leq P_{\theta,K}[V_\theta](\nu)$.

It follows that $\gamma_j\nearrow P_{\theta,K}[V_\theta](\nu)$ almost everywhere as well. Testing against non-negative continuous functions supported in B and using Theorem 2.4.2, we find that $\theta^n_{P_{\theta,K}[V_\theta](\nu)}$ does not charge B, as desired.

We next prove the second equality in (14.7). The proof is essentially the same as that of Lemma 2.4.3.

It suffices to show that $\theta^n_{P_{\theta,K}[V_\theta](\nu)}$ does not charge the set $K\cap\{P_{\theta,K}[V_\theta](\nu)|_K<\nu\}$.

Let $x\in(X\setminus S)\cap K$ be a point such that $P_{\theta,K}[V_\theta](\nu)(x)<\nu(x)-\epsilon$ for some $\epsilon>0$. We may assume that $x\in B$ and B' is small enough, so that

(1) $P_{\theta,K}[V_\theta](\nu)|_{\overline{B'}}<\nu(x)-\epsilon$.
(2) $\nu|_{\overline{B'}\cap K}>\nu(x)-\epsilon$.

By (1), we have $\gamma_j|_{\partial B}\leq\nu(x)-\epsilon$, hence by Proposition 1.2.10, $\gamma_j|_B\leq\nu(x)-\epsilon$. By (2), we have $\gamma_j|_{B\cap K}\leq\nu|_{B\cap K}$. On the other hand, $\gamma_j|_{K\setminus B}=\eta_j|_{K\setminus B}\leq\nu|_{K\setminus B}$. Therefore, $\gamma_j|_K\leq\nu$. It follows again that $\gamma_j\leq P_{\theta,K}[V_\theta](\nu)$. Therefore, we conclude as in the previous paragraph that $\gamma_j \nearrow P_{\theta,K}[V_\theta](v)$. Thus, testing against non-negative continuous functions supported in B, we conclude that $\theta^n_{P_{\theta,K}[V_\theta](v)}$ does not charge B again by Theorem 2.4.2.

Step 2. We handle the general case.

By Corollary 14.1.4,

$$P_{\theta,K}[\varphi](v) = P_{\theta,X}[\varphi]\left(P_{\theta,K}[V_\theta](v)\right).$$

Now Corollary 3.1.1 gives

$$\begin{aligned}\theta^n_{P_{\theta,K}[\varphi](v)} \leq & \mathbb{1}_{\{P_{\theta,K}[\varphi](v)=P_{\theta,K}[V_\theta](v)\}}\theta^n_{P_{\theta,K}[V_\theta](v)} \\ \leq & \mathbb{1}_{\{P_{\theta,K}[\varphi](v)|_K=P_{\theta,K}[V_\theta](v)|_K=v\}}\,\theta^n_{P_{\theta,K}[V_\theta](v)},\end{aligned}$$

where in the second inequality we have used Step 1. Hence (14.8) follows, and *a fortiori* (14.7) follows as well. □

Corollary 14.2.1 *Let $v \in C^0(K)$ and $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$. Then*

$$\begin{aligned}\int_{(X\setminus K)\cup\{P_{\theta,K}[\varphi](v)<v\}} \theta^n_{P_{\theta,K}[\varphi](v)} =&0, \\ \int_{(X\setminus K)\cup\{P_{\theta,K}[\varphi]_{\mathcal{I}}(v)<v\}} \theta^n_{P_{\theta,K}[\varphi]_{\mathcal{I}}(v)} =&0.\end{aligned} \tag{14.9}$$

Proof The first equation in (14.9) follows from Lemma 14.2.1. For the second, it suffices to observe that

$$P_{\theta,K}[\varphi]_{\mathcal{I}}(v) = P_{\theta,K}\left[P_\theta[\varphi]_{\mathcal{I}}\right]_{\mathcal{I}}(v) = P_{\theta,K}\left[P_\theta[\varphi]_{\mathcal{I}}\right](v)$$

by Corollary 14.1.3. □

Definition 14.2.2 Given $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$, we define the *partial equilibrium energy* functional $\mathcal{E}^\varphi_{\theta,K} : C^0(K) \to \mathbb{R}$ as follows: given $v \in C^0(K)$, we set

$$\mathcal{E}^\varphi_{\theta,K}(v) := E^{P_\theta[\varphi]_{\mathcal{I}}}_\theta\left(P_{\theta,K}[\varphi]_{\mathcal{I}}(v)\right). \tag{14.10}$$

Recall that the energy $E^{P_\theta[\varphi]_{\mathcal{I}}}_\theta$ functional is defined in Definition 3.1.6.

Note that by Lemma 14.1.3, we have

$$P_{\theta,K}[\varphi]_{\mathcal{I}}(v) \in \mathcal{E}^\infty\left(X,\theta;P_\theta[\varphi]_{\mathcal{I}}\right),$$

so $\mathcal{E}^\varphi_{\theta,K}(v) \in \mathbb{R}$.

Proposition 14.2.1 *Let $v, f \in C^0(K)$ and $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$. Then $\mathbb{R} \ni t \mapsto \mathcal{E}^\varphi_{\theta,K}(v+tf)$ is differentiable and*

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{E}^\varphi_{\theta,K}(v+tf) = \int_K f\,\theta^n_{P_{\theta,K}[\varphi]_{\mathcal{I}}(v+tf)} \tag{14.11}$$

for all $t \in \mathbb{R}$.

Proof We may assume that φ is $\mathcal{I}$-model by replacing φ by $P_\theta[\varphi]_{\mathcal{I}}$.

Note that it suffices to prove (14.11) at $t = 0$, which is equivalent to

$$\lim_{t\to 0} \frac{E_\theta^\varphi\left(P_{\theta,K}[\varphi]_{\mathcal{I}}(v+tf)\right) - E_\theta^\varphi\left(P_{\theta,K}[\varphi]_{\mathcal{I}}(v)\right)}{t} = \int_K f\,\theta^n_{P_{\theta,K}[\varphi]_{\mathcal{I}}(v)}. \tag{14.12}$$

It suffices to prove the right derivative at 0 for an arbitrary f; applying the same argument to $-f$ then gives the left derivative.

Fix $t > 0$. By the comparison principle (3.25) and Proposition 3.1.18, we find

$$\begin{aligned}
&E_\theta^\varphi\left(P_{\theta,K}[\varphi]_{\mathcal{I}}(v+tf)\right) - E_\theta^\varphi\left(P_{\theta,K}[\varphi]_{\mathcal{I}}(v)\right)\\
=&\frac{1}{n+1}\sum_{i=0}^n \int_X \left(P_{\theta,K}[\varphi]_{\mathcal{I}}(v+tf) - P_{\theta,K}[\varphi]_{\mathcal{I}}(v)\right)\,\theta^i_{P_{\theta,K}[\varphi]_{\mathcal{I}}(v+tf)} \wedge \theta^{n-i}_{P_{\theta,K}[\varphi]_{\mathcal{I}}(v)}\\
\leq&\int_X \left(P_{\theta,K}[\varphi]_{\mathcal{I}}(v+tf) - P_{\theta,K}[\varphi]_{\mathcal{I}}(v)\right)\,\theta^n_{P_{\theta,K}[\varphi]_{\mathcal{I}}(v)}.
\end{aligned}$$

By Lemma 14.2.1, we have

$$\int_X \left(P_{\theta,K}[\varphi]_{\mathcal{I}}(v+tf) - P_{\theta,K}[\varphi]_{\mathcal{I}}(v)\right)\,\theta^n_{P_{\theta,K}[\varphi]_{\mathcal{I}}(v)} \leq t\int_K f\,\theta^n_{P_{\theta,K}[\varphi]_{\mathcal{I}}(v)}.$$

Thus, letting $t \to 0+$, we get the inequality

$$\varlimsup_{t\to 0+} \frac{E_\theta^\varphi\left(P_{\theta,K}[\varphi]_{\mathcal{I}}(v+tf)\right) - E_\theta^\varphi\left(P_{\theta,K}[\varphi]_{\mathcal{I}}(v)\right)}{t} \leq \int_K f\,\theta^n_{P_{\theta,K}[\varphi]_{\mathcal{I}}(v)}.$$

Similarly, for $t > 0$, we have

$$\begin{aligned}
&E_\theta^\varphi\left(P_{\theta,K}[\varphi]_{\mathcal{I}}(v+tf)\right) - E_\theta^\varphi\left(P_{\theta,K}[\varphi]_{\mathcal{I}}(v)\right)\\
\geq&\int_X \left(P_{\theta,K}[\varphi]_{\mathcal{I}}(v+tf) - P_{\theta,K}[\varphi]_{\mathcal{I}}(v)\right)\,\theta^n_{P_{\theta,K}[\varphi]_{\mathcal{I}}(v+tf)}\\
\geq& t\int_K f\,\theta^n_{P_{\theta,K}[\varphi]_{\mathcal{I}}(v+tf)}.
\end{aligned}$$

Letting $t \to 0+$, we find

$$\varliminf_{t\to 0+} \frac{E_\theta^\varphi\left(P_{\theta,K}[\varphi]_{\mathcal{I}}(v+tf)\right) - E_\theta^\varphi\left(P_{\theta,K}[\varphi]_{\mathcal{I}}(v)\right)}{t} \geq \int_K f\,\theta^n_{P_{\theta,K}[\varphi]_{\mathcal{I}}(v)}.$$

Here we used the following elementary continuity consequence: since

$$P_{\theta,K}[\varphi]_{\mathcal{I}}(v) - t\|f\|_{C^0(K)} \leq P_{\theta,K}[\varphi]_{\mathcal{I}}(v+tf) \leq P_{\theta,K}[\varphi]_{\mathcal{I}}(v) + t\|f\|_{C^0(K)},$$

the envelopes converge in d_S as $t \to 0+$, and hence their Monge–Ampère measures converge weakly by Theorem 6.2.1. We conclude (14.12). □

Lemma 14.2.2 *Fix a Kähler form ω on X and a weighted subset (K, v) of X. There exists a uniformly bounded increasing sequence $(v_j^-)_j$ in $C^\infty(X)$ and a uniformly bounded decreasing sequence $(v_j^+)_j$ in $C^\infty(X)$, such that*

$$\sup_X |v_j^-| \leq C, \quad \sup_X |v_j^+| \leq C \tag{14.13}$$

for some constant C depending only on $\|v\|_{C^0(K)}$, X, K and $\theta + \omega$. Furthermore, for any $\varphi \in \mathrm{PSH}(X, \theta)_{>0}$ and any $\delta \in [0, 1]$ we have

(1) *We have*

$$P_{\theta+\delta\omega,X}[\varphi](v_j^+) \searrow P_{\theta+\delta\omega,K}[\varphi](v),$$

and

$$P_{\theta+\delta\omega,X}[\varphi](v_j^-) \nearrow P_{\theta+\delta\omega,K}[\varphi](v)$$

almost everywhere;

(2)

$$\lim_{j\to\infty} \mathcal{E}^\varphi_{\theta,X}(v_j^-) = \mathcal{E}^\varphi_{\theta,K}(v), \quad \lim_{j\to\infty} \mathcal{E}^\varphi_{\theta,X}(v_j^+) = \mathcal{E}^\varphi_{\theta,K}(v).$$

Proof **Step 1**. First we construct $(v_j^-)_j$.

Let

$$C_{K,v} := \sup\left\{\sup_X \eta : \eta \in \mathrm{PSH}(X, \theta + \omega), \eta|_K \leq v\right\}.$$

Since K is non-pluripolar, we have $C_{K,v} \in \mathbb{R}$ by Remark 1.5.2. Set

$$M_{K,v} := \max\{C_{K,v} + 1, \sup_K v\}.$$

Now define $\widetilde{v}\colon X \to \mathbb{R}$ as

$$\widetilde{v}(x) = \begin{cases} v(x), & x \in K; \\ M_{K,v}, & x \in X \setminus K. \end{cases}$$

Since $M_{K,v} \geq \sup_K v$, the function $\widetilde{v}$ is lower semicontinuous. Therefore there exists an increasing and uniformly bounded sequence $(v_j^-)_j$ in $C^\infty(X)$ such that $v_j^- \nearrow \widetilde{v}$, and such that

$$\|v_j^-\|_{C^0(X)} \leq \|\tilde{v}\|_{L^\infty(X)} + 1.$$

The right-hand side can clearly be bounded by a constant as in (14.13).

Step 2. We construct $(v_j^+)_j$.

Since

$$h := P_{\theta+\omega,K}[V_{\theta+\omega}](v) \vee \left(\inf_K v - 1\right)$$

is usc, there exists a decreasing and uniformly bounded sequence $(v_j^+)_j$ in $C^\infty(X)$ such that $v_j^+ \searrow h$. We may guarantee that

$$\|v_j^+\|_{C^0(X)} \leq \|h\|_{L^\infty(X)} + 1.$$

Note that $\inf_K v - 1 \leq h \leq C_{K,v} \vee (\inf_K v - 1)$. So (14.13) is satisfied.

Step 3. We finish the proof. Fix $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$ and $\delta \in [0,1]$.

Note that (2) follows from Lemma 14.1.3, Proposition 3.1.20 and (1). Let us prove (1).

Step 3.1. We prove (1) for $(v_j^-)_j$.

Observe that $P_{\theta+\delta\omega,X}[\varphi](v_j^-)$ is increasing in $j \geq 1$, and

$$P_{\theta+\delta\omega,X}[\varphi]\left(v_j^-\right) \leq P_{\theta+\delta\omega,K}[\varphi](v).$$

We claim that

$$P_{\theta+\delta\omega,X}[\varphi]\left(v_j^-\right) \nearrow P_{\theta+\delta\omega,K}[\varphi](v) \quad \text{a.e.}. \tag{14.14}$$

Let $\eta \in \mathrm{PSH}(X,\theta+\delta\omega)$ be a potential such that $\eta \leq \varphi$ and $\eta|_K \leq v$. Take some $\epsilon > 0$. Then, since η is upper semicontinuous, $\widetilde{v}$ is lower semicontinuous and $\eta < \epsilon + \widetilde{v}$, compactness gives $j_0 > 0$ such that $\eta < \epsilon + v_j^-$ for $j \geq j_0$. Hence,

$$\eta \leq \epsilon + P_{\theta+\delta\omega,X}[\varphi](v_j^-)$$

for $j \geq j_0$. Therefore,

$$\eta \leq \sup_{j\geq 1}{}^* P_{\theta+\delta\omega,X}[\varphi]\left(v_j^-\right) + \epsilon.$$

Letting $\epsilon \to 0+$, we find

$$\eta \leq \sup_{j\geq 1}{}^* P_{\theta+\delta\omega,X}[\varphi]\left(v_j^-\right).$$

Since η is arbitrary, (14.14) follows.

Step 3.2. We prove (1) for $(v_j^+)_j$.

Note that $P_{\theta+\delta\omega,X}[\varphi](v_j^+)$ is decreasing in j, and

$$\chi \coloneqq \lim_{j\to\infty} P_{\theta+\delta\omega,X}[\varphi]\left(v_j^+\right) \geq P_{\theta+\delta\omega,K}[\varphi](v). \tag{14.15}$$

In particular, $\chi \in \mathrm{PSH}(X,\theta+\delta\omega)_{>0}$, since $P_{\theta+\delta\omega,K}[\varphi](v)$ has the same singularity types as $P_{\theta+\delta\omega}[\varphi]$ by Lemma 14.1.3. Observe that

$$P_{\theta+\delta\omega,X}[\varphi]\left(v_j^+\right) \leq v_j^+$$

for all $j \geq 1$, since v_j^+ is continuous. Thus, $\chi \leq h$. On the other hand, $h|_K \leq v$ quasi-everywhere. So in particular, $\chi|_K \leq v$ quasi-everywhere. On the other hand,

$$\chi \leq P_{\theta+\delta\omega,X}[\varphi]\left(v_1^+\right) \sim_P \varphi$$

by Lemma 14.1.3. Hence, by Corollary 14.1.2,

$$\chi \le P_{\theta+\delta\omega,K}[\varphi](v).$$

Therefore, (14.15) is an equality. □

Proposition 14.2.2 *Let (K, v) be a weighted subset of X. Let $(\varphi_j)_j$ be a sequence in* $\mathrm{PSH}(X, \theta)_{>0}$ *and $\varphi \in \mathrm{PSH}(X, \theta)_{>0}$. Then the following hold:*

(1) *If $(\varphi_j)_j$ is decreasing, and $\varphi_j \xrightarrow{d_S} \varphi$, then*

$$P_{\theta,K}[\varphi_j]_{\mathcal{I}}(v) \searrow P_{\theta,K}[\varphi]_{\mathcal{I}}(v), \quad P_{\theta,K}[\varphi_j](v) \searrow P_{\theta,K}[\varphi](v). \tag{14.16}$$

(2) *If $\varphi_j \nearrow \varphi$ almost everywhere then*

$$P_{\theta,K}[\varphi_j]_{\mathcal{I}}(v) \nearrow P_{\theta,K}[\varphi]_{\mathcal{I}}(v) \text{ a.e.}, \quad P_{\theta,K}[\varphi_j](v) \nearrow P_{\theta,K}[\varphi](v) \text{ a.e.}.$$

Proof We only give the proof for the relative $\mathcal{I}$-envelopes; the proof for the relative P-envelopes is identical, replacing $P_\theta[\bullet]_{\mathcal{I}}$ by $P_\theta[\bullet]$ and using Corollary 14.1.2 for the final candidate condition.

Set

$$u_j := P_{\theta,K}[\varphi_j]_{\mathcal{I}}(v), \quad u := P_{\theta,K}[\varphi]_{\mathcal{I}}(v).$$

By Lemma 14.1.3, $d_S(u_j, P_\theta[\varphi_j]_{\mathcal{I}}) = 0$ and $d_S(u, P_\theta[\varphi]_{\mathcal{I}}) = 0$ for all j. Moreover, Theorem 6.2.3 gives

$$P_\theta[\varphi_j]_{\mathcal{I}} \xrightarrow{d_S} P_\theta[\varphi]_{\mathcal{I}}$$

in case (1), and the same convergence holds in case (2) by Corollary 6.2.3 and Theorem 6.2.3. Consequently, $u_j \xrightarrow{d_S} u$ in both cases.

Assume first that $(\varphi_j)_j$ is decreasing. Then $(u_j)_j$ is decreasing and $u_j \ge u$. Let $u_\infty := \inf_j u_j$. The Monge–Ampère masses of u_j converge to the mass of u, by the preceding d_S-convergence and Theorem 6.2.1. Since $u_j \ge u_\infty \ge u$, the monotonicity theorem Theorem 2.4.3 gives

$$\int_X \theta^n_{u_\infty} = \int_X \theta^n_u.$$

Hence $u_j \xrightarrow{d_S} u_\infty$ by Corollary 6.2.5. The uniqueness of the d_S-limit gives $d_S(u_\infty, u) = 0$. Since $u_\infty \ge u$, Proposition 6.2.2 implies $u_\infty \sim_P u$. In particular, $u_\infty \preceq_{\mathcal{I}} \varphi$ by Corollary 6.1.1. The condition $u_\infty \le v$ on K quasi-everywhere is inherited from the u_j's, so u_∞ is a candidate for $P_{\theta,K}[\varphi]_{\mathcal{I}}(v)$. Thus $u_\infty \le u$, and the reverse inequality was already observed. This proves the $\mathcal{I}$-part of (1).

Assume next that $\varphi_j \nearrow \varphi$ almost everywhere. Then $(u_j)_j$ is increasing and $u_j \le u$. Put $\tilde{u} := \sup^*_j u_j$. By Corollary 14.1.1 and Corollary 6.2.3, we have $u_j \xrightarrow{d_S} \tilde{u}$. The d_S-convergence above therefore gives $d_S(\tilde{u}, u) = 0$, hence $\tilde{u} \sim_P u$ by Proposition 6.2.2. In particular, $\tilde{u} \preceq_{\mathcal{I}} \varphi$ by Corollary 6.1.1. Also $\tilde{u} \le v$ on K quasi-everywhere by Proposition 1.2.5. Hence $\tilde{u}$ is a candidate for $P_{\theta,K}[\varphi]_{\mathcal{I}}(v)$, so $\tilde{u} \le u$. Since $u_j \le u$ for all j, this proves $u_j \nearrow u$ almost everywhere. □

14.3 Quantization of partial equilibrium measures

Let X be a connected compact Kähler manifold of dimension n, let L be a pseudo-effective line bundle on X, and let h be a Hermitian metric on L. Set $\theta = c_1(L, h)$. Let (T, h_T) be a Hermitian line bundle on X.

14.3.1 Bernstein–Markov measures

Let $K \subseteq X$ be a closed non-pluripolar subset, and $v \colon K \to [-\infty, \infty)$ be a measurable function.

Definition 14.3.1 Let μ be a Borel probability measure on K. We introduce the following functions on $\mathrm{H}^0(X, L^k \otimes T)$ $(k \geq 1)$, with values possibly equal to ∞:

$$\begin{aligned} N^k_{v,\mu}(s) &:= \left(\int_K h^k \otimes h_T(s,s) \mathrm{e}^{-kv} \,\mathrm{d}\mu \right)^{1/2}, \\ N^k_{v,K}(s) &:= \sup_{K \setminus \{v=-\infty\}} \left(h^k \otimes h_T(s,s) \mathrm{e}^{-kv} \right)^{1/2}. \end{aligned} \tag{14.17}$$

Observe that if μ puts no mass on $\{v = -\infty\}$ then we always have

$$N^k_{v,\mu}(s) \leq N^k_{v,K}(s). \tag{14.18}$$

We start with the following elementary observation:

Lemma 14.3.1 *Let $v_1, v_2 \colon K \to [-\infty, \infty)$ be two measurable functions such that*

(1) $v_1 \leq v_2$;
(2) $\{v_1 = -\infty\} = \{v_2 = -\infty\}$.

Then for any $s \in \mathrm{H}^0(X, L^k \otimes T)$ $(k \geq 1)$, we have

$$N^k_{v_1,K}(s) \geq N^k_{v_2,K}(s).$$

Proof This follows immediately from the definition (14.17). □

Definition 14.3.2 Let (K, v) be a weighted subset of X. A Borel probability measure μ on K is *Bernstein–Markov* with respect to (K, v) if for each $\epsilon > 0$, there is a constant $C_\epsilon > 0$ such that

$$N^k_{v,K}(s) \leq C_\epsilon \mathrm{e}^{\epsilon k} N^k_{v,\mu}(s) \tag{14.19}$$

for any $s \in \mathrm{H}^0(X, L^k \otimes T)$ and any $k \in \mathbb{Z}_{>0}$. We write $\mathrm{BM}(K, v)$ for the set of Bernstein–Markov measures with respect to (K, v).

As pointed out in [BBWN11], any volume form on X is Bernstein–Markov with respect to (X, v), with $v \in C^\infty(X)$.

Proposition 14.3.1 *Assume that (K, v) is a weighted subset of X. Then*

(1) $N^k_{v,K}$ *is a norm on* $\mathrm{H}^0(X, L^k \otimes T)$.
(2) *For any* $\mu \in \mathrm{BM}(K, v)$, $N^k_{v,\mu}$ *is a norm on* $\mathrm{H}^0(X, L^k \otimes T)$.

Proof (1) As v is bounded, $N^k_{v,K}$ is clearly finite on $\mathrm{H}^0(X, L^k \otimes T)$. In order to show that it is a norm, it suffices to show that for any $s \in \mathrm{H}^0(X, L^k \otimes T)$, $N^k_{v,K}(s) = 0$ implies that $s = 0$. In fact, we have $s|_K = 0$, hence $s = 0$ by the connectedness of X and the assumption that K is non-pluripolar.

(2) As v is bounded, clearly $N^k_{v,\mu}$ is finite and satisfies the triangle inequality. Non-degeneracy follows from the fact that $N^k_{v,K}$ is a norm and (14.19). □

14.3.2 Partial Bergman kernels

In this section, we fix a weighted subset (K, v) of X and $\mu \in \mathrm{BM}(K, v)$.

Definition 14.3.3 For any $\varphi \in \mathrm{PSH}(X, \theta)_{>0}$, we introduce the *partial Bergman kernels* of φ (with respect to (K, v)) as follows: For any $k \geq 1$, we introduce

$$B^k_{v,\varphi,\mu}(x) \coloneqq \sup\left\{h^k \otimes h_T(s,s)\mathrm{e}^{-kv}(x) : s \in \mathrm{H}^0\left(X, L^k \otimes T \otimes \mathcal{I}(k\varphi)\right),\right. \\ \left. N^k_{v,\mu}(s) \leq 1\right\}, \quad x \in K. \tag{14.20}$$

We extend $B^k_{v,\varphi,\mu}$ to the whole X by setting it to be 0 outside K.

The *partial Bergman measures* of φ (with respect to (K, v)) are defined as

$$\beta^k_{v,\varphi,\mu} \coloneqq \frac{n!}{k^n} B^k_{v,\varphi,\mu}\,\mathrm{d}\mu \tag{14.21}$$

for each $k \geq 1$.

Remark 14.3.1 Recall that in the definition of $\beta^k_{v,\varphi,\mu}$, the singular metric φ only appears through the conditions $s \in \mathrm{H}^0(X, L^k \otimes T \otimes \mathcal{I}(k\varphi))$. The pointwise norm used to define $\beta^k_{v,\varphi,\mu}$ and $N^k_{v,\mu}$ is defined via v not φ.

This leads to the following seemingly paradoxical consequence, if $\psi \in \mathrm{PSH}(X, \theta)$ and $\varphi \preceq_{\mathcal{I}} \psi$, then

$$\beta^k_{v,\varphi,\mu} \leq \beta^k_{v,\psi,\mu} \quad \forall\, k \geq 1.$$

We shall apply this consequence without further mention in the sequel.

Observe that

$$\int_X \beta^k_{v,\varphi,\mu} = \frac{n!}{k^n} h^0\left(X, T \otimes L^k \otimes \mathcal{I}(k\varphi)\right).$$

In particular,

$$\lim_{k\to\infty}\int_X \beta^k_{v,\varphi,\mu} = \mathrm{vol}\left(\theta_\varphi\right) = \int_X \theta^n_{P_{\theta,K}[\varphi]_{\mathcal{I}}(v)} \tag{14.22}$$

by Theorem 7.4.1 and Lemma 14.1.3. The goal of this section is to improve this result to the convergence of the Bergman kernels themselves:

Theorem 14.3.1 *Suppose that $\varphi \in \mathrm{PSH}(X,\theta)_{>0}$. Then*

$$\beta^k_{v,\varphi,\mu} \rightharpoonup \theta^n_{P_{\theta,K}[\varphi]_{\mathcal{I}}(v)}, \quad \text{as } k \to \infty. \tag{14.23}$$

The proof will be given at the end of this subsection. We shall prove a few preliminaries at first.

Take a Kähler form ω on X so that

$$\int_X \omega^n = 1.$$

Proposition 14.3.2 *Assume here that $K = X$. Let $\varphi \in \mathrm{PSH}(X,\theta)$ be a potential with analytic singularities such that θ_φ is a Kähler current. If $v \in C^\infty(X)$, then*

$$\beta^k_{v,\varphi,\omega^n} \rightharpoonup \theta^n_{P_{\theta,X}[\varphi]_{\mathcal{I}}(v)} = \theta^n_{P_{\theta,X}[\varphi](v)} \quad \text{as } k \to \infty. \tag{14.24}$$

Proof The equality part in (14.24) follows from Corollary 14.1.3. It remains to prove the weak convergence.

Let τ be the weak limit of a subsequence of $(\beta^k_{v,\varphi,\omega^n})_k$. In view of (14.22), it suffices to show that

$$\tau \leq \theta^n_{P_{\theta,X}[\varphi](v)}. \tag{14.25}$$

We can obtain an upper bound of τ via the observation:

$$\beta^k_{v,\varphi,\omega^n} \leq \beta^k_{v,V_\theta,\omega^n}.$$

After replacing φ by V_θ, the corresponding problem is handled by [Ber09, Theorem 1.2], giving us

$$\beta^k_{v,V_\theta,\omega^n} \rightharpoonup \theta^n_{P_{\theta,X}[V_\theta](v)} = \mathbb{1}_{\{v=P_{\theta,X}[V_\theta](v)\}}\,\theta^n_v \quad \text{as } k \to \infty,$$

where the last identity is due to Proposition 3.1.7.

We obtain that

$$\tau \leq \mathbb{1}_{\{v=P_{\theta,X}[V_\theta](v)\}}\,\theta^n_v. \tag{14.26}$$

Now observe that it suffices to show that

$$\tau\left\{P_{\theta,X}[\varphi](v) < v\right\} = 0. \tag{14.27}$$

Assuming this, then (14.26) further gives

$$\tau \leq \mathbb{1}_{\{P_{\theta,X}[\varphi](v)=v\}}\,\theta^n_v = \theta^n_{P_{\theta,X}[\varphi](v)},$$

where the equality follows from Proposition 3.1.7, and (14.25) follows.

It now remains to argue (14.27). Fix a Kähler form ω on X. Observe that

$$P_{\theta+\epsilon\omega,X}[\varphi](v) \sim \varphi$$

for all $\epsilon \geq 0$ by Lemma 14.1.3 and Proposition 3.2.10. As a result,

$$P_{\theta+\epsilon\omega,X}[\varphi](v) \searrow P_{\theta,X}[\varphi](v) \quad \text{as } \epsilon \to 0+,$$

and hence,

$$\left\{P_{\theta,X}[\varphi](v) < v\right\} = \bigcup_{\epsilon\in\mathbb{Q}_{>0}} \left\{P_{\theta+\epsilon\omega,X}[\varphi](v) < v\right\}.$$

So in order to show (14.27), we may fix $\epsilon > 0$ and argue that

$$\tau\left\{P_{\theta+\epsilon\omega,X}[\varphi](v) < v\right\} = 0. \tag{14.28}$$

Let $k \geq 1$, and $s \in \mathrm{H}^0(X, L^k \otimes T \otimes \mathcal{I}(k\varphi))$ be a section such that $N^k_{v,\omega^n}(s) \leq 1$. Then by [Ber09, Lemma 4.1], there exists $C > 0$ such that

$$h^k \otimes h_T(s,s)\mathrm{e}^{-kv} \leq B^k_{v,\varphi,\omega^n} \leq B^k_{v,V_\theta,\omega^n} \leq k^n C.$$

This implies that

$$\frac{1}{k}\log h^k \otimes h_T(s,s) \leq v + \frac{\log C}{k} + n\frac{\log k}{k}.$$

We define φ_k as in Proposition 1.8.2. Take $\alpha_k \nearrow 1$ as in Proposition 1.8.2. After adding a constant to φ, we may assume that $\varphi \leq 0$ on X; this does not change the multiplier ideals or the envelopes that depend only on the singularity type of φ. For the fixed $\epsilon > 0$, we also have $\alpha_k\varphi \in \mathrm{PSH}(X, \theta + \epsilon\omega)$ for all sufficiently large k, and $\alpha_k\varphi \searrow \varphi$. Then

$$\frac{1}{k}\log h^k \otimes h_T(s,s) \leq \varphi_k \leq \alpha_k\varphi.$$

We notice that $\frac{1}{k}\log h^k \otimes h_T(s,s) \in \mathrm{PSH}(X, \theta + \epsilon\omega)$ for all $k \geq k_0(\epsilon)$. In particular,

$$\frac{1}{k}\log h^k \otimes h_T(s,s) - \frac{\log C}{k} - n\frac{\log k}{k} \leq P_{\theta+\epsilon\omega,X}[\alpha_k\varphi](v).$$

Now taking supremum over all candidates s, we obtain that

$$B^k_{v,\varphi,\omega^n} \leq Ck^n\mathrm{e}^{k\left(P_{\theta+\epsilon\omega,X}[\alpha_k\varphi](v)-v\right)}, \quad k \geq k_0. \tag{14.29}$$

Now we prove (14.28). Since $\alpha_k\varphi \searrow \varphi$ and the Monge–Ampère masses stay bounded from below, $\alpha_k\varphi \xrightarrow{d_S} \varphi$ by Corollary 6.2.6. Thus, by Proposition 14.2.2,

$$P_{\theta+\epsilon\omega,X}[\alpha_k\varphi](v) \searrow P_{\theta+\epsilon\omega,X}[\varphi](v)$$

and we get that

$$\bigcup_{k=1}^{\infty}\left\{P_{\theta+\epsilon\omega,X}[\alpha_k\varphi](v)<v\right\}=\left\{P_{\theta+\epsilon\omega,X}[\varphi](v)<v\right\}.$$

It suffices to show that τ does not put mass on the set $\{P_{\theta+\epsilon\omega,X}[\alpha_m\varphi](v)<v\}$ for each fixed m. Note that the latter set is open. If F is a compact subset of it, then

$$\sup_F\left(P_{\theta+\epsilon\omega,X}[\alpha_m\varphi](v)-v\right)<0.$$

Since $\alpha_\ell\varphi\leq\alpha_m\varphi$ for $\ell\geq m$, (14.29) gives $\beta^\ell_{v,\varphi,\omega^n}(F)\to 0$ along the chosen subsequence. By inner regularity, τ puts no mass on $\{P_{\theta+\epsilon\omega,X}[\alpha_m\varphi](v)<v\}$, proving (14.28). □

Definition 14.3.4 Take $k\geq 1$ and $\varphi\in\mathrm{PSH}(X,\theta)$, let $\mathrm{Norm}\,\mathrm{H}^0(X,L^k\otimes T\otimes\mathcal{I}(k\varphi))$ be the space of Hermitian norms on the vector space $\mathrm{H}^0(X,L^k\otimes T\otimes\mathcal{I}(k\varphi))$.

Let $\mathcal{L}_{k,\varphi}\colon\ \mathrm{Norm}\,\mathrm{H}^0(X,L^k\otimes T\otimes\mathcal{I}(k\varphi))\to\mathbb{R}$ be the *partial Donaldson functional*:

$$\mathcal{L}_{k,\varphi}(H)=\frac{n!}{k^{n+1}}\log\frac{\mathrm{vol}\{s\in\mathrm{H}^0\left(X,L^k\otimes T\otimes\mathcal{I}(k\varphi)\right):H(s)\leq 1\}}{\mathrm{vol}\left\{s\in\mathrm{H}^0\left(X,L^k\otimes T\otimes\mathcal{I}(k\varphi)\right):N^k_{0,\omega^n}(s)\leq 1\right\}},\quad(14.30)$$

where vol is a Lebesgue measure on $\mathrm{H}^0(X,L^k\otimes T\otimes\mathcal{I}(k\varphi))$. The normalization does not matter.

Proposition 14.3.3 *Let $w,w'\in C^0(X)$ and $\varphi\in\mathrm{PSH}(X,\theta)$ be a potential with analytic singularities such that θ_φ is a Kähler current. Then*

$$\lim_{k\to\infty}\left(\mathcal{L}_{k,\varphi}(N^k_{w,\omega^n})-\mathcal{L}_{k,\varphi}(N^k_{w',\omega^n})\right)=\mathcal{E}^\varphi_{\theta,X}(w)-\mathcal{E}^\varphi_{\theta,X}(w').\qquad(14.31)$$

In particular,

$$\lim_{k\to\infty}\mathcal{L}_{k,\varphi}(N^k_{w,\omega^n})=\mathcal{E}^\varphi_{\theta,X}(w).\qquad(14.32)$$

Proof First observe that by Proposition 14.3.1, for any $k\geq 1$, N^k_{w,ω^n} and N^k_{w',ω^n} are both norms, hence the expressions inside the limit in (14.31) make sense.

We first assume that $w,w'\in C^\infty(X)$. To start, we make the following observation:

$$\begin{aligned}\mathcal{L}_{k,\varphi}(N^k_{w,\omega^n})-\mathcal{L}_{k,\varphi}(N^k_{w',\omega^n})&=\int_0^1\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{L}_{k,\varphi}(N^k_{w'+t(w-w'),\omega^n})\,\mathrm{d}t\\&=\int_0^1\int_X(w-w')\,\beta^k_{w'+t(w-w'),\varphi,\omega^n}\,\mathrm{d}t.\end{aligned}$$

By Proposition 14.3.2, we have

$$\lim_{k\to\infty}\int_X(w-w')\,\beta^k_{w'+t(w-w'),\varphi,\omega^n}=\int_X(w-w')\,\theta^n_{P_{\theta,X}[\varphi](w'+t(w-w'))}.$$

By Theorem 7.4.1, we have $\left|\int_X (w-w')\beta^k_{w'+t(w-w'),\varphi,\omega^n}\right| \le C\sup_X|w-w'|$. Hence, by the dominated convergence theorem we obtain that

$$\begin{aligned}\lim_{k\to\infty}\left(\mathcal{L}_{k,\varphi}(N^k_{w,\omega^n}) - \mathcal{L}_{k,\varphi}(N^k_{w',\omega^n})\right) &= \int_0^1\int_X (w-w')\,\theta^n_{P_{\theta,X}[\varphi](w'+t(w-w'))}\,\mathrm{d}t \\ &= \mathcal{E}^{\varphi}_{\theta,X}(w) - \mathcal{E}^{\varphi}_{\theta,X}(w'),\end{aligned}$$

where in the last line we have used Proposition 14.2.1.

This proves (14.31) for smooth weights.

For general $w, w' \in C^0(X)$, choose smooth functions w_j, w'_j converging uniformly to w, w', respectively. If $\|w_j - w\|_{C^0} \le \delta_j$, then

$$\mathrm{e}^{-k\delta_j/2}N^k_{w,\omega^n} \le N^k_{w_j,\omega^n} \le \mathrm{e}^{k\delta_j/2}N^k_{w,\omega^n},$$

and hence

$$\left|\mathcal{L}_{k,\varphi}(N^k_{w_j,\omega^n}) - \mathcal{L}_{k,\varphi}(N^k_{w,\omega^n})\right| \le \frac{n!}{k^n}h^0\left(X, L^k\otimes T\otimes \mathcal{I}(k\varphi)\right)\delta_j.$$

The same estimate holds for w'_j and w'. Moreover, Proposition 14.2.1 shows that the energy side is Lipschitz with respect to the uniform norm of the weight. Letting first $k\to\infty$ in the smooth case, and then $j\to\infty$, proves (14.31) for continuous weights. Equation (14.32) again follows by taking $w'=0$. □

Lemma 14.3.2 *Let $\varphi \in \mathrm{PSH}(X,\theta)$. Let (K,v) be a weighted subset of X. Let $\mu \in \mathrm{BM}(K,v)$. Then*

$$\lim_{k\to\infty}\left(\mathcal{L}_{k,\varphi}(N^k_{v,K}) - \mathcal{L}_{k,\varphi}(N^k_{v,\mu})\right) = 0\,. \tag{14.33}$$

Proof By (14.18) and (14.19), for every $\epsilon>0$ and all $k\ge 1$,

$$N^k_{v,\mu} \le N^k_{v,K} \le C_\epsilon \mathrm{e}^{\epsilon k}N^k_{v,\mu}.$$

The corresponding unit balls therefore satisfy inclusions with radii differing by at most the factor $C_\epsilon\mathrm{e}^{\epsilon k}$. Taking logarithmic volumes and multiplying by $n!k^{-n-1}$ gives

$$\left|\mathcal{L}_{k,\varphi}(N^k_{v,K}) - \mathcal{L}_{k,\varphi}(N^k_{v,\mu})\right| \le \frac{2n!}{k^{n+1}}h^0\left(X, L^k\otimes T\otimes\mathcal{I}(k\varphi)\right)(\log C_\epsilon + \epsilon k).$$

The right-hand side has limsup at most $2\epsilon\,\mathrm{vol}(\theta_\varphi)$ by Theorem 7.4.1. Letting $\epsilon\to 0+$ proves (14.33). □

Corollary 14.3.1 *Let $w \in C^0(X)$, $\varphi \in \mathrm{PSH}(X,\theta)$ be a potential with analytic singularities such that θ_φ is a Kähler current. Then*

$$\lim_{k\to\infty}\mathcal{L}_{k,\varphi}(N^k_{w,X}) = \mathcal{E}^{\varphi}_{\theta,X}(w).$$

Proof This follows from Lemma 14.3.2 and Proposition 14.3.3: indeed $\omega^n \in \mathrm{BM}(X,0)$, hence $\omega^n \in \mathrm{BM}(X,w)$ as w is bounded. ◻

It would be helpful to consider the following auxiliary functions:

$$\begin{aligned} P'_{\theta,K}[\varphi](v) &:= \sup\Big\{\eta \in \mathrm{PSH}(X,\theta) : \eta|_K \leq v \text{ and } \eta \leq \varphi\Big\},\\ P'_{\theta,K}[\varphi]_{\mathcal{I}}(v) &:= \sup\Big\{\eta \in \mathrm{PSH}(X,\theta) : \eta|_K \leq v \text{ and } \eta \leq_{\mathcal{I}} \varphi\Big\}. \end{aligned}$$

We note the following maximum principles, that follow from the above definitions:

Lemma 14.3.3 *Let $v \in C^0(K)$, $\varphi \in \mathrm{PSH}(X,\theta)$, and $\eta \in \mathrm{PSH}(X,\theta)$. Assume that $\eta \leq \varphi$. Then*

$$\sup_K(\eta - v) = \sup_{\{\eta \neq -\infty\}}\Big(\eta - P'_{\theta,K}[\varphi](v)\Big) = \sup_{\left\{P'_{\theta,K}[\varphi](v)\neq -\infty\right\}}\Big(\eta - P'_{\theta,K}[\varphi](v)\Big). \tag{14.34}$$

Proof We prove the first equality first. We write $S = \{\eta = -\infty\}$.

By definition, $P'_{\theta,K}[\varphi](v)|_K \leq v$, so

$$\Big(\eta - P'_{\theta,K}[\varphi](v)\Big)\Big|_{K\setminus S} \geq \eta|_{K\setminus S} - v|_{K\setminus S}\,.$$

This implies that

$$\sup_K(\eta - v) \leq \sup_{X\setminus S}\Big(\eta - P'_{\theta,K}[\varphi](v)\Big)\,.$$

Conversely, observe that $\sup_K(\eta - v) > -\infty$ as K is non-pluripolar. It follows that $\eta - \sup_K(\eta - v)$ is a candidate in the definition of $P'_{\theta,K}[\varphi](v)$. Hence,

$$\eta - \sup_K(\eta - v) \leq P'_{\theta,K}[\varphi](v), \tag{14.35}$$

This implies that

$$\sup_K(\eta - v) \geq \sup_{X\setminus S}\Big(\eta - P'_{\theta,K}[\varphi](v)\Big)\,,$$

finishing the proof of the first identity in (14.34).

In order to prove the latter identity, observe that

$$\Big\{P'_{\theta,K}[\varphi](v) = -\infty\Big\} \subseteq S$$

as a consequence of (14.35). On $\{P'_{\theta,K}[\varphi](v) \neq -\infty\} \cap S$, we have $\eta - P'_{\theta,K}[\varphi](v) = -\infty$. Hence the last equality of (14.34) also follows. ◻

Proposition 14.3.4 *Let $\varphi \in \mathrm{PSH}(X,\theta)$ be a potential with analytic singularities such that θ_φ is a Kähler current. Let (K,v), (K',v') be two weighted subsets of X. Then*

$$\lim_{k\to\infty}\Big(\mathcal{L}_{k,\varphi}(N^k_{v,K})-\mathcal{L}_{k,\varphi}(N^k_{v',K'})\Big)=\mathcal{E}^{\varphi}_{\theta,K}(v)-\mathcal{E}^{\varphi}_{\theta,K'}(v'). \tag{14.36}$$

In particular,

$$\lim_{k\to\infty}\mathcal{L}_{k,\varphi}(N^k_{v,K})=\mathcal{E}^{\varphi}_{\theta,K}(v). \tag{14.37}$$

Proof First observe that by Proposition 14.3.1, for any $k>0$, $N^k_{v,K}$ and $N^k_{v',K'}$ are both norms, hence the expressions inside the limit in (14.36) make sense. Moreover, (14.37) is just a special case of (14.36) for $K'=X$ and $v'=0$.

To prove (14.36) it is enough to show that for any fixed $w\in C^\infty(X)$ we have

$$\lim_{k\to\infty}\big(\mathcal{L}_{k,\varphi}(N^k_{v,K})-\mathcal{L}_{k,\varphi}(N^k_{w,X})\big)=\mathcal{E}^{\varphi}_{\theta,K}(v)-\mathcal{E}^{\varphi}_{\theta,X}(w)\,. \tag{14.38}$$

For $\epsilon\in(0,1)$ small enough, $\theta_{(1-\epsilon)\varphi}$ is still a Kähler current. Let us fix such ϵ, along with an arbitrary $\epsilon'\in(0,1)$.

Let $(v^-_j)_j,(v^+_j)_j$ be the sequences of smooth functions constructed in Lemma 14.2.2 for the data (K,v).

By Proposition 1.8.2 there exists $k_0(\epsilon,\epsilon')\in\mathbb{N}$ such that

$$\frac{1}{k}\log h^k\otimes h_T(s,s)\le(1-\epsilon)\varphi,$$

and $\frac{1}{k}\log h^k\otimes h_T(s,s)\in\mathrm{PSH}(X,\theta+\epsilon'\omega)$ for any $s\in\mathrm{H}^0(X,T\otimes L^k\otimes\mathcal{I}(k\varphi))$, as long as $k\ge k_0(\epsilon,\epsilon')$.

In particular, Lemma 14.3.3 gives that

$$\begin{aligned}N^k_{P'_{\theta+\epsilon'\omega,K}[(1-\epsilon)\varphi](v),X}(s)=&N^k_{v,K}(s),\\ N^k_{P'_{\theta+\epsilon'\omega,X}[(1-\epsilon)\varphi](v^-_j),X}(s)=&N^k_{v^-_j,X}(s),\\ N^k_{P'_{\theta+\epsilon'\omega,X}[(1-\epsilon)\varphi](v^+_j),X}(s)=&N^k_{v^+_j,X}(s).\end{aligned}$$

As

$$P'_{\theta+\epsilon'\omega,X}[(1-\epsilon)\varphi](v^-_j)\le P'_{\theta+\epsilon'\omega,K}[(1-\epsilon)\varphi](v)\le P'_{\theta+\epsilon'\omega,X}[(1-\epsilon)\varphi](v^+_j),$$

where the second inequality follows from Lemma 14.2.2 and the fact that $P'=P$ for the global continuous obstacles $v^\pm_j$. By Lemma 14.3.1 we have

$$N^k_{v^+_j,X}(s)\le N^k_{v,K}(s)\le N^k_{v^-_j,X}(s),\quad s\in\mathrm{H}^0(X,T\otimes L^k\otimes\mathcal{I}(k\varphi)),k\ge k_0(\epsilon,\epsilon').$$

Composing with $\mathcal{L}_{k,\varphi}$ we arrive at

$$\mathcal{L}_{k,\varphi}(N^k_{v^-_j,X})\le\mathcal{L}_{k,\varphi}(N^k_{v,K})\le\mathcal{L}_{k,\varphi}(N^k_{v^+_j,X})\,,\ k\ge k_0(\epsilon,\epsilon').$$

For any $j>0$, by Corollary 14.3.1 we get

$$\begin{aligned}
\mathcal{E}^{\varphi}_{\theta,X}(v_j^-) - \mathcal{E}^{\varphi}_{\theta,X}(w) &= \lim_{k\to\infty}\left(\mathcal{L}_{k,\varphi}(N^k_{v_j^-,X}) - \mathcal{L}_{k,\varphi}(N^k_{w,X})\right)\\
&\leq \varliminf_{k\to\infty}\left(\mathcal{L}_{k,\varphi}(N^k_{v,K}) - \mathcal{L}_{k,\varphi}(N^k_{w,X})\right)\\
&\leq \varlimsup_{k\to\infty}\left(\mathcal{L}_{k,\varphi}(N^k_{v,K}) - \mathcal{L}_{k,\varphi}(N^k_{w,X})\right)\\
&\leq \lim_{k\to\infty}\left(\mathcal{L}_{k,\varphi}(N^k_{v_j^+,X}) - \mathcal{L}_{k,\varphi}(N^k_{w,X})\right)\\
&=\mathcal{E}^{\varphi}_{\theta,X}(v_j^+) - \mathcal{E}^{\varphi}_{\theta,X}(w).
\end{aligned}$$

Using Lemma 14.2.2, we can let $j \to \infty$ to arrive at

$$\begin{aligned}
\mathcal{E}^{\varphi}_{\theta,K}(v) - \mathcal{E}^{\varphi}_{\theta,X}(w) &\leq \varliminf_{k\to\infty}\left(\mathcal{L}_{k,\varphi}(N^k_{v,K}) - \mathcal{L}_{k,\varphi}(N^k_{w,X})\right)\\
&\leq \varlimsup_{k\to\infty}\left(\mathcal{L}_{k,\varphi}(N^k_{v,K}) - \mathcal{L}_{k,\varphi}(N^k_{w,X})\right)\\
&\leq \mathcal{E}^{\varphi}_{\theta,K}(v) - \mathcal{E}^{\varphi}_{\theta,X}(w).
\end{aligned}$$

Hence, (14.38) follows. □

Corollary 14.3.2 *Let $\varphi \in \mathrm{PSH}(X,\theta)$ be a potential with analytic singularities such that θ_φ is a Kähler current. Then*

$$\lim_{k\to\infty} \mathcal{L}_{k,\varphi}(N^k_{v,\mu}) = \mathcal{E}^{\varphi}_{\theta,K}(v).$$

Proof Our claim follows from Proposition 14.3.4 and Lemma 14.3.2. □

Lemma 14.3.4 *Let $I \subseteq \mathbb{R}$ be an open interval containing 0. Let $(f_k : I \to \mathbb{R})_{k\geq 1}$ be a sequence of concave functions, and let $g : I \to \mathbb{R}$ be a function. Assume that*

$$\liminf_{k\to\infty} f_k(t) \geq g(t) \qquad \textit{for every } t \in I,$$

and that

$$\lim_{k\to\infty} f_k(0) = g(0).$$

If each f_k and g is differentiable at 0, then

$$\lim_{k\to\infty} f_k'(0) = g'(0).$$

Proof Since f_k is concave and differentiable at 0, for every $t \in I$,

$$f_k(t) \leq f_k(0) + t f_k'(0).$$

Taking the lower limit as $k \to \infty$ and using the assumptions gives

$$\varliminf_{k\to\infty} t f_k'(0) \geq g(t) - g(0).$$

For $t > 0$, this yields

$$\varliminf_{k\to\infty} f_k'(0) \geq \frac{g(t) - g(0)}{t}.$$

Letting $t \to 0+$ and using the differentiability of g at 0, we get

$$\varliminf_{k\to\infty} f_k'(0) \geq g'(0).$$

For $t < 0$, the same inequality gives,

$$\varlimsup_{k\to\infty} f_k'(0) \leq \frac{g(t) - g(0)}{t}.$$

Letting $t \to 0-$ and again using the differentiability of g at 0, we obtain

$$\varlimsup_{k\to\infty} f_k'(0) \leq g'(0).$$

Therefore

$$\lim_{k\to\infty} f_k'(0) = g'(0),$$

as claimed. □

Proposition 14.3.5 *Let $\varphi \in \mathrm{PSH}(X,\theta)$ be a potential with analytic singularities such that θ_φ is a Kähler current. Then*

$$\beta^k_{v,\varphi,\mu} \rightharpoonup \theta^n_{P_{\theta,K}[\varphi]_{\mathcal{I}}(v)} = \theta^n_{P_{\theta,K}[\varphi](v)} \quad \text{as } k \to \infty.$$

Proof For $w \in C^0(X)$, let $f_k, g \colon \mathbb{R} \to \mathbb{R}$ $(k \geq 1)$ be defined as

$$f_k(t) \coloneqq \mathcal{L}_{k,\varphi}(N^k_{v+tw,\mu}), \quad g(t) \coloneqq \mathcal{E}^\varphi_{\theta,K}(v + tw).$$

By Corollary 14.3.2, for all $t \in \mathbb{R}$, we have

$$\lim_{k\to\infty} f_k(t) = g(t).$$

Here we use that $\mu \in \mathrm{BM}(K, v+tw)$ for all fixed t, which follows from $\mu \in \mathrm{BM}(K, v)$ and the boundedness of tw on K. Note that f_k is concave by Hölder's inequality, so by Lemma 14.3.4,

$$\lim_{k\to\infty} f_k'(0) = g'(0),$$

which is equivalent to $\beta^k_{v,\varphi,\mu} \rightharpoonup \theta^n_{P_{\theta,K}[\varphi](v)}$, by Proposition 14.2.1. □

Proposition 14.3.6 *Suppose that $\varphi \in \mathrm{PSH}(X,\theta)$ is such that θ_φ is a Kähler current. Let (K, v) be a weighted subset of X and $\mu \in \mathrm{BM}(K, v)$. Then*

$$\beta^k_{v,\varphi,\mu} \rightharpoonup \theta^n_{P_{\theta,K}[\varphi]_{\mathcal{I}}(v)} \quad \text{as } k \to \infty. \tag{14.39}$$

Proof Let τ be the weak limit of a subsequence of $\beta^k_{v,\varphi,\mu}$. We claim that

$$\tau \leq \theta^n_{P_{\theta,K}[\varphi]_{\mathcal{I}}(v)}. \tag{14.40}$$

Observe that this claim implies the conclusion. In fact, by Theorem 7.4.1, we have equality of the total masses, so equality holds in (14.40). As τ is an arbitrary cluster point of the sequence $(\beta^k_{v,\varphi,\mu})_k$, we get (14.39).

It remains to prove (14.40). Let (φ_j) be a quasi-equisingular approximation of φ in $\mathrm{PSH}(X,\theta)$. We may assume that θ_{φ_j} is a Kähler current for all $j \geq 1$. By Corollary 7.1.2, we know that

$$\varphi_j \xrightarrow{d_S} P_\theta[\varphi]_{\mathcal{I}}.$$

The quasi-equisingular approximation is decreasing, so applying the decreasing case of Proposition 14.2.2 to the limit $P_\theta[\varphi]_{\mathcal{I}}$, and using the definition of the relative $\mathcal{I}$-envelope, we get

$$P_{\theta,K}[\varphi_j]_{\mathcal{I}}(v) \searrow P_{\theta,K}[\varphi]_{\mathcal{I}}(v).$$

In particular,

$$\lim_{j\to\infty}\int_X \theta^n_{P_{\theta,K}[\varphi_j]_{\mathcal{I}}(v)} = \int_X \theta^n_{P_{\theta,K}[\varphi]_{\mathcal{I}}(v)}. \tag{14.41}$$

Observe that

$$\beta^k_{v,\varphi,\mu} \leq \beta^k_{v,\varphi_j,\mu}$$

for any $k \geq 1$. As $\mu \in \mathrm{BM}(K,v)$, by Proposition 14.3.5,

$$\tau \leq \theta^n_{P_{\theta,K}[\varphi_j]_{\mathcal{I}}(v)},$$

for any $j \geq 1$ fixed. By (14.41) and Theorem 2.4.2, we can let $j \to \infty$ in the preceding inequality and obtain (14.40). □

Proof (Proof of Theorem 14.3.1) By Lemma 14.1.3 and Proposition 3.2.6, the three potentials below have the same multiplier ideals. Hence

$$\begin{aligned}\mathrm{H}^0\left(X, L^k \otimes T \otimes \mathcal{I}(k\varphi)\right) &= \mathrm{H}^0\left(X, L^k \otimes T \otimes \mathcal{I}(kP_\theta[\varphi]_{\mathcal{I}})\right)\\ &= \mathrm{H}^0\left(X, L^k \otimes T \otimes \mathcal{I}\left(kP_{\theta,K}[\varphi]_{\mathcal{I}}(v)\right)\right).\end{aligned}$$

This allows us to replace φ with $P_{\theta,K}[\varphi]_{\mathcal{I}}(v)$.

By Lemma 2.4.5, there exists a sequence $(\varphi_j)_j$ in $\mathrm{PSH}(X,\theta)$ such that $\varphi_j \nearrow \varphi$ a.e. and θ_{φ_j} is a Kähler current for each $j \geq 1$. This gives

$$\beta^k_{v,\varphi_j,\mu} \leq \beta^k_{v,\varphi,\mu}.$$

Let τ be the weak limit of a subsequence of $(\beta^k_{v,\varphi,\mu})_k$. Then by Proposition 14.3.6,

$$\theta^n_{P_{\theta,K}[\varphi_j]_{\mathcal{I}}(v)} \leq \tau.$$

By Proposition 14.2.2 and Theorem 2.4.2, we have

$$\theta^n_{P_{\theta,K}[\varphi_j]_{\mathcal{I}}(v)} \rightharpoonup \theta^n_{P_{\theta,K}[\varphi]_{\mathcal{I}}(v)}.$$

Hence,

$$\theta^n_{P_{\theta,K}[\varphi]_{\mathcal{I}}(v)} \leq \tau. \tag{14.42}$$

A comparison of total masses using (14.22) gives that equality holds in (14.42). As τ is arbitrary, we obtain (14.23). □

Remark 14.3.2 The results in this chapter can also be reformulated as the large deviation principle of a determinantal point process on X using the Gärtner–Ellis theorem exactly as in [Ber14]. We do not pursue this reformulation here.

Comments

A brief history

Here we recall the origin of various results.

Chapter 1.

The notion of plurisubharmonic functions was introduced by Lelong [Lel45], based on F. Riesz's theory of subharmonic functions [Rie26]. See [Bre72] for an excellent introduction to the early history of the subject. We refer to [Bre65] for the foundations of potential theory and [GZ17] for the pluripotential theory.

The study of Josefson-type theorems was initiated by Josefson [Jos78], who proved Theorem 1.1.4. The global Josefson theorem Theorem 1.1.5 was due to Vu [Vu19]. In the projective setting, it was due to Dinh–Sibony [DS06] and in the Kähler setting, it was established by Guedj–Zeriahi [GZ05].

The extension theorem Theorem 1.2.1 was proved in [GR56]. In fact, they proved a more general version for complex spaces, see Theorem B.2.2. For some related important extension theorems, see [Shi72, Wan24].

Proposition 1.2.8 was due to Kiselman [Kis78].

The idea of fine topology was due to H. Cartan [Car41]. The plurifine topology was introduced by Fuglede during the Séminaire d'analyse de Lelong–Dolbeault–Skoda of the year 1983/1984 [LDS86], but often wrongly attributed to Bedford–Taylor. The key result Theorem 1.3.2 was claimed in Bedford–Taylor's work [BT87, Theorem 2.3] without proof. The first rigorous proof was given by El Marzguioui–Wiegerinck [EMW06]. A weaker result was proved earlier in [Kli91, Theorem 4.8.7].

Results in Section 1.3.2 are certainly well-known and are already implicitly used in the literature. I could not find the proofs in the literature and hence all details are presented.

The semicontinuity theorem Theorem 1.4.1 was due to Siu [Siu74].

Multiplier ideal sheaves were introduced by Nadel [Nad90]. The idea of Theorem 1.4.3 first appeared in the ground-breaking work of Boucksom–Favre–Jonsson [BFJ08].

The strong openness Theorem 1.4.4 was first established by Guan–Zhou [GZ15]. A more elegant proof was due to Hiep [Hie14].

Lemma 1.6.3 was due to [Dem15, Proposition 4.1.6].

The notions of positivity of currents Definition 1.7.1 is due to Lelong [Lel68] and Harvey–Knapp [HK74], although it is often mistakenly attributed to Demailly.

The L^2 extension theorems such as Theorem 1.4.5 were first studied by Ohsawa–Takegoshi [OT87]. Since then, it has become a very active area. It is almost impossible to give a complete list of the various generalizations and improvements.

Chapter 2

The Monge–Ampère operators for bounded plurisubharmonic functions were introduced by Bedford–Taylor [BT76, BT82]. The non-pluripolar product was due to Bedford–Taylor [BT87], Guedj–Zeriahi [GZ07] and Boucksom–Eyssidieux–Guedj–Zeriahi [BEGZ10].

The key lemma Lemma 2.4.4 was proved in [DDNL21b]. Theorem 2.4.4 was due to [DDNL25].

Chapter 3

Lemma 3.1.1 was proved in [DDNL18b, Lemma 3.7].

The notion of the P-envelope is due to Ross–Witt Nyström [RWN14] based on the ideas of Rashkovskii–Sigurdsson [RS05].

Theorem 3.1.1 is due to [DDNL18b, Theorem 3.8]. The diamond inequality Theorem 3.1.3 and Proposition 3.1.5 are due to [DDNL21b, Theorem 5.4]. Most results in Section 3.1.3 are simple generalizations of the corresponding results in [DDNL18a, DDNL18c].

The study of finite energy potentials was initiated by Cegrell [Ceg98] in the local setup. The global setup was first studied by Guedj–Zeriahi [GZ07].

The $\mathcal{I}$-envelope was introduced by Darvas–Xia [DX22], inspired by the works of Dano Kim [Kim15] and Boucksom–Favre–Jonsson [BFJ08]. The notion of $\mathcal{I}$-model singularities was first formulated in the explicit way in [DX22] in 2020, although it was already essentially known in Boucksom–Jonsson's work. In fact, they correspond exactly to the homogeneous non-Archimedean potentials assuming that the relevant masses do not vanish. A less explicit equivalent formulation of $\mathcal{I}$-model potentials also appeared in [Dem15]. A few months later, the same notion was rediscovered by Trusiani [Tru22].

Theorem 2.4.7 is due to [DNT21].

Chapter 4

The study of geodesics in the space of Kähler metrics was initiated by Mabuchi [Mab87], Semmes [Sem92] and Donaldson [Don99]. The existence of geodesics with smooth boundary data was established by Chen [Che00].

The notion of weak geodesics was studied in detail by Darvas [Dar17] in the Kähler case.

The case of general big classes was partly handled in [DDNL18c], [DDNL18a]. However, the key fact that the geodesics between two full mass potentials have the correct limit at the end points does not seem to have been proved in any references. We give a proof in Proposition 4.2.1. We also extend the relevant results to the relative setting.

Previously, Proposition 4.2.2 and Proposition 4.2.4 were only known in the Kähler case.

Most results in Section 4.3 are simple extensions of [DDNL18a].

Chapter 5

The toric framework was first written down by Berman–Berndtsson [BB13] and Coman–Guedj–Sahin–Zeriahi in [CGSZ19].

The beautiful theorem Theorem 5.2.2 was first proved by Yi Yao, who did not publish the result. Later on, a new proof was found by Botero–Burgos Gil–Holmes–de Jong [BBGHdJ22]. We chose to present the approach of Yao, which integrates naturally with our framework.

Chapter 6

The notion of P-preorder is new, as well as most results in Section 6.1.

The d_S-pseudometric was introduced in [DDNL21b]. The basic properties are proved in [DDNL21b] and [Xia25b].

Example 6.1.3 was due to Berman–Boucksom–Jonsson [BBJ21].

Theorem 6.2.4 is proved in [Xia24]. Theorem 6.2.6 and Theorem 6.2.5 appear to be new. These results appeared previously in the form of lecture notes.

Chapter 7

The notion of $\mathcal{I}$-good singularities was due to [DX24b]. The name *$\mathcal{I}$-good* was chosen in [Xia24].

The key results Theorem 7.1.1 and Theorem 7.4.1 are due to [DX24b, DX22]. Previously, Rashkovskii [Ras13] also studied the local approximation problem with a different motivation.

There are some further examples of $\mathcal{I}$-good singularities provided by [BBGHdJ22] with applications in the theory of modular forms in [BBGHdJ24].

Chapter 8

The trace operator was introduced in [DX24a]. Here we present a different point of view. Theorem 8.4.1 was proved in [DX24a].

The analytic Bertini theorem Theorem 8.5.1 was proved in [Xia22], based on the works of Matsumura–Fujino [FM21] and [Fuj23]. A weaker result was established by Meng–Zhou [MZ23].

Chapter 9

The technique of test curves originates from [RWN14]. It was generalized by Darvas–Di Nezza–Lu [DDNL18a], [DX24b], [DZ24] and [DXZ25]. We give the full details of the proofs.

Test curves in Definition 9.1.1 are called *maximal test curves* in the literature, a terminology which I do not like. I prefer to call the usual notion of test curves in the literature *sub-test curves*.

Proposition 9.2.2 was first proved by He–Testorf–Wang in [HTW23]. Proposition 9.2.3 was due to Hisamoto [His16].

Remark 9.3.2 was a folklore result. I am unaware of any written proof in the literature before our paper [DX22]. Finski [Fin25] also gave a different proof with different techniques later on.

Definition 9.3.3 was not the original definition of maximal geodesic rays of Berman–Boucksom–Jonsson in [BBJ21]. One of the first major applications of our theory was this pluripotential-theoretical characterization of maximal geodesic rays, as proved in our very first paper [DX22].

Example 9.3.1 is essentially due to Ross–Witt Nyström [RWN14]. The Phong–Sturm geodesics were first constructed in [PS07].

Results in Section 9.4 are easy generalizations of the results proved in [Xia25a].

Chapter 10

The algebraic theory of partial Okounkov bodies was developed in [Xia25b]. The transcendental Okounkov body was first defined by Deng [Den17] as suggested by Demailly. The volume identity was proved in [DRWN⁺26]. The transcendental theory of partial Okounkov bodies is new.

Chapter 11

The notion of b-divisors was introduced by Shokurov [Sho93] in the context of minimal model programs.

The applications of b-divisors in pluripotential theory began with [BFJ09]. The intersection theory of nef b-divisors was introduced by Dang–Favre [DF22]. The technique of singularity b-divisors was introduced in [Xia23b] in 2020. The general form first appeared in [Xia24]. One year later, a special case was rediscovered in [BBGHdJ22].

The current chapter reproduces a large part of [Xia25c]. For further developments, we refer to [Xia25d].

Example 11.1.1 is reproduced from [DF22, Example 3.3].

Chapter 12

The whole chapter appears to be new. The study of toric pluripotential theory on big line bundles was made possible by the development of partial Okounkov bodies. The key result is Theorem 12.2.2.

Most results in this chapter resulted from discussions with Yi Yao.

Chapter 13

Most results in this chapter come from [Xia25a]. Results from Section 13.3 are new, although the main idea was already contained in [Xia25b].

Theorem 13.4.3 is due to [DXZ25]. An alternative approach to the transcendental theory is due to Mesquita-Piccione [MP25]. For further progress in this direction, we refer to [MPWN25, WN25].

Special cases of the results in this section have been applied to study K-stability, see [Xia23b], [DZ24], [DXZ25] and [DR24]. In [DX22], we established the bijective correspondence between a class of $\mathcal{I}$-model test curves and the maximal geodesic rays in the sense of [BBJ21].

Chapter 14

The study of asymptotic expansions of Bergman kernels in Kähler geometry was initiated by Tian [Tia90] with later improvements due to Bouche [Bou90], Ruan [Rua98], Catlin [Cat99], Zeldítch [Zel98], Lu [Lu00] among many others.

The special case of Theorem 14.3.1 without the prescribed singularity φ was due to Berman–Boucksom–Witt Nyström, see [BB10], [BBWN11]. The general case is due to [DX24b].

Open problems

We give a list of important open problems in this theory.

We do not repeat the conjectures mentioned in the main text.

Conjecture 14.3.1 Let X be a connected compact Kähler manifold and Y be a submanifold. Fix a Kähler class α on X. For each Kähler current $S \in \alpha|_Y$, we can find a Kähler current $T \in \alpha$ such that

$$\mathrm{Tr}_Y(T) \sim_{\mathcal{I}} S.$$

If we formally view Tr_Y as an analogue of the trace operator in the theory of Sobolev spaces, then this conjecture corresponds exactly to the Dirichlet problem.

Using Proposition 8.2.2, one could also reduce this conjecture to a strong version of the extension theorem Theorem 1.6.3.

Conjecture 14.3.2 Let X be a connected compact Kähler manifold and Y be a submanifold. Fix a Kähler class α on X. Consider Kähler currents $R \in \alpha$, $S \in \alpha|_Y$ with gentle analytic singularities such that $S \preceq R|_Y$. Then there is a Kähler current $T \in \alpha$ with analytic singularities such that

$$\mathrm{Tr}_Y(T) \sim_{\mathcal{I}} S, \quad T \preceq R.$$

This conjecture was also proposed by Darvas for different purposes.

Conjecture 14.3.3 Let X be a connected smooth projective variety of dimension n. Assume that (L_i, h_i) is a Hermitian big line bundle on X for each $i = 1, \dots, n$ with the h_i's being $\mathcal{I}$-good. Then

$$\int_X c_1(L_1, h_1) \wedge \cdots \wedge c_1(L_n, h_n) = \sup_\nu \mathrm{vol}\left(\Delta_\nu(L_1, h_1), \dots, \Delta_\nu(L_n, h_n)\right),$$

where $\nu\colon \mathbb{C}(X)^\times \to \mathbb{Z}^n$ runs over all (surjective) valuations of rank n.

See [Sch14, Section 5.1] for the notion of mixed volumes.

This conjecture seems reasonable in view of Corollary 10.3.3 and Corollary 10.3.2.

Even when $h_1, \dots, h_n$ have minimal singularities, this conjecture remains open:

Conjecture 14.3.4 Let X be a connected smooth projective variety of dimension n. Assume that $L_1, \dots, L_n$ are big line bundles on X. Then

$$\langle L_1, \dots, L_n \rangle = \sup_\nu \mathrm{vol}\left(\Delta_\nu(L_1), \dots, \Delta_\nu(L_n)\right), \tag{14.43}$$

where $\nu\colon \mathbb{C}(X)^\times \to \mathbb{Z}^n$ runs over all (surjective) valuations of rank n.

Here on the left-hand side, we are using the movable intersection theory [BDPP13].

In [Wil25], Wilms proved the $\leq$ direction of (14.43).

Problem 14.3.1 Is it possible to extend the definition of the trace operator Tr_Y to the case where the ambient variety is only unibranch?

The difficulty lies in the lack of Demailly type regularization theorems.

Problem 14.3.2 Is there a natural definition of the transcendental Okounkov body of a closed positive $(1,1)$-current T with 0-mass so that its dimension is equal to the numerical dimension of T?

See [Cao14] for the definition of the numerical dimension of a current.

The following two problems are proposed by Witt Nyström.

Problem 14.3.3 Consider a compact Kähler manifold X and a connected submanifold Y. We have defined the trace operator Tr_Y from a subset of $\mathrm{QPSH}(X)/\sim_{\mathcal{I}}$ to $\mathrm{QPSH}(Y)/\sim_{\mathcal{I}}$. Is it possible to refine this operator to one from a subset of $\mathrm{QPSH}(X)/\sim_P$ to $\mathrm{QPSH}(Y)/\sim_P$?

Problem 14.3.4 Consider a connected compact Kähler manifold X of dimension n and a smooth flag $Y_\bullet$ on X. Consider a closed smooth real $(1,1)$-form θ on X representing a big cohomology class and $\varphi \in \mathrm{PSH}(X,\theta)$ with $\int_X \theta_\varphi^n > 0$.

Can one define a refined notion of partial Okounkov bodies $\Delta'_{Y_\bullet}(\theta+\mathrm{dd^c}\varphi)$ contained in $\Delta_{Y_\bullet}(\theta+\mathrm{dd^c}\varphi)$ with volume given by $\frac{1}{n!}\int_X \theta_\varphi^n$?

Note that a satisfactory solution to the latter problem is not very likely, as can be easily seen from examples on $\mathbb{P}^1$.

We also look for generalizations of our theory to more general settings.

Problem 14.3.5 To what extent can the results in the current book be generalized to the non-Kähler setting?

The non-pluripolar products in the non-Kähler setting were recently studied by Boucksom–Guedj–Lu in [BGL24]. See also the references therein.

Problem 14.3.6 To what extent can the results in the current book about closed positive $(1,1)$-currents be generalized to closed positive currents of higher bidegree?

A fundamental issue is the lack of a strong enough Demailly type approximation for general currents. The regularization theorem of Dinh–Sibony [DS04] seems too weak for our purposes.

Appendix A
Convex functions and convex bodies

This appendix collects the basic facts about convex functions and convex bodies that are used throughout the book. Our standard reference is [Roc70]. All results stated for convex functions apply equally to concave functions, after replacing the function by its negative.

A.1 The notion of convex functions

Let N be a real vector space of finite dimension.

Definition A.1.1 Let $F\colon N \to [-\infty, \infty]$ be a function. The *epigraph* of F is defined as the following set

$$\operatorname{epi} F := \{(n, r) \in N \times \mathbb{R} : r \geq F(n)\}\,.$$

Definition A.1.2 A *convex function* on N is a function $F\colon N \to [-\infty, \infty]$ such that the epigraph $\operatorname{epi} F$ is a convex subset of $N \times \mathbb{R}$.

The *effective domain* of F is the set

$$\operatorname{Dom} F := \{n \in N : F(n) < \infty\}\,.$$

A convex function F on N such that $\operatorname{Dom} F \neq \emptyset$ and $F(n) \neq -\infty$ for all $n \in N$ is said to be *proper*.

The set of convex functions on N is denoted by $\operatorname{Conv}(N)$. The subset of proper convex functions is denoted by $\operatorname{Conv}^{\mathrm{prop}}(N)$.

The following characterization of convex functions is well-known.

Lemma A.1.1 *Let $F\colon N \to [-\infty, \infty]$. Then F is convex if and only if the following condition holds: suppose that $n, r \in N$ and $a, b \in \mathbb{R}$ such that $a > F(n)$, $b > F(r)$, then for any $t \in (0, 1)$, we have*

$$F(tn + (1 - t)r) < ta + (1 - t)b.$$

See [Roc70, Theorem 4.2] for the proof.

Example A.1.1 Let $A \subseteq N$ be a convex subset. Then the *characteristic function* $\chi_A \colon N \to \{0, \infty\}$ of A is defined by

$$\chi_A(n) := \begin{cases} 0, & n \in A; \\ \infty, & n \notin A. \end{cases}$$

The function χ_A lies in $\mathrm{Conv}(N)$.

Example A.1.2 Let M be the dual vector space of N and $P \subseteq M$ be a convex subset. The *support function* $\mathrm{Supp}_P \in \mathrm{Conv}(N)$ of P is defined as follows:

$$\mathrm{Supp}_P(n) := \sup\{\langle m, n\rangle : m \in P\}.$$

It is well-known that convexity is preserved by a number of natural operations. We recall a few to fix the notation.

Definition A.1.3 Let $F_1, \dots, F_m \in \mathrm{Conv}^{\mathrm{prop}}(N)$ $(m \in \mathbb{Z}_{>0})$. We define their *infimal convolution* $F_1 \square \cdots \square F_m \in \mathrm{Conv}(N)$ as follows:

$$F_1 \square \cdots \square F_m(n) := \inf\left\{\sum_{i=1}^m F_i(n_i) : n_i \in N, \sum_{i=1}^m n_i = n\right\}.$$

The fact $F_1 \square \cdots \square F_m \in \mathrm{Conv}(N)$ is proved in [Roc70, Theorem 5.4]. One should note that $F_1 \square \cdots \square F_m$ is not always proper.

Proposition A.1.1 *Let* $\{F_i\}_{i \in I}$ *be a non-empty family in* $\mathrm{Conv}(N)$. *Then* $\sup_{i \in I} F_i \in \mathrm{Conv}(N)$.

This follows from [Roc70, Theorem 5.5]. In particular, this allows us to introduce

Definition A.1.4 Let $f \colon N \to [-\infty, \infty]$. The *lower convex envelope* of f is defined as

$$\mathrm{CE}\, f := \sup\{F \in \mathrm{Conv}(N) : F \leq f\}.$$

It follows from Proposition A.1.1 that $\mathrm{CE}\, f \in \mathrm{Conv}(N)$.

Definition A.1.5 Given a non-empty family $\{F_i\}_{i \in I}$ in $\mathrm{Conv}(N)$, we define

$$\bigwedge_{i \in I} F_i := \mathrm{CE}\left(\inf_{i \in I} F_i\right).$$

When the family I is finite, say $I = \{1, \dots, m\}$, we also write

$$F_1 \wedge \cdots \wedge F_m = \bigwedge_{i \in I} F_i.$$

Definition A.1.6 Given a non-empty family $\{F_i\}_{i\in I}$ in $\mathrm{Conv}(N)$, we define

$$\bigvee_{i\in I} F_i \coloneqq \sup_{i\in I} F_i.$$

When the family I is finite, say $I = \{1,\ldots,m\}$, we also write

$$F_1 \vee \cdots \vee F_m = \bigvee_{i\in I} F_i.$$

Recall that $\bigvee_{i\in I} F_i \in \mathrm{Conv}(N)$ by Proposition A.1.1.

Proposition A.1.2 *Let $F_1,\ldots,F_m \in \mathrm{Conv}^{\mathrm{prop}}(N)$. Then*

$$F_1 \wedge \cdots \wedge F_m(x) = \inf\left\{\sum_{i=1}^m \lambda_i F_i(x_i) : x_i \in \mathrm{Dom}(F_i),\right.$$
$$\left.\lambda_i \in [0,1], \sum_{i=1}^m \lambda_i = 1, \sum_{i=1}^m \lambda_i x_i = x\right\}.$$

See [Roc70, Theorem 5.6] for the more general result.

Lemma A.1.2 *Let $\{F_i\}_{i\in I}$ be a decreasing net in $\mathrm{Conv}(N)$. Then $\inf_{i\in I} F_i \in \mathrm{Conv}(N)$.*

Proof Write $F = \inf_{i\in I} F_i$. We shall apply the characterization in Lemma A.1.1. Take $n, r \in N$, $a, b \in \mathbb{R}$ such that $a > F(n)$, $b > F(r)$ and $t \in (0,1)$. We need to show that

$$F(tn + (1-t)r) < ta + (1-t)b. \tag{A.1}$$

By definition, there are $j_n, j_r \in I$ such that $a > F_i(n)$ for $i \geq j_n$ and $b > F_i(r)$ for $i \geq j_r$. Since I is directed, we can find $j \in I$ dominating both j_n and j_r, and then for any $i \geq j$, we have

$$a > F_i(n), \quad b > F_i(r).$$

It follows from Lemma A.1.1 that

$$F_i(tn + (1-t)r) < ta + (1-t)b$$

for any $i \geq j$. Since F_i is decreasing in i, we conclude (A.1). □

Definition A.1.7 Let $F \in \mathrm{Conv}(N)$. The *closure* $\mathrm{cl}\, F \in \mathrm{Conv}(N)$ of F is defined as follows: If $F(n) = -\infty$ for some $n \in N$, then $\mathrm{cl}\, F \coloneqq -\infty$. Otherwise, we define $\mathrm{cl}\, F$ as the lower semicontinuity regularization of F.

A convex function $F \in \mathrm{Conv}(N)$ is *closed* if $F = \mathrm{cl}\, F$. In other words, $F \in \mathrm{Conv}(N)$ is closed if one of the following conditions holds:

(1) $F \equiv -\infty$;
(2) $F \equiv \infty$;

(3) F is proper and lower semicontinuous.

Proposition A.1.3 *Let $F \in \mathrm{Conv}(N)$ be a closed convex function. Then F is the supremum of all affine functions lying below F.*

See [Roc70, Theorem 12.1].

Theorem A.1.1 *Let $F \in \mathrm{Conv}^{\mathrm{prop}}(N)$. Then $\mathrm{cl}\, F$ is a closed proper convex function. Moreover, $\mathrm{cl}\, F$ agrees with F except possibly on the relative boundary of $\mathrm{Dom}\, F$.*

See [Roc70, Theorem 7.4].

Proposition A.1.4 *Let $F, F' \in \mathrm{Conv}(N)$ be closed convex functions. Assume that*

(1) $\mathrm{RelInt}\,\mathrm{Dom}\, F = \mathrm{RelInt}\,\mathrm{Dom}\, F'$, *and*
(2) $F = F'$ *on* $\mathrm{RelInt}\,\mathrm{Dom}\, F$.

Then $F = F'$.

This is a special case of [Roc70, Corollary 7.3.4].

Definition A.1.8 Given $F, F' \in \mathrm{Conv}(N)$, we write $F \preceq F'$ if there is $C \in \mathbb{R}$ such that

$$F \leq F' + C.$$

We say $F \sim F'$ if $F \preceq F'$ and $F' \preceq F$ both hold.

Theorem A.1.2 *Let $C \subseteq N$ be an open subset. Let $(f_i)_{i>0}$ be a sequence of real-valued convex functions on C. Suppose that the sequence converges on a dense subset of C and the limit is finite. Then the limit*

$$f(x) \coloneqq \lim_{i\to\infty} f_i(x)$$

exists for all $x \in C$ and is convex on C. Moreover, the sequence $(f_i)_i$ converges uniformly to f on each compact subset of C.

This is a special case of [Roc70, Theorem 10.8].

A.2 Legendre transform

Let N be a real vector space of finite dimension and M be the dual vector space. The pairing $M \times N \to \mathbb{R}$ will be denoted by $\langle \bullet, \bullet \rangle$.

Definition A.2.1 Let $F \in \mathrm{Conv}(N)$ be a convex function. We define the *Legendre transform* of F as the function $F^* \in \mathrm{Conv}(M)$:

$$F^*(m) \coloneqq \sup_{n\in N} \left(\langle m, n\rangle - F(n)\right) = \sup_{n\in \mathrm{RelInt}\,\mathrm{Dom}\, F} \left(\langle m, n\rangle - F(n)\right). \tag{A.2}$$

The latter equality follows from [Roc70, Corollary 12.2.2].

Recall the well-known Legendre–Fenchel duality [Roc70, Theorem 12.2].

Theorem A.2.1 *Let $F \in \operatorname{Conv}(N)$. Then F^* is a closed convex function. The function F^* is proper if and only if F is.*

Moreover, we have $(\operatorname{cl} F)^* = F^*$ and

$$F^{**} = \operatorname{cl} F.$$

Example A.2.1 Let $P \subseteq M$ be a closed convex subset. Then

$$\operatorname{Supp}_P^* = \chi_P, \quad \chi_P^* = \operatorname{Supp}_P .$$

See [Roc70, Theorem 13.2].

The following special case will be useful to us in the sequel.

Corollary A.2.1 *Let $F\colon (0, \infty) \to [-\infty, \infty)$ be a convex function. If we define $G\colon \mathbb{R} \to (-\infty, \infty]$ by*

$$G(\tau) = \sup_{t>0} (t\tau - F(t)) ,$$

then G is a convex function and

$$F(t) = G^*(t), \quad \forall t > 0. \tag{A.3}$$

Moreover,

$$G(\tau) = \sup_{t\in\mathbb{R}_{>0}} (t\tau - F(t)) = \sup_{t\in\mathbb{Q}_{>0}} (t\tau - F(t)) . \tag{A.4}$$

Proof We distinguish two cases.

First suppose that $F(t) = -\infty$ for some $t > 0$. Then $F(t) = -\infty$ for all $t > 0$ by the convexity of F. Our assertions are clear in this case.

Next assume that $F(t) \neq -\infty$ for all $t > 0$. In this case, Theorem A.1.1 guarantees that F admits a closed proper extension $\tilde{F} \in \operatorname{Conv}(\mathbb{R})$ with

$$\tilde{F}(t) = \infty, \quad \forall t < 0.$$

It follows from (A.2) that

$$G(\tau) = \tilde{F}^*(\tau), \quad \forall \tau \in \mathbb{R}.$$

Now Theorem A.2.1 implies (A.3). Finally (A.4) follows from the continuity of F. □

Proposition A.2.1 *Let $F\colon N \to [-\infty, \infty]$. Define $F^*\colon M \to [-\infty, \infty]$ by*

$$F^*(m) := \sup_{n\in N} (\langle m, n\rangle - F(n)) .$$

Then

$$F^* = (\operatorname{cl} \operatorname{CE} F)^*.$$

See [Roc70, Corollary 12.1.1].

Definition A.2.2 Let $F \in \mathrm{Conv}(N)$ and $n \in N$. An element $m \in M$ is a *subgradient* of F at n if

$$F(n') \geq F(n) + \langle n' - n, m\rangle, \quad \forall n' \in N. \tag{A.5}$$

The set of subgradients of F at n is denoted by $\nabla F(n)$.

More generally, for any subset $E \subseteq N$, we write

$$\nabla F(E) = \bigcup_{n\in E} \nabla F(n).$$

Definition A.2.3 Given $F, F' \in \mathrm{Conv}(N)$, we write $F \preceq_P F'$ if

$$\overline{\nabla F(N)} \subseteq \overline{\nabla F'(N)}.$$

We write $F \sim_P F'$ if $F \preceq_P F'$ and $F' \preceq_P F$.

Theorem A.2.2 *Suppose that* $F \in \mathrm{Conv}^{\mathrm{prop}}(N)$. *Then the following hold:*

(1) *For any* $n \notin \mathrm{Dom}\, F$, $\nabla F(n) = \varnothing$*;*
(2) *for any* $n \in \mathrm{RelInt}\,\mathrm{Dom}\, F$, $\nabla F(n) \neq \varnothing$*; moreover, for any* $n' \in N$*, we have*

$$\partial_{n'} F(n) = \sup\left\{\langle n', m\rangle : m \in \nabla F(n)\right\};$$

(3) *for* $n \in N$*, the set* $\nabla F(n)$ *is non-empty and bounded if and only if* $n \in \mathrm{Int}\,\mathrm{Dom}\, F$.

For the proof, we refer to [Roc70, Theorem 23.4].

Proposition A.2.2 *Let* $F \in \mathrm{Conv}^{\mathrm{prop}}(N)$. *Then*

$$\nabla F(N) \subseteq \mathrm{Dom}\, F^*.$$

If moreover F *is closed, we have*

$$\mathrm{RelInt}\,\mathrm{Dom}\, F^* \subseteq \nabla F(N). \tag{A.6}$$

In particular, if F *is a proper closed convex function on* N*, then*

$$\overline{\nabla F(N)} = \overline{\mathrm{Dom}\, F^*}.$$

Proof Suppose that $m \in \nabla F(n)$ for some $n \in N$. Then (A.5) holds. In particular,

$$\langle m, n'\rangle - F(n') \leq \langle m, n\rangle - F(n).$$

It follows that

$$F^*(m) \leq \langle m, n\rangle - F(n) < \infty.$$

(A.6) is proved in [Roc70, Corollary 23.5.1]. For the last assertion, it suffices to observe that $\overline{\mathrm{RelInt}\,\mathrm{Dom}\, F^*} = \overline{\mathrm{Dom}\, F^*}$. □

Proposition A.2.3 *Let $\{F_i\}_{i\in I}$ be a non-empty family in $\mathrm{Conv}^{\mathrm{prop}}(N)$. Then*

$$\left(\bigwedge_{i\in I} F_i\right)^* = \bigvee_{i\in I} F_i^*, \quad \left(\bigvee_{i\in I} \mathrm{cl}\, F_i\right)^* = \mathrm{cl} \bigwedge_{i\in I} F_i^*.$$

If I is finite and $\overline{\mathrm{Dom}\, F_i}$ is independent of the choice of $i \in I$, then

$$\left(\bigvee_{i\in I} F_i\right)^* = \bigwedge_{i\in I} F_i^*.$$

Recall that $\wedge$ is defined in Definition A.1.5 and $\vee$ in Definition A.1.6. See [Roc70, Theorem 16.5] for the proof.

Proposition A.2.4 *Let $F_1, \dots, F_r \in \mathrm{Conv}^{\mathrm{prop}}(N)$ ($r \in \mathbb{Z}_{>0}$). Assume that*

$$\bigcap_{i=1}^{r} \mathrm{RelInt}\, \mathrm{Dom}(F_i) \neq \varnothing,$$

then for any $m \in M$,

$$\left(\sum_{i=1}^{r} F_i\right)^* (m) = \inf \left\{ \sum_{i=1}^{r} F_i^*(m_i) : m_1, \dots, m_r \in M, \sum_{i=1}^{r} m_i = m \right\}.$$

See [Roc70, Theorem 16.4] for the proof.

Proposition A.2.5 *Let $P \subseteq M$ be a convex body*[1] *and $F \in \mathrm{Conv}^{\mathrm{prop}}(N)$. The following are equivalent:*

(1) *$F \preceq \mathrm{Supp}_P$;*
(2) *$\mathrm{Dom}\, F = N$ and $F^*|_{M\setminus P} \equiv \infty$;*
(3) *$\mathrm{Dom}\, F = N$ and $\nabla F(N) \subseteq P$.*

Moreover, under these conditions,

$$F(n) - \mathrm{Supp}_P(n) \leq F(0), \quad \forall n \in N. \tag{A.7}$$

Proof (1) $\implies$ (2). It is clear that $\mathrm{Dom}\, F = N$ since $\mathrm{Dom}\,\mathrm{Supp}_P = N$. From $F \preceq \mathrm{Supp}_P$ and Example A.2.1, we know that

$$\chi_P = \mathrm{Supp}_P^* \preceq F^*.$$

So (2) follows.

(2) $\implies$ (3). This follows from Proposition A.2.2.

(3) $\implies$ (1). Taking $n \in N$, we know that F is locally Lipschitz [Roc70, Theorem 10.4], so we can compute

[1] Here a convex body refers to a non-empty compact convex subset, not necessarily having non-empty interior.

$$F(n) - F(0) = \int_0^1 \partial_n F(tn)\,\mathrm{d}t$$
$$\leq \int_0^1 \mathrm{Supp}_P(n)\,\mathrm{d}t = \mathrm{Supp}_P(n).$$

In particular, (A.7) also follows. □

A.3 Classes of convex functions

Let N be a real vector space of finite dimension and M be the dual vector space.

We shall fix a convex body $P \subseteq M$.

The following classes are introduced in [BB13].

Definition A.3.1 We define the set $\mathcal{P}(N, P)$ as the set of proper convex functions $F \in \mathrm{Conv}(N)$ such that $F \leq \mathrm{Supp}_P$.

We define the set $\mathcal{E}^\infty(N, P)$ as the set of closed convex functions $F \in \mathrm{Conv}(N)$ such that $F \sim \mathrm{Supp}_P$.

We define the set $\mathcal{E}(N, P)$ as follows: Suppose that $\mathrm{Int}\, P = \varnothing$. Then $\mathcal{E}(N, P) := \mathcal{P}(N, P)$; otherwise, let

$$\mathcal{E}(N, P) = \left\{ F \in \mathcal{P}(N, P) : P = \overline{\nabla F(N)} \right\}.$$

We define the set $\mathcal{E}^1(N, P)$ as the subset of $\mathcal{E}(N, P)$ consisting of $F \in \mathcal{E}(N, P)$ with

$$\int_P F^*\,\mathrm{d}\,\mathrm{vol} < \infty,$$

where $\mathrm{d}\,\mathrm{vol}$ is any Lebesgue measure on M.

Observe that for any $F \in \mathcal{P}(N, P)$, we have $\mathrm{Dom}\, F = N$ and F is necessarily closed.

Proposition A.3.1 *We have*

$$\mathcal{E}^\infty(N, P) \subseteq \mathcal{E}^1(N, P) \subseteq \mathcal{E}(N, P) \subseteq \mathcal{P}(N, P).$$

Proof When $\mathrm{Int}\, P = \varnothing$, the assertion is clear. We assume that $\mathrm{Int}\, P \neq \varnothing$. The second inclusion follows from the definition. We only handle the first inequality. Take $F \in \mathcal{E}^\infty(N, P)$. By definition, $F \sim \mathrm{Supp}_P$ and hence $F^* \sim \chi_P$. It follows that $P = \mathrm{Dom}\, F^*$.

By Proposition A.2.5, we already know that

$$\nabla F(N) \subseteq P = \mathrm{Dom}\, F^*.$$

On the other hand, by Proposition A.2.2, we have

$$\mathrm{Int}\, P \subseteq \nabla F(N).$$

So it follows that
$$P = \overline{\nabla F(N)}.$$

It is clear that $F^* \sim \chi_P$ is integrable. □

Proposition A.3.2 *For any $F \in \mathcal{E}^\infty(N, P)$, we have $F^*|_{M\setminus P} \equiv \infty$ and F^* is bounded on P.*

Proof From $F \sim \mathrm{Supp}_P$, we take the Legendre transform to get $F^* \sim \mathrm{Supp}_P^* = \chi_P$, where we applied Example A.2.1. □

Definition A.3.2 We endow the topology of pointwise convergence on $\mathcal{P}(N, P)$. Note that this topology coincides with the compact-open topology.

Proposition A.3.3 *Let $F \in \mathcal{P}(N, P)$. Then there is a decreasing sequence $F_j \in \mathcal{E}^\infty(N, P) \cap C^\infty(N)$ converging to F.*

See [BB13, Lemma 2.2].

We observe that the point $0 \in N$ plays a special role since it does in the definition of the support function.

Proposition A.3.4 *For any $F \in \mathcal{P}(N, P)$, we have*

$$\max_N (F - \mathrm{Supp}_P) = F(0).$$

Proof It follows from (A.7) that

$$\sup_N (F - \mathrm{Supp}_P) \leq F(0).$$

The equality is clearly obtained at $0 \in N$. □

Lemma A.3.1 *Let $P' \subseteq M$ be another convex body. Then for any $F \in \mathcal{P}(N, P)$ and $F' \in \mathcal{P}(N, P')$, we have*
$$F + F' \in \mathcal{P}(N, P + P').$$

Moreover, let F and F' be real-valued convex functions on N. Assume that Dom F^* *is bounded. Then*
$$\overline{\nabla F(N)} + \overline{\nabla F'(N)} = \overline{\nabla (F + F')(N)}.$$

In particular, if $F \in \mathcal{P}(N, P)$, $F' \in \mathcal{P}(N, P')$ and

$$\overline{\nabla F(N)} = P, \quad \overline{\nabla F'(N)} = P',$$

then $F + F' \in \mathcal{E}(N, P + P')$.

Proof The former assertion follows immediately from the observation

$$\mathrm{Supp}_{P+P'} = \mathrm{Supp}_P + \mathrm{Supp}_{P'}.$$

We prove the gradient-image equality. In view of Proposition A.2.2, this means

$$\overline{\mathrm{Dom}(F+F')^*} = \overline{\mathrm{Dom}\, F^*} + \overline{\mathrm{Dom}\, F'^*}. \tag{A.8}$$

It follows from Proposition A.2.4 that

$$\mathrm{Dom}(F+F')^* = \mathrm{Dom}\, F^* + \mathrm{Dom}\, F'^*.$$

Since $\overline{\mathrm{Dom}\, F^*}$ is compact, (A.8) follows[2].

Finally, under the assumptions in the last assertion, the former assertion gives $F+F' \in \mathcal{P}(N, P+P')$. If $\mathrm{Int}(P+P') = \varnothing$, this is exactly the definition of $\mathcal{E}(N, P+P')$. Otherwise, the gradient-image equality gives

$$\overline{\nabla(F+F')(N)} = P+P',$$

so again $F+F' \in \mathcal{E}(N, P+P')$. □

A.4 Monge–Ampère measures

Let N be a free Abelian group of finite rank (i.e., a lattice) and M be its dual lattice. There is a canonical Lebesgue type measure on $M_{\mathbb{R}}$, denoted by $\mathrm{d\,vol}$, normalized so that the smallest cubes in M have volume 1. Similarly, the canonical measure on $N_{\mathbb{R}}$ is normalized in the same way and is denoted by $\mathrm{d\,vol}$ as well.

We will write

$$N_{\mathbb{R}} = N \otimes_{\mathbb{Z}} \mathbb{R}, \quad M_{\mathbb{R}} = M \otimes_{\mathbb{Z}} \mathbb{R}.$$

Definition A.4.1 Let F be a proper closed convex function on $N_{\mathbb{R}}$. We define the *real Monge–Ampère measure* $\mathrm{MA}_{\mathbb{R}}\, F$ as the Alexandrov measure on $N_{\mathbb{R}}$ given as follows: for each Borel measurable set $E \subseteq N_{\mathbb{R}}$, define

$$\mathrm{MA}_{\mathbb{R}}\, F(E) \coloneqq n! \int_{\nabla F(E)} \mathrm{d\,vol}\,.$$

The measurability and countable additivity in this definition are standard properties of the subgradient map; see for example [Fig17, Section 2.1].

Proposition A.4.1 *Suppose that $F \in C^{1,1}(N_{\mathbb{R}}) \cap \mathrm{Conv}(N_{\mathbb{R}})$. Fix an identification $N = \mathbb{Z}^n$, then*

$$\mathrm{MA}_{\mathbb{R}}\, F = n! \cdot \det \nabla^2 F \,\mathrm{d\,vol}\,.$$

See [Fig17, Example 2.2].

Proposition A.4.2 *Let $P \subseteq M_{\mathbb{R}}$ be a convex body and $F \in \mathcal{P}(N_{\mathbb{R}}, P)$. Then $F \in \mathcal{E}(N_{\mathbb{R}}, P)$ if and only if*

$$\int_{N_{\mathbb{R}}} \mathrm{MA}_{\mathbb{R}}\, F = n! \,\mathrm{vol}\, P. \tag{A.9}$$

[2] In general, the Minkowski sum does not commute with the closure.

Proof Since $F \in \mathcal{P}(N_{\mathbb{R}}, P)$, the function F is closed and has domain $N_{\mathbb{R}}$. By Proposition A.2.2, $\overline{\nabla F(N_{\mathbb{R}})} = \overline{\mathrm{Dom}\, F^*}$, so this closure is convex. As $\nabla F(N_{\mathbb{R}})$ contains the relative interior of Dom F^*, the definition of $\mathrm{MA}_{\mathbb{R}}$ shows that (A.9) is equivalent to

$$\mathrm{vol}\, \overline{\nabla F(N_{\mathbb{R}})} = \mathrm{vol}\, P.$$

We first handle the case where $\mathrm{Int}\, P \neq \emptyset$. By Proposition A.2.5, $\overline{\nabla F(N_{\mathbb{R}})} \subseteq P$. Hence the latter is equivalent to

$$\overline{\nabla F(N_{\mathbb{R}})} = P.$$

Indeed, a proper closed subset of the full-dimensional convex body P has complement containing a non-empty relatively open subset of P, hence has strictly smaller volume.

Now assume that $\mathrm{Int}\, P = \emptyset$, then $\mathrm{vol}\, \overline{\nabla F(N_{\mathbb{R}})} = \mathrm{vol}\, P = 0$ by Proposition A.2.5. The assertion is clear. □

Theorem A.4.1 *Let $F, F_j \in \mathcal{P}(N_{\mathbb{R}}, P)$ ($j \in \mathbb{Z}_{>0}$). Assume that $F_j \to F$. Then $\mathrm{MA}_{\mathbb{R}}(F_j)$ converges to $\mathrm{MA}_{\mathbb{R}}(F)$ weakly.*

See [Fig17, Proposition 2.6].

There is a well-known comparison principle.

Theorem A.4.2 *Let $F, F' \in \mathcal{P}(N_{\mathbb{R}}, P)$. Assume that $F \leq F'$. Then*

$$\overline{\nabla F(N_{\mathbb{R}})} \subseteq \overline{\nabla F'(N_{\mathbb{R}})}, \quad \int_{N_{\mathbb{R}}} \mathrm{MA}_{\mathbb{R}}(F) \leq \int_{N_{\mathbb{R}}} \mathrm{MA}_{\mathbb{R}}(F').$$

Proof It suffices to observe that $F'^* \leq F^*$, and hence the first assertion follows from Proposition A.2.2. The second assertion follows from the first. □

A.5 Separation lemmata

Lemma A.5.1 *Let $\alpha, \beta_1, \ldots, \beta_m \in \mathbb{Z}^n$. Let Δ be the polytope generated by $\beta_1, \ldots, \beta_m$. Then the following are equivalent:*

(1)
$$|z^\alpha|^2 \left(\sum_{i=1}^m |z^{\beta_i}|^2 \right)^{-1} \tag{A.10}$$

is a bounded function on $\mathbb{C}^{*n}$.

(2) $\alpha \in \Delta$.

Proof (2) $\Longrightarrow$ (1). Write $\alpha = \sum_i t_i \beta_i$, where $t_i \in [0, 1]$, $\sum_i t_i = 1$. Then

$$\begin{aligned}|z^\alpha|^2 \left(\sum_{i=1}^m |z^{\beta_i}|^2\right)^{-1} &= \prod_i |z^{\beta_i}|^{2t_i} \left(\sum_{i=1}^m |z^{\beta_i}|^2\right)^{-1} \\ &\le \prod_i \left(\sum_{j=1}^m |z^{\beta_j}|^2\right)^{t_i} \left(\sum_{i=1}^m |z^{\beta_i}|^2\right)^{-1} = 1.\end{aligned}$$

(1) $\implies$ (2). Assume that $\alpha \notin \Delta$. By the strict separation theorem, there is $a = (a_1, \dots, a_n) \in \mathbb{R}^n$ such that

$$a \cdot \alpha > \max_i a \cdot \beta_i .$$

Set

$$z(t) := (t^{a_1}, \dots, t^{a_n}).$$

Then (A.10) evaluated at $z(t)$ grows like

$$\frac{t^{2a\cdot\alpha}}{\sum_i t^{2a\cdot\beta_i}}$$

as $t \to \infty$, and hence is not bounded. □

Lemma A.5.2 *Let $\beta_1, \dots, \beta_m \in \mathbb{N}^n$ and $\beta \in \mathbb{R}^n$. Then the following are equivalent:*

(1) *$\log \sum_{i=1}^m e^{x\cdot\beta_i} - x \cdot \beta$ is bounded from below.*
(2) *β is in the convex hull of the β_i's.*

Proof If $\beta = \sum_i t_i \beta_i$ with $t_i \in [0, 1]$ and $\sum_i t_i = 1$, then by the weighted arithmetic–geometric mean inequality,

$$x \cdot \beta = \sum_i t_i\, x \cdot \beta_i \le \log \sum_i t_i e^{x\cdot\beta_i} \le \log \sum_i e^{x\cdot\beta_i}.$$

This proves that the function in (1) is bounded from below.

Conversely, if β does not lie in the convex hull of the β_i's, choose $a \in \mathbb{R}^n$ such that

$$a \cdot \beta > \max_i a \cdot \beta_i .$$

Evaluating at $x = ta$ and letting $t \to \infty$, we find

$$\log \sum_i e^{t\,a\cdot\beta_i} - t\, a \cdot \beta \to -\infty,$$

contradicting (1). □

Appendix B
Pluripotential theory on unibranch spaces

This appendix explains how much of the pluripotential theory developed in the body of the book extends from compact Kähler manifolds to compact unibranch Kähler spaces. The unibranch hypothesis is the natural generality in which plurisubharmonic functions descend through resolutions and the basic envelope formalism remains meaningful. Some constructions, especially the trace operator of Chapter 8, require extra care in the singular setting; we record here the precise statements needed in the main text.

B.1 Complex spaces

A complex space is assumed to be reduced, Hausdorff and paracompact in the whole book.

Definition B.1.1 A *prime divisor* over an irreducible complex space Z is a connected smooth hypersurface $E \subseteq X'$, where $X' \to Z$ is a proper bimeromorphic morphism with X' smooth. Such a morphism $X' \to Z$ is also called a *resolution* of Z. The *center* of the prime divisor is defined as the image of E in Z.

Two prime divisors $E_1 \subseteq X'_1$ and $E_2 \subseteq X'_2$ over Z are *equivalent* if there is a common resolution $X'' \to Z$ dominating both X'_1 and X'_2 such that the strict transforms of E_1 and E_2 coincide.

The set Z^{div} is the set of pairs (c, E), where $c \in \mathbb{Q}_{>0}$ and E is an equivalence class of a prime divisor over Z. For simplicity, we will denote the pair (c, E) by $c \operatorname{ord}_E$, although one should not really think of this object as a valuation unless Z is projective and irreducible.

Note that a prime divisor on Z does not always define a prime divisor over Z if Z is singular.

Definition B.1.2 A complex space X is *unibranch* if for all $x \in X$, the local ring $\mathcal{O}_{X,x}$ is unibranch.

See [Stacks, Tag 0BPZ] for the notion of unibranch local rings. In [CAS], the same notion is known as *local irreducibility*. We prefer the name *unibranchness*, because the corresponding terminology in algebraic geometry is standard and goes back to Grothendieck [EGA IV_1, (0.23.2.1)].

The notion of unibranchness coincides with the usual one when $\mathcal{X}$ is an algebraic variety:

Proposition B.1.1 *Let $\mathcal{X}$ be a reduced scheme of finite type over $\mathbb{C}$, and let $\mathcal{X}^{\mathrm{An}}$ be the associated complex space. Then $\mathcal{X}^{\mathrm{An}}$ is unibranch in the sense above if and only if $\mathcal{X}$ is unibranch in the algebraic sense (see [Stacks, Tag 0BQ2]).*

This is [Xia23a, Remark 2.7] in the arXiv version.

Proof Let $x \in \mathcal{X}(\mathbb{C})$. Then we claim that $\mathcal{X}$ is unibranch at x if and only if $\mathcal{X}^{\mathrm{An}}$ is unibranch at x. By GAGA, there is a natural morphism of ringed spaces $(\mathcal{X}^{\mathrm{An}}, \mathcal{O}_{\mathcal{X}^{\mathrm{An}}}) \to (\mathcal{X}, \mathcal{O}_{\mathcal{X}})$. Then since $\mathcal{O}_{\mathcal{X},x}$ is excellent, $\mathcal{X}$ is unibranch at x if and only if $\widehat{\mathcal{O}_{\mathcal{X},x}} = \widehat{\mathcal{O}_{\mathcal{X}^{\mathrm{An}},x}}$ is integral ([EGA IV_4, Théorème 18.9.1]). The latter implies that $\mathcal{O}_{\mathcal{X}^{\mathrm{An}},x}$ is unibranch by [AC, Chapitre 5, Exercise 2.8]. Running the same argument the other way round, we conclude that $\mathcal{O}_{\mathcal{X},x}$ is unibranch if and only if $\mathcal{O}_{\mathcal{X}^{\mathrm{An}},x}$ is. □

Theorem B.1.1 (Zariski's main theorem) *Let $\pi\colon Y \to X$ be a proper bimeromorphic morphism between complex spaces. Assume that X is unibranch. Then π has connected fibers.*

We follow [Dem85, Proof of Théorème 1.7].

Proof Let $Z \subsetneq X$ be a nowhere dense analytic subset such that π is an isomorphism over $X \setminus Z$. Assume that the fiber $F = \pi^{-1}(x)$ over some $x \in X$ is disconnected. Since π is proper, F is compact. Choose a decomposition $F = F_1 \sqcup F_2$ into two non-empty disjoint compact open subsets of F, and choose disjoint open neighborhoods W_1, W_2 of F_1, F_2 in Y. Since proper maps are closed, after shrinking an open neighborhood U of x, we may assume that

$$\pi^{-1}(U) \subseteq W_1 \cup W_2.$$

Thus $\pi^{-1}(U)$ is disconnected.

Since X is unibranch, we may shrink U further so that U is irreducible. Then $U \setminus Z$ is connected (cf. [CAS, Chapter 9, §1.2]). Moreover, by bimeromorphicity, $\pi^{-1}(Z)$ contains no irreducible component of Y, hence is nowhere dense. On the other hand, write

$$\pi^{-1}(U) = V_1 \sqcup V_2$$

as a union of two non-empty disjoint open subsets. Each V_i meets $\pi^{-1}(U \setminus Z)$: otherwise V_i would be a non-empty open subset of Y contained in the nowhere dense analytic subset $\pi^{-1}(Z)$, which is impossible. Hence

$$\pi^{-1}(U \setminus Z) = \left(V_1 \cap \pi^{-1}(U \setminus Z)\right) \sqcup \left(V_2 \cap \pi^{-1}(U \setminus Z)\right)$$

is disconnected. This contradicts the fact that $\pi\colon \pi^{-1}(U \setminus Z) \to U \setminus Z$ is an isomorphism and $U \setminus Z$ is connected. □

Definition B.1.3 A *modification* of a compact complex space X is a finite composition of blow-ups with smooth centers.

We say a modification $\pi' : Z \to X$ *dominates* another $\pi : Y \to X$ if there is a morphism $g : Z \to Y$ of complex spaces making the following diagram commutative:

$$\begin{tikzcd} Z \arrow[rr, "g"] \arrow[dr, "\pi'"'] & & Y \arrow[dl, "\pi"] \\ & X. & \end{tikzcd} \tag{B.1}$$

The modifications of X together with the domination relation form a directed set.

Theorem B.1.2 (Hironaka's Chow lemma) *Suppose that X is a compact complex space. Then every proper bimeromorphic morphism to X can be dominated by a modification.*

This follows from the proof of [Hir75, Corollary 2].

Theorem B.1.3 *Let X be a compact complex space. Then there is a modification $\pi : Y \to X$ such that Y is smooth.*

See [BM97, Wł09].

Corollary B.1.1 *Let X be an irreducible compact complex space and E be a prime divisor over X. Then there is a modification $\pi : Y \to X$ such that Y is smooth and E can be realized as a prime divisor on Y.*

Proof Choose a smooth model $X' \to X$ on which E is realized. By Theorem B.1.2, this proper bimeromorphic morphism is dominated by a modification $Y \to X$. After further resolving the pair if necessary, we may assume that Y is smooth and that the strict transform of E on Y is smooth. This strict transform realizes the same prime divisor over X. □

B.2 Plurisubharmonic functions

Let X be a complex space.

Definition B.2.1 A function $\varphi : X \to [-\infty, \infty)$ is *plurisubharmonic* if

(1) φ is not identically $-\infty$ on any irreducible component of X, and
(2) for any $x \in X$, there is an open neighborhood V of x in X, a domain $\Omega \subseteq \mathbb{C}^N$, a closed immersion $V \hookrightarrow \Omega$ and a plurisubharmonic function $\tilde{\varphi} \in \mathrm{PSH}(\Omega)$ such that $\varphi|_V = \tilde{\varphi}|_V$ under this immersion.

The set of plurisubharmonic functions on X is denoted by $\mathrm{PSH}(X)$.

Similarly, if θ is a smooth closed[1] real $(1,1)$-form on X, then a function $\varphi\colon X \to [-\infty,\infty)$ is *θ-plurisubharmonic* if for any $x \in X$, there is an open neighborhood V of x in X, a domain $\Omega \subseteq \mathbb{C}^N$, a closed immersion $V \hookrightarrow \Omega$ and a smooth function g on Ω such that $\theta = (\mathrm{dd^c} g)|_V$ and $g|_V + \varphi|_V \in \mathrm{PSH}(V)$.

Theorem B.2.1 (Fornaess–Narasimhan) *Let $\varphi\colon X \to [-\infty,\infty)$ be a function. Assume that φ is not identically $-\infty$ on any irreducible component of X. Then the following are equivalent:*

(1) *φ is psh;*
(2) *φ is usc and for any morphism $f\colon \Delta \to X$ from the open unit disk Δ in $\mathbb{C}$ to X such that $f^*\varphi$ is not identically $-\infty$, the pull-back $f^*\varphi$ is subharmonic.*

See [FsN80].

Theorem B.2.2 (Grauert–Remmert) *Let X be a unibranch[2] complex space, let Z be an analytic subset in X, and let $\varphi \in \mathrm{PSH}(X \setminus Z)$. Then the function φ admits an extension to $\mathrm{PSH}(X)$ in the following two cases:*

(1) *The set Z has codimension at least 2 everywhere.*
(2) *The set Z has codimension at least 1 everywhere and is locally bounded from above on an open neighborhood of Z.*

In both cases, the extension is unique and is given by

$$\varphi(x) = \varlimsup_{X\setminus Z \ni y \to x} \varphi(y), \quad x \in X. \tag{B.2}$$

The proof given below combines [Dem85, Théorème 1.7] and [GR56].[3]

Proof We first prove the uniqueness, which is a local problem on X. Let ψ denote the function defined by the right-hand side of (B.2). If $\eta \in \mathrm{PSH}(X)$ is an extension of φ, then upper semicontinuity gives $\eta \geq \psi$. Conversely, take $z \in Z$ and a holomorphic map $f\colon \Delta \to X$ such that $f(0) = z$ and $f(\Delta) \not\subseteq Z$. From the subharmonicity of $f^*\eta$ and (B.2), we find that

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[1] Here *closed* means that locally θ is defined by a closed form under a local embedding.

[2] Unibranchness is very important here. Otherwise, consider the case where X is the union of two copies of $\mathbb{C}$ intersecting only at their origins, Z is the common origin. If we set $\varphi \equiv 0$ on one punctured plane and $\varphi \equiv 1$ on the other, then it is clear that φ cannot be extended to X. This leads to a few misleading statements in the modern literature. The problem is that in the early German literature, *komplexer Raum* is assumed to be either normal or unibranch!

[3] This theorem has a quite entangled history. The corresponding results for subharmonic functions are generally attributed to Brelot. In [GR56], they cited a paper of Brelot written in 1934, which I cannot find. But in 1949, on the very first issue of *Annales de l'Institut Fourier*, Brelot published a paper [Bre49] with a very similar title, studying the behavior of a subharmonic function on the punctured neighborhood of a point. The general theorem was due to Grauert and Remmert, see [GR56]. Their original proof was quite complicated, due to the fact that resolution of singularities was not available at that time. Later on, in 1985, Demailly published the influential paper [Dem85] and gave a simpler proof. Oddly enough, Demailly did not cite either Grauert–Remmert or Brelot. He did not even mention that this result was already proved by Grauert–Remmert. The paper [Dem85] is so influential that in France few people know the existence of [GR56].

$$\eta(z) = f^*\eta(0) = \varlimsup_{\Delta^* \ni w \to 0} f^*\eta(w) \le \varlimsup_{X \setminus Z \ni y \to z} \varphi(y) = \psi(z).$$

Thus $\eta = \psi$, and uniqueness follows.

Having established the uniqueness of the extension, the existence also becomes a local problem. So we can freely shrink X around any point in Z in the proof.

(2) Let $\pi : Y \to X$ be a resolution of singularities. By the smooth version of the extension theorem Theorem 1.2.1, we know that $\pi^*\varphi$ admits a unique extension to a psh function on Y, which we denote by η. Note that all fibers of π are connected since X is unibranch, thanks to Theorem B.1.1. The restriction of η to each fiber is psh on a compact connected complex space, hence is constant by the maximum principle. It therefore descends to an upper semicontinuous function ψ on X, and $\psi|_{X \setminus Z} = \varphi$.

We verify that ψ is plurisubharmonic using Theorem B.2.1. Let $f : \Delta \to X$ be a holomorphic map. We assume that $f^*\psi \not\equiv -\infty$. It suffices to show that $f^*\psi$ is subharmonic at $0 \in \Delta$. After replacing Δ by a finite ramified cover if necessary, the germ of f lifts to Y. Indeed, consider the proper projection

$$\Delta \times_X Y \longrightarrow \Delta.$$

Choose an irreducible analytic curve in $\Delta \times_X Y$ passing through a point over 0 and dominating Δ; after shrinking Δ and normalizing this curve, we obtain a finite ramified cover of the disk. Thus there are an integer $k > 0$ and a holomorphic map $f' : \Delta \to Y$ such that

$$f(t^k) = \pi(f'(t))$$

for all t close to 0. Therefore, $\psi(f(t^k)) = \eta(f'(t))$ near 0. Since subharmonicity descends under the finite map $t \mapsto t^k$, it follows that $f^*\psi$ is subharmonic near 0.

(1) By the local description of complex spaces [CAS, Section 3.4], we may assume that there is a domain $\Omega \subseteq \mathbb{C}^n$, a finite s-sheet branched covering $\Phi : X \to \Omega$ with branched locus contained in a proper analytic subset $V \subsetneq \Omega$. We may assume that X is connected, $n \ge 1$.

It suffices to show that φ is locally bounded from above near Z, since then (2) applies. Suppose that this fails. Then we can find $z \in Z$ and $x_i \in X \setminus Z$ $(i \ge 1)$ converging to z such that

$$\lim_{i \to \infty} \varphi(x_i) = \infty.$$

Using (B.2) around each x_i, we may further assume that $x_i \in X \setminus (\Phi^{-1}(\Phi(Z) \cup V))$.

Let L be a complex line passing through $\Phi(z)$ intersecting $(\Phi(Z) \cup V) \cap \overline{B}$ only at $\Phi(z)$, where $B \Subset B' \Subset \Omega$ are two small open balls centered at $\Phi(z)$. Since Φ is finite, $\Phi(Z)$ still has codimension at least 2 in Ω. Hence, for each i, the set of directions of lines through $\Phi(x_i)$ meeting $B' \cap \Phi(Z)$ is contained in a proper analytic subset of the projective space of directions. We can therefore choose lines L_i through $\Phi(x_i)$ converging to L such that $L_i \cap (B' \cap \Phi(Z)) = \varnothing$. Moreover, since $\Phi(x_i) \notin V$, no such L_i is contained in V, and hence $L_i \cap (B' \cap V)$ is discrete. Then Φ restricts to a branched covering over $B' \cap L_i$ for all $i \ge 1$. Adding a constant to φ, we may assume that $\varphi|_{\Phi^{-1}(L \cap \partial B)} < 0$. We can then find an open neighborhood U of $\Phi^{-1}(L \cap \partial B)$ so that $\varphi|_U < 0$. For large i, we have $\Phi^{-1}(L_i \cap \partial B) \subseteq U$ and $\Phi(x_i) \in B$. Put

$$C_i \coloneqq \Phi^{-1}(L_i \cap B').$$

By the choice of L_i, the complex curve C_i is contained in $X \setminus Z$, and $\Phi|_{C_i}$ is finite over $L_i \cap B'$. Hence $C_i \cap \Phi^{-1}(\overline{B})$ is compact and its boundary is contained in $\Phi^{-1}(L_i \cap \partial B)$. Passing to the normalization of each irreducible component of C_i, the pull-back of φ is subharmonic. The one-dimensional maximum principle therefore gives

$$\varphi(x_i) \leq \sup_{\Phi^{-1}(L_i \cap \partial B)} \varphi < 0,$$

which is a contradiction. □

Corollary B.2.1 *Let $\pi\colon Y \to X$ be a proper bimeromorphic morphism between compact Kähler spaces. Let θ be a smooth closed real $(1,1)$-form on X. Assume that X is unibranch. Then the pull-back induces a bijection*

$$\pi^*\colon \mathrm{PSH}(X,\theta) \xrightarrow{\sim} \mathrm{PSH}(Y,\pi^*\theta).$$

Proof Injectivity follows from the surjectivity of π. For surjectivity, let $u \in \mathrm{PSH}(Y,\pi^*\theta)$. The assertion is local on X, so after subtracting a local potential of θ we may assume that $\theta = 0$.

By Theorem B.1.1, the fibers of π are connected. The restriction of u to any compact fiber is psh, hence is constant by the maximum principle. Thus u descends to a function v on X. Since π is proper, v is upper semicontinuous.

It remains to check that v is psh. By Theorem B.2.1, it suffices to test on holomorphic disks. Let $f\colon \Delta \to X$ be a holomorphic disk such that $f^*v \not\equiv -\infty$. After a finite ramified cover $\rho\colon \Delta \to \Delta$, the germ of $f \circ \rho$ lifts to Y. The pull-back $(f^*v) \circ \rho$ is therefore subharmonic, so f^*v is subharmonic. Hence $v \in \mathrm{PSH}(X)$, and the general θ-psh case follows by adding back the local potentials. □

B.3 Extensions of the results in the smooth setting

Let X be an irreducible unibranch compact Kähler space of dimension n. Let θ be a closed real smooth $(1,1)$-form on X. We say *the cohomology class* $[\theta]$ is big if for any proper bimeromorphic morphism $\pi\colon Y \to X$ from a compact Kähler manifold Y, $[\pi^*\theta]$ is big.

The non-pluripolar products can be defined exactly as in Chapter 2 and the results in that chapter hold *mutatis mutandis*.

The results in Chapter 3 can also be easily extended. The definition of the P-envelope remains unchanged. As for the $\mathcal{I}$-envelope, we define

Definition B.3.1 Given $\varphi \in \mathrm{PSH}(X,\theta)$, we define $P_\theta[\varphi]_\mathcal{I} \in \mathrm{PSH}(X,\theta)$ as the unique element with the following property: If $\pi\colon Y \to X$ is a proper bimeromorphic morphism from a compact Kähler manifold Y, then

$$\pi^* P_\theta[\varphi]_\mathcal{I} = P_{\pi^*\theta}[\pi^*\varphi]_\mathcal{I}.$$

It follows from Corollary B.2.1 and Proposition 3.2.5 that $P_\theta[\varphi]_{\mathcal{I}}$ is independent of the choice of π and is well-defined. The other results can be easily extended.

Chapter 4 and Chapter 6 can be extended without big changes. The only exception is Theorem 6.2.6, where we do not have the notion of multiplier ideal sheaves. So we do not know how to extend this theorem.

Chapter 7 can be extended except for Section 7.4 for the same reason as above.

The trace operator defined in Chapter 8 can be extended as long as Y is not contained in X^{Sing} using the embedded resolution. In general, due to the lack of Demailly regularization, we do not know how to define the trace operator.

Chapter 9 can be easily extended.

Chapter 10 is easy to extend since the partial Okounkov bodies are bimeromorphically invariant in the sense of Theorem 10.4.2.

Chapter 11 is unchanged, since we always take projective limits with respect to all models in that section.

Chapter 12 can be extended using the toric resolution [CLS11, Theorem 11.1.9].

Chapter 13 can be extended except for the parts involving the trace operator.

Chapter 14 can be easily extended by considering a resolution.

Appendix C
Almost semigroups

This appendix develops the theory of *almost semigroups* in $\mathbb{Z}^{n+1}$ that underpins the partial Okounkov body construction of Chapter 10. Almost semigroups generalize the integral semigroups arising in the classical Lazarsfeld–Mustaţă and Kaveh–Khovanskii constructions to the setting where the underlying graded family is no longer multiplicative. The main result is the construction of a convex body $\Delta(S) \subseteq \mathbb{R}^n$ out of an almost semigroup S. The corresponding statements were stated without proof in Chapter 10; the proofs are given here.

C.1 Convex bodies

Fix $n \in \mathbb{N}$.

Definition C.1.1 A *convex body* in $\mathbb{R}^n$ is a non-empty compact convex set.

We allow a convex body to have empty interior.

We write $\mathcal{K}_n$ for the set of convex bodies in $\mathbb{R}^n$.

Definition C.1.2 The *Hausdorff metric* between $K_1, K_2 \in \mathcal{K}_n$ is given by

$$d_{\mathrm{Haus}}(K_1, K_2) := \max\left\{ \sup_{x_1 \in K_1} \inf_{x_2 \in K_2} |x_1 - x_2|, \sup_{x_2 \in K_2} \inf_{x_1 \in K_1} |x_1 - x_2| \right\}.$$

It is well-known that the metric space $(\mathcal{K}_n, d_{\mathrm{Haus}})$ is complete. We will need the following fundamental theorem:

Theorem C.1.1 (Blaschke selection theorem) *The metric space $(\mathcal{K}_n, d_{\mathrm{Haus}})$ is locally compact.*

We refer to [Sch14, Theorem 1.8.7] for details.

Theorem C.1.2 *The Lebesgue volume* $\mathrm{vol}\colon \mathcal{K}_n \to \mathbb{R}_{\geq 0}$ *is continuous.*

See [Sch14, Theorem 1.8.20].

Theorem C.1.3 *Let $K_i, K \in \mathcal{K}_n$ ($i \in \mathbb{N}$). Then $K_i \xrightarrow{d_{\mathrm{Haus}}} K$ if and only if the following conditions hold:*

(1) *each point $x \in K$ is the limit of a sequence $x_i \in K_i$, and*
(2) *the limit of any convergent sequence $(x_{i_j})_{j\in\mathbb{N}}$ with $x_{i_j} \in K_{i_j}$ lies in K, where i_j is a strictly increasing sequence in $\mathbb{Z}_{>0}$.*

See [Sch14, Theorem 1.8.8].

For a convex body $K \subseteq \mathbb{R}^n$ and $\epsilon > 0$, we write $K_\epsilon := \{x \in K : B(x, \epsilon) \subseteq K\}$ for the *inner ϵ-shrinking* of K. (Compare with the outer thickening K^ϵ used in Corollary 5.2.1; the two are different operations.)

Lemma C.1.1 *Let $K \in \mathcal{K}_n$ be a convex body with positive volume and $K' \in \mathcal{K}_n$. Assume that, for some sufficiently large $k \in \mathbb{Z}_{>0}$, K' contains $K \cap (k^{-1}\mathbb{Z})^n$. Then $K' \supseteq K_{n^{1/2}k^{-1}}$.*

Proof Let $x \in K_{n^{1/2}k^{-1}}$. By assumption, the closed ball B with center x and radius $n^{1/2}k^{-1}$ is contained in K. Observe that x can be written as a convex combination of points in $B \cap (k^{-1}\mathbb{Z})^n$, which are contained in K' by assumption. It follows that $x \in K'$. □

Given a sequence of convex bodies K_i ($i \in \mathbb{N}$), we set

$$\varliminf_{i\to\infty} K_i = \overline{\bigcup_{i=0}^{\infty} \bigcap_{j\geq i} K_j}.$$

Suppose K is the limit of a subsequence of K_i, we have

$$\varliminf_{i\to\infty} K_i \subseteq K. \tag{C.1}$$

This is a simple consequence of Theorem C.1.3.

Lemma C.1.2 *Assume that $n > 0$. Let $K \subseteq \mathbb{R}^n$ be a convex body. Let*

$$t_{\min} := \min\{t \in \mathbb{R} : \{x_1 = t\} \cap K \neq \varnothing\}, \quad t_{\max} := \max\{t \in \mathbb{R} : \{x_1 = t\} \cap K \neq \varnothing\}.$$

Then for $t \in [t_{\min}, t_{\max}]$, the map

$$t \mapsto \{x_1 = t\} \cap K$$

is continuous with respect to the Hausdorff metric.

Here x_1 denotes the first coordinate in $\mathbb{R}^n$.

Proof We may assume that $t_{\min} < t_{\max}$ as otherwise there is nothing to prove.

For each $t \in [t_{\min}, t_{\max}]$, we write $K_t = \{x_1 = t\} \cap K$. Let $t_j \to t$ be a convergent sequence in $[t_{\min}, t_{\max}]$, we want to show that K_{t_j} converges to K_t with respect to the Hausdorff metric. Recall that this amounts to the following two assertions:

(1) For each convergent sequence $x_j \in K_{t_j}$ with limit x, we have $x \in K_t$;
(2) Given any $x \in K_t$, up to replacing t_j by a subsequence, we can find $x_j \in K_{t_j}$ converging to x.

The first assertion is obvious. Let us prove the second. Take $x = (t, x') \in K_t$. Up to replacing t_j by a subsequence and taking the symmetry into account, we may assume that $t_j > t$ for all j. In particular, $t < t_{\max}$.

We can find a point $y = (y^1, y') \in K$ such that $y^1 > t$ (for example, there is always such a point with $y^1 = t_{\max}$). Replacing t_j by a subsequence, we may assume that $t_j \in (t, y^1)$ for all j. Then it suffices to take

$$x_j = \frac{y^1 - t_j}{y^1 - t} x + \frac{t_j - t}{y^1 - t} y.$$

This completes the proof. □

Lemma C.1.3 *Let $D_j \subseteq \mathbb{R}^n$ $(j \geq 1)$ be a decreasing sequence of bounded convex sets. Assume that* $\operatorname{vol} \bigcap_j D_j > 0$. *Then*

$$\overline{\bigcap_{j=1}^{\infty} D_j} = \bigcap_{j=1}^{\infty} \overline{D_j}.$$

Proof The $\subseteq$ direction is clear. By convexity, it suffices to show that both sides have the same positive volume: indeed, a proper inclusion of closed convex sets with positive volume would cut off a non-empty relatively open convex piece by a separating hyperplane, hence would strictly decrease the volume. As the boundary of convex sets has zero Lebesgue measure, it follows that the volumes of both sides are equal to $\lim_{j\to\infty} \operatorname{vol} D_j$. □

Definition C.1.3 Let $K, K' \in \mathcal{K}_n$, their *Minkowski sum* is given by

$$K + K' := \{x + x' : x \in K, x' \in K'\}.$$

Proposition C.1.1 *The Minkowski sum $\mathcal{K}_n \times \mathcal{K}_n \to \mathcal{K}_n$ is continuous.*

See [Sch14, Page 139].

Theorem C.1.4 (Brunn–Minkowski) *Let $K, K' \in \mathcal{K}_n$. Then for any $t \in (0, 1)$, we have*

$$\operatorname{vol}\big((1-t)K' + tK\big) \geq \big(\operatorname{vol} K'\big)^{1-t} \big(\operatorname{vol} K\big)^{t}.$$

In other words, the volume is log concave. See [Sch14, Page 372].

C.2 The Okounkov bodies of almost semigroups

Fix an integer $n \geq 0$. Fix a closed convex cone $C \subseteq \mathbb{R}^n \times \mathbb{R}_{\geq 0}$ such that $C \cap \{x_{n+1} = 0\} = \{0\}$. Here x_{n+1} is the last coordinate of $\mathbb{R}^{n+1}$.

C.2.1 Generalities on semigroups

Write $\hat{\mathcal{S}}(C)$ for the set of subsets of $C \cap \mathbb{Z}^{n+1}$ and $\mathcal{S}(C)$ for the set of subsemigroups $S \subseteq C \cap \mathbb{Z}^{n+1}$. For each $k \in \mathbb{N}$ and $S \in \hat{\mathcal{S}}(C)$, we write

$$S_k := \{x \in \mathbb{Z}^n : (x, k) \in S\} .$$

Note that S_k is a finite set by our assumption on C.

We introduce a pseudometric on $\hat{\mathcal{S}}(C)$ as follows:

$$d_{\mathrm{sg}}(S, S') := \varlimsup_{k \to \infty} k^{-n} \left(|S_k| + |S'_k| - 2|(S \cap S')_k|\right) . \tag{C.2}$$

Here $|\bullet|$ denotes the cardinality of a finite set.

Lemma C.2.1 *The above defined d_{sg} is a pseudometric on $\hat{\mathcal{S}}(C)$.*

Proof Only the triangle inequality needs to be argued. Take $S, S', S'' \in \hat{\mathcal{S}}(C)$. We claim that for any $k \in \mathbb{N}$,

$$|S_k| + |S'_k| - 2|S_k \cap S'_k| + |S''_k| + |S'_k| - 2|S''_k \cap S'_k| \geq |S_k| + |S''_k| - 2|S_k \cap S''_k|.$$

From this the triangle inequality follows. To argue the claim, we rearrange it to the following form:

$$|S'_k| - |S_k \cap S'_k| \geq |S'_k \cap S''_k| - |S_k \cap S''_k|,$$

which is obvious. □

Given $S, S' \in \hat{\mathcal{S}}(C)$, we say S is equivalent to S' and write $S \sim S'$ if $d_{\mathrm{sg}}(S, S') = 0$. This is an equivalence relation by Lemma C.2.1.

Lemma C.2.2 *Given $S, S', S'' \in \hat{\mathcal{S}}(C)$, we have*

$$d_{\mathrm{sg}}(S \cap S'', S' \cap S'') \leq d_{\mathrm{sg}}(S, S').$$

In particular, if $S^i, S'^i \in \hat{\mathcal{S}}(C)$ ($i \in \mathbb{N}$) and $S^i \to S$, $S'^i \to S'$, then

$$S^i \cap S'^i \to S \cap S'.$$

Proof Observe that for any $k \in \mathbb{N}$,

$$|S_k \cap S''_k| - |S_k \cap S'_k \cap S''_k| \leq |S_k| - |S_k \cap S'_k|.$$

The same holds if we interchange S with S'. It follows that

$$|S_k \cap S''_k| + |S'_k \cap S''_k| - 2|S_k \cap S'_k \cap S''_k| \leq |S_k| + |S'_k| - 2|S_k \cap S'_k|.$$

The first assertion follows.

Next we compute

$$\begin{aligned} d_{\mathrm{sg}}(S^i \cap S'^i, S \cap S') \leq & d_{\mathrm{sg}}(S^i \cap S'^i, S^i \cap S') + d_{\mathrm{sg}}(S^i \cap S', S \cap S') \\ \leq & d_{\mathrm{sg}}(S'^i, S') + d_{\mathrm{sg}}(S^i, S) \end{aligned}$$

and the second assertion follows. □

The volume of $S \in \mathcal{S}(C)$ is defined as

$$\operatorname{vol} S := \lim_{k\to\infty} (ka)^{-n} |S_{ka}| = \overline{\lim_{k\to\infty}}\, k^{-n} |S_k|,$$

where a is a sufficiently divisible positive integer. The existence of the limit and its independence from a both follow from the more precise result [KK12, Theorem 2].

Lemma C.2.3 *Let $S, S' \in \mathcal{S}(C)$. Then*

$$|\operatorname{vol} S - \operatorname{vol} S'| \leq d_{\mathrm{sg}}(S, S').$$

Proof Choose a sufficient divisible positive integer a. Taking the limsup in the definition of d_{sg} along $k = ma$ gives

$$d_{\mathrm{sg}}(S, S') \geq \operatorname{vol} S + \operatorname{vol} S' - 2 \operatorname{vol}(S \cap S').$$

It follows that $\operatorname{vol} S - \operatorname{vol} S' \leq d_{\mathrm{sg}}(S, S')$ and $\operatorname{vol} S' - \operatorname{vol} S \leq d_{\mathrm{sg}}(S, S')$. □

We define $\overline{\mathcal{S}}(C)$ as the closure of $\mathcal{S}(C)$ in $\hat{\mathcal{S}}(C)$ with respect to the topology defined by the pseudometric d_{sg}. By Lemma C.2.3, $\operatorname{vol}\colon \mathcal{S}(C) \to \mathbb{R}$ admits a unique 1-Lipschitz extension to

$$\operatorname{vol}\colon \overline{\mathcal{S}}(C) \to \mathbb{R}. \tag{C.3}$$

Lemma C.2.4 *Suppose that $S, S' \in \overline{\mathcal{S}}(C)$ and $S \subseteq S'$. Then*

$$\operatorname{vol} S \leq \operatorname{vol} S'.$$

Proof Take sequences S^j, S'^j in $\mathcal{S}(C)$ such that $S^j \to S$, $S'^j \to S'$. By Lemma C.2.2, after replacing S^j by $S^j \cap S'^j$, we may assume that $S^j \subseteq S'^j$ for each j. Then our assertion follows easily. □

C.2.2 Okounkov bodies of semigroups

Given $S \in \hat{\mathcal{S}}(C)$, we will write $C(S) \subseteq C$ for the closed convex cone generated by $S \cup \{0\}$. Moreover, for each $k \in \mathbb{Z}_{>0}$, we define

$$\Delta_k(S) := \operatorname{Conv}\left\{k^{-1}x \in \mathbb{R}^n : x \in S_k\right\} \subseteq \mathbb{R}^n.$$

Here Conv denotes the convex hull.

Definition C.2.1 Let $\mathcal{S}'(C)$ be the subset of $\mathcal{S}(C)$ consisting of semigroups S such that S generates $\mathbb{Z}^{n+1}$ (as an Abelian group).

Note that for any $S \in \mathcal{S}'(C)$, the cone $C(S)$ has full dimension (i.e., the topological interior is non-empty). Given a full-dimensional subcone $C' \subseteq C$, the semigroup $C' \cap \mathbb{Z}^{n+1}$ lies in $\mathcal{S}'(C)$.

This class behaves well under intersections:

Lemma C.2.5 *Let $S, S' \in \mathcal{S}'(C)$. Assume that* $\mathrm{vol}(S \cap S') > 0$. *Then $S \cap S' \in \mathcal{S}'(C)$, and*

$$C(S \cap S') = C(S) \cap C(S').$$

The lemma obviously fails if $\mathrm{vol}(S \cap S') = 0$.

Proof We first observe that the cone $C(S) \cap C(S')$ has full dimension since otherwise $\mathrm{vol}(S \cap S') = 0$. Take a full-dimensional subcone C' in $C(S) \cap C(S')$ such that C' intersects the boundary of $C(S) \cap C(S')$ only at 0. It follows from [KK12, Theorem 1] that there is an integer $N > 0$ such that any $x \in \mathbb{Z}^{n+1} \cap C'$ with Euclidean norm at least N lies in $S \cap S'$. Therefore, $S \cap S' \in \mathcal{S}'(C)$. Since C' can be chosen around any ray in the interior of $C(S) \cap C(S')$, the same argument also shows that

$$C(S \cap S') = C(S) \cap C(S').$$

This completes the proof. □

We recall the following definition from [KK12].

Definition C.2.2 Given $S \in \mathcal{S}'(C)$, its *Okounkov body* is defined as follows

$$\Delta(S) := \{x \in \mathbb{R}^n : (x, 1) \in C(S)\}.$$

Theorem C.2.1 *For each $S \in \mathcal{S}'(C)$, we have*

$$\mathrm{vol}\, S = \lim_{k \to \infty} k^{-n} |S_k| = \mathrm{vol}\, \Delta(S) > 0. \tag{C.4}$$

Moreover, as $k \to \infty$,

$$\Delta_k(S) \xrightarrow{d_{\mathrm{Haus}}} \Delta(S). \tag{C.5}$$

This is essentially proved in [WN14, Lemma 4.8], which itself follows from a theorem of Khovanskii [Kho92]. Recall that (C.4) fails for a general $S \in \mathcal{S}(C)$, see [KK12, Theorem 2].

Proof The equalities (C.4) follow from the general theorem [KK12, Theorem 2].

It remains to prove (C.5). By the argument of [WN14, Lemma 4.8], for any compact set $K \subseteq \mathrm{Int}\, \Delta(S)$, there is $k_0 > 0$ such that for any $k \geq k_0$, $\alpha \in K \cap (k^{-1}\mathbb{Z})^n$ implies that $\alpha \in \Delta_k(S)$.

To conclude, fix $\epsilon > 0$. Choose $\delta > 0$ such that $K := \Delta(S)_\delta$ has positive volume and $d_{\mathrm{Haus}}(K, \Delta(S)) < \epsilon/2$. Applying Lemma C.1.1 to K, we find that $\Delta_k(S)$ contains $K_{n^{1/2}k^{-1}}$ for all large k. After increasing k, we may also assume that $d_{\mathrm{Haus}}(K, K_{n^{1/2}k^{-1}}) < \epsilon/2$. Since $\Delta_k(S) \subseteq \Delta(S)$, this implies (C.5). □

Corollary C.2.1 *Let* $S, S' \in \mathcal{S}'(C)$. *Assume that* $\operatorname{vol}(S \cap S') > 0$. *Then we have*

$$d_{\mathrm{sg}}(S, S') = \operatorname{vol}(S) + \operatorname{vol}(S') - 2\operatorname{vol}(S \cap S').$$

Proof This is a direct consequence of Lemma C.2.5 and (C.4). □

Lemma C.2.6 *Given* $S \in \mathcal{S}'(C)$, *we have* $S \sim \operatorname{Reg}(S)$.

Recall that the regularization $\operatorname{Reg}(S)$ of S is defined as $C(S) \cap \mathbb{Z}^{n+1}$.

Proof Since S and $\operatorname{Reg}(S)$ have the same Okounkov body, we have $\operatorname{vol} S = \operatorname{vol}\operatorname{Reg}(S)$ by Theorem C.2.1. By Corollary C.2.1 again,

$$d_{\mathrm{sg}}(\operatorname{Reg}(S), S) = \operatorname{vol}\operatorname{Reg}(S) - \operatorname{vol} S = 0.$$

This completes the proof. □

Lemma C.2.7 *Let* $S, S' \in \mathcal{S}'(C)$. *Assume that* $d_{\mathrm{sg}}(S, S') = 0$. *Then* $\Delta(S) = \Delta(S')$.

Proof Observe that $\operatorname{vol}(S \cap S') > 0$, as otherwise

$$d_{\mathrm{sg}}(S, S') \geq \operatorname{vol} S + \operatorname{vol} S' > 0,$$

which is a contradiction.

It follows from Lemma C.2.5 that $S \cap S' \in \mathcal{S}'(C)$. It suffices to show that $\Delta(S) = \Delta(S \cap S')$. In fact, suppose that this holds, since $\operatorname{vol}\Delta(S') = \operatorname{vol} S' = \operatorname{vol} S = \operatorname{vol}\Delta(S)$, the inclusion $\Delta(S') \supseteq \Delta(S \cap S') = \Delta(S)$ is an equality.

After replacing S' by $S \cap S'$, we still have $d_{\mathrm{sg}}(S, S') = 0$ by Lemma C.2.2, and now $S' \subseteq S$. Hence Corollary C.2.1 gives

$$0 = d_{\mathrm{sg}}(S, S') = \operatorname{vol} S - \operatorname{vol} S',$$

so $\operatorname{vol} S = \operatorname{vol} S'$. Therefore

$$\operatorname{vol}\Delta(S) = \operatorname{vol}\Delta(S') > 0$$

by (C.4). Therefore, they are equal. □

Lemma C.2.8 *Suppose that* $S^i \in \mathcal{S}'(C)$ *is a decreasing sequence such that*

$$\lim_{i\to\infty} \operatorname{vol} S^i > 0.$$

Then there is $S \in \mathcal{S}'(C)$ *such that* $S^i \to S$.

In general, one cannot simply take $S = \bigcap_i S^i$. For example, consider the sequence $S^i = S^1 \cap \{x_{n+1} \geq i\}$.

Proof By Lemma C.2.6, we may replace S^i by its regularization and assume that $S^i = C(S^i) \cap \mathbb{Z}^{n+1}$. We define

$$S = \left(\bigcap_{i=1}^{\infty} C(S^i)\right) \cap \mathbb{Z}^{n+1}.$$

The convex bodies $\Delta(S^i)$ form a decreasing sequence and satisfy

$$\lim_{i\to\infty} \operatorname{vol} \Delta(S^i) = \lim_{i\to\infty} \operatorname{vol} S^i > 0.$$

By continuity from above for Lebesgue measure,

$$\operatorname{vol} \bigcap_i \Delta(S^i) = \lim_{i\to\infty} \operatorname{vol} \Delta(S^i) > 0.$$

Set

$$K \coloneqq \bigcap_{i=1}^{\infty} C(S^i).$$

The cone K is full-dimensional. As noted above, $K \cap \mathbb{Z}^{n+1}$ generates $\mathbb{Z}^{n+1}$ as an Abelian group; hence $S = K \cap \mathbb{Z}^{n+1}$ lies in $\mathcal{S}'(C)$. Moreover, the closed convex cone generated by S is exactly K. Therefore

$$\Delta(S) = \{x : (x, 1) \in K\} = \bigcap_{i=1}^{\infty} \{x : (x, 1) \in C(S^i)\} = \bigcap_{i=1}^{\infty} \Delta(S^i).$$

By Corollary C.2.1 and Theorem C.2.1, we can compute the distance

$$d_{\mathrm{sg}}(S, S^i) = \operatorname{vol} S^i - \operatorname{vol} S = \operatorname{vol} \Delta(S^i) - \operatorname{vol} \Delta(S),$$

which tends to 0 by construction. □

C.2.3 Okounkov bodies of almost semigroups

Definition C.2.3 We define $\overline{\mathcal{S}'(C)}_{>0}$ to be the set of elements in the closure of $\mathcal{S}'(C)$ in $\hat{\mathcal{S}}(C)$ with positive volume. An element in $\overline{\mathcal{S}'(C)}_{>0}$ is called an *almost semigroup* in C.

Recall that the volume here is defined in (C.3).

Our goal is to prove the following theorem:

Theorem C.2.2 *The Okounkov body map $\Delta\colon \mathcal{S}'(C) \to \mathcal{K}_n$ as defined in Definition C.2.2 admits a unique continuous extension*

$$\Delta\colon \overline{\mathcal{S}'(C)}_{>0} \to \mathcal{K}_n. \tag{C.6}$$

Moreover, for any $S \in \overline{\mathcal{S}'(C)}_{>0}$, we have

$$\operatorname{vol} S = \operatorname{vol} \Delta(S). \tag{C.7}$$

Proof The uniqueness of the extension is clear once existence is known. Moreover, (C.7) follows easily from Theorem C.2.1 and Theorem C.1.2 by continuity. It remains to argue the existence of the continuous extension. We first construct an extension and prove its continuity.

Step 1. We construct the desired map (C.6). Let $S \in \overline{\mathcal{S}'(C)}_{>0}$. We wish to construct a convex body $\Delta(S) \in \mathcal{K}_n$.

Let $S^i \in \mathcal{S}'(C)$ be a sequence that converges to S such that

$$d_{\mathrm{sg}}(S^i, S^{i+1}) \leq 2^{-i}.$$

For each $i, j \geq 0$, we introduce

$$S^{i,j} = S^i \cap S^{i+1} \cap \cdots \cap S^{i+j}.$$

Then by Lemma C.2.2,

$$d_{\mathrm{sg}}(S^{i,j}, S^{i,j+1}) \leq 2^{-i-j}.$$

Take $i_0 > 0$ large enough so that for $i \geq i_0$, $\operatorname{vol} S^i > 2^{-1} \operatorname{vol} S$ and $2^{2-i} < \operatorname{vol} S$ and hence

$$\operatorname{vol} S^i - \operatorname{vol} S^{i,j} \leq d_{\mathrm{sg}}(S^{i,0}, S^{i,1}) + d_{\mathrm{sg}}(S^{i,1}, S^{i,2}) + \cdots + d_{\mathrm{sg}}(S^{i,j-1}, S^{i,j}) \leq 2^{1-i}.$$

It follows that $\operatorname{vol} S^{i,j} > 2^{-1} \operatorname{vol} S - 2^{1-i} > 0$ whenever $i \geq i_0$. In particular, by Lemma C.2.5, $S^{i,j} \in \mathcal{S}'(C)$ for $i \geq i_0$.

By Lemma C.2.8, for $i \geq i_0$, there exists $T^i \in \mathcal{S}'(C)$ such that $S^{i,j} \to T^i$ as $j \to \infty$. Moreover,

$$d_{\mathrm{sg}}(T^i, S) = \lim_{j\to\infty} d_{\mathrm{sg}}(S^{i,j}, S) \leq \varlimsup_{j\to\infty} d_{\mathrm{sg}}(S^{i,j}, S^i) + d_{\mathrm{sg}}(S^i, S) \leq 2^{1-i} + d_{\mathrm{sg}}(S^i, S).$$

Therefore, $T^i \to S$. We then define

$$\Delta(S) \coloneqq \overline{\bigcup_{i=i_0}^{\infty} \Delta(T^i)}.$$

In other words, we have defined

$$\Delta(S) \coloneqq \varliminf_{i\to\infty} \Delta(S^i).$$

Indeed, by the construction in the proof of Lemma C.2.8 and the cone equality in Lemma C.2.5,

$$\Delta(T^i) = \bigcap_{j\geq 0} \Delta(S^{i,j}) = \bigcap_{j\geq i} \Delta(S^j).$$

Thus the bodies $\Delta(T^i)$ are increasing in i and are all contained in the compact slice $\{x \in \mathbb{R}^n : (x, 1) \in C\}$. In particular, $\Delta(S)$ is a convex body and

$$\operatorname{vol}\Delta(S) = \lim_{i\to\infty} \operatorname{vol}\Delta(T^i) = \lim_{i\to\infty} \operatorname{vol} T^i = \operatorname{vol} S.$$

This is an honest limit: if Δ is the limit of a subsequence of $\Delta(S^i)$, then $\Delta(S) \subseteq \Delta$ by (C.1). Moreover,

$$\operatorname{vol}\Delta = \lim_{m\to\infty} \operatorname{vol}\Delta(S^{i_m}) = \lim_{m\to\infty} \operatorname{vol} S^{i_m} = \operatorname{vol} S.$$

Comparing the volumes, we find that equality holds. So by Theorem C.1.1,

$$\Delta(S) = \lim_{i\to\infty} \Delta(S^i). \tag{C.8}$$

Next we claim that $\Delta(S)$ as defined above does not depend on the choice of the sequence S^i. In fact, suppose that $S'^i \in \mathcal{S}'(C)$ is another sequence satisfying the same conditions as S^i. By Lemma C.2.2, we have

$$R^i = S^{i+1} \cap S'^{i+1} \longrightarrow S \cap S = S.$$

Since $\operatorname{vol} S > 0$ and the volume is continuous on $\overline{\mathcal{S}}(C)$, we have $\operatorname{vol} R^i > 0$ for all large i. Thus $R^i \in \mathcal{S}'(C)$ by Lemma C.2.5. Moreover,

$$d_{\mathrm{sg}}(R^i, R^{i+1}) \le d_{\mathrm{sg}}(S^{i+1}, S^{i+2}) + d_{\mathrm{sg}}(S'^{i+1}, S'^{i+2}),$$

so, after passing to a slightly faster subsequence if necessary, the same fast Cauchy condition holds. It follows that

$$\lim_{i\to\infty} \Delta(R^i) \subseteq \lim_{i\to\infty} \Delta(S^i).$$

Comparing the volumes, we find that equality holds. The same is true with S'^i in place of S^i. So we conclude that $\Delta(S)$ as in (C.8) does not depend on the choices we made.

Step 2. It remains to prove the continuity of Δ defined in Step 1. Suppose that $S^i \in \overline{\mathcal{S}'(C)}_{>0}$ is a sequence with limit $S \in \overline{\mathcal{S}'(C)}_{>0}$. We want to show that

$$\Delta(S^i) \xrightarrow{d_{\mathrm{Haus}}} \Delta(S). \tag{C.9}$$

We first reduce to the case where $S^i \in \mathcal{S}'(C)$. By (C.8), for each i, we can choose $T^i \in \mathcal{S}'(C)$ such that $d_{\mathrm{sg}}(S^i, T^i) < 2^{-i}$ and $d_{\mathrm{Haus}}(\Delta(S^i), \Delta(T^i)) < 2^{-i}$. If we have shown $\Delta(T^i) \xrightarrow{d_{\mathrm{Haus}}} \Delta(S)$, then (C.9) follows immediately.

Next we reduce to the case where $d_{\mathrm{sg}}(S^i, S^{i+1}) \le 2^{-i}$. Thanks to Theorem C.1.1, in order to prove (C.9), it suffices to show that each subsequence of $\Delta(S^i)$ admits a subsequence that converges to $\Delta(S)$. Hence, we may pass to a sufficiently fast subsequence and reduce to the required case.

After these reductions, (C.9) is nothing but (C.8). □

Corollary C.2.2 *Suppose that* $S, S' \in \overline{\mathcal{S}'(C)}_{>0}$ *with* $S \subseteq S'$. *Then*

$$\Delta(S) \subseteq \Delta(S'). \tag{C.10}$$

Proof Let $S^j, S'^j \in \mathcal{S}'(C)$ be elements such that $S^j \to S$, $S'^j \to S'$. Then it follows from Lemma C.2.2 that $S^j \cap S'^j \to S$. Since vol is continuous, for large j, $S^j \cap S'^j$ has positive volume and hence lies in $\mathcal{S}'(C)$ by Lemma C.2.5. We may therefore replace S^j by $S^j \cap S'^j$ and assume that $S^j \subseteq S'^j$. Hence, (C.10) follows from the continuity of Δ proved in Theorem C.2.2. □

Remark C.2.1 The construction of Δ is independent of the choice of C in the following sense. If C' is another cone satisfying the same assumptions as C and $C' \supseteq C$, then the Okounkov body map $\Delta \colon \overline{\mathcal{S}'(C')}_{>0} \to \mathcal{K}_n$ extends the corresponding map (C.6). We will use this fact without further comment.

Index

Symbols

A

B

C

D

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